

E34719240_1

The equation $2x^4 + x + y^3 + y + 3 = 0$ has no integer solutions

Size $h = 47$, length $l = 11.58$

Theorem 1. *The equation*

$$2x^4 + x + y^3 + y + 3 = 0 \tag{1}$$

has no integer solutions.

The proof combines quadratic Hilbert symbols over $\mathbb{Q}$ and cubic Hilbert symbols over $K = \mathbb{Q}(\omega)$, where $\omega^2 + \omega + 1 = 0$. To a hypothetical integer solution we attach the values of auxiliary polynomials H_i with rational coefficients, linked by a graph, and the values of auxiliary polynomials J_i with coefficients in $\mathbb{Z}[\omega]$, linked by 2 weighted graphs. By the product formula and the reciprocity law, the products of the quadratic symbols over all places are 1, and the sums of the cubic symbols over all places are 0. We show that the local contributions vanish at the primes outside a finite set S , determine the sign contributions at the real place, compute the local contributions jointly at the primes of S , and obtain a contradiction. Apart from polynomial identities, which can be checked by expanding both sides, finite computations with residues, and basic properties of the Legendre symbol and of the quadratic Hilbert symbol, the proof uses two standard facts from class field theory: the explicit formula for the cubic Hilbert symbol at the places not above 3, and the Hilbert reciprocity law.

1 Legendre symbols and Hilbert symbols

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. It is multiplicative in a , and it depends only on a modulo p [1, Chapter 5].

For a prime p , let $\mathbb{Q}_p$ be the field of p -adic numbers, and put $\mathbb{Q}_\infty = \mathbb{R}$. For v a prime or $v = \infty$, and non-zero $a, b \in \mathbb{Q}$, the Hilbert symbol $(a, b)_v$ is 1 if the equation $z^2 = ar^2 + bs^2$ has a solution $(r, s, z) \neq (0, 0, 0)$ in $\mathbb{Q}_v$, and -1 otherwise. In particular, $(a, b)_v = 1$ for every v if this equation has a non-zero solution in $\mathbb{Q}$. For an odd integer t , we put $\varepsilon(t) = (t - 1)/2$ and $\omega(t) = (t^2 - 1)/8$. We use the following facts [2, Chapter III, Proposition 2 and Theorems 1 and 3].

Theorem 2. *Let a, a' and b be non-zero integers, and let v be a prime or $v = \infty$.*

- (i) $(a, b)_\infty = -1$ if and only if $a < 0$ and $b < 0$.
- (ii) Let p be an odd prime, and write $a = p^\alpha u$ and $b = p^\beta w$ with $p \nmid uw$. Then $(a, b)_p = (-1)^{\alpha\beta\varepsilon(p)}(u/p)^\beta(w/p)^\alpha$.
- (iii) Write $a = 2^\alpha u$ and $b = 2^\beta w$ with u and w odd. Then $(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w) + \alpha\omega(w) + \beta\omega(u)}$.
- (iv) $(a, b)_v = 1$ for all but finitely many v , and $\prod_v (a, b)_v = 1$, the product over all primes v and $v = \infty$.
- (v) $(a, b)_v = (b, a)_v$ and $(aa', b)_v = (a, b)_v(a', b)_v$.

Lemma 3. Let a and b be non-zero integers. If p is an odd prime with $p \nmid a$ and $(a/p) = 1$, then $(a, b)_p = 1$. If $a \equiv 1 \pmod{8}$, then $(a, b)_2 = 1$.

Proof. The first claim is Theorem 2(ii) with $\alpha = 0$. For the second, $\varepsilon(a)$ and $\omega(a)$ are even, and Theorem 2(iii) with $\alpha = 0$ gives $(a, b)_2 = 1$. $\square$

Lemma 4. Let p be a prime. For a non-zero integer X , write $X = p^j u$ with $p \nmid u$. If p is odd, put $c_p(X) = (j, e) \in \mathbb{F}_2^2$, where $(u/p) = (-1)^e$; if $p = 2$, put $c_2(X) = (j, e_1, e_2) \in \mathbb{F}_2^3$, where $u \equiv (-1)^{e_1} 5^{e_2} \pmod{8}$; all entries are taken modulo 2. Then $c_p(XX') = c_p(X) + c_p(X')$ and $(X, X')_p = (-1)^{\beta_p(c_p(X), c_p(X'))}$ for non-zero integers X, X' , where β_p is the symmetric bilinear form

$$\begin{aligned} \beta_p((j, e), (j', e')) &= jj' \varepsilon(p) + je' + j'e && \text{if } p \text{ is odd,} \\ \beta_2((j, e_1, e_2), (j', e'_1, e'_2)) &= e_1 e'_1 + je'_2 + j'e_2. \end{aligned}$$

Proof. For odd p , this is Theorem 2(ii), as the Legendre symbol is multiplicative. For $p = 2$, the odd residues 1, 3, 5, 7 modulo 8 are $(-1)^{e_1} 5^{e_2}$ with $(e_1, e_2) = (0, 0), (1, 1), (0, 1), (1, 0)$, respectively; in each case $\varepsilon(u) \equiv e_1$ and $\omega(u) \equiv e_2 \pmod{2}$, so Theorem 2(iii) gives the formula, and c_2 is additive because $(e_1, e_2) \mapsto (-1)^{e_1} 5^{e_2}$ is an isomorphism from $\mathbb{F}_2^2$ to the group of odd residues modulo 8. $\square$

Lemma 5. Let p be a prime, $k \geq 1$, and let X be a non-zero integer with $X \equiv r \pmod{p^k}$, where $0 \leq r < p^k$. Then $c_p(X) \in b + D$ for the vector b and the subspace D given as follows. If $r = 0$, then $b = 0$ and D is the whole space. Otherwise write $r = p^j r'$ with $p \nmid r'$, so that $j < k$. If p is odd, or if $p = 2$ and $k - j \geq 3$, then $b = c_p(r)$ and $D = 0$. If $p = 2$ and $k - j = 2$, then $b = (j, e_1, 0)$ with $r' \equiv (-1)^{e_1} \pmod{4}$, and D is spanned by $(0, 0, 1)$. If $p = 2$ and $k - j = 1$, then $b = (j, 0, 0)$, and D is spanned by $(0, 1, 0)$ and $(0, 0, 1)$.

Proof. If $r \neq 0$, then $v_p(X) = j$ and $X/p^j \equiv r' \pmod{p^{k-j}}$. For odd p , the Legendre symbol of X/p^j is therefore that of r' . For $p = 2$, the residue of $X/2^j$ modulo 8 is that of r' if $k - j \geq 3$, and its residue modulo 4, which determines e_1 , is that of r' if $k - j = 2$. $\square$

Lemma 6. Let p be a prime, and let E be a set of edges on a finite set of vertices. For every vertex i , let b_i be a vector and D_i a subspace of $\mathbb{F}_2^2$ (of $\mathbb{F}_2^3$ if $p = 2$), and put $\sigma_i = \sum_{j \sim i} b_j$, the sum over the edges $\{i, j\} \in E$. Suppose that $\beta_p(d, \sigma_i) = 0$ for every vertex i and every $d \in D_i$, and that $\beta_p(d, d') = 0$ for every edge $\{i, j\} \in E$ and all $d \in D_i$ and $d' \in D_j$. Then

$$\sum_{\{i, j\} \in E} \beta_p(c_i, c_j) = \sum_{\{i, j\} \in E} \beta_p(b_i, b_j) \quad \text{whenever } c_i \in b_i + D_i \text{ for every } i.$$

By bilinearity, it suffices to check the hypotheses for d and d' in spanning sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As β_p is bilinear and symmetric, the left side equals the right side plus $\sum_i \beta_p(d_i, \sigma_i) + \sum_{\{i, j\} \in E} \beta_p(d_i, d_j)$, and both sums vanish. $\square$

2 Cubic Hilbert symbols over $\mathbb{Q}(\omega)$

The field. Let $K = \mathbb{Q}(\omega)$, where $\omega^2 + \omega + 1 = 0$. Its ring of integers is $\mathbb{Z}[\omega]$, and every element of K can be written uniquely as $A + B\omega$ with $A, B \in \mathbb{Q}$. We write

$$\langle A, B \rangle = A + B\omega,$$

also when A and B are polynomials in x and y . Then $\langle A, B \rangle \langle C, D \rangle = \langle AC - BD, AD + BC - BD \rangle$, and the norm is $N(\langle A, B \rangle) = A^2 - AB + B^2$. In particular, a value $\langle A, B \rangle$ with $A, B \in \mathbb{Z}$ is zero if and only if $A = B = 0$.

Places. The finite places of K correspond to the non-zero prime ideals of $\mathbb{Z}[\omega]$. A prime $p \equiv 2 \pmod{3}$ is inert: there is one place above p , with uniformizer p and residue field $\mathbb{Z}[\omega]/p\mathbb{Z}[\omega]$ with p^2 elements. If $p \equiv 1 \pmod{3}$, the polynomial $t^2 + t + 1$ has two roots r modulo p , and for each of them there is a place v_r above p , at which ω is the root of $t^2 + t + 1$ in $\mathbb{Z}_p$ congruent to r modulo p ; the completion is $\mathbb{Q}_p$, the uniformizer is p , the residue field is $\mathbb{F}_p$, and ω reduces to r . The prime 3 is ramified: $\pi = 1 - \omega$ generates the unique prime ideal above 3, $\pi^2 = -3\omega$, and $\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega] = \mathbb{F}_3$, where $\langle A, B \rangle$ reduces to $A + B$. The only infinite place of K is complex. For a finite place v we write K_v for the completion and $v(a)$ for the normalized valuation, and we call a a *unit* at v if $v(a) = 0$.

Hilbert symbols. For a place v of K and $a, b \in K_v^\times$, the cubic Hilbert symbol $(a, b)_v$ is a cube root of unity, see [3, Chapter V, §3]. We write $(a, b)_v = \omega^{\theta_v(a, b)}$ with $\theta_v(a, b) \in \mathbb{Z}/3\mathbb{Z}$, and we use the following three facts [3, Chapter V, §3 and Chapter VI, §8].

- (H1) $\theta_v(a, bc) = \theta_v(a, b) + \theta_v(a, c)$, $\theta_v(a, b) = -\theta_v(b, a)$, and $\theta_v(a, c^3) = 0$. In particular, $\theta_v(a, b)$ depends only on the classes of a and b in $K_v^\times/K_v^{\times 3}$.
- (H2) Let v be a finite place not above 3, let q be the number of elements of its residue field, and let ϖ be a uniformizer at v . Write $a = \varpi^e u$ and $b = \varpi^f w$ with units u and w , and define $\chi(u) \in \mathbb{Z}/3\mathbb{Z}$ by $\bar{u}^{(q-1)/3} = \bar{\omega}^{\chi(u)}$, where the bar denotes the reduction to the residue field. Then $\theta_v(a, b) = e\chi(w) - f\chi(u)$. In particular, $\theta_v(a, b) = 0$ if a and b are both units.
- (H3) (Hilbert reciprocity.) For $a, b \in K^\times$, $\theta_v(a, b) = 0$ for all but finitely many places v , and $\sum_v \theta_v(a, b) = 0$, where the sum is over all places of K . At the complex place, $\theta_v(a, b) = 0$.

The general formula in (H2) contains an additional term $ef\chi(-1)$, which vanishes because $-1 = (-1)^3$ is a cube. Formula (H2) fixes the normalization of the symbols; with the opposite sign convention all values θ_v change sign, and the proof below is the same.

Classes at the places not above 3. Let $p \neq 3$ be a prime and v a place above p . For $a = p^e u \in K_v^\times$ with a unit u , put $\text{cl}_v(a) = (e \bmod 3, \chi(u)) \in \mathbb{F}_3^2$. By (H2) and (H1), cl_v is additive, $\text{cl}_v(a)$ determines the class of a modulo cubes, and $\theta_v(a, b) = e\chi(w) - f\chi(u)$ for $\text{cl}_v(a) = (e, \chi(u))$ and $\text{cl}_v(b) = (f, \chi(w))$. We put $\text{cl}_p(a) = \text{cl}_v(a) \in \mathbb{F}_3^2$ if $p \equiv 2 \pmod{3}$, and $\text{cl}_p(a) = (\text{cl}_{v_r}(a), \text{cl}_{v_{r'}}(a)) \in \mathbb{F}_3^4$ if $p \equiv 1 \pmod{3}$, where $r < r'$ are the roots of $t^2 + t + 1$ in $\{1, \dots, p-1\}$; accordingly, θ_p is the sum of the forms θ_v over the places v above p . For $p \equiv 2 \pmod{3}$ and a rational integer u prime to p , we have $\bar{u} \in \mathbb{F}_p$, so that $\bar{u}^{(p^2-1)/3} = (\bar{u}^{p-1})^{(p+1)/3} = 1$ and $\chi(u) = 0$.

The place above 3.

Lemma 7. *Let v be the place above 3. Every non-zero element of K_v is equal to $\pi^a \omega^b 2^c (1 + 3\omega)^d$ times a cube, for unique $a, b, c, d \in \mathbb{Z}/3\mathbb{Z}$, and every unit congruent to 1 modulo 9 is a cube. Put $\text{cl}_3(A) = (a, b, c, d) \in \mathbb{F}_3^4$. Then cl_3 is additive, and $\theta_v(A, B) = \text{cl}_3(A)^\top M_3 \text{cl}_3(B)$, where*

$$M_3 = \begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \pmod{3}.$$

Proof. For $\alpha \in \mathbb{Z}[\omega]$, the polynomial $g(s) = s + 3s^2 + 3s^3 - \alpha$ satisfies $g'(s) \equiv 1 \pmod{\pi}$, so by Hensel's lemma it has a root s in the completion of $\mathbb{Z}[\omega]$, and then $(1 + 3s)^3 = 1 + 9(s + 3s^2 + 3s^3) = 1 + 9\alpha$. Hence every unit congruent to 1 modulo 9 is a cube in K_v , and the class of a unit modulo cubes depends only on its residue modulo 9. There are 54 units in $\mathbb{Z}[\omega]/(9)$,

and a direct enumeration shows that the 27 products $\omega^b 2^c (1 + 3\omega)^d$ with $b, c, d \in \{0, 1, 2\}$ are pairwise distinct modulo 9, also after multiplication by $-1 = (-1)^3$. Hence they represent all 27 classes of units modulo cubes. Since π^3 is a cube times a unit, the exponent of π is the valuation modulo 3, and the first statement follows.

By (H1), it remains to compute θ_v on the 16 ordered pairs of the elements $\pi, \omega, 2$ and $1 + 3\omega$. Their norms are 3, 1, 4 and 7, hence, for each such pair, all the symbols at the places not above 2, 3 and 7 vanish by (H2), and, by (H3), θ_v at the place above 3 is minus the sum of the values at the place above 2 and at the two places above 7. These places correspond to $\bar{\omega} = 2$ and $\bar{\omega} = 4$ in $\mathbb{F}_7$. The classes (e, χ) of the four elements at these places are

	2	$\bar{\omega} = 2$	$\bar{\omega} = 4$
π	(0, 2)	(0, 0)	(0, 2)
ω	(0, 1)	(0, 2)	(0, 2)
2	(1, 0)	(0, 2)	(0, 1)
$1 + 3\omega$	(0, 2)	(1, 0)	(0, 0)

where, at the places above 7, $\chi(u)$ is defined by $\bar{u}^2 = \bar{\omega}^{\chi(u)}$ in $\mathbb{F}_7$, and 7 is a uniformizer. Formula (H2) and the sum over these three places give M_3 . $\square$

We write θ_3 for the form of Lemma 7. Since $3 = -\omega^2 \pi^2$ and $4 = 2^2$, we have $\text{cl}_3(3) = (2, 2, 0, 0)$ and $\text{cl}_3(4) = (0, 0, 2, 0)$.

Residue classes. The local computations use the following description of the classes of the values of a polynomial on a residue class.

Lemma 8. *Let p be a prime, $k \geq 1$, and let $z \in \mathbb{Z}[\omega]$ be non-zero with $z \equiv \zeta \pmod{p^k}$, where $\zeta = \langle A, B \rangle$ with $0 \leq A, B < p^k$. Then $\text{cl}_p(z) \in b + D$ for the vector b and the subspace D given as follows.*

- (i) $p \equiv 2 \pmod{3}$. If $\zeta \neq 0$, then $b = \text{cl}_p(\zeta)$ and $D = 0$. If $\zeta = 0$, then $b = 0$, and D is $\mathbb{F}_3^2$, or the span of $(1, 0)$ if $z \in \mathbb{Z}$.
- (ii) $p \equiv 1 \pmod{3}$. For each root r , let ρ be the root of $t^2 + t + 1$ modulo p^k congruent to r . If $A + B\rho \not\equiv 0 \pmod{p^k}$, the two coordinates of $\text{cl}_{v_r}(z)$ are those of $\text{cl}_{v_r}(\zeta)$; otherwise these two coordinates are arbitrary.
- (iii) $p = 3$, $z \notin \mathbb{Z}$. If $\zeta \neq 0$, let $a = v(\zeta) < 2k$ be its valuation at π and $m = 2k - a$. Then $b = \text{cl}_3(\zeta)$, and $D = 0$ if $m \geq 4$, while $D = D_m$ if $m \leq 3$, where D_1 is spanned by $(0, 1, 0, 0)$, $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$, D_2 by $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$, and D_3 by $(0, 0, 1, 1)$. If $\zeta = 0$, then $b = 0$ and $D = \mathbb{F}_3^4$.
- (iv) $p = 3$, $z \in \mathbb{Z}$, so that $B = 0$. If $3^k \nmid A$, let $j = v_3(A)$. Then $b = \text{cl}_3(A)$ and $D = 0$ if $k - j \geq 2$, while $b = j \text{cl}_3(3)$ and D is spanned by $\text{cl}_3(4)$ if $k - j = 1$. If $3^k \mid A$, then $b = 0$ and D is spanned by $\text{cl}_3(3)$ and $\text{cl}_3(4)$.

Proof. (i) If $\zeta \neq 0$, then $e = v(z) = v(\zeta) < k$ and $z/p^e \equiv \zeta/p^e \pmod{p}$, so that χ is determined. If $z \in \mathbb{Z}$, its unit part is a rational integer, and $\chi = 0$. (ii) At v_r , z and ζ are the p -adic numbers $A' + B'\omega$ and $A + B\omega$ with $\omega \equiv \rho \pmod{p^k}$, and the argument of (i) applies. (iii) If $\zeta \neq 0$, then $v(z) = a$, and $z/\pi^a \equiv \zeta/\pi^a \pmod{\pi^m}$. By Lemma 7, the class of a unit depends only on its residue modulo $9 = \omega\pi^4$; the units congruent to 1 modulo π^m form a subgroup, and a direct enumeration of their residues modulo 9 shows that their classes form D_m . (iv) If $3^k \nmid A$, then $z = 3^j u$ with a rational unit $u \equiv A/3^j \pmod{3^{k-j}}$. The class of a rational unit depends only on its residue modulo 9, the units ± 1 are cubes, and the rational units congruent to 1 modulo 3 are 4^s times a unit congruent to 1 modulo 9; hence the classes are as stated, and if $3^k \mid A$ the valuation j is also unknown. $\square$

Lemma 9. *Let p be a prime, and consider vertices i with vectors b_i , subspaces D_i and vectors γ_i , and a set of edges (i, j) with $i < j$ and weights $w_{ij} \in \{1, 2\}$. Put $\rho_{ij} = w_{ij}$ and $\rho_{ji} = -w_{ij}$ for the edges, and $\sigma_i = \gamma_i + \sum_j \rho_{ij} b_j$. Suppose that $\theta_p(d, \sigma_i) = 0$ for every i and every $d \in D_i$, and that $\theta_p(d, d') = 0$ for every edge (i, j) and all $d \in D_i, d' \in D_j$. Then*

$$\sum_{(i,j)} w_{ij} \theta_p(c_i, c_j) + \sum_i \theta_p(c_i, \gamma_i) = \sum_{(i,j)} w_{ij} \theta_p(b_i, b_j) + \sum_i \theta_p(b_i, \gamma_i)$$

whenever $c_i \in b_i + D_i$ for every i . By bilinearity, it suffices to check the hypotheses for d and d' in spanning sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As θ_p is bilinear and $\theta_p(u, v) = -\theta_p(v, u)$, the left side equals the right side plus $\sum_i \theta_p(d_i, \sigma_i) + \sum_{(i,j)} w_{ij} \theta_p(d_i, d_j)$, and both sums vanish. $\square$

3 The quadratic certificate

Put $F(x, y) = 2x^4 + x + y^3 + y + 3$, so that (1) is the equation $F(x, y) = 0$. The vertices are the polynomials

$$\begin{aligned} H_0 &= -y - 1, \\ H_1 &= x^2 + x + 1, \\ H_2 &= 3x^2 + 3xy + y^2 + y + 1, \\ H_3 &= x^2 + xy + x + y^2 - y + 1, \\ H_4 &= 4x^2 + 2xy + 4x + y^2 + y + 1, \\ H_5 &= x^2 - xy - 2x + y^2 + y + 1, \\ H_6 &= 9x^2 - 9xy + 3y^2 + 3y + 3, \\ H_7 &= x^2 - 2x - y - 1, \\ H_8 &= x - 1, \\ H_9 &= xy + x, \\ H_{10} &= x^2 - xy + y^2, \\ H_{11} &= -xy + 2x - y, \\ H_{12} &= x^2 - xy - x + y^2 - y + 1, \\ H_{13} &= -y + 1, \\ H_{14} &= xy + 2x + 2, \\ H_{15} &= x^2 - xy + 3x + y^2 + 3, \\ H_{16} &= y + 1, \\ H_{17} &= -2xy + 3x - y, \\ H_{18} &= 9x^2 + 3y^2, \\ H_{19} &= -2y, \\ H_{20} &= -x^3 - x^2 - y, \\ H_{21} &= x^4 + x^2y + y^2, \\ H_{22} &= x^3 + x^2 + x - y, \\ H_{23} &= x^4 - x^2y + x^2 + y^2 + y + 1, \\ H_{24} &= x^2 + 1, \\ H_{25} &= x^3 + x^2 - y, \end{aligned}$$

$$\begin{aligned}
H_{26} &= x^4 + x^3 - x^2y - 2xy + x + y^2 - y + 1, \\
H_{27} &= x^2 - 1, \\
H_{28} &= -x^3 - x^2 - 2x - y - 1, \\
H_{29} &= x^4 + x^3 - x^2y + 2x^2 + xy + 2x + y^2 + y + 1, \\
H_{30} &= -x^2 - x - 1, \\
H_{31} &= x^3 - x - y - 2, \\
H_{32} &= x^4 + x^2y + y^2 + 3y + 3, \\
H_{33} &= x^3 + x^2 + x - y - 1, \\
H_{34} &= x^4 - x^2y + y^2 + 3y + 3, \\
H_{35} &= -xy + x - y - 1, \\
H_{36} &= -y, \\
H_{37} &= -x - y, \\
H_{38} &= y^2 - y + 1, \\
H_{39} &= -1, \\
H_{40} &= x - y + 1, \\
H_{41} &= x + 1, \\
H_{42} &= x^2 + xy - 2x + y^2 - y + 1, \\
H_{43} &= x^2 + xy - x + y^2 + y + 1, \\
H_{44} &= x^2 - x + 1, \\
H_{45} &= x + y + 1, \\
H_{46} &= x + y - 1, \\
H_{47} &= xy + 2x - y + 1, \\
H_{48} &= 2x + y + 1, \\
H_{49} &= -xy + 2x + y + 1, \\
H_{50} &= -2xy + 3x + y, \\
H_{51} &= x^3 - y + 1, \\
H_{52} &= -x^3 - x - y - 1, \\
H_{53} &= x^4 + x^3 - x^2y + x^2 + xy + 3x + y^2 + 3y + 3, \\
H_{54} &= x - y, \\
H_{55} &= -x^2 - x - y + 1, \\
H_{56} &= -x^2 + 2x + y + 1, \\
H_{57} &= -x + 1, \\
H_{58} &= -x + y, \\
H_{59} &= y^2 + y + 1, \\
H_{60} &= xy - x + y + 1, \\
H_{61} &= x^2 + y, \\
H_{62} &= 3x^2 - 3xy + y^2 + y + 1, \\
H_{63} &= -xy + y, \\
H_{64} &= xy - 2, \\
H_{65} &= -x^3 - x^2 + y + 1, \\
H_{66} &= -x^3 - x^2 + y + 2,
\end{aligned}$$

$$\begin{aligned}
H_{67} &= 2xy - 3x - y, \\
H_{68} &= y^2 + 3y + 3, \\
H_{69} &= xy - y - 1, \\
H_{70} &= x^2 + xy + x + y^2 + 2y + 1, \\
H_{71} &= -xy + y - 2, \\
H_{72} &= x^2 - xy + 3x + y^2 - 3y + 3, \\
H_{73} &= x^2 + 3x + 3, \\
H_{74} &= xy + x + y, \\
H_{75} &= 2x - y + 2, \\
H_{76} &= x^2 - 2xy + x + y^2 - y + 1, \\
H_{77} &= xy + 2x + y + 3, \\
H_{78} &= -x^3 - 2x^2 - 3x - y - 2, \\
H_{79} &= -x^3 - 2x^2 - x - y - 1, \\
H_{80} &= x^4 + x^3 + x^2y + 2x^2 + 2xy + 2x + y^2 + 2y + 1, \\
H_{81} &= x + 2, \\
H_{82} &= 2x + 1, \\
H_{83} &= x^2 + xy + 3x + y^2 + 3, \\
H_{84} &= 9x^2 - 9x + 3, \\
H_{85} &= -3xy + 2x + 2y - 1, \\
H_{86} &= -3xy - 4x + y, \\
H_{87} &= x^2 + xy + 3x + y^2 + 3y + 3, \\
H_{88} &= -3x^3 - 2x^2 - x + y + 3, \\
H_{89} &= -x^2 - xy - 2x - y^2 - 2y + 1, \\
H_{90} &= x^3 + x^2y + x^2 + xy + 2x, \\
H_{91} &= 2xy + 4x, \\
H_{92} &= x^2 + xy + 2x + y^2 + 2y - 1, \\
H_{93} &= 2x^2 + 2xy - x + y^2 - 1, \\
H_{94} &= x^2 + xy - x + y^2 - 2, \\
H_{95} &= -2x^2 - 2xy + x + 2y + 1, \\
H_{96} &= -xy - 2x - y^2 + y, \\
H_{97} &= 9x^2 + 3y^2 - 6y + 3, \\
H_{98} &= x^3 - x^2y + x + y^2, \\
H_{99} &= -2xy + y, \\
H_{100} &= -x^2 + xy - x + y^2, \\
H_{101} &= 2xy - x + y, \\
H_{102} &= y - 1, \\
H_{103} &= 2xy - x + 2y, \\
H_{104} &= 2xy + x - y^2 + y, \\
H_{105} &= x^2 - x - y, \\
H_{106} &= -x^3 + x^2y + xy + x - y^2,
\end{aligned}$$

$$\begin{aligned}
H_{107} &= -x^2 + 1, \\
H_{108} &= -x^3 - xy - y^2 - y, \\
H_{109} &= -x^3 + x^2y - x^2 - x - y^2 - y, \\
H_{110} &= xy - 2x + y, \\
H_{111} &= x^2 + 3, \\
H_{112} &= -2, \\
H_{113} &= x - y^2 - 2y, \\
H_{114} &= x^2 - xy - x - y^2, \\
H_{115} &= -2xy + 2x, \\
H_{116} &= x^3 + x^2 + x + y^2, \\
H_{117} &= x^3 + x^2y - x + y^2 - 2, \\
H_{118} &= x^2 - xy + 2x + y^2 - y + 1, \\
H_{119} &= -x^2y + x^2 - xy + x + y^2, \\
H_{120} &= x^4 + x^3 - 2x^2y + x^2 - xy + y^2, \\
H_{121} &= x^3 - x^2y + x^2 - xy - x + y^2 - 1, \\
H_{122} &= x^3 - x^2y + x^2 + x + y^2 + y + 1, \\
H_{123} &= x^4 - 2x^3 + x^2y + x^2 - xy + y^2, \\
H_{124} &= x^3 - x^2y + x + 1, \\
H_{125} &= xy - y^2 - y, \\
H_{126} &= -2xy - x - y^2 - y + 1, \\
H_{127} &= x + y^2 + 2y - 1, \\
H_{128} &= -x^3 + 2x^2 - x + y^2, \\
H_{129} &= -x^2 + x, \\
H_{130} &= -2xy + 2x + 2y^2 - 2, \\
H_{131} &= -2x + y^2 + 2y - 1, \\
H_{132} &= xy - 2x + y^2 + y - 1, \\
H_{133} &= -x^3 + x^2 - xy + x - y^2 + 1, \\
H_{134} &= -x^3 + x^2y - x^2 - y^2 + 2, \\
H_{135} &= xy - 3x + y + 1, \\
H_{136} &= -x^2 - y^2 + 2y, \\
H_{137} &= -x^2 + 2xy - 2x - y^2 + y + 1, \\
H_{138} &= -x^3 - 2x^2 + y^2 - y, \\
H_{139} &= x + y^2 + y, \\
H_{140} &= 5xy + x - y^2 + 4y - 3, \\
H_{141} &= -x^2 + 2xy - x - y^2 - y - 2, \\
H_{142} &= 2xy + 2x + 3y + 3, \\
H_{143} &= -x^3 - x^2y - 2x^2 - xy - x - y^2 - 2y - 1, \\
H_{144} &= 5xy + 3x + y^2 + 3y, \\
H_{145} &= -2xy - x - y + 1, \\
H_{146} &= -xy - 3x - y^2 + 3y,
\end{aligned}$$

$$\begin{aligned}
H_{147} &= 2x^3 + x^2y + x^2 + 2x - y^2 - y + 1, \\
H_{148} &= -xy - 2y, \\
H_{149} &= 5xy + 4x - y^2 + 2y + 2, \\
H_{150} &= -3x^3 + x^2y - 3x^2 - xy - 3x - 2y^2 - y, \\
H_{151} &= 2xy - 3x - 2y^2 + y, \\
H_{152} &= x^4 - x^2y - xy, \\
H_{153} &= -x^4 - 2x^2y - x^2 + y, \\
H_{154} &= x^3 + x^2 + 2y, \\
H_{155} &= -x^2y + x^2 + y, \\
H_{156} &= -x^4 - x^3 - x^2 - x - y - 1, \\
H_{157} &= -x^4 + x^3 - xy - y, \\
H_{158} &= -x^3 - x^2y + xy + 1, \\
H_{159} &= x^3 - x^2y - x^2 + xy + 2x, \\
H_{160} &= x^3 - x^2y + 2x^2 + x + y^2 - y + 1, \\
H_{161} &= -x^4 + x^2y + x^2 + xy + x, \\
H_{162} &= x^2y - x^2 + xy + x - y^2, \\
H_{163} &= -2x^2 + xy - x + y, \\
H_{164} &= x^3 + 3x^2 + x - 2y - 1, \\
H_{165} &= -x^3 - x^2y + x + 1, \\
H_{166} &= -x^2y - x^2 - x + 2y, \\
H_{167} &= -x^2y + x + y + 1, \\
H_{168} &= -x^3 - x^2y - x^2 - xy + y + 1, \\
H_{169} &= x^2y + x^2 + 1, \\
H_{170} &= -x^3 + x^2y + x + 2, \\
H_{171} &= -x^3 + x^2y + x^2 - x - y^2 - y + 1, \\
H_{172} &= x^3 + 2x + 2y + 3, \\
H_{173} &= -x^4 + x^3 + x^2 - 2xy - x - y, \\
H_{174} &= -y^2, \\
H_{175} &= 2xy + 6x, \\
H_{176} &= 2, \\
H_{177} &= xy + 8x - 3y, \\
H_{178} &= 3x + 2y^2 + 3y, \\
H_{179} &= xy + 3x - y^2 - y, \\
H_{180} &= -x + y^2 - 1, \\
H_{181} &= -x^3 - x^2y + x^2 + y, \\
H_{182} &= 2x^2 + x - 1, \\
H_{183} &= -2xy + 2x + y^2 - 2y - 2, \\
H_{184} &= -x^2 + 2xy - y^2 + 1, \\
H_{185} &= 75, \\
H_{186} &= -5x - 4,
\end{aligned}$$

$$\begin{aligned}
H_{187} &= 21, \\
H_{188} &= -4x - 1, \\
H_{189} &= -5x + 4, \\
H_{190} &= -9x + 4, \\
H_{191} &= 7, \\
H_{192} &= -3, \\
H_{193} &= -30x - 10y + 10, \\
H_{194} &= 300, \\
H_{195} &= 10, \\
H_{196} &= -4x - y, \\
H_{197} &= 2x + 10y - 8, \\
H_{198} &= 84, \\
H_{199} &= 2x - 2y + 3, \\
H_{200} &= x - 5y + 4, \\
H_{201} &= x + y + 3, \\
H_{202} &= 3, \\
H_{203} &= 2x + 4y + 6, \\
H_{204} &= 12, \\
H_{205} &= 2x - y + 3, \\
H_{206} &= 4x + 4y + 2, \\
H_{207} &= 3x - y + 1, \\
H_{208} &= -2y + 2, \\
H_{209} &= -x^2 - 2y, \\
H_{210} &= x^2 - y, \\
H_{211} &= -x - y - 1, \\
H_{212} &= 5x^2 + 4y + 3, \\
H_{213} &= -y - 2, \\
H_{214} &= x^2 - x - 2y - 3, \\
H_{215} &= x + 2y - 1, \\
H_{216} &= 14x + 14, \\
H_{217} &= -6x + 10y, \\
H_{218} &= 28, \\
H_{219} &= -2y - 3, \\
H_{220} &= -4x - 5y - 5, \\
H_{221} &= -x^2 + x - 2y, \\
H_{222} &= 2x - y - 2, \\
H_{223} &= -2x + 2y - 3, \\
H_{224} &= 3x - 2y - 1, \\
H_{225} &= -2x - y - 1, \\
H_{226} &= 1.
\end{aligned}$$

The edges are $\{0, 1\}, \{0, 3\}, \{0, 4\}, \{0, 6\}, \{0, 23\}, \{0, 87\}, \{0, 185\}, \{1, 7\}, \{1, 9\}, \{1, 11\}, \{1, 14\}, \{1, 17\}, \{1, 20\}, \{1, 22\}, \{1, 25\}, \{1, 28\}, \{1, 31\}, \{1, 33\}, \{1, 35\}, \{1, 40\}, \{1, 48\}, \{1, 53\}, \{1, 56\},$

$\{1, 58\}, \{1, 60\}, \{1, 61\}, \{1, 186\}, \{2, 7\}, \{2, 8\}, \{2, 41\}, \{2, 45\}, \{2, 48\}, \{2, 56\}, \{2, 57\}, \{2, 82\},$
 $\{2, 89\}, \{2, 90\}, \{2, 91\}, \{2, 92\}, \{2, 93\}, \{2, 94\}, \{2, 95\}, \{2, 193\}, \{2, 195\}, \{3, 46\}, \{3, 55\},$
 $\{3, 100\}, \{3, 101\}, \{3, 102\}, \{3, 115\}, \{3, 116\}, \{3, 117\}, \{3, 136\}, \{3, 158\}, \{3, 165\}, \{3, 182\},$
 $\{3, 197\}, \{4, 36\}, \{4, 49\}, \{4, 93\}, \{4, 132\}, \{4, 147\}, \{4, 148\}, \{4, 149\}, \{4, 206\}, \{5, 8\}, \{5, 35\},$
 $\{5, 48\}, \{5, 60\}, \{5, 162\}, \{5, 171\}, \{5, 174\}, \{5, 176\}, \{5, 183\}, \{5, 184\}, \{5, 222\}, \{6, 37\},$
 $\{6, 54\}, \{6, 130\}, \{6, 149\}, \{6, 177\}, \{6, 183\}, \{6, 213\}, \{6, 224\}, \{8, 12\}, \{8, 43\}, \{8, 62\}, \{9, 10\},$
 $\{10, 36\}, \{10, 46\}, \{10, 74\}, \{10, 96\}, \{10, 98\}, \{10, 99\}, \{10, 100\}, \{10, 101\}, \{10, 103\}, \{10, 104\},$
 $\{10, 105\}, \{10, 106\}, \{10, 108\}, \{10, 109\}, \{10, 110\}, \{10, 113\}, \{10, 114\}, \{10, 196\}, \{11, 12\},$
 $\{12, 41\}, \{12, 63\}, \{12, 85\}, \{12, 115\}, \{12, 131\}, \{12, 132\}, \{12, 133\}, \{12, 134\}, \{12, 135\},$
 $\{12, 136\}, \{12, 137\}, \{12, 200\}, \{13, 23\}, \{13, 43\}, \{13, 76\}, \{13, 80\}, \{14, 15\}, \{15, 47\}, \{15, 48\},$
 $\{15, 64\}, \{15, 77\}, \{15, 89\}, \{15, 92\}, \{15, 104\}, \{15, 105\}, \{15, 140\}, \{15, 141\}, \{15, 142\},$
 $\{15, 145\}, \{15, 146\}, \{15, 201\}, \{15, 205\}, \{16, 70\}, \{17, 18\}, \{18, 19\}, \{18, 37\}, \{18, 46\}, \{18, 50\},$
 $\{18, 67\}, \{18, 105\}, \{18, 144\}, \{18, 146\}, \{18, 150\}, \{18, 151\}, \{18, 175\}, \{18, 176\}, \{18, 178\},$
 $\{18, 179\}, \{18, 217\}, \{20, 21\}, \{21, 138\}, \{21, 152\}, \{21, 153\}, \{21, 154\}, \{21, 155\}, \{21, 209\},$
 $\{22, 23\}, \{23, 24\}, \{23, 90\}, \{23, 98\}, \{23, 116\}, \{23, 124\}, \{23, 156\}, \{23, 157\}, \{23, 158\},$
 $\{23, 159\}, \{23, 160\}, \{25, 26\}, \{26, 41\}, \{26, 45\}, \{26, 46\}, \{26, 51\}, \{26, 106\}, \{26, 107\},$
 $\{26, 121\}, \{26, 164\}, \{26, 166\}, \{27, 32\}, \{28, 29\}, \{29, 65\}, \{29, 78\}, \{29, 150\}, \{29, 165\},$
 $\{29, 210\}, \{30, 53\}, \{31, 32\}, \{32, 107\}, \{32, 117\}, \{32, 153\}, \{32, 169\}, \{32, 170\}, \{32, 212\},$
 $\{33, 34\}, \{34, 122\}, \{34, 147\}, \{34, 164\}, \{34, 171\}, \{34, 213\}, \{35, 42\}, \{35, 73\}, \{36, 118\},$
 $\{37, 38\}, \{37, 53\}, \{37, 62\}, \{37, 76\}, \{37, 84\}, \{37, 87\}, \{37, 202\}, \{38, 58\}, \{38, 155\}, \{38, 180\},$
 $\{38, 208\}, \{39, 68\}, \{40, 72\}, \{41, 42\}, \{41, 43\}, \{41, 62\}, \{42, 99\}, \{42, 126\}, \{42, 127\}, \{42, 128\},$
 $\{42, 130\}, \{42, 145\}, \{42, 148\}, \{42, 174\}, \{42, 215\}, \{43, 57\}, \{43, 61\}, \{43, 95\}, \{43, 114\},$
 $\{43, 168\}, \{43, 181\}, \{44, 47\}, \{44, 48\}, \{44, 49\}, \{44, 50\}, \{44, 51\}, \{44, 52\}, \{44, 54\}, \{44, 55\},$
 $\{44, 58\}, \{44, 63\}, \{44, 64\}, \{44, 65\}, \{44, 66\}, \{44, 67\}, \{44, 69\}, \{44, 71\}, \{44, 188\}, \{44, 189\},$
 $\{45, 53\}, \{45, 68\}, \{45, 123\}, \{46, 59\}, \{46, 80\}, \{46, 187\}, \{52, 53\}, \{53, 66\}, \{53, 88\}, \{53, 105\},$
 $\{53, 109\}, \{53, 134\}, \{53, 172\}, \{53, 173\}, \{53, 214\}, \{54, 87\}, \{54, 118\}, \{54, 120\}, \{58, 59\},$
 $\{59, 166\}, \{59, 167\}, \{59, 169\}, \{62, 91\}, \{62, 137\}, \{62, 141\}, \{62, 159\}, \{62, 182\}, \{62, 184\},$
 $\{62, 207\}, \{68, 81\}, \{68, 105\}, \{68, 113\}, \{68, 127\}, \{68, 139\}, \{68, 178\}, \{68, 219\}, \{69, 70\},$
 $\{70, 125\}, \{70, 179\}, \{70, 180\}, \{70, 220\}, \{71, 72\}, \{72, 94\}, \{72, 102\}, \{72, 160\}, \{72, 225\},$
 $\{73, 74\}, \{73, 75\}, \{73, 77\}, \{73, 78\}, \{73, 79\}, \{73, 81\}, \{75, 76\}, \{76, 81\}, \{76, 124\}, \{76, 125\},$
 $\{76, 151\}, \{76, 163\}, \{76, 170\}, \{76, 199\}, \{76, 223\}, \{79, 80\}, \{80, 108\}, \{80, 133\}, \{80, 143\},$
 $\{80, 156\}, \{80, 161\}, \{80, 167\}, \{80, 168\}, \{80, 211\}, \{82, 83\}, \{83, 102\}, \{83, 103\}, \{83, 138\},$
 $\{83, 139\}, \{84, 85\}, \{84, 86\}, \{84, 88\}, \{84, 190\}, \{86, 87\}, \{87, 105\}, \{87, 126\}, \{87, 131\},$
 $\{87, 142\}, \{87, 143\}, \{87, 144\}, \{87, 203\}, \{96, 97\}, \{97, 105\}, \{97, 140\}, \{97, 207\}, \{97, 208\},$
 $\{105, 123\}, \{105, 204\}, \{110, 111\}, \{111, 135\}, \{111, 154\}, \{111, 172\}, \{111, 175\}, \{111, 176\},$
 $\{111, 177\}, \{111, 216\}, \{112, 218\}, \{118, 119\}, \{118, 121\}, \{118, 122\}, \{118, 176\}, \{119, 120\},$
 $\{120, 152\}, \{120, 161\}, \{120, 162\}, \{120, 163\}, \{120, 210\}, \{123, 128\}, \{123, 129\}, \{123, 157\},$
 $\{123, 173\}, \{123, 181\}, \{123, 221\}, \{176, 191\}, \{176, 198\}, \{186, 187\}, \{187, 188\}, \{187, 189\},$
 $\{187, 196\}, \{187, 200\}, \{187, 212\}, \{187, 220\}, \{190, 191\}, \{191, 192\}, \{191, 199\}, \{191, 223\},$
 $\{191, 225\}, \{193, 194\}, \{194, 195\}, \{197, 198\}, \{201, 202\}, \{202, 205\}, \{202, 207\}, \{202, 209\},$
 $\{202, 214\}, \{202, 215\}, \{202, 219\}, \{202, 221\}, \{202, 222\}, \{203, 204\}, \{204, 206\}, \{217, 218\}.$

For a vertex i , we write $j \sim i$ if $\{i, j\}$ is an edge. For every non-constant vertex i , Table 1 gives a positive integer m_i and polynomials W_i , A_i and B_i with integer coefficients such that

$$m_i^2 \prod_{j \sim i} H_j - W_i^2 = A_i F + B_i H_i. \quad (2)$$

For every edge $\{i, j\}$ between two non-constant vertices, Table 2 gives a positive integer k_{ij} and polynomials α_{ij} , β_{ij} and γ_{ij} with integer coefficients such that

$$\alpha_{ij} F + \beta_{ij} H_i + \gamma_{ij} H_j = k_{ij}. \quad (3)$$

These identities can be checked by expanding both sides. For the signs, we use

$$Q_0 = y^2 - y + 2,$$

$$\begin{aligned}
M_0 &= 2x^4 + x + 1, \\
E_0 &= 7, \\
Q_7 &= x^4 - 4x^3 + x^2y + 2x^2 - 2xy + 4x + y^2 - y + 2, \\
M_7 &= x^6 - 6x^5 + 11x^4 + 4x^3 - 8x^2 - 7x + 1, \\
E_7 &= 3x^4 - 12x^3 + 6x^2 + 12x + 7, \\
Q_{13} &= y^2 + y + 2, \\
M_{13} &= 2x^4 + x + 5, \\
E_{13} &= 7, \\
Q_{16} &= y^2 - y + 2, \\
M_{16} &= 2x^4 + x + 1, \\
E_{16} &= 7, \\
Q_{19} &= 4y^2 + 4, \\
M_{19} &= 16x^4 + 8x + 24, \\
E_{19} &= 64, \\
Q_{36} &= y^2 + 1, \\
M_{36} &= 2x^4 + x + 3, \\
E_{36} &= 4, \\
Q_{37} &= x^2 - xy + y^2 + 1, \\
M_{37} &= 2x^4 - x^3 + 3, \\
E_{37} &= 3x^2 + 4, \\
Q_{40} &= x^2 + xy + 2x + y^2 + y + 2, \\
M_{40} &= 2x^4 + x^3 + 3x^2 + 5x + 5, \\
E_{40} &= 3x^2 + 6x + 7, \\
Q_{45} &= x^2 - xy + 2x + y^2 - y + 2, \\
M_{45} &= 2x^4 - x^3 - 3x^2 - 3x + 1, \\
E_{45} &= 3x^2 + 6x + 7, \\
Q_{46} &= x^2 - xy - 2x + y^2 + y + 2, \\
M_{46} &= 2x^4 - x^3 + 3x^2 - 3x + 5, \\
E_{46} &= 3x^2 - 6x + 7, \\
Q_{48} &= 4x^2 - 2xy + 4x + y^2 - y + 2, \\
M_{48} &= 2x^4 - 8x^3 - 12x^2 - 7x + 1, \\
E_{48} &= 12x^2 + 12x + 7, \\
Q_{54} &= x^2 + xy + y^2 + 1, \\
M_{54} &= 2x^4 + x^3 + 2x + 3, \\
E_{54} &= 3x^2 + 4, \\
Q_{55} &= x^4 + 2x^3 - x^2y - x^2 - xy - 2x + y^2 + y + 2, \\
M_{55} &= x^6 + 3x^5 - 2x^4 - 5x^3 + x^2 + 3x - 5, \\
E_{55} &= 3x^4 + 6x^3 - 3x^2 - 6x + 7, \\
Q_{56} &= x^4 - 4x^3 + x^2y + 2x^2 - 2xy + 4x + y^2 - y + 2, \\
M_{56} &= x^6 - 6x^5 + 11x^4 + 4x^3 - 8x^2 - 7x + 1,
\end{aligned}$$

$$\begin{aligned}
E_{56} &= 3x^4 - 12x^3 + 6x^2 + 12x + 7, \\
Q_{58} &= x^2 + xy + y^2 + 1, \\
M_{58} &= 2x^4 + x^3 + 2x + 3, \\
E_{58} &= 3x^2 + 4, \\
Q_{61} &= x^4 - x^2y + y^2 + 1, \\
M_{61} &= x^6 - 2x^4 + x^2 - x - 3, \\
E_{61} &= 3x^4 + 4, \\
Q_{75} &= 4x^2 + 2xy + 8x + y^2 + 2y + 5, \\
M_{75} &= 2x^4 + 8x^3 + 24x^2 + 27x + 13, \\
E_{75} &= 12x^2 + 24x + 16, \\
Q_{102} &= y^2 + y + 2, \\
M_{102} &= 2x^4 + x + 5, \\
E_{102} &= 7, \\
Q_{105} &= x^4 - 2x^3 + x^2y + x^2 - xy + y^2 + 1, \\
M_{105} &= x^6 - 3x^5 + 5x^4 - x^3 + x^2 + 3, \\
E_{105} &= 3x^4 - 6x^3 + 3x^2 + 4, \\
Q_{156} &= x^8 + 2x^7 + 3x^6 + 4x^5 - x^4y + 5x^4 - x^3y + 4x^3 - x^2y + 3x^2 \\
&\quad - xy + 2x + y^2 - y + 2, \\
M_{156} &= x^{12} + 3x^{11} + 6x^{10} + 10x^9 + 15x^8 + 18x^7 + 19x^6 + 18x^5 + 14x^4 \\
&\quad + 11x^3 + 7x^2 + 3x - 1, \\
E_{156} &= 3x^8 + 6x^7 + 9x^6 + 12x^5 + 15x^4 + 12x^3 + 9x^2 + 6x + 7, \\
Q_{193} &= 900x^2 - 300xy - 600x + 100y^2 + 100y + 200, \\
M_{193} &= 2000x^4 - 27000x^3 + 27000x^2 - 11000x + 5000, \\
E_{193} &= 270000x^2 - 180000x + 70000, \\
Q_{196} &= 16x^2 - 4xy + y^2 + 1, \\
M_{196} &= 2x^4 - 64x^3 - 3x + 3, \\
E_{196} &= 48x^2 + 4, \\
Q_{197} &= 4x^2 - 20xy - 32x + 100y^2 + 80y + 164, \\
M_{197} &= 2000x^4 - 8x^3 + 96x^2 + 416x + 4312, \\
E_{197} &= 1200x^2 - 9600x + 59200, \\
Q_{199} &= 4x^2 + 4xy + 12x + 4y^2 + 6y + 13, \\
M_{199} &= 16x^4 + 8x^3 + 36x^2 + 70x + 63, \\
E_{199} &= 48x^2 + 144x + 172, \\
Q_{200} &= x^2 + 5xy + 8x + 25y^2 + 20y + 41, \\
M_{200} &= 250x^4 + x^3 + 12x^2 + 198x + 539, \\
E_{200} &= 75x^2 + 600x + 3700, \\
Q_{201} &= x^2 - xy + 6x + y^2 - 3y + 10, \\
M_{201} &= 2x^4 - x^3 - 9x^2 - 27x - 27, \\
E_{201} &= 3x^2 + 18x + 31, \\
Q_{203} &= 4x^2 - 8xy + 24x + 16y^2 - 24y + 52,
\end{aligned}$$

$$\begin{aligned}
M_{203} &= 128x^4 - 8x^3 - 72x^2 - 184x - 120, \\
E_{203} &= 192x^2 + 1152x + 2752, \\
Q_{205} &= 4x^2 + 2xy + 12x + y^2 + 3y + 10, \\
M_{205} &= 2x^4 + 8x^3 + 36x^2 + 57x + 33, \\
E_{205} &= 12x^2 + 36x + 31, \\
Q_{206} &= 16x^2 - 16xy + 16x + 16y^2 - 8y + 20, \\
M_{206} &= 128x^4 - 64x^3 - 96x^2 - 48x + 152, \\
E_{206} &= 768x^2 + 768x + 1216, \\
Q_{207} &= 9x^2 + 3xy + 6x + y^2 + y + 2, \\
M_{207} &= 2x^4 + 27x^3 + 27x^2 + 13x + 5, \\
E_{207} &= 27x^2 + 18x + 7, \\
Q_{208} &= 4y^2 + 4y + 8, \\
M_{208} &= 16x^4 + 8x + 40, \\
E_{208} &= 112, \\
Q_{209} &= x^4 - 2x^2y + 4y^2 + 4, \\
M_{209} &= x^6 - 16x^4 + 4x^2 - 8x - 24, \\
E_{209} &= 12x^4 + 64, \\
Q_{210} &= x^4 + x^2y + y^2 + 1, \\
M_{210} &= x^6 + 2x^4 + x^2 + x + 3, \\
E_{210} &= 3x^4 + 4, \\
Q_{211} &= x^2 - xy + 2x + y^2 - y + 2, \\
M_{211} &= 2x^4 - x^3 - 3x^2 - 3x + 1, \\
E_{211} &= 3x^2 + 6x + 7, \\
Q_{212} &= 25x^4 - 20x^2y + 30x^2 + 16y^2 - 12y + 25, \\
M_{212} &= 125x^6 + 97x^4 + 215x^2 - 64x - 117, \\
E_{212} &= 1200x^4 + 1440x^2 + 1456, \\
Q_{213} &= y^2 - 2y + 5, \\
M_{213} &= 2x^4 + x - 7, \\
E_{213} &= 16, \\
Q_{214} &= x^4 - 2x^3 + 2x^2y - 5x^2 - 2xy + 6x + 4y^2 - 6y + 13, \\
M_{214} &= x^6 - 3x^5 + 10x^4 + 17x^3 + 22x^2 - 23x - 15, \\
E_{214} &= 12x^4 - 24x^3 - 60x^2 + 72x + 172, \\
Q_{215} &= x^2 - 2xy - 2x + 4y^2 + 2y + 5, \\
M_{215} &= 16x^4 - x^3 + 3x^2 + x + 29, \\
E_{215} &= 12x^2 - 24x + 76, \\
Q_{217} &= 36x^2 + 60xy + 100y^2 + 100, \\
M_{217} &= 2000x^4 + 216x^3 + 1600x + 3000, \\
E_{217} &= 10800x^2 + 40000, \\
Q_{219} &= 4y^2 - 6y + 13, \\
M_{219} &= 16x^4 + 8x - 15,
\end{aligned}$$

$$\begin{aligned}
E_{219} &= 172, \\
Q_{220} &= 16x^2 - 20xy + 40x + 25y^2 - 25y + 50, \\
M_{220} &= 250x^4 - 64x^3 - 240x^2 - 275x + 125, \\
E_{220} &= 1200x^2 + 3000x + 4375, \\
Q_{221} &= x^4 - 2x^3 - 2x^2y + x^2 + 2xy + 4y^2 + 4, \\
M_{221} &= x^6 - 3x^5 - 13x^4 - x^3 + 4x^2 - 12x - 24, \\
E_{221} &= 12x^4 - 24x^3 + 12x^2 + 64, \\
Q_{222} &= 4x^2 + 2xy - 8x + y^2 - 2y + 5, \\
M_{222} &= 2x^4 + 8x^3 - 24x^2 + 27x - 7, \\
E_{222} &= 12x^2 - 24x + 16, \\
Q_{223} &= 4x^2 + 4xy + 12x + 4y^2 + 6y + 13, \\
M_{223} &= 16x^4 + 8x^3 + 36x^2 + 70x + 63, \\
E_{223} &= 48x^2 + 144x + 172, \\
Q_{224} &= 9x^2 + 6xy - 6x + 4y^2 - 2y + 5, \\
M_{224} &= 16x^4 + 27x^3 - 27x^2 + 29x + 19, \\
E_{224} &= 108x^2 - 72x + 76, \\
Q_{225} &= 4x^2 - 2xy + 4x + y^2 - y + 2, \\
M_{225} &= 2x^4 - 8x^3 - 12x^2 - 7x + 1, \\
E_{225} &= 12x^2 + 12x + 7,
\end{aligned}$$

and the identities

$$-F = Q_0H_0 - M_0, \quad (4)$$

$$4Q_0 = (2y - 1)^2 + E_0. \quad (5)$$

$$4H_1 = (2x + 1)^2 + 3. \quad (6)$$

$$4H_2 = (3x + 2y + 1)^2 + 3(x - 1)^2. \quad (7)$$

$$4H_3 = (x + 2y - 1)^2 + 3(x + 1)^2. \quad (8)$$

$$4H_4 = (2x + 2y + 1)^2 + 3(2x + 1)^2. \quad (9)$$

$$4H_5 = (x - 2y - 1)^2 + 3(x - 1)^2. \quad (10)$$

$$4H_6 = 3(3x - 2y - 1)^2 + 9(x + 1)^2. \quad (11)$$

$$-F = Q_7H_7 - M_7, \quad (12)$$

$$4Q_7 = (x^2 - 2x + 2y - 1)^2 + E_7. \quad (13)$$

$$4H_{10} = (x - 2y)^2 + 3x^2. \quad (14)$$

$$4H_{12} = (x - 2y + 1)^2 + 3(x - 1)^2. \quad (15)$$

$$-F = Q_{13}H_{13} - M_{13}, \quad (16)$$

$$4Q_{13} = (2y + 1)^2 + E_{13}. \quad (17)$$

$$4H_{15} = (x - 2y)^2 + 3(x + 2)^2. \quad (18)$$

$$-F = Q_{16}(-H_{16}) - M_{16}, \quad (19)$$

$$4Q_{16} = (2y - 1)^2 + E_{16}. \quad (20)$$

$$H_{18} = 3y^2 + 9x^2. \quad (21)$$

$$-8F = Q_{19}H_{19} - M_{19}, \quad (22)$$

$$16Q_{19} = (8y)^2 + E_{19}. \quad (23)$$

$$4H_{21} = (x^2 + 2y)^2 + 3(x^2)^2. \quad (24)$$

$$4H_{23} = (x^2 - 2y - 1)^2 + 3(x^2 + 1)^2. \quad (25)$$

$$H_{24} = x^2 + 1. \quad (26)$$

$$4H_{26} = (x^2 + 2x - 2y + 1)^2 + 3(x^2 - 1)^2. \quad (27)$$

$$4H_{29} = (x^2 - x - 2y - 1)^2 + 3(x^2 + x + 1)^2. \quad (28)$$

$$4(-H_{30}) = (2x + 1)^2 + 3. \quad (29)$$

$$4H_{32} = (x^2 + 2y + 3)^2 + 3(x^2 - 1)^2. \quad (30)$$

$$4H_{34} = (x^2 - 2y - 3)^2 + 3(x^2 + 1)^2. \quad (31)$$

$$-F = Q_{36}H_{36} - M_{36}, \quad (32)$$

$$4Q_{36} = (2y)^2 + E_{36}. \quad (33)$$

$$-F = Q_{37}H_{37} - M_{37}, \quad (34)$$

$$4Q_{37} = (-x + 2y)^2 + E_{37}. \quad (35)$$

$$4H_{38} = (2y - 1)^2 + 3. \quad (36)$$

$$-F = Q_{40}H_{40} - M_{40}, \quad (37)$$

$$4Q_{40} = (x + 2y + 1)^2 + E_{40}. \quad (38)$$

$$4H_{42} = (x + 2y - 1)^2 + 3(x - 1)^2. \quad (39)$$

$$4H_{43} = (x + 2y + 1)^2 + 3(x - 1)^2. \quad (40)$$

$$4H_{44} = (2x - 1)^2 + 3. \quad (41)$$

$$-F = Q_{45}(-H_{45}) - M_{45}, \quad (42)$$

$$4Q_{45} = (-x + 2y - 1)^2 + E_{45}. \quad (43)$$

$$-F = Q_{46}(-H_{46}) - M_{46}, \quad (44)$$

$$4Q_{46} = (-x + 2y + 1)^2 + E_{46}. \quad (45)$$

$$-F = Q_{48}(-H_{48}) - M_{48}, \quad (46)$$

$$4Q_{48} = (-2x + 2y - 1)^2 + E_{48}. \quad (47)$$

$$4H_{53} = (x^2 - x - 2y - 3)^2 + 3(x^2 + x + 1)^2. \quad (48)$$

$$-F = Q_{54}H_{54} - M_{54}, \quad (49)$$

$$4Q_{54} = (x + 2y)^2 + E_{54}. \quad (50)$$

$$F = Q_{55}(-H_{55}) - M_{55}, \quad (51)$$

$$4Q_{55} = (-x^2 - x + 2y + 1)^2 + E_{55}. \quad (52)$$

$$-F = Q_{56}(-H_{56}) - M_{56}, \quad (53)$$

$$4Q_{56} = (x^2 - 2x + 2y - 1)^2 + E_{56}. \quad (54)$$

$$-F = Q_{58}(-H_{58}) - M_{58}, \quad (55)$$

$$4Q_{58} = (x + 2y)^2 + E_{58}. \quad (56)$$

$$4H_{59} = (2y + 1)^2 + 3. \quad (57)$$

$$F = Q_{61}H_{61} - M_{61}, \quad (58)$$

$$4Q_{61} = (-x^2 + 2y)^2 + E_{61}. \quad (59)$$

$$4H_{62} = (3x - 2y - 1)^2 + 3(x + 1)^2. \quad (60)$$

$$4H_{68} = (2y + 3)^2 + 3. \quad (61)$$

$$4H_{70} = (x + 2y + 2)^2 + 3x^2. \quad (62)$$

$$4H_{72} = (x - 2y + 3)^2 + 3(x + 1)^2. \quad (63)$$

$$\begin{aligned}
4H_{73} &= (2x + 3)^2 + 3. & (64) \\
-F &= Q_{75}H_{75} - M_{75}, & (65) \\
4Q_{75} &= (2x + 2y + 2)^2 + E_{75}. & (66) \\
4H_{76} &= (2x - 2y + 1)^2 + 3. & (67) \\
4H_{80} &= (x^2 + 2x + 2y + 2)^2 + 3(x^2)^2. & (68) \\
4H_{83} &= (x + 2y)^2 + 3(x + 2)^2. & (69) \\
4H_{84} &= 9(2x - 1)^2 + 3. & (70) \\
4H_{87} &= (x + 2y + 3)^2 + 3(x + 1)^2. & (71) \\
12(-H_{89}) &= 3(x + 2y + 2)^2 + (3x + 2)^2 - 28. & (72) \\
12H_{92} &= 3(x + 2y + 2)^2 + (3x + 2)^2 - 28. & (73) \\
4H_{93} &= 4(x + y)^2 + (2x - 1)^2 - 5. & (74) \\
12H_{94} &= 3(x + 2y)^2 + (3x - 2)^2 - 28. & (75) \\
(-H_{96}) &= (x + y - 3)(y + 2) + 6, & (76) \\
H_{97} &= 3(y - 1)^2 + 9x^2. & (77) \\
-F &= Q_{102}(-H_{102}) - M_{102}, & (78) \\
4Q_{102} &= (2y + 1)^2 + E_{102}. & (79) \\
4(-H_{104}) &= (2y + 1)(-4x + 2y - 3) + 3, & (80) \\
-F &= Q_{105}H_{105} - M_{105}, & (81) \\
4Q_{105} &= (x^2 - x + 2y)^2 + E_{105}. & (82) \\
H_{111} &= x^2 + 3. & (83) \\
4H_{118} &= (x - 2y + 1)^2 + 3(x + 1)^2. & (84) \\
4H_{120} &= (2x^2 + x - 2y)^2 + 3x^2. & (85) \\
4H_{123} &= (x^2 - x + 2y)^2 + 3(x^2 - x)^2. & (86) \\
H_{132} &= (x + y + 3)(y - 2) + 5, & (87) \\
(-H_{136}) &= (y - 1)^2 + x^2 - 1. & (88) \\
(-H_{146}) &= (x + y - 6)(y + 3) + 18, & (89) \\
(-H_{151}) &= (2y - 3)(-x + y + 1) + 3, & (90) \\
F &= Q_{156}(-H_{156}) - M_{156}, & (91) \\
4Q_{156} &= (-x^4 - x^3 - x^2 - x + 2y - 1)^2 + E_{156}. & (92) \\
(-H_{174}) &= y^2. & (93) \\
(-H_{179}) &= (y + 3)(-x + y - 2) + 6, & (94) \\
-1000F &= Q_{193}H_{193} - M_{193}, & (95) \\
400Q_{193} &= (-300x + 200y + 100)^2 + E_{193}. & (96) \\
-F &= Q_{196}H_{196} - M_{196}, & (97) \\
4Q_{196} &= (-4x + 2y)^2 + E_{196}. & (98) \\
-1000F &= Q_{197}(-H_{197}) - M_{197}, & (99) \\
400Q_{197} &= (-20x + 200y + 80)^2 + E_{197}. & (100) \\
-8F &= Q_{199}H_{199} - M_{199}, & (101) \\
16Q_{199} &= (4x + 8y + 6)^2 + E_{199}. & (102) \\
-125F &= Q_{200}H_{200} - M_{200}, & (103)
\end{aligned}$$

$$100Q_{200} = (5x + 50y + 20)^2 + E_{200}. \quad (104)$$

$$-F = Q_{201}(-H_{201}) - M_{201}, \quad (105)$$

$$4Q_{201} = (-x + 2y - 3)^2 + E_{201}. \quad (106)$$

$$-64F = Q_{203}(-H_{203}) - M_{203}, \quad (107)$$

$$64Q_{203} = (-8x + 32y - 24)^2 + E_{203}. \quad (108)$$

$$-F = Q_{205}H_{205} - M_{205}, \quad (109)$$

$$4Q_{205} = (2x + 2y + 3)^2 + E_{205}. \quad (110)$$

$$-64F = Q_{206}(-H_{206}) - M_{206}, \quad (111)$$

$$64Q_{206} = (-16x + 32y - 8)^2 + E_{206}. \quad (112)$$

$$-F = Q_{207}H_{207} - M_{207}, \quad (113)$$

$$4Q_{207} = (3x + 2y + 1)^2 + E_{207}. \quad (114)$$

$$-8F = Q_{208}H_{208} - M_{208}, \quad (115)$$

$$16Q_{208} = (8y + 4)^2 + E_{208}. \quad (116)$$

$$8F = Q_{209}(-H_{209}) - M_{209}, \quad (117)$$

$$16Q_{209} = (-2x^2 + 8y)^2 + E_{209}. \quad (118)$$

$$-F = Q_{210}H_{210} - M_{210}, \quad (119)$$

$$4Q_{210} = (x^2 + 2y)^2 + E_{210}. \quad (120)$$

$$-F = Q_{211}H_{211} - M_{211}, \quad (121)$$

$$4Q_{211} = (-x + 2y - 1)^2 + E_{211}. \quad (122)$$

$$64F = Q_{212}H_{212} - M_{212}, \quad (123)$$

$$64Q_{212} = (-20x^2 + 32y - 12)^2 + E_{212}. \quad (124)$$

$$-F = Q_{213}H_{213} - M_{213}, \quad (125)$$

$$4Q_{213} = (2y - 2)^2 + E_{213}. \quad (126)$$

$$-8F = Q_{214}H_{214} - M_{214}, \quad (127)$$

$$16Q_{214} = (2x^2 - 2x + 8y - 6)^2 + E_{214}. \quad (128)$$

$$-8F = Q_{215}(-H_{215}) - M_{215}, \quad (129)$$

$$16Q_{215} = (-2x + 8y + 2)^2 + E_{215}. \quad (130)$$

$$-1000F = Q_{217}(-H_{217}) - M_{217}, \quad (131)$$

$$400Q_{217} = (60x + 200y)^2 + E_{217}. \quad (132)$$

$$-8F = Q_{219}H_{219} - M_{219}, \quad (133)$$

$$16Q_{219} = (8y - 6)^2 + E_{219}. \quad (134)$$

$$-125F = Q_{220}H_{220} - M_{220}, \quad (135)$$

$$100Q_{220} = (-20x + 50y - 25)^2 + E_{220}. \quad (136)$$

$$8F = Q_{221}(-H_{221}) - M_{221}, \quad (137)$$

$$16Q_{221} = (-2x^2 + 2x + 8y)^2 + E_{221}. \quad (138)$$

$$-F = Q_{222}H_{222} - M_{222}, \quad (139)$$

$$4Q_{222} = (2x + 2y - 2)^2 + E_{222}. \quad (140)$$

$$-8F = Q_{223}(-H_{223}) - M_{223}, \quad (141)$$

$$16Q_{223} = (4x + 8y + 6)^2 + E_{223}. \quad (142)$$

$$-8F = Q_{224}H_{224} - M_{224}, \quad (143)$$

$$16Q_{224} = (6x + 8y - 2)^2 + E_{224}. \quad (144)$$

$$-F = Q_{225}H_{225} - M_{225}, \quad (145)$$

$$4Q_{225} = (-2x + 2y - 1)^2 + E_{225}. \quad (146)$$

4 The cubic certificates

There are 2 certificates. The g -th certificate consists of polynomials $J_i^{(g)}$ with coefficients in $\mathbb{Z}[\omega]$ and constants $c_i^{(g)} \in \mathbb{Z}[\omega]$, listed in Table 3 and Table 4, and of *edges* (i, j) with $i < j$ and weights $w_{ij}^{(g)} \in \{1, 2\}$. For an edge we put $\rho_{ij}^{(g)} = w_{ij}^{(g)}$ and $\rho_{ji}^{(g)} = -w_{ij}^{(g)} \bmod 3$, and $\rho_{ij}^{(g)} = 0$ otherwise. In the rest of the proof we usually suppress the index g . The edges $(i, j; w_{ij})$ of the first certificate are

(0, 20; 1), (0, 22; 2), (0, 29; 2), (0, 30; 2), (0, 43; 2), (0, 48; 1), (0, 49; 1), (0, 60; 1), (0, 71; 1),
(1, 11; 2), (1, 24; 2), (1, 25; 2), (1, 30; 2), (1, 35; 2), (1, 37; 1), (1, 45; 1), (1, 48; 1), (1, 56; 2), (2, 3; 2),
(2, 6; 1), (2, 9; 2), (2, 11; 1), (2, 12; 2), (2, 17; 2), (2, 28; 2), (2, 33; 1), (2, 36; 2), (2, 41; 2), (2, 43; 1),
(2, 48; 1), (2, 49; 2), (2, 50; 2), (2, 65; 1), (2, 70; 2), (3, 13; 1), (3, 33; 1), (3, 35; 2), (3, 37; 2),
(3, 41; 2), (3, 43; 2), (3, 45; 1), (3, 53; 1), (3, 55; 2), (3, 58; 2), (4, 13; 1), (4, 18; 2), (4, 38; 2),
(4, 41; 1), (4, 46; 2), (4, 54; 1), (5, 17; 1), (5, 25; 2), (5, 26; 1), (5, 30; 1), (5, 35; 1), (5, 36; 1),
(5, 37; 2), (5, 38; 2), (5, 40; 2), (5, 46; 2), (5, 48; 2), (5, 53; 1), (5, 58; 1), (5, 68; 1), (6, 17; 2),
(6, 22; 1), (6, 24; 2), (6, 40; 2), (6, 47; 1), (6, 49; 1), (7, 16; 1), (7, 17; 2), (7, 19; 2), (7, 21; 2),
(7, 26; 2), (7, 29; 2), (7, 33; 1), (7, 37; 2), (7, 39; 1), (7, 46; 2), (7, 47; 2), (7, 49; 2), (7, 53; 2),
(7, 63; 1), (7, 69; 2), (8, 12; 2), (8, 13; 2), (8, 20; 1), (8, 22; 1), (8, 24; 1), (8, 25; 1), (8, 26; 1),
(8, 28; 2), (8, 31; 2), (8, 33; 2), (8, 53; 1), (8, 60; 1), (8, 63; 2), (8, 71; 2), (9, 24; 1), (9, 28; 1),
(9, 40; 2), (10, 12; 2), (10, 16; 1), (10, 24; 2), (10, 25; 2), (10, 29; 1), (10, 30; 2), (10, 47; 2), (10, 54; 2),
(10, 62; 2), (10, 66; 1), (10, 71; 2), (11, 24; 2), (11, 29; 2), (11, 30; 1), (11, 50; 1), (11, 60; 1),
(11, 71; 2), (12, 13; 1), (12, 20; 1), (12, 23; 2), (12, 27; 2), (12, 29; 1), (12, 31; 2), (12, 36; 1),
(12, 38; 1), (12, 44; 1), (12, 45; 2), (12, 48; 1), (12, 50; 1), (12, 54; 1), (12, 60; 1), (12, 68; 2),
(12, 69; 1), (13, 17; 1), (13, 21; 2), (13, 30; 1), (13, 34; 1), (13, 37; 2), (13, 40; 2), (13, 46; 2),
(13, 48; 2), (13, 49; 1), (13, 51; 1), (13, 61; 2), (13, 65; 1), (13, 68; 2), (14, 19; 2), (14, 20; 1),
(14, 23; 2), (14, 25; 1), (14, 26; 1), (14, 41; 2), (14, 42; 1), (14, 43; 2), (14, 47; 2), (14, 57; 2),
(14, 58; 2), (14, 61; 2), (14, 67; 1), (14, 69; 1), (14, 70; 2), (15, 30; 1), (15, 38; 2), (15, 39; 2),
(15, 41; 1), (15, 42; 1), (15, 45; 2), (15, 54; 1), (15, 56; 2), (15, 59; 1), (15, 67; 2), (15, 70; 1),
(16, 20; 1), (16, 29; 1), (16, 30; 2), (16, 36; 1), (16, 41; 1), (16, 46; 2), (16, 47; 1), (16, 48; 2),
(16, 58; 1), (16, 65; 1), (16, 70; 2), (17, 36; 1), (17, 39; 2), (17, 58; 2), (18, 26; 1), (18, 27; 2),
(18, 37; 1), (18, 41; 2), (18, 45; 2), (18, 63; 2), (19, 25; 2), (19, 30; 1), (19, 36; 1), (19, 37; 1),
(19, 40; 2), (19, 41; 1), (19, 42; 2), (19, 45; 1), (19, 47; 2), (19, 56; 1), (19, 58; 2), (20, 22; 2),
(20, 28; 2), (20, 31; 1), (20, 37; 1), (20, 39; 2), (20, 49; 2), (20, 52; 2), (20, 53; 1), (20, 64; 2),
(20, 66; 1), (20, 71; 1), (21, 29; 2), (21, 30; 2), (21, 33; 1), (21, 41; 1), (21, 48; 1), (21, 49; 2),
(21, 55; 1), (21, 57; 2), (21, 59; 2), (22, 44; 2), (22, 51; 1), (22, 58; 2), (22, 60; 2), (22, 66; 1),
(22, 68; 1), (22, 71; 2), (23, 29; 1), (23, 37; 2), (23, 41; 1), (23, 46; 1), (23, 49; 2), (23, 61; 2),
(23, 71; 1), (24, 30; 2), (24, 32; 1), (24, 37; 2), (24, 42; 2), (24, 44; 2), (24, 45; 1), (24, 50; 1),
(24, 60; 2), (24, 61; 2), (24, 67; 1), (24, 70; 2), (25, 26; 2), (25, 34; 1), (25, 38; 2), (25, 41; 2),
(25, 43; 1), (25, 45; 1), (25, 46; 1), (25, 47; 1), (25, 58; 2), (25, 59; 1), (25, 60; 1), (25, 66; 1),
(25, 71; 1), (26, 29; 2), (26, 30; 1), (26, 31; 1), (26, 32; 1), (26, 33; 2), (26, 35; 2), (26, 37; 1),
(26, 42; 2), (26, 44; 2), (26, 47; 1), (26, 54; 1), (26, 70; 2), (27, 34; 1), (27, 36; 2), (27, 48; 1),
(27, 53; 1), (27, 55; 1), (27, 71; 1), (28, 33; 2), (28, 35; 1), (28, 63; 2), (28, 67; 1), (29, 32; 2),
(29, 33; 1), (29, 35; 1), (29, 41; 2), (29, 49; 2), (29, 54; 1), (29, 57; 2), (29, 59; 2), (29, 61; 2),
(29, 63; 1), (29, 66; 1), (30, 31; 2), (30, 37; 2), (30, 41; 2), (30, 42; 1), (30, 48; 2), (30, 51; 2),
(30, 57; 2), (30, 61; 1), (30, 63; 1), (30, 65; 1), (31, 37; 2), (31, 41; 1), (31, 45; 1), (31, 48; 1),
(31, 57; 2), (31, 60; 1), (31, 70; 1), (32, 51; 2), (32, 56; 1), (32, 62; 1), (32, 64; 1), (32, 71; 1),
(33, 36; 2), (33, 42; 1), (33, 47; 1), (33, 58; 1), (34, 38; 1), (34, 66; 1), (34, 67; 2), (35, 38; 1),
(35, 39; 2), (35, 41; 1), (35, 54; 2), (35, 59; 2), (35, 63; 1), (36, 48; 1), (36, 55; 2), (36, 58; 2),
(36, 62; 1), (36, 66; 1), (36, 67; 2), (36, 69; 2), (37, 40; 2), (37, 46; 1), (37, 47; 2), (37, 49; 2),

(37, 52; 2), (37, 60; 1), (38, 46; 1), (38, 57; 1), (38, 58; 1), (38, 60; 2), (38, 64; 1), (38, 65; 2),
 (38, 68; 1), (39, 43; 1), (39, 61; 1), (39, 63; 1), (39, 66; 1), (40, 46; 1), (40, 49; 1), (40, 55; 1),
 (41, 42; 1), (41, 44; 2), (41, 50; 2), (41, 52; 2), (41, 54; 1), (41, 56; 2), (41, 66; 2), (41, 67; 2),
 (41, 69; 2), (42, 48; 2), (42, 56; 2), (42, 58; 2), (42, 59; 1), (42, 61; 2), (43, 45; 2), (43, 47; 2),
 (43, 52; 1), (43, 62; 1), (44, 51; 2), (44, 53; 1), (44, 63; 2), (45, 46; 2), (45, 48; 1), (45, 55; 2),
 (45, 61; 1), (45, 65; 1), (46, 65; 2), (46, 71; 1), (47, 58; 1), (47, 64; 2), (48, 60; 1), (48, 67; 2),
 (49, 50; 1), (49, 62; 1), (49, 71; 2), (50, 52; 1), (50, 60; 1), (50, 65; 1), (50, 69; 1), (51, 53; 2),
 (51, 56; 1), (51, 59; 2), (51, 61; 2), (52, 64; 2), (53, 55; 2), (53, 60; 2), (53, 71; 2), (54, 56; 1),
 (54, 67; 2), (55, 60; 1), (55, 68; 2), (55, 71; 1), (56, 59; 2), (56, 61; 1), (58, 64; 2), (58, 66; 2),
 (59, 61; 2), (59, 71; 2), (60, 68; 2), (60, 71; 1), (61, 66; 2), (61, 67; 2), (63, 64; 1), (64, 67; 1). The
 edges $(i, j; w_{ij})$ of the second certificate are (0, 4; 1), (0, 10; 2), (0, 30; 1), (0, 34; 2), (0, 38; 2),
 (0, 40; 2), (0, 53; 2), (0, 56; 2), (0, 59; 1), (0, 61; 2), (0, 68; 1), (1, 5; 2), (1, 17; 1), (1, 20; 1), (1, 27; 1),
 (1, 35; 2), (1, 38; 1), (1, 43; 2), (1, 59; 1), (2, 6; 2), (2, 8; 2), (2, 12; 2), (2, 13; 2), (2, 18; 1), (2, 39; 1),
 (2, 43; 1), (2, 45; 2), (2, 46; 1), (2, 49; 1), (2, 62; 1), (2, 64; 1), (3, 13; 1), (3, 19; 1), (3, 24; 2),
 (3, 26; 1), (3, 28; 2), (3, 29; 2), (3, 30; 2), (3, 37; 1), (3, 39; 1), (3, 41; 2), (3, 45; 1), (3, 47; 1),
 (3, 52; 2), (3, 55; 1), (3, 57; 1), (3, 61; 1), (3, 63; 2), (4, 15; 1), (4, 17; 1), (4, 26; 1), (4, 38; 2),
 (4, 43; 1), (4, 44; 1), (4, 49; 2), (4, 56; 2), (5, 12; 1), (5, 14; 2), (5, 20; 1), (5, 31; 1), (5, 36; 1),
 (5, 42; 1), (5, 47; 2), (5, 50; 2), (5, 52; 2), (5, 58; 2), (5, 68; 2), (6, 17; 2), (6, 19; 1), (6, 22; 2),
 (6, 37; 1), (6, 46; 1), (6, 48; 1), (6, 52; 2), (6, 66; 2), (6, 67; 2), (6, 68; 1), (7, 15; 1), (7, 20; 2),
 (7, 35; 1), (7, 54; 2), (7, 58; 2), (8, 14; 2), (8, 15; 2), (8, 17; 1), (8, 25; 2), (8, 27; 1), (8, 29; 1),
 (8, 30; 2), (8, 34; 1), (8, 37; 1), (8, 45; 1), (8, 47; 2), (8, 52; 2), (8, 53; 1), (8, 56; 1), (8, 63; 2),
 (9, 15; 1), (9, 24; 2), (9, 29; 1), (9, 31; 2), (9, 44; 2), (9, 48; 2), (9, 50; 1), (9, 54; 2), (9, 59; 1),
 (10, 12; 1), (10, 13; 1), (10, 18; 2), (10, 22; 2), (10, 23; 2), (10, 24; 2), (10, 31; 1), (10, 32; 1),
 (10, 49; 2), (10, 50; 1), (10, 54; 2), (11, 16; 1), (11, 22; 1), (11, 24; 2), (11, 26; 2), (11, 28; 2),
 (11, 30; 1), (11, 35; 1), (11, 40; 1), (11, 42; 1), (11, 44; 2), (11, 46; 1), (11, 50; 2), (11, 53; 1),
 (11, 54; 2), (11, 55; 1), (11, 59; 1), (11, 65; 2), (12, 13; 1), (12, 20; 2), (12, 21; 2), (12, 26; 1),
 (12, 28; 2), (12, 38; 2), (12, 46; 2), (12, 49; 1), (12, 55; 1), (12, 56; 1), (13, 23; 1), (13, 30; 2),
 (13, 37; 2), (13, 38; 1), (13, 44; 2), (13, 52; 1), (13, 55; 2), (13, 57; 1), (13, 64; 2), (13, 67; 2),
 (14, 17; 1), (14, 19; 1), (14, 24; 2), (14, 26; 2), (14, 31; 2), (14, 32; 2), (14, 43; 1), (14, 44; 1),
 (14, 45; 1), (14, 51; 1), (14, 64; 1), (14, 67; 1), (14, 68; 1), (15, 22; 1), (15, 28; 1), (15, 41; 1),
 (15, 42; 1), (15, 48; 1), (15, 52; 2), (15, 64; 2), (16, 21; 1), (16, 22; 2), (16, 24; 1), (16, 29; 2),
 (16, 30; 1), (16, 31; 1), (16, 36; 2), (16, 38; 2), (16, 39; 2), (16, 40; 1), (16, 47; 2), (16, 50; 1),
 (16, 61; 1), (16, 62; 2), (16, 63; 2), (16, 64; 1), (16, 66; 2), (16, 68; 1), (17, 23; 1), (17, 25; 2),
 (17, 27; 1), (17, 37; 1), (17, 38; 2), (17, 47; 2), (17, 49; 1), (17, 51; 2), (17, 60; 1), (17, 64; 1),
 (18, 47; 1), (18, 61; 2), (18, 62; 1), (18, 63; 2), (19, 30; 1), (19, 31; 1), (19, 35; 2), (19, 37; 2),
 (19, 42; 2), (19, 46; 1), (19, 49; 1), (19, 53; 2), (19, 60; 1), (19, 65; 1), (19, 67; 2), (20, 26; 2),
 (20, 28; 2), (20, 30; 1), (20, 34; 1), (20, 36; 1), (20, 39; 2), (20, 41; 1), (20, 43; 1), (20, 44; 1),
 (20, 59; 1), (21, 30; 2), (21, 43; 1), (21, 46; 2), (21, 49; 1), (21, 52; 2), (21, 60; 2), (21, 68; 1),
 (22, 23; 2), (22, 28; 2), (22, 30; 2), (22, 37; 1), (22, 40; 2), (22, 41; 2), (22, 47; 1), (22, 54; 2),
 (22, 57; 2), (22, 59; 2), (22, 63; 2), (22, 64; 1), (23, 32; 1), (23, 36; 2), (23, 38; 1), (23, 39; 1),
 (23, 43; 2), (23, 51; 1), (23, 52; 1), (23, 61; 2), (23, 62; 1), (23, 65; 2), (23, 66; 1), (23, 68; 2),
 (24, 27; 1), (24, 28; 1), (24, 34; 1), (24, 37; 2), (24, 39; 1), (24, 40; 2), (24, 41; 2), (24, 44; 2),
 (24, 56; 2), (24, 59; 2), (24, 61; 2), (24, 63; 1), (24, 64; 2), (25, 27; 2), (25, 31; 1), (25, 37; 1),
 (25, 45; 1), (25, 49; 1), (25, 62; 2), (26, 27; 2), (26, 29; 1), (26, 30; 1), (26, 34; 2), (26, 36; 2),
 (26, 40; 2), (26, 43; 1), (26, 47; 2), (26, 57; 1), (26, 59; 2), (26, 61; 2), (26, 62; 1), (26, 64; 1),
 (26, 65; 2), (27, 39; 2), (27, 47; 2), (27, 49; 2), (27, 53; 1), (27, 55; 1), (27, 56; 1), (27, 60; 1),
 (27, 63; 1), (28, 31; 2), (28, 32; 1), (28, 33; 1), (28, 40; 1), (28, 47; 1), (28, 54; 2), (28, 55; 1),
 (28, 56; 1), (28, 61; 2), (28, 67; 1), (29, 36; 2), (29, 43; 2), (29, 59; 2), (29, 64; 1), (30, 32; 1),
 (30, 33; 1), (30, 40; 1), (30, 45; 2), (30, 49; 1), (30, 56; 1), (30, 59; 1), (30, 61; 1), (30, 63; 1),
 (30, 65; 2), (30, 66; 1), (30, 67; 1), (31, 34; 2), (31, 39; 2), (31, 53; 1), (31, 60; 1), (31, 62; 2),
 (32, 33; 2), (32, 35; 1), (32, 39; 2), (32, 41; 1), (32, 60; 1), (33, 40; 1), (33, 45; 1), (33, 53; 2),

(33, 54; 2), (34, 51; 1), (34, 58; 2), (35, 38; 2), (35, 42; 1), (35, 47; 2), (35, 50; 2), (35, 59; 2), (35, 65; 1), (36, 37; 2), (36, 39; 1), (36, 40; 1), (36, 47; 1), (36, 50; 2), (36, 51; 1), (36, 54; 2), (36, 55; 2), (36, 62; 1), (37, 41; 2), (37, 43; 1), (37, 46; 1), (37, 52; 2), (37, 61; 2), (37, 64; 2), (37, 65; 1), (37, 66; 2), (38, 47; 1), (38, 51; 1), (38, 61; 1), (38, 65; 1), (39, 40; 1), (39, 41; 1), (39, 47; 1), (39, 50; 1), (40, 49; 2), (40, 51; 2), (40, 54; 2), (40, 63; 1), (40, 65; 1), (41, 67; 1), (41, 68; 2), (42, 47; 1), (43, 49; 2), (43, 56; 2), (43, 63; 2), (43, 65; 2), (43, 68; 1), (44, 49; 2), (44, 50; 2), (44, 51; 1), (44, 54; 1), (44, 67; 1), (44, 68; 1), (45, 49; 1), (45, 52; 1), (45, 54; 2), (45, 57; 2), (45, 58; 2), (45, 63; 1), (46, 54; 2), (46, 57; 2), (46, 66; 1), (46, 67; 1), (47, 50; 2), (47, 61; 1), (47, 62; 1), (48, 51; 2), (49, 58; 2), (49, 61; 2), (49, 62; 1), (49, 65; 2), (49, 68; 1), (51, 66; 2), (51, 67; 1), (51, 68; 2), (52, 54; 2), (53, 58; 1), (53, 68; 1), (54, 55; 2), (54, 63; 1), (54, 67; 2), (55, 57; 2), (55, 63; 1), (55, 64; 1), (56, 59; 2), (58, 62; 2), (58, 63; 2), (61, 63; 1), (61, 65; 1), (63, 64; 1), (64, 66; 1), (64, 67; 1), (66, 67; 1). Let (ξ, η) denote (x, y) or (y, x) . We call a vertex i *linear* if $J_i = \varepsilon_i(\lambda_i \eta - s_i(\xi))$ for a unit ε_i of $\mathbb{Z}[\omega]$, an element $\lambda_i \in \mathbb{Z}[\omega]$ whose norm is a power of 3, and a polynomial s_i with coefficients in $\mathbb{Z}[\omega]$; the last column of Table 3 and its analogues gives $\lambda_i \eta = s_i(\xi)$ for the linear vertices, where $\lambda_i = 1$ if no factor is shown. For a linear vertex i , we put $f_i(t) = \lambda_i^d F$ and $g_{ij}(t) = \lambda_i^{d_j} J_j$ evaluated at $\xi = t$, $\eta = s_i(t)/\lambda_i$, where d and d_j are the degrees of F and J_j in η ; these are polynomials in t with coefficients in $\mathbb{Z}[\omega]$. For every linear vertex i , Table 5 gives an integer $D_i \geq 1$ and a polynomial $w_i(t)$ with coefficients in $\mathbb{Z}[\omega]$ such that

$$D_i^3 \lambda_i^{e_i} c_i \prod_j g_{ij}^{\rho_{ij}} - w_i^3 = f_i Q_i \quad (147)$$

for a polynomial $Q_i(t)$ with coefficients in $\mathbb{Z}[\omega]$, where $\rho_{ij} \in \{0, 1, 2\}$; the table also gives an integer $e_i \geq 0$ with $e_i + \sum_j \rho_{ij} d_j \equiv 0 \pmod{3}$, and $e_i = 0$ if $\lambda_i = 1$; Q_i is the quotient of the division of the left side by f_i , and the remainder is zero. For every other vertex i , Table 5 gives an integer $D_i \geq 1$ and polynomials T_i , U_i and V_i in x and y with coefficients in $\mathbb{Z}[\omega]$ such that

$$D_i^3 c_i \prod_j J_j^{\rho_{ij}} - T_i^3 = U_i F + V_i J_i. \quad (148)$$

For the edges (i, j) listed in Table 6, the table gives a linear endpoint $k \in \{i, j\}$ and the factorization of the norm of the resultant $R_{ij} = \text{Res}_t(f_k, g_{kl})$, where $\{k, l\} = \{i, j\}$; it is non-zero. These identities can be checked by expanding both sides and by polynomial division, using $\omega^2 = -1 - \omega$. They hold for each certificate, with its own data.

5 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) .

Step 1: the values are non-zero. For $i \in \{21, 29, 47, 49, 53, 69, 74, 77, 80, 81, 82, 103, 120, 138, 152, 155, 161, 163, 186, 188, 189, 190\}$, checking the 2^2 residue pairs modulo 2, we find that F and H_i have no common zero modulo 2, so $H_i \neq 0$. For $i \in \{14, 26, 51, 65, 78, 79, 85, 122, 133, 135, 143, 164, 168\}$, checking the 3^2 residue pairs modulo 3, we find that F and H_i have no common zero modulo 3, so $H_i \neq 0$. For $i \in \{66, 71, 145, 147\}$, checking the 4^2 residue pairs modulo 4, we find that F and H_i have no common zero modulo 4, so $H_i \neq 0$. For $i \in \{8, 11, 17, 20, 27, 28, 32, 33, 34, 35, 41, 50, 57, 60, 63, 64, 67, 88, 95, 100, 101, 106, 107, 108, 109, 110, 114, 117, 119, 123, 125, 128, 131, 137, 144, 153, 165, 166, 167, 169, 170, 171, 174, 177, 181, 183, 216\}$, checking the 5^2 residue pairs modulo 5, we find that F and H_i have no common zero modulo 5, so $H_i \neq 0$. For $i \in \{52, 116, 154, 158, 162, 173, 184\}$, checking the 7^2 residue pairs modulo 7, we find that F and H_i have no common zero modulo 7, so $H_i \neq 0$. Checking the 8^2 residue pairs modulo 8, we find that F and H_{23} have no common zero modulo 8, so $H_{23} \neq 0$. Checking the 9^2 residue pairs

modulo 9, we find that F and H_{140} have no common zero modulo 9, so $H_{140} \neq 0$. For $i \in \{31, 86, 98, 130, 134, 149, 150, 160, 172, 182\}$, checking the 11^2 residue pairs modulo 11, we find that F and H_i have no common zero modulo 11, so $H_i \neq 0$. For $i \in \{25, 124, 141, 178\}$, checking the 13^2 residue pairs modulo 13, we find that F and H_i have no common zero modulo 13, so $H_i \neq 0$. For $i \in \{9, 22, 90, 126, 159, 175\}$, checking the 17^2 residue pairs modulo 17, we find that F and H_i have no common zero modulo 17, so $H_i \neq 0$. For $i \in \{142, 148, 157\}$, checking the 19^2 residue pairs modulo 19, we find that F and H_i have no common zero modulo 19, so $H_i \neq 0$. For $i \in \{115, 129\}$, checking the 23^2 residue pairs modulo 23, we find that F and H_i have no common zero modulo 23, so $H_i \neq 0$. Checking the 25^2 residue pairs modulo 25, we find that F and H_{121} have no common zero modulo 25, so $H_{121} \neq 0$. Checking the 41^2 residue pairs modulo 41, we find that F and H_{91} have no common zero modulo 41, so $H_{91} \neq 0$. Checking the 49^2 residue pairs modulo 49, we find that F and H_{99} have no common zero modulo 49, so $H_{99} \neq 0$. The other non-constant H_i are non-zero by Step 2. For $i \in \{1, 3, 5, 6, 7, 8, 9, 10, 11, 14, 15, 20, 27, 29, 30, 33, 39, 42, 46, 47, 50, 51, 60, 62, 65, 69, 70\}$ in the first certificate, there is no residue pair (a, b) modulo 2 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{2}$, as one checks for the 2^2 residue pairs; hence $J_i \neq 0$. For $i \in \{2, 13, 17, 18, 19, 28, 38, 45, 52, 54, 58, 59, 66, 67, 68\}$ in the first certificate, there is no residue pair (a, b) modulo 3 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{3}$, as one checks for the 3^2 residue pairs; hence $J_i \neq 0$. For $i \in \{16, 21, 22, 23, 24, 25, 26, 34, 35, 36, 41, 53, 55, 56\}$ in the first certificate, there is no residue pair (a, b) modulo 4 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{4}$, as one checks for the 4^2 residue pairs; hence $J_i \neq 0$. For $i \in \{0, 4, 12, 32, 40, 43, 48, 49, 61, 63, 71\}$ in the first certificate, there is no residue pair (a, b) modulo 5 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{5}$, as one checks for the 5^2 residue pairs; hence $J_i \neq 0$. For $i \in \{31, 37, 64\}$ in the first certificate, there is no residue pair (a, b) modulo 11 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{11}$, as one checks for the 11^2 residue pairs; hence $J_i \neq 0$. For $i \in \{57\}$ in the first certificate, there is no residue pair (a, b) modulo 13 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{13}$, as one checks for the 13^2 residue pairs; hence $J_i \neq 0$. For $i \in \{44\}$ in the first certificate, there is no residue pair (a, b) modulo 17 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{17}$, as one checks for the 17^2 residue pairs; hence $J_i \neq 0$. For $i \in \{1, 3, 5, 6, 8, 9, 10, 11, 14, 16, 17, 19, 25, 27, 28, 30, 31, 34, 40, 43, 44, 47, 48, 53, 60, 63, 64, 65, 66\}$ in the second certificate, there is no residue pair (a, b) modulo 2 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{2}$, as one checks for the 2^2 residue pairs; hence $J_i \neq 0$. For $i \in \{2, 13, 18, 26, 29, 38, 42, 50, 51, 54, 56, 62\}$ in the second certificate, there is no residue pair (a, b) modulo 3 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{3}$, as one checks for the 3^2 residue pairs; hence $J_i \neq 0$. For $i \in \{12, 20, 21, 22, 23, 24, 35, 36, 39, 55, 58, 59, 61, 67\}$ in the second certificate, there is no residue pair (a, b) modulo 4 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{4}$, as one checks for the 4^2 residue pairs; hence $J_i \neq 0$. For $i \in \{0, 4, 7, 15, 33, 45, 46, 49, 52, 57, 68\}$ in the second certificate, there is no residue pair (a, b) modulo 5 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{5}$, as one checks for the 5^2 residue pairs; hence $J_i \neq 0$. For $i \in \{32, 37\}$ in the second certificate, there is no residue pair (a, b) modulo 11 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{11}$, as one checks for the 11^2 residue pairs; hence $J_i \neq 0$. For $i \in \{41\}$ in the second certificate, there is no residue pair (a, b) modulo 17 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{17}$, as one checks for the 17^2 residue pairs; hence $J_i \neq 0$. Also, all c_i are non-zero.

Step 2: signs. *The sign of H_0 .* Since $F(x, y) = 0$, (4) gives $Q_0 H_0 = M_0$, and M_0 and E_0 depend only on x . We show that $M_0 > 0$ and $E_0 > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_0 and E_0 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_0 > 0$ and $E_0 > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_0 > 0$ and $E_0 > 0$. Hence $M_0 > 0$ and $E_0 > 0$, so (5) gives $4Q_0 \geq E_0 > 0$, and $Q_0 H_0 = M_0 > 0$ gives $H_0 > 0$.

The sign of H_1 . By (6), $H_1 > 0$.

The sign of H_2 . If $H_2 \leq 0$, then (7) gives $(3x + 2y + 1)^2 \leq 0$ and $3(x - 1)^2 \leq 0$, so $|3x + 2y + 1| \leq 0$ and $|x - 1| \leq 0$. Solving these two linear forms for x and y , we find that

the only integer point satisfying both inequalities is $(1, -2)$, and it is not a solution of (1) with $H_2 \leq 0$. Hence $H_2 > 0$.

The sign of H_3 . If $H_3 \leq 0$, then (8) gives $(x + 2y - 1)^2 \leq 0$ and $3(x + 1)^2 \leq 0$, so $|x + 2y - 1| \leq 0$ and $|x + 1| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(-1, 1)$, and it is not a solution of (1) with $H_3 \leq 0$. Hence $H_3 > 0$.

The sign of H_4 . If $H_4 \leq 0$, then (9) gives $(2x + 2y + 1)^2 \leq 0$ and $3(2x + 1)^2 \leq 0$, so $|2x + 2y + 1| \leq 0$ and $|2x + 1| \leq 0$. Solving these two linear forms for x and y , we find that no integer point (x, y) satisfies both inequalities. Hence $H_4 > 0$.

The sign of H_5 . If $H_5 \leq 0$, then (10) gives $(x - 2y - 1)^2 \leq 0$ and $3(x - 1)^2 \leq 0$, so $|x - 2y - 1| \leq 0$ and $|x - 1| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(1, 0)$, and it is not a solution of (1) with $H_5 \leq 0$. Hence $H_5 > 0$.

The sign of H_6 . If $H_6 \leq 0$, then (11) gives $3(3x - 2y - 1)^2 \leq 0$ and $9(x + 1)^2 \leq 0$, so $|3x - 2y - 1| \leq 0$ and $|x + 1| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(-1, -2)$, and it is not a solution of (1) with $H_6 \leq 0$. Hence $H_6 > 0$.

The sign of H_7 . Since $F(x, y) = 0$, (12) gives $Q_7 H_7 = M_7$, and M_7 and E_7 depend only on x . We show that $M_7 > 0$ and $E_7 > 0$. If $x \geq 3$, write $x = 3 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_7 and E_7 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_7 > 0$ and $E_7 > 0$. It remains to consider $0 \leq x \leq 2$. For $x = 0$ and $x = 2$ we have $M_7 > 0$ and $E_7 > 0$. For $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with this x . Hence $M_7 > 0$ and $E_7 > 0$, so (13) gives $4Q_7 \geq E_7 > 0$, and $Q_7 H_7 = M_7 > 0$ gives $H_7 > 0$.

The sign of H_{10} . If $H_{10} \leq 0$, then (14) gives $(x - 2y)^2 \leq 0$ and $3x^2 \leq 0$, so $|x - 2y| \leq 0$ and $|x| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(0, 0)$, and it is not a solution of (1) with $H_{10} \leq 0$. Hence $H_{10} > 0$.

The sign of H_{12} . If $H_{12} \leq 0$, then (15) gives $(x - 2y + 1)^2 \leq 0$ and $3(x - 1)^2 \leq 0$, so $|x - 2y + 1| \leq 0$ and $|x - 1| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(1, 1)$, and it is not a solution of (1) with $H_{12} \leq 0$. Hence $H_{12} > 0$.

The sign of H_{13} . Since $F(x, y) = 0$, (16) gives $Q_{13} H_{13} = M_{13}$, and M_{13} and E_{13} depend only on x . We show that $M_{13} > 0$ and $E_{13} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{13} and E_{13} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{13} > 0$ and $E_{13} > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_{13} > 0$ and $E_{13} > 0$. Hence $M_{13} > 0$ and $E_{13} > 0$, so (17) gives $4Q_{13} \geq E_{13} > 0$, and $Q_{13} H_{13} = M_{13} > 0$ gives $H_{13} > 0$.

The sign of H_{15} . If $H_{15} \leq 0$, then (18) gives $(x - 2y)^2 \leq 0$ and $3(x + 2)^2 \leq 0$, so $|x - 2y| \leq 0$ and $|x + 2| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(-2, -1)$, and it is not a solution of (1) with $H_{15} \leq 0$. Hence $H_{15} > 0$.

The sign of H_{16} . Since $F(x, y) = 0$, (19) gives $Q_{16}(-H_{16}) = M_{16}$, and M_{16} and E_{16} depend only on x . We show that $M_{16} > 0$ and $E_{16} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{16} and E_{16} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{16} > 0$ and $E_{16} > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_{16} > 0$ and $E_{16} > 0$. Hence $M_{16} > 0$ and $E_{16} > 0$, so (20) gives $4Q_{16} \geq E_{16} > 0$, and $Q_{16}(-H_{16}) = M_{16} > 0$ gives $(-H_{16}) > 0$. Hence $H_{16} < 0$.

The sign of H_{18} . If $H_{18} \leq 0$, then (21) gives $3y^2 \leq 0$ and $9x^2 \leq 0$, so $|y| \leq 0$ and $|x| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(0, 0)$, and it is not a solution of (1) with $H_{18} \leq 0$. Hence $H_{18} > 0$.

The sign of H_{19} . Since $F(x, y) = 0$, (22) gives $Q_{19}H_{19} = M_{19}$, and M_{19} and E_{19} depend only on x . We show that $M_{19} > 0$ and $E_{19} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{19} and E_{19} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{19} > 0$ and $E_{19} > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_{19} > 0$ and $E_{19} > 0$. Hence $M_{19} > 0$ and $E_{19} > 0$, so (23) gives $16Q_{19} \geq E_{19} > 0$, and $Q_{19}H_{19} = M_{19} > 0$ gives $H_{19} > 0$.

The sign of H_{21} . By (24), $H_{21} \geq 0$, and $H_{21} \neq 0$ by Step 1, so $H_{21} > 0$.

The sign of H_{23} . By (25), $H_{23} \geq 0$, and $H_{23} \neq 0$ by Step 1, so $H_{23} > 0$.

The sign of H_{24} . By (26), $H_{24} > 0$.

The sign of H_{26} . By (27), $H_{26} \geq 0$, and $H_{26} \neq 0$ by Step 1, so $H_{26} > 0$.

The sign of H_{29} . By (28), $H_{29} \geq 0$, and $H_{29} \neq 0$ by Step 1, so $H_{29} > 0$.

The sign of H_{30} . By (29), $(-H_{30}) > 0$. Hence $H_{30} < 0$.

The sign of H_{32} . By (30), $H_{32} \geq 0$, and $H_{32} \neq 0$ by Step 1, so $H_{32} > 0$.

The sign of H_{34} . By (31), $H_{34} \geq 0$, and $H_{34} \neq 0$ by Step 1, so $H_{34} > 0$.

The sign of H_{36} . Since $F(x, y) = 0$, (32) gives $Q_{36}H_{36} = M_{36}$, and M_{36} and E_{36} depend only on x . We show that $M_{36} > 0$ and $E_{36} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{36} and E_{36} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{36} > 0$ and $E_{36} > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_{36} > 0$ and $E_{36} > 0$. Hence $M_{36} > 0$ and $E_{36} > 0$, so (33) gives $4Q_{36} \geq E_{36} > 0$, and $Q_{36}H_{36} = M_{36} > 0$ gives $H_{36} > 0$.

The sign of H_{37} . Since $F(x, y) = 0$, (34) gives $Q_{37}H_{37} = M_{37}$, and M_{37} and E_{37} depend only on x . We show that $M_{37} > 0$ and $E_{37} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{37} and E_{37} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{37} > 0$ and $E_{37} > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_{37} > 0$ and $E_{37} > 0$. Hence $M_{37} > 0$ and $E_{37} > 0$, so (35) gives $4Q_{37} \geq E_{37} > 0$, and $Q_{37}H_{37} = M_{37} > 0$ gives $H_{37} > 0$.

The sign of H_{38} . By (36), $H_{38} > 0$.

The sign of H_{40} . Since $F(x, y) = 0$, (37) gives $Q_{40}H_{40} = M_{40}$, and M_{40} and E_{40} depend only on x . We show that $M_{40} > 0$ and $E_{40} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{40} and E_{40} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{40} > 0$ and $E_{40} > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_{40} > 0$ and $E_{40} > 0$. Hence $M_{40} > 0$ and $E_{40} > 0$, so (38) gives $4Q_{40} \geq E_{40} > 0$, and $Q_{40}H_{40} = M_{40} > 0$ gives $H_{40} > 0$.

The sign of H_{42} . If $H_{42} \leq 0$, then (39) gives $(x + 2y - 1)^2 \leq 0$ and $3(x - 1)^2 \leq 0$, so $|x + 2y - 1| \leq 0$ and $|x - 1| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(1, 0)$, and it is not a solution of (1) with $H_{42} \leq 0$. Hence $H_{42} > 0$.

The sign of H_{43} . If $H_{43} \leq 0$, then (40) gives $(x + 2y + 1)^2 \leq 0$ and $3(x - 1)^2 \leq 0$, so $|x + 2y + 1| \leq 0$ and $|x - 1| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(1, -1)$, and it is not a solution of (1) with $H_{43} \leq 0$. Hence $H_{43} > 0$.

The sign of H_{44} . By (41), $H_{44} > 0$.

The sign of H_{45} . Since $F(x, y) = 0$, (42) gives $Q_{45}(-H_{45}) = M_{45}$, and M_{45} and E_{45} depend only on x . We show that $M_{45} > 0$ and $E_{45} > 0$. If $x \geq 2$, write $x = 2 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{45} and E_{45} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{45} > 0$ and $E_{45} > 0$. It remains to consider $0 \leq x \leq 1$. For $x = 0$ we have $M_{45} > 0$ and $E_{45} > 0$. For $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with this x . Hence $M_{45} > 0$ and $E_{45} > 0$, so (43) gives $4Q_{45} \geq E_{45} > 0$, and $Q_{45}(-H_{45}) = M_{45} > 0$ gives $(-H_{45}) > 0$. Hence $H_{45} < 0$.

The sign of H_{46} . Since $F(x, y) = 0$, (44) gives $Q_{46}(-H_{46}) = M_{46}$, and M_{46} and E_{46} depend

only on x . We show that $M_{46} > 0$ and $E_{46} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{46} and E_{46} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{46} > 0$ and $E_{46} > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_{46} > 0$ and $E_{46} > 0$. Hence $M_{46} > 0$ and $E_{46} > 0$, so (45) gives $4Q_{46} \geq E_{46} > 0$, and $Q_{46}(-H_{46}) = M_{46} > 0$ gives $(-H_{46}) > 0$. Hence $H_{46} < 0$.

The sign of H_{48} . Since $F(x, y) = 0$, (46) gives $Q_{48}(-H_{48}) = M_{48}$, and M_{48} and E_{48} depend only on x . We show that $M_{48} > 0$ and $E_{48} > 0$. If $x \geq 6$, write $x = 6 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{48} and E_{48} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{48} > 0$ and $E_{48} > 0$. It remains to consider $0 \leq x \leq 5$. For $x = 0$ we have $M_{48} > 0$ and $E_{48} > 0$. For $x = 2$ and $x = 4$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with such x . For $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with this x . For $x = 5$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every y , as one checks for the 11 residues of y modulo 11, so (1) has no solution with this x . For $x = 3$, we have $F(x, y) \not\equiv 0 \pmod{47}$ for every y , as one checks for the 47 residues of y modulo 47, so (1) has no solution with this x . Hence $M_{48} > 0$ and $E_{48} > 0$, so (47) gives $4Q_{48} \geq E_{48} > 0$, and $Q_{48}(-H_{48}) = M_{48} > 0$ gives $(-H_{48}) > 0$. Hence $H_{48} < 0$.

The sign of H_{53} . By (48), $H_{53} \geq 0$, and $H_{53} \neq 0$ by Step 1, so $H_{53} > 0$.

The sign of H_{54} . Since $F(x, y) = 0$, (49) gives $Q_{54}H_{54} = M_{54}$, and M_{54} and E_{54} depend only on x . We show that $M_{54} > 0$ and $E_{54} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{54} and E_{54} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{54} > 0$ and $E_{54} > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_{54} > 0$ and $E_{54} > 0$. Hence $M_{54} > 0$ and $E_{54} > 0$, so (50) gives $4Q_{54} \geq E_{54} > 0$, and $Q_{54}H_{54} = M_{54} > 0$ gives $H_{54} > 0$.

The sign of H_{55} . Since $F(x, y) = 0$, (51) gives $Q_{55}(-H_{55}) = M_{55}$, and M_{55} and E_{55} depend only on x . We show that $M_{55} > 0$ and $E_{55} > 0$. If $x \geq 2$, write $x = 2 + z$ with $z \geq 0$; if $x \leq -4$, write $x = -4 - z$ with $z \geq 0$. In both cases, expanding M_{55} and E_{55} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{55} > 0$ and $E_{55} > 0$. It remains to consider $-3 \leq x \leq 1$. For $x = -2$ and $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with such x . For $x = -1$ and $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with such x . For $x = -3$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , as one checks for the 7 residues of y modulo 7, so (1) has no solution with this x . Hence $M_{55} > 0$ and $E_{55} > 0$, so (52) gives $4Q_{55} \geq E_{55} > 0$, and $Q_{55}(-H_{55}) = M_{55} > 0$ gives $(-H_{55}) > 0$. Hence $H_{55} < 0$.

The sign of H_{56} . Since $F(x, y) = 0$, (53) gives $Q_{56}(-H_{56}) = M_{56}$, and M_{56} and E_{56} depend only on x . We show that $M_{56} > 0$ and $E_{56} > 0$. If $x \geq 3$, write $x = 3 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{56} and E_{56} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{56} > 0$ and $E_{56} > 0$. It remains to consider $0 \leq x \leq 2$. For $x = 0$ and $x = 2$ we have $M_{56} > 0$ and $E_{56} > 0$. For $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with this x . Hence $M_{56} > 0$ and $E_{56} > 0$, so (54) gives $4Q_{56} \geq E_{56} > 0$, and $Q_{56}(-H_{56}) = M_{56} > 0$ gives $(-H_{56}) > 0$. Hence $H_{56} < 0$.

The sign of H_{58} . Since $F(x, y) = 0$, (55) gives $Q_{58}(-H_{58}) = M_{58}$, and M_{58} and E_{58} depend only on x . We show that $M_{58} > 0$ and $E_{58} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{58} and E_{58} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{58} > 0$ and $E_{58} > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_{58} > 0$ and $E_{58} > 0$. Hence $M_{58} > 0$ and $E_{58} > 0$, so (56) gives $4Q_{58} \geq E_{58} > 0$, and $Q_{58}(-H_{58}) = M_{58} > 0$ gives $(-H_{58}) > 0$. Hence $H_{58} < 0$.

The sign of H_{59} . By (57), $H_{59} > 0$.

The sign of H_{61} . Since $F(x, y) = 0$, (58) gives $Q_{61}H_{61} = M_{61}$, and M_{61} and E_{61} depend

only on x . We show that $M_{61} > 0$ and $E_{61} > 0$. If $x \geq 2$, write $x = 2 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_{61} and E_{61} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{61} > 0$ and $E_{61} > 0$. It remains to consider $-1 \leq x \leq 1$. For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with this x . For $x = -1$ and $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with such x . Hence $M_{61} > 0$ and $E_{61} > 0$, so (59) gives $4Q_{61} \geq E_{61} > 0$, and $Q_{61}H_{61} = M_{61} > 0$ gives $H_{61} > 0$.

The sign of H_{62} . If $H_{62} \leq 0$, then (60) gives $(3x - 2y - 1)^2 \leq 0$ and $3(x + 1)^2 \leq 0$, so $|3x - 2y - 1| \leq 0$ and $|x + 1| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(-1, -2)$, and it is not a solution of (1) with $H_{62} \leq 0$. Hence $H_{62} > 0$.

The sign of H_{68} . By (61), $H_{68} > 0$.

The sign of H_{70} . If $H_{70} \leq 0$, then (62) gives $(x + 2y + 2)^2 \leq 0$ and $3x^2 \leq 0$, so $|x + 2y + 2| \leq 0$ and $|x| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(0, -1)$, and it is not a solution of (1) with $H_{70} \leq 0$. Hence $H_{70} > 0$.

The sign of H_{72} . If $H_{72} \leq 0$, then (63) gives $(x - 2y + 3)^2 \leq 0$ and $3(x + 1)^2 \leq 0$, so $|x - 2y + 3| \leq 0$ and $|x + 1| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(-1, 1)$, and it is not a solution of (1) with $H_{72} \leq 0$. Hence $H_{72} > 0$.

The sign of H_{73} . By (64), $H_{73} > 0$.

The sign of H_{75} . Since $F(x, y) = 0$, (65) gives $Q_{75}H_{75} = M_{75}$, and M_{75} and E_{75} depend only on x . We show that $M_{75} > 0$ and $E_{75} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{75} and E_{75} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{75} > 0$ and $E_{75} > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_{75} > 0$ and $E_{75} > 0$. Hence $M_{75} > 0$ and $E_{75} > 0$, so (66) gives $4Q_{75} \geq E_{75} > 0$, and $Q_{75}H_{75} = M_{75} > 0$ gives $H_{75} > 0$.

The sign of H_{76} . By (67), $H_{76} > 0$.

The sign of H_{80} . By (68), $H_{80} \geq 0$, and $H_{80} \neq 0$ by Step 1, so $H_{80} > 0$.

The sign of H_{83} . If $H_{83} \leq 0$, then (69) gives $(x + 2y)^2 \leq 0$ and $3(x + 2)^2 \leq 0$, so $|x + 2y| \leq 0$ and $|x + 2| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(-2, 1)$, and it is not a solution of (1) with $H_{83} \leq 0$. Hence $H_{83} > 0$.

The sign of H_{84} . By (70), $H_{84} > 0$.

The sign of H_{87} . If $H_{87} \leq 0$, then (71) gives $(x + 2y + 3)^2 \leq 0$ and $3(x + 1)^2 \leq 0$, so $|x + 2y + 3| \leq 0$ and $|x + 1| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(-1, -1)$, and it is not a solution of (1) with $H_{87} \leq 0$. Hence $H_{87} > 0$.

The sign of H_{89} . If $(-H_{89}) \leq 0$, then (72) gives $3(x + 2y + 2)^2 \leq 28$ and $(3x + 2)^2 \leq 28$, so $|x + 2y + 2| \leq 3$ and $|3x + 2| \leq 5$. Solving these two linear forms for x and y , we find that the only integer points satisfying both inequalities are $(-2, -1)$, $(-2, 0)$, $(-2, 1)$, $(-1, -2)$, $(-1, -1)$, $(-1, 0)$, $(-1, 1)$, $(0, -2)$, $(0, -1)$, $(0, 0)$, $(1, -3)$, $(1, -2)$, $(1, -1)$, $(1, 0)$, and none of them is a solution of (1) with $(-H_{89}) \leq 0$. Hence $(-H_{89}) > 0$. Hence $H_{89} < 0$.

The sign of H_{92} . If $H_{92} \leq 0$, then (73) gives $3(x + 2y + 2)^2 \leq 28$ and $(3x + 2)^2 \leq 28$, so $|x + 2y + 2| \leq 3$ and $|3x + 2| \leq 5$. Solving these two linear forms for x and y , we find that the only integer points satisfying both inequalities are $(-2, -1)$, $(-2, 0)$, $(-2, 1)$, $(-1, -2)$, $(-1, -1)$, $(-1, 0)$, $(-1, 1)$, $(0, -2)$, $(0, -1)$, $(0, 0)$, $(1, -3)$, $(1, -2)$, $(1, -1)$, $(1, 0)$, and none of them is a solution of (1) with $H_{92} \leq 0$. Hence $H_{92} > 0$.

The sign of H_{93} . If $H_{93} \leq 0$, then (74) gives $4(x + y)^2 \leq 5$ and $(2x - 1)^2 \leq 5$, so $|x + y| \leq 1$ and $|2x - 1| \leq 2$. Solving these two linear forms for x and y , we find that the only integer points

satisfying both inequalities are $(0, -1)$, $(0, 0)$, $(0, 1)$, $(1, -2)$, $(1, -1)$, $(1, 0)$, and none of them is a solution of (1) with $H_{93} \leq 0$. Hence $H_{93} > 0$.

The sign of H_{94} . If $H_{94} \leq 0$, then (75) gives $3(x + 2y)^2 \leq 28$ and $(3x - 2)^2 \leq 28$, so $|x + 2y| \leq 3$ and $|3x - 2| \leq 5$. Solving these two linear forms for x and y , we find that the only integer points satisfying both inequalities are $(-1, -1)$, $(-1, 0)$, $(-1, 1)$, $(-1, 2)$, $(0, -1)$, $(0, 0)$, $(0, 1)$, $(1, -2)$, $(1, -1)$, $(1, 0)$, $(1, 1)$, $(2, -2)$, $(2, -1)$, $(2, 0)$, and none of them is a solution of (1) with $H_{94} \leq 0$. Hence $H_{94} > 0$.

The sign of H_{96} . Suppose that $(-H_{96}) \leq 0$. Then (76) gives $(x + y - 3)(y + 2) < 0$, so the two factors have opposite signs. As these factors are integers, this means that $x + y \geq 4$ and $y \leq -3$, or $x + y \leq 2$ and $y \geq -1$. In the first case, write $x + y = 4 + X$ and $y = -3 - Y$ with $X, Y \geq 0$, so that $x = X + Y + 7$ and $y = -Y - 3$. Then $F(x, y) = G_{96,1}(X, Y)$, where

$$G_{96,1}(X, Y) = 2X^4 + 8X^3Y + 56X^3 + 12X^2Y^2 + 168X^2Y + 588X^2 + 8XY^3 \\ + 168XY^2 + 1176XY + 2745X + 2Y^4 + 55Y^3 + 579Y^2 + 2717Y + 4782.$$

All coefficients of $G_{96,1}$ have the same sign and its constant term is non-zero, so $G_{96,1}(X, Y) \neq 0$. In the second case, write $x + y = 2 - X$ and $y = -1 + Y$ with $X, Y \geq 0$, so that $x = -X - Y + 3$ and $y = Y - 1$. Then $F(x, y) = G_{96,2}(X, Y)$, where

$$G_{96,2}(X, Y) = 2X^4 + 8X^3Y - 24X^3 + 12X^2Y^2 - 72X^2Y + 108X^2 + 8XY^3 \\ - 72XY^2 + 216XY - 217X + 2Y^4 - 23Y^3 + 105Y^2 - 213Y + 166.$$

Expanding $G_{96,2}(4 + X, 4 + Y)$, and, for $0 \leq i < 4$, $G_{96,2}(i, 4 + Y)$ and $G_{96,2}(4 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{96,2}(i, j) \neq 0$ for $0 \leq i, j < 4$. Hence $G_{96,2}(X, Y) \neq 0$ for all $X, Y \geq 0$. In both cases $F(x, y) \neq 0$, a contradiction. Hence $(-H_{96}) > 0$. Hence $H_{96} < 0$.

The sign of H_{97} . If $H_{97} \leq 0$, then (77) gives $3(y - 1)^2 \leq 0$ and $9x^2 \leq 0$, so $|y - 1| \leq 0$ and $|x| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(0, 1)$, and it is not a solution of (1) with $H_{97} \leq 0$. Hence $H_{97} > 0$.

The sign of H_{102} . Since $F(x, y) = 0$, (78) gives $Q_{102}(-H_{102}) = M_{102}$, and M_{102} and E_{102} depend only on x . We show that $M_{102} > 0$ and $E_{102} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{102} and E_{102} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{102} > 0$ and $E_{102} > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_{102} > 0$ and $E_{102} > 0$. Hence $M_{102} > 0$ and $E_{102} > 0$, so (79) gives $4Q_{102} \geq E_{102} > 0$, and $Q_{102}(-H_{102}) = M_{102} > 0$ gives $(-H_{102}) > 0$. Hence $H_{102} < 0$.

The sign of H_{104} . Suppose that $(-H_{104}) \leq 0$. Then (80) gives $(2y + 1)(-4x + 2y - 3) < 0$, so the two factors have opposite signs. As these factors are integers, this means that $y \geq 0$ and $-2x + y \leq 1$, or $y \leq -1$ and $-2x + y \geq 2$. In the first case, write $y = X$ and $-2x + y = -Y + 1$ with integers $X, Y \geq 0$, so that $x = \frac{X+Y-1}{2}$ and $y = X$. Then $8F(x, y) = G_{104,1}(X, Y)$, where

$$G_{104,1}(X, Y) = X^4 + 4X^3Y + 4X^3 + 6X^2Y^2 - 12X^2Y + 6X^2 + 4XY^3 \\ - 12XY^2 + 12XY + 8X + Y^4 - 4Y^3 + 6Y^2 + 21.$$

Expanding $G_{104,1}(1 + X, 1 + Y)$, and, for $i = 0$, $G_{104,1}(i, 1 + Y)$ and $G_{104,1}(1 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{104,1}(i, j) \neq 0$ for $i = j = 0$. Hence $G_{104,1}(X, Y) \neq 0$ for all $X, Y \geq 0$. In the second case, write $y = -X - 1$ and $-2x + y = Y + 2$ with integers $X, Y \geq 0$, so that $x = \frac{-X-Y-3}{2}$ and $y = -X - 1$. Then $8F(x, y) = G_{104,2}(X, Y)$, where

$$G_{104,2}(X, Y) = X^4 + 4X^3Y + 4X^3 + 6X^2Y^2 + 36X^2Y + 30X^2 + 4XY^3 \\ + 36XY^2 + 108XY + 72X + Y^4 + 12Y^3 + 54Y^2 + 104Y + 77.$$

All coefficients of $G_{104,2}$ have the same sign and its constant term is non-zero, so $G_{104,2}(X, Y) \neq 0$ for all $X, Y \geq 0$. In both cases $F(x, y) \neq 0$, a contradiction. Hence $(-H_{104}) > 0$. Hence $H_{104} < 0$.

The sign of H_{105} . Since $F(x, y) = 0$, (81) gives $Q_{105}H_{105} = M_{105}$, and M_{105} and E_{105} depend only on x . We show that $M_{105} > 0$ and $E_{105} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{105} and E_{105} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{105} > 0$ and $E_{105} > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_{105} > 0$ and $E_{105} > 0$. Hence $M_{105} > 0$ and $E_{105} > 0$, so (82) gives $4Q_{105} \geq E_{105} > 0$, and $Q_{105}H_{105} = M_{105} > 0$ gives $H_{105} > 0$.

The sign of H_{111} . By (83), $H_{111} > 0$.

The sign of H_{113} . Suppose that $(-H_{113}) \leq 0$, and put $Y = H_{113}$, so that $Y \geq 0$. Since $(-H_{113}) = -x + y^2 + 2y$, we have $x = y^2 + 2y + Y$. If $y \geq 0$, put $X = y$; if $y \leq -1$, put $X = -1 - y$. In both cases $X \geq 0$, and substituting into F gives $F(x, y) = G_{113,1}(X, Y)$ in the first case and $F(x, y) = G_{113,2}(X, Y)$ in the second, where

$$\begin{aligned} G_{113,1}(X, Y) &= 2X^8 + 16X^7 + 8X^6Y + 48X^6 + 48X^5Y + 64X^5 + 12X^4Y^2 \\ &\quad + 96X^4Y + 32X^4 + 48X^3Y^2 + 64X^3Y + X^3 + 8X^2Y^3 \\ &\quad + 48X^2Y^2 + X^2 + 16XY^3 + 3X + 2Y^4 + Y + 3, \\ G_{113,2}(X, Y) &= 2X^8 + 8X^6Y - 8X^6 + 12X^4Y^2 - 24X^4Y + 12X^4 - X^3 \\ &\quad + 8X^2Y^3 - 24X^2Y^2 + 24X^2Y - 10X^2 - 4X + 2Y^4 - 8Y^3 \\ &\quad + 12Y^2 - 7Y + 2. \end{aligned}$$

All coefficients of $G_{113,1}$ have the same sign and its constant term is non-zero, so $G_{113,1}(X, Y) \neq 0$ for all $X, Y \geq 0$. Expanding $G_{113,2}(2 + X, 2 + Y)$, and, for $0 \leq i < 2$, $G_{113,2}(i, 2 + Y)$ and $G_{113,2}(2 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{113,2}(i, j) \neq 0$ for $0 \leq i, j < 2$. Hence $G_{113,2}(X, Y) \neq 0$ for all $X, Y \geq 0$. In both cases $F(x, y) \neq 0$, a contradiction. Hence $(-H_{113}) > 0$. Hence $H_{113} < 0$.

The sign of H_{118} . If $H_{118} \leq 0$, then (84) gives $(x - 2y + 1)^2 \leq 0$ and $3(x + 1)^2 \leq 0$, so $|x - 2y + 1| \leq 0$ and $|x + 1| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(-1, 0)$, and it is not a solution of (1) with $H_{118} \leq 0$. Hence $H_{118} > 0$.

The sign of H_{120} . By (85), $H_{120} \geq 0$, and $H_{120} \neq 0$ by Step 1, so $H_{120} > 0$.

The sign of H_{123} . By (86), $H_{123} \geq 0$, and $H_{123} \neq 0$ by Step 1, so $H_{123} > 0$.

The sign of H_{127} . Suppose that $H_{127} \leq 0$, and put $Y = -H_{127}$, so that $Y \geq 0$. Since $H_{127} = x + y^2 + 2y - 1$, we have $x = -y^2 - 2y - Y + 1$. If $y \geq 0$, put $X = y$; if $y \leq -1$, put $X = -1 - y$. In both cases $X \geq 0$, and substituting into F gives $F(x, y) = G_{127,1}(X, Y)$ in the first case and $F(x, y) = G_{127,2}(X, Y)$ in the second, where

$$\begin{aligned} G_{127,1}(X, Y) &= 2X^8 + 16X^7 + 8X^6Y + 40X^6 + 48X^5Y + 16X^5 + 12X^4Y^2 \\ &\quad + 72X^4Y - 52X^4 + 48X^3Y^2 - 32X^3Y - 15X^3 + 8X^2Y^3 \\ &\quad + 24X^2Y^2 - 72X^2Y + 39X^2 + 16XY^3 - 48XY^2 + 48XY - 17X \\ &\quad + 2Y^4 - 8Y^3 + 12Y^2 - 9Y + 6, \\ G_{127,2}(X, Y) &= 2X^8 + 8X^6Y - 16X^6 + 12X^4Y^2 - 48X^4Y + 48X^4 - X^3 \\ &\quad + 8X^2Y^3 - 48X^2Y^2 + 96X^2Y - 68X^2 - 4X + 2Y^4 - 16Y^3 \\ &\quad + 48Y^2 - 65Y + 35. \end{aligned}$$

Expanding $G_{127,1}(2 + X, 2 + Y)$, and, for $0 \leq i < 2$, $G_{127,1}(i, 2 + Y)$ and $G_{127,1}(2 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{127,1}(i, j) \neq 0$ for $0 \leq i, j < 2$. Hence $G_{127,1}(X, Y) \neq 0$ for all $X, Y \geq 0$. Expanding $G_{127,2}(3 + X, 3 + Y)$, and, for

$0 \leq i < 3$, $G_{127,2}(i, 3 + Y)$ and $G_{127,2}(3 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{127,2}(i, j) \neq 0$ for $0 \leq i, j < 3$. Hence $G_{127,2}(X, Y) \neq 0$ for all $X, Y \geq 0$. In both cases $F(x, y) \neq 0$, a contradiction. Hence $H_{127} > 0$.

The sign of H_{132} . Suppose that $H_{132} \leq 0$. Then (87) gives $(x + y + 3)(y - 2) < 0$, so the two factors have opposite signs. As these factors are integers, this means that $x + y \geq -2$ and $y \leq 1$, or $x + y \leq -4$ and $y \geq 3$. In the first case, write $x + y = -2 + X$ and $y = 1 - Y$ with $X, Y \geq 0$, so that $x = X + Y - 3$ and $y = -Y + 1$. Then $F(x, y) = G_{132,1}(X, Y)$, where

$$\begin{aligned} G_{132,1}(X, Y) = & 2X^4 + 8X^3Y - 24X^3 + 12X^2Y^2 - 72X^2Y + 108X^2 + 8XY^3 \\ & - 72XY^2 + 216XY - 215X + 2Y^4 - 25Y^3 + 111Y^2 - 219Y + 164. \end{aligned}$$

Expanding $G_{132,1}(6 + X, 6 + Y)$, and, for $0 \leq i < 6$, $G_{132,1}(i, 6 + Y)$ and $G_{132,1}(6 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{132,1}(i, j) \neq 0$ for $0 \leq i, j < 6$. Hence $G_{132,1}(X, Y) \neq 0$ for all $X, Y \geq 0$. In the second case, write $x + y = -4 - X$ and $y = 3 + Y$ with $X, Y \geq 0$, so that $x = -X - Y - 7$ and $y = Y + 3$. Then $F(x, y) = G_{132,2}(X, Y)$, where

$$\begin{aligned} G_{132,2}(X, Y) = & 2X^4 + 8X^3Y + 56X^3 + 12X^2Y^2 + 168X^2Y + 588X^2 + 8XY^3 \\ & + 168XY^2 + 1176XY + 2743X + 2Y^4 + 57Y^3 + 597Y^2 + 2771Y + 4828. \end{aligned}$$

All coefficients of $G_{132,2}$ have the same sign and its constant term is non-zero, so $G_{132,2}(X, Y) \neq 0$. In both cases $F(x, y) \neq 0$, a contradiction. Hence $H_{132} > 0$.

The sign of H_{136} . If $(-H_{136}) \leq 0$, then (88) gives $(y - 1)^2 \leq 1$ and $x^2 \leq 1$, so $|y - 1| \leq 1$ and $|x| \leq 1$. Solving these two linear forms for x and y , we find that the only integer points satisfying both inequalities are $(-1, 0)$, $(-1, 1)$, $(-1, 2)$, $(0, 0)$, $(0, 1)$, $(0, 2)$, $(1, 0)$, $(1, 1)$, $(1, 2)$, and none of them is a solution of (1) with $(-H_{136}) \leq 0$. Hence $(-H_{136}) > 0$. Hence $H_{136} < 0$.

The sign of H_{139} . Suppose that $H_{139} \leq 0$, and put $Y = -H_{139}$, so that $Y \geq 0$. Since $H_{139} = x + y^2 + y$, we have $x = -y^2 - y - Y$. If $y \geq 0$, put $X = y$; if $y \leq -1$, put $X = -1 - y$. In both cases $X \geq 0$, and substituting into F gives $F(x, y) = G_{139,1}(X, Y)$ in the first case and $F(x, y) = G_{139,2}(X, Y)$ in the second, where

$$\begin{aligned} G_{139,1}(X, Y) = & 2X^8 + 8X^7 + 8X^6Y + 12X^6 + 24X^5Y + 8X^5 + 12X^4Y^2 \\ & + 24X^4Y + 2X^4 + 24X^3Y^2 + 8X^3Y + X^3 + 8X^2Y^3 \\ & + 12X^2Y^2 - X^2 + 8XY^3 + 2Y^4 - Y + 3, \\ G_{139,2}(X, Y) = & 2X^8 + 8X^7 + 8X^6Y + 12X^6 + 24X^5Y + 8X^5 + 12X^4Y^2 \\ & + 24X^4Y + 2X^4 + 24X^3Y^2 + 8X^3Y - X^3 + 8X^2Y^3 \\ & + 12X^2Y^2 - 4X^2 + 8XY^3 - 5X + 2Y^4 - Y + 1. \end{aligned}$$

Expanding $G_{139,1}(1 + X, 1 + Y)$, and, for $i = 0$, $G_{139,1}(i, 1 + Y)$ and $G_{139,1}(1 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{139,1}(i, j) \neq 0$ for $i = j = 0$. Hence $G_{139,1}(X, Y) \neq 0$ for all $X, Y \geq 0$. Expanding $G_{139,2}(1 + X, 1 + Y)$, and, for $i = 0$, $G_{139,2}(i, 1 + Y)$ and $G_{139,2}(1 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{139,2}(i, j) \neq 0$ for $i = j = 0$. Hence $G_{139,2}(X, Y) \neq 0$ for all $X, Y \geq 0$. In both cases $F(x, y) \neq 0$, a contradiction. Hence $H_{139} > 0$.

The sign of H_{146} . Suppose that $(-H_{146}) \leq 0$. Then (89) gives $(x + y - 6)(y + 3) < 0$, so the two factors have opposite signs. As these factors are integers, this means that $x + y \geq 7$ and $y \leq -4$, or $x + y \leq 5$ and $y \geq -2$. In the first case, write $x + y = 7 + X$ and $y = -4 - Y$ with $X, Y \geq 0$, so that $x = X + Y + 11$ and $y = -Y - 4$. Then $F(x, y) = G_{146,1}(X, Y)$, where

$$\begin{aligned} G_{146,1}(X, Y) = & 2X^4 + 8X^3Y + 88X^3 + 12X^2Y^2 + 264X^2Y + 1452X^2 \\ & + 8XY^3 + 264XY^2 + 2904XY + 10649X + 2Y^4 + 87Y^3 + 1440Y^2 \end{aligned}$$

$$+ 10600Y + 29228.$$

All coefficients of $G_{146,1}$ have the same sign and its constant term is non-zero, so $G_{146,1}(X, Y) \neq 0$. In the second case, write $x + y = 5 - X$ and $y = -2 + Y$ with $X, Y \geq 0$, so that $x = -X - Y + 7$ and $y = Y - 2$. Then $F(x, y) = G_{146,2}(X, Y)$, where

$$\begin{aligned} G_{146,2}(X, Y) = & 2X^4 + 8X^3Y - 56X^3 + 12X^2Y^2 - 168X^2Y + 588X^2 + 8XY^3 \\ & - 168XY^2 + 1176XY - 2745X + 2Y^4 - 55Y^3 + 582Y^2 - 2732Y + 4802. \end{aligned}$$

Expanding $G_{146,2}(9 + X, 9 + Y)$, and, for $0 \leq i < 9$, $G_{146,2}(i, 9 + Y)$ and $G_{146,2}(9 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{146,2}(i, j) \neq 0$ for $0 \leq i, j < 9$. Hence $G_{146,2}(X, Y) \neq 0$ for all $X, Y \geq 0$. In both cases $F(x, y) \neq 0$, a contradiction. Hence $(-H_{146}) > 0$. Hence $H_{146} < 0$.

The sign of H_{151} . Suppose that $(-H_{151}) \leq 0$. Then (90) gives $(2y - 3)(-x + y + 1) < 0$, so the two factors have opposite signs. As these factors are integers, this means that $y \geq 2$ and $-x + y \leq -2$, or $y \leq 1$ and $-x + y \geq 0$. In the first case, write $y = 2 + X$ and $-x + y = -2 - Y$ with $X, Y \geq 0$, so that $x = X + Y + 4$ and $y = X + 2$. Then $F(x, y) = G_{151,1}(X, Y)$, where

$$\begin{aligned} G_{151,1}(X, Y) = & 2X^4 + 8X^3Y + 33X^3 + 12X^2Y^2 + 96X^2Y + 198X^2 + 8XY^3 \\ & + 96XY^2 + 384XY + 526X + 2Y^4 + 32Y^3 + 192Y^2 + 513Y + 529. \end{aligned}$$

All coefficients of $G_{151,1}$ have the same sign and its constant term is non-zero, so $G_{151,1}(X, Y) \neq 0$. In the second case, write $y = 1 - X$ and $-x + y = Y$ with $X, Y \geq 0$, so that $x = -X - Y + 1$ and $y = -X + 1$. Then $F(x, y) = G_{151,2}(X, Y)$, where

$$\begin{aligned} G_{151,2}(X, Y) = & 2X^4 + 8X^3Y - 9X^3 + 12X^2Y^2 - 24X^2Y + 15X^2 + 8XY^3 \\ & - 24XY^2 + 24XY - 13X + 2Y^4 - 8Y^3 + 12Y^2 - 9Y + 8. \end{aligned}$$

Expanding $G_{151,2}(2 + X, 2 + Y)$, and, for $0 \leq i < 2$, $G_{151,2}(i, 2 + Y)$ and $G_{151,2}(2 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{151,2}(i, j) \neq 0$ for $0 \leq i, j < 2$. Hence $G_{151,2}(X, Y) \neq 0$ for all $X, Y \geq 0$. In both cases $F(x, y) \neq 0$, a contradiction. Hence $(-H_{151}) > 0$. Hence $H_{151} < 0$.

The sign of H_{156} . Since $F(x, y) = 0$, (91) gives $Q_{156}(-H_{156}) = M_{156}$, and M_{156} and E_{156} depend only on x . We show that $M_{156} > 0$ and $E_{156} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_{156} and E_{156} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{156} > 0$ and $E_{156} > 0$. It remains to consider $-1 \leq x \leq 0$. For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with this x . For $x = -1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with this x . Hence $M_{156} > 0$ and $E_{156} > 0$, so (92) gives $4Q_{156} \geq E_{156} > 0$, and $Q_{156}(-H_{156}) = M_{156} > 0$ gives $(-H_{156}) > 0$. Hence $H_{156} < 0$.

The sign of H_{174} . By (93), $(-H_{174}) \geq 0$, and $H_{174} \neq 0$ by Step 1, so $(-H_{174}) > 0$. Hence $H_{174} < 0$.

The sign of H_{179} . Suppose that $(-H_{179}) \leq 0$. Then (94) gives $(y + 3)(-x + y - 2) < 0$, so the two factors have opposite signs. As these factors are integers, this means that $y \geq -2$ and $-x + y \leq 1$, or $y \leq -4$ and $-x + y \geq 3$. In the first case, write $y = -2 + X$ and $-x + y = 1 - Y$ with $X, Y \geq 0$, so that $x = X + Y - 3$ and $y = X - 2$. Then $F(x, y) = G_{179,1}(X, Y)$, where

$$\begin{aligned} G_{179,1}(X, Y) = & 2X^4 + 8X^3Y - 23X^3 + 12X^2Y^2 - 72X^2Y + 102X^2 + 8XY^3 \\ & - 72XY^2 + 216XY - 202X + 2Y^4 - 24Y^3 + 108Y^2 - 215Y + 152. \end{aligned}$$

Expanding $G_{179,1}(5 + X, 5 + Y)$, and, for $0 \leq i < 5$, $G_{179,1}(i, 5 + Y)$ and $G_{179,1}(5 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{179,1}(i, j) \neq 0$ for

$0 \leq i, j < 5$. Hence $G_{179,1}(X, Y) \neq 0$ for all $X, Y \geq 0$. In the second case, write $y = -4 - X$ and $-x + y = 3 + Y$ with $X, Y \geq 0$, so that $x = -X - Y - 7$ and $y = -X - 4$. Then $F(x, y) = G_{179,2}(X, Y)$, where

$$\begin{aligned} G_{179,2}(X, Y) = & 2X^4 + 8X^3Y + 55X^3 + 12X^2Y^2 + 168X^2Y + 576X^2 + 8XY^3 \\ & + 168XY^2 + 1176XY + 2694X + 2Y^4 + 56Y^3 + 588Y^2 + 2743Y + 4730. \end{aligned}$$

All coefficients of $G_{179,2}$ have the same sign and its constant term is non-zero, so $G_{179,2}(X, Y) \neq 0$. In both cases $F(x, y) \neq 0$, a contradiction. Hence $(-H_{179}) > 0$. Hence $H_{179} < 0$.

The sign of H_{180} . Suppose that $H_{180} \leq 0$, and put $Y = -H_{180}$, so that $Y \geq 0$. Since $H_{180} = -x + y^2 - 1$, we have $x = y^2 + Y - 1$. If $y \geq 0$, put $X = y$; if $y \leq -1$, put $X = -1 - y$. In both cases $X \geq 0$, and substituting into F gives $F(x, y) = G_{180,1}(X, Y)$ in the first case and $F(x, y) = G_{180,2}(X, Y)$ in the second, where

$$\begin{aligned} G_{180,1}(X, Y) = & 2X^8 + 8X^6Y - 8X^6 + 12X^4Y^2 - 24X^4Y + 12X^4 + X^3 \\ & + 8X^2Y^3 - 24X^2Y^2 + 24X^2Y - 7X^2 + X + 2Y^4 - 8Y^3 \\ & + 12Y^2 - 7Y + 4, \\ G_{180,2}(X, Y) = & 2X^8 + 16X^7 + 8X^6Y + 48X^6 + 48X^5Y + 64X^5 + 12X^4Y^2 \\ & + 96X^4Y + 32X^4 + 48X^3Y^2 + 64X^3Y - X^3 + 8X^2Y^3 \\ & + 48X^2Y^2 - 2X^2 + 16XY^3 - 2X + 2Y^4 + Y + 1. \end{aligned}$$

Expanding $G_{180,1}(1 + X, 1 + Y)$, and, for $i = 0$, $G_{180,1}(i, 1 + Y)$ and $G_{180,1}(1 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{180,1}(i, j) \neq 0$ for $i = j = 0$. Hence $G_{180,1}(X, Y) \neq 0$ for all $X, Y \geq 0$. Expanding $G_{180,2}(1 + X, 1 + Y)$, and, for $i = 0$, $G_{180,2}(i, 1 + Y)$ and $G_{180,2}(1 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{180,2}(i, j) \neq 0$ for $i = j = 0$. Hence $G_{180,2}(X, Y) \neq 0$ for all $X, Y \geq 0$. In both cases $F(x, y) \neq 0$, a contradiction. Hence $H_{180} > 0$.

The sign of H_{193} . Since $F(x, y) = 0$, (95) gives $Q_{193}H_{193} = M_{193}$, and M_{193} and E_{193} depend only on x . We show that $M_{193} > 0$ and $E_{193} > 0$. If $x \geq 13$, write $x = 13 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{193} and E_{193} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{193} > 0$ and $E_{193} > 0$. It remains to consider $0 \leq x \leq 12$. For $x = 0$ we have $M_{193} > 0$ and $E_{193} > 0$. For $x \in \{2, 4, 6, 8, 10, 12\}$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with such x . For $x \in \{1, 9, 11\}$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with such x . For $x = 5$ and $x = 7$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every y , as one checks for the 11 residues of y modulo 11, so (1) has no solution with such x . For $x = 3$, we have $F(x, y) \not\equiv 0 \pmod{47}$ for every y , as one checks for the 47 residues of y modulo 47, so (1) has no solution with this x . Hence $M_{193} > 0$ and $E_{193} > 0$, so (96) gives $400Q_{193} \geq E_{193} > 0$, and $Q_{193}H_{193} = M_{193} > 0$ gives $H_{193} > 0$.

The sign of H_{196} . Since $F(x, y) = 0$, (97) gives $Q_{196}H_{196} = M_{196}$, and M_{196} and E_{196} depend only on x . We show that $M_{196} > 0$ and $E_{196} > 0$. If $x \geq 33$, write $x = 33 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{196} and E_{196} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{196} > 0$ and $E_{196} > 0$. It remains to consider $0 \leq x \leq 32$. For $x = 0$ we have $M_{196} > 0$ and $E_{196} > 0$. For $x \in \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32\}$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with such x . For $x \in \{1, 9, 11, 19, 21, 29, 31\}$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with such x . For $x = 15$ and $x = 25$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , as one checks for the 7 residues of y modulo 7, so (1) has no solution with such x . For $x \in \{5, 7, 13, 17, 23, 27\}$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every y , as one checks for the 11 residues of

y modulo 11, so (1) has no solution with such x . For $x = 3$, we have $F(x, y) \not\equiv 0 \pmod{47}$ for every y , as one checks for the 47 residues of y modulo 47, so (1) has no solution with this x . Hence $M_{196} > 0$ and $E_{196} > 0$, so (98) gives $4Q_{196} \geq E_{196} > 0$, and $Q_{196}H_{196} = M_{196} > 0$ gives $H_{196} > 0$.

The sign of H_{197} . Since $F(x, y) = 0$, (99) gives $Q_{197}(-H_{197}) = M_{197}$, and M_{197} and E_{197} depend only on x . We show that $M_{197} > 0$ and $E_{197} > 0$. If $x \geq 4$, write $x = 4 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{197} and E_{197} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{197} > 0$ and $E_{197} > 0$. It remains to consider $0 \leq x \leq 3$. For $x \in \{0, 1, 2, 3\}$ we have $M_{197} > 0$ and $E_{197} > 0$. Hence $M_{197} > 0$ and $E_{197} > 0$, so (100) gives $400Q_{197} \geq E_{197} > 0$, and $Q_{197}(-H_{197}) = M_{197} > 0$ gives $(-H_{197}) > 0$. Hence $H_{197} < 0$.

The sign of H_{199} . Since $F(x, y) = 0$, (101) gives $Q_{199}H_{199} = M_{199}$, and M_{199} and E_{199} depend only on x . We show that $M_{199} > 0$ and $E_{199} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_{199} and E_{199} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{199} > 0$ and $E_{199} > 0$. It remains to consider $-1 \leq x \leq 0$. For $x = -1$ and $x = 0$ we have $M_{199} > 0$ and $E_{199} > 0$. Hence $M_{199} > 0$ and $E_{199} > 0$, so (102) gives $16Q_{199} \geq E_{199} > 0$, and $Q_{199}H_{199} = M_{199} > 0$ gives $H_{199} > 0$.

The sign of H_{200} . Since $F(x, y) = 0$, (103) gives $Q_{200}H_{200} = M_{200}$, and M_{200} and E_{200} depend only on x . We show that $M_{200} > 0$ and $E_{200} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -4$, write $x = -4 - z$ with $z \geq 0$. In both cases, expanding M_{200} and E_{200} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{200} > 0$ and $E_{200} > 0$. It remains to consider $-3 \leq x \leq 0$. For $x \in \{-3, -2, -1, 0\}$ we have $M_{200} > 0$ and $E_{200} > 0$. Hence $M_{200} > 0$ and $E_{200} > 0$, so (104) gives $100Q_{200} \geq E_{200} > 0$, and $Q_{200}H_{200} = M_{200} > 0$ gives $H_{200} > 0$.

The sign of H_{201} . Since $F(x, y) = 0$, (105) gives $Q_{201}(-H_{201}) = M_{201}$, and M_{201} and E_{201} depend only on x . We show that $M_{201} > 0$ and $E_{201} > 0$. If $x \geq 4$, write $x = 4 + z$ with $z \geq 0$; if $x \leq -3$, write $x = -3 - z$ with $z \geq 0$. In both cases, expanding M_{201} and E_{201} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{201} > 0$ and $E_{201} > 0$. It remains to consider $-2 \leq x \leq 3$. For $x = -2$ we have $M_{201} > 0$ and $E_{201} > 0$. For $x = 0$ and $x = 2$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with such x . For $x = -1$ and $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with such x . For $x = 3$, we have $F(x, y) \not\equiv 0 \pmod{47}$ for every y , as one checks for the 47 residues of y modulo 47, so (1) has no solution with this x . Hence $M_{201} > 0$ and $E_{201} > 0$, so (106) gives $4Q_{201} \geq E_{201} > 0$, and $Q_{201}(-H_{201}) = M_{201} > 0$ gives $(-H_{201}) > 0$. Hence $H_{201} < 0$.

The sign of H_{203} . Since $F(x, y) = 0$, (107) gives $Q_{203}(-H_{203}) = M_{203}$, and M_{203} and E_{203} depend only on x . We show that $M_{203} > 0$ and $E_{203} > 0$. If $x \geq 2$, write $x = 2 + z$ with $z \geq 0$; if $x \leq -3$, write $x = -3 - z$ with $z \geq 0$. In both cases, expanding M_{203} and E_{203} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{203} > 0$ and $E_{203} > 0$. It remains to consider $-2 \leq x \leq 1$. For $x = -2$ and $x = -1$ we have $M_{203} > 0$ and $E_{203} > 0$. For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with this x . For $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with this x . Hence $M_{203} > 0$ and $E_{203} > 0$, so (108) gives $64Q_{203} \geq E_{203} > 0$, and $Q_{203}(-H_{203}) = M_{203} > 0$ gives $(-H_{203}) > 0$. Hence $H_{203} < 0$.

The sign of H_{205} . Since $F(x, y) = 0$, (109) gives $Q_{205}H_{205} = M_{205}$, and M_{205} and E_{205} depend only on x . We show that $M_{205} > 0$ and $E_{205} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_{205} and E_{205} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{205} > 0$ and $E_{205} > 0$. It remains to consider $-1 \leq x \leq 0$. For $x = -1$ and $x = 0$ we have $M_{205} > 0$ and $E_{205} > 0$. Hence

$M_{205} > 0$ and $E_{205} > 0$, so (110) gives $4Q_{205} \geq E_{205} > 0$, and $Q_{205}H_{205} = M_{205} > 0$ gives $H_{205} > 0$.

The sign of H_{206} . Since $F(x, y) = 0$, (111) gives $Q_{206}(-H_{206}) = M_{206}$, and M_{206} and E_{206} depend only on x . We show that $M_{206} > 0$ and $E_{206} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{206} and E_{206} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{206} > 0$ and $E_{206} > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_{206} > 0$ and $E_{206} > 0$. Hence $M_{206} > 0$ and $E_{206} > 0$, so (112) gives $64Q_{206} \geq E_{206} > 0$, and $Q_{206}(-H_{206}) = M_{206} > 0$ gives $(-H_{206}) > 0$. Hence $H_{206} < 0$.

The sign of H_{207} . Since $F(x, y) = 0$, (113) gives $Q_{207}H_{207} = M_{207}$, and M_{207} and E_{207} depend only on x . We show that $M_{207} > 0$ and $E_{207} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -13$, write $x = -13 - z$ with $z \geq 0$. In both cases, expanding M_{207} and E_{207} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{207} > 0$ and $E_{207} > 0$. It remains to consider $-12 \leq x \leq 0$. For $x = 0$ we have $M_{207} > 0$ and $E_{207} > 0$. For $x \in \{-12, -10, -8, -6, -4, -2\}$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with such x . For $x \in \{-11, -9, -1\}$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with such x . For $x = -3$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , as one checks for the 7 residues of y modulo 7, so (1) has no solution with this x . For $x = -5$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every y , as one checks for the 11 residues of y modulo 11, so (1) has no solution with this x . For $x = -7$, we have $F(x, y) \not\equiv 0 \pmod{37}$ for every y , as one checks for the 37 residues of y modulo 37, so (1) has no solution with this x . Hence $M_{207} > 0$ and $E_{207} > 0$, so (114) gives $4Q_{207} \geq E_{207} > 0$, and $Q_{207}H_{207} = M_{207} > 0$ gives $H_{207} > 0$.

The sign of H_{208} . Since $F(x, y) = 0$, (115) gives $Q_{208}H_{208} = M_{208}$, and M_{208} and E_{208} depend only on x . We show that $M_{208} > 0$ and $E_{208} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{208} and E_{208} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{208} > 0$ and $E_{208} > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_{208} > 0$ and $E_{208} > 0$. Hence $M_{208} > 0$ and $E_{208} > 0$, so (116) gives $16Q_{208} \geq E_{208} > 0$, and $Q_{208}H_{208} = M_{208} > 0$ gives $H_{208} > 0$.

The sign of H_{209} . Since $F(x, y) = 0$, (117) gives $Q_{209}(-H_{209}) = M_{209}$, and M_{209} and E_{209} depend only on x . We show that $M_{209} > 0$ and $E_{209} > 0$. If $x \geq 4$, write $x = 4 + z$ with $z \geq 0$; if $x \leq -4$, write $x = -4 - z$ with $z \geq 0$. In both cases, expanding M_{209} and E_{209} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{209} > 0$ and $E_{209} > 0$. It remains to consider $-3 \leq x \leq 3$. For $x \in \{-2, 0, 2\}$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with such x . For $x = -1$ and $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with such x . For $x = -3$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , as one checks for the 7 residues of y modulo 7, so (1) has no solution with this x . For $x = 3$, we have $F(x, y) \not\equiv 0 \pmod{47}$ for every y , as one checks for the 47 residues of y modulo 47, so (1) has no solution with this x . Hence $M_{209} > 0$ and $E_{209} > 0$, so (118) gives $16Q_{209} \geq E_{209} > 0$, and $Q_{209}(-H_{209}) = M_{209} > 0$ gives $(-H_{209}) > 0$. Hence $H_{209} < 0$.

The sign of H_{210} . Since $F(x, y) = 0$, (119) gives $Q_{210}H_{210} = M_{210}$, and M_{210} and E_{210} depend only on x . We show that $M_{210} > 0$ and $E_{210} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{210} and E_{210} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{210} > 0$ and $E_{210} > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_{210} > 0$ and $E_{210} > 0$. Hence $M_{210} > 0$ and $E_{210} > 0$, so (120) gives $4Q_{210} \geq E_{210} > 0$, and $Q_{210}H_{210} = M_{210} > 0$ gives $H_{210} > 0$.

The sign of H_{211} . Since $F(x, y) = 0$, (121) gives $Q_{211}H_{211} = M_{211}$, and M_{211} and E_{211} depend only on x . We show that $M_{211} > 0$ and $E_{211} > 0$. If $x \geq 2$, write $x = 2 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{211} and E_{211} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{211} > 0$ and $E_{211} > 0$. It

remains to consider $0 \leq x \leq 1$. For $x = 0$ we have $M_{211} > 0$ and $E_{211} > 0$. For $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with this x . Hence $M_{211} > 0$ and $E_{211} > 0$, so (122) gives $4Q_{211} \geq E_{211} > 0$, and $Q_{211}H_{211} = M_{211} > 0$ gives $H_{211} > 0$.

The sign of H_{212} . Since $F(x, y) = 0$, (123) gives $Q_{212}H_{212} = M_{212}$, and M_{212} and E_{212} depend only on x . We show that $M_{212} > 0$ and $E_{212} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{212} and E_{212} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{212} > 0$ and $E_{212} > 0$. It remains to consider $x = 0$. For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with this x . Hence $M_{212} > 0$ and $E_{212} > 0$, so (124) gives $64Q_{212} \geq E_{212} > 0$, and $Q_{212}H_{212} = M_{212} > 0$ gives $H_{212} > 0$.

The sign of H_{213} . Since $F(x, y) = 0$, (125) gives $Q_{213}H_{213} = M_{213}$, and M_{213} and E_{213} depend only on x . We show that $M_{213} > 0$ and $E_{213} > 0$. If $x \geq 2$, write $x = 2 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_{213} and E_{213} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{213} > 0$ and $E_{213} > 0$. It remains to consider $-1 \leq x \leq 1$. For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with this x . For $x = -1$ and $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with such x . Hence $M_{213} > 0$ and $E_{213} > 0$, so (126) gives $4Q_{213} \geq E_{213} > 0$, and $Q_{213}H_{213} = M_{213} > 0$ gives $H_{213} > 0$.

The sign of H_{214} . Since $F(x, y) = 0$, (127) gives $Q_{214}H_{214} = M_{214}$, and M_{214} and E_{214} depend only on x . We show that $M_{214} > 0$ and $E_{214} > 0$. If $x \geq 3$, write $x = 3 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_{214} and E_{214} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{214} > 0$ and $E_{214} > 0$. It remains to consider $-1 \leq x \leq 2$. For $x \in \{-1, 1, 2\}$ we have $M_{214} > 0$ and $E_{214} > 0$. For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with this x . Hence $M_{214} > 0$ and $E_{214} > 0$, so (128) gives $16Q_{214} \geq E_{214} > 0$, and $Q_{214}H_{214} = M_{214} > 0$ gives $H_{214} > 0$.

The sign of H_{215} . Since $F(x, y) = 0$, (129) gives $Q_{215}(-H_{215}) = M_{215}$, and M_{215} and E_{215} depend only on x . We show that $M_{215} > 0$ and $E_{215} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{215} and E_{215} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{215} > 0$ and $E_{215} > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_{215} > 0$ and $E_{215} > 0$. Hence $M_{215} > 0$ and $E_{215} > 0$, so (130) gives $16Q_{215} \geq E_{215} > 0$, and $Q_{215}(-H_{215}) = M_{215} > 0$ gives $(-H_{215}) > 0$. Hence $H_{215} < 0$.

The sign of H_{217} . Since $F(x, y) = 0$, (131) gives $Q_{217}(-H_{217}) = M_{217}$, and M_{217} and E_{217} depend only on x . We show that $M_{217} > 0$ and $E_{217} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{217} and E_{217} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{217} > 0$ and $E_{217} > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_{217} > 0$ and $E_{217} > 0$. Hence $M_{217} > 0$ and $E_{217} > 0$, so (132) gives $400Q_{217} \geq E_{217} > 0$, and $Q_{217}(-H_{217}) = M_{217} > 0$ gives $(-H_{217}) > 0$. Hence $H_{217} < 0$.

The sign of H_{219} . Since $F(x, y) = 0$, (133) gives $Q_{219}H_{219} = M_{219}$, and M_{219} and E_{219} depend only on x . We show that $M_{219} > 0$ and $E_{219} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_{219} and E_{219} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{219} > 0$ and $E_{219} > 0$. It remains to consider $-1 \leq x \leq 0$. For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with this x . For $x = -1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with this x . Hence $M_{219} > 0$ and $E_{219} > 0$, so (134) gives $16Q_{219} \geq E_{219} > 0$, and $Q_{219}H_{219} = M_{219} > 0$ gives $H_{219} > 0$.

The sign of H_{220} . Since $F(x, y) = 0$, (135) gives $Q_{220}H_{220} = M_{220}$, and M_{220} and E_{220} depend only on x . We show that $M_{220} > 0$ and $E_{220} > 0$. If $x \geq 2$, write $x = 2 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_{220} and E_{220} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{220} > 0$ and $E_{220} > 0$. It remains to consider $-1 \leq x \leq 1$. For $x = -1$ and $x = 0$ we have $M_{220} > 0$ and $E_{220} > 0$. For $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with this x . Hence $M_{220} > 0$ and $E_{220} > 0$, so (136) gives $100Q_{220} \geq E_{220} > 0$, and $Q_{220}H_{220} = M_{220} > 0$ gives $H_{220} > 0$.

The sign of H_{221} . Since $F(x, y) = 0$, (137) gives $Q_{221}(-H_{221}) = M_{221}$, and M_{221} and E_{221} depend only on x . We show that $M_{221} > 0$ and $E_{221} > 0$. If $x \geq 6$, write $x = 6 + z$ with $z \geq 0$; if $x \leq -3$, write $x = -3 - z$ with $z \geq 0$. In both cases, expanding M_{221} and E_{221} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{221} > 0$ and $E_{221} > 0$. It remains to consider $-2 \leq x \leq 5$. For $x \in \{-2, 0, 2, 4\}$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with such x . For $x = -1$ and $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with such x . For $x = 5$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every y , as one checks for the 11 residues of y modulo 11, so (1) has no solution with this x . For $x = 3$, we have $F(x, y) \not\equiv 0 \pmod{47}$ for every y , as one checks for the 47 residues of y modulo 47, so (1) has no solution with this x . Hence $M_{221} > 0$ and $E_{221} > 0$, so (138) gives $16Q_{221} \geq E_{221} > 0$, and $Q_{221}(-H_{221}) = M_{221} > 0$ gives $(-H_{221}) > 0$. Hence $H_{221} < 0$.

The sign of H_{222} . Since $F(x, y) = 0$, (139) gives $Q_{222}H_{222} = M_{222}$, and M_{222} and E_{222} depend only on x . We show that $M_{222} > 0$ and $E_{222} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -7$, write $x = -7 - z$ with $z \geq 0$. In both cases, expanding M_{222} and E_{222} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{222} > 0$ and $E_{222} > 0$. It remains to consider $-6 \leq x \leq 0$. For $x \in \{-6, -4, -2, 0\}$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with such x . For $x = -1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with this x . For $x = -3$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , as one checks for the 7 residues of y modulo 7, so (1) has no solution with this x . For $x = -5$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every y , as one checks for the 11 residues of y modulo 11, so (1) has no solution with this x . Hence $M_{222} > 0$ and $E_{222} > 0$, so (140) gives $4Q_{222} \geq E_{222} > 0$, and $Q_{222}H_{222} = M_{222} > 0$ gives $H_{222} > 0$.

The sign of H_{223} . Since $F(x, y) = 0$, (141) gives $Q_{223}(-H_{223}) = M_{223}$, and M_{223} and E_{223} depend only on x . We show that $M_{223} > 0$ and $E_{223} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_{223} and E_{223} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{223} > 0$ and $E_{223} > 0$. It remains to consider $-1 \leq x \leq 0$. For $x = -1$ and $x = 0$ we have $M_{223} > 0$ and $E_{223} > 0$. Hence $M_{223} > 0$ and $E_{223} > 0$, so (142) gives $16Q_{223} \geq E_{223} > 0$, and $Q_{223}(-H_{223}) = M_{223} > 0$ gives $(-H_{223}) > 0$. Hence $H_{223} < 0$.

The sign of H_{224} . Since $F(x, y) = 0$, (143) gives $Q_{224}H_{224} = M_{224}$, and M_{224} and E_{224} depend only on x . We show that $M_{224} > 0$ and $E_{224} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -3$, write $x = -3 - z$ with $z \geq 0$. In both cases, expanding M_{224} and E_{224} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{224} > 0$ and $E_{224} > 0$. It remains to consider $-2 \leq x \leq 0$. For $x = 0$ we have $M_{224} > 0$ and $E_{224} > 0$. For $x = -2$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with this x . For $x = -1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with this x . Hence $M_{224} > 0$ and $E_{224} > 0$, so (144) gives $16Q_{224} \geq E_{224} > 0$, and $Q_{224}H_{224} = M_{224} > 0$ gives $H_{224} > 0$.

The sign of H_{225} . Since $F(x, y) = 0$, (145) gives $Q_{225}H_{225} = M_{225}$, and M_{225} and E_{225} depend only on x . We show that $M_{225} > 0$ and $E_{225} > 0$. If $x \geq 6$, write $x = 6 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{225} and E_{225} as polynomials

in z gives non-negative coefficients and positive constant terms, so $M_{225} > 0$ and $E_{225} > 0$. It remains to consider $0 \leq x \leq 5$. For $x = 0$ we have $M_{225} > 0$ and $E_{225} > 0$. For $x = 2$ and $x = 4$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with such x . For $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with this x . For $x = 5$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every y , as one checks for the 11 residues of y modulo 11, so (1) has no solution with this x . For $x = 3$, we have $F(x, y) \not\equiv 0 \pmod{47}$ for every y , as one checks for the 47 residues of y modulo 47, so (1) has no solution with this x . Hence $M_{225} > 0$ and $E_{225} > 0$, so (146) gives $4Q_{225} \geq E_{225} > 0$, and $Q_{225}H_{225} = M_{225} > 0$ gives $H_{225} > 0$.

For a place v , put $B_v = \prod_{\{i,j\}} (H_i, H_j)_v$, the product over the edges. By Theorem 2(i), $(H_i, H_j)_\infty = -1$ exactly when $H_i < 0$ and $H_j < 0$. Here $H_i > 0$ for $i \in \{0, 1, 2, 3, 4, 5, 6, 7, 10, 12, 13, 15, 18, 19, 21, 23, 24, 26, 29, 32, 34, 36, 37, 38, 40, 42, 43, 44, 53, 54, 59, 61, 62, 68, 70, 72, 73, 75, 76, 80, 83, 84, 87, 92, 93, 94, 97, 105, 111, 118, 120, 123, 127, 132, 139, 176, 180, 185, 187, 191, 193, 194, 195, 196, 198, 199, 200, 202, 204, 205, 207, 208, 210, 211, 212, 213, 214, 218, 219, 220, 222, 224, 225, 226\}$ and $H_i < 0$ for $i \in \{16, 30, 39, 45, 46, 48, 55, 56, 58, 89, 96, 102, 104, 112, 113, 136, 146, 151, 156, 174, 179, 192, 197, 201, 203, 206, 209, 215, 217, 221, 223\}$. For $i \in \{8, 9, 11, 14, 17, 20, 22, 25, 27, 28, 31, 33, 35, 41, 47, 49, 50, 51, 52, 57, 60, 63, 64, 65, 66, 67, 69, 71, 74, 77, 78, 79, 81, 82, 85, 86, 88, 90, 91, 95, 98, 99, 100, 101, 103, 106, 107, 108, 109, 110, 114, 115, 116, 117, 119, 121, 122, 124, 125, 126, 128, 129, 130, 131, 133, 134, 135, 137, 138, 140, 141, 142, 143, 144, 145, 147, 148, 149, 150, 152, 153, 154, 155, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 175, 177, 178, 181, 182, 183, 184, 186, 188, 189, 190, 216\}$, the sign of H_i is not needed: $H_j > 0$ for every edge $\{i, j\}$, so $(H_i, H_j)_\infty = 1$. Counting the edges whose two values are negative, we find $B_\infty = 1$.

Step 3: the quadratic graphs at the odd primes outside S . Let $S = \{2, 3, 5, 7, 11, 13, 19, 31, 37, 43\}$. It contains 2 and the primes dividing the m_i , the k_{ij} and the constant vertices. Let $p \notin S$ be an odd prime; we show that $B_p = 1$. For an edge $\{i, j\}$ such that p divides neither H_i nor H_j , $(H_i, H_j)_p = 1$ by Theorem 2(ii). Now let i be a vertex with $p \mid H_i$. Then H_i is not constant, as $p \notin S$. For every edge $\{i, j\}$, $p \nmid H_j$: if H_j is constant, because $p \notin S$, and otherwise by (3), because $p \nmid k_{ij}$. In particular, no edge has both values divisible by p . By Theorem 2(v), the edges at i contribute $\prod_{j \sim i} (H_i, H_j)_p = (H_i, \Pi_i)_p$, where $\Pi_i = \prod_{j \sim i} H_j$ is not divisible by p . By (2), $m_i^2 \Pi_i \equiv W_i^2 \pmod{p}$, and $p \nmid m_i$, so Π_i is a non-zero square modulo p , and $(H_i, \Pi_i)_p = (\Pi_i, H_i)_p = 1$ by Lemma 3. Hence $B_p = 1$.

Step 4: cubic reciprocity. For every place v of K , put

$$\mathcal{B}_v^{(g)} = \sum_{(i,j)} w_{ij} \theta_v(J_i, J_j) + \sum_i \theta_v(J_i, c_i) \in \mathbb{Z}/3\mathbb{Z}, \quad (149)$$

the first sum over the edges of the g -th certificate, and, for a rational prime p , put $\mathcal{B}_p^{(g)} = \sum_{v|p} \mathcal{B}_v^{(g)}$. Applying (H3) to every pair (J_i, J_j) with an edge, multiplied by w_{ij} , and to every pair (J_i, c_i) , and adding, we find that $\mathcal{B}_v^{(g)} = 0$ for all but finitely many v and $\sum_p \mathcal{B}_p^{(g)} = 0$ for every g , as the complex place contributes 0. By (H1) and the definition of cl_p , $\mathcal{B}_p^{(g)} = \sum_{(i,j)} w_{ij} \theta_p(\text{cl}_p(J_i), \text{cl}_p(J_j)) + \sum_i \theta_p(\text{cl}_p(J_i), \text{cl}_p(c_i))$, where θ_p is the form of Lemma 7 if $p = 3$.

Step 5: the cubic certificates at the primes outside S . The set S also contains 3 and the primes dividing the norms of the constants c_i , of the D_i , of the R_{ij} . Let $p \notin S$, and let v be a place above p , with residue field k_v ; then all these numbers are units at v . First, no edge (i, j) has both values divisible by v . Indeed, suppose that v divides J_i and J_j . If R_{ij} is defined, let k be the linear endpoint used for it and $\{k, l\} = \{i, j\}$, and put $t = \xi$. As $J_k = \varepsilon_k(\lambda_k \eta - s_k(\xi))$ and λ_k is a unit at v , we have $\eta \equiv s_k(t)/\lambda_k$ modulo v , hence $f_k(t) \equiv \lambda_k^d F(x, y) = 0$ and $g_{kl}(t) \equiv \lambda_k^{d_l} J_l \equiv 0$

modulo v . The resultant can be written as $R_{ij} = Af_k + Cg_{kl}$ with polynomials A and C with coefficients in $\mathbb{Z}[\omega]$, so v divides R_{ij} , a contradiction. By (H2), a term of (149) in which all arguments are units vanishes. Now let v divide J_i . Collecting the terms of (149) which contain J_i , and using (H1) and $\rho_{ji} = -w_{ij}$ for the edges (j, i) with $j < i$, we obtain $\theta_v(J_i, \Pi_i)$ with $\Pi_i = c_i \prod_j J_j^{\rho_{ij}}$, which is a unit. We claim that $D_i^3 \Pi_i \equiv W^3$ modulo v for some $W \in K$ integral at v . If i is linear, put $t = \xi$; then $g_{ij}(t) \equiv \lambda_i^{d_j} J_j$ and $f_i(t) \equiv 0$ modulo v as above, and (147) gives $D_i^3 \lambda_i^{E_i} \Pi_i \equiv w_i(t)^3$ modulo v , where $E_i = e_i + \sum_j \rho_{ij} d_j$ is divisible by 3; this gives the claim with $W = w_i(t)/\lambda_i^{E_i/3}$. Otherwise, (148) gives the claim with $W = T_i(x, y)$. Hence the reduction of Π_i is a non-zero cube in k_v , $\chi(\Pi_i) = 0$, and $\theta_v(J_i, \Pi_i) = v(J_i)\chi(\Pi_i) = 0$ by (H2). Hence $\mathcal{B}_v^{(g)} = 0$, and $\mathcal{B}_p^{(g)} = 0$.

Step 6: the primes in S . Let $p \in S$, and suppose that $x \equiv a$ and $y \equiv b \pmod{p^k}$. Lemma 5 gives, for every vertex H_i of the quadratic graph, a vector b_i and a subspace D_i with $c_p(H_i) \in b_i + D_i$, and Lemma 8 gives the analogous data for the values J_i of the cubic certificates. If the hypotheses of Lemma 6 hold for every quadratic graph and those of Lemma 9 hold for every cubic certificate, then B_p and $(\mathcal{B}_p^{(g)})_g$ are determined by a, b and k . Starting from the solutions of (1) modulo p , we refine a residue class $x \equiv a, y \equiv b \pmod{p^k}$ to the classes modulo p^{k+1} that contain solutions of (1) modulo p^{k+1} , until all these hypotheses hold, or until no solution remains. Table 7 lists the resulting terminal classes together with the joint value $((\mathcal{B}_p^{(g)})_g; B_p)$; every integer solution lies in one of them. The possible joint values are: for $p = 2$, $(1, 0; -1)$; for $p = 3$, $(0, 0; 1), (0, 1; 1), (0, 2; 1), (1, 0; 1), (2, 2; 1)$; for $p = 5$, $(0, 0; 1)$; for $p = 7$, $(1, 0; -1), (1, 0; 1), (2, 0; 1)$; for $p = 11$, $(0, 0; 1)$; for $p = 13$, $(0, 1; 1)$; for $p = 19$, $(0, 0; 1)$; for $p = 31$, $(0, 0; 1)$; for $p = 37$, $(0, 0; 1)$; for $p = 43$, $(0, 0; 1)$.

Step 7: contradiction. By Steps 4 and 5, $(\sum_{p \in S} \mathcal{B}_p^{(g)})_g = (0, 0)$. By Theorem 2(iv) and Step 3, $\prod_{p \in S} B_p = B_\infty = 1$ by Step 2. But by Step 6, the pair $((\sum_{p \in S} \mathcal{B}_p^{(g)})_g; \prod_{p \in S} B_p)$ lies in the set consisting of the pairs $(0, 0; -1), (0, 1; -1), (0, 1; 1), (0, 2; -1), (1, 0; -1), (1, 0; 1), (1, 1; -1), (2, 0; -1), (2, 0; 1), (2, 1; -1), (2, 1; 1), (2, 2; -1)$, and $(2, 2; 1)$, which does not contain $(0, 0; 1)$. This contradiction proves the theorem. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.
- [2] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, New York, 1973.
- [3] J. Neukirch, *Algebraic Number Theory*, Grundlehren der mathematischen Wissenschaften 322, Springer, Berlin, 1999.

Table 1: The data of the identities (2).

$i = 0, m = 4$, neighbours 1, 3, 4, 6, 23, 87, 185	
W	$930x^3 + 210x^2 - 720x - 540$
A	$21600x^{10} + 97200x^9 + 315000x^8 + 748800x^7 + 1368000x^6 + 1986300x^5$ $+ 2245500x^4 + 1882800x^3 + 489600x^2 - 410400x - 280800$
B	$-64800x^{13} - 90000x^{12}y - 172800x^{12} - 25200x^{11}y^2 - 248400x^{11}y$ $-133200x^{11} - 7200x^{10}y^3 + 68400x^{10}y^2 - 518400x^{10}y + 291600x^{10}$ $+18000x^9y^4 - 25200x^9y^3 + 176400x^9y^2 - 1180800x^9y + 1227600x^9$ $-7200x^8y^5 - 244800x^8y^3 + 282600x^8y^2 - 2431800x^8y + 2412000x^8$ $-3600x^7y^6 - 61200x^7y^5 - 36000x^7y^4 - 532800x^7y^3$ $+471600x^7y^2 - 4060800x^7y + 3639600x^7 - 3600x^6y^7 - 7200x^6y^6$ $-68400x^6y^5 + 18000x^6y^4 - 824400x^6y^3 + 950400x^6y^2 - 5209200x^6y$ $+4658400x^6 + 43200x^5y^6 - 50400x^5y^5 - 57600x^5y^4 - 1076400x^5y^3$ $+1518300x^5y^2 - 5427900x^5y + 5351400x^5 + 3600x^4y^8 + 21600x^4y^6$ $-169200x^4y^5 - 327600x^4y^4 - 1288800x^4y^3 + 1673100x^4y^2$ $-4668300x^4y + 5279400x^4 - 36000x^3y^7 - 86400x^3y^6 - 374400x^3y^5$ $-637200x^3y^4 - 1245600x^3y^3 + 1267200x^3y^2 - 3265200x^3y + 4107600x^3$ $-3600x^2y^9 - 14400x^2y^8 - 72000x^2y^7 - 183600x^2y^6$ $-442800x^2y^5 - 597600x^2y^4 - 831600x^2y^3 + 25200x^2y^2$ $-1112400x^2y + 1076400x^2 - 3600xy^9 - 18000xy^8 - 72000xy^7 - 158400xy^6$ $-270000xy^5 - 291600xy^4 - 324000xy^3 - 597600xy^2 + 212400xy - 810000x - 3600y^9$ $-14400y^8 - 32400y^7 - 46800y^6 - 57600y^5 - 54000y^4 - 57600y^3 - 316800y^2$ $+248400y - 561600$
$i = 1, m = 64$, neighbours 0, 7, 9, 11, 14, 17, 20, 22, 25, 28, 31, 33, 35, 40, 48, 53, 56, 58, 60, 61, 186	
W	$-11x^2y^2 - 15x^2y - 14x^2 + 44789xy^2 + 60017xy + 57202x + 10293y^2 + 56881y$ $+88498$
A	$24576xy^{18} + 147456xy^{17} + 491520xy^{16} + 724992xy^{15} - 1929216xy^{14}$ $-15765504xy^{13} - 58134528xy^{12} - 143192064xy^{11} - 253992960xy^{10}$ $-301510656xy^9 - 125190144xy^8 + 391237632xy^7 + 1095806976xy^6$ $+1431797760xy^5 + 679526400xy^4 - 1381490688xy^3 - 3702190080xy^2 - 2858938368xy$ $-764743680x + 36864y^{18} + 221184y^{17} + 1081344y^{16} + 3624960y^{15} + 9019392y^{14}$ $+14585856y^{13} + 5136384y^{12} - 54067200y^{11} - 205774848y^{10} - 446976000y^9$ $-654077952y^8 - 561328128y^7 + 76455936y^6 + 1138728960y^5 + 1904996352y^4$ $+1260539904y^3 - 1409765376y^2 - 3158544384y - 2284978176$
B	$-81920x^{36}y^7 + 122880x^{36}y^6 + 491520x^{36}y^5 - 737280x^{36}y^4$ $-737280x^{36}y^3 + 1105920x^{36}y^2 + 327680x^{36}y - 491520x^{36}$ $+122880x^{35}y^8 - 741376x^{35}y^7 - 618496x^{35}y^6 + 5103616x^{35}y^5$ $-753664x^{35}y^4 - 8474624x^{35}y^3 + 2379776x^{35}y^2 + 4112384x^{35}y$ $-1130496x^{35} + 937984x^{34}y^8 - 1564672x^{34}y^7 - 9113600x^{34}y^6$ $+13025280x^{34}y^5 + 19099648x^{34}y^4 - 20766720x^{34}y^3 - 14512128x^{34}y^2$ $+8323072x^{34}y + 2605056x^{34} - 40960x^{33}y^{10} - 184320x^{33}y^9$ $+3559424x^{33}y^8 + 5058560x^{33}y^7 - 33107968x^{33}y^6 - 17108992x^{33}y^5$ $+77623296x^{33}y^4 + 28348416x^{33}y^3 - 61616128x^{33}y^2 - 16572416x^{33}y$ $+13123584x^{33} - 217088x^{32}y^{10} - 2154496x^{32}y^9 + 5996544x^{32}y^8$ $+41791488x^{32}y^7 - 37138432x^{32}y^6 - 176652288x^{32}y^5$ $+74817536x^{32}y^4 + 243757056x^{32}y^3 - 49029120x^{32}y^2 - 96976896x^{32}y$ $+15335424x^{32} + 81920x^{31}y^{11} - 270336x^{31}y^{10} - 9838592x^{31}y^9$ $-10403840x^{31}y^8 + 121196544x^{31}y^7 + 121147392x^{31}y^6$ $-436396032x^{31}y^5 - 285990912x^{31}y^4 + 498225152x^{31}y^3$ $+211283968x^{31}y^2 - 153870336x^{31}y - 16367616x^{31} + 598016x^{30}y^{11}$ $+2113536x^{30}y^{10} - 23154688x^{30}y^9 - 87625728x^{30}y^8$ $+148320256x^{30}y^7 + 589680640x^{30}y^6 - 330633216x^{30}y^5$ $-1126494208x^{30}y^4 + 175992832x^{30}y^3 + 686370816x^{30}y^2 + 13737984x^{30}y$ $-79183872x^{30} - 40960x^{29}y^{12} + 1880064x^{29}y^{11} + 12521472x^{29}y^{10}$ $-18599936x^{29}y^9 - 238440448x^{29}y^8 - 142979072x^{29}y^7$ $+1112076288x^{29}y^6 + 898682880x^{29}y^5 - 1668255744x^{29}y^4$ $-1331347456x^{29}y^3 + 817397760x^{29}y^2 + 499630080x^{29}y - 127991808x^{29}$ $-421888x^{28}y^{12} + 2576384x^{28}y^{11} + 34242560x^{28}y^{10} + 56610816x^{28}y^9$ $-337256448x^{28}y^8 - 951951360x^{28}y^7 + 805662720x^{28}y^6$ $+3237048320x^{28}y^5 - 456019968x^{28}y^4 - 3514109952x^{28}y^3$ $-31838208x^{28}y^2 + 1068441600x^{28}y - 115752960x^{28} - 40960x^{27}y^{13}$ $-1744896x^{27}y^{12} - 2330624x^{27}y^{11} + 57376768x^{27}y^{10}$

$$\begin{aligned}
& +230797312x^{27}y^9 - 162619392x^{27}y^8 - 1953751040x^{27}y^7 \\
& -949309440x^{27}y^6 + 5165953024x^{27}y^5 + 2768478208x^{27}y^4 \\
& -4777242624x^{27}y^3 - 1717932032x^{27}y^2 + 1321279488x^{27}y - 9584640x^{27} \\
& -94208x^{26}y^{13} - 5140480x^{26}y^{12} - 18190336x^{26}y^{11} + 71106560x^{26}y^{10} \\
& +442712064x^{26}y^9 + 230170624x^{26}y^8 - 2475978752x^{26}y^7 \\
& -3115405312x^{26}y^6 + 5208838144x^{26}y^5 + 6058160128x^{26}y^4 \\
& -4152123392x^{26}y^3 - 3518943232x^{26}y^2 + 954007552x^{26}y + 239222784x^{26} \\
& +81920x^{25}y^{14} + 258048x^{25}y^{13} - 12574720x^{25}y^{12} - 46784512x^{25}y^{11} \\
& +112324608x^{25}y^{10} + 655302656x^{25}y^9 + 302575616x^{25}y^8 \\
& -2690248704x^{25}y^7 - 3829972992x^{25}y^6 + 4137562112x^{25}y^5 \\
& +7819808768x^{25}y^4 - 1838419968x^{25}y^3 - 5155983360x^{25}y^2 \\
& -345726976x^{25}y + 635682816x^{25} + 557056x^{24}y^{14} + 2170880x^{24}y^{13} \\
& -24686592x^{24}y^{12} - 92110848x^{24}y^{11} + 243654656x^{24}y^{10} \\
& +1091956736x^{24}y^9 - 114831360x^{24}y^8 - 3919876096x^{24}y^7 \\
& -3910430720x^{24}y^6 + 3243110400x^{24}y^5 + 9653280768x^{24}y^4 \\
& +2912358400x^{24}y^3 - 6465957888x^{24}y^2 - 2908028928x^{24}y + 947994624x^{24} \\
& -40960x^{23}y^{15} + 1171456x^{23}y^{14} + 7602176x^{23}y^{13} - 28123136x^{23}y^{12} \\
& -160677888x^{23}y^{11} + 370606080x^{23}y^{10} + 2057306112x^{23}y^9 \\
& +412418048x^{23}y^8 - 6191788032x^{23}y^7 - 7822544896x^{23}y^6 \\
& -400490496x^{23}y^5 + 13688127488x^{23}y^4 + 12704559104x^{23}y^3 \\
& -5606944768x^{23}y^2 - 6327541760x^{23}y + 674217984x^{23} - 503808x^{22}y^{15} \\
& -1282048x^{22}y^{14} + 13574144x^{22}y^{13} + 14036992x^{22}y^{12} \\
& -213876736x^{22}y^{11} + 155758592x^{22}y^{10} + 3270094848x^{22}y^9 \\
& +4286038016x^{22}y^8 - 5852438528x^{22}y^7 - 18474381312x^{22}y^6 \\
& -13703467008x^{22}y^5 + 16648622080x^{22}y^4 + 28116422656x^{22}y^3 \\
& +931434496x^{22}y^2 - 8504524800x^{22}y - 434946048x^{22} - 2158592x^{21}y^{15} \\
& -14569472x^{21}y^{14} - 1417216x^{21}y^{13} + 100855808x^{21}y^{12} \\
& -195383296x^{21}y^{11} - 623562752x^{21}y^{10} + 3947044864x^{21}y^9 \\
& +12413640704x^{21}y^8 + 2211868672x^{21}y^7 - 31627038720x^{21}y^6 \\
& -39750426624x^{21}y^5 + 9754423296x^{21}y^4 + 42616852480x^{21}y^3 \\
& +14069690368x^{21}y^2 - 6430253056x^{21}y - 1677312000x^{21} + 81920x^{20}y^{16} \\
& -4579328x^{20}y^{15} - 40296448x^{20}y^{14} - 75571200x^{20}y^{13} \\
& +54140928x^{20}y^{12} - 376225792x^{20}y^{11} - 1917845504x^{20}y^{10} \\
& +4000116736x^{20}y^9 + 23592296448x^{20}y^8 + 19974643712x^{20}y^7 \\
& -37699407872x^{20}y^6 - 71164276736x^{20}y^5 - 12489170944x^{20}y^4 \\
& +44406530048x^{20}y^3 + 27484770304x^{20}y^2 + 431013888x^{20}y - 1792917504x^{20} \\
& +557056x^{19}y^{16} - 2154496x^{19}y^{15} - 47890432x^{19}y^{14} \\
& -191315968x^{19}y^{13} - 436662272x^{19}y^{12} - 1687773184x^{19}y^{11} \\
& -4412125184x^{19}y^{10} + 3539042304x^{19}y^9 + 36100431872x^{19}y^8 \\
& +45727346688x^{19}y^7 - 28264222720x^{19}y^6 - 92634152960x^{19}y^5 \\
& -43415654400x^{19}y^4 + 27025403904x^{19}y^3 + 30455537664x^{19}y^2 \\
& +7680499712x^{19}y - 523075584x^{19} + 2011136x^{18}y^{16} + 17764352x^{18}y^{15} \\
& +33370112x^{18}y^{14} - 171347968x^{18}y^{13} - 1295429632x^{18}y^{12} \\
& -4900397056x^{18}y^{11} - 10217717760x^{18}y^{10} - 188145664x^{18}y^9 \\
& +46336081920x^{18}y^8 + 74873782272x^{18}y^7 - 1217302528x^{18}y^6 \\
& -90183380992x^{18}y^5 - 65185570816x^{18}y^4 - 472535040x^{18}y^3 \\
& +19134070784x^{18}y^2 + 9464496128x^{18}y + 850722816x^{18} + 40960x^{17}y^{17} \\
& +6107136x^{17}y^{16} + 64376832x^{17}y^{15} + 252284928x^{17}y^{14} \\
& +273121280x^{17}y^{13} - 1556992000x^{17}y^{12} - 8761204736x^{17}y^{11} \\
& -20371468288x^{17}y^{10} - 12291284992x^{17}y^9 + 46598049792x^{17}y^8 \\
& +97776689152x^{17}y^7 + 34821595136x^{17}y^6 - 63428534272x^{17}y^5 \\
& -63892623360x^{17}y^4 - 18824933376x^{17}y^3 + 3129335808x^{17}y^2 \\
& +5291802624x^{17}y + 1113784320x^{17} + 544768x^{16}y^{17} + 14540800x^{16}y^{16} \\
& +125640704x^{16}y^{15} + 549621760x^{16}y^{14} + 1258917888x^{16}y^{13} \\
& +83734528x^{16}y^{12} - 9863925760x^{16}y^{11} - 31320797184x^{16}y^{10} \\
& -33246654464x^{16}y^9 + 30622298112x^{16}y^8 + 103226077184x^{16}y^7 \\
& +63986757632x^{16}y^6 - 28561035264x^{16}y^5 - 45558505472x^{16}y^4 \\
& -19412819968x^{16}y^3 - 4949569536x^{16}y^2 + 414334976x^{16}y + 512999424x^{16} \\
& +1433600x^{15}y^{17} + 21434368x^{15}y^{16} + 156954624x^{15}y^{15} \\
& +780247040x^{15}y^{14} + 2578665472x^{15}y^{13} + 3965239296x^{15}y^{12} \\
& -5644001280x^{15}y^{11} - 36718944256x^{15}y^{10} - 55427067904x^{15}y^9 \\
& +1718726656x^{15}y^8 + 88533278720x^{15}y^7 + 77219110912x^{15}y^6 \\
& -1126187008x^{15}y^5 - 27524956160x^{15}y^4 - 12451131392x^{15}y^3
\end{aligned}$$

$$\begin{aligned}
& -4079165440x^{15}y^2 - 1256177664x^{15}y - 15925248x^{15} - 163840x^{14}y^{18} \\
& -1167360x^{14}y^{17} + 577536x^{14}y^{16} + 82878464x^{14}y^{15} \\
& + 800530432x^{14}y^{14} + 3858436096x^{14}y^{13} + 9114054656x^{14}y^{12} \\
& + 3416428544x^{14}y^{11} - 32753750016x^{14}y^{10} - 69470826496x^{14}y^9 \\
& - 30420500480x^{14}y^8 + 59235344384x^{14}y^7 + 76203479040x^{14}y^6 \\
& + 18410278912x^{14}y^5 - 14964752384x^{14}y^4 - 8758022144x^{14}y^3 \\
& - 1817321472x^{14}y^2 - 742924288x^{14}y - 131727360x^{14} - 1032192x^{13}y^{18} \\
& - 14573568x^{13}y^{17} - 83169280x^{13}y^{16} - 184389632x^{13}y^{15} \\
& + 434626560x^{13}y^{14} + 4531888128x^{13}y^{13} + 13843972096x^{13}y^{12} \\
& + 14846615552x^{13}y^{11} - 19345944576x^{13}y^{10} - 69178445824x^{13}y^9 \\
& - 56100220928x^{13}y^8 + 20956528640x^{13}y^7 + 60548726784x^{13}y^6 \\
& + 31428227072x^{13}y^5 - 1846476800x^{13}y^4 - 6179344384x^{13}y^3 \\
& - 1422143488x^{13}y^2 - 242409472x^{13}y - 64241664x^{13} - 2646016x^{12}y^{18} \\
& - 37953536x^{12}y^{17} - 218333184x^{12}y^{16} - 640331776x^{12}y^{15} \\
& - 516354048x^{12}y^{14} + 3654066176x^{12}y^{13} + 15805624320x^{12}y^{12} \\
& + 24820666368x^{12}y^{11} + 505241600x^{12}y^{10} - 51404124160x^{12}y^9 \\
& - 63878496256x^{12}y^8 - 15249842176x^{12}y^7 + 29435473920x^{12}y^6 \\
& + 28982235136x^{12}y^5 + 8527597568x^{12}y^4 - 1293778944x^{12}y^3 \\
& - 908115968x^{12}y^2 - 94806016x^{12}y - 13860864x^{12} + 40960x^{11}y^{19} \\
& - 3588096x^{11}y^{18} - 50663424x^{11}y^{17} - 309346304x^{11}y^{16} \\
& - 1066000384x^{11}y^{15} - 1873424384x^{11}y^{14} + 828153856x^{11}y^{13} \\
& + 12928491520x^{11}y^{12} + 27841507328x^{11}y^{11} + 17833680896x^{11}y^{10} \\
& - 22455455744x^{11}y^9 - 46975045632x^{11}y^8 - 29231960064x^{11}y^7 \\
& + 89960448x^{11}y^6 + 11951611904x^{11}y^5 + 7790202880x^{11}y^4 \\
& + 1596391424x^{11}y^3 - 74235904x^{11}y^2 - 8110080x^{11}y - 1179648x^{11} \\
& + 94208x^{10}y^{19} - 1044480x^{10}y^{18} - 29954048x^{10}y^{17} \\
& - 259248128x^{10}y^{16} - 1166753792x^{10}y^{15} - 2892103680x^{10}y^{14} \\
& - 2479108096x^{10}y^{13} + 6667632640x^{10}y^{12} + 22069604352x^{10}y^{11} \\
& + 22357700608x^{10}y^{10} - 1232699392x^{10}y^9 - 20919119872x^{10}y^8 \\
& - 18076987392x^{10}y^7 - 7017480192x^{10}y^6 - 111558656x^{10}y^5 \\
& + 1807826944x^{10}y^4 + 805474304x^{10}y^3 + 14139392x^{10}y^2 - 19808256x^{10}y \\
& + 561152x^9y^{19} + 8032256x^9y^{18} + 24084480x^9y^{17} - 97820672x^9y^{16} \\
& - 961896448x^9y^{15} - 3135741952x^9y^{14} - 4311588864x^9y^{13} \\
& + 1694744576x^9y^{12} + 14215856128x^9y^{11} + 17096368128x^9y^{10} \\
& + 2742128640x^9y^9 - 9361649664x^9y^8 - 6695055360x^9y^7 \\
& - 1074515968x^9y^6 + 196972544x^9y^5 + 284676096x^9y^4 \\
& + 238223360x^9y^3 + 15433728x^9y^2 - 19759104x^9y + 81920x^8y^{20} \\
& + 2949120x^8y^{19} + 26238976x^8y^{18} + 100159488x^8y^{17} + 92635136x^8y^{16} \\
& - 698167296x^8y^{15} - 3091906560x^8y^{14} - 5127643136x^8y^{13} \\
& - 801718272x^8y^{12} + 10337443840x^8y^{11} + 14603108352x^8y^{10} \\
& + 3304792064x^8y^9 - 8653938688x^8y^8 - 7526531072x^8y^7 \\
& - 1313636352x^8y^6 + 802844672x^8y^5 + 520314880x^8y^4 \\
& + 294621184x^8y^3 + 137158656x^8y^2 + 27869184x^8y + 393216x^7y^{20} \\
& + 6742016x^7y^{19} + 48717824x^7y^{18} + 188973056x^7y^{17} \\
& + 335745024x^7y^{16} - 266461184x^7y^{15} - 2825957376x^7y^{14} \\
& - 6113759232x^7y^{13} - 4176195584x^7y^{12} + 6020382720x^7y^{11} \\
& + 14420799488x^7y^{10} + 9540186112x^7y^9 - 1936842752x^7y^8 \\
& - 6234173440x^7y^7 - 3476242432x^7y^6 - 986275840x^7y^5 \\
& - 403210240x^7y^4 - 230707200x^7y^3 - 71589888x^7y^2 - 9289728x^7y \\
& + 405504x^6y^{20} + 6266880x^6y^{19} + 51077120x^6y^{18} + 235749376x^6y^{17} \\
& + 607432704x^6y^{16} + 578891776x^6y^{15} - 1392099328x^6y^{14} \\
& - 5571129344x^6y^{13} - 7558770688x^6y^{12} - 2077913088x^6y^{11} \\
& + 7182491648x^6y^{10} + 10148646912x^6y^9 + 5264396288x^6y^8 \\
& + 348176384x^6y^7 - 580927488x^6y^6 + 8036352x^6y^5 + 116850688x^6y^4 \\
& - 16834560x^6y^3 - 49545216x^6y^2 - 18579456x^6y - 40960x^5y^{21} \\
& - 765952x^5y^{20} - 3080192x^5y^{19} + 11337728x^5y^{18} + 151343104x^5y^{17} \\
& + 626065408x^5y^{16} + 1280143360x^5y^{15} + 860512256x^5y^{14} \\
& - 2039250944x^5y^{13} - 5872988160x^5y^{12} - 6257803264x^5y^{11} \\
& - 2039152640x^5y^{10} + 2182864896x^5y^9 + 2681987072x^5y^8 \\
& + 1108303872x^5y^7 + 238452736x^5y^6 + 277917696x^5y^5 \\
& + 372523008x^5y^4 + 272019456x^5y^3 + 122314752x^5y^2 + 27869184x^5y \\
& - 135168x^4y^{21} - 2158592x^4y^{20} - 13729792x^4y^{19} - 42233856x^4y^{18} \\
& - 28270592x^4y^{17} + 250150912x^4y^{16} + 987156480x^4y^{15}
\end{aligned}$$

$$\begin{aligned}
& +1702690816x^4y^{14} + 1201012736x^4y^{13} - 829247488x^4y^{12} \\
& -2678673408x^4y^{11} - 2607132672x^4y^{10} - 1271472128x^4y^9 \\
& -462905344x^4y^8 - 572682240x^4y^7 - 828620800x^4y^6 - 776859648x^4y^5 \\
& -516562944x^4y^4 - 245846016x^4y^3 - 72769536x^4y^2 - 9289728x^4y \\
& -122880x^3y^{21} - 1757184x^3y^{20} - 12668928x^3y^{19} - 51032064x^3y^{18} \\
& -117248000x^3y^{17} - 115302400x^3y^{16} + 142254080x^3y^{15} \\
& +668696576x^3y^{14} + 1081921536x^3y^{13} + 1006809088x^3y^{12} \\
& +709992448x^3y^{11} + 780275712x^3y^{10} + 1164959744x^3y^9 \\
& +1118896128x^3y^8 + 101863424x^3y^7 - 1515675648x^3y^6 \\
& -2476277760x^3y^5 - 1236590592x^3y^4 + 2737987584x^3y^3 \\
& +7354834944x^3y^2 + 5699297280x^3y + 1529487360x^3 + 8192x^2y^{21} \\
& -270336x^2y^{20} - 3293184x^2y^{19} - 17408000x^2y^{18} - 53833728x^2y^{17} \\
& -104079360x^2y^{16} - 128421888x^2y^{15} - 107692032x^2y^{14} \\
& -125976576x^2y^{13} - 300679168x^2y^{12} - 562814976x^2y^{11} \\
& -604520448x^2y^{10} - 122200064x^2y^9 + 885227520x^2y^8 \\
& +1985716224x^2y^7 + 2317000704x^2y^6 + 979660800x^2y^5 \\
& -2060181625x^2y^4 - 5013221706x^2y^3 - 4462535189x^2y^2 + 627080796x^2y \\
& +3040468796x^2 + 12288xy^{21} + 40960xy^{20} + 176128xy^{19} + 483328xy^{18} \\
& +2985984xy^{17} + 15163392xy^{16} + 52948992xy^{15} + 122855424xy^{14} \\
& +188153856xy^{13} + 149471232xy^{12} - 132128768xy^{11} - 728858624xy^{10} \\
& -1529827328xy^9 - 2176352256xy^8 - 2090053632xy^7 - 785547264xy^6 \\
& +1499959296xy^5 + 3297757575xy^4 + 2277898166xy^3 - 2887986709xy^2 \\
& -6322981540xy - 4568354500x - 36864y^{21} - 221184y^{20} - 1118208y^{19} - 3956736y^{18} \\
& -10764288y^{17} - 21454848y^{16} - 25030656y^{15} + 12423168y^{14} + 156880896y^{13} \\
& +485634048y^{12} + 1022054400y^{11} + 1625628672y^{10} + 1918550016y^9 \\
& +1384833024y^8 - 297467904y^7 - 2628636672y^6 - 3911417856y^5 - 3922930425y^4 \\
& -1257828426y^3 + 2330572523y^2 + 1692901852y - 976961476 \\
i = 2, m = 31104, \text{ neighbours } 7, 8, 41, 45, 48, 56, 57, 82, 89, 90, 91, 92, 93, 94, 95, 193, 195 \\
W \quad x^6 - x^5y + 2463x^5 + x^4y^2 - 3680x^4y - 8960x^4 - 3x^3y^3 \\
\quad +4481x^3y^2 - 3381613x^3y - 118143368x^3 + 4x^2y^4 - 4889x^2y^3 \\
\quad +10125411x^2y^2 + 189519656x^2y + 555320833x^2 - 6xy^5 + 6163xy^4 \\
\quad -20179118xy^3 - 213988645xy^2 - 574063640xy - 1034482137x + 5y^6 - 11183y^5 \\
\quad +30143202y^4 + 73393249y^3 + 56352445y^2 - 507863788y - 410487352 \\
A \quad -9287604633600x^{22} + 60369430118400x^{21} - 4643802316800x^{20}y \\
\quad -97519848652800x^{20} + 65013232435200x^{19}y - 142796921241600x^{19} \\
\quad -285593842483200x^{18}y + 718047933235200x^{18} + 582216715468800x^{17}y \\
\quad -1410554953728000x^{17} - 698892248678400x^{16}y + 2344539694694400x^{16} \\
\quad +765066431692800x^{15}y - 3105833037004800x^{15} - 878839588454400x^{14}y \\
\quad +2971017650995200x^{14} + 719789359104000x^{13}y - 1384651243929600x^{13} \\
\quad +73285005312000x^{12}y - 2473368929280000x^{12} - 2078536893235200x^{11}y \\
\quad +5743676011622400x^{11} + 4404392539545600x^{10}y - 6958402184448000x^{10} \\
\quad -7291386392371200x^9y - 4751058731443200x^9 + 7196016116275200x^8y \\
\quad +41946398343760414x^8 + 6157690941977440x^7y - 85656220647467085x^7 \\
\quad -32085678730984246x^6y + 133299993516435137x^6 + 84461592967252926x^5y \\
\quad -38558929344217843x^5 - 116992574093602691x^4y - 357152437636731415x^4 \\
\quad -19755060728157536x^3y + 410391338391903668x^3 + 135640915027858711x^2y \\
\quad +59783910037743660x^2 - 36678052347724176xy - 216061360135696320x \\
\quad -51576382190408256y + 19713573200109312 \\
B \quad 6191736422400x^{24} - 9287604633600x^{23}y - 46438023168000x^{23} \\
\quad -17027275161600x^{22}y^2 + 65013232435200x^{22} - 74300837068800x^{21}y^3 \\
\quad -155567377612800x^{21}y^2 - 9287604633600x^{21}y + 215936807731200x^{21} \\
\quad -160211179929600x^{20}y^4 - 22445044531200x^{20}y^3 \\
\quad +570800701440000x^{20}y^2 + 741460436582400x^{20}y - 517396974796800x^{20} \\
\quad -256957061529600x^{19}y^5 - 240703753420800x^{19}y^4 \\
\quad +1657063460044800x^{19}y^3 + 2100740073062400x^{19}y^2 - 478118146867200x^{19}y \\
\quad +236446934630400x^{19} - 307264919961600x^{18}y^6 + 2708884684800x^{18}y^5 \\
\quad +3605719007232000x^{18}y^4 + 3946844985753600x^{18}y^3 \\
\quad -3680987303116800x^{18}y^2 - 4567373070336000x^{18}y - 1479825004953600x^{18} \\
\quad -278628139008000x^{17}y^7 + 426842829619200x^{17}y^6 \\
\quad +5365720085299200x^{17}y^5 + 4106862673920000x^{17}y^4 \\
\quad -11921994989568000x^{17}y^3 - 13476314323353600x^{17}y^2 \\
\quad +986614500556800x^{17}y + 3419386439270400x^{17} - 191943829094400x^{16}y^8
\end{aligned}$$

$$\begin{aligned}
& +773967052800000x^{16}y^7 + 5809009714790400x^{16}y^6 \\
& +1351346474188800x^{16}y^5 - 24222459867955200x^{16}y^4 \\
& -24934896540057600x^{16}y^3 + 8169222242304000x^{16}y^2 \\
& +11927412758937600x^{16}y + 159920942284800x^{16} - 97519848652800x^{15}y^9 \\
& +828918713548800x^{15}y^8 + 4533125028249600x^{15}y^7 \\
& -3479949361152000x^{15}y^6 - 34621481189376000x^{15}y^5 \\
& -29757485246054400x^{15}y^4 + 29633650517606400x^{15}y^3 \\
& +40372249883443200x^{15}y^2 + 8650919986790400x^{15}y - 881209862553600x^{15} \\
& -34828517376000x^{14}y^{10} + 600598432972800x^{14}y^9 \\
& +2388849308467200x^{14}y^8 - 7141780979712000x^{14}y^7 \\
& -35792686832025600x^{14}y^6 - 21576846989721600x^{14}y^5 \\
& +60487266602188800x^{14}y^4 + 77065640873164800x^{14}y^3 \\
& +9508185243648000x^{14}y^2 - 19329004803686400x^{14}y - 5055576975360000x^{14} \\
& -7739670528000x^{13}y^{11} + 301073183539200x^{13}y^{10} \\
& +669481500672000x^{13}y^9 - 7440145278566400x^{13}y^8 \\
& -27187140663705600x^{13}y^7 - 5067549278208000x^{13}y^6 \\
& +83912927389286400x^{13}y^5 + 98709629504716800x^{13}y^4 \\
& -10075309601587200x^{13}y^3 - 66814447258828800x^{13}y^2 \\
& -28649091867033600x^{13}y - 2975044698316800x^{13} - 773967052800x^{12}y^{12} \\
& +99454766284800x^{12}y^{11} - 79525114675200x^{12}y^{10} \\
& -5053424379494400x^{12}y^9 - 14639780295475200x^{12}y^8 \\
& +9008009035776000x^{12}y^7 + 84090746319667200x^{12}y^6 \\
& +86583307213209600x^{12}y^5 - 51291183572582400x^{12}y^4 \\
& -132665886012211200x^{12}y^3 - 58540884583219200x^{12}y^2 \\
& +10674323820748800x^{12}y + 11049703887974400x^{12} + 18962192793600x^{11}y^{12} \\
& -160791655219200x^{11}y^{11} - 2304293407948800x^{11}y^{10} \\
& -5023433156198400x^{11}y^9 + 13491987156172800x^{11}y^8 \\
& +62548147372032000x^{11}y^7 + 50932449843609600x^{11}y^6 \\
& -86387880532377600x^{11}y^5 - 169762900819968000x^{11}y^4 \\
& -58505523963494400x^{11}y^3 + 61713278786764800x^{11}y^2 \\
& +48055898499379200x^{11}y + 10975587472742400x^{11} + 1547934105600x^{10}y^{13} \\
& -64626248908800x^{10}y^{12} - 668127058329600x^{10}y^{11} \\
& -643553604403200x^{10}y^{10} + 10090401959116800x^{10}y^9 \\
& +34234304171212800x^{10}y^8 + 15934240191283200x^{10}y^7 \\
& -91063608990105600x^{10}y^6 - 150609538164326400x^{10}y^5 \\
& -19893662141644800x^{10}y^4 + 125753973247180800x^{10}y^3 \\
& +84994800303513600x^{10}y^2 - 7580931699456000x^{10}y - 36078971500529343x^{10} \\
& -11609505792000x^9y^{13} - 104872535654400x^9y^{12} + 296816364748800x^9y^{11} \\
& +4809044282572800x^9y^{10} + 13127061690777600x^9y^9 \\
& -2915340396134400x^9y^8 - 67600023817420800x^9y^7 \\
& -93988043499110400x^9y^6 + 26351449738444800x^9y^5 \\
& +156136049904844800x^9y^4 + 95366478820147200x^9y^3 \\
& -21099466530201600x^9y^2 - 17889856907005483x^9y + 45605669298680948x^9 \\
& -773967052800x^8y^{14} - 5030785843200x^8y^{13} + 163113556377600x^8y^{12} \\
& +1428162704179200x^8y^{11} + 2984997430886400x^8y^{10} \\
& -6515448142233600x^8y^9 - 36061833874636800x^8y^8 \\
& -39726567869644800x^8y^7 + 44401328868556800x^8y^6 \\
& +127160464873881600x^8y^5 + 57720527880192000x^8y^4 \\
& -65468712045772800x^8y^3 - 71422511042294337x^8y^2 - 24890401784090641x^8y \\
& -66733804681807660x^8 + 386983526400x^7y^{14} + 30378206822400x^7y^{13} \\
& +226191871180800x^7y^{12} + 116482041446400x^7y^{11} \\
& -3855903857049600x^7y^{10} - 13735593286041600x^7y^9 \\
& -9814289213030400x^7y^8 + 34434374654361600x^7y^7 \\
& +71356279416422400x^7y^6 + 14729753965363200x^7y^5 \\
& -73124794132070400x^7y^4 - 59136113619765166x^7y^3 \\
& +351178489467470x^7y^2 + 16348298822583283x^7y + 3339481204558540x^7 \\
& +1934917632000x^6y^{14} + 11029030502400x^6y^{13} - 124221711974400x^6y^{12} \\
& -1235831891558400x^6y^{11} - 3474918575308800x^6y^{10} \\
& -268953550848000x^6y^9 + 16737617992089600x^6y^8 \\
& +27457255165132800x^6y^7 - 5027883466752000x^6y^6 \\
& -49195474285363200x^6y^5 - 31547477547418860x^6y^4 \\
& +18554118667679144x^6y^3 + 55670771638447085x^6y^2 + 88709064091133133x^6y \\
& +246873150812202345x^6 - 773967052800x^5y^{14} - 27475830374400x^5y^{13}
\end{aligned}$$

$$\begin{aligned}
& -203553334886400x^5y^{12} - 440000269516800x^5y^{11} \\
& + 881935456665600x^5y^{10} + 5543732507443200x^5y^9 \\
& + 7061868881510400x^5y^8 - 6148200775680000x^5y^7 \\
& - 21795105698611200x^5y^6 - 10327235877271676x^5y^5 \\
& + 14838302838773458x^5y^4 + 13957546968285247x^5y^3 \\
& - 86915224212354380x^5y^2 - 207889022978815175x^5y - 233960813170055134x^5 \\
& - 1547934105600x^4y^{14} - 11609505792000x^4y^{13} + 11416014028800x^4y^{12} \\
& + 360862138368000x^4y^{11} + 1290203077017600x^4y^{10} \\
& + 1263888197222400x^4y^9 - 2445542395084800x^4y^8 \\
& - 6504612603494400x^4y^7 - 2249922222491117x^4y^6 \\
& + 6435923032457898x^4y^5 + 5422783146359979x^4y^4 \\
& - 2094013024865071x^4y^3 + 86707914207550323x^4y^2 - 10285082495413483x^4y \\
& - 247296303493681777x^4 + 386983526400x^3y^{14} + 8707129344000x^3y^{13} \\
& + 63271806566400x^3y^{12} + 195233189068800x^3y^{11} + 168144342220800x^3y^{10} \\
& - 483729408000000x^3y^9 - 1212999863500800x^3y^8 - 301653658827909x^3y^7 \\
& + 1615075744389020x^3y^6 + 1233714394077960x^3y^5 - 918940880582146x^3y^4 \\
& + 18220132542390042x^3y^3 + 158189228285196482x^3y^2 \\
& + 402518392394465762x^3y + 423421607591328506x^3 + 386983526400x^2y^{14} \\
& + 4063327027200x^2y^{13} + 15092357529600x^2y^{12} + 15479341056000x^2y^{11} \\
& - 42374696140800x^2y^{10} - 117062516736000x^2y^9 - 21671077478806x^2y^8 \\
& + 199490009330929x^2y^7 + 140276129701599x^2y^6 - 144612006251776x^2y^5 \\
& - 10225554888001981x^2y^4 - 68187140025097554x^2y^3 \\
& - 340497409799380554x^2y^2 - 178183269046153700x^2y + 105380904070380923x^2 \\
& + 135xy^9 - 531526xy^8 + 1981692670xy^7 - 2843458061524xy^6 \\
& + 3936619015350143xy^5 + 22464457610400524xy^4 + 54988762447420336xy^3 \\
& + 18809309227007602xy^2 - 240636944720413048xy - 220813159009882800x - 25y^{10} \\
& + 111855y^9 - 426603339y^8 + 673875414926y^7 - 907645125741707y^6 \\
& - 3516378352337065y^5 - 4371186562963897y^4 + 30223887400236872y^3 \\
& + 121842436536753888y^2 - 54287163902203456y - 227640585752299840
\end{aligned}$$

$i = 3, m = 12288$, neighbours 0, 46, 55, 100, 101, 102, 115, 116, 117, 136, 158, 165, 182, 197

W $-58x^4y - 1454x^4 + 175x^3y^2 + 476309x^3y + 11909402x^3 - 349x^2y^3$
 $-479635x^2y^2 - 7663489x^2y + 17443577x^2 + 293xy^4 + 4159xy^3 - 4250819xy^2$
 $-2650726xy + 19282351x + 172y^5 + 479423y^4 + 11911343y^3 - 14786320y^2 + 46031787y$
 -8826597

A $-12079595520x^{24} - 9663676416x^{23}y - 123211874304x^{23} - 45902462976x^{22}y$
 $-471708205056x^{22} + 45298483200x^{21}y - 853423423488x^{21} + 454796771328x^{20}y$
 $-1007136276480x^{20} - 318297341952x^{19}y - 2391306928128x^{19}$
 $-4931494871040x^{18}y - 6093778452480x^{18} - 9669263228928x^{17}y$
 $-6052217094144x^{17} - 6705383473152x^{16}y + 6373364465664x^{16}$
 $-967915339776x^{15}y + 28280475353088x^{15} + 2440531279872x^{14}y$
 $+44021655797760x^{14} + 15213674889216x^{13}y + 29872988356608x^{13}$
 $+30720940572672x^{12}y - 25245409738752x^{12} + 34090428137472x^{11}y$
 $-89513533440000x^{11} + 16042637131776x^{10}y - 122008060919808x^{10}$
 $-63248661282816x^9y - 71609096355840x^9 - 112604623405056x^8y$
 $+118440652972032x^8 - 25226582704128x^7y + 287734950309888x^7$
 $+15916671369216x^6y + 230885452877408x^6 + 169464419686400x^5y$
 $+779162601264x^5 + 384146354899584x^4y - 539554731726464x^4$
 $-195351400396880x^3y - 914605247259408x^3 - 415042610976656x^2y$
 $-536939444989856x^2 + 10780483199680xy - 43615421103744x - 295970136704832y$
 $+307574504959424$

B $24159191040x^{26} - 4831838208x^{25}y + 222264557568x^{25} - 19327352832x^{24}y^2$
 $-101468602368x^{24}y + 696992661504x^{24} + 24159191040x^{23}y^3$
 $-106300440576x^{23}y^2 - 459024629760x^{23}y + 799669223424x^{23}$
 $-7247757312x^{22}y^4 + 179985973248x^{22}y^3 - 212600881152x^{22}y^2$
 $-433657479168x^{22}y + 678269288448x^{22} - 36238786560x^{21}y^5$
 $-1207959552x^{21}y^4 + 489827598336x^{21}y^3 - 526066384896x^{21}y^2$
 $+1862673629184x^{21}y + 4144207233024x^{21} + 2415919104x^{20}y^6$
 $-117172076544x^{20}y^5 + 45902462976x^{20}y^4 + 463252488192x^{20}y^3$
 $-2223853535232x^{20}y^2 + 5585302978560x^{20}y + 9619736887296x^{20}$
 $+16911433728x^{19}y^7 - 8455716864x^{19}y^6 - 125627793408x^{19}y^5$
 $+393794813952x^{19}y^4 - 491639537664x^{19}y^3 - 4997932646400x^{19}y^2$
 $+6663859863552x^{19}y + 1900648857600x^{19} + 36238786560x^{18}y^8$

$$\begin{aligned}
& -138915348480x^{18}y^7 + 269374980096x^{18}y^6 - 322525200384x^{18}y^5 \\
& + 1606586204160x^{18}y^4 - 1565515579392x^{18}y^3 - 3652265705472x^{18}y^2 \\
& + 8635778334720x^{18}y - 18812120334336x^{18} + 70061654016x^{17}y^9 \\
& - 223472517120x^{17}y^8 + 135291469824x^{17}y^7 + 244007829504x^{17}y^6 \\
& - 1104679010304x^{17}y^5 + 3993514278912x^{17}y^4 - 2763207475200x^{17}y^3 \\
& + 5065125396480x^{17}y^2 + 15371776032768x^{17}y - 26305074561024x^{17} \\
& + 50734301184x^{16}y^{10} - 3623878656x^{16}y^9 - 616059371520x^{16}y^8 \\
& + 1642824990720x^{16}y^7 - 1919447728128x^{16}y^6 - 808124940288x^{16}y^5 \\
& + 5466620952576x^{16}y^4 - 6628678041600x^{16}y^3 + 15003461615616x^{16}y^2 \\
& + 9314670477312x^{16}y - 18602227924992x^{16} + 12079595520x^{15}y^{11} \\
& + 199313326080x^{15}y^{10} - 1001398468608x^{15}y^9 + 1303992336384x^{15}y^8 \\
& + 1215811289088x^{15}y^7 - 5106649006080x^{15}y^6 + 4658496012288x^{15}y^5 \\
& + 181193932800x^{15}y^4 - 10652391309312x^{15}y^3 + 21269487550464x^{15}y^2 \\
& - 20386874916864x^{15}y - 3210610212864x^{15} + 183609851904x^{14}y^{11} \\
& - 589484261376x^{14}y^{10} - 309237645312x^{14}y^9 + 3167269945344x^{14}y^8 \\
& - 3449328500736x^{14}y^7 - 2338609692672x^{14}y^6 + 10216921890816x^{14}y^5 \\
& - 10779227062272x^{14}y^4 - 5370588168192x^{14}y^3 + 18550785835008x^{14}y^2 \\
& - 50509427245056x^{14}y + 24758737698816x^{14} + 74893492224x^{13}y^{12} \\
& - 13891534848x^{13}y^{11} - 1287080902656x^{13}y^{10} + 2786762686464x^{13}y^9 \\
& + 292930191360x^{13}y^8 - 7462170132480x^{13}y^7 + 7857776885760x^{13}y^6 \\
& + 5267307626496x^{13}y^5 - 17247246483456x^{13}y^4 + 11884510052352x^{13}y^3 \\
& - 5230493171712x^{13}y^2 - 53327250063360x^{13}y + 28679879393280x^{13} \\
& + 12079595520x^{12}y^{13} + 187233730560x^{12}y^{12} - 1169908826112x^{12}y^{11} \\
& + 1364390313984x^{12}y^{10} + 2653283155968x^{12}y^9 - 6320648355840x^{12}y^8 \\
& - 589484261376x^{12}y^7 + 11911085162496x^{12}y^6 - 9271089561600x^{12}y^5 \\
& - 8990238965760x^{12}y^4 + 24290858631168x^{12}y^3 - 36862810914816x^{12}y^2 \\
& + 380050735104x^{12}y + 28855169777664x^{12} + 113548197888x^{11}y^{13} \\
& - 497679335424x^{11}y^{12} - 79121350656x^{11}y^{11} + 2807297998848x^{11}y^{10} \\
& - 2373640519680x^{11}y^9 - 5417698590720x^{11}y^8 + 8947356401664x^{11}y^7 \\
& + 3082108796928x^{11}y^6 - 17078132146176x^{11}y^5 + 7756308283392x^{11}y^4 \\
& + 16873383002112x^{11}y^3 - 54331602370560x^{11}y^2 + 72406138945536x^{11}y \\
& + 21422532427776x^{11} + 24159191040x^{10}y^{14} - 24763170816x^{10}y^{13} \\
& - 614247432192x^{10}y^{12} + 1708658786304x^{10}y^{11} + 471708205056x^{10}y^{10} \\
& - 5758947164160x^{10}y^9 + 3123783401472x^{10}y^8 + 8785489821696x^{10}y^7 \\
& - 9169016979456x^{10}y^6 - 8243719962624x^{10}y^5 + 17001426714624x^{10}y^4 \\
& - 2081314308096x^{10}y^3 - 39534430519296x^{10}y^2 + 58963214499840x^{10}y \\
& - 121990178783232x^{10} + 50734301184x^9y^{14} - 376279400448x^9y^{13} \\
& + 584652423168x^9y^{12} + 1142729736192x^9y^{11} - 3067613282304x^9y^{10} \\
& - 1233326702592x^9y^9 + 6886577405952x^9y^8 - 605791715328x^9y^7 \\
& - 10056263270400x^9y^6 + 5272139464704x^9y^5 + 12115230326784x^9y^4 \\
& - 12727061839872x^9y^3 + 51368983068672x^9y^2 + 34234281050112x^9y \\
& - 84479220817920x^9 + 12079595520x^8y^{15} - 82141249536x^8y^{14} \\
& + 16307453952x^8y^{13} + 674041430016x^8y^{12} - 647466319872x^8y^{11} \\
& - 2251032625152x^8y^{10} + 3243371397120x^8y^9 + 3150358511616x^8y^8 \\
& - 6997105704960x^8y^7 - 2270359977984x^8y^6 + 9890772811776x^8y^5 \\
& + 449360953344x^8y^4 - 9663676416000x^8y^3 + 117417738240000x^8y^2 \\
& - 25044557783040x^8y + 182301573972800x^8 - 60397977600x^7y^{14} \\
& + 254275485696x^7y^{13} + 40466644992x^7y^{12} - 1285268963328x^7y^{11} \\
& + 632970805248x^7y^{10} + 2990907850752x^7y^9 - 2065610833920x^7y^8 \\
& - 4312415600640x^7y^7 + 3308601212928x^7y^6 + 4856601378816x^7y^5 \\
& - 6076036546560x^7y^4 - 3152774430720x^7y^3 + 37203501662208x^7y^2 \\
& - 240920930448192x^7y - 2956595515808x^7 - 15099494400x^6y^{15} \\
& + 57378078720x^6y^{14} + 44090523648x^6y^{13} - 283870494720x^6y^{12} \\
& - 229512314880x^6y^{11} + 989922852864x^6y^{10} + 641426522112x^6y^9 \\
& - 2612816510976x^6y^8 + 1811939328x^6y^7 + 3666157240320x^6y^6 \\
& - 676457349120x^6y^5 - 5301734473728x^6y^4 + 6643777536x^6y^3 \\
& - 8412222799140x^6y^2 - 72593672134584x^6y + 256732336826652x^6 \\
& + 9663676416x^5y^{14} + 6039797760x^5y^{13} - 173946175488x^5y^{12} \\
& + 131667591168x^5y^{11} + 783361769472x^5y^{10} - 1058776547328x^5y^9 \\
& - 1156017291264x^5y^8 + 2147752083456x^5y^7 + 1104679010304x^5y^6 \\
& - 1874753224704x^5y^5 - 1615645900800x^5y^4 + 328565493680x^5y^3 \\
& - 168359710469468x^5y^2 + 216469975057712x^5y + 481377873982364x^5 \\
& + 3019898880x^4y^{15} + 3623878656x^4y^{14} - 56170119168x^4y^{13}
\end{aligned}$$

$$\begin{aligned}
& +24159191040x^4y^{12} + 196897406976x^4y^{11} + 3623878656x^4y^{10} \\
& -587672322048x^4y^9 + 10267656192x^4y^8 + 903553744896x^4y^7 \\
& -37446746112x^4y^6 - 753766760448x^4y^5 + 316484982703x^4y^4 \\
& +299548474146x^4y^3 - 384884134124561x^4y^2 + 2084834962428x^4y \\
& -499454803976688x^4 + 5435817984x^3y^{13} - 42882564096x^3y^{12} \\
& +104488501248x^3y^{11} - 31406948352x^3y^{10} - 245215789056x^3y^9 \\
& +366615724032x^3y^8 - 21139292160x^3y^7 - 422785843200x^3y^6 \\
& +152202984075x^3y^5 + 474018018383x^3y^4 - 995329373397x^3y^3 \\
& +135495042442577x^3y^2 + 13317978190894x^3y + 227095947579284x^3 \\
& -3019898880x^2y^{14} + 8455716864x^2y^{13} + 12683575296x^2y^{12} \\
& -42278584320x^2y^{11} - 39862665216x^2y^{10} + 121399934976x^2y^9 \\
& +61605937152x^2y^8 - 158846681088x^2y^7 - 114152042665x^2y^6 \\
& +85940963341x^2y^5 + 675740899863x^2y^4 + 1467673242365x^2y^3 \\
& +415121270721989x^2y^2 - 255277994320186x^2y + 1426892703490147x^2 \\
& -3019898880xy^{13} + 14495514624xy^{12} - 12683575296xy^{11} - 36842766336xy^{10} \\
& +64625836032xy^9 + 12683575296xy^8 - 61606008360xy^7 - 3137361982xy^6 \\
& +237334365155xy^5 + 16120624098485xy^4 + 270564255227288xy^3 \\
& +120738625726426xy^2 + 257868038242413xy + 1164299170809583x - 29584y^8 \\
& -164951096y^7 - 234108836433y^6 - 11650000981435y^5 - 139134015356659y^4 \\
& +180631485591862y^3 - 691037081981867y^2 + 392311641754069y - 1000632329478681
\end{aligned}$$

$i = 4, m = 12$, neighbours 0, 36, 49, 93, 132, 147, 148, 149, 206

W $10x^3y - 212x^3 - 19x^2y^2 + 971x^2y - 3157x^2 + xy^3 - 1083xy^2 + 5103xy + 656x$
 $+81y^4 - 1717y^3 + 2418y^2 + 1275y - 7745$

A $1548288x^{12} + 552960x^{11}y + 10063872x^{11} + 3594240x^{10}y + 28975104x^{10}$
 $+9096192x^9y + 51867648x^9 + 13478400x^8y + 63230976x^8 + 15752448x^7y$
 $+57908736x^7 + 15002496x^6y + 52683264x^6 + 15391296x^5y + 32921856x^5$
 $+7966944x^4y + 27098712x^4 - 1388448x^3y + 17433360x^3 + 22061808x^2y$
 $-63415872x^2 + 9688464xy - 42491196x + 3351132y - 34158672$

B $-774144x^{14} + 110592x^{13}y - 4257792x^{13} + 138240x^{12}y^2 + 414720x^{12}y$
 $-10036224x^{12} - 96768x^{11}y^3 + 691200x^{11}y^2 + 1092096x^{11}y - 15220224x^{11}$
 $+13824x^{10}y^4 - 774144x^{10}y^3 + 1133568x^{10}y^2 + 1658880x^{10}y$
 $-17563392x^{10} + 17280x^9y^5 + 86400x^9y^4 - 2730240x^9y^3$
 $+1257984x^9y^2 - 1050624x^9y - 22377600x^9 - 9216x^8y^6 + 123264x^8y^5$
 $+279936x^8y^4 - 5635584x^8y^3 + 2070720x^8y^2 - 3499200x^8y$
 $-34271424x^8 + 9792x^7y^7 - 84672x^7y^6 + 344160x^7y^5 + 612576x^7y^4$
 $-7663392x^7y^3 + 2984256x^7y^2 - 4362336x^7y - 31303584x^7 + 9792x^6y^8$
 $+15840x^6y^7 - 280800x^6y^6 + 554112x^6y^5 + 776736x^6y^4$
 $-7702272x^6y^3 + 4805280x^6y^2 - 4381344x^6y - 35578332x^6 + 576x^5y^9$
 $+24768x^5y^8 - 26208x^5y^7 - 444384x^5y^6 + 693792x^5y^5$
 $+391392x^5y^4 - 6729984x^5y^3 + 1656864x^5y^2 + 1739070x^5y$
 $-21914820x^5 - 576x^4y^{10} - 4896x^4y^9 + 14688x^4y^8 - 40032x^4y^7$
 $-400608x^4y^6 + 779328x^4y^5 + 5472x^4y^4 - 5336064x^4y^3$
 $+2517887x^4y^2 - 20092097x^4y + 14763191x^4 - 2304x^3y^{10} - 15264x^3y^9$
 $+29376x^3y^8 - 25344x^3y^7 - 321696x^3y^6 + 741024x^3y^5$
 $+184320x^3y^4 - 3301664x^3y^3 + 8269782x^3y^2 - 8736754x^3y$
 $-19839600x^3 + 576x^2y^{11} - 1440x^2y^{10} + 3456x^2y^9 + 67104x^2y^8$
 $-68256x^2y^7 - 301248x^2y^6 + 450432x^2y^5 + 338633x^2y^4$
 $-1915343x^2y^3 - 5649249x^2y^2 + 7980802x^2y + 6123638x^2 + 576xy^{11}$
 $+864xy^{10} + 24480xy^9 + 60768xy^8 - 63360xy^7 - 175392xy^6 + 150912xy^5$
 $-206482xy^4 + 1645284xy^3 + 3112722xy^2 - 4820516xy + 1829736x - 1152y^{11}$
 $-1728y^{10} + 9792y^9 + 19008y^8 - 14976y^7 - 44001y^6 + 300267y^5 - 3589735y^4$
 $+11407066y^3 - 11378115y^2 + 1364035y + 42490991$

$i = 5, m = 144$, neighbours 8, 35, 48, 60, 162, 171, 174, 176, 183, 184, 222

W $-922x^3y + 729x^3 - 922x^2y^2 + 529x^2y - 1224x^2 - xy^4 + xy^3 - 200xy^2$
 $-112xy - 516x - y^5 + 922y^4 - 730y^3 + 1111y^2 + 152y - 1791$

A $186624x^{15} - 124416x^{14}y - 1586304x^{14} + 1741824x^{13}y + 4214592x^{13}$
 $-8398080x^{12}y + 1438560x^{12} + 18849024x^{11}y - 33067440x^{11} - 18802368x^{10}y$
 $+85460184x^{10} + 808704x^9y - 120723372x^9 + 13954032x^8y + 124327062x^8$
 $-12577680x^7y - 12333435x^7 + 15516036x^6y + 123089343x^6 - 27311256x^5y$
 $-103818658x^5 + 31074062x^4y + 68993368x^4 - 27127528x^3y - 32781109x^3$
 $+17216808x^2y + 6763977x^2 - 6037068xy + 2360593x + 1106293y - 2832519$

B $-373248x^{17} - 124416x^{16}y + 2426112x^{16} + 248832x^{15}y^2 - 933120x^{15}y$

$$\begin{aligned}
& -3203712x^{15} + 373248x^{14}y^3 - 2737152x^{14}y^2 + 9424512x^{14}y \\
& -11897280x^{14} + 124416x^{13}y^4 - 1492992x^{13}y^3 + 7838208x^{13}y^2 \\
& -26671680x^{13}y + 46570464x^{13} - 248832x^{12}y^5 + 1244160x^{12}y^4 \\
& -622080x^{12}y^3 - 5660928x^{12}y^2 + 33522336x^{12}y - 65337840x^{12} \\
& -705024x^{11}y^6 + 3442176x^{11}y^5 - 8335872x^{11}y^4 + 10533888x^{11}y^3 \\
& -7610112x^{11}y^2 - 20851344x^{11}y + 50118264x^{11} + 705024x^{10}y^7 \\
& -3939840x^{10}y^6 + 3981312x^{10}y^5 + 8128512x^{10}y^4 - 20196864x^{10}y^3 \\
& +11959488x^{10}y^2 + 17229672x^{10}y - 54327996x^{10} + 2363904x^9y^7 \\
& -5018112x^9y^6 - 2239488x^9y^5 - 1119744x^9y^4 + 25297920x^9y^3 \\
& -5163264x^9y^2 - 28657476x^9y + 101634750x^9 - 165888x^8y^9 \\
& -870912x^8y^8 + 6262272x^8y^7 + 705024x^8y^6 - 17459712x^8y^5 \\
& +17832960x^8y^4 - 35582976x^8y^3 + 13085712x^8y^2 + 20524266x^8y \\
& -124238370x^8 + 41472x^7y^{10} - 290304x^7y^9 - 3027456x^7y^8 \\
& +1036800x^7y^7 + 20653056x^7y^6 - 8045568x^7y^5 - 25132032x^7y^4 \\
& +42550272x^7y^3 - 27401328x^7y^2 + 2798628x^7y + 95368880x^7 \\
& +373248x^6y^{10} + 331776x^6y^9 - 4354560x^6y^8 - 9331200x^6y^7 \\
& +10285056x^6y^6 + 20113920x^6y^5 + 4064256x^6y^4 - 37780992x^6y^3 \\
& +23675004x^6y^2 - 11079362x^6y - 72658357x^6 - 82944x^5y^{11} \\
& +414720x^5y^{10} + 2695680x^5y^9 + 1368576x^5y^8 - 16962048x^5y^7 \\
& -9497088x^5y^6 + 14349312x^5y^5 - 6759936x^5y^4 + 36080643x^5y^3 \\
& -14950558x^5y^2 + 6820906x^5y + 71787586x^5 - 248832x^4y^{11} \\
& -373248x^4y^{10} + 1783296x^4y^9 + 9040896x^4y^8 + 2820096x^4y^7 \\
& -25505280x^4y^6 - 2695680x^4y^5 + 12607491x^4y^4 - 32730641x^4y^3 \\
& +11658433x^4y^2 - 6248418x^4y - 63275237x^4 + 41472x^3y^{12} \\
& -124416x^3y^{11} - 1327104x^3y^{10} - 1410048x^3y^9 + 4022784x^3y^8 \\
& +12068352x^3y^7 + 5349888x^3y^6 - 14805504x^3y^5 + 1610024x^3y^4 \\
& +21857642x^3y^3 - 10986510x^3y^2 + 9343597x^3y + 41187952x^3 \\
& +41472x^2y^{12} + 373248x^2y^{11} - 414720x^2y^{10} - 3608064x^2y^9 \\
& -2032128x^2y^8 + 4893696x^2y^7 + 5018109x^2y^6 + 1783298x^2y^5 \\
& -2525637x^2y^4 - 11656510x^2y^3 + 10548696x^2y^2 - 7395027x^2y \\
& -23628664x^2 - 41472xy^{12} + 207360xy^{11} + 1036800xy^{10} - 497664xy^9 \\
& -4686336xy^8 - 2156547xy^7 + 3819116xy^6 - 193539xy^5 + 1511406xy^4 \\
& +1678440xy^3 - 2538520xy^2 + 4744150xy + 4482180x - 41472y^{12} - 124416y^{11} \\
& +248832y^{10} + 1161216y^9 + 539135y^8 - 1698507y^7 - 2014604y^6 + 3361101y^5 \\
& -3761889y^4 + 2319586y^3 + 2792101y^2 - 5231772y + 5289876
\end{aligned}$$

$i = 6, m = 720$, neighbours 0, 37, 54, 130, 149, 177, 183, 213, 224

$$\begin{aligned}
W \quad & 480xy^5 - 2400xy^4 - 1680xy^3 + 60000xy^2 + 144960xy + 90240x + 240y^6 + 3840y^5 \\
& +10320y^4 + 8400y^3 + 26400y^2 + 73920y + 51840 \\
A \quad & -230947200x^6y^3 - 2697883200x^6y^2 - 18029628000x^6y - 94407541200x^6 \\
& +57736800x^4y^3 + 674470800x^4y^2 + 65820542490x^4y - 33041392830x^4 \\
& -41212800x^3y^6 - 580478400x^3y^5 - 4038271200x^3y^4 \\
& -24313024800x^3y^3 - 166392991260x^3y^2 - 106415474895x^3y \\
& -138290076495x^3 + 36806400x^2y^7 + 411609600x^2y^6 + 552096000x^2y^5 \\
& -1068940800x^2y^4 - 1737676800x^2y^3 + 292377600x^2y^2 + 1148774400x^2y \\
& +364953600x^2 - 14515200xy^8 - 249350400xy^7 - 295293600xy^6 + 1335106800xy^5 \\
& +10015229610xy^4 + 33840048780xy^3 + 77762833785xy^2 - 81914323950xy \\
& -48916788165x - 6739200y^9 - 65232000y^8 - 915278400y^7 - 6660802800y^6 \\
& -17868740400y^5 - 44862987915y^4 - 94744270620y^3 - 83927994750y^2 \\
& -56377633635y - 2256930 \\
B \quad & 51321600x^8y^3 + 599529600x^8y^2 + 4006584000x^8y + 20979453600x^8 \\
& +51321600x^7y^4 + 599529600x^7y^3 + 4006584000x^7y^2 + 20979453600x^7y \\
& +34214400x^6y^5 + 382579200x^6y^4 + 2441275200x^6y^3 \\
& +12301048800x^6y^2 - 22955466420x^6y + 349380540x^6 + 22809600x^5y^6 \\
& +260755200x^5y^5 + 1744027200x^5y^4 + 9273657600x^5y^3 \\
& +6081695460x^5y^2 + 18968697450x^5y + 41176618110x^5 + 11404800x^4y^7 \\
& +147484800x^4y^6 + 1091102400x^4y^5 + 6260976000x^4y^4 \\
& +19685883600x^4y^3 + 25306296210x^4y^2 + 65211582870x^4y + 31352720220x^4 \\
& +3801600x^3y^8 + 52963200x^3y^7 + 413812800x^3y^6 + 2477714400x^3y^5 \\
& +12258378000x^3y^4 + 6338520000x^3y^3 + 25374837600x^3y^2 \\
& +21976573590x^3y + 794005470x^3 - 4492800x^2y^9 - 37228800x^2y^8 \\
& +113337600x^2y^7 + 1457469600x^2y^6 + 7232492400x^2y^5 \\
& -7620928410x^2y^4 + 16085954670x^2y^3 + 10999678920x^2y^2
\end{aligned}$$

$$\begin{aligned}
& -25356785835x^2y + 15906838065x^2 + 691200xy^{10} + 29414400xy^9 + 127190400xy^8 \\
& + 649600800xy^7 + 4402449600xy^6 + 4354842930xy^5 + 17696361525xy^4 \\
& + 35467246185xy^3 + 25967110020xy^2 + 59076158865xy + 45798846075x + 153600y^{10} \\
& + 22012800y^9 + 302452800y^8 + 1905853200y^7 + 3977203200y^6 + 11972156505y^5 \\
& + 27858873435y^4 + 20257714815y^3 + 46163055750y^2 + 54717249015y - 893538270
\end{aligned}$$

$i = 7, m = 2$, neighbours 1, 2

$$\begin{aligned}
W & -3x^2 - 2xy - 4x - y + 1 \\
A & 0 \\
B & 3x^2 - 6x - 3y - 3
\end{aligned}$$

$i = 8, m = 10$, neighbours 2, 5, 12, 43, 62

$$\begin{aligned}
W & -12xy^2 - 21xy + 12x - 456y^{10} - 48760y^2 - 87591y + 49116 \\
A & -207936y^{17} + 207936y^{15} + 1247616y^{14} - 207936y^{13} - 2495232y^{12} \\
& -7277760y^{11} + 3742848y^{10} - 22230912y^9 - 39978432y^8 + 44578660y^7 \\
& + 173364104y^6 + 195291732y^5 - 440835664y^4 - 1235474656y^3 - 730916128y^2 \\
& + 1501777456y - 402260064 \\
B & 900x^9 - 900x^8y - 2700x^8 + 1500x^7y^2 + 2400x^7y + 5100x^7 - 600x^6y^3 \\
& -5100x^6y^2 - 5100x^6y - 6300x^6 + 1300x^5y^4 + 4400x^5y^3 \\
& + 7500x^5y^2 + 6200x^5y + 5800x^5 - 700x^4y^5 - 2900x^4y^4 - 5100x^4y^3 \\
& -5900x^4y^2 - 6400x^4y - 4200x^4 + 415872x^3y^{17} - 415872x^3y^{15} \\
& -2495232x^3y^{14} + 415872x^3y^{13} + 4990464x^3y^{12} + 14555520x^3y^{11} \\
& -7485696x^3y^{10} + 44461824x^3y^9 + 79956864x^3y^8 - 89157320x^3y^7 \\
& -346727708x^3y^6 - 390582864x^3y^5 + 881672528x^3y^4 \\
& + 2470957312x^3y^3 + 1461839756x^3y^2 - 3003550712x^3y + 804522428x^3 \\
& + 415872x^2y^{17} - 415872x^2y^{15} - 2495232x^2y^{14} + 415872x^2y^{13} \\
& + 4990464x^2y^{12} + 14555520x^2y^{11} - 7485696x^2y^{10} + 44461824x^2y^9 \\
& + 79956864x^2y^8 - 89157120x^2y^7 - 346727508x^2y^6 - 390587964x^2y^5 \\
& + 881663728x^2y^4 + 2470945212x^2y^3 + 1461829556x^2y^2 - 3003557412x^2y \\
& + 804519028x^2 + 415872xy^{17} - 415872xy^{15} - 2495232xy^{14} + 415872xy^{13} \\
& + 4990464xy^{12} + 14555520xy^{11} - 7485696xy^{10} + 44461824xy^9 + 79956764xy^8 \\
& -89155920xy^7 - 346724008xy^6 - 390581464xy^5 + 881671684xy^4 + 2470952408xy^3 \\
& + 1461835603xy^2 - 3003553608xy + 804520284x + 623808y^{17} - 623808y^{15} \\
& -3742848y^{14} + 623808y^{13} + 7474752y^{12} + 21814128y^{11} - 11217600y^{10} \\
& + 66692636y^9 + 119934396y^8 - 133736580y^7 - 520091912y^6 - 585877896y^5 \\
& + 1321332108y^4 + 3702273160y^3 + 2191419333y^2 - 4501167108y + 1205601164
\end{aligned}$$

$i = 9, m = 2$, neighbours 1, 10

$$\begin{aligned}
W & 2x^2 - xy + x - 2y \\
A & 0 \\
B & 3xy + 3x
\end{aligned}$$

$i = 10, m = 128$, neighbours 9, 36, 46, 74, 96, 98, 99, 100, 101, 103, 104, 105, 106, 108, 109, 110, 113, 114, 196

$$\begin{aligned}
W & -x^8y^5 + 5x^7y^6 + 5x^6y^7 - x^6y^4 - x^6y^3 + x^6y^2 + x^6y \\
& -x^5y + 6x^4y^9 - 7x^4y^8 - x^4y^4 - x^4y^3 - 26x^4y - 28x^4 \\
& + 2496x^3y^{10} - 38977x^3y^9 - 3312x^3y^8 - 26x^3y^2 - 220933x^3y \\
& -229998x^3 - 38970x^2y^{10} - 3312x^2y^9 - 220905x^2y^2 - 285343x^2y \\
& -422948x^2 + 27xy^4 + 28xy^3 - 55345xy^2 - 149370xy - 448172x - 45237y^{12} \\
& + 59995y^{11} + 27y^5 + 220933y^4 + 229999y^3 + 273578y^2 + 498505y + 113520 \\
A & -2359296x^{35} - 4718592x^{34}y - 393216x^{34} - 6291456x^{33}y + 21233664x^{33} \\
& -27721728x^{32}y - 43548672x^{32} + 7176192x^{31}y - 31604736x^{31} + 13516800x^{30}y \\
& -57556992x^{30} - 137478144x^{29}y - 183189504x^{29} + 24133632x^{28}y - 45047808x^{28} \\
& -21970944x^{27}y - 358477824x^{27} + 8503296x^{26}y - 517650432x^{26} + 472645632x^{25}y \\
& + 60552192x^{25} + 125792256x^{24}y + 216261120x^{24} - 432992256x^{23}y \\
& + 285111552x^{23} - 750197760x^{22}y + 110565904x^{22} - 882194694x^{21}y \\
& -616749792x^{21} + 31405759x^{20}y - 64574236x^{20} + 685069644x^{19}y \\
& -2877301541x^{19} + 1588839329x^{18}y - 1499300470x^{18} + 4546501195x^{17}y \\
& + 515950837x^{17} + 1588360164x^{16}y + 2583928462x^{16} - 368309415x^{15}y \\
& + 8635449950x^{15} - 6359763894x^{14}y + 7105629710x^{14} - 11845345248x^{13}y \\
& + 2949775789x^{13} - 10329173479x^{12}y - 33490505846x^{12} + 8651344492x^{11}y \\
& -28043797736x^{11} + 63819770678x^{10}y + 16446931343x^{10} + 57245910353x^9y \\
& + 80028853037x^9 + 9144375447x^8y + 135189148496x^8 - 92648119711x^7y \\
& + 122496396966x^7 - 200334401166x^6y + 1136293684x^6 - 235193567153x^5y \\
& -233333621348x^5 - 106271417544x^4y - 285561878191x^4 + 151599989064x^3y \\
& -129502211162x^3 + 192110571645x^2y + 44714613707x^2 + 69121669805xy \\
& + 35349522560x - 36294992800y - 4295596800
\end{aligned}$$

$$\begin{aligned}
B \quad & 4718592x^{37} + 14155776x^{36}y + 786432x^{36} + 9437184x^{35}y^2 + 13369344x^{35}y \\
& - 42467328x^{35} - 4718592x^{34}y^3 + 12582912x^{34}y^2 + 12976128x^{34}y \\
& + 89456640x^{34} - 14155776x^{33}y^4 + 1572864x^{33}y^3 + 55443456x^{33}y^2 \\
& + 82182144x^{33}y + 70680576x^{33} - 9437184x^{32}y^5 - 6291456x^{32}y^4 \\
& + 42860544x^{32}y^3 - 2555904x^{32}y^2 + 64487424x^{32}y + 95059968x^{32} \\
& + 4718592x^{31}y^6 - 7864320x^{31}y^5 - 6291456x^{31}y^4 - 105971712x^{31}y^3 \\
& + 98304x^{31}y^2 + 395378688x^{31}y + 346226688x^{31} + 14155776x^{30}y^7 \\
& - 1572864x^{30}y^6 - 49152000x^{30}y^5 - 75694080x^{30}y^4 - 20840448x^{30}y^3 \\
& + 328040448x^{30}y^2 + 417497088x^{30}y + 252346368x^{30} + 9437184x^{29}y^8 \\
& + 6291456x^{29}y^7 - 42860544x^{29}y^6 + 30277632x^{29}y^5 - 28114944x^{29}y^4 \\
& - 35733504x^{29}y^3 + 64094208x^{29}y^2 + 292847616x^{29}y + 869326848x^{29} \\
& - 4718592x^{28}y^9 + 7864320x^{28}y^8 + 6291456x^{28}y^7 + 105971712x^{28}y^6 \\
& - 7274496x^{28}y^5 - 377290752x^{28}y^4 - 295845888x^{28}y^3 \\
& + 26984448x^{28}y^2 + 1006804992x^{28}y + 1391161344x^{28} - 14155776x^{27}y^{10} \\
& + 2621440x^{27}y^9 + 50724864x^{27}y^8 + 71237632x^{27}y^7 + 13631488x^{27}y^6 \\
& - 340508672x^{27}y^5 - 218660864x^{27}y^4 - 82673664x^{27}y^3 \\
& + 274432000x^{27}y^2 + 1017360384x^{27}y + 473511936x^{27} - 10485760x^{26}y^{11} \\
& - 12320768x^{26}y^{10} + 40370176x^{26}y^9 + 3211264x^{26}y^8 + 63668224x^{26}y^7 \\
& + 18890752x^{26}y^6 + 49037312x^{26}y^5 - 133988352x^{26}y^4 \\
& - 683261952x^{26}y^3 - 397148160x^{26}y^2 + 216545280x^{26}y + 61099008x^{26} \\
& + 7077888x^{25}y^{12} - 3145728x^{25}y^{11} - 16580608x^{25}y^{10} \\
& - 171245568x^{25}y^9 - 27033600x^{25}y^8 + 445235200x^{25}y^7 \\
& + 359907328x^{25}y^6 - 46776320x^{25}y^5 - 955973632x^{25}y^4 \\
& - 1062649856x^{25}y^3 - 234733568x^{25}y^2 + 1342970880x^{25}y + 1022860800x^{25} \\
& + 17956864x^{24}y^{13} + 15204352x^{24}y^{12} - 35946496x^{24}y^{11} \\
& - 88997888x^{24}y^{10} - 100499456x^{24}y^9 + 216481792x^{24}y^8 \\
& + 141164544x^{24}y^7 + 102514688x^{24}y^6 - 199917568x^{24}y^5 \\
& - 65722776x^{24}y^4 + 65126400x^{24}y^3 + 1273106432x^{24}y^2 \\
& + 2542751232x^{24}y + 1271267296x^{24} + 786432x^{23}y^{14} - 6684672x^{23}y^{13} \\
& - 19169280x^{23}y^{12} + 158121984x^{23}y^{11} + 125288448x^{23}y^{10} \\
& + 82690048x^{23}y^9 - 62128128x^{23}y^8 - 14483456x^{23}y^7 \\
& + 587235328x^{23}y^6 + 387956736x^{23}y^5 - 164528128x^{23}y^4 \\
& - 126550016x^{23}y^3 + 1047244800x^{23}y^2 + 1431375340x^{23}y + 835581888x^{23} \\
& - 12845056x^{22}y^{15} - 20447232x^{22}y^{14} - 8552448x^{22}y^{13} \\
& + 83607552x^{22}y^{12} + 27770880x^{22}y^{11} - 414826496x^{22}y^{10} \\
& - 205520896x^{22}y^9 + 376274944x^{22}y^8 + 1093959680x^{22}y^7 \\
& + 1035206656x^{22}y^6 - 287473664x^{22}y^5 - 1537900544x^{22}y^4 \\
& - 1712816128x^{22}y^3 + 33267212x^{22}y^2 + 612124738x^{22}y - 804746440x^{22} \\
& - 2490368x^{21}y^{16} + 11272192x^{21}y^{15} + 33456128x^{21}y^{14} \\
& - 24117248x^{21}y^{13} - 232390656x^{21}y^{12} - 473169920x^{21}y^{11} \\
& - 453394432x^{21}y^{10} - 545079296x^{21}y^9 + 89325568x^{21}y^8 \\
& + 808337408x^{21}y^7 + 44843008x^{21}y^6 - 1388101632x^{21}y^5 \\
& - 2313830400x^{21}y^4 - 1678238944x^{21}y^3 + 210059394x^{21}y^2 \\
& - 410822752x^{21}y + 4788702522x^{21} + 4325376x^{20}y^{17} + 22806528x^{20}y^{16} \\
& + 19988480x^{20}y^{15} - 28655616x^{20}y^{14} + 178601984x^{20}y^{13} \\
& + 278642688x^{20}y^{12} + 449593344x^{20}y^{11} + 180420608x^{20}y^{10} \\
& - 936263680x^{20}y^9 - 820117504x^{20}y^8 + 44564480x^{20}y^7 \\
& + 101793792x^{20}y^6 - 628310016x^{20}y^5 - 970483436x^{20}y^4 \\
& - 515645904x^{20}y^3 + 1144121448x^{20}y^2 + 4633245934x^{20}y + 3283653020x^{20} \\
& + 1048576x^{19}y^{18} - 7340032x^{19}y^{17} - 26148864x^{19}y^{16} \\
& - 92504064x^{19}y^{15} + 102612992x^{19}y^{14} + 522387456x^{19}y^{13} \\
& + 466092032x^{19}y^{12} + 434978816x^{19}y^{11} - 720896x^{19}y^{10} \\
& - 415956992x^{19}y^9 + 316391424x^{19}y^8 + 1461944320x^{19}y^7 \\
& + 1682571264x^{19}y^6 + 723811828x^{19}y^5 + 153081524x^{19}y^4 \\
& + 2169858984x^{19}y^3 + 727262394x^{19}y^2 - 2577421255x^{19}y + 882921938x^{19} \\
& - 524288x^{18}y^{19} - 12451840x^{18}y^{18} - 14712832x^{18}y^{17} \\
& - 22396928x^{18}y^{16} - 153468928x^{18}y^{15} - 301613056x^{18}y^{14} \\
& - 681951232x^{18}y^{13} - 618774528x^{18}y^{12} - 776404992x^{18}y^{11} \\
& - 931086336x^{18}y^{10} + 249610240x^{18}y^9 + 1653981184x^{18}y^8 \\
& + 2370191360x^{18}y^7 + 1716266208x^{18}y^6 + 711981188x^{18}y^5 \\
& + 999312513x^{18}y^4 - 3841802520x^{18}y^3 - 5892217890x^{18}y^2 \\
& - 3008511075x^{18}y - 2096832675x^{18} - 131072x^{17}y^{20} + 1835008x^{17}y^{19} \\
& + 23429120x^{17}y^{18} + 81592320x^{17}y^{17} + 43712512x^{17}y^{16}
\end{aligned}$$

$$\begin{aligned}
& -35094528x^{17}y^{15} - 290160640x^{17}y^{14} - 459735040x^{17}y^{13} \\
& + 354320384x^{17}y^{12} + 280018944x^{17}y^{11} - 699744256x^{17}y^{10} \\
& - 440877056x^{17}y^9 + 532676608x^{17}y^8 + 869639916x^{17}y^7 \\
& + 465805776x^{17}y^6 - 1195925287x^{17}y^5 - 5257739038x^{17}y^4 \\
& - 437167414x^{17}y^3 - 4577289089x^{17}y^2 - 2126960565x^{17}y - 7139694807x^{17} \\
& + 2621440x^{16}y^{20} + 10125312x^{16}y^{19} + 37142528x^{16}y^{18} \\
& + 56197120x^{16}y^{17} + 279068672x^{16}y^{16} + 719831040x^{16}y^{15} \\
& + 887275520x^{16}y^{14} + 1254588416x^{16}y^{13} + 1805942784x^{16}y^{12} \\
& + 949157888x^{16}y^{11} - 1071054848x^{16}y^{10} - 1672724480x^{16}y^9 \\
& - 690650612x^{16}y^8 - 100751028x^{16}y^7 - 2148969384x^{16}y^6 \\
& - 1424407046x^{16}y^5 + 3867718475x^{16}y^4 - 67314856x^{16}y^3 \\
& - 1618705075x^{16}y^2 - 2233885731x^{16}y - 10229308847x^{16} - 131072x^{15}y^{21} \\
& - 12156928x^{15}y^{20} - 37240832x^{15}y^{19} - 92831744x^{15}y^{18} \\
& - 137003008x^{15}y^{17} + 12812288x^{15}y^{16} - 1130496x^{15}y^{15} \\
& + 30048256x^{15}y^{14} - 259948544x^{15}y^{13} - 911998976x^{15}y^{12} \\
& - 1430568960x^{15}y^{11} - 1985511424x^{15}y^{10} - 1550640352x^{15}y^9 \\
& - 615970948x^{15}y^8 - 906906753x^{15}y^7 + 3867869464x^{15}y^6 \\
& + 4302559361x^{15}y^5 - 35740530x^{15}y^4 - 8678387x^{15}y^3 \\
& + 359307881x^{15}y^2 - 2282432937x^{15}y - 10031332551x^{15} - 131072x^{14}y^{22} \\
& - 4325376x^{14}y^{21} - 16908288x^{14}y^{20} - 51216384x^{14}y^{19} \\
& - 170180608x^{14}y^{18} - 317276160x^{14}y^{17} - 828375040x^{14}y^{16} \\
& - 1402191872x^{14}y^{15} - 1130971136x^{14}y^{14} - 1043480576x^{14}y^{13} \\
& - 1450704896x^{14}y^{12} - 1546158080x^{14}y^{11} - 1172809453x^{14}y^{10} \\
& - 664379856x^{14}y^9 + 1050140455x^{14}y^8 + 5211405086x^{14}y^7 \\
& + 412017974x^{14}y^6 + 31164726x^{14}y^5 + 20683484x^{14}y^4 + 8740862x^{14}y^3 \\
& + 6358515746x^{14}y^2 + 3646314868x^{14}y + 50593776356x^{14} + 1835008x^{13}y^{22} \\
& + 15794176x^{13}y^{21} + 55607296x^{13}y^{20} + 64487424x^{13}y^{19} \\
& + 187891712x^{13}y^{18} + 271941632x^{13}y^{17} - 213516288x^{13}y^{16} \\
& + 227229696x^{13}y^{15} + 1670365184x^{13}y^{14} + 2091565056x^{13}y^{13} \\
& + 1308409856x^{13}y^{12} + 457342461x^{13}y^{11} + 257971892x^{13}y^{10} \\
& + 2272570280x^{13}y^9 + 1441888774x^{13}y^8 - 3873796939x^{13}y^7 \\
& + 67642536x^{13}y^6 + 37308111x^{13}y^5 + 18922044x^{13}y^4 + 5760877x^{13}y^3 \\
& + 14045956834x^{13}y^2 + 32120329561x^{13}y + 23075615912x^{13} + 524288x^{12}y^{23} \\
& + 4882432x^{12}y^{22} + 38846464x^{12}y^{21} + 84475904x^{12}y^{20} \\
& + 175112192x^{12}y^{19} + 543817728x^{12}y^{18} + 732921856x^{12}y^{17} \\
& + 822820864x^{12}y^{16} + 1855520768x^{12}y^{15} + 2637266944x^{12}y^{14} \\
& + 2203041792x^{12}y^{13} + 1289938139x^{12}y^{12} + 751040644x^{12}y^{11} \\
& + 1211845761x^{12}y^{10} - 3725213978x^{12}y^9 - 4267333763x^{12}y^8 \\
& + 24435572x^{12}y^7 + 15903733x^{12}y^6 + 9984574x^{12}y^5 + 6550273x^{12}y^4 \\
& + 3294012256x^{12}y^3 - 12113682901x^{12}y^2 - 80744918224x^{12}y \\
& - 57160527605x^{12} - 3014656x^{11}y^{23} - 9895936x^{11}y^{22} - 21741568x^{11}y^{21} \\
& - 69566464x^{11}y^{20} - 86966272x^{11}y^{19} - 125698048x^{11}y^{18} \\
& - 322961408x^{11}y^{17} - 221724672x^{11}y^{16} + 313786368x^{11}y^{15} \\
& + 234602496x^{11}y^{14} - 348756308x^{11}y^{13} - 241376816x^{11}y^{12} \\
& - 1315200807x^{11}y^{11} - 5294783254x^{11}y^{10} - 481404206x^{11}y^9 \\
& - 77072702x^{11}y^8 - 39590628x^{11}y^7 - 11689984x^{11}y^6 - 586860x^{11}y^5 \\
& + 1093400670x^{11}y^4 - 47183788251x^{11}y^3 - 9197518888x^{11}y^2 \\
& - 128736914877x^{11}y - 135416527595x^{11} - 786432x^{10}y^{24} - 7077888x^{10}y^{23} \\
& - 35323904x^{10}y^{22} - 111820800x^{10}y^{21} - 199868416x^{10}y^{20} \\
& - 438206464x^{10}y^{19} - 953761792x^{10}y^{18} - 1152172032x^{10}y^{17} \\
& - 816644096x^{10}y^{16} - 527646720x^{10}y^{15} - 693534268x^{10}y^{14} \\
& - 767350466x^{10}y^{13} - 2297047976x^{10}y^{12} - 1284258290x^{10}y^{11} \\
& + 3909038943x^{10}y^{10} - 60318910x^{10}y^9 - 44074726x^{10}y^8 \\
& - 15383094x^{10}y^7 - 5416864x^{10}y^6 - 2200611638x^{10}y^5 \\
& - 24740931870x^{10}y^4 + 22260235180x^{10}y^3 - 61247213794x^{10}y^2 \\
& - 97878596698x^{10}y - 141862981718x^{10} + 1310720x^9y^{24} + 2654208x^9y^{23} \\
& - 24756224x^9y^{22} - 39813120x^9y^{21} - 28131328x^9y^{20} \\
& - 301072384x^9y^{19} - 587268096x^9y^{18} - 368345088x^9y^{17} \\
& - 37715968x^9y^{16} + 45936220x^9y^{15} - 207284430x^9y^{14} \\
& - 794764897x^9y^{13} + 4166467876x^9y^{12} + 4492449933x^9y^{11} \\
& + 44049546x^9y^{10} + 7525384x^9y^9 - 55852x^9y^8 - 1831524x^9y^7 \\
& - 3294453894x^9y^6 + 22442396383x^9y^5 + 105584079574x^9y^4 \\
& + 95533498821x^9y^3 + 28886586407x^9y^2 - 18296748714x^9y - 177308332067x^9
\end{aligned}$$

$$\begin{aligned}
&+524288x^8y^{25} + 6488064x^8y^{24} + 16318464x^8y^{23} + 34766848x^8y^{22} \\
&+119259136x^8y^{21} + 148701184x^8y^{20} + 16564224x^8y^{19} \\
&+59424768x^8y^{18} + 241090560x^8y^{17} + 276105948x^8y^{16} \\
&-31313840x^8y^{15} + 1115434183x^8y^{14} + 5267389218x^8y^{13} \\
&+501818668x^8y^{12} + 41879864x^8y^{11} + 21306106x^8y^{10} + 5054488x^8y^9 \\
&+1718082x^8y^8 - 1092074460x^8y^7 + 47185057547x^8y^6 \\
&+83322998450x^8y^5 + 92960941831x^8y^4 + 110318251758x^8y^3 \\
&+59746462433x^8y^2 - 41791683409x^8y - 131642234434x^8 + 524288x^7y^{25} \\
&+3735552x^7y^{24} + 17858560x^7y^{23} + 66273280x^7y^{22} + 109887488x^7y^{21} \\
&+60391424x^7y^{20} + 55377920x^7y^{19} + 148029440x^7y^{18} \\
&+117029632x^7y^{17} - 360980x^7y^{16} + 1765774824x^7y^{15} \\
&+1021480318x^7y^{14} - 4000976995x^7y^{13} - 3750196x^7y^{12} \\
&+2230804x^7y^{11} + 672946x^7y^{10} + 902322x^7y^9 + 2206797870x^7y^8 \\
&+24747703884x^7y^7 - 22257150436x^7y^6 - 2573895254x^7y^5 \\
&+24184853186x^7y^4 - 1985659968x^7y^3 + 78271640115x^7y^2 \\
&+77834400269x^7y + 91391535089x^7 - 131072x^6y^{26} - 1703936x^6y^{25} \\
&-3899392x^6y^{24} + 10813440x^6y^{23} + 33947648x^6y^{22} - 212992x^6y^{21} \\
&-38764544x^6y^{20} + 13565952x^6y^{19} + 26259456x^6y^{18} \\
&+152968906x^6y^{17} + 748093190x^6y^{16} - 4134694308x^6y^{15} \\
&-4447938051x^6y^{14} - 9505972x^6y^{13} - 3832404x^6y^{12} - 556448x^6y^{11} \\
&-149714x^6y^{10} + 3301516314x^6y^9 - 22433487178x^6y^8 \\
&-105573878552x^6y^7 - 95529800166x^6y^6 - 86134184080x^6y^5 \\
&-70877472728x^6y^4 - 15125658480x^6y^3 + 200333233296x^6y^2 \\
&+233960599495x^6y + 43059913144x^6 - 393216x^5y^{26} - 2981888x^5y^{25} \\
&-3817472x^5y^{24} + 4636672x^5y^{23} - 1081344x^5y^{22} - 28901376x^5y^{21} \\
&-24690688x^5y^{20} + 4374528x^5y^{19} + 199701372x^5y^{18} \\
&-1018525374x^5y^{17} - 5153026114x^5y^{16} - 432215456x^5y^{15} \\
&-393216x^5y^{14} - 1204586x^5y^{13} - 68788x^5y^{12} - 171452x^5y^{11} \\
&+1093266014x^5y^{10} - 47185625624x^5y^9 - 83321375586x^5y^8 \\
&-92956408132x^5y^7 - 110320283206x^5y^6 - 68892664218x^5y^5 \\
&-751165196x^5y^4 + 1045989x^5y^3 + 235206318543x^5y^2 + 95618016653x^5y \\
&-109633942146x^5 - 360448x^4y^{26} - 1130496x^4y^{25} + 344064x^4y^{24} \\
&+688128x^4y^{23} - 7618560x^4y^{22} - 10223616x^4y^{21} + 933888x^4y^{20} \\
&+230527484x^4y^{19} - 3298758079x^4y^{18} - 1305971102x^4y^{17} \\
&+3987770873x^4y^{16} + 851968x^4y^{15} + 368278x^4y^{14} + 29516x^4y^{13} \\
&-149902x^4y^{12} - 1108964478x^4y^{11} - 40831094824x^4y^{10} \\
&+2857938058x^4y^9 + 1039420880x^4y^8 - 24186332516x^4y^7 \\
&+1984931934x^4y^6 + 14374123144x^4y^5 + 1013553x^4y^4 + 84176076x^4y^3 \\
&+204040422984x^4y^2 + 239537999375x^4y + 87572747790x^4 - 32768x^3y^{26} \\
&+458752x^3y^{25} + 1081344x^3y^{24} - 1605632x^3y^{23} - 3719168x^3y^{22} \\
&+606208x^3y^{21} + 228575616x^3y^{20} - 5305220213x^3y^{19} \\
&+3325360132x^3y^{18} + 4394557849x^3y^{17} + 131072x^3y^{16} + 131072x^3y^{15} \\
&-134784x^3y^{13} - 1101379140x^3y^{12} - 26660491026x^3y^{11} \\
&+45195280416x^3y^{10} + 57615008720x^3y^9 + 83326068784x^3y^8 \\
&+70877596262x^3y^7 + 15125177760x^3y^6 + 1404x^3y^5 + 59205508x^3y^4 \\
&+24412755553x^3y^3 + 356634859410x^3y^2 + 676340726311x^3y \\
&+720284211275x^3 + 98304x^2y^{26} + 376832x^2y^{25} + 65536x^2y^{24} \\
&-1032192x^2y^{23} - 688128x^2y^{22} + 933888x^2y^{21} - 3524985348x^2y^{20} \\
&+4376081884x^2y^{19} + 397144736x^2y^{18} + 4547178x^2y^{13} \\
&-2767168560x^2y^{12} + 15069207370x^2y^{11} + 11945437084x^2y^{10} \\
&+68790183664x^2y^9 + 65926004200x^2y^8 + 751956480x^2y^7 - 2187x^2y^6 \\
&-4536x^2y^5 + 97607001534x^2y^4 + 252168719314x^2y^3 \\
&+335314047180x^2y^2 + 737572084154x^2y + 609743058439x^2 + 32768xy^{26} \\
&+65536xy^{25} - 98304xy^{24} - 196608xy^{23} + 98304xy^{22} - 2046189561xy^{21} \\
&+5427954862xy^{20} - 3599465561xy^{19} + 4885596xy^{14} + 19984746054xy^{13} \\
&-10711464394xy^{12} - 9719138868xy^{11} - 10349890078xy^{10} + 4231151810xy^9 \\
&-13621264800xy^8 - 2187xy^7 - 23862276xy^6 - 48845613999xy^5 - 77208024752xy^4 \\
&-82369200439xy^3 - 118255370387xy^2 - 10311306448xy + 16903863587x \\
&-2046386169y^{22} + 5427987630y^{21} - 3599400025y^{20} + 2442798y^{15} \\
&+19985452512y^{14} - 5700821144y^{13} - 2845884038y^{12} + 12275117150y^{11} \\
&-49545006470y^{10} - 13621264800y^9 - 729y^8 - 11930382y^7 - 48823810435y^6 \\
&-101643511346y^5 - 173811275819y^4 - 346123873254y^3 - 318021860594y^2 \\
&-320683377940y - 274325391345
\end{aligned}$$

$i = 11, m = 2$, neighbours 1, 12

$W \quad 2x^2 - xy + y - 2$

$A \quad 0$

$B \quad -3xy + 6x - 3y$

$i = 12, m = 192$, neighbours 8, 11, 41, 63, 85, 115, 131, 132, 133, 134, 135, 136, 137, 200

$W \quad x^5y^2 - x^4y - 2x^4 - 3x^3y^2 - 8472x^3y - 14360x^3 - 6x^2y^3$
 $-8474x^2y^2 + 1232x^2y - 503x^2 - 6xy^4 + 3xy^3 + 15583xy^2 - 38443xy - 5014x$
 $+3y^5 + 8484y^4 + 14332y^3 - 37918y^2 + 100260y - 16224$

$A \quad 663552x^{23} - 995328x^{22}y - 4644864x^{22} + 9787392x^{21}y + 17307648x^{21}$
 $-45370368x^{20}y - 74096640x^{20} + 144281088x^{19}y + 345835008x^{19}$
 $-377761536x^{18}y - 1250629632x^{18} + 831517056x^{17}y + 3381008256x^{17}$
 $-1434905280x^{16}y - 7108646400x^{16} + 1727693280x^{15}y + 11771261280x^{15}$
 $-771222384x^{14}y - 15131546496x^{14} - 2554692264x^{13}y + 14687016120x^{13}$
 $+8294399028x^{12}y - 9970000992x^{12} - 13972385682x^{11}y + 2612517030x^{11}$
 $+15789670497x^{10}y + 5446801800x^{10} - 12304588486x^9y - 12741257919x^9$
 $+5729081824x^8y + 17679172277x^8 + 1467166421x^7y - 18038126676x^7$
 $-8664342734x^6y + 13265432242x^6 + 13658551067x^5y - 6371035700x^5$
 $-14391948482x^4y + 516489413x^4 + 12628428671x^3y + 1447220999x^3$
 $-8060724556x^2y - 622655627x^2 + 3584215726xy - 387833202x - 1461915450y + 692101175$

$B \quad -1327104x^{25} + 663552x^{24}y + 7962624x^{24} + 1990656x^{23}y^2 - 12275712x^{23}y$
 $-25325568x^{23} + 1327104x^{22}y^3 - 17584128x^{22}y^2 + 60438528x^{22}y$
 $+114241536x^{22} - 663552x^{21}y^4 - 2654208x^{21}y^3 + 53913600x^{21}y^2$
 $-126600192x^{21}y - 549448704x^{21} - 1990656x^{20}y^5 + 16588800x^{20}y^4$
 $-23445504x^{20}y^3 - 107910144x^{20}y^2 + 131120640x^{20}y + 1834195968x^{20}$
 $-1327104x^{19}y^6 + 16588800x^{19}y^5 - 72548352x^{19}y^4 + 34504704x^{19}y^3$
 $+382358016x^{19}y^2 - 121865472x^{19}y - 4356198144x^{19} + 663552x^{18}y^7$
 $-3760128x^{18}y^6 - 10100736x^{18}y^5 + 62668800x^{18}y^4 + 294543360x^{18}y^3$
 $-1302868224x^{18}y^2 + 164287872x^{18}y + 7903353600x^{18} + 2211840x^{17}y^8$
 $-19390464x^{17}y^7 + 47849472x^{17}y^6 + 52494336x^{17}y^5$
 $-301916160x^{17}y^4 - 654557184x^{17}y^3 + 2463829632x^{17}y^2 - 4709952x^{17}y$
 $-11069846208x^{17} - 5234688x^{16}y^8 + 69525504x^{16}y^7 - 278470656x^{16}y^6$
 $+310984704x^{16}y^5 + 70557696x^{16}y^4 + 1752367104x^{16}y^3$
 $-3676140864x^{16}y^2 - 246251808x^{16}y + 11660773824x^{16} + 221184x^{15}y^{10}$
 $-7667712x^{15}y^9 + 21897216x^{15}y^8 + 115089408x^{15}y^7$
 $-701964288x^{15}y^6 + 1637056512x^{15}y^5 - 1482301440x^{15}y^4$
 $-3253764096x^{15}y^3 + 4462509600x^{15}y^2 + 903247920x^{15}y - 9677790576x^{15}$
 $-884736x^{14}y^{11} + 21676032x^{14}y^{10} - 119439360x^{14}y^9$
 $+157040640x^{14}y^8 + 129466368x^{14}y^7 + 674906112x^{14}y^6$
 $-4570767360x^{14}y^5 + 6949601280x^{14}y^4 - 55148544x^{14}y^3$
 $-554996880x^{14}y^2 - 3842604648x^{14}y + 8156115504x^{14} - 1327104x^{13}y^{12}$
 $+7004160x^{13}y^{11} - 11722752x^{13}y^{10} + 134995968x^{13}y^9$
 $-472006656x^{13}y^8 - 645414912x^{13}y^7 + 4612055040x^{13}y^6$
 $-6431072256x^{13}y^5 + 4459290624x^{13}y^4 - 6012813312x^{13}y^3$
 $-771399000x^{13}y^2 + 5706903996x^{13}y - 7573365324x^{13} + 1105920x^{12}y^{13}$
 $-6930432x^{12}y^{12} - 2285568x^{12}y^{11} - 33251328x^{12}y^{10}$
 $+629194752x^{12}y^9 - 1463721984x^{12}y^8 + 277880832x^{12}y^7$
 $+1627987968x^{12}y^6 + 1373626368x^{12}y^5 - 10884390912x^{12}y^4$
 $+17546895360x^{12}y^3 - 8028388404x^{12}y^2 - 1156393026x^{12}y + 4084538940x^{12}$
 $-2727936x^{11}y^{13} + 44236800x^{11}y^{12} - 155271168x^{11}y^{11}$
 $-86040576x^{11}y^{10} + 726884352x^{11}y^9 + 672841728x^{11}y^8$
 $-2496872448x^{11}y^7 - 5077647360x^{11}y^6 + 18038292480x^{11}y^5$
 $-14148108288x^{11}y^4 - 510418944x^{11}y^3 + 7220449170x^{11}y^2$
 $-1238706550x^{11}y + 3049372734x^{11} - 1327104x^{10}y^{14} + 884736x^{10}y^{13}$
 $+28090368x^{10}y^{12} + 57655296x^{10}y^{11} - 320421888x^{10}y^{10}$
 $-1052024832x^{10}y^9 + 3953516544x^{10}y^8 - 1009262592x^{10}y^7$
 $-6475014144x^{10}y^6 + 4510973952x^{10}y^5 + 3551772672x^{10}y^4$
 $-10057605120x^{10}y^3 + 2644909982x^{10}y^2 - 5708053284x^{10}y - 9096024814x^{10}$
 $+1105920x^9y^{15} - 6266880x^9y^{14} - 8110080x^9y^{13} + 4128768x^9y^{12}$
 $+360824832x^9y^{11} - 424747008x^9y^{10} - 1399947264x^9y^9$
 $+1484218368x^9y^8 + 3324837888x^9y^7 - 2121523200x^9y^6$
 $-11781881856x^9y^5 + 20987265024x^9y^4 - 6936625152x^9y^3$
 $+508403398x^9y^2 + 10086461351x^9y + 10646502914x^9 - 516096x^8y^{15}$

$$\begin{aligned}
& +18432000x^8y^{14} - 62816256x^8y^{13} - 106905600x^8y^{12} \\
& +362446848x^8y^{11} + 689946624x^8y^{10} - 1018478592x^8y^9 \\
& -3798392832x^8y^8 + 5199593472x^8y^7 + 5563293696x^8y^6 \\
& -10608279552x^8y^5 - 498180096x^8y^4 + 3429236736x^8y^3 \\
& -5058820552x^8y^2 - 5640545386x^8y - 10387484237x^8 - 1105920x^7y^{16} \\
& +3907584x^7y^{15} + 8036352x^7y^{14} + 9953280x^7y^{13} - 79257600x^7y^{12} \\
& -479232000x^7y^{11} + 1156276224x^7y^{10} + 1199259648x^7y^9 \\
& -3560325120x^7y^8 - 967311360x^7y^7 + 4631445504x^7y^6 \\
& +1019731968x^7y^5 - 7639990272x^7y^4 + 10216194072x^7y^3 \\
& -290671753x^7y^2 + 1206764191x^7y + 12252685729x^7 - 368640x^6y^{16} \\
& -2506752x^6y^{15} - 12312576x^6y^{14} + 119513088x^6y^{13} - 4276224x^6y^{12} \\
& -660750336x^6y^{11} + 102924288x^6y^{10} + 1876672512x^6y^9 \\
& +33103872x^6y^8 - 4337123328x^6y^7 + 969596928x^6y^6 \\
& +6634045440x^6y^5 - 3073572858x^6y^4 - 7642071636x^6y^3 \\
& +4862521360x^6y^2 + 1097648307x^6y - 13392199019x^6 + 2949120x^5y^{16} \\
& -11354112x^5y^{15} - 21528576x^5y^{14} + 66207744x^5y^{13} \\
& +129392640x^5y^{12} - 16588800x^5y^{11} - 995696640x^5y^{10} \\
& +264019968x^5y^9 + 2598469632x^5y^8 - 1420886016x^5y^7 \\
& -2215010304x^5y^6 + 463085562x^5y^5 - 533586510x^5y^4 \\
& +4271459626x^5y^3 - 6099544793x^5y^2 - 4233332897x^5y + 12309563600x^5 \\
& +737280x^4y^{16} + 6561792x^4y^{15} - 10616832x^4y^{14} - 106831872x^4y^{13} \\
& +144801792x^4y^{12} + 481517568x^4y^{11} - 715972608x^4y^{10} \\
& -889159680x^4y^9 + 1683726336x^4y^8 + 498843648x^4y^7 \\
& -2612846592x^4y^6 + 1440350208x^4y^5 + 3779886957x^4y^4 \\
& -4446404406x^4y^3 + 7979732214x^4y^2 + 8546836504x^4y - 6684398765x^4 \\
& -2580480x^3y^{16} + 6561792x^3y^{15} + 29270016x^3y^{14} - 69009408x^3y^{13} \\
& -143990784x^3y^{12} + 300662784x^3y^{11} + 478715904x^3y^{10} \\
& -946741248x^3y^9 - 911794176x^3y^8 + 2106187776x^3y^7 \\
& +954408960x^3y^6 - 2940715053x^3y^5 - 1291437093x^3y^4 \\
& +3118106100x^3y^3 - 7778943081x^3y^2 - 8945825795x^3y + 363855510x^3 \\
& -368640x^2y^{16} - 3538944x^2y^{15} + 5824512x^2y^{14} + 48144384x^2y^{13} \\
& -54779904x^2y^{12} - 237699072x^2y^{11} + 255098880x^2y^{10} \\
& +573235200x^2y^9 - 560037888x^2y^8 - 829587456x^2y^7 + 534306852x^2y^6 \\
& +921412191x^2y^5 + 169538886x^2y^4 - 915808616x^2y^3 \\
& +4683443205x^2y^2 + 7952940868x^2y + 2523042702x^2 + 737280xy^{16} \\
& -221184xy^{15} - 9510912xy^{14} + 3686400xy^{13} + 46817280xy^{12} - 17399808xy^{11} \\
& -117006336xy^{10} + 29196288xy^9 + 164487168xy^8 + 3760155xy^7 - 128235825xy^6 \\
& -145668159xy^5 - 699815679xy^4 - 473647014xy^3 - 893555148xy^2 - 8907868999xy \\
& -2030817542x - 9y^8 - 50913y^7 - 72115152y^6 - 315022107y^5 + 194477885y^4 \\
& -104732792y^3 - 2873633759y^2 + 4607359954y - 2339521701
\end{aligned}$$

$i = 13, m = 4$, neighbours 23, 43, 76, 80

$$W \quad 2x^3 + 2x^2 + 6x + 6$$

$$A \quad 8x^8 + 48x^6 + 12x^5 + 100x^4 + 24x^3 + 120x^2 - 36x + 108$$

$$\begin{aligned}
B \quad & 16x^{11} - 32x^{10} + 16x^9y^2 + 64x^9y + 32x^9 - 16x^8y^3 + 8x^8y^2 \\
& +40x^8y - 96x^8 - 16x^7y^3 + 48x^7y^2 + 176x^7y + 16x^7 - 32x^6y^3 \\
& +48x^6y^2 - 240x^6 + 32x^5y^3 + 220x^5y^2 + 300x^5y + 56x^5 - 16x^4y^5 \\
& -64x^4y^4 - 160x^4y^3 - 60x^4y^2 - 236x^4y - 440x^4 + 16x^3y^5 \\
& +48x^3y^4 + 128x^3y^3 + 184x^3y^2 + 24x^3y - 160x^3 + 48x^2y^4 \\
& +32x^2y^3 + 8x^2y^2 - 248x^2y - 320x^2 - 16xy^6 + 16xy^5 + 112xy^4 \\
& +208xy^3 + 172xy^2 + 76xy - 40x - 16y^7 - 64y^6 - 144y^5 - 240y^4 - 336y^3 \\
& -324y^2 - 404y - 344
\end{aligned}$$

$i = 14, m = 2$, neighbours 1, 15

$$W \quad 2x^2 - xy + 4x - 2y$$

$$A \quad 0$$

$$B \quad 3xy + 6x + 6$$

$i = 15, m = 96$, neighbours 14, 47, 48, 64, 77, 89, 92, 104, 105, 140, 141, 142, 145, 146, 201, 205

$$\begin{aligned}
W \quad & x^8y^2 + 6x^7y^5 - x^7y^4 + 6x^6y^6 - x^6y^4 - 18x^6 + 4x^5y^8 \\
& +x^5y^4 + x^5y^3 - 41x^5y + 1459x^5 - 4x^4y^9 - 14x^4y^8 + x^4y^3 \\
& -24x^4y^2 - 25086x^4y + 1032727x^4 + 2278x^3y^9 - 51222x^3y^8 + x^3y^4 \\
& +73x^3y^3 - 28817x^3y^2 - 208743359x^3y + 8470692870x^3 + 2330x^2y^{10} \\
& -51222x^2y^9 + 157x^2y^4 + 148007x^2y^3 - 215978942x^2y^2 \\
& +6440666861x^2y - 7077895934x^2 - 32254xy^{11} - 22xy^5 + 335373xy^4
\end{aligned}$$

$$\begin{aligned}
& -11951548xy^3 - 2024837951xy^2 - 7378933983xy - 22660415951x + 33118y^{12} \\
& + 119212y^{11} - 461y^6 + 197633y^5 + 205510480y^4 - 8460097689y^3 - 284438698y^2 \\
& - 14714511629y - 13240742106 \\
A \quad & -7299072x^{25} - 1327104x^{24}y - 214990848x^{24} - 50761728x^{23}y - 2972381184x^{23} \\
& - 867760128x^{22}y - 25679545336x^{22} - 8952062984x^{21}y - 155964874676x^{21} \\
& - 63198548516x^{20}y - 711953073750x^{20} - 328912863506x^{19}y - 2553822598783x^{19} \\
& - 1341332160811x^{18}y - 7360539175201x^{18} - 4616528798199x^{17}y \\
& - 16097392806518x^{17} - 14917010571307x^{16}y - 12551129288308x^{16} \\
& - 49719715432709x^{15}y + 143057212479043x^{15} - 163982091158688x^{14}y \\
& + 1222091277958290x^{14} - 412221455930070x^{13}y + 6368736899222151x^{13} \\
& + 12854682281791x^{12}y + 24607929040210671x^{12} + 8060407768086681x^{11}y \\
& + 69784341352300510x^{11} + 55738701211114885x^{10}y + 124681895957867410x^{10} \\
& + 239219490233590083x^9y + 5531758998263362x^9 + 728999961355201237x^8y \\
& - 821715713624997021x^8 + 1478095291706284525x^7y - 2636213675245119879x^7 \\
& + 1242815043422388895x^6y - 2121326340662319005x^6 - 2883354963473967622x^5y \\
& + 13650089438603560536x^5 - 11915076883470836879x^4y + 70244481036556045488x^4 \\
& - 17492419118735879183x^3y + 194334816321104134338x^3 \\
& + 20521689679521106235x^2y + 211809703124627336816x^2 + 85320022718186684393xy \\
& + 94126718204747883149x + 53767012482087411098y - 31200188917680944214 \\
B \quad & 14598144x^{27} + 17252352x^{26}y + 386187264x^{26} + 2654208x^{25}y^2 \\
& + 435953664x^{25}y + 4742406144x^{25} - 14598144x^{24}y^3 + 41803776x^{24}y^2 \\
& + 5118308352x^{24}y + 35980609520x^{24} - 17252352x^{23}y^4 - 343056384x^{23}y^3 \\
& + 242528256x^{23}y^2 + 37230575616x^{23}y + 189997590424x^{23} - 2654208x^{22}y^5 \\
& - 331776000x^{22}y^4 - 3587825664x^{22}y^3 + 398297104x^{22}y^2 \\
& + 189617769440x^{22}y + 749588901396x^{22} + 14598144x^{21}y^6 + 19243008x^{21}y^5 \\
& - 2732507136x^{21}y^4 - 22067251184x^{21}y^3 - 2251535336x^{21}y^2 \\
& + 730822019736x^{21}y + 2323482410994x^{21} + 17252352x^{20}y^7 \\
& + 307224576x^{20}y^6 + 805552128x^{20}y^5 - 12404938752x^{20}y^4 \\
& - 89224528896x^{20}y^3 - 12299406836x^{20}y^2 + 2281942253792x^{20}y \\
& + 5734867923916x^{20} + 2654208x^{19}y^8 + 236224512x^{19}y^7 \\
& + 2572591104x^{19}y^6 + 7187927024x^{19}y^5 - 32608592912x^{19}y^4 \\
& - 253321211656x^{19}y^3 + 11064732298x^{19}y^2 + 6175532311874x^{19}y \\
& + 9199582306084x^{19} - 13123584x^{18}y^9 - 62447616x^{18}y^8 \\
& + 1082474496x^{18}y^7 + 11138254832x^{18}y^6 + 32893325344x^{18}y^5 \\
& - 42574969536x^{18}y^4 - 531194181352x^{18}y^3 + 507587898840x^{18}y^2 \\
& + 14891643103976x^{18}y - 15011410293351x^{18} - 13860864x^{17}y^{10} \\
& - 228114432x^{17}y^9 - 1073627136x^{17}y^8 + 970278912x^{17}y^7 \\
& + 26643704928x^{17}y^6 + 93514679320x^{17}y^5 + 14373751824x^{17}y^4 \\
& - 759840198427x^{17}y^3 + 4465066640304x^{17}y^2 + 26108362607109x^{17}y \\
& - 253656937563679x^{17} - 1548288x^{16}y^{11} - 111624192x^{16}y^{10} \\
& - 1235957760x^{16}y^9 - 5473880048x^{16}y^8 - 7448234944x^{16}y^7 \\
& + 33935458792x^{16}y^6 + 180134501968x^{16}y^5 + 166308083784x^{16}y^4 \\
& + 807525950450x^{16}y^3 + 27541863590143x^{16}y^2 - 32692134956301x^{16}y \\
& - 159998535633337x^{16} + 8146944x^{15}y^{12} + 78833664x^{15}y^{11} \\
& + 130369536x^{15}y^{10} - 1521349616x^{15}y^9 - 10440430672x^{15}y^8 \\
& - 25842212592x^{15}y^7 + 14137184104x^{15}y^6 + 136914072619x^{15}y^5 \\
& + 423538446641x^{15}y^4 + 17374164173547x^{15}y^3 + 129550747350268x^{15}y^2 \\
& - 710942405667439x^{15}y - 7115664164927408x^{15} + 6119424x^{14}y^{13} \\
& + 112140288x^{14}y^{12} + 825246720x^{14}y^{11} + 3322303488x^{14}y^{10} \\
& + 6991395648x^{14}y^9 - 428541976x^{14}y^8 - 34389471960x^{14}y^7 \\
& - 120695603213x^{14}y^6 - 1299871023666x^{14}y^5 + 2944665487787x^{14}y^4 \\
& + 120182610279398x^{14}y^3 + 432664263048580x^{14}y^2 - 4803452271613888x^{14}y \\
& - 23174273721072778x^{14} - 221184x^{13}y^{14} + 23851008x^{13}y^{13} \\
& + 372077568x^{13}y^{12} + 2549735408x^{13}y^{11} + 10158079888x^{13}y^{10} \\
& + 22736524520x^{13}y^9 + 20402112248x^{13}y^8 - 19416145644x^{13}y^7 \\
& - 1351336718706x^{13}y^6 - 10959850076971x^{13}y^5 + 30099615869787x^{13}y^4 \\
& + 587722986070333x^{13}y^3 + 675236498367861x^{13}y^2 - 22581822059226563x^{13}y \\
& - 50350130944770665x^{13} - 2469888x^{12}y^{15} - 33527808x^{12}y^{14} \\
& - 188743680x^{12}y^{13} - 433548304x^{12}y^{12} + 712592512x^{12}y^{11} \\
& + 6136399404x^{12}y^{10} + 12249455972x^{12}y^9 + 96685528180x^{12}y^8 \\
& + 123859751436x^{12}y^7 - 9566494320727x^{12}y^6 - 53498024420468x^{12}y^5 \\
& + 219607352472903x^{12}y^4 + 2132670069672250x^{12}y^3 \\
& - 2567295290130784x^{12}y^2 - 79989620293359814x^{12}y - 38825586401749640x^{12}
\end{aligned}$$

$$\begin{aligned}
& -847872x^{11}y^{16} - 17565696x^{11}y^{15} - 164551680x^{11}y^{14} \\
& -959993856x^{11}y^{13} - 3635306208x^{11}y^{12} - 9714378836x^{11}y^{11} \\
& -24322234052x^{11}y^{10} + 25143094408x^{11}y^9 + 1020549714232x^{11}y^8 \\
& + 783799244148x^{11}y^7 - 51266260108918x^{11}y^6 - 175135979534403x^{11}y^5 \\
& + 1120237638094993x^{11}y^4 + 5484519321139913x^{11}y^3 \\
& - 23551029280379692x^{11}y^2 - 214695137184066231x^{11}y + 212749495910289355x^{11} \\
& + 442368x^{10}y^{17} + 4349952x^{10}y^{16} + 8994816x^{10}y^{15} \\
& - 121835504x^{10}y^{14} - 1189299088x^{10}y^{13} - 5954149836x^{10}y^{12} \\
& - 26009767076x^{10}y^{11} - 89290661372x^{10}y^{10} + 546262270300x^{10}y^9 \\
& + 6460974864572x^{10}y^8 - 1225032048768x^{10}y^7 - 212777333931615x^{10}y^6 \\
& - 326612677930812x^{10}y^5 + 4019623087971028x^{10}y^4 \\
& + 8979349705202334x^{10}y^3 - 97527274753236472x^{10}y^2 \\
& - 393903031588951213x^{10}y + 978051568220198786x^{10} + 147456x^9y^{18} \\
& + 3170304x^9y^{17} + 31730688x^9y^{16} + 163860496x^9y^{15} \\
& + 368434976x^9y^{14} - 234403368x^9y^{13} - 8398986224x^9y^{12} \\
& - 116700659048x^9y^{11} - 400077543804x^9y^{10} + 3460348018064x^9y^9 \\
& + 26905283634336x^9y^8 - 36765358111208x^9y^7 - 654342460667388x^9y^6 \\
& + 41414714030228x^9y^5 + 9051401671154652x^9y^4 + 3992439463752782x^9y^3 \\
& - 251477923992660401x^9y^2 - 302048790102702878x^9y + 1365989232574144295x^9 \\
& - 36864x^8y^{19} - 497664x^8y^{18} + 1013760x^8y^{17} + 52402160x^8y^{16} \\
& + 331333296x^8y^{15} + 1493544948x^8y^{14} + 6632524968x^8y^{13} \\
& - 38377874620x^8y^{12} - 661660532880x^8y^{11} - 1782756408516x^8y^{10} \\
& + 15881856022780x^8y^9 + 75993736363656x^8y^8 - 212925727774786x^8y^7 \\
& - 1375027618569180x^8y^6 + 929577180716408x^8y^5 + 6569171555731795x^8y^4 \\
& - 21171874241568374x^8y^3 - 288823529798655263x^8y^2 \\
& + 437097070747598724x^8y - 3169047171731643814x^8 - 184320x^7y^{19} \\
& - 3419136x^7y^{18} - 16229376x^7y^{17} + 27852848x^7y^{16} \\
& + 1319606748x^7y^{15} + 12268928928x^7y^{14} + 59687216848x^7y^{13} \\
& - 138748909848x^7y^{12} - 2940914588760x^7y^{11} - 4400978754936x^7y^{10} \\
& + 55198898006888x^7y^9 + 133786251238970x^7y^8 - 665698994322188x^7y^7 \\
& - 2031823910100170x^7y^6 - 334527006706487x^7y^5 \\
& - 55774672556674243x^7y^4 + 59231610206216947x^7y^3 \\
& + 452792553007804664x^7y^2 + 740327682857942538x^7y - 21085884319026157742x^7 \\
& + 36864x^6y^{20} - 9216x^6y^{19} - 4819968x^6y^{18} - 31110688x^6y^{17} \\
& + 70016036x^6y^{16} + 1982963296x^6y^{15} + 62228291656x^6y^{14} \\
& + 335974674368x^6y^{13} - 611463925688x^6y^{12} - 9409801362044x^6y^{11} \\
& - 1943031061364x^6y^{10} + 134200005257684x^6y^9 + 80932960586310x^6y^8 \\
& - 324601927689280x^6y^7 + 3317942703451065x^6y^6 \\
& - 36387178731843820x^6y^5 - 233328632055469435x^6y^4 \\
& + 723231626812229070x^6y^3 + 2688467717159937584x^6y^2 \\
& - 3631382674240879627x^6y - 62622832818883267891x^6 + 73728x^5y^{20} \\
& + 626688x^5y^{19} - 764912x^5y^{18} - 835764912x^5y^{17} - 10356624888x^5y^{16} \\
& - 8718033888x^5y^{15} + 249209438956x^5y^{14} + 1142505057008x^5y^{13} \\
& - 2764122987192x^5y^{12} - 20065868506008x^5y^{11} + 19645417514264x^5y^{10} \\
& + 178805737834544x^5y^9 + 1063752681074486x^5y^8 + 6623291843586316x^5y^7 \\
& + 15529557831866452x^5y^6 - 182395052916995587x^5y^5 \\
& - 340346675909967821x^5y^4 + 2236963788154708186x^5y^3 \\
& + 6552616644058644010x^5y^2 - 22004880325383219165x^5y \\
& - 127530989849664465579x^5 - 9216x^4y^{20} + 790256x^4y^{19} \\
& - 892948456x^4y^{18} - 9138961624x^4y^{17} - 42254754260x^4y^{16} \\
& - 69222600960x^4y^{15} + 812530587912x^4y^{14} + 2203292189004x^4y^{13} \\
& - 8018322414932x^4y^{12} - 26164719961188x^4y^{11} + 45810478623280x^4y^{10} \\
& + 389766541592080x^4y^9 + 738674958129984x^4y^8 + 33020321075175384x^4y^7 \\
& - 5608814186008642x^4y^6 - 407231562882455826x^4y^5 \\
& + 26706872007242458x^4y^4 + 3424714925059878726x^4y^3 \\
& + 11609573008745698696x^4y^2 - 86866581134355101077x^4y \\
& + 67817998355641013131x^4 - 73728x^3y^{20} - 98467304x^3y^{19} \\
& + 4337936552x^3y^{18} - 8533330724x^3y^{17} - 159907026756x^3y^{16} \\
& - 98235906396x^3y^{15} + 1776937892944x^3y^{14} + 2013191594304x^3y^{13} \\
& - 13467013277640x^3y^{12} + 7761295744332x^3y^{11} - 982771395249280x^3y^{10} \\
& - 6030830898846012x^3y^9 + 11357479901792502x^3y^8 \\
& + 57855142078161899x^3y^7 - 96646937243832989x^3y^6 \\
& - 441376810465700133x^3y^5 + 696415755193131995x^3y^4
\end{aligned}$$

$$\begin{aligned}
& +2886971700140945011x^3y^3 - 4587527226435203480x^3y^2 \\
& +118652786153249459680x^3y - 353003351646713724x^3 + 941697900x^2y^{20} \\
& +10698968584x^2y^{19} + 20705055372x^2y^{18} - 37533428304x^2y^{17} \\
& -303754576668x^2y^{16} + 74483779824x^2y^{15} + 2259495371504x^2y^{14} \\
& +55578338956x^2y^{13} + 13943180233000x^2y^{12} - 815518782790204x^2y^{11} \\
& -3439733943230040x^2y^{10} - 6117969698692060x^2y^9 \\
& +24853465429237206x^2y^8 + 37958189109294908x^2y^7 \\
& -161316150397128103x^2y^6 - 182262615516197180x^2y^5 \\
& +962920661493064245x^2y^4 - 4742125152288158470x^2y^3 \\
& +28365681219158499383x^2y^2 - 13501062961415667433x^2y \\
& -211233281661360646519x^2 + 1039574020xy^{21} + 3084407436xy^{20} \\
& +9648560864xy^{19} + 47198846808xy^{18} - 67247973732xy^{17} - 310966571772xy^{16} \\
& +316656439904xy^{15} + 1418629598952xy^{14} - 946374966428xy^{13} \\
& +137823108995032xy^{12} + 1499743648477504xy^{11} - 4820336430383316xy^{10} \\
& -323808432002817xy^9 + 24615747143150931xy^8 - 7379600689295663xy^7 \\
& -104536297537332581xy^6 + 19948182239198691xy^5 + 4632432993432174885xy^4 \\
& -113117529203151136244xy^3 - 11716211829950509823xy^2 - 257869589891272538650xy \\
& -256514909391921229417x - 1096801924y^{22} - 7896126032y^{21} - 10921095172y^{20} \\
& +23688378096y^{19} + 32763396108y^{18} - 71065134288y^{17} - 98262971288y^{16} \\
& +200214675756y^{15} - 13364487194448y^{14} + 510763760905048y^{13} \\
& +2076023700514400y^{12} - 489843924552204y^{11} - 1842764276005305y^{10} \\
& +4626442906337966y^9 + 5528443183299402y^8 - 13968361167646132y^7 \\
& -55476160172398064y^6 + 3519281420705836862y^5 - 71284110222467332715y^4 \\
& -9317396294925388058y^3 - 83525776310304328839y^2 - 173254318639745063476y \\
& -27238894921519494198
\end{aligned}$$

$i = 16, m = 1$, neighbours 70

$$\begin{aligned}
W & x \\
A & 0 \\
B & x + y + 1
\end{aligned}$$

$i = 17, m = 2$, neighbours 1, 18

$$\begin{aligned}
W & 6x^2 + 3x + 3y \\
A & 0 \\
B & -6xy + 9x - 3y
\end{aligned}$$

$i = 18, m = 486$, neighbours 17, 19, 37, 46, 50, 67, 105, 144, 146, 150, 151, 175, 176, 178, 179, 217

$$\begin{aligned}
W & -30485376x^4 - 3084480x^3y^7 - 464832x^3y^6 + 2650176x^3y^5 \\
& -15113088x^3y^4 + 28567296x^3y^3 - 7095168x^3y^2 - 38563776x^3y \\
& +119746080x^3 + 10161792xy^2 + 112669920x + 3568320y^9 - 2753280y^8 + 8616288y^7 \\
& -3312864y^6 - 5211360y^5 + 25928064y^4 - 30248640y^3 - 13524192y^2 + 18930240y \\
& +5329152 \\
A & -4081466880x^{13}y^9 - 14693280768x^{13}y^8 + 47753162496x^{13}y^7 \\
& +143259487488x^{13}y^6 - 285141479904x^{13}y^5 - 396718580736x^{13}y^4 \\
& +818232072768x^{13}y^3 - 334731302496x^{13}y^2 - 2720977920x^{12}y^{11} \\
& -10611813888x^{12}y^{10} + 53467216128x^{12}y^9 + 154891668096x^{12}y^8 \\
& -427941802368x^{12}y^7 - 575792940096x^{12}y^6 + 1508357103840x^{12}y^5 \\
& +99179645184x^{12}y^4 - 967001540544x^{12}y^3 - 223154201664x^{12}y^2 \\
& +334731302496x^{12}y + 2176782336x^{11}y^{12} + 42356556288x^{11}y^{11} \\
& +47072918016x^{11}y^{10} - 449981723520x^{11}y^9 - 454267263744x^{11}y^8 \\
& +1895433219072x^{11}y^7 + 1195665722496x^{11}y^6 - 4016775629952x^{11}y^5 \\
& +347128758144x^{11}y^4 + 1487694677760x^{11}y^3 + 9916452864x^{10}y^{13} \\
& +9251324928x^{10}y^{12} - 206839671552x^{10}y^{11} - 250942188672x^{10}y^{10} \\
& +1420350474240x^{10}y^9 + 1146994229952x^{10}y^8 - 4627924276896x^{10}y^7 \\
& -41324852160x^{10}y^6 + 3246067137168x^{10}y^5 + 1562079411648x^{10}y^4 \\
& -2677850419968x^{10}y^3 + 223154201664x^{10}y^2 + 502096953744x^{10}y \\
& -5079158784x^9y^{14} - 73224539136x^9y^{13} - 30248204544x^9y^{12} \\
& +743847338880x^9y^{11} + 434540173824x^9y^{10} - 2808916525152x^9y^9 \\
& -2084226571440x^9y^8 + 6560090697888x^9y^7 + 1732200053040x^9y^6 \\
& -5368098295584x^9y^5 - 874020623184x^9y^4 + 1227348109152x^9y^3 \\
& +725251155408x^9y^2 - 11972302848x^8y^{15} + 17746822656x^8y^{14} \\
& +234049450752x^8y^{13} - 29817383040x^8y^{12} - 1353561803712x^8y^{11} \\
& -658646717760x^8y^{10} + 4759143437088x^8y^9 + 1827400268016x^8y^8 \\
& -7538571364032x^8y^7 - 2746036426032x^8y^6 + 5833002882384x^8y^5 \\
& +576481687632x^8y^4 - 334731302496x^8y^3 - 278942752080x^8y^2
\end{aligned}$$

$$\begin{aligned}
& -502096953744x^8y + 3627970560x^7y^{16} + 42265857024x^7y^{15} \\
& -20724781824x^7y^{14} - 554746931712x^7y^{13} - 88896616128x^7y^{12} \\
& + 2809613775744x^7y^{11} + 1508901299424x^7y^{10} - 7810141966560x^7y^9 \\
& - 335435355328x^7y^8 + 7080095087568x^7y^7 + 6215257764864x^7y^6 \\
& - 1875115166760x^7y^5 - 4686238234944x^7y^4 + 223154201664x^7y^3 \\
& - 111577100832x^7y^2 - 251048476872x^7y + 5079158784x^6y^{17} \\
& - 19228243968x^6y^{16} - 134930271744x^6y^{15} + 136086687360x^6y^{14} \\
& + 963532293696x^6y^{13} + 229786585344x^6y^{12} - 3492097393824x^6y^{11} \\
& - 3820601624976x^6y^{10} + 6848522860464x^6y^9 + 7621527178368x^6y^8 \\
& - 2940033648672x^6y^7 - 6434279481312x^6y^6 - 3077668364616x^6y^5 \\
& + 1115771008320x^6y^4 + 3291524474544x^6y^3 - 948405357072x^6y^2 \\
& - 753145430616x^6y - 725594112x^5y^{18} - 10611813888x^5y^{17} \\
& + 30051689472x^5y^{16} + 253421301888x^5y^{15} - 14402287296x^5y^{14} \\
& - 1419422696352x^5y^{13} - 1171208099088x^5y^{12} + 3069383553720x^5y^{11} \\
& + 4660644036384x^5y^{10} + 56732389632x^5y^9 - 9117410299056x^5y^8 \\
& - 3980960758080x^5y^7 + 4824676489680x^5y^6 + 4751324877096x^5y^5 \\
& + 409116036384x^5y^4 - 2036282090184x^5y^3 - 948405357072x^5y^2 \\
& + 251048476872x^5y - 302330880x^4y^{19} + 4323331584x^4y^{18} \\
& + 43172849664x^4y^{17} - 43369364736x^4y^{16} - 454724539200x^4y^{15} \\
& + 142256126880x^4y^{14} + 1220634474048x^4y^{13} + 314343808560x^4y^{12} \\
& - 340458110712x^4y^{11} - 2642953878144x^4y^{10} - 4320130655808x^4y^9 \\
& + 3967568444880x^4y^8 + 11410136055144x^4y^7 - 3688243055280x^4y^6 \\
& - 7960716207972x^4y^5 + 1896810714144x^4y^4 + 529991228952x^4y^3 \\
& + 725251155408x^4y^2 + 878669669052x^4y - 464679074930688x^4 \\
& + 2932609536x^3y^{19} - 30399369984x^3y^{18} + 84818928384x^3y^{17} \\
& - 324185623488x^3y^{16} + 894440239776x^3y^{15} - 5389328536848x^3y^{14} \\
& + 28246976302176x^3y^{13} - 20224300372164x^3y^{12} + 26354203857288x^3y^{11} \\
& + 49527576521676x^3y^{10} - 278098650007560x^3y^9 + 185146659925488x^3y^8 \\
& + 117089003726280x^3y^7 - 770765224027644x^3y^6 + 1535888462926860x^3y^5 \\
& - 43715268097632x^3y^4 - 1624114829232414x^3y^3 + 7242921065926080x^3y^2 \\
& - 7385934170362248x^3y - 2384329602896028x^3 - 1632586752x^2y^{20} \\
& + 1496537856x^2y^{19} + 29242954368x^2y^{18} + 14492986560x^2y^{17} \\
& - 189446198112x^2y^{16} - 378830040480x^2y^{15} - 4053124514736x^2y^{14} \\
& - 130800638376x^2y^{13} + 7376125063104x^2y^{12} - 42812332884036x^2y^{11} \\
& + 78069490295040x^2y^{10} + 19518281915400x^2y^9 - 315510027844788x^2y^8 \\
& + 811031999120928x^2y^7 - 350280342329292x^2y^6 - 695319005901888x^2y^5 \\
& + 2883077740037988x^2y^4 - 3697883124622236x^2y^3 + 108309528681804x^2y^2 \\
& + 4618865199905568x^2y - 7169561837683200x^2 + 362797056xy^{21} + 680244480xy^{20} \\
& - 11465898624xy^{19} - 57847234752xy^{18} - 38528291520xy^{17} + 366654609072xy^{16} \\
& + 585371632680xy^{15} - 525675691836xy^{14} - 386567349048xy^{13} - 363837971772xy^{12} \\
& - 2597641092720xy^{11} - 3532586112144xy^{10} + 4401785502576xy^9 \\
& + 11307149166636xy^8 - 2350523153484xy^7 - 6750414600336xy^6 \\
& - 2876984551314xy^5 - 669462604992xy^4 + 3263630199336xy^3 \\
& + 310720508173260xy^2 - 1268167831916544xy - 397134546223104x + 151165440y^{22} \\
& - 5895452160y^{21} + 9198417024y^{20} + 33097673088y^{19} + 97184261376y^{18} \\
& - 862956257760y^{17} + 3023609241312y^{16} - 20789958063168y^{15} \\
& + 39457597753584y^{14} - 101893191476796y^{13} + 129233177288064y^{12} \\
& - 44719888038552y^{11} - 148425755060988y^{10} + 489816847962180y^9 \\
& - 366553780392852y^8 + 44147437420896y^7 + 924695264903292y^6 \\
& - 152303452615952y^5 - 172172802648636y^4 + 1241234090771040y^3 \\
& - 1073515407155712y^2 - 64099210788864y - 9466620346368 \\
B \quad & 1844218368x^5y^{18} - 33044765184x^5y^{17} + 115754935680x^5y^{16} \\
& - 153364897152x^5y^{15} + 1067554901664x^5y^{14} - 6352631248032x^5y^{13} \\
& + 4562819193960x^5y^{12} - 6180879442896x^5y^{11} - 10367718671448x^5y^{10} \\
& + 63508805754624x^5y^9 - 41696796237768x^5y^8 - 49226269659552x^5y^7 \\
& + 166958862068160x^5y^6 - 321854763666744x^5y^5 - 91228329581688x^5y^4 \\
& + 554016640912524x^5y^3 - 1657433963915784x^5y^2 + 1380234689878464x^5y \\
& + 1341074194716024x^5 - 614739456x^3y^{20} + 11014921728x^3y^{19} \\
& - 38584978560x^3y^{18} + 51121632384x^3y^{17} - 355851633888x^3y^{16} \\
& + 2117543749344x^3y^{15} - 1520939731320x^3y^{14} + 2060293147632x^3y^{13} \\
& + 3455906223816x^3y^{12} - 21169601918208x^3y^{11} + 13898932079256x^3y^{10} \\
& + 16408756553184x^3y^9 - 55652954022720x^3y^8 + 107284921222248x^3y^7 \\
& + 30409443193896x^3y^6 - 184672213637508x^3y^5 + 552477987971928x^3y^4
\end{aligned}$$

	$ \begin{aligned} & -460078229959488x^3y^3 - 447024731572008x^3y^2 + 281633760970380x^3y \\ & + 903168657506304x^3 + 695361024x^2y^{21} - 6394298112x^2y^{20} \\ & + 125971200x^2y^{19} + 23686364736x^2y^{18} + 141762319776x^2y^{17} \\ & - 694573704000x^2y^{16} + 4641391612944x^2y^{15} - 8373393091512x^2y^{14} \\ & + 19319300055144x^2y^{13} - 26009409421620x^2y^{12} + 7005983507712x^2y^{11} \\ & + 27562079158884x^2y^{10} - 45400348576224x^2y^9 + 41576252867748x^2y^8 \\ & + 103190428838736x^2y^7 - 94468931001504x^2y^6 + 3991118029776x^2y^5 \\ & + 612399490556292x^2y^4 - 877096399886442x^2y^3 - 750930036093324x^2y^2 \\ & + 1980035748871092x^2y - 2540148892720452x^2 - 120932352xy^{22} - 226748160xy^{21} \\ & + 3650645376xy^{20} + 20607628608xy^{19} + 12691598400xy^{18} - 102212716752xy^{17} \\ & - 154504464120xy^{16} + 380932106148xy^{15} - 1443400054560xy^{14} \\ & + 6638624713152xy^{13} - 11631610562076xy^{12} + 35650282719312xy^{11} \\ & - 43315269260868xy^{10} - 9260934332220xy^9 + 70976256134616xy^8 \\ & - 495592483280400xy^7 + 341870928352506xy^6 - 621785145784932xy^5 \\ & - 228420536569974xy^4 + 1003567412915262xy^3 - 1376226782981604xy^2 \\ & + 1755627477239412xy + 1449584502546996x - 50388480y^{23} + 1965150720y^{22} \\ & - 3116527488y^{21} - 9218572416y^{20} - 29565440640y^{19} + 267421111200y^{18} \\ & - 1073362173984y^{17} + 2876151304800y^{16} - 6747723347472y^{15} \\ & + 14846808488436y^{14} - 11744025421968y^{13} - 9017005600596y^{12} \\ & + 56075135891472y^{11} - 131772769379172y^{10} + 32584716934344y^9 \\ & + 122546303794800y^8 - 552835136223036y^7 + 401510566715940y^6 \\ & + 192081770355168y^5 - 824446415684844y^4 + 1356836756228388y^3 \\ & + 8452866738204y^2 - 602096214457440y + 1023478798496256 \end{aligned} $
$i = 19, m = 2$, neighbours 18	
W	$6x$
A	0
B	$-6y$
$i = 20, m = 2$, neighbours 1, 21	
W	$2x^2 - 2xy$
A	0
B	$-4x^3 - 4xy - 4y$
$i = 21, m = 1$, neighbours 20, 138, 152, 153, 154, 155, 209	
W	$ \begin{aligned} & 2x^3y^2 - 3x^3y - 8x^3 + x^2y^2 + 6x^2y + 13x^2 + xy^4 - 5xy^3 - xy + 8x \\ & + 2y^4 + 14y^3 + 4y^2 + 8y + 15 \end{aligned} $
A	$ \begin{aligned} & -3x^{18} - 12x^{17} + 3x^{16}y - 9x^{16} - 3x^{15}y - 3x^{15} - 6x^{14}y + 9x^{14} - 3x^{13}y \\ & + 24x^{13} + 7x^{12} + 12x^{11}y - 32x^{11} + 19x^{10}y - 11x^{10} - 8x^9y + 57x^9 - 30x^8y \\ & - 65x^8 - 4x^7y - 23x^7 - 26x^6y - 39x^6 + 19x^5y + 103x^5 + 19x^4y + 61x^4 \\ & + 61x^3y + 55x^3 - 38x^2y - 133x^2 + 4xy - 55x - 55y - 75 \end{aligned} $
B	$ \begin{aligned} & 6x^{18} - x^{17}y + 25x^{17} - 16x^{16}y + 22x^{16} - 2x^{15}y^2 - 21x^{15}y + 15x^{15} \\ & + 4x^{14}y^3 - 6x^{14}y^2 + 6x^{14}y + 9x^{14} + 17x^{13}y^3 - 25x^{13}y^2 \\ & + 23x^{13}y + 2x^{13} - 4x^{12}y^4 + 21x^{12}y^3 - 21x^{12}y^2 + 9x^{12}y + 18x^{12} \\ & + 22x^{11}y^3 - 5x^{11}y^2 + 2x^{11}y + 64x^{11} + 12x^{10}y^4 + 5x^{10}y^3 \\ & - 20x^{10}y^2 - 40x^{10}y - 29x^{10} - 12x^9y^5 + 25x^9y^4 - 32x^9y^3 \\ & - 27x^9y^2 - 65x^9y - 193x^9 - 16x^8y^5 + 36x^8y^4 - 55x^8y^3 \\ & + 9x^8y^2 + 70x^8y + 141x^8 + 6x^7y^5 + 4x^7y^4 - 28x^7y^3 - 22x^7y^2 \\ & + 178x^7y + 153x^7 - 8x^6y^6 + 34x^6y^5 - 52x^6y^4 + 6x^6y^3 \\ & - 60x^6y^2 - 127x^6y + 54x^6 - 8x^5y^6 + 22x^5y^5 - 27x^5y^4 \\ & + 11x^5y^3 + 23x^5y^2 - 194x^5y - 312x^5 + 12x^4y^6 - 17x^4y^5 \\ & + 4x^4y^4 + 55x^4y^3 + 16x^4y^2 + 67x^4y + 96x^4 + 12x^3y^6 \\ & - 12x^3y^5 + 10x^3y^4 - 178x^3y^3 + 45x^3y^2 + 251x^3y - 2x^3 \\ & - 5x^2y^6 + 18x^2y^5 + 27x^2y^4 + 162x^2y^3 - 72x^2y^2 + 30x^2y \\ & + 216x^2 - 8xy^6 - 4xy^5 + 132xy^4 + 28xy^3 + 42xy^2 - 11xy - 52x - 4y^6 - 56y^5 \\ & - 212y^4 - 144y^3 - 245y^2 - 409y - 129 \end{aligned} $
$i = 22, m = 4$, neighbours 1, 23	
W	$-4x^2 - 4xy - 4x - 4$
A	0
B	$16x^3 - 16xy - 16y - 16$
$i = 23, m = 128$, neighbours 0, 13, 22, 24, 90, 98, 116, 124, 156, 157, 158, 159, 160	
W	$ \begin{aligned} & 2x^5y + 3x^5 + x^4y + 7214x^3y^2 + 14043x^3y - 47629x^3 + 4043x^2y^2 \\ & + 1221x^2y + 56914x^2 - 4xy^5 - 14436xy^4 - 28124xy^3 - 25537xy^2 + 2857xy - 35219x \\ & - 2y^5 - 8086y^4 - 2381y^3 + 67849y^2 + 10971y - 46204 \end{aligned} $

$$\begin{aligned}
A \quad & -16384x^{39} + 49152x^{38} + 16384x^{37}y - 278528x^{37} - 32768x^{36}y + 524288x^{36} \\
& + 65536x^{35}y - 1228800x^{35} + 32768x^{34}y + 1441792x^{34} - 1097728x^{33}y \\
& + 294912x^{33} + 2703360x^{32}y - 3571712x^{32} - 9469952x^{31}y + 19955712x^{31} \\
& + 15302656x^{30}y - 33996800x^{30} - 29605888x^{29}y + 78921728x^{29} + 35422208x^{28}y \\
& - 102465536x^{28} - 34095104x^{27}y + 158875648x^{27} + 20676608x^{26}y - 167034880x^{26} \\
& + 39124992x^{25}y + 174407680x^{25} - 80936960x^{24}y - 136773632x^{24} \\
& + 186646528x^{23}y + 53329920x^{23} - 220053504x^{22}y + 20496384x^{22} \\
& + 267059200x^{21}y - 143769600x^{21} - 250626048x^{20}y + 194445312x^{20} \\
& + 164003840x^{19}y - 255868928x^{19} - 130875392x^{18}y + 213532672x^{18} \\
& - 21577728x^{17}y - 200474624x^{17} + 9273344x^{16}y + 15646720x^{16} - 109920256x^{15}y \\
& - 78462976x^{15} + 96452592x^{14}y - 236176064x^{14} - 87343120x^{13}y - 110960096x^{13} \\
& + 241057660x^{12}y - 292788512x^{12} - 8044672x^{11}y - 649824824x^{11} \\
& + 672702800x^{10}y - 429496436x^{10} + 529099288x^9y - 1995028688x^9 \\
& + 1357901096x^8y - 1174235612x^8 + 1804775492x^7y - 3626301084x^7 \\
& + 1556017784x^6y - 3231898120x^6 + 2174385820x^5y - 4289551768x^5 \\
& + 1408147640x^4y - 3325541452x^4 - 72324160x^3y - 5654919620x^3 - 176563320x^2y \\
& - 1362648800x^2 + 329091688xy - 1823517700x - 2637896132y - 2993561328 \\
B \quad & 32768x^{39} - 98304x^{38} + 524288x^{37} - 32768x^{36}y - 933888x^{36} + 16384x^{35}y^3 \\
& - 32768x^{35}y^2 + 376832x^{35}y + 1900544x^{35} - 49152x^{34}y^3 + 65536x^{34}y^2 \\
& - 933888x^{34}y - 1720320x^{34} + 229376x^{33}y^3 - 131072x^{33}y^2 + 3457024x^{33}y \\
& - 2703360x^{33} - 16384x^{32}y^4 - 376832x^{32}y^3 - 5718016x^{32}y + 9453568x^{32} \\
& - 16384x^{31}y^5 + 180224x^{31}y^4 + 507904x^{31}y^3 + 1277952x^{31}y^2 \\
& + 11501568x^{31}y - 36864000x^{31} + 32768x^{30}y^5 - 409600x^{30}y^4 \\
& - 147456x^{30}y^3 - 3178496x^{30}y^2 - 13221888x^{30}y + 55656448x^{30} \\
& - 16384x^{29}y^6 - 32768x^{29}y^5 + 1327104x^{29}y^4 - 3047424x^{29}y^3 \\
& + 10633216x^{29}y^2 + 10354688x^{29}y - 115523584x^{29} + 49152x^{28}y^6 \\
& - 49152x^{28}y^5 - 2031616x^{28}y^4 + 6750208x^{28}y^3 - 16728064x^{28}y^2 \\
& - 884736x^{28}y + 130809856x^{28} - 180224x^{27}y^6 + 622592x^{27}y^5 \\
& + 3129344x^{27}y^4 - 19070976x^{27}y^3 + 32702464x^{27}y^2 - 39600128x^{27}y \\
& - 190660608x^{27} + 16384x^{26}y^7 + 294912x^{26}y^6 - 1556480x^{26}y^5 \\
& - 2850816x^{26}y^4 + 27934720x^{26}y^3 - 39878656x^{26}y^2 + 64831488x^{26}y \\
& + 171884544x^{26} + 16384x^{25}y^8 - 98304x^{25}y^7 - 360448x^{25}y^6 \\
& + 3883008x^{25}y^5 - 1720320x^{25}y^4 - 43319296x^{25}y^3 + 49725440x^{25}y^2 \\
& - 151093248x^{25}y - 174718976x^{25} - 49152x^{24}y^8 + 196608x^{24}y^7 \\
& + 475136x^{24}y^6 - 5849088x^{24}y^5 + 7045120x^{24}y^4 + 46727168x^{24}y^3 \\
& - 47202304x^{24}y^2 + 168050688x^{24}y + 122716160x^{24} + 16384x^{23}y^9 \\
& + 32768x^{23}y^8 - 802816x^{23}y^7 + 835584x^{23}y^6 + 9486336x^{23}y^5 \\
& - 23396352x^{23}y^4 - 39174144x^{23}y^3 + 25313280x^{23}y^2 - 244285440x^{23}y \\
& - 46350336x^{23} - 16384x^{22}y^8 + 966656x^{22}y^7 - 2146304x^{22}y^6 \\
& - 9650176x^{22}y^5 + 36044800x^{22}y^4 + 16072704x^{22}y^3 - 9371648x^{22}y^2 \\
& + 227590144x^{22}y - 4145152x^{22} - 16384x^{21}y^{10} + 81920x^{21}y^9 \\
& - 81920x^{21}y^8 - 1343488x^{21}y^7 + 5488640x^{21}y^6 + 6815744x^{21}y^5 \\
& - 51675136x^{21}y^4 + 25575424x^{21}y^3 - 43646976x^{21}y^2 - 210993152x^{21}y \\
& + 126320640x^{21} - 212992x^{20}y^9 + 245760x^{20}y^8 + 2686976x^{20}y^7 \\
& - 8912896x^{20}y^6 - 4423680x^{20}y^5 + 64962560x^{20}y^4 - 75431936x^{20}y^3 \\
& + 62210048x^{20}y^2 + 192495616x^{20}y - 149012480x^{20} + 49152x^{19}y^{10} \\
& + 147456x^{19}y^9 - 1540096x^{19}y^8 + 540672x^{19}y^7 + 12632064x^{19}y^6 \\
& - 14385152x^{19}y^5 - 51412992x^{19}y^4 + 108937216x^{19}y^3 \\
& - 105529344x^{19}y^2 - 60456960x^{19}y + 249397248x^{19} + 65536x^{18}y^{10} \\
& - 196608x^{18}y^9 + 983040x^{18}y^8 + 2179072x^{18}y^7 - 15384576x^{18}y^6 \\
& + 12156928x^{18}y^5 + 56918016x^{18}y^4 - 149536768x^{18}y^3 \\
& + 120750080x^{18}y^2 + 113311744x^{18}y - 198098944x^{18} - 81920x^{17}y^{11} \\
& + 180224x^{17}y^{10} + 360448x^{17}y^9 - 3080192x^{17}y^8 + 5521408x^{17}y^7 \\
& + 12369920x^{17}y^6 - 41451520x^{17}y^5 - 8486912x^{17}y^4 + 126484480x^{17}y^3 \\
& - 111738880x^{17}y^2 + 81461248x^{17}y + 252723200x^{17} - 32768x^{16}y^{11} \\
& - 147456x^{16}y^{10} + 245760x^{16}y^9 + 2195456x^{16}y^8 - 2621440x^{16}y^7 \\
& - 13959168x^{16}y^6 + 26722304x^{16}y^5 + 10551296x^{16}y^4 \\
& - 124698624x^{16}y^3 + 125206528x^{16}y^2 + 68419584x^{16}y - 22773760x^{16} \\
& + 16384x^{15}y^{12} + 442368x^{15}y^{10} - 507904x^{15}y^9 - 3244032x^{15}y^8 \\
& + 9502720x^{15}y^7 + 1048576x^{15}y^6 - 43810816x^{15}y^5 + 37601280x^{15}y^4 \\
& + 66600960x^{15}y^3 - 63963136x^{15}y^2 + 135168000x^{15}y + 199442432x^{15} \\
& + 16384x^{14}y^{12} + 32768x^{14}y^{11} - 196608x^{14}y^{10} + 901120x^{14}y^9 \\
& + 1720320x^{14}y^8 - 8142848x^{14}y^7 - 3260416x^{14}y^6 + 26443776x^{14}y^5
\end{aligned}$$

$$\begin{aligned}
& -30441472x^{14}y^4 - 33095680x^{14}y^3 + 37978112x^{14}y^2 + 20070432x^{14}y \\
& + 246744448x^{14} - 65536x^{13}y^{12} + 16384x^{13}y^{11} + 491520x^{13}y^{10} \\
& - 1720320x^{13}y^9 - 1130496x^{13}y^8 + 7651328x^{13}y^7 - 9175040x^{13}y^6 \\
& - 16547840x^{13}y^5 + 41893888x^{13}y^4 + 1867776x^{13}y^3 + 4259840x^{13}y^2 \\
& + 157663264x^{13}y + 353958848x^{13} + 16384x^{12}y^{12} - 32768x^{12}y^{11} \\
& - 81920x^{12}y^{10} + 1130496x^{12}y^9 - 458752x^{12}y^8 - 8224768x^{12}y^7 \\
& + 8175616x^{12}y^6 + 15400960x^{12}y^5 - 37060608x^{12}y^4 + 29589504x^{12}y^3 \\
& - 183910368x^{12}y^2 - 259963288x^{12}y + 395340992x^{12} + 16384x^{11}y^{13} \\
& + 196608x^{11}y^{11} - 16384x^{11}y^{10} - 2031616x^{11}y^9 + 1671168x^{11}y^8 \\
& + 2523136x^{11}y^7 - 7356416x^{11}y^6 + 9371648x^{11}y^5 + 12386304x^{11}y^4 \\
& - 17104896x^{11}y^3 + 10698784x^{11}y^2 + 149404848x^{11}y + 1222728560x^{11} \\
& - 16384x^{10}y^{12} - 229376x^{10}y^{11} - 163840x^{10}y^{10} + 720896x^{10}y^9 \\
& - 933888x^{10}y^8 - 2850816x^{10}y^7 + 10895360x^{10}y^6 + 4374528x^{10}y^5 \\
& - 23642096x^{10}y^4 + 32686784x^{10}y^3 - 457899848x^{10}y^2 - 970628840x^{10}y \\
& + 1041884360x^{10} - 16384x^9y^{13} - 16384x^9y^{12} + 163840x^9y^{11} \\
& - 360448x^9y^{10} - 491520x^9y^9 + 2899968x^9y^8 - 1458176x^9y^7 \\
& - 966656x^9y^6 + 13107200x^9y^5 - 10125296x^9y^4 - 6316576x^9y^3 \\
& - 271957824x^9y^2 - 410653372x^9y + 3043806512x^9 - 16384x^8y^{12} \\
& - 49152x^8y^{11} + 229376x^8y^{10} + 557056x^8y^9 - 311296x^8y^8 \\
& - 294912x^8y^7 + 5324800x^8y^6 - 131056x^8y^5 - 10140908x^8y^4 \\
& + 222580880x^8y^3 - 691262052x^8y^2 - 1299787420x^8y + 2442729416x^8 \\
& - 16384x^7y^{13} + 16384x^7y^{11} - 262144x^7y^{10} + 720896x^7y^9 \\
& + 1572864x^7y^8 - 2834432x^7y^7 + 1294336x^7y^6 + 5947408x^7y^5 \\
& - 10314192x^7y^4 + 235827272x^7y^3 - 1503046460x^7y^2 - 1554050644x^7y \\
& + 5366873012x^7 + 81920x^6y^{11} + 212992x^6y^{10} + 131072x^6y^9 \\
& - 262144x^6y^8 - 720896x^6y^7 + 983040x^6y^6 + 507956x^6y^5 \\
& - 2578268x^6y^4 + 901402864x^6y^3 - 887981092x^6y^2 - 1576245772x^6y \\
& + 6263536035x^6 + 16384x^5y^{13} + 81920x^5y^{12} + 32768x^5y^{11} \\
& - 163840x^5y^{10} + 196608x^5y^9 - 196608x^5y^8 - 1622016x^5y^7 \\
& + 1310720x^5y^6 + 1720368x^5y^5 - 3290800x^5y^4 + 935072104x^5y^3 \\
& - 2936924264x^5y^2 - 1020772426x^5y + 7328007832x^5 - 16384x^4y^{11} \\
& - 131072x^4y^{10} - 147456x^4y^9 - 49152x^4y^8 - 393232x^4y^7 \\
& + 81168x^4y^6 - 207383364x^4y^5 + 21084644x^4y^4 + 1146049664x^4y^3 \\
& - 2495422041x^4y^2 - 827884987x^4y + 5094176683x^4 - 16384x^3y^{12} \\
& - 81920x^3y^{11} - 81920x^3y^{10} + 81920x^3y^9 - 65536x^3y^8 \\
& - 278544x^3y^7 + 254448x^3y^6 - 232883560x^3y^5 + 410121368x^3y^4 \\
& + 2571873932x^3y^3 - 2196965846x^3y^2 + 1334866830x^3y + 12725418284x^3 \\
& - 16384x^2y^{11} - 49152x^2y^{10} + 16384x^2y^9 + 81904x^2y^8 \\
& - 213780x^2y^7 - 208572700x^2y^6 - 668660472x^2y^5 - 493034440x^2y^4 \\
& + 949393489x^2y^3 - 3196960674x^2y^2 - 685318221x^2y + 3084520683x^2 \\
& - 16xy^8 - 122416xy^7 - 233468104xy^6 - 289534396xy^5 + 1935127172xy^4 \\
& + 2411618190xy^3 - 2483378938xy^2 - 698673354xy + 5209597076x - 4y^8 - 32340y^7 \\
& - 65360576y^6 + 27158780y^5 + 1129830547y^4 - 656654193y^3 - 3133734209y^2 \\
& + 5075183524y + 6845874368
\end{aligned}$$

$i = 24, m = 1$, neighbours 23
 $W \quad -y - 1$
 $A \quad 0$
 $B \quad x^2 - y$

$i = 25, m = 2$, neighbours 1, 26
 $W \quad x^3 + x^2 - 2xy + 2x - y + 2$
 $A \quad 0$
 $B \quad 3x^3 + 3x^2 - 3y$

$i = 26, m = 2430000$, neighbours 25, 41, 45, 46, 51, 106, 107, 121, 164, 166
 $W \quad -256x^5y + 19348x^5 + 265x^4y^2 + 12625x^4y - 59664x^4 - 71x^3y^3$
 $\quad - 1137396x^3y^2 + 79124263x^3y + 69887264x^3 - x^2y^5 + 51x^2y^4$
 $\quad + 301x^2y^3 + 131257233x^2y^2 - 174821044x^2y - 77007534x^2 + xy^6$
 $\quad + 1730xy^5 + 2094569xy^4 - 158497104xy^3 - 614201704xy^2 + 378774574xy - 359125207x$
 $\quad - 3339y^6 - 2171235y^5 - 103427355y^4 + 488768356y^3 - 569913421y^2 + 372324525y$
 $\quad + 193168431$
 $A \quad -5904900000000x^{22} - 41334300000000x^{21} + 41334300000001x^{20}$
 $\quad + 47239200000000x^{19}y + 507821399993329x^{19} + 124002900000000x^{18}y$
 $\quad + 194861711098966x^{18} - 436962600000007x^{17}y - 2291101115618487x^{17}$

$$\begin{aligned}
& -1423080899950066x^{16}y - 1747835793382156x^{16} + 1334507315777870x^{15}y \\
& + 5615573731288612x^{15} + 6406801614880564x^{14}y + 5001208635158893x^{14} \\
& - 832685376961905x^{13}y - 8959080707949006x^{13} - 16001820134368894x^{12}y \\
& - 9284663915428267x^{12} - 3553873221090682x^{11}y + 8630469441843227x^{11} \\
& + 23591480006988542x^{10}y - 9104548257135411x^{10} + 15921806078134020x^9y \\
& - 1540616168674952x^9 - 5700952224125841x^8y + 173053982075210186x^8 \\
& + 65648774418445837x^7y + 222196167823442159x^7 - 81583182059753792x^6y \\
& + 82005635048004436x^6 - 440067206138240589x^5y + 158946542540055879x^5 \\
& - 263064823587166775x^4y + 228303076191669572x^4 + 86344820086645134x^3y \\
& + 25453348812758974x^3 + 62108188903687667x^2y - 36593204797978837x^2 \\
& - 41753460060095600xy + 42288243127216656x + 21233195358651672y - 22128946455348388 \\
B \quad & 11809800000000x^{22} + 70858800000000x^{21} + 11809800000000x^{20}y \\
& - 153527400000002x^{20} - 11809800000000x^{19}y - 868020299986656x^{19} \\
& + 5904900000000x^{18}y^3 - 236196000000002x^{18}y + 448772377788724x^{18} \\
& + 35429400000000x^{17}y^3 - 59049000000000x^{17}y^2 + 13356x^{17}y \\
& + 4257432753448251x^{17} + 59049000000000x^{16}y^4 - 767637000000001x^{16}y^3 \\
& - 17714700000000x^{16}y^2 + 1281363277702185x^{16}y - 460611266690613x^{16} \\
& - 5904900000000x^{15}y^4 - 419247899993328x^{15}y^3 + 490106700000008x^{15}y^2 \\
& + 11809799787791x^{15}y - 12057804384767632x^{15} - 5904900000000x^{14}y^5 \\
& - 141717600000001x^{14}y^4 + 348389088894364x^{14}y^3 \\
& + 360198899940163x^{14}y^2 - 3672847422666619x^{14}y - 660866834465540x^{14} \\
& - 53144100000000x^{13}y^5 - 94478399993322x^{13}y^4 + 2078524726710776x^{13}y^3 \\
& - 1505749398182890x^{13}y^2 + 454808949742855x^{13}y + 23651351703843531x^{13} \\
& - 29524500000000x^{12}y^5 + 543250788851095x^{12}y^4 - 714507411097297x^{12}y^3 \\
& - 366074480565291x^{12}y^2 + 8573306648268024x^{12}y + 6804509827329104x^{12} \\
& + 23619600000000x^{11}y^6 + 336579300000004x^{11}y^5 + 188956799877204x^{11}y^4 \\
& - 5975758007182698x^{11}y^3 + 2539208753812290x^{11}y^2 - 4609678426565799x^{11}y \\
& - 33560811120807115x^{11} + 1062882000000001x^{10}y^6 + 395628299970082x^{10}y^5 \\
& - 1458510083485989x^{10}y^4 - 596124797905624x^{10}y^3 \\
& - 124916310897472x^{10}y^2 - 8642380916775575x^{10}y + 22912024568172273x^{10} \\
& - 6675x^9y^6 - 531440949094786x^9y^5 + 17780567106495x^9y^4 \\
& + 8793516834997114x^9y^3 - 5798896717522869x^9y^2 - 55186826605105x^9y \\
& - 13912171322719719x^9 - 29524499999999x^8y^7 - 442867488874330x^8y^6 \\
& - 578665534688407x^8y^5 + 3200158935731074x^8y^4 + 3112863966604941x^8y^3 \\
& + 3280730929676511x^8y^2 + 30678462046667340x^8y - 286558453429781068x^8 \\
& - 70858800006678x^7y^7 - 324769460157493x^7y^6 + 625970241912740x^7y^5 \\
& + 780426847159271x^7y^4 - 4968837562299570x^7y^3 + 7609990382584414x^7y^2 \\
& - 172063122432959651x^7y - 164219961532862678x^7 + 64953911148917x^6y^7 \\
& + 454691810742259x^6y^6 + 93992491575663x^6y^5 - 2690678388177302x^6y^4 \\
& + 6088205682852367x^6y^3 - 26984372252708869x^6y^2 - 24279204543258377x^6y \\
& + 20393773157137333x^6 + 11809799999999x^5y^8 + 248005800033040x^5y^7 \\
& + 383818334473391x^5y^6 - 1068984257061670x^5y^5 - 1349685569363992x^5y^4 \\
& - 6337426564624118x^5y^3 - 83233052484879590x^5y^2 + 82206408323846552x^5y \\
& - 205929500145175490x^5 - x^4y^9 + 11809800009893x^4y^8 \\
& + 41334236214987x^4y^7 - 171314680459906x^4y^6 - 115328574895290x^4y^5 \\
& - 2435319194325432x^4y^4 - 202227615505712895x^4y^3 \\
& + 26561874160544802x^4y^2 - 231779458526554162x^4y - 541329918041953265x^4 \\
& + 6678x^3y^9 - 35429417528204x^3y^8 - 212598293469550x^3y^7 \\
& - 13850149282247x^3y^6 - 812870194750430x^3y^5 - 69548795724112496x^3y^4 \\
& + 21749553153937798x^3y^3 - 46784856960976613x^3y^2 \\
& - 411982497919406978x^3y - 114486937735195995x^3 - x^2y^{10} - 11145704x^2y^9 \\
& - 35443945784070x^2y^8 - 94029010057893x^2y^7 - 340525404358825x^2y^6 \\
& - 10289287701448240x^2y^5 - 27857757861815067x^2y^4 \\
& - 246676784895763835x^2y^3 + 121385797762991331x^2y^2 \\
& - 111014039426725198x^2y - 36663419980290137x^2 + 6678xy^{10} - 6395754xy^9 \\
& + 23612290293579xy^8 + 32991750223600xy^7 - 1237105943738806xy^6 \\
& - 56393220306419744xy^5 + 165947434538023698xy^4 + 115152600732917996xy^3 \\
& - 202615468371244908xy^2 + 402247456459401066xy + 4934725980135451x - 11148921y^{10} \\
& - 14510656251y^9 + 18200151190755y^8 - 439461329338476y^7 - 9095272201771374y^6 \\
& + 89999499715011942y^5 - 256023672077363362y^4 + 288932219857963768y^3 \\
& - 125093512849049573y^2 - 156340531623703775y + 29072796631043403
\end{aligned}$$

$i = 27, m = 8, \text{ neighbours } 32$

W	$-8y - 16$
A	0
B	$64x^2 + 64y + 64$
$i = 28, m = 2, \text{neighbours } 1, 29$	
W	$-2xy - 2x - 2y$
A	0
B	$-4x^3 - 4x^2 + 4xy - 4x - 4$
$i = 29, m = 12, \text{neighbours } 28, 65, 78, 150, 165, 210$	
W	$2x^3y^2 - 2x^3y - 1084x^3 - 26x^2y^2 - 1050x^2y - 940x^2 - 6xy^4 + 4xy^3$ $-919xy^2 - 733xy + 259x + 5y^5 + 47y^4 - 66y^3 - 270y^2 + 328y + 761$
A	$-78x^{13} - 25x^{12}y - 1277x^{12} - 6x^{11}y - 1582x^{11} + 1354x^{10}y + 3337x^{10}$ $+1727x^9y + 25035x^9 + 3497x^8y + 66138x^8 - 11322x^7y + 19803x^7 - 49078x^6y$ $-145409x^6 - 102415x^5y - 416259x^5 - 148728x^4y - 363442x^4 - 114584x^3y$ $-210449x^3 - 249476x^2y - 229565x^2 - 300832xy - 332416x - 212544y - 181136$
B	$-276x^{13} - 238x^{12}y + 670x^{12} + 144x^{11}y^2 - 1322x^{11}y - 3002x^{11}$ $+50x^{10}y^2 - 4140x^{10}y - 14750x^{10} - 210x^9y^3 - 138x^9y^2 - 7983x^9y$ $-38805x^9 + 25x^8y^4 + 479x^8y^3 - 595x^8y^2 - 10861x^8y - 52360x^8$ $-85x^7y^4 + 209x^7y^3 + 9466x^7y^2 + 27466x^7y + 144579x^7 + 25x^6y^5$ $+40x^6y^4 - 2179x^6y^3 + 25184x^6y^2 + 98107x^6y + 344993x^6$ $+75x^5y^5 + 735x^5y^4 - 4400x^5y^3 + 69688x^5y^2 + 277479x^5y$ $+237369x^5 - 216x^4y^5 - 1399x^4y^4 - 25487x^4y^3 + 20772x^4y^2$ $-49401x^4y - 636478x^4 + 135x^3y^6 + 1595x^3y^5 + 5840x^3y^4$ $+28077x^3y^3 - 189957x^3y^2 - 719559x^3y - 168519x^3 - 25x^2y^7$ $-491x^2y^6 - 757x^2y^5 + 3598x^2y^4 - 3138x^2y^3 - 742365x^2y^2$ $-196270x^2y + 744912x^2 + 85xy^7 + 934xy^6 + 8972xy^5 + 73116xy^4 - 157647xy^3$ $-143760xy^2 + 863613xy + 855612x - 25y^8 - 445y^7 - 1079y^6 + 10428y^5 + 8683y^4$ $-91897y^3 + 196636y^2 + 355265y - 35713$
$i = 30, m = 1024, \text{neighbours } 53$	
W	$-1024x - 1024y - 2048$
A	0
B	$-1048576x^2 + 1048576y + 1048576$
$i = 31, m = 2, \text{neighbours } 1, 32$	
W	$-x^3 - 2x^2 - 2xy - 3x - y$
A	0
B	$3x^3 - 3x - 3y - 6$
$i = 32, m = 11, \text{neighbours } 27, 31, 107, 117, 153, 169, 170, 212$	
W	$330x^3y^2 + 762x^3y + 539x^3 - 279x^2y^2 - 1334x^2y + 290x^2 + 660xy^4$ $+1522xy^3 - 217xy^2 - 1778xy + 347x - 559y^4 - 2670y^3 - 7603y^2 - 7730y - 2628$
A	$-1089x^{19} - 363x^{17}y + 9438x^{17} + 363x^{16} + 2178x^{15}y - 41019x^{15} - 726x^{14}y$ $-7623x^{14} - 16335x^{13}y + 69333x^{13} - 1089x^{12}y + 459921x^{12} + 102366x^{11}y$ $-378873x^{11} + 475530x^{10}y - 2160275x^{10} - 1025376x^9y + 1245651x^9$ $-107345x^8y + 2657745x^8 + 1649390x^7y + 1654198x^7 - 3941846x^6y - 7173797x^6$ $+1955222x^5y + 2602282x^5 + 1282251x^4y + 1643519x^4 + 4852021x^3y$ $+5969584x^3 - 2034110x^2y - 3153206x^2 - 2045681xy - 2022481x - 2275047y - 3016935$
B	$2178x^{19} - 2057x^{17}y - 19481x^{17} + 363x^{16} + 1694x^{15}y^3 - 1210x^{15}y^2$ $+10043x^{15}y + 80223x^{15} - 605x^{14}y^3 + 5687x^{14}y + 9438x^{14} + 968x^{13}y^4$ $-5687x^{13}y^3 + 22627x^{13}y^2 + 13310x^{13}y - 108416x^{13} - 1694x^{12}y^4$ $-726x^{12}y^3 + 3993x^{12}y^2 - 10285x^{12}y - 886083x^{12} - 2420x^{11}y^4$ $+11495x^{11}y^3 - 132979x^{11}y^2 - 349085x^{11}y + 637109x^{11} - 968x^{10}y^5$ $+2783x^{10}y^4 + 242x^{10}y^3 - 50457x^{10}y^2 - 111683x^{10}y + 4237786x^{10}$ $+968x^9y^4 - 18150x^9y^3 + 344124x^9y^2 + 1669074x^9y - 2829860x^9$ $+2420x^8y^5 - 3025x^8y^4 - 433301x^8y^3 + 1045198x^8y^2 - 1864819x^8y$ $-3640949x^8 + 1936x^7y^4 + 754336x^7y^3 - 907544x^7y^2 - 1678667x^7y$ $-1901533x^7 - 1936x^6y^5 + 5929x^6y^4 + 1429193x^6y^3 - 2328802x^6y^2$ $+946896x^6y + 6877396x^6 - 1936x^5y^4 - 2956547x^5y^3 - 508474x^5y^2$ $+3409091x^5y - 3116942x^5 + 436084x^4y^5 - 1045605x^4y^4 - 328976x^4y^3$ $+5226018x^4y^2 + 3036945x^4y - 2010742x^4 - 737880x^3y^5 - 61012x^3y^4$ $+951669x^3y^3 + 4559698x^3y^2 + 1434656x^3y - 4039822x^3 - 435600x^2y^6$ $-389759x^2y^5 + 2183180x^2y^4 + 3195822x^2y^3 - 2792564x^2y^2$ $-1196526x^2y + 4296278x^2 + 737880xy^6 + 3012356xy^5 + 6670126xy^4$ $+1153586xy^3 - 6830352xy^2 - 1486181xy + 3636070x - 312481y^6 - 2047617y^5$

$$-8548760y^4 - 17452061y^3 - 21741741y^2 - 10978043y + 714807$$

$i = 33, m = 4$, neighbours 1, 34

$W \quad -4x^2 - 4xy - 8x - 4$

$A \quad 0$

$B \quad 16x^3 - 16xy - 16x - 16y - 32$

$i = 34, m = 1$, neighbours 33, 122, 147, 164, 171, 213

$W \quad -x^3y^2 - 2x^3y - 11x^3 - x^2y^2 - 2x^2y - 18x^2 + xy^4 + 2xy^3 - 24xy^2$
 $-48xy - 22x + 4y^4 + 5y^3 - 43y^2 - 71y - 35$

$A \quad -3x^{13} - x^{12}y - 6x^{12} - x^{11}y - 28x^{11} - 2x^{10}y - 48x^{10} + 3x^9y - 113x^9$
 $+16x^8y - 214x^8 - 10x^7y - 445x^7 + 2x^6y - 557x^6 - 31x^5y - 616x^5$
 $-123x^4y - 567x^4 - 216x^3y - 520x^3 - 474x^2y - 600x^2 - 377xy - 504x - 295y - 217$

$B \quad 6x^{13} + 2x^{12}y + 12x^{12} + 10x^{11}y + 60x^{11} - x^{10}y^2 + 19x^{10}y + 117x^{10}$
 $+3x^9y^3 - 10x^9y^2 + 30x^9y + 259x^9 + 2x^8y^4 + 9x^8y^3 - 23x^8y^2$
 $+14x^8y + 492x^8 + 8x^7y^4 + 43x^7y^3 - 52x^7y^2 + 32x^7y + 890x^7$
 $-x^6y^5 + 6x^6y^4 + 64x^6y^3 - 89x^6y^2 + 54x^6y + 1042x^6$
 $-10x^5y^5 - 19x^5y^4 + 73x^5y^3 - 147x^5y^2 + 123x^5y + 984x^5$
 $-16x^4y^5 - 45x^4y^4 + 106x^4y^3 - 281x^4y^2 - 61x^4y + 683x^4$
 $+3x^3y^6 - 10x^3y^5 + 6x^3y^4 + 259x^3y^3 - 539x^3y^2 - 807x^3y$
 $+202x^3 + 8x^2y^6 + 12x^2y^5 + 126x^2y^4 + 271x^2y^3 - 977x^2y^2$
 $-1054x^2y + 190x^2 - 5xy^6 + 6xy^5 + 289xy^4 + 82xy^3 - 1778xy^2 - 1583xy + 63x$
 $-2y^7 - 23y^6 + 3y^5 + 344y^4 - 52y^3 - 1421y^2 - 1099y - 192$

$i = 35, m = 4$, neighbours 1, 5, 42, 73

$W \quad 4x^4 - x^2y^2 + 6x^2y - 9x^2 - 2xy^2 + 2x + 3y^2 - 6y + 3$

$A \quad 12x^2y^2 - 24x^2y + 12x^2 + 24xy^2 - 24x + 12y^2 + 24y + 12$

$B \quad -15x^3y^3 - 27x^3y^2 + 75x^3y - 33x^3 - 45x^2y^3 - 27x^2y^2 + 33x^2y$
 $-9x^2 + 12xy^4 - 81xy^3 + 63xy^2 - 15xy + 45x + 12y^4 - 27y^3 + 15y^2 + 51y - 3$

$i = 36, m = 2$, neighbours 4, 10, 118

$W \quad 4x^3 + 6x^2 + 2x$

$A \quad 0$

$B \quad 24x^5 - 36x^4y + 60x^4 + 16x^3y^2 - 72x^3y + 56x^3 - 12x^2y^3 + 24x^2y^2$
 $-60x^2y + 24x^2 - 12xy^3 + 8xy^2 - 24xy + 4x - 4y^5 - 4y^3 - 4y$

$i = 37, m = 4$, neighbours 6, 18, 38, 53, 62, 76, 84, 87, 202

$W \quad 23688x^8 + 54396x^4 - 687x - 4683y + 34560$

$A \quad 52488x^{12}y^2 - 52488x^{12}y - 280508184x^{12} - 157464x^{11}y^3$
 $+367416x^{11}y^2 - 367416x^{11}y + 209952x^{11} + 209952x^{10}y^4$
 $-804816x^{10}y^3 + 1224720x^{10}y^2 - 1014768x^{10}y + 419904x^{10}$
 $-227448x^9y^5 + 892296x^9y^4 - 1977048x^9y^3 + 2370708x^9y^2$
 $-1705860x^9y + 140901444x^9 + 262440x^8y^6 - 615276x^8y^5$
 $+2096604x^8y^4 + 137437236x^8y^3 + 3356316x^8y^2 + 138405348x^8y$
 $-866825388x^8 - 256608x^7y^7 + 317844x^7y^6 - 1845828x^7y^5$
 $+2274480x^7y^4 - 3335904x^7y^3 + 2571912x^7y^2 - 1886652x^7y + 763992x^7$
 $+188568x^6y^8 - 202176x^6y^7 + 1304424x^6y^6 - 2040228x^6y^5$
 $+2283228x^6y^4 - 1567836x^6y^3 + 813078x^6y^2 - 556470x^6y - 69970554x^6$
 $-106920x^5y^9 + 212868x^5y^8 - 574452x^5y^7 + 2172420x^5y^6$
 $-1470636x^5y^5 + 1523124x^5y^4 - 138849066x^5y^3 - 1578042x^5y^2$
 $-138288222x^5y + 238956264x^5 + 46656x^4y^{10} - 232308x^4y^9$
 $+84726x^4y^8 - 1923426x^4y^7 - 69263586x^4y^6 - 2001672x^4y^5$
 $-141587838x^4y^4 + 225701802x^4y^3 - 74836224x^4y^2 + 337628358x^4y$
 $-998024490x^4 - 13608x^3y^{11} + 204120x^3y^{10} + 40662x^3y^9$
 $+1290006x^3y^8 - 909306x^3y^7 + 1086696x^3y^6 + 590004x^3y^5$
 $-1625184x^3y^4 + 3095982x^3y^3 - 3926313x^3y^2 + 2700945x^3y$
 $+33907329x^3 + 1944x^2y^{12} - 135108x^2y^{11} + 13608x^2y^{10}$
 $-610740x^2y^9 + 1504332x^2y^8 + 384264x^2y^7 + 561978x^2y^6$
 $+3850659x^2y^5 - 1039635x^2y^4 + 107945865x^2y^3 + 844911x^2y^2$
 $+105166269x^2y - 14444541x^2 + 67068xy^{12} - 93474xy^{11} + 109350xy^{10}$
 $-1810998xy^9 - 1152873xy^8 - 2327373xy^7 + 99185958xy^6 + 764721xy^5$
 $+204777477xy^4 - 24903909xy^3 + 108479574xy^2 - 81258498xy + 177942069x$
 $-33048y^{13} + 361746y^{12} - 486y^{11} + 3053214y^{10} + 34271991y^9 - 12764709y^8$
 $+120501837y^7 - 27847314y^6 + 79887141y^5 - 43316505y^4 + 169962138y^3$
 $-72121077y^2 + 270009909y - 398131200$

$B \quad -15552x^3y^{13} + 462672x^3y^{12} - 224046x^3y^{11} + 3033774x^3y^{10}$

$$\begin{aligned}
& -3914244x^3y^9 - 31570236x^3y^8 + 19388970x^3y^7 - 43221600x^3y^6 \\
& -66351960x^3y^5 + 46924596x^3y^4 - 54460026x^3y^3 - 36383661x^3y^2 \\
& +47910123x^3y + 3259683x^3 + 1944x^2y^{14} - 239112x^2y^{13} + 421524x^2y^{12} \\
& -1088316x^2y^{11} + 5045328x^2y^{10} + 35508456x^2y^9 - 11953170x^2y^8 \\
& +42989778x^2y^7 + 76542732x^2y^6 - 44428014x^2y^5 + 55679238x^2y^4 \\
& +184649544x^2y^3 - 55719414x^2y^2 + 144208350x^2y + 87277446x^2 \\
& +100116xy^{14} - 451332xy^{13} + 209304xy^{12} - 4543290xy^{11} - 36442710xy^{10} \\
& +9301878xy^9 - 36625932xy^8 - 79476714xy^7 + 255919662xy^6 - 44343936xy^5 \\
& +229467978xy^4 + 340824078xy^3 + 71789490xy^2 + 132448338xy + 135045423x \\
& -33048y^{15} + 361746y^{14} - 34830y^{13} + 3308040y^{12} + 35334711y^{11} - 9751833y^{10} \\
& +163879038y^9 + 62133966y^8 + 162010611y^7 + 290256156y^6 + 166232169y^5 \\
& +124169409y^4 + 309979764y^3 + 39610809y^2 + 75565503y + 88209567
\end{aligned}$$

$i = 38, m = 1$, neighbours 37, 58, 155, 180, 208

$W \quad -xy + x - 2y + 1$

$A \quad xy + y + 1$

$B \quad -2x^5 + 2x^4y^2 - 2x^4y - 4x^4 + 2x^3y^2 - 2x^3y - 2x^2y^4 + 2x^2y^3$
 $+2x^2y^2 - 2x^2y - x^2 - 3xy^2 - xy - 3x + 2y^4 - 5y^2 - 4y - 4$

$i = 39, m = 1$, neighbours 68

$W \quad 1$

$A \quad 0$

$B \quad -y^2 - 3y - 2$

$i = 40, m = 100$, neighbours 1, 72

$W \quad 33x^2 + 34xy + 33y^2 + 67$

$A \quad 0$

$B \quad 8911x^3 - 3333x^2y + 31089x^2 + 3333xy^2 - 5578xy + 34489x + 1089y^3 + 1089y^2$
 $-4489y + 25511$

$i = 41, m = 20$, neighbours 2, 12, 26, 42, 43, 62

$W \quad 40y^2 - 240y + 320$

$A \quad 400y^9 + 2000y^7 + 4800y^6 + 1600y^5 + 25600y^4 + 4800y^3 + 25600y^2 + 44800y$
 -25600

$B \quad 3600x^{13} + 3600x^{12}y - 14400x^{12} + 6000x^{11}y^2 - 12000x^{11}y + 31200x^{11}$
 $+2400x^{10}y^3 - 25200x^{10}y^2 + 21600x^{10}y - 42000x^{10} + 5200x^9y^4$
 $-14800x^9y^3 + 44400x^9y^2 - 18400x^9y + 34000x^9 + 2800x^8y^5$
 $-12800x^8y^4 + 12400x^8y^3 - 42400x^8y^2 + 2000x^8y - 8800x^8$
 $+2000x^7y^6 + 2400x^7y^5 + 2000x^7y^3 + 46800x^7y^2 + 16800x^7y$
 $-19600x^7 - 800x^6y^7 + 4400x^6y^6 - 13600x^6y^5 - 8400x^6y^4$
 $-12800x^6y^3 - 68000x^6y^2 - 22000x^6y + 34800x^6 - 400x^5y^8$
 $+1200x^5y^7 - 4000x^5y^6 + 22000x^5y^5 + 45600x^5y^4 - 19200x^5y^3$
 $+122400x^5y^2 + 14400x^5y - 34400x^5 + 400x^4y^9 - 800x^4y^8$
 $-8000x^4y^7 - 5600x^4y^6 + 22400x^4y^5 - 118400x^4y^4 + 39600x^4y^3$
 $-164400x^4y^2 - 5600x^4y + 24400x^4 + 400x^3y^{10} + 800x^3y^8$
 $-7600x^3y^7 + 16400x^3y^6 - 11600x^3y^5 + 72000x^3y^4 - 36400x^3y^3$
 $+144800x^3y^2 - 89600x^3y + 38000x^3 - 400x^2y^{10} + 2000x^2y^9$
 $+3200x^2y^8 - 14400x^2y^7 + 2000x^2y^6 - 34000x^2y^5 - 80400x^2y^4$
 $+22400x^2y^3 - 162000x^2y^2 + 90000x^2y - 45600x^2 - 400xy^{11} - 1200xy^{10}$
 $+4000xy^9 - 4800xy^8 + 31600xy^7 + 3600xy^6 + 51200xy^5 + 91600xy^4$
 $-11600xy^3 + 173200xy^2 - 90400xy + 49600x - 1200y^{10} - 6000y^9 - 1200y^8$
 $-36400y^7 - 18000y^6 - 56000y^5 - 125600y^4 + 4800y^3 - 203600y^2 + 44800y - 25200$

$i = 42, m = 8064$, neighbours 35, 41, 99, 126, 127, 128, 130, 145, 148, 174, 215

$W \quad x^4y^2 + 150x^4y - 266x^4 - 452x^3y^2 - 1232036x^3y + 2176282x^3$
 $+x^2y^3 + 1233098x^2y^2 - 1088367x^2y + 809526x^2 + xy^5 + 1053xy^4$
 $-1860xy^3 - 1087454xy^2 + 270033xy - 2955949x - 1355y^5 - 1231238y^4 + 2175688y^3$
 $-1081222y^2 - 892241y + 5975597$

$A \quad -1560674304x^{17} - 780337152x^{16}y + 16777248768x^{16} + 7413202944x^{15}y$
 $-75692703744x^{15} - 23800283136x^{14}y + 177526702080x^{14} + 11314888704x^{13}y$
 $-186695663616x^{13} + 128950714368x^{12}y - 126512160768x^{12} - 411920474112x^{11}y$
 $+808965771264x^{11} + 595836186624x^{10}y - 1571720951808x^{10} - 311817848832x^9y$
 $+1995651302400x^9 - 633719116800x^8y - 1828848139776x^8 + 2156297117182x^7y$
 $+1276860192668x^7 - 4055165271763x^6y - 1253611418920x^6 + 6382857295906x^5y$
 $+3252331225972x^5 - 9674940860756x^4y - 3049612303784x^4 + 17049908001764x^3y$
 $-3524986351185x^3 - 13659970098274x^2y + 2466267271335x^2 + 4096446555188xy$

$$\begin{aligned}
& +4629502460908x + 5234833533900y - 9015026296840 \\
B \quad & 3121348608x^{19} - 1560674304x^{18}y - 27311800320x^{18} - 1560674304x^{17}y^2 \\
& +12485394432x^{17}y + 93640458240x^{17} + 3121348608x^{16}y^3 \\
& +10144382976x^{16}y^2 - 46820229120x^{16}y - 138900013056x^{16} \\
& -1560674304x^{15}y^4 - 16387080192x^{15}y^3 - 12485394432x^{15}y^2 \\
& +106125852672x^{15}y - 10144382976x^{15} - 1560674304x^{14}y^5 \\
& +7023034368x^{14}y^4 + 16777248768x^{14}y^3 - 48380903424x^{14}y^2 \\
& -149434564608x^{14}y + 396996526080x^{14} + 3121348608x^{13}y^6 \\
& +4682022912x^{13}y^5 - 12485394432x^{13}y^4 + 55403937792x^{13}y^3 \\
& +174015184896x^{13}y^2 + 88958435328x^{13}y - 764242698240x^{13} \\
& -1560674304x^{12}y^7 - 7023034368x^{12}y^6 + 13655900160x^{12}y^5 \\
& +1560674304x^{12}y^4 - 156457598976x^{12}y^3 - 215177969664x^{12}y^2 \\
& +179477544960x^{12}y + 872075538432x^{12} - 1560674304x^{11}y^8 \\
& +2341011456x^{11}y^7 - 13655900160x^{11}y^6 - 46820229120x^{11}y^5 \\
& +42138206208x^{11}y^4 + 118611247104x^{11}y^3 + 58037575680x^{11}y^2 \\
& -718885601280x^{11}y - 796309678080x^{11} + 3121348608x^{10}y^9 \\
& +1040449536x^{10}y^8 - 1820786688x^{10}y^7 + 37846351872x^{10}y^6 \\
& +9103933440x^{10}y^5 - 105865740288x^{10}y^4 + 67499163648x^{10}y^3 \\
& +228590014464x^{10}y^2 + 1470155194368x^{10}y + 763572096000x^{10} \\
& -520224768x^9y^{10} + 3901685760x^9y^9 + 16257024000x^9y^8 \\
& -11444944896x^9y^7 - 3901685760x^9y^6 + 93380345856x^9y^5 \\
& +149694676992x^9y^4 - 244245528576x^9y^3 - 581668190208x^9y^2 \\
& -2382320553980x^9y - 1085442877240x^9 - 1040449536x^8y^{11} \\
& -2601123840x^8y^{10} - 10794663936x^8y^9 + 1430618112x^8y^8 \\
& +21849440256x^8y^7 - 30693261312x^8y^6 - 149174452224x^8y^5 \\
& -105215459328x^8y^4 + 355833741312x^8y^3 + 1101141070844x^8y^2 \\
& +3820579455206x^8y + 2292276540384x^8 - 5202247680x^7y^{11} \\
& -7413202944x^7y^{10} - 7153090560x^7y^9 - 23280058368x^7y^8 \\
& -19898597376x^7y^7 + 21979496448x^7y^6 + 119391584256x^7y^5 \\
& -64377815040x^7y^4 - 340226998272x^7y^3 - 2021688951218x^7y^2 \\
& -6546433184148x^7y - 4992772261360x^7 + 520224768x^6y^{12} \\
& -6242697216x^6y^{11} - 9103933440x^6y^{10} + 12095225856x^6y^9 \\
& +10274439168x^6y^8 + 15346630656x^6y^7 - 42398318592x^6y^6 \\
& -10014326784x^6y^5 + 192613220355x^6y^4 + 94810968476x^6y^3 \\
& +3563936220522x^6y^2 + 11295193152548x^6y - 1968912299632x^6 \\
& +1040449536x^5y^{13} + 5202247680x^5y^{12} + 3771629568x^5y^{11} \\
& +1300561920x^5y^{10} + 14956462080x^5y^9 - 8063483904x^5y^8 \\
& +41097756672x^5y^7 + 51112083456x^5y^6 - 87137648643x^5y^5 \\
& -103654795866x^5y^4 + 213799354636x^5y^3 - 5855224782484x^5y^2 \\
& -11678751076215x^5y + 5529108987406x^5 + 4161798144x^4y^{13} \\
& +16257024000x^4y^{12} + 12615450624x^4y^{11} - 2991292416x^4y^{10} \\
& +18988204032x^4y^9 - 10534551552x^4y^8 - 54883713024x^4y^7 \\
& +6242697216x^4y^6 + 77253384377x^4y^5 - 33147177870x^4y^4 \\
& -291114502504x^4y^3 + 12208671567484x^4y^2 - 2432832785905x^4y \\
& +3867326086870x^4 + 4942135296x^3y^{13} + 18858147840x^3y^{12} \\
& +10794663936x^3y^{11} - 25491013632x^3y^{10} - 21329215488x^3y^9 \\
& +2210955264x^3y^8 - 21979496447x^3y^7 - 46820227016x^3y^6 \\
& -48645450811x^3y^5 + 6398215788x^3y^4 - 7395726134031x^3y^3 \\
& +1065808918108x^3y^2 - 5711039143253x^3y - 17285929399146x^3 \\
& +1040449536x^2y^{13} + 3771629568x^2y^{12} - 1170505728x^2y^{11} \\
& -19898597376x^2y^{10} - 26011238400x^2y^9 + 5852528639x^2y^8 \\
& +28352245039x^2y^7 + 34198357905x^2y^6 + 53703938843x^2y^5 \\
& +3098106720159x^2y^4 + 2139353239669x^2y^3 + 1223377109719x^2y^2 \\
& +10019718139735x^2y + 4478494847219x^2 - 1300561920xy^{13} - 5332303872xy^{12} \\
& -4682022912xy^{11} + 4031741952xy^{10} + 9624158208xy^9 + 14436240022xy^8 \\
& +9891425433xy^7 - 5909547134xy^6 + 1465666880991xy^5 - 8073417155368xy^4 \\
& +5588018159230xy^3 - 1962462570898xy^2 - 18242391231849xy + 13128277635444x \\
& -520224768y^{13} - 1820786688y^{12} + 520224768y^{11} + 6502809600y^{10} \\
& +2601123840y^9 - 8065319929y^8 - 10621637757y^7 - 1511566766456y^6 \\
& +3852403681249y^5 - 2034288912037y^4 - 3362817312660y^3 + 10864837323981y^2 \\
& -4688809634995y - 8662680615889
\end{aligned}$$

$i = 43, m = 72$, neighbours 8, 13, 41, 57, 61, 95, 114, 168, 181

W	$-x^4y + 1076x^3y + 279x^3 - 1075x^2y^2 - 499x^2y - 285x^2 - xy^4 + 219xy^2$ $+40xy + 52x + 1076y^4 + 279y^3 + 246y^2 + 1911y + 315$
A	$10368x^{11}y - 31104x^{11} - 77760x^{10}y + 220320x^{10} + 194400x^9y - 664848x^9$ $-16848x^8y + 992088x^8 - 1046520x^7y - 128628x^7 + 2970108x^6y - 3242752x^6$ $-4431508x^5y + 9689480x^5 + 3002012x^4y - 12864478x^4 + 6345368x^3y$ $+7968291x^3 - 11009828x^2y + 285056x^2 + 4168949xy - 4074501x + 1152519y + 2132236$
B	$-31104x^{13}y + 72576x^{13} + 10368x^{12}y^2 + 98496x^{12}y - 393984x^{12}$ $+31104x^{11}y^3 - 93312x^{11}y^2 + 20736x^{11}y + 857952x^{11} + 10368x^{10}y^4$ $-46656x^{10}y^3 + 222912x^{10}y^2 - 606528x^{10}y - 654480x^{10} + 20736x^9y^4$ $-129600x^9y^3 - 132192x^9y^2 + 1474848x^9y - 1371816x^9 + 10368x^8y^5$ $-98496x^8y^4 + 264384x^8y^3 - 408240x^8y^2 - 1946592x^8y + 5704664x^8$ $-10368x^7y^5 - 25920x^7y^4 + 41472x^7y^3 + 1502712x^7y^2 + 1067256x^7y$ $-11300024x^7 - 41472x^6y^5 + 248832x^6y^4 - 326592x^6y^3$ $-3027129x^6y^2 + 2658320x^6y + 5933656x^6 + 51840x^5y^5 - 88128x^5y^4$ $+150333x^5y^3 + 4031071x^5y^2 - 5351722x^5y + 4904998x^5 + 41472x^4y^5$ $-196992x^4y^4 + 50081x^4y^3 - 243544x^4y^2 + 1375390x^4y - 1648203x^4$ $-72575x^3y^5 + 127447x^3y^4 - 5833939x^3y^3 + 959029x^3y^2$ $-3408971x^3y - 14443947x^3 - x^2y^6 - 2151x^2y^5 + 2350840x^2y^4$ $+3351121x^2y^3 + 769478x^2y^2 + 6082249x^2y + 13454686x^2 + 2152xy^6$ $+1187286xy^5 - 3380562xy^4 + 2495985xy^3 + 1262560xy^2 - 7173240xy + 3562574x$ $-1157776y^6 + 547000y^5 - 1641y^4 - 4784731y^3 + 1834303y^2 - 297790y - 6495933$
$i = 44, m = 16, \text{neighbours } 47, 48, 49, 50, 51, 52, 54, 55, 58, 63, 64, 65, 66, 67, 69, 71, 188, 189$	
W	$-x^2y^2 - 2x^2y + x^2 - 2639xy^2 - 6142xy + 4031x - 9313y^2 + 11230y - 10415$
A	$-18432xy^{13} - 27648xy^{12} + 129024xy^{11} + 370944xy^{10} + 820224xy^9$ $+2308608xy^8 - 2617344xy^7 + 2979072xy^6 + 30832128xy^5 - 86863104xy^4$ $+42015744xy^3 - 4792320xy^2 - 47736576xy + 15853824x + 11520y^{13} + 73728y^{12}$ $+145152y^{11} - 161280y^{10} - 117504y^9 - 2882304y^8 + 506880y^7 + 635904y^6$ $-30813696y^5 + 2681856y^4 + 89627904y^3 - 125819136y^2 + 89402112y - 36029952$
B	$-40960x^{25}y^8 + 122880x^{25}y^7 + 71680x^{25}y^6 - 491520x^{25}y^5$ $+368640x^{25}y^4 + 61440x^{24}y^9 - 100352x^{24}y^8 - 257024x^{24}y^7$ $+866816x^{24}y^6 - 1337344x^{24}y^5 + 1210368x^{24}y^4 - 368640x^{24}y^3$ $-156672x^{23}y^9 + 444416x^{23}y^8 - 744704x^{23}y^7 + 1033472x^{23}y^6$ $-589312x^{23}y^5 - 369664x^{23}y^4 + 1615872x^{23}y^3 - 1474560x^{23}y^2$ $-20480x^{22}y^{11} + 102400x^{22}y^{10} - 98304x^{22}y^9 - 193536x^{22}y^8$ $+854784x^{22}y^7 - 1779200x^{22}y^6 + 4887808x^{22}y^5 - 9317888x^{22}y^4$ $+10243072x^{22}y^3 - 7053312x^{22}y^2 + 1474560x^{22}y + 52224x^{21}y^{11}$ $-353280x^{21}y^{10} + 856576x^{21}y^9 - 802560x^{21}y^8 + 2105088x^{21}y^7$ $-6185472x^{21}y^6 + 9352960x^{21}y^5 - 13724928x^{21}y^4 + 15851008x^{21}y^3$ $-12828672x^{21}y^2 + 5824512x^{21}y - 20480x^{20}y^{12} + 78848x^{20}y^{11}$ $+342528x^{20}y^{10} - 687616x^{20}y^9 - 364544x^{20}y^8 - 963072x^{20}y^7$ $+2803712x^{20}y^6 - 3160320x^{20}y^5 + 2180096x^{20}y^4 + 2429696x^{20}y^3$ $-5664768x^{20}y^2 + 6746112x^{20}y + 113664x^{19}y^{12} - 476416x^{19}y^{11}$ $+226048x^{19}y^{10} - 763904x^{19}y^9 + 2697472x^{19}y^8 - 5552128x^{19}y^7$ $+11581184x^{19}y^6 - 16255744x^{19}y^5 + 22565120x^{19}y^4 - 20262400x^{19}y^3$ $+12902400x^{19}y^2 - 3704832x^{19}y - 200704x^{18}y^{12} + 572160x^{18}y^{11}$ $-581120x^{18}y^{10} + 661248x^{18}y^9 - 921856x^{18}y^8 + 5101056x^{18}y^7$ $-2340352x^{18}y^6 - 8058880x^{18}y^5 + 16914944x^{18}y^4 - 23181056x^{18}y^3$ $+21437952x^{18}y^2 - 15307776x^{18}y + 40960x^{17}y^{13} - 41216x^{17}y^{12}$ $-137472x^{17}y^{11} + 168960x^{17}y^{10} + 3760384x^{17}y^9 - 5611008x^{17}y^8$ $+2387456x^{17}y^7 - 9892608x^{17}y^6 + 12347904x^{17}y^5 - 4053760x^{17}y^4$ $-14910208x^{17}y^3 + 19106304x^{17}y^2 - 11473920x^{17}y - 186368x^{16}y^{13}$ $+541696x^{16}y^{12} + 440320x^{16}y^{11} - 199168x^{16}y^{10} - 6152704x^{16}y^9$ $+5156096x^{16}y^8 - 4971264x^{16}y^7 + 9581824x^{16}y^6 - 5736192x^{16}y^5$ $-95232x^{16}y^4 - 4734976x^{16}y^3 + 14419968x^{16}y^2 - 4829184x^{16}y$ $+358400x^{15}y^{13} - 532224x^{15}y^{12} - 1777152x^{15}y^{11} + 659200x^{15}y^{10}$ $-227840x^{15}y^9 + 9974528x^{15}y^8 - 16061184x^{15}y^7 + 27444480x^{15}y^6$ $-25185280x^{15}y^5 + 9643008x^{15}y^4 + 9054976x^{15}y^3 - 7451136x^{15}y^2$ $-3990528x^{15}y - 436736x^{14}y^{13} + 80128x^{14}y^{12} + 1341952x^{14}y^{11}$ $+617472x^{14}y^{10} + 2683136x^{14}y^9 + 722944x^{14}y^8 - 2950912x^{14}y^7$ $-3069440x^{14}y^6 - 3537664x^{14}y^5 - 1529856x^{14}y^4 + 11504128x^{14}y^3$ $-31830528x^{14}y^2 + 14146560x^{14}y + 318976x^{13}y^{13} - 19200x^{13}y^{12}$ $+1263616x^{13}y^{11} + 759040x^{13}y^{10} + 2057728x^{13}y^9 - 26998528x^{13}y^8$

$$\begin{aligned}
& +30863360x^{13}y^7 - 37544192x^{13}y^6 + 34256640x^{13}y^5 - 26911744x^{13}y^4 \\
& +10460416x^{13}y^3 - 10690560x^{13}y^2 + 35813376x^{13}y - 122880x^{12}y^{14} \\
& +368384x^{12}y^{13} + 635136x^{12}y^{12} - 623104x^{12}y^{11} - 10027264x^{12}y^{10} \\
& +2999296x^{12}y^9 - 7693312x^{12}y^8 + 41725696x^{12}y^7 - 62998272x^{12}y^6 \\
& +70246144x^{12}y^5 - 36339712x^{12}y^4 + 31824896x^{12}y^3 + 13836288x^{12}y^2 \\
& +12699648x^{12}y - 40960x^{11}y^{15} + 641024x^{11}y^{14} - 826112x^{11}y^{13} \\
& -2338816x^{11}y^{12} - 4259328x^{11}y^{11} + 9614592x^{11}y^{10} + 2110208x^{11}y^9 \\
& +16517888x^{11}y^8 - 15051264x^{11}y^7 + 530176x^{11}y^6 + 51936768x^{11}y^5 \\
& -39629824x^{11}y^4 + 37076224x^{11}y^3 - 19216896x^{11}y^2 - 29021184x^{11}y \\
& +206848x^{10}y^{15} - 1079296x^{10}y^{14} - 1127680x^{10}y^{13} + 3244032x^{10}y^{12} \\
& +5066240x^{10}y^{11} + 7447808x^{10}y^{10} - 16982272x^{10}y^9 + 36251136x^{10}y^8 \\
& -75559168x^{10}y^7 + 108455680x^{10}y^6 - 81053952x^{10}y^5 - 3755264x^{10}y^4 \\
& -13176576x^{10}y^3 - 39931392x^{10}y^2 - 24579072x^{10}y - 328704x^9y^{15} \\
& +83456x^9y^{14} + 3420160x^9y^{13} - 410880x^9y^{12} + 1387776x^9y^{11} \\
& -9302016x^9y^{10} - 497408x^9y^9 + 7578112x^9y^8 - 44084992x^9y^7 \\
& +82260224x^9y^6 - 177435904x^9y^5 + 106177024x^9y^4 - 67103232x^9y^3 \\
& +9062400x^9y^2 + 4995072x^9y + 20480x^8y^{16} - 14848x^8y^{15} \\
& +1393152x^8y^{14} - 817920x^8y^{13} - 2052608x^8y^{12} - 4104448x^8y^{11} \\
& -7727104x^8y^{10} + 22859008x^8y^9 - 52080128x^8y^8 + 48469248x^8y^7 \\
& -53207808x^8y^6 - 35127808x^8y^5 + 130954752x^8y^4 - 70410496x^8y^3 \\
& +62765568x^8y^2 + 11123712x^8y - 113664x^7y^{16} + 526080x^7y^{15} \\
& -146432x^7y^{14} - 3578112x^7y^{13} - 389120x^7y^{12} - 484352x^7y^{11} \\
& +10898944x^7y^{10} - 8491520x^7y^9 - 39840000x^7y^8 + 107659264x^7y^7 \\
& -128052992x^7y^6 + 168973056x^7y^5 - 28470016x^7y^4 - 1817088x^7y^3 \\
& +54010368x^7y^2 + 2202624x^7y + 241664x^6y^{16} - 250368x^6y^{15} \\
& -2502400x^6y^{14} + 2179072x^6y^{13} + 4016384x^6y^{12} - 896768x^6y^{11} \\
& -1821952x^6y^{10} - 12418816x^6y^9 + 45208064x^6y^8 + 25299712x^6y^7 \\
& -75835904x^6y^6 + 182561536x^6y^5 - 174477056x^6y^4 + 86125056x^6y^3 \\
& -11796480x^6y^2 - 1400832x^6y - 227072x^5y^{16} - 522240x^5y^{15} \\
& +2157312x^5y^{14} + 2080768x^5y^{13} - 5421824x^5y^{12} + 3268096x^5y^{11} \\
& +2075392x^5y^{10} + 6708736x^5y^9 + 48142080x^5y^8 - 92768256x^5y^7 \\
& +68386304x^5y^6 + 4776192x^5y^5 - 128712704x^5y^4 + 91725312x^5y^3 \\
& -71350272x^5y^2 - 645120x^5y + 66560x^4y^{16} + 563712x^4y^{15} \\
& -226048x^4y^{14} - 2558976x^4y^{13} + 3482368x^4y^{12} + 4677376x^4y^{11} \\
& -4613376x^4y^{10} + 6563584x^4y^9 + 993792x^4y^8 - 94370048x^4y^7 \\
& +142824448x^4y^6 - 180716544x^4y^5 + 52250112x^4y^4 + 887808x^4y^3 \\
& -61507584x^4y^2 - 73728x^4y + 9472x^3y^{16} - 104704x^3y^{15} \\
& -23808x^3y^{14} + 1132032x^3y^{13} + 928000x^3y^{12} - 5678336x^3y^{11} \\
& -630528x^3y^{10} - 253184x^3y^9 - 43420160x^3y^8 + 5501440x^3y^7 \\
& +60112896x^3y^6 - 245845248x^3y^5 + 353068032x^3y^4 - 176430080x^3y^3 \\
& +20520960x^3y^2 + 95473152x^3y - 31707648x^3 + 13056x^2y^{16} \\
& +28160x^2y^{15} - 14336x^2y^{14} - 318976x^2y^{13} - 1461760x^2y^{12} \\
& -3181056x^2y^{11} + 1562112x^2y^{10} - 9056256x^2y^9 - 35820544x^2y^8 \\
& +94595584x^2y^7 - 84554240x^2y^6 - 1142016x^2y^5 + 294133247x^2y^4 \\
& -356297220x^2y^3 + 334379518x^2y^2 - 83331068x^2y + 40352255x^2 \\
& +7936xy^{16} - 33280xy^{15} - 266240xy^{14} - 327424xy^{13} - 1185792xy^{12} \\
& -382464xy^{11} + 898048xy^{10} - 5387264xy^9 + 6728192xy^8 + 86697472xy^7 \\
& -143106048xy^6 + 245446656xy^5 - 59366047xy^4 - 179730236xy^3 + 313945634xy^2 \\
& -178775812xy + 72051841x - 11520y^{16} - 75776y^{15} - 160768y^{14} + 65280y^{13} \\
& -220160y^{12} + 2591744y^{11} + 33024y^{10} + 2594816y^9 + 39006976y^8 - 4819968y^7 \\
& -60738304y^6 + 215570176y^5 - 273807553y^4 + 102135356y^3 - 32047394y^2 \\
& +1744516y - 382369
\end{aligned}$$

$i = 45, m = 864, \text{neighbours } 2, 26, 53, 68, 123$

$$W \quad -x^3y + 1892x^3 + x^2y^2 - 1892x^2y + 519x^2 - 2xy^3 + 1891xy^2 - 518xy - 3018x + 2y^4 - 1892y^3 + 518y^2 + 3017y + 851$$

$$A \quad 373248x^{12} - 559872x^{11} + 653184x^{10} + 419904x^9 + 163296x^8 + 6823440x^7 + 10305144x^6 + 9077508x^5 + 25670988x^4 + 44424216x^3 + 31015008x^2 + 11732148x + 2178972$$

$$B \quad -746496x^{15} + 746496x^{14}y + 1866240x^{14} + 1492992x^{13}y^2 + 4105728x^{13}y + 3545856x^{13} + 746496x^{12}y^3 + 1119744x^{12}y^2 - 933120x^{12}y - 4758912x^{12} - 1492992x^{11}y^3 - 3919104x^{11}y^2 - 3639168x^{11}y - 606528x^{11} - 746496x^{10}y^4 + 1492992x^{10}y^3 + 14276736x^{10}y^2 + 31679424x^{10}y$$

	$+8141472x^{10} + 4478976x^9y^4 + 17915904x^9y^3 + 18242496x^9y^2$ $-163296x^9y - 33370704x^9 + 1492992x^8y^5 + 3732480x^8y^4 + 746496x^8y^3$ $-20318688x^8y^2 - 16154640x^8y - 19799640x^8 - 2985984x^7y^5$ $-13436928x^7y^4 - 15676416x^7y^3 - 3837456x^7y^2 + 24593544x^7y$ $-25418736x^7 - 746496x^6y^6 - 1492992x^6y^5 + 6718464x^6y^4$ $+34338816x^6y^3 + 53147016x^6y^2 + 45509976x^6y - 87486696x^6$ $+2985984x^5y^6 + 7464960x^5y^5 + 18662400x^5y^4 + 25380864x^5y^3$ $+10331423x^5y^2 + 22717408x^5y - 47229268x^5 + 1492992x^4y^7$ $+3732480x^4y^6 - 5971968x^4y^5 - 5971968x^4y^4 + 26127327x^4y^3$ $+7254385x^4y^2 + 199838x^4y - 24383972x^4 - 5971968x^3y^6$ $-27620352x^3y^5 - 40310756x^3y^4 - 14247806x^3y^3 - 1768593x^3y^2$ $-22203250x^3y - 81302449x^3 - 746496x^2y^8 - 3732480x^2y^7$ $-12690432x^2y^6 - 15676436x^2y^5 + 8999604x^2y^4 + 19503523x^2y^3$ $+36016735x^2y^2 + 30065965x^2y - 83072707x^2 - 2239488xy^7 - 13436916xy^6$ $-24657084xy^5 - 2665216xy^4 - 11246160xy^3 - 19425638xy^2 + 25672088xy - 24977663x$ $+746496y^9 + 3732480y^8 + 8211452y^7 + 9712020y^6 + 5368644y^5 + 14495304y^4$ $+14565200y^3 - 3212672y^2 - 52789y - 7261117$
$i = 46, m = 4$, neighbours 3, 10, 18, 26, 59, 80, 187	
W	$-53x^3 - x^2y^3 + 53x^2y - 110x^2 - xy^3 - 54xy^2 + 110xy + 324x + 53y^3$ $-111y^2 - 324y + 121$
A	$6048x^{12} - 6048x^{11} + 504x^{10} - 17388x^9 + 82278x^8 - 90783x^7 + 86593x^6$ $-42728x^5 + 186984x^4 - 206202x^3 - 15284x^2 + 56313x - 11205$
B	$-12096x^{15} + 12096x^{14}y - 9072x^{13}y^2 + 15120x^{13}y + 2016x^{13}$ $+9072x^{12}y^3 - 15120x^{12}y^2 + 22176x^{12}y + 39816x^{12} - 5040x^{11}y^4$ $+16128x^{11}y^3 - 21168x^{11}y^2 - 11592x^{11}y - 118692x^{11} + 5040x^{10}y^5$ $-6048x^{10}y^4 + 43344x^{10}y^3 + 20664x^{10}y^2 + 128268x^{10}y + 110754x^{10}$ $-1008x^9y^6 + 10080x^9y^5 - 17136x^9y^4 + 23184x^9y^3 - 62244x^9y^2$ $+29106x^9y - 7244x^9 + 1008x^8y^7 - 1008x^8y^6 + 26208x^8y^5$ $+5040x^8y^4 + 83664x^8y^3 - 57078x^8y^2 + 41642x^8y + 90434x^8$ $+2016x^7y^7 - 6048x^7y^6 + 14112x^7y^5 - 29232x^7y^4 + 14112x^7y^3$ $-76544x^7y^2 - 164334x^7y - 400273x^7 + 8064x^6y^7 + 7056x^6y^6$ $+26208x^6y^5 + 10080x^6y^4 + 144143x^6y^3 + 126094x^6y^2 + 296482x^6y$ $+231151x^6 - 1008x^5y^8 - 4032x^5y^7 - 13104x^5y^6 - 24192x^5y^5$ $-85679x^5y^4 - 76608x^5y^3 - 114948x^5y^2 - 42430x^5y + 66051x^5$ $+1008x^4y^9 + 2016x^4y^8 - 6048x^4y^7 - 4032x^4y^6 + 14111x^4y^5$ $+17137x^4y^4 + 79954x^4y^3 - 28166x^4y^2 - 75255x^4y - 104939x^4$ $-2016x^3y^9 - 6048x^3y^8 - 22176x^3y^7 - 27216x^3y^6 - 23186x^3y^5$ $-42551x^3y^4 - 80198x^3y^3 + 64764x^3y^2 - 143052x^3y - 412011x^3$ $+3024x^2y^8 + 2016x^2y^7 + 8064x^2y^6 + 28329x^2y^5 + 55109x^2y^4$ $+74018x^2y^3 + 185614x^2y^2 + 427649x^2y + 305985x^2 - 1008xy^{10}$ $-4032xy^9 - 4032xy^8 - 7056xy^7 - 12096xy^6 - 19046xy^5 - 18799xy^4$ $-60116xy^3 - 205187xy^2 - 18438xy + 217168x + 1008y^{11} + 2016y^{10} + 3024y^9$ $+5040y^8 + 7056y^7 + 9072y^6 + 7271y^5 + 20045y^4 + 43076y^3 - 30473y^2$ $-108587y - 18974$
$i = 47, m = 2$, neighbours 15, 44	
W	$2x^2 - xy + 2x - y - 3$
A	0
B	$3xy + 6x - 3y + 3$
$i = 48, m = 16$, neighbours 1, 2, 5, 15, 44	
W	$-30x^3 + 59x^2y + 7x^2 - 117xy^2 - 12xy - 45x - y^4 + 235y^3 + 25y^2 + 90y + 581$
A	$6272x^6 + 26880x^5 + 155008x^4 + 768704x^3 + 863040x^2 + 383296x - 77376$
B	$-5888x^9 + 2560x^8y - 23552x^8 - 768x^7y^2 + 11008x^7y - 143488x^7$ $+512x^6y^3 - 5248x^6y^2 + 64576x^6y - 700736x^6 - 256x^5y^4$ $-27872x^5y^2 + 315200x^5y - 535330x^5 + 256x^4y^5 + 256x^4y^4$ $+1776x^4y^3 - 143152x^4y^2 + 97371x^4y - 234269x^4 - 248x^3y^4$ $-5024x^3y^3 + 20712x^3y^2 - 8804x^3y - 423088x^3 + 252x^2y^5$ $+2734x^2y^4 - 376963x^2y^3 - 3165x^2y^2 - 163607x^2y - 1356695x^2$ $+2xy^6 - 1178xy^5 + 167180xy^4 - 244188xy^3 + 75124xy^2 + 293694xy - 811660x - y^7$ $+471y^6 - 55390y^5 + 44332y^4 - 84559y^3 - 113587y^2 + 78997y - 104665$
$i = 49, m = 2$, neighbours 4, 44	
W	$4x^2 + xy + y - 1$
A	0

$$\begin{aligned}
B & -3xy + 6x + 3y + 3 \\
i = 50, m = 2, \text{ neighbours } 18, 44 \\
W & 6x^2 - 3x + 3y \\
A & 0 \\
B & -6xy + 9x + 3y \\
i = 51, m = 2, \text{ neighbours } 26, 44 \\
W & -2xy + 2y \\
A & 0 \\
B & 4x^3 - 4xy + 4 \\
i = 52, m = 2, \text{ neighbours } 44, 53 \\
W & -2xy - 2x + 2 \\
A & 0 \\
B & -4x^3 + 4xy - 4y - 8 \\
i = 53, m = 293534171136, \text{ neighbours } 1, 30, 37, 45, 52, 66, 88, 105, 109, 134, 172, 173, 214 \\
W & -3x^9 + 10x^8y + 10940x^8 + 5x^7y^2 - 7097x^7y + 39354x^7 - 3x^6y^3 \\
& + 14523x^6y^2 + 44598239x^6y + 436567993x^6 + 27x^5y^4 + 25600x^5y^3 \\
& - 29290628x^5y^2 - 60637084960x^5y - 595776961572x^5 - 36x^4y^5 \\
& - 79811x^4y^4 - 89583913x^4y^3 + 59622887840x^4y^2 - 21852001173x^4y \\
& - 678651361399x^4 - 108x^3y^6 - 37657x^3y^5 + 73050590x^3y^4 \\
& - 176801186x^3y^3 - 248412012629068x^3y^2 - 2440473640223824x^3y \\
& - 1885938988587311x^3 + 64x^2y^7 + 43040x^2y^6 - 103843905x^2y^5 \\
& - 365052445755x^2y^4 - 3575975041225x^2y^3 - 2527653328140772x^2y^2 \\
& - 4659464549923394x^2y + 4142484870650772x^2 - 58xy^8 - 231841xy^7 \\
& - 239322719xy^6 + 239808202877xy^5 + 496738763872581xy^4 + 4880603904372167xy^3 \\
& + 8612051006132567xy^2 + 12033335573661177xy + 15309742862423784x + 258807y^8 \\
& + 653306412y^7 + 733569593363y^6 - 488431542282774y^5 + 179010320090498y^4 \\
& + 5559510399833789y^3 + 9324882527810476y^2 + 10733919751761690y + 9671719947726009 \\
A & -516973857746991213182976x^{30} - 3618817004228938492280832x^{29} \\
& + 258486928873495606591488x^{28}y - 17577111163397701248221184x^{28} \\
& + 1033947715493982426365952x^{27}y - 59451993640903989515994448x^{27} - 3364x^{26}y \\
& - 160520382830440772113308108x^{26} - 19903493523259161677563332x^{25}y \\
& - 339393337610898734505684846x^{25} - 120196421926175523685746841x^{24}y \\
& - 574874929814986548484473463x^{24} - 437876857511702362087497756x^{23}y \\
& - 690160100091725675727949232x^{23} - 1215405539563173956922439249x^{22}y \\
& - 345855511612633758213534110x^{22} - 2680250965493801743618968917x^{21}y \\
& + 1093399701202853742232687385x^{21} - 5022659514159917337655944270x^{20}y \\
& + 4181543042658183661452559767x^{20} - 7896258704492109164946115483x^{19}y \\
& + 9511020832384375918995108755x^{19} - 11092965902236671651174196425x^{18}y \\
& + 17024069396189818415258583898x^{18} - 12923564893267852417478870885x^{17}y \\
& + 27499513566540239573575585665x^{17} - 14363078444453725268592214677x^{16}y \\
& + 41739637228900423361313082183x^{16} - 11187508386545766197396247704x^{15}y \\
& + 56892401945235741598660405592x^{15} - 13627741417501633969067741246x^{14}y \\
& + 70955344135717425558866806980x^{14} - 11363142338286905557061860574x^{13}y \\
& + 1989198788577143521971667960080x^{13} - 314685656546598403087751550832x^{12}y \\
& + 6405806391131908532790876422072x^{12} - 1560753026296437073822459214102x^{11}y \\
& + 5678935058433854299754052802972x^{11} + 2509337680622608546109812798056x^{10}y \\
& - 12876088692430391841868192091383x^{10} + 25692898953652909007752435363396x^9y \\
& - 58849623752622542352963984636766x^9 + 66157262738646293618702696347788x^8y \\
& - 138970571484642511563427936602613x^8 + 84032848634377982630105873396199x^7y \\
& - 260555647410443859185777869919945x^7 + 34703466585854082760257707001918x^6y \\
& - 403455949848511280205609863185932x^6 - 72995949978138743128937629856349x^5y \\
& - 497474290721532421881041554956463x^5 - 160116045784972732747701619262467x^4y \\
& - 477082953720897719507234170567947x^4 - 175052923567831394311145052987555x^3y \\
& - 350915036416426375113663911136427x^3 - 118898572986078173782820879434036x^2y \\
& - 191275083766825217760166162288647x^2 - 53652814585629863017089333401139xy \\
& - 74743116993087334107642486789201x - 13862159127518249003564923011603y \\
& - 18224294345800771061258746770924 \\
B & 1033947715493982426365952x^{30} + 6203686292963894558195712x^{29} \\
& + 516973857746991213182976x^{28}y + 28175075247211021118472192x^{28} \\
& + 2584869288734956065914880x^{27}y + 82888141858767591180241568x^{27} \\
& + 516973857746991213182976x^{26}y^3 - 1033947715493982426365952x^{26}y^2 \\
& + 15767702661283232002087496x^{26}y + 194554495132117694067943672x^{26}
\end{aligned}$$

$$\begin{aligned}
& +3101843146481947279097856x^{25}y^3 - 3704979313853437027811328x^{25}y^2 \\
& +57901072067663015816428512x^{25}y + 324918069593981982747634500x^{25} \\
& +516973857746991213182976x^{24}y^4 + 14216781088042258362531840x^{24}y^3 \\
& -10856451012686815476835768x^{24}y^2 + 197397851349726279042013538x^{24}y \\
& +400310090516086828822713890x^{24} + 689298476995988284243968x^{23}y^4 \\
& +37480604686656862955717968x^{23}y^3 - 29725996820451994818003792x^{23}y^2 \\
& +497587338081478522072338686x^{23}y + 141823161640244761630271726x^{23} \\
& -1292434644367478032957440x^{22}y^5 + 6806822460335384306912548x^{22}y^4 \\
& +78063052519795673610671228x^{22}y^3 - 51783548084323486432240674x^{22}y^2 \\
& +1169739515462856359831008956x^{22}y - 618128407685341039598246838x^{22} \\
& -258486928873495606591488x^{21}y^5 + 29553672201202997656927728x^{21}y^4 \\
& +102791635382025755519024170x^{21}y^3 - 72634827013451058683676906x^{21}y^2 \\
& +2181715842007973486008348029x^{21}y - 2315956706093627001122498267x^{21} \\
& +516973857746991213182976x^{20}y^6 - 4997413958220915060765404x^{20}y^5 \\
& +96070975231315934187319913x^{20}y^4 + 53937605825269398955956437x^{20}y^3 \\
& +10770288703058770464126003x^{20}y^2 + 3997155705782275531577957015x^{20}y \\
& -4608563459289395277144617144x^{20} - 3360330075355442885689344x^{19}y^6 \\
& -14216781088042258392523176x^{19}y^5 + 240220519233101656810972005x^{19}y^4 \\
& -257194494229966447891677321x^{19}y^3 + 49112516493030125471155714x^{19}y^2 \\
& +5725054678275724341806388826x^{19}y - 8120355435127522682524605581x^{19} \\
& -9564016368319337443885056x^{18}y^6 - 27658101389463963194503915x^{18}y^5 \\
& +524814627923152566313845408x^{18}y^4 - 966396463968640338429666777x^{18}y^3 \\
& +346717132345227010629576668x^{18}y^2 + 8678266486747363515172738849x^{18}y \\
& -12381345071005131470329032699x^{18} + 1895570811738967781670912x^{17}y^7 \\
& -15853864970907730537573564x^{17}y^6 - 6203686292963291098912727x^{17}y^5 \\
& +1013785735044873790942777683x^{17}y^4 - 2187144060355112205489351844x^{17}y^3 \\
& -81768029290079316318095421x^{17}y^2 + 8250903317854955409763289859x^{17}y \\
& -21080476421633239332835134115x^{17} + 5083576267845413596295900x^{16}y^7 \\
& -45062887933612734412537436x^{16}y^6 - 2929518527234890568812679x^{16}y^5 \\
& +1717904129288497733376027375x^{16}y^4 - 3687229881856788795668468052x^{16}y^3 \\
& +349042165046167900566239178x^{16}y^2 + 10728138667453214294754329051x^{16}y \\
& -34008179738979689873047238777x^{16} - 3791141623477935533309424x^{15}y^7 \\
& -131225197558110472196372545x^{15}y^6 + 14130618781950032256779044x^{15}y^5 \\
& +2578234798528042303276831256x^{15}y^4 - 5335250666516321936420592366x^{15}y^3 \\
& -2482861353861805756883804203x^{15}y^2 + 738863979072392986804216551x^{15}y \\
& -43094464399884397319418497039x^{15} + 86162309624498535301596x^{14}y^8 \\
& +4911251648596349454298207x^{14}y^7 - 205755595383638030283431358x^{14}y^6 \\
& -12235048758720419819659970x^{14}y^5 + 3320091607057315483975454952x^{14}y^4 \\
& -6707172504529398941660196431x^{14}y^3 + 2165075758025123778348027837x^{14}y^2 \\
& +18132125368986355688098933199x^{14}y - 43401590647360350912579369662x^{14} \\
& +14216781088042258392533268x^{13}y^8 + 48681704937841933902912803x^{13}y^7 \\
& -264690615165616389617755868x^{13}y^6 - 34637247203839038695071604x^{13}y^5 \\
& +3901774289015193050319746094x^{13}y^4 - 7450637552681987060348786333x^{13}y^3 \\
& +60201191045983146094678994x^{13}y^2 + 26903937437936838987729100862x^{13}y \\
& -3851095168689459611412044083041x^{13} + 25245556719978004109562919x^{12}y^8 \\
& +102705473072067676502881752x^{12}y^7 - 298207754401713417916600015x^{12}y^6 \\
& -175340974107011800774942282x^{12}y^5 + 4053111166979793294057664315x^{12}y^4 \\
& -6511608724239570832240147174x^{12}y^3 \\
& +39317717563378704064064819635x^{12}y^2 + 665164291476734996742116098387x^{12}y \\
& -8867737773993653410449552998042x^{12} - 2584869288734956065942488x^{11}y^9 \\
& +32397028418811047022800134x^{11}y^8 + 131397522175338758617054777x^{11}y^7 \\
& -417197907997855285589549198x^{11}y^6 - 508620239553820683799403388x^{11}y^5 \\
& +3688961146622776035209728214x^{11}y^4 - 1972553875355824752751950565x^{11}y^3 \\
& +23763569803851339120062859684x^{11}y^2 - 1313637111697969166738276158693x^{11}y \\
& +1379390214072382188978175765380x^{11} + 3364x^{10}y^{10} \\
& -2412544669485958754749960x^{10}y^9 + 48854029557092028165110873x^{10}y^8 \\
& +204118511503673645923107698x^{10}y^7 - 428054351835429298419460685x^{10}y^6 \\
& -1035502293130788010879731956x^{10}y^5 + 2462259368114080415263725719x^{10}y^4 \\
& -7261004389215233391303286868x^{10}y^3 \\
& +571141103033906308029370514559x^{10}y^2 \\
& -9409078426423941148433425795960x^{10}y + 42722543284660745154828030541905x^{10} \\
& -30032400x^9y^{10} + 2154057740611866194205280x^9y^9 \\
& +101585363044752191786456772x^9y^8 + 343442960890590907714701229x^9y^7
\end{aligned}$$

$$\begin{aligned}
& -392731518074345782466111272x^9y^6 - 1541969457857952481080674686x^9y^5 \\
& + 6012255956727418998799735134x^9y^4 \\
& - 1985061139254438661687917931979x^9y^3 \\
& + 1087437254644362002556187113491x^9y^2 \\
& - 22579535505595303573312061959675x^9y + 99380541819869386184428117469490x^9 \\
& + 3364x^8y^{11} - 1378596953991909497454955x^8y^{10} \\
& - 7668445556243233600105418x^8y^9 + 67895900784222319807601628x^8y^8 \\
& + 305359914155956750281808294x^8y^7 - 253433257819845194687128344x^8y^6 \\
& - 2021049158213570531813742680x^8y^5 + 254884115570032426975425024565x^8y^4 \\
& - 4629819230814208894531095104217x^8y^3 \\
& - 2761408541568939582441534051276x^8y^2 \\
& - 34337340301949617054301800897080x^8y + 133406328425487306570048287164210x^8 \\
& - 30011520x^7y^{11} - 7754607866205130336167175x^7y^{10} \\
& - 62036862930485710216604630x^7y^9 - 119076313172444706266154143x^7y^8 \\
& + 37135698913984936409933297x^7y^7 - 53043113050393114980275363x^7y^6 \\
& + 867751915927938132893991964x^7y^5 - 718737275887554485741556404494x^7y^4 \\
& + 2715640859650928491458501432304x^7y^3 \\
& - 9164561712428337703324881804414x^7y^2 \\
& - 29969548791759460538622789127635x^7y + 151857371236644596944932192220998x^7 \\
& + 66890959517x^6y^{11} - 13958294158832193581929269x^6y^{10} \\
& - 118903986481888194993438925x^6y^9 - 334050582204714973222806649x^6y^8 \\
& - 267643095679968751933613671x^6y^7 - 471305009457201997908929539x^6y^6 \\
& + 251798093057993583088092781478x^6y^5 \\
& - 5270978752599149548749723274299x^6y^4 \\
& + 24609686880058383588790307173153x^6y^3 \\
& - 21814799054303548005019230789682x^6y^2 \\
& - 23621966785450736182119694479964x^6y + 192816636540575961918420916124324x^6 \\
& + 17516x^5y^{12} + 689298476996189497984683x^5y^{11} \\
& - 10339477153926859932836675x^5y^{10} - 112269487032054432055116923x^5y^9 \\
& - 326466903255106397108447546x^5y^8 - 220902484260545487228519930x^5y^7 \\
& - 1694189204197445983431203146x^5y^6 + 744194415829509889047682008638x^5y^5 \\
& - 8275731182663178512814107614153x^5y^4 \\
& + 64284797572788578607895626766053x^5y^3 \\
& - 16963306648221016014107014472555x^5y^2 \\
& + 6258129408554268124109384599668x^5y + 267433568408634670061689135736147x^5 \\
& - 3364x^4y^{13} - 150089552x^4y^{12} + 1723246192489196733160105x^4y^{11} \\
& + 6031361672003615861871061x^4y^{10} - 17490949827064043376812052x^4y^9 \\
& - 85298332014136420092708653x^4y^8 + 2502225915771521405698246x^4y^7 \\
& + 1172670571242301003490245320x^4y^6 \\
& - 1681121081465491999396383367567x^4y^5 \\
& - 12668677588272281423901033988287x^4y^4 \\
& + 85798543126490374929992760703250x^4y^3 \\
& - 15643022042054656158305166033131x^4y^2 \\
& + 27721406905432263401739665255391x^4y + 315589934755297878440231321435670x^4 \\
& + 30032400x^3y^{13} + 396701310329x^3y^{12} \\
& + 2757193909508271524556278x^3y^{11} + 22057554146236421025061269x^3y^{10} \\
& + 69103824537242270750588009x^3y^9 + 110806754463888077868801528x^3y^8 \\
& - 2531147987444925857414019448x^3y^7 \\
& + 1212369606394598850370695099264x^3y^6 \\
& - 3376937199288550604834525411205x^3y^5 \\
& - 4413874383575013560741544476882x^3y^4 \\
& + 90821039822310978089404933962595x^3y^3 \\
& + 9443434332424100573080205183592x^3y^2 \\
& + 69691797553383442300961184494613x^3y + 269909880522152514243738688710752x^3 \\
& - 3364x^2y^{14} - 67071095621x^2y^{13} - 338141589357413x^2y^{12} \\
& + 2412543864307872776079925x^2y^{11} + 22573824245733342959885700x^2y^{10} \\
& + 82304496780312707753148322x^2y^9 + 852959886591960530513221696x^2y^8 \\
& - 243168133365234390405496385518x^2y^7 \\
& + 890235766694719375418012743009x^2y^6 \\
& - 11107463926939132789348056220696x^2y^5 \\
& - 7531025795469664143283481776471x^2y^4 \\
& + 49339161321666818293446366030245x^2y^3 \\
& - 19541436129998763119660072821743x^2y^2
\end{aligned}$$

$$\begin{aligned}
& +25328241076210098942073151474788x^2y + 120608941469757474832195095935323x^2 \\
& +30021612xy^{14} + 262678689579xy^{13} + 849269705438953xy^{12} \\
& +1378598230017345813135979xy^{11} + 12579953254151072396919153xy^{10} \\
& +40278541846467524299163509xy^9 - 1161815242370358685800507757xy^8 \\
& +720451186126294909587217685444xy^7 + 2252085936508310778147618653630xy^6 \\
& -13906204652173855874911843677019xy^5 - 24366492470670260273963471358287xy^4 \\
& -37871393678086842310309867325692xy^3 - 93521597645315833479165039069255xy^2 \\
& -58380673426346594604343218751376xy - 4940053946406114902569884034092x \\
& -66981063249y^{14} - 337959601951221y^{13} - 805501079710266216y^{12} \\
& +343945983581454673468419y^{11} + 2859279999475000919747450y^{10} \\
& +723379928664159089362790247y^9 - 240978836910099815784036269747y^8 \\
& +887493234228471873134556340450y^7 + 3445564401898759980598191553141y^6 \\
& -5896319008126060396122204802013y^5 - 16423112056505353804655213078783y^4 \\
& -31120336305843380979014712226816y^3 - 53274291597567130394745965923528y^2 \\
& -3631695870753769888422318343126y - 12956427903279627026584631585103
\end{aligned}$$

$i = 54, m = 32$, neighbours 6, 44, 87, 118, 120

$$\begin{aligned}
W & 9x^3 + 9x^2y + 32x^2 + 9xy^2 + 32xy + 24x + 9y^3 + 32y^2 + 24y + 108 \\
A & 4608x^8 + 6912x^7 + 10368x^6 + 8640x^5 + 4896x^4 + 4464x^3 - 4032x^2 - 864x - 3888 \\
B & -9216x^{10}y + 32256x^{10} + 3072x^9y^2 - 19968x^9y + 83712x^9 - 6144x^8y^3 \\
& +13824x^8y^2 - 76032x^8y + 122496x^8 + 6144x^7y^4 + 40704x^7y^2 \\
& -118656x^7y + 132288x^7 - 3072x^6y^5 + 3072x^6y^4 - 6144x^6y^3 \\
& +77952x^6y^2 - 105024x^6y + 132000x^6 - 12288x^5y^5 - 9216x^5y^4 \\
& -3072x^5y^3 + 88512x^5y^2 - 111840x^5y + 131055x^5 + 6144x^4y^6 \\
& +15360x^4y^5 + 15360x^4y^4 - 33792x^4y^3 + 32544x^4y^2 - 132723x^4y \\
& +96288x^4 + 3072x^3y^6 - 3072x^3y^5 - 3072x^3y^4 - 6144x^3y^3 \\
& +72858x^3y^2 - 97152x^3y + 54752x^3 - 3072x^2y^7 - 12288x^2y^6 \\
& -24576x^2y^5 - 12288x^2y^4 - 8730x^2y^3 + 57792x^2y^2 - 26352x^2y \\
& +14808x^2 + 3072xy^7 + 18432xy^6 + 27648xy^5 + 12531xy^4 - 19776xy^3 \\
& +9648xy^2 - 13824xy + 14688x - 3072y^7 - 9216y^6 - 12207y^5 - 8640y^4 - 10832y^3 \\
& -9624y^2 - 1728y + 1296
\end{aligned}$$

$i = 55, m = 2$, neighbours 3, 44

$$\begin{aligned}
W & -2xy + 4x + 2y \\
A & 0 \\
B & -4x^2 - 4xy + 4x + 4
\end{aligned}$$

$i = 56, m = 2$, neighbours 1, 2

$$\begin{aligned}
W & -3x^2 - 2xy - 4x - y + 1 \\
A & 0 \\
B & -3x^2 + 6x + 3y + 3
\end{aligned}$$

$i = 57, m = 2$, neighbours 2, 43

$$\begin{aligned}
W & 2y^2 + 6y + 4 \\
A & 0 \\
B & -12x^3 - 24x^2y - 28xy^2 - 28xy - 16x - 16y^3 - 40y^2 - 40y - 12
\end{aligned}$$

$i = 58, m = 4$, neighbours 1, 38, 44, 59

$$\begin{aligned}
W & 4x^2y^2 - x^2 + 6xy - y^2 + 4 \\
A & 0 \\
B & -24x^3y^2 - 15x^3 + 24x^2y^3 - 27x^2y + 27xy^2 - 24x + 15y^3 + 24y
\end{aligned}$$

$i = 59, m = 2$, neighbours 46, 58, 166, 167, 169

$$\begin{aligned}
W & -2x^3y - 2x^2 - 3xy - 2x - y - 1 \\
A & 2x^4y + 2x^4 + xy + x + 4y + 3 \\
B & -4x^8y - 4x^8 + 4x^7y + 4x^7 + 4x^6y^3 + 4x^6y + 4x^6 - 16x^5y - 4x^5 \\
& -12x^4y^3 + 6x^4y^2 - 16x^4 + 4x^3y^2 - 12x^3 + 8x^2y^3 - 12x^2y^2 \\
& -13x^2 - xy^2 + 4xy - 10x + 4y^2 - 7y - 10
\end{aligned}$$

$i = 60, m = 2$, neighbours 1, 5

$$\begin{aligned}
W & 2x^2 - xy - x + y - 1 \\
A & 0 \\
B & 3xy - 3x + 3y + 3
\end{aligned}$$

$i = 61, m = 2$, neighbours 1, 43

$$\begin{aligned}
W & -x^2 - 2xy - y - 2 \\
A & 0 \\
B & 3x^2 + 3y
\end{aligned}$$

$i = 62, m = 160$, neighbours 8, 37, 41, 91, 137, 141, 159, 182, 184, 207

W $84x^3y - 1413x^3 + 251x^2y^2 - 1703x^2y + 813x^2 + 505xy^3 - 871xy^2 + 1600xy$
 $+ 7397x + 757y^4 + 2497y^3 + 2364y^2 + 5735y + 7019$

A $614400x^{13} - 460800x^{12}y + 1843200x^{12} - 1228800x^{11}y + 3456000x^{11}$
 $- 1075200x^{10}y + 3417600x^{10} + 153600x^9y - 1708800x^9 + 2803200x^8y$
 $+ 1814400x^8 + 1113600x^7y - 5592000x^7 + 4924800x^6y - 5258400x^6$
 $+ 11318400x^5y + 37266000x^5 - 6928800x^4y + 36207312x^4 + 20994720x^3y$
 $+ 37333536x^3 + 15985776x^2y + 11857728x^2 - 17615616xy - 21827712x - 10301184y$
 $- 11102976$

B $-409600x^{15} - 1024000x^{14} - 512000x^{13}y^2 - 1075200x^{13}y - 1587200x^{13}$
 $+ 1126400x^{12}y^3 + 1331200x^{12}y^2 - 2816000x^{12}y - 1868800x^{12}$
 $- 614400x^{11}y^4 - 563200x^{11}y^3 + 1638400x^{11}y^2 - 640000x^{11}y$
 $+ 678400x^{11} - 512000x^{10}y^5 - 307200x^{10}y^4 - 921600x^{10}y^3$
 $- 3891200x^{10}y^2 - 499200x^{10}y - 3718400x^{10} + 1126400x^9y^6$
 $+ 1177600x^9y^5 - 819200x^9y^4 + 2406400x^9y^3 + 614400x^9y^2$
 $- 5164800x^9y - 1980800x^9 - 614400x^8y^7 - 870400x^8y^6 + 1177600x^8y^5$
 $+ 665600x^8y^4 + 1075200x^8y^3 + 2163200x^8y^2 - 355200x^8y + 804800x^8$
 $+ 102400x^7y^8 + 307200x^7y^7 - 1484800x^7y^6 - 3276800x^7y^5$
 $- 153600x^7y^4 - 2457600x^7y^3 - 5363200x^7y^2 - 6072000x^7y$
 $- 25605600x^7 - 51200x^6y^8 + 1024000x^6y^7 + 2457600x^6y^6$
 $- 307200x^6y^5 - 2611200x^6y^4 - 3481600x^6y^3 - 7433600x^6y^2$
 $- 24744800x^6y - 23879008x^6 - 204800x^5y^8 - 716800x^5y^7 + 102400x^5y^6$
 $+ 3123200x^5y^5 + 3072000x^5y^4 + 768000x^5y^3 - 7478400x^5y^2$
 $- 23548288x^5y - 5015424x^5 + 102400x^4y^8 - 307200x^4y^7 - 1792000x^4y^6$
 $- 768000x^4y^5 + 2304000x^4y^4 + 2611200x^4y^3 - 7854704x^4y^2$
 $- 6432744x^4y - 7501539x^4 + 102400x^3y^8 + 409600x^3y^7 + 256000x^3y^6$
 $- 1024000x^3y^5 - 1484800x^3y^4 - 12184264x^3y^3 - 812230x^3y^2$
 $- 11348045x^3y - 33881642x^3 - 51200x^2y^8 - 102400x^2y^7 + 204800x^2y^6$
 $+ 409600x^2y^5 - 6418243x^2y^4 - 7241130x^2y^3 - 4897643x^2y^2$
 $- 33003975x^2y - 32001676x^2 - 2483717xy^5 - 8341786xy^4 - 7735704xy^3$
 $- 20913264xy^2 - 42947971xy - 27252974x - 573049y^6 - 3207409y^5 - 6033647y^4$
 $- 11247550y^3 - 17273471y^2 - 22543969y - 15957433$

$i = 63, m = 2$, neighbours 12, 44

W $2x^2 - xy - 2x - y + 2$

A 0

B $-3xy + 3y$

$i = 64, m = 2$, neighbours 15, 44

W $2x^2 - xy + 2x + 2y$

A 0

B $3xy - 6$

$i = 65, m = 2$, neighbours 29, 44

W $x^3 - 3x^2 - 2xy - 2x + y - 1$

A 0

B $-3x^3 - 3x^2 + 3y + 3$

$i = 66, m = 2$, neighbours 44, 53

W $x^3 - 3x^2 - 2xy - 4x + y$

A 0

B $-3x^3 - 3x^2 + 3y + 6$

$i = 67, m = 2$, neighbours 18, 44

W $6x^2 - 3x + 3y$

A 0

B $6xy - 9x - 3y$

$i = 68, m = 2$, neighbours 39, 45, 81, 105, 113, 127, 139, 178, 219

W $-6x^3y - 6x^3 - 12x^2y - 12x^2 + 12xy + 15x - 39y - 54$

A $12x^4y + 18x^4 - 12x^3y - 24x^2y + 36xy + 27x - 99y - 108$

B $40x^7y + 12x^7 - 8x^6y^3 - 4x^6y^2 + 52x^6y - 48x^6 - 40x^5y^5$
 $- 156x^5y^4 - 136x^5y^3 - 28x^5y^2 - 132x^5y - 84x^5 - 16x^4y^7$
 $- 120x^4y^6 - 328x^4y^5 - 364x^4y^4 - 100x^4y^3 - 76x^4y^2 - 22x^4y$
 $+ 90x^4 - 16x^3y^8 - 112x^3y^7 - 260x^3y^6 - 144x^3y^5 + 256x^3y^4$
 $+ 276x^3y^3 + 36x^3y^2 - 16x^3y - 96x^3 + 56x^2y^7 + 428x^2y^6$
 $+ 1012x^2y^5 + 696x^2y^4 - 32x^2y^3 + 84x^2y^2 - 336x^2y - 516x^2$

	$+16xy^9 + 160xy^8 + 612xy^7 + 1028xy^6 + 600xy^5 - 68xy^4 - 72xy^3 - 144xy^2$ $+261xy + 549x + 32y^9 + 192y^8 + 360y^7 + 184y^6 - 32y^5 - 40y^4 - 48y^3 + 219y^2$ $-405y - 864$
$i = 69, m = 2$, neighbours 44, 70	
W	$2x^2 + xy + y + 1$
A	0
B	$3xy - 3y - 3$
$i = 70, m = 2$, neighbours 16, 69, 125, 179, 180, 220	
W	$x^3y + 3x^3 - 2x^2y^2 + 4x^2y + 4x^2 - 6xy^2 - 9xy - 5x + 2y^4 + 3y^3 + 5y^2 - 6y$ -14
A	$-30x^6 - 6x^5y - 69x^5 - 63x^4y - 27x^4 - 5x^3y + 32x^3 + 97x^2y + 62x^2 + 5xy$ $+5x - 56y - 72$
B	$60x^8 - 48x^7y + 78x^7 - 12x^6y^2 - 24x^6y - 84x^6 + 60x^5y^3 + 54x^5y^2$ $+10x^5y - 28x^5 - 48x^4y^4 - 36x^4y^3 + 79x^4y^2 + 46x^4y + 138x^4$ $-12x^3y^5 - 68x^3y^4 - 84x^3y^3 - 113x^3y^2 - 30x^3y + 90x^3$ $+44x^2y^6 + 136x^2y^5 + 192x^2y^4 + 80x^2y^3 - 101x^2y^2 - 65x^2y$ $-21x^2 - 20xy^7 - 84xy^6 - 164xy^5 - 116xy^4 + 89xy^3 + 152xy^2 - 62xy - 103x$ $+20y^7 + 56y^6 + 36y^5 - 57y^4 - 28y^3 + 56y^2 + 32y + 20$
$i = 71, m = 2$, neighbours 44, 72	
W	$2x^2 - xy + 2x - y$
A	0
B	$-3xy + 3y - 6$
$i = 72, m = 6$, neighbours 40, 71, 94, 102, 160, 225	
W	$-x^4 - 2x^3y - 12x^3 - 2x^2y + 8x^2 + 3xy^3 + 5xy^2 - 11xy + 25x + 4y^4 + 5y^3$ $-19y^2 + 31y + 6$
A	$-36x^6 + 108x^5y - 216x^5 + 468x^4y - 404x^4 + 840x^3y - 32x^3 + 648x^2y$ $+708x^2 + 60xy + 920x - 172y + 372$
B	$72x^8 - 144x^7y + 216x^7 - 144x^6y^2 - 144x^6y - 57x^6 - 36x^5y^3$ $-216x^5y^2 - 193x^5y - 545x^5 + 72x^4y^4 - 108x^4y^3 - 32x^4y^2$ $-441x^4y + 226x^4 + 36x^3y^5 - 144x^3y^4 + 383x^3y^3 - 305x^3y^2$ $-241x^3y + 555x^3 - 144x^2y^5 + 111x^2y^4 - 171x^2y^3 - 421x^2y^2$ $-411x^2y - 143x^2 - 36xy^6 + 32xy^5 - 410xy^4 - 387xy^3 + 57xy^2 - 1494xy - 592x$ $+20y^6 - 124y^5 - 197y^4 - 25y^3 - 535y^2 - 532y - 336$
$i = 73, m = 12$, neighbours 35, 74, 75, 77, 78, 79, 81	
W	$12xy + 132x + 12y^2 + 60y + 360$
A	$144xy^3 + 720xy^2 + 1152xy + 1440x + 288y^3 + 1440y^2 + 3024y + 4752$
B	$-288x^9y^3 - 576x^9y^2 + 288x^9y + 576x^9 + 144x^8y^4 - 1728x^8y^3$ $-4752x^8y^2 + 288x^8y + 2592x^8 + 864x^7y^4 - 4320x^7y^3 - 16272x^7y^2$ $-3024x^7y + 4320x^7 + 1728x^6y^4 - 7776x^6y^3 - 32832x^6y^2 - 8928x^6y$ $+4320x^6 + 288x^5y^5 + 1728x^5y^4 - 13824x^5y^3 - 49104x^5y^2$ $-17424x^5y + 864x^5 + 1152x^4y^5 + 2016x^4y^4 - 18720x^4y^3$ $-54432x^4y^2 - 25344x^4y - 4896x^4 + 1440x^3y^5 + 1296x^3y^4$ $-19296x^3y^3 - 43200x^3y^2 - 21456x^3y - 5184x^3 + 144x^2y^6$ $+1296x^2y^5 - 432x^2y^4 - 16848x^2y^3 - 29808x^2y^2 - 18864x^2y$ $-6624x^2 + 288xy^6 + 1440xy^5 + 864xy^4 - 5760xy^3 - 9072xy^2 + 1296xy + 12096x$ $-864y^4 - 4464y^3 - 9216y^2 - 20160y - 47952$
$i = 74, m = 2$, neighbours 10, 73	
W	$2x^2 - xy + 3x - 3y$
A	0
B	$3xy + 3x + 3y$
$i = 75, m = 2$, neighbours 73, 76	
W	$2x^2 - 2xy + 4x - 3y$
A	0
B	$6x - 3y + 6$
$i = 76, m = 384$, neighbours 13, 37, 75, 81, 124, 125, 151, 163, 170, 199, 223	
W	$390x^3y + 2174x^3 + 781x^2y^2 + 1704x^2y - 12690x^2 + 1172xy^3 + 1237xy^2$ $-1146xy - 19218x + 1561y^4 + 773y^3 + 10411y^2 + 21998y - 10448$
A	$7077888x^9 - 7077888x^8y + 21233664x^8 - 16809984x^7y + 20348928x^7$ $-4423680x^6y + 31629312x^6 - 23445504x^5y + 59940864x^5 - 64253952x^4y$ $+77197312x^4 - 20951040x^3y + 240803840x^3 - 209080320x^2y + 418062336x^2$

$$\begin{aligned}
& -376012800xy + 427409408x + 42106880y + 253149184 \\
B \quad & -4718592x^{13}y^3 + 11796480x^{13}y^2 - 7077888x^{13}y + 18874368x^{12}y^4 \\
& -63700992x^{12}y^3 + 76677120x^{12}y^2 - 31850496x^{12}y - 24772608x^{11}y^5 \\
& +126812160x^{11}y^4 - 219414528x^{11}y^3 + 147456000x^{11}y^2 - 30081024x^{11}y \\
& -14155776x^{11} + 3538944x^{10}y^6 - 91422720x^{10}y^5 + 255983616x^{10}y^4 \\
& -245956608x^{10}y^3 + 30965760x^{10}y^2 + 32735232x^{10}y - 28311552x^{10} \\
& +21233664x^9y^7 - 44826624x^9y^6 - 51609600x^9y^5 + 134332416x^9y^4 \\
& +26247168x^9y^3 - 188891136x^9y^2 + 66355200x^9y + 1769472x^9 \\
& -21233664x^8y^8 + 126222336x^8y^7 - 202604544x^8y^6 + 102776832x^8y^5 \\
& -85082112x^8y^4 + 259817472x^8y^3 - 265715712x^8y^2 + 75202560x^8y \\
& -43794432x^8 + 8257536x^7y^9 - 90243072x^7y^8 + 248020992x^7y^7 \\
& -219561984x^7y^6 + 42909696x^7y^5 - 121061376x^7y^4 + 357433344x^7y^3 \\
& -304496640x^7y^2 + 57507840x^7y - 120324096x^7 - 1179648x^6y^{10} \\
& +28901376x^6y^9 - 127696896x^6y^8 + 165593088x^6y^7 + 33177600x^6y^6 \\
& -152027136x^6y^5 - 109559808x^6y^4 + 410222592x^6y^3 - 335462400x^6y^2 \\
& -24993792x^6y - 74326016x^6 - 3538944x^5y^{10} + 30081024x^5y^9 \\
& -62226432x^5y^8 - 39960576x^5y^7 + 267632640x^5y^6 - 269991936x^5y^5 \\
& -76087296x^5y^4 + 318504960x^5y^3 - 173260800x^5y^2 - 141918208x^5y \\
& -379633664x^5 - 2359296x^4y^{10} + 9437184x^4y^9 + 16515072x^4y^8 \\
& -152764416x^4y^7 + 358170624x^4y^6 - 335757312x^4y^5 + 18874368x^4y^4 \\
& +109707264x^4y^3 + 38137308x^4y^2 - 458350568x^4y - 541720068x^4 \\
& -3538944x^3y^9 + 34799616x^3y^8 - 161316864x^3y^7 + 378814464x^3y^6 \\
& -338116608x^3y^5 + 32882688x^3y^4 + 38793244x^3y^3 - 149571160x^3y^2 \\
& -353030312x^3y - 135308868x^3 - 2359296x^2y^9 + 25952256x^2y^8 \\
& -140378112x^2y^7 + 290193408x^2y^6 - 148193280x^2y^5 - 91430717x^2y^4 \\
& -16283332x^2y^3 - 85688856x^2y^2 - 55304044x^2y - 379141444x^2 \\
& +14155776xy^8 - 82575360xy^7 + 126222336xy^6 + 9752118xy^5 - 122329489xy^4 \\
& -121317252xy^3 + 35174048xy^2 - 537066696xy - 1068348480x + 4718592y^8 \\
& -21233664y^7 + 17617295y^6 + 21692053y^5 - 69133945y^4 - 134901304y^3 \\
& -228270244y^2 - 788407872y - 868608256 \\
i = 77, m = 2, \text{ neighbours } 15, 73 \\
W \quad & 2x^2 - xy + 6x - 3y + 3 \\
A \quad & 0 \\
B \quad & 3xy + 6x + 3y + 9 \\
i = 78, m = 2, \text{ neighbours } 29, 73 \\
W \quad & -2xy - 4x - 4y - 2 \\
A \quad & 0 \\
B \quad & -4x^3 - 8x^2 + 4xy - 4x + 4y - 4 \\
i = 79, m = 2, \text{ neighbours } 73, 80 \\
W \quad & -2xy - 4x - 2y - 2 \\
A \quad & 0 \\
B \quad & -4x^3 - 8x^2 - 4xy - 12x - 8y - 8 \\
i = 80, m = 243, \text{ neighbours } 13, 46, 79, 108, 133, 143, 156, 161, 167, 168, 211 \\
W \quad & 192x^5y - 1109x^5 + 191x^4y^2 - 1845x^4y + 56x^4 + x^3y^4 - 4x^3y^3 \\
& -790131x^3y^2 + 4541919x^3y + 11148847x^3 + x^2y^5 - 7x^2y^4 + 2x^2y^3 \\
& +3019058x^2y^2 - 11334083x^2y - 23123487x^2 + xy^6 - 1164xy^5 - 1572132xy^4 \\
& +9084901xy^3 + 2338794xy^2 - 41994792xy - 44534492x - 2316y^6 - 1561056y^5 \\
& +15112934y^4 - 455532y^3 - 43947009y^2 - 30016412y - 9255337 \\
A \quad & -59049x^{31} - 590490x^{30} + 59049x^{29}y - 2066715x^{29} - 236196x^{28}y - 3365793x^{28} \\
& -2243862x^{27}y - 767637x^{27} - 5550606x^{26}y + 9093546x^{26} - 4369626x^{25}y \\
& +22025277x^{25} + 6436341x^{24}y + 29052108x^{24} + 20430954x^{23}y + 46176318x^{23} \\
& +28048275x^{22}y + 74933181x^{22} + 32122656x^{21}y - 95009841x^{21} - 11632653x^{20}y \\
& -389723401x^{20} - 139001346x^{19}y + 929967326x^{19} + 140477571x^{18}y \\
& +2490140647x^{18} + 960904370x^{17}y - 9009930494x^{17} - 1708902413x^{16}y \\
& -8439048237x^{16} - 6944717299x^{15}y + 87741133816x^{15} + 24371284916x^{14}y \\
& -4711017378x^{14} + 93224320897x^{13}y + 8250074356874x^{13} - 2487264081388x^{12}y \\
& +83145843295380x^{12} - 532209078364x^{11}y + 300102152933109x^{11} \\
& +28387617022872x^{10}y + 762425909485976x^{10} + 297442127511064x^9y \\
& +745934850736692x^9 + 1013857839106212x^8y - 749648079547867x^8 \\
& +1933136312879571x^7y - 4796474534584353x^7 + 1004192831259779x^6y \\
& -10171533204181081x^6 - 3052360237084961x^5y - 14995681629225872x^5 \\
& -7991656082510907x^4y - 15915006743850623x^4 - 8693758437574516x^3y
\end{aligned}$$

$$\begin{aligned}
& -12019548483640088x^3 - 4907968149232036x^2y - 5542374093979348x^2 \\
& -1049405036163536xy - 1144253332432496x + 4669413199776y - 32038771778208 \\
B \quad & 118098x^{31} + 1062882x^{30} - 236196x^{29}y + 2834352x^{29} - 590490x^{28}y \\
& + 1594323x^{28} + 59049x^{27}y^3 + 118098x^{27}y^2 + 413343x^{27}y - 7203978x^{27} \\
& + 531441x^{26}y^3 - 118098x^{26}y^2 + 3483891x^{26}y - 17065161x^{26} \\
& - 118098x^{25}y^4 + 1535274x^{25}y^3 - 1771470x^{25}y^2 + 6318243x^{25}y \\
& - 9034497x^{25} - 295245x^{24}y^4 + 1121931x^{24}y^3 - 2716254x^{24}y^2 \\
& + 8207811x^{24}y + 8739252x^{24} + 59049x^{23}y^5 - 59049x^{23}y^4 \\
& - 3188646x^{23}y^3 + 1771470x^{23}y^2 + 1476225x^{23}y - 48479229x^{23} \\
& - 59049x^{22}y^5 + 1299078x^{22}y^4 - 8503056x^{22}y^3 + 2184813x^{22}y^2 \\
& - 35075106x^{22}y - 132978348x^{22} + 59049x^{21}y^6 - 708588x^{21}y^5 \\
& + 3897234x^{21}y^4 - 9211644x^{21}y^3 - 16710867x^{21}y^2 - 21611934x^{21}y \\
& + 316738836x^{21} + 177147x^{20}y^6 - 767637x^{20}y^5 + 4605822x^{20}y^4 \\
& - 6672537x^{20}y^3 + 8503056x^{20}y^2 + 248950584x^{20}y + 684259814x^{20} \\
& - 118098x^{19}y^7 + 531441x^{19}y^6 + 1003833x^{19}y^5 - 2539107x^{19}y^4 \\
& - 1240029x^{19}y^3 + 133509789x^{19}y^2 + 57395628x^{19}y - 3076698348x^{19} \\
& - 118098x^{18}y^7 + 1358127x^{18}y^6 + 354294x^{18}y^5 - 10865016x^{18}y^4 \\
& + 7735419x^{18}y^3 - 139355640x^{18}y^2 - 1993671389x^{18}y - 3904989756x^{18} \\
& + 59049x^{17}y^8 + 649539x^{17}y^7 + 413343x^{17}y^6 - 6022998x^{17}y^5 \\
& - 5432508x^{17}y^4 + 15884181x^{17}y^3 - 960018642x^{17}y^2 + 563927214x^{17}y \\
& + 27062368441x^{17} + 236196x^{16}y^8 + 1003833x^{16}y^7 - 5255361x^{16}y^6 \\
& - 7381125x^{16}y^5 + 10983114x^{16}y^4 + 15352741x^{16}y^3 \\
& + 1700375004x^{16}y^2 + 15710478077x^{16}y + 3329146237x^{16} + 59049x^{15}y^9 \\
& + 177147x^{15}y^8 - 2243862x^{15}y^7 - 8266860x^{15}y^6 + 4605822x^{15}y^5 \\
& + 22084326x^{15}y^4 + 2121138x^{15}y^3 + 6879798982x^{15}y^2 - 12806872298x^{15}y \\
& - 227329338369x^{15} + 59049x^{14}y^9 - 1358127x^{14}y^8 - 6790635x^{14}y^7 \\
& - 590490x^{14}y^6 + 21789081x^{14}y^5 + 18482336x^{14}y^4 - 15510529x^{14}y^3 \\
& - 17379496241x^{14}y^2 - 120037687348x^{14}y + 181420459399x^{14} - 59049x^{13}y^{10} \\
& - 649539x^{13}y^9 - 4664871x^{13}y^8 - 5019165x^{13}y^7 + 14171760x^{13}y^6 \\
& + 30351186x^{13}y^5 - 4364994x^{13}y^4 - 789123x^{13}y^3 - 46698119186x^{13}y^2 \\
& + 101601044180x^{13}y - 16225148972531x^{13} - 236196x^{12}y^{10} - 2421009x^{12}y^9 \\
& - 5727753x^{12}y^8 + 4369626x^{12}y^7 + 30469284x^{12}y^6 + 21080493x^{12}y^5 \\
& - 49001055x^{12}y^4 - 7228727962x^{12}y^3 + 178522340106x^{12}y^2 \\
& + 5410042747112x^{12}y - 150040459237214x^{12} - 590490x^{11}y^{10} \\
& - 3601989x^{11}y^9 - 2361960x^{11}y^8 + 19427121x^{11}y^7 + 40035222x^{11}y^6 \\
& - 16238479x^{11}y^5 - 75468011x^{11}y^4 - 237416335x^{11}y^3 \\
& + 341133941079x^{11}y^2 + 11929826759535x^{11}y - 418107564913022x^{11} \\
& - 59049x^{10}y^{11} - 1180980x^{10}y^{10} - 3188646x^{10}y^9 + 5137263x^{10}y^8 \\
& + 36019890x^{10}y^7 + 23088160x^{10}y^6 - 69893198x^{10}y^5 + 30128841x^{10}y^4 \\
& + 88914992332x^{10}y^3 - 6166206844282x^{10}y^2 + 102358064557187x^{10}y \\
& - 782630401470294x^{10} - 236196x^9y^{11} - 1594323x^9y^{10} - 1417176x^9y^9 \\
& + 18659484x^9y^8 + 41334300x^9y^7 - 24982356x^9y^6 - 125680701x^9y^5 \\
& - 32099695120x^9y^4 - 9030385078263x^9y^3 - 1647345386353x^9y^2 \\
& + 12572690125421x^9y + 335385820511287x^9 - 295245x^8y^{11} - 1476225x^8y^{10} \\
& + 4782969x^8y^9 + 31473117x^8y^8 + 17478503x^8y^7 - 65806384x^8y^6 \\
& + 7070905532x^8y^5 + 2266679193165x^8y^4 - 74568411761691x^8y^3 \\
& + 28609128162567x^8y^2 - 430499587810055x^8y + 3165887034778264x^8 \\
& - 354294x^7y^{11} + 13463172x^7y^9 + 26217756x^7y^8 - 23969262x^7y^7 \\
& - 47749397x^7y^6 + 25058812137x^7y^5 + 6841703771486x^7y^4 \\
& - 206282441697394x^7y^3 + 164168271313132x^7y^2 - 1938367448294691x^7y \\
& + 6076926383370237x^7 - 59049x^6y^{12} - 472392x^6y^{11} + 3306744x^6y^{10} \\
& + 15588936x^6y^9 + 3070548x^6y^8 - 48882966x^6y^7 - 7233316351x^6y^6 \\
& - 2330736545230x^6y^5 + 58919468342642x^6y^4 - 504374497083930x^6y^3 \\
& + 742955222028152x^6y^2 - 2752951298828406x^6y + 5013012131915676x^6 \\
& - 118098x^5y^{12} + 413343x^5y^{11} + 5550606x^5y^{10} + 6908733x^5y^9 \\
& - 15470839x^5y^8 - 27648810x^5y^7 - 19127918x^5y^6 - 13808527644x^5y^5 \\
& + 581639960339x^5y^4 + 98507371597381x^5y^3 + 1014530523454434x^5y^2 \\
& - 2148758054860531x^5y + 4669181998311709x^5 + 59049x^4y^{12} \\
& + 1358127x^4y^{11} + 3070548x^4y^{10} - 4015331x^4y^9 - 15168662x^4y^8 \\
& + 18680064x^4y^7 + 14169873234x^4y^6 + 2050652152299x^4y^5 \\
& - 110761074099587x^4y^4 + 1449012822625724x^4y^3 + 453550252946125x^4y^2 \\
& + 221512360512374x^4y + 8860510928182214x^4 + 177147x^3y^{12} + 590490x^3y^{11} \\
& - 1180980x^3y^{10} - 6086679x^3y^9 - 17698422x^3y^8 - 10668120514x^3y^7
\end{aligned}$$

$$\begin{aligned}
& -2216043580592x^3y^6 + 76792223412136x^3y^5 - 789580530944275x^3y^4 \\
& + 3475062399544332x^3y^3 - 611333588856178x^3y^2 + 51628778620239x^3y \\
& + 14298175668885927x^3 - 413343x^2y^{11} - 1771471x^2y^{10} + 3531035x^2y^9 \\
& + 7210422213x^2y^8 + 2341749577512x^2y^7 - 54085790670469x^2y^6 \\
& + 417598958233080x^2y^5 - 1434039193532118x^2y^4 + 2897446589880497x^2y^3 \\
& + 493895235829310x^2y^2 - 1743268445593604x^2y + 10100062799958172x^2 \\
& - 59049xy^{12} - 295245xy^{11} - 408711xy^{10} + 8744157xy^9 + 3488812034xy^8 \\
& - 118755193450xy^7 - 22956642547944xy^6 + 336142692022962xy^5 \\
& - 1293294920158497xy^4 + 1930211098609985xy^3 + 992055383278177xy^2 \\
& - 3306037524497069xy + 2619525201205978x - 5363856y^{10} - 7220083680y^9 \\
& - 2352447193632y^8 + 51894277043328y^7 - 331462665450652y^6 \\
& + 487453383776360y^5 + 590927410700592y^4 - 830970834751248y^3 \\
& - 612591032368805y^2 - 558503589704918y + 10455052351055
\end{aligned}$$

$i = 81, m = 1, \text{ neighbours } 68, 73, 76$

$$\begin{aligned}
W & -y^2 - 3y - 3 \\
A & 0 \\
B & x^3y^2 + 3x^3y + 3x^3 - 2x^2y^3 - 4x^2y^2 + 6x^2 + xy^4 - 3xy^2 + 9x + y^4 \\
& - 6y^2 - 9y
\end{aligned}$$

$i = 82, m = 64, \text{ neighbours } 2, 83$

$$\begin{aligned}
W & -64y^6 + 368y + 329 \\
A & -4096y^9 + 4096y^7 + 10752y^6 - 4096y^5 + 25600y^4 + 17984y^3 - 14848y^2 \\
& - 81088y - 36456 \\
B & 4096x^3y^9 - 4096x^3y^7 - 10752x^3y^6 + 4096x^3y^5 - 25600x^3y^4 \\
& - 17984x^3y^3 + 14848x^3y^2 + 81088x^3y + 42600x^3 - 2048x^2y^9 \\
& + 2048x^2y^7 + 5376x^2y^6 - 2048x^2y^5 + 12800x^2y^4 + 8992x^2y^3 \\
& - 7424x^2y^2 - 28256x^2y - 2868x^2 + 1024xy^9 - 1024xy^7 - 2688xy^6 \\
& + 1024xy^5 - 6400xy^4 - 4496xy^3 + 18048xy^2 + 34608xy + 21914x + 1536y^9 - 1536y^7 \\
& - 4032y^6 + 1536y^5 - 9600y^4 + 1448y^3 + 6592y^2 + 49864y + 13415
\end{aligned}$$

$i = 83, m = 6, \text{ neighbours } 82, 102, 103, 138, 139$

$$\begin{aligned}
W & -48xy^2 + 54xy - 6x - 12y^4 + 6y^3 - 42y^2 + 30y + 18 \\
A & 144x^4 + 30x^3y + 501x^3 - 72x^2y^2 + 108x^2y - 36x^2 - 72xy^4 + 36xy^3 \\
& - 180xy^2 + 306xy - 1854x - 174y^4 + 597y^3 - 1197y^2 + 1215y - 2025 \\
B & -288x^6 + 228x^5y - 138x^5 + 60x^4y^2 - 546x^4y + 1278x^4 - 288x^3y^3 \\
& + 504x^3y^2 - 324x^3y - 36x^3 + 72x^2y^4 - 252x^2y^3 - 252x^2y^2 \\
& + 438x^2y - 681x^2 + 216xy^5 - 858xy^4 + 1407xy^3 - 468xy^2 - 360xy + 684x - 144y^6 \\
& + 318y^5 - 1137y^4 + 1569y^3 + 357y^2 - 900y + 1917
\end{aligned}$$

$i = 84, m = 18, \text{ neighbours } 37, 85, 86, 88, 190$

$$\begin{aligned}
W & 18xy + 210x - 18y^2 + 18y - 12 \\
A & 13122x^3y^3 - 15309x^3y^2 - 62694x^3y + 2916x^3 - 39366x^2y^3 \\
& + 4374x^2y^2 + 42282x^2y - 9720x^2 + 30618xy^3 + 5832xy^2 - 3402xy - 2592x \\
& - 8748y^3 - 3645y^2 - 5832y + 1512 \\
B & -2916x^5y^3 - 5346x^5y^2 + 8100x^5y + 7128x^5 - 2916x^4y^3 \\
& - 5346x^4y^2 + 8100x^4y + 7128x^4 - 1944x^3y^3 - 3564x^3y^2 + 5400x^3y \\
& + 4752x^3 + 2106x^2y^3 + 14013x^2y^2 + 14814x^2y - 7356x^2 - 1458xy^6 \\
& + 1701xy^5 + 8424xy^4 + 11799xy^3 + 15372xy^2 + 16362xy - 912x + 2916y^6 + 1215y^5 \\
& + 3888y^4 + 7515y^3 + 6633y^2 + 5472y - 1560
\end{aligned}$$

$i = 85, m = 2, \text{ neighbours } 12, 84$

$$\begin{aligned}
W & 6x^2 - 3xy - 6x + 3 \\
A & 0 \\
B & -9xy + 6x + 6y - 3
\end{aligned}$$

$i = 86, m = 2, \text{ neighbours } 84, 87$

$$\begin{aligned}
W & 6x^2 + 3xy + 6x - 3y - 6 \\
A & 0 \\
B & -9xy - 12x + 3y
\end{aligned}$$

$i = 87, m = 12, \text{ neighbours } 0, 37, 54, 86, 105, 126, 131, 142, 143, 144, 203$

$$\begin{aligned}
W & x^4y^2 + 2x^4 + 568x^3y - 12361x^3 - x^2y^4 - 556x^2y^2 - 5587x^2y \\
& + 17121x^2 - 21xy^3 + 17917xy^2 + 24460xy + 54945x - y^5 + 572y^4 - 12317y^3 \\
& - 41597y^2 - 34566y + 16425 \\
A & -15552x^{15} + 7776x^{14}y - 186624x^{14} + 46656x^{13}y - 1071792x^{13} + 15552x^{12}y \\
& - 3674808x^{12} - 519696x^{11}y - 7768548x^{11} - 1301184x^{10}y - 9703638x^{10} \\
& + 26244x^9y - 7025373x^9 + 1617408x^8y - 5314206x^8 - 10021887x^7y - 1470154x^7
\end{aligned}$$

$$\begin{aligned}
& -30327845x^6y + 17614320x^6 - 9512369x^5y - 33904517x^5 + 21166693x^4y \\
& -298214356x^4 - 108236684x^3y - 356647752x^3 - 72900008x^2y - 18559764x^2 \\
& + 89931300xy + 157732596x - 17706012y - 33270768 \\
B \quad & 31104x^{17} - 46656x^{16}y + 279936x^{16} + 15552x^{15}y^2 - 326592x^{15}y \\
& + 1210464x^{15} + 31104x^{14}y^3 + 139968x^{14}y^2 - 961632x^{14}y + 2893968x^{14} \\
& - 46656x^{13}y^4 + 62208x^{13}y^3 + 264384x^{13}y^2 - 1613520x^{13}y \\
& + 3457080x^{13} + 15552x^{12}y^5 - 163296x^{12}y^4 + 184032x^{12}y^3 \\
& + 383616x^{12}y^2 - 1694520x^{12}y + 1985796x^{12} + 31104x^{11}y^6 \\
& + 228960x^{11}y^5 + 101376x^{11}y^4 + 1033632x^{11}y^3 + 1307376x^{11}y^2 \\
& - 1480428x^{11}y + 4626126x^{11} - 63936x^{10}y^7 - 188928x^{10}y^6 \\
& - 26208x^{10}y^5 - 232992x^{10}y^4 + 1419840x^{10}y^3 + 1008000x^{10}y^2 \\
& + 377370x^{10}y + 9654738x^{10} - 31104x^9y^8 - 344160x^9y^7 - 966816x^9y^6 \\
& - 1110240x^9y^5 - 1505376x^9y^4 + 1508256x^9y^3 - 4213764x^9y^2 \\
& + 11205228x^9y - 6827338x^9 + 39744x^8y^9 + 158400x^8y^8 + 46368x^8y^7 \\
& - 465120x^8y^6 - 801792x^8y^5 - 1655424x^8y^4 + 2595744x^8y^3 \\
& - 10712736x^8y^2 + 17134238x^8y - 7667865x^8 + 25920x^7y^{10} \\
& + 200160x^7y^9 + 649152x^7y^8 + 959328x^7y^7 + 667584x^7y^6 \\
& - 715392x^7y^5 - 2077056x^7y^4 + 1761693x^7y^3 - 1745200x^7y^2 \\
& - 33416852x^7y + 137553640x^7 + 3456x^6y^{11} + 38304x^6y^{10} + 218880x^6y^9 \\
& + 823392x^6y^8 + 1535328x^6y^7 + 1656000x^6y^6 - 501696x^6y^5 \\
& - 1007710x^6y^4 + 920729x^6y^3 + 22291423x^6y^2 - 98144136x^6y \\
& + 224159963x^6 + 576x^5y^{11} - 41184x^5y^{10} + 53568x^5y^9 \\
& + 1022976x^5y^8 + 2796768x^5y^7 + 3780864x^5y^6 + 1469953x^5y^5 \\
& + 702137x^5y^4 + 326301x^5y^3 + 6305344x^5y^2 + 35867077x^5y \\
& - 384992807x^5 - 71424x^4y^{11} - 428544x^4y^{10} - 739584x^4y^9 \\
& + 14976x^4y^8 + 1590048x^4y^7 + 2072735x^4y^6 + 456197x^4y^5 \\
& - 96983x^4y^4 + 294751x^4y^3 - 21446002x^4y^2 + 142442651x^4y \\
& + 347879020x^4 - 29376x^3y^{12} - 305856x^3y^{11} - 1133856x^3y^{10} \\
& - 1983168x^3y^9 - 1649952x^3y^8 - 290880x^3y^7 + 508322x^3y^6 \\
& + 230359x^3y^5 - 299767x^3y^4 - 1051078x^3y^3 + 182856070x^3y^2 \\
& - 23819586x^3y + 618880762x^3 - 3456x^2y^{13} - 68256x^2y^{12} \\
& - 376992x^2y^{11} - 950976x^2y^{10} - 1204704x^2y^9 - 589824x^2y^8 \\
& + 215422x^2y^7 + 386206x^2y^6 + 109665x^2y^5 + 259532x^2y^4 \\
& + 12238316x^2y^3 - 853490401x^2y^2 - 906238682x^2y - 517291041x^2 \\
& - 4032xy^{13} - 26496xy^{12} - 65664xy^{11} - 83520xy^{10} - 46944xy^9 + 43488xy^8 \\
& + 59617xy^7 + 36251xy^6 + 431941xy^5 - 36301455xy^4 + 622458053xy^3 \\
& + 1210747711xy^2 + 1108278029xy - 691633983x + 1728y^{13} + 14688y^{12} + 44064y^{11} \\
& + 50976y^{10} + 6048y^9 - 26785y^8 - 16997y^7 - 350936y^6 + 15074101y^5 \\
& - 148344596y^4 - 585304883y^3 - 381945004y^2 + 463950075y - 56656107
\end{aligned}$$

$i = 88, m = 2$, neighbours 53, 84

$$W \quad 3x^3 - 6x^2 - 6xy - 9x + 3y + 3$$

$$A \quad 0$$

$$B \quad -9x^3 - 6x^2 - 3x + 3y + 9$$

$i = 89, m = 2$, neighbours 2, 15

$$W \quad -2x^2 + 2xy - 2x + 2y^2 - 2y - 4$$

$$A \quad 0$$

$$B \quad -8x^2 - 12x - 12y - 4$$

$i = 90, m = 2$, neighbours 2, 23

$$W \quad -x^3 + x^2y + 3x^2 + 5xy + 4x + 2y^2 + 2y + 2$$

$$A \quad 0$$

$$B \quad 11x^3 + 3x^2y - 5x^2 - 13xy - 2x - 8y^2 - 8y - 8$$

$i = 91, m = 2$, neighbours 2, 62

$$W \quad -6x^2 + 2y^2 + 2y + 2$$

$$A \quad 0$$

$$B \quad 6xy + 12x$$

$i = 92, m = 2$, neighbours 2, 15

$$W \quad -2x^2 + 2xy - 2x + 2y^2 - 2y - 4$$

$$A \quad 0$$

$$B \quad 8x^2 + 12x + 12y + 4$$

$i = 93, m = 2$, neighbours 2, 4

$$W \quad -6x^2 - 4xy - 5x - y^2 - 4y - 1$$

A	0
B	$6x^2 + 6xy - 3x + 3y^2 - 3$
$i = 94, m = 2, \text{ neighbours } 2, 72$	
W	$-3x^2 + xy - 7x + y^2 - 4y$
A	0
B	$3x^2 + 3xy - 3x + 3y^2 - 6$
$i = 95, m = 2, \text{ neighbours } 2, 43$	
W	$4xy + 5x + 2y^2 + 2y - 1$
A	0
B	$-6x^2 - 6xy + 3x + 6y + 3$
$i = 96, m = 2, \text{ neighbours } 10, 97$	
W	$6x^2 - 3xy + 3y^2 - 3y$
A	0
B	$-3xy - 6x - 3y^2 + 3y$
$i = 97, m = 9, \text{ neighbours } 96, 105, 140, 207, 208$	
W	$24xy^4 - 6xy^3 + 54xy^2 - 24xy - 48x - 6y^4 + 24y^3 - 54y^2 + 60y - 24$
A	$-3888x^3y^2 + 5589x^3y + 13608x^3 + 1215xy^3 + 1458xy^2 - 6561xy + 3888x$ $+ 864y^6 - 2592y^5 + 1944y^4 + 864y^3 - 1296y^2 + 216$
B	$864x^5y^2 - 1242x^5y - 3024x^5 - 288x^3y^4 + 990x^3y^3 - 108x^3y^2$ $- 630x^3y + 36x^3 - 192x^2y^6 + 576x^2y^5 - 306x^2y^4 - 804x^2y^3$ $+ 990x^2y^2 - 801x^2y - 1164x^2 + 96xy^6 - 216xy^5 + 279xy^4 - 2118xy^3$ $+ 999xy^2 - 2817xy - 4728x - 288y^7 + 276y^6 - 54y^5 - 702y^4 + 1158y^3 - 54y^2 + 72y$ $- 408$
$i = 98, m = 10, \text{ neighbours } 10, 23$	
W	$5x^3 + 5x^2y - 10xy - 5x + 5y^2 + 10y$
A	0
B	$75x^3 - 75x^2y + 75x + 75y^2$
$i = 99, m = 2, \text{ neighbours } 10, 42$	
W	$2x^2 - 2x - 2y^2 + y$
A	0
B	$-6xy + 3y$
$i = 100, m = 2, \text{ neighbours } 3, 10$	
W	$-x^2 - 3xy + x + y^2 - 2y$
A	0
B	$-3x^2 + 3xy - 3x + 3y^2$
$i = 101, m = 2, \text{ neighbours } 3, 10$	
W	$2x^2 + x - 2y^2 + y$
A	0
B	$6xy - 3x + 3y$
$i = 102, m = 2, \text{ neighbours } 3, 72, 83$	
W	$2x^3 + 8x^2 + 10x + 4$
A	$2xy - 2x + 4y^2 + 4y - 8$
B	$4x^3y^2 + 32x^3y + 20x^3 + 8x^2y^3 - 8x^2y^2 + 68x^2y - 6x^2 + 4xy^4$ $+ 6xy^3 - 28xy^2 + 74xy - 42x + 4y^5 - 16y^4 + 20y^3 - 48y^2 + 32y - 44$
$i = 103, m = 2, \text{ neighbours } 10, 83$	
W	$2x^2 + 3x - 2y^2$
A	0
B	$6xy - 3x + 6y$
$i = 104, m = 2, \text{ neighbours } 10, 15$	
W	$2x^2 + 4x - 2y^2 - 2y$
A	0
B	$-4x^2 + 8xy - 4x + 8y$
$i = 105, m = 40960, \text{ neighbours } 10, 15, 18, 53, 68, 87, 97, 123, 204$	
W	$-79405056x^4 + 50651136x^3y^3 - 292921344x^3y^2 - 76455936x^3y$ $- 394076160x^3 + 158810112xy + 18616320x - 56401920y^5 - 203710464y^4$ $+ 234270720y^3 - 387145728y^2 + 158846976y + 217534464$
A	$815372697600x^{14}y^2 + 2446118092800x^{14}y + 2446118092800x^{14}$ $- 815372697600x^{13}y^3 + 1630745395200x^{13}y^2 + 9784472371200x^{13}y$ $+ 12230590464000x^{13} + 2174327193600x^{12}y^4 + 4348654387200x^{12}y^3$

$$\begin{aligned}
& +7610145177600x^{12}y^2 + 16307453952000x^{12}y + 22830435532800x^{12} \\
& -1358954496000x^{11}y^5 + 2446118092800x^{11}y^4 + 12502381363200x^{11}y^3 \\
& +15627976704000x^{11}y^2 + 6115295232000x^{11}y + 15084394905600x^{11} \\
& +2808505958400x^{10}y^6 + 8923801190400x^{10}y^5 + 19795437158400x^{10}y^4 \\
& +36284085043200x^{10}y^3 + 45117289267200x^{10}y^2 + 15492081254400x^{10}y \\
& -1449551462400x^9y^7 - 2219625676800x^9y^6 + 634178764800x^9y^5 \\
& +4937534668800x^9y^4 + 32614907904000x^9y^3 + 90234578534400x^9y^2 \\
& +74606601830400x^9y - 11007531417600x^9 + 2083730227200x^8y^8 \\
& +8969099673600x^8y^7 + 21471481036800x^8y^6 + 43984827187200x^8y^5 \\
& +69714365644800x^8y^4 + 98388305510400x^8y^3 + 114763707187200x^8y^2 \\
& +58910677401600x^8y - 62987540889600x^8 - 634178764800x^7y^9 \\
& -2038431744000x^7y^8 - 1223059046400x^7y^7 + 12864769228800x^7y^6 \\
& +64006756761600x^7y^5 + 118636727500800x^7y^4 + 85546185523200x^7y^3 \\
& -41176321228800x^7y^2 - 115782923059200x^7y - 127809670348800x^7 \\
& +724775731200x^6y^{10} + 2400819609600x^6y^9 + 2423468851200x^6y^8 \\
& +2740558233600x^6y^7 + 28651290624000x^6y^6 + 80404807680000x^6y^5 \\
& +95058866995200x^6y^4 + 22490696908800x^6y^3 - 71345111040000x^6y^2 \\
& -126586611302400x^6y - 67879777075200x^6 - 90596966400x^5y^{11} \\
& -90596966400x^5y^{10} + 3193543065600x^5y^9 + 25910732390400x^5y^8 \\
& +79227047116800x^5y^7 + 154626372403200x^5y^6 + 153335365632000x^5y^5 \\
& +63667018137600x^5y^4 - 1834588569600x^5y^3 + 74912366592000x^5y^2 \\
& -13555571097600x^5y + 77358484684800x^5 + 90596966400x^4y^{12} \\
& -588880281600x^4y^{11} - 3578580172800x^4y^{10} - 7383652761600x^4y^9 \\
& +12910067712000x^4y^8 + 50281316352000x^4y^7 + 95919538176000x^4y^6 \\
& +50315290214400x^4y^5 - 64788155596800x^4y^4 - 177105744691200x^4y^3 \\
& +46476243763200x^4y^2 + 8357570150400x^4y - 2950165187002368x^4 \\
& -45298483200x^3y^{13} + 79272345600x^3y^{12} + 1322149478400x^3y^{11} \\
& +17473182105600x^3y^{10} + 38710915891200x^3y^9 + 77825934950400x^3y^8 \\
& +70666109337600x^3y^7 + 9786946867200x^3y^6 + 6885802002981888x^3y^5 \\
& -32347043557620480x^3y^4 + 27099284018490432x^3y^3 \\
& -4842870503302320x^3y^2 + 60498341000439012x^3y - 94814256379177899x^3 \\
& -362387865600x^2y^{13} - 747424972800x^2y^{12} + 554906419200x^2y^{11} \\
& +16318778572800x^2y^{10} + 50926819737600x^2y^9 + 144230370508800x^2y^8 \\
& +248292311040000x^2y^7 - 961206181429248x^2y^6 + 15132382759747584x^2y^5 \\
& -38673442324611072x^2y^4 - 2212471687348224x^2y^3 \\
& -117929684540325888x^2y^2 - 29942894202716160x^2y - 77562854454067200x^2 \\
& +45298483200xy^{14} + 237817036800xy^{13} + 342569779200xy^{12} \\
& -11989234483200xy^{11} - 48630575923200xy^{10} - 122572873728000xy^9 \\
& -185025270988800xy^8 - 268815473971200xy^7 - 373373366400000xy^6 \\
& -2204196436207872xy^5 + 7840144622243520xy^4 - 6032926636882704xy^3 \\
& +6314841998117964xy^2 - 4060034819766585xy + 181853135866669995x \\
& +317089382400y^{14} + 1166435942400y^{13} + 4872418099200y^{12} - 384329318400y^{11} \\
& -13287142195200y^{10} - 75357167616000y^9 - 170719134105600y^8 \\
& -2281839276183552y^7 - 35428772629029888y^6 + 79769603437837056y^5 \\
& -59837318635646016y^4 + 96148233842143536y^3 - 33170932748313828y^2 \\
& +161516468807117739y - 15773747675922432 \\
B \quad & 90596966400x^5y^{13} + 294440140800x^5y^{12} + 1749653913600x^5y^{11} \\
& -10258690867200x^5y^{10} - 34388272742400x^5y^9 - 117804987187200x^5y^8 \\
& -225519449702400x^5y^7 - 207019017216000x^5y^6 - 14033429238859776x^5y^5 \\
& +64475567232284160x^5y^4 - 46467011146799232x^5y^3 \\
& +50786178229756512x^5y^2 - 132830446880410056x^5y + 127540572566017878x^5 \\
& +90596966400x^4y^{13} + 294440140800x^4y^{12} + 1749653913600x^4y^{11} \\
& -10258690867200x^4y^{10} - 34388272742400x^4y^9 - 117804987187200x^4y^8 \\
& -225519449702400x^4y^7 - 207019017216000x^4y^6 - 14033429238859776x^4y^5 \\
& +64475567232284160x^4y^4 - 46467011146799232x^4y^3 \\
& +50786178229756512x^4y^2 - 132830446880410056x^4y + 127540572566017878x^4 \\
& +90596966400x^3y^{14} + 362387865600x^3y^{13} + 1970484019200x^3y^{12} \\
& -8946450432000x^3y^{11} - 42082290892800x^3y^{10} - 143596191744000x^3y^9 \\
& -313873190092800x^3y^8 - 376158604492800x^3y^7 - 14188693501771776x^3y^6 \\
& +53950495303139328x^3y^5 + 1889664277413888x^3y^4 \\
& +15935919869657088x^3y^3 - 94740813208092672x^3y^2 + 27917737405710336x^3y \\
& -230491149570146304x^3 - 90596966400x^2y^{15} + 317089382400x^2y^{14} \\
& +2400819609600x^2y^{13} + 12830795366400x^2y^{12} + 33738876518400x^2y^{11}
\end{aligned}$$

$$\begin{aligned}
& +103587014246400x^2y^{10} + 182481577574400x^2y^9 + 232298805657600x^2y^8 \\
& +4466587153711104x^2y^7 + 43206656565325824x^2y^6 \\
& -56540845081469952x^2y^5 + 59701011895892736x^2y^4 \\
& -20999634429602880x^2y^3 - 139094178730528464x^2y^2 \\
& -71908155563762916x^2y - 46059733888554069x^2 + 45298483200xy^{16} \\
& -79272345600xy^{15} - 416179814400xy^{14} - 14494806835200xy^{13} \\
& -43430451609600xy^{12} - 119397910118400xy^{11} - 207252891648000xy^{10} \\
& -358884971827200xy^9 - 1775808663118848xy^8 + 9622972640422656xy^7 \\
& -87717359890976832xy^6 + 91186472215108656xy^5 - 243773537264264484xy^4 \\
& +144764157168004059xy^3 - 61858222802214948xy^2 - 200677962779191269xy \\
& +487396182184603308x + 317089382400y^{16} + 985242009600y^{15} + 3921149952000y^{14} \\
& -2252891750400y^{13} - 11981979648000y^{12} - 68190806016000y^{11} \\
& -185159264256000y^{10} + 798795417858048y^9 - 12821779497673728y^8 \\
& +92056707559238400y^7 - 153901555189416000y^6 + 264312335841460272y^5 \\
& -124349563200872484y^4 + 213832859130796059y^3 + 218430075786145452y^2 \\
& -81199044514218753y + 537885042309792513 \\
i = 106, m = 2, \text{ neighbours } 10, 26 \\
W & x^3 + x^2y + 2x^2 - 3xy + x + y^2 - 2y \\
A & 0 \\
B & -3x^3 + 3x^2y + 3xy + 3x - 3y^2 \\
i = 107, m = 10, \text{ neighbours } 26, 32 \\
W & -10xy - 20x + 10y^2 + 10y - 20 \\
A & 0 \\
B & -100x^6 - 100x^5 - 100x^4 + 100x^3y - 200x^3 - 100x^2y^2 - 200x^2y - 500x^2 \\
& +100xy^2 - 300xy - 500x + 400y^2 + 400y - 100 \\
i = 108, m = 2, \text{ neighbours } 10, 80 \\
W & -x^3 + 2x^2y - 2x^2 - xy - 2x + y^2 + y \\
A & 0 \\
B & -3x^3 - 3xy - 3y^2 - 3y \\
i = 109, m = 22, \text{ neighbours } 10, 53 \\
W & 11x^3 + 11x^2y - 11x^2 - 33x + 11y^2 + 33y \\
A & 0 \\
B & -363x^3 + 363x^2y - 363x^2 - 363x - 363y^2 - 363y \\
i = 110, m = 2, \text{ neighbours } 10, 111 \\
W & 2x^2 - xy + 3y \\
A & 0 \\
B & 3xy - 6x + 3y \\
i = 111, m = 3, \text{ neighbours } 110, 135, 154, 172, 175, 176, 177, 216 \\
W & 720xy^2 + 1824xy + 1152x - 432y^2 + 1728y + 11808 \\
A & 48384xy^3 - 193536xy^2 - 1990656xy - 1734912x + 48384y^3 + 870912y^2 + 400896y \\
& -6448896 \\
B & 504x^9y^4 + 3024x^9y^3 - 12600x^9y^2 - 27216x^9y + 72576x^9 \\
& +504x^8y^4 + 18144x^8y^3 - 10584x^8y^2 - 141120x^8y + 120960x^8 \\
& -3528x^7y^4 + 27216x^7y^3 + 68040x^7y^2 - 152208x^7y - 48384x^7 \\
& +2016x^6y^5 + 6048x^6y^4 - 47376x^6y^3 - 48384x^6y^2 + 305424x^6y \\
& +72576x^6 + 1008x^5y^5 + 67032x^5y^4 + 7056x^5y^3 - 605304x^5y^2 \\
& +34272x^5y + 556416x^5 - 16128x^4y^5 + 59976x^4y^4 + 418320x^4y^3 \\
& -116424x^4y^2 - 1252944x^4y - 193536x^4 + 2016x^3y^6 - 10080x^3y^5 \\
& -186984x^3y^4 - 32256x^3y^3 + 2150568x^3y^2 + 4368384x^3y + 1728000x^3 \\
& +82656x^2y^5 - 70560x^2y^4 - 1570464x^2y^3 - 2022048x^2y^2 \\
& +3214080x^2y + 13478400x^2 - 18144xy^6 + 53424xy^5 + 836640xy^4 + 265104xy^3 \\
& -6870528xy^2 - 13250304xy - 5184000x - 16128y^6 - 290304y^5 - 211968y^4 \\
& +2308608y^3 + 1400832y^2 - 11854080y - 40027392 \\
i = 112, m = 2, \text{ neighbours } 218 \\
W & 2 \\
A & 0 \\
B & -54 \\
i = 113, m = 2, \text{ neighbours } 10, 68 \\
W & -2xy - 3x + y^2 \\
A & 0
\end{aligned}$$

B	$3x - 3y^2 - 6y$
$i = 114, m = 2, \text{neighbours } 10, 43$	
W	$-x^2 - 3xy - x + y^2 + 2y$
A	0
B	$3x^2 - 3xy - 3x - 3y^2$
$i = 115, m = 2, \text{neighbours } 3, 12$	
W	$2x^2 - 2y^2 + 2y - 2$
A	0
B	$-6xy + 6x$
$i = 116, m = 4, \text{neighbours } 3, 23$	
W	$4x^2y - 4x^2 - 4xy - 4x - 4y^2 - 4$
A	0
B	$16x^3 + 16x^2y - 16xy - 16$
$i = 117, m = 2, \text{neighbours } 3, 32$	
W	$-x^3 + x^2y - 2x^2 - 2xy - 3x - y^2 - 4y$
A	0
B	$3x^3 + 3x^2y - 3x + 3y^2 - 6$
$i = 118, m = 6, \text{neighbours } 36, 54, 119, 121, 122, 176$	
W	$2x^3y + 9x^3 - 5x^2y - 15x^2 - 2xy^3 - 13xy^2 - 13xy - 22x - y^4 - 7y^3 + 2y^2 - 23y - 18$
A	$36x^7 - 36x^6y + 90x^6 - 126x^5y - 81x^5 - 45x^4y - 389x^4 + 246x^3y - 215x^3 + 111x^2y + 51x^2 - 138xy + 59x - 193y - 53$
B	$-72x^9 - 36x^8 + 144x^7y^2 + 72x^7y + 306x^7 - 72x^6y^3 + 252x^6y^2 + 166x^6 - 432x^5y^3 - 90x^5y^2 - 308x^5y - 406x^5 + 216x^4y^4 - 216x^4y^3 - 10x^4y^2 + 190x^4y + 274x^4 + 432x^3y^4 + 505x^3y^3 + 544x^3y^2 + 798x^3y + 642x^3 - 216x^2y^5 - 152x^2y^4 - 45x^2y^3 - 202x^2y^2 + 182x^2y + 101x^2 - 149xy^5 - 360xy^4 - 728xy^3 - 728xy^2 - 961xy - 586x + 71y^6 + 129y^5 + 13y^4 - 206y^3 - 460y^2 - 361y - 165$
$i = 119, m = 4, \text{neighbours } 118, 120$	
W	$4x^3 + 4x^2 - 4y$
A	0
B	$16x^3 - 16x^2y + 32x^2 - 16xy + 16x + 16y^2 - 16y$
$i = 120, m = 48, \text{neighbours } 54, 119, 152, 161, 162, 163, 210$	
W	$-30x^3y^2 - 34x^3y + 226x^3 + 204x^2y^2 + 113x^2y + 2035x^2 + 119xy^4 + 135xy^3 + 1510xy^2 + 15xy + 2098x - 59y^5 - 887y^4 - 2y^3 - 1786y^2 - 3254y + 2739$
A	$20886x^{14} + 2891x^{13} - 24367x^{12}y + 433079x^{12} + 11151x^{11}y + 587890x^{11} - 464931x^{10}y + 1987229x^{10} - 485881x^9y + 5403219x^9 - 1188905x^8y + 2234804x^8 - 5479353x^7y - 5766062x^7 + 3987798x^6y - 7330963x^6 + 13609978x^5y - 1811622x^5 - 323656x^4y + 6394957x^4 - 8214760x^3y + 3330031x^3 - 1032336x^2y - 1657258x^2 + 737813xy - 2997379x + 6775373y - 2500707$
B	$-41772x^{14} - 4608x^{13}y^2 + 9216x^{13}y + 31382x^{13} + 6912x^{12}y^3 - 20736x^{12}y^2 - 23290x^{12}y - 858072x^{12} - 2304x^{11}y^4 + 29952x^{11}y^3 - 23040x^{11}y^2 + 26588x^{11}y - 360760x^{11} - 25344x^{10}y^4 + 27498x^{10}y^3 - 20936x^{10}y^2 - 799820x^{10}y - 2811959x^{10} + 6912x^9y^5 - 62208x^9y^4 + 47947x^9y^3 + 49392x^9y^2 + 133886x^9y - 8077775x^9 + 36864x^8y^5 - 81917x^8y^4 - 457986x^8y^3 - 608477x^8y^2 - 3348927x^8y + 4526087x^8 - 6912x^7y^6 + 64512x^7y^5 + 7534x^7y^4 - 225663x^7y^3 + 514886x^7y^2 - 4945566x^7y + 11330609x^7 - 25344x^6y^6 + 32156x^6y^5 - 346943x^6y^4 - 1890202x^6y^3 - 3119759x^6y^2 + 1207477x^6y - 12559676x^6 + 2304x^5y^7 - 34560x^5y^6 + 23544x^5y^5 + 50906x^5y^4 - 3186569x^5y^3 - 2076138x^5y^2 + 960287x^5y - 13592150x^5 + 6912x^4y^7 - 28875x^4y^6 - 316322x^4y^5 - 1693768x^4y^4 + 392534x^4y^3 - 699545x^4y^2 - 8495627x^4y + 12423562x^4 + 6912x^3y^7 + 17641x^3y^6 + 118942x^3y^5 - 1533899x^3y^4 + 6238236x^3y^3 + 38654x^3y^2 + 2437207x^3y + 19137675x^3 - 4658x^2y^7 - 207147x^2y^6 - 1355032x^2y^5 - 12821x^2y^4 - 3968424x^2y^3 - 6788098x^2y^2 + 1151025x^2y - 4493286x^2 + 10561xy^7 + 122370xy^6 - 368859xy^5 + 2891822xy^4 - 2015454xy^3 + 3147979xy^2 + 7027702xy - 10529542x - 3481y^8 - 104666y^7 - 787005y^6 - 214296y^5 - 3552340y^4 - 5456538y^3 - 5119199y^2 - 9111625y - 7580181$
$i = 121, m = 2, \text{neighbours } 26, 118$	

W	$x^3 + x^2y + 3x^2 - 3xy + 3x + y^2 - 4y + 1$
A	0
B	$3x^3 - 3x^2y + 3x^2 - 3xy - 3x + 3y^2 - 3$
$i = 122, m = 22, \text{neighbours } 34, 118$	
W	$11x^3 + 11x^2y + 11x^2 - 22xy - 33x + 11y^2 + 11y - 33$
A	0
B	$363x^3 - 363x^2y + 363x^2 + 363x + 363y^2 + 363y + 363$
$i = 123, m = 383328, \text{neighbours } 45, 105, 128, 129, 157, 173, 181, 221$	
W	$-4x^5y + 253x^5 + 9x^4y^2 + 722x^4y - 1627x^4 + 17024x^3y^2 - 1036859x^3y$ $+456229x^3 + 26x^2y^4 - 1521x^2y^3 - 5035094x^2y^2 + 7575522x^2y$ $-9656577x^2 - 13xy^5 + 32102xy^4 - 2070752xy^3 - 133216xy^2 - 6612226xy$ $-18426160x - 75474y^5 - 5912478y^4 + 13325096y^3 + 932804y^2 - 1151095y + 16619046$
A	$440821066752x^{18} - 3526568534016x^{17} + 11461347735552x^{16} - 18514484802908x^{15}$ $+881642133335x^{14}y + 13224636758480x^{14} - 6171496895838x^{13}y$ $+4412875834246x^{13} + 13218947450307x^{12}y - 22078980139987x^{12}$ $-3492414387426x^{11}y + 19008844666765x^{11} - 13297655185320x^{10}y$ $+30433189953479x^{10} + 2215070359415x^9y - 139543278013377x^9$ $+24110487751810x^8y + 209187805354019x^8 - 154549561309836x^7y$ $+44264042331583x^7 + 409077309508879x^6y - 561666539226774x^6$ $-435817305270021x^5y + 776044301919887x^5 + 157798607018439x^4y$ $-401194436340000x^4 + 100806834793173x^3y - 65718548681448x^3$ $-80448963385474x^2y - 59208088668061x^2 - 58897851599791xy + 234838210423364x$ $+43441477164704y - 92064229983372$
B	$-881642133504x^{18} + 5436793156608x^{17} + 1175522844672x^{16}y$ $-12196049513472x^{16} + 146940355584x^{15}y^2 - 5877614223360x^{15}y$ $+9551123111608x^{15} - 440821066752x^{14}y^3 - 881642133504x^{14}y^2$ $+7787838846290x^{14}y + 3967380086224x^{14} + 2351045689344x^{13}y^3$ $+2057164978176x^{13}y^2 + 7787842770600x^{13}y - 12205398874660x^{13}$ $+293880711168x^{12}y^4 - 6171494934528x^{12}y^3 + 440821066414x^{12}y^2$ $-19237800114848x^{12}y + 6081718975210x^{12} - 2204105333760x^{11}y^4$ $+7787838845276x^{11}y^3 - 9991948103346x^{11}y^2 - 3562767972395x^{11}y$ $+22498470207266x^{11} + 5142912445609x^{10}y^4 - 1910229379864x^{10}y^3$ $+4690710504463x^{10}y^2 + 3375372648981x^{10}y - 65156302443818x^{10}$ $+881642133504x^9y^5 - 2791864793772x^9y^4 - 8086394239066x^9y^3$ $+20165008117725x^9y^2 - 3773097054208x^9y + 138201581036365x^9$ $-2057164978345x^8y^5 - 6459682412030x^8y^4 + 11489920415254x^8y^3$ $+5580049438910x^8y^2 + 13251259721887x^8y - 32379917194390x^8$ $+146938393767x^7y^5 + 8504444830598x^7y^4 - 3674588483242x^7y^3$ $-14250179355276x^7y^2 + 140635968962720x^7y - 377920641552997x^7$ $+587761422336x^6y^6 + 4843341296737x^6y^5 + 1772556142131x^6y^4$ $-35417774904871x^6y^3 + 23528743458381x^6y^2 - 373777140894200x^6y$ $+447968480275291x^6 + 293880710492x^5y^6 - 1746199471375x^5y^5$ $-3652361501108x^5y^4 + 97308105735599x^5y^3 - 205905932941017x^5y^2$ $+439682887575095x^5y - 68788972226821x^5 + 169x^4y^7 - 1175527603972x^4y^6$ $-4424290986818x^4y^5 - 2910299957167x^4y^4 - 32535047079642x^4y^3$ $+87194972202487x^4y^2 - 16194238604019x^4y - 454985243941576x^4$ $+1962831x^3y^7 - 592450380154x^3y^6 + 380779713004x^3y^5$ $+27587185989548x^3y^4 - 116814997731020x^3y^3 + 100044906287559x^3y^2$ $-146481252623347x^3y - 280863635264282x^3 - 169x^2y^8 + 5699121652x^2y^7$ $+1483923638740x^2y^6 + 33983949803624x^2y^5 - 245266219351676x^2y^4$ $+441144365316510x^2y^3 - 289979329847404x^2y^2 + 41565731325460x^2y$ $+920399222744530x^2 - 1962324xy^8 - 1004316408xy^7 - 531220700864xy^6$ $-58014278984668xy^5 + 210262121873122xy^4 - 240184465679124xy^3$ $-24321773812293xy^2 + 519964260070004xy - 266209956538697x - 5696324676y^8$ $-892476729144y^7 - 32945999509476y^6 + 157709478793968y^5 - 166701572640652y^4$ $-35962441669580y^3 + 182884790005096y^2 - 348689044732700y - 75771122033697$
$i = 124, m = 2, \text{neighbours } 23, 76$	
W	$2x^2 - 2xy - 2x + 2y^2$
A	0
B	$4x^3 - 4x^2y + 4x^2 - 4xy + 4y^2 + 4$
$i = 125, m = 2, \text{neighbours } 70, 76$	
W	$2x^2 - xy + 2x - y^2 + y + 2$

A	0
B	$3xy - 3y^2 - 3y$
$i = 126, m = 2, \text{ neighbours } 42, 87$	
W	$2x^2 - 2y^2 - 2y - 2$
A	0
B	$-4x^2 - 8xy - 4x + 8$
$i = 127, m = 2, \text{ neighbours } 42, 68$	
W	$-2xy - 2x + 4y + 2$
A	0
B	$4xy + 8x + 4y^2 - 8$
$i = 128, m = 2, \text{ neighbours } 42, 123$	
W	$-2x^2y - 2y^2 + 2y$
A	0
B	$-4x^3 - 4x^2y + 8x^2 - 4x + 4y$
$i = 129, m = 1, \text{ neighbours } 123$	
W	$-y$
A	0
B	$-x^2 + x - y$
$i = 130, m = 2, \text{ neighbours } 6, 42$	
W	$6x^2 - 6x + 6y$
A	0
B	$-6xy + 6x + 6y^2 - 6$
$i = 131, m = 2, \text{ neighbours } 12, 87$	
W	$2x^2 + 4x - 4y - 2$
A	0
B	$4x^2 + 4y^2 - 8$
$i = 132, m = 2, \text{ neighbours } 4, 12$	
W	$4x^2 - xy + y^2 - 3y - 1$
A	0
B	$3xy - 6x + 3y^2 + 3y - 3$
$i = 133, m = 2, \text{ neighbours } 12, 80$	
W	$2x^2y - 4x^2 - 2x + 2y^2 + 2y$
A	0
B	$-4x^3 + 4x^2y - 4x^2 - 4xy + 4y + 4$
$i = 134, m = 14, \text{ neighbours } 12, 53$	
W	$7x^3 + 7x^2y - 21x^2 - 28x + 7y^2 + 28y$
A	0
B	$-147x^3 + 147x^2y - 147x^2 - 147y^2 + 294$
$i = 135, m = 2, \text{ neighbours } 12, 111$	
W	$2x^2 - xy - x + 3y - 3$
A	0
B	$3xy - 9x + 3y + 3$
$i = 136, m = 2, \text{ neighbours } 3, 12$	
W	$-x^2 + y^2 + 2y - 2$
A	0
B	$-3x^2 - 3y^2 + 6y$
$i = 137, m = 2, \text{ neighbours } 12, 62$	
W	$2x^2 - 6x + 4y$
A	0
B	$-8x^2 + 8xy + 4x - 4y^2 - 4y + 4$
$i = 138, m = 2, \text{ neighbours } 21, 83$	
W	$-x^3 - 2x^2y - 2xy - y^2 - 3y$
A	0
B	$-3x^3 - 6x^2 + 3y^2 - 3y$
$i = 139, m = 2, \text{ neighbours } 68, 83$	
W	$-2xy - 3x - y^2 - 3y - 6$
A	0
B	$3x + 3y^2 + 3y$
$i = 140, m = 2, \text{ neighbours } 15, 97$	

W	$6x^2 - 3xy + 9x - 3y^2 + 3$
A	0
B	$15xy + 3x - 3y^2 + 12y - 9$
$i = 141, m = 2, \text{neighbours } 15, 62$	
W	$2x^2 + 4x - 2y - 2$
A	0
B	$-8x^2 + 8xy - 12x - 4y^2 - 4$
$i = 142, m = 2, \text{neighbours } 15, 87$	
W	$2x^2 + 6x - 2y^2 - 3y + 3$
A	0
B	$6xy + 6x + 9y + 9$
$i = 143, m = 2, \text{neighbours } 80, 87$	
W	$-x^3 + x^2y - 2x^2 - 3xy - 5x - y^2 - 4y - 3$
A	0
B	$-3x^3 - 3x^2y - 6x^2 - 3xy - 3x - 3y^2 - 6y - 3$
$i = 144, m = 2, \text{neighbours } 18, 87$	
W	$6x^2 + 3xy + 9x - 3y^2 - 3y$
A	0
B	$15xy + 9x + 3y^2 + 9y$
$i = 145, m = 2, \text{neighbours } 15, 42$	
W	$2x^2 + x - 2y^2 + y - 3$
A	0
B	$-6xy - 3x - 3y + 3$
$i = 146, m = 2, \text{neighbours } 15, 18$	
W	$6x^2 - 3xy + 9x + 3y^2 + 3y$
A	0
B	$-3xy - 9x - 3y^2 + 9y$
$i = 147, m = 2, \text{neighbours } 4, 34$	
W	$-2x^2y - 4xy - 8x - 4y - 4$
A	0
B	$8x^3 + 4x^2 - 8xy - 8x - 4y^2 - 12y - 4$
$i = 148, m = 2, \text{neighbours } 4, 42$	
W	$4x^2 + 3xy - 2x + 2y^2 - 2$
A	0
B	$-3xy - 6y$
$i = 149, m = 2, \text{neighbours } 4, 6$	
W	$12x^2 - 3xy + 6x - 3y^2 - 6y$
A	0
B	$15xy + 12x - 3y^2 + 6y + 6$
$i = 150, m = 2, \text{neighbours } 18, 29$	
W	$3x^3 + 3x^2y - 3x^2 - 3xy - 3x + 3y$
A	0
B	$-9x^3 + 3x^2y - 9x^2 - 3xy - 9x - 6y^2 - 3y$
$i = 151, m = 2, \text{neighbours } 18, 76$	
W	$6x^2 - 6xy + 3x + 3y$
A	0
B	$6xy - 9x - 6y^2 + 3y$
$i = 152, m = 4, \text{neighbours } 21, 120$	
W	$-2x^4 - 4x^3 - 2x^2y - 2xy + 4y^2$
A	$6x^4 - 6x^2y - 6xy$
B	$-12x^2y - 12xy - 6x - 6y^3 - 6y - 18$
$i = 153, m = 4, \text{neighbours } 21, 32$	
W	$-2x^4 + 4x^2y + 6x^2 + 4y^2 + 6y$
A	$6x^4 + 12x^2y + 6x^2 - 6y$
B	$-24x^2y - 12x^2 + 6x + 6y^3 + 18y + 18$
$i = 154, m = 2, \text{neighbours } 21, 111$	
W	$-x^3 + 3x^2 - 2xy$
A	0
B	$3x^3 + 3x^2 + 6y$

$i = 155, m = 2$, neighbours 21, 38	
W	$x^2y + x^2 + 2y^2 - y$
A	0
B	$-3x^2y + 3x^2 + 3y$
$i = 156, m = 4$, neighbours 23, 80	
W	$2x^4 - 2x^3 + 2x^2 + 4xy + 2x + 4y^2 + 6y + 2$
A	$6x^4 + 6x^3 + 6x^2 + 6x + 6y + 6$
B	$-12x^3 - 12x^2 - 6x + 6y^3 - 6y + 6$
$i = 157, m = 4$, neighbours 23, 123	
W	$4x^2 - 4xy - 4x + 4y^2$
A	0
B	$-16x^4 + 16x^3 - 16x^2 + 16x - 16y^2 - 16y$
$i = 158, m = 2$, neighbours 3, 23	
W	$x^3 - x^2y + 2x^2 + xy + 2x + 2y^2 + 1$
A	0
B	$-3x^3 - 3x^2y + 3xy + 3$
$i = 159, m = 2$, neighbours 23, 62	
W	$x^3 + x^2y + 3x^2 - 5xy - 4x + 2y^2 + 2y + 2$
A	0
B	$11x^3 - 3x^2y + 5x^2 - 13xy - 2x + 8y^2 + 8y + 8$
$i = 160, m = 22$, neighbours 23, 72	
W	$11x^3 + 11x^2y - 22xy - 11x + 11y^2 - 11y - 33$
A	0
B	$363x^3 - 363x^2y + 726x^2 + 363x + 363y^2 - 363y + 363$
$i = 161, m = 4$, neighbours 80, 120	
W	$-2x^4 - 8x^3 - 2x^2y - 6x^2 + 2xy - 2x + 4y^2 + 4y$
A	$6x^4 - 6x^2y - 6x^2 - 6xy - 6x$
B	$12x^2y + 12x^2 + 12xy + 18x + 6y^3 + 6y + 18$
$i = 162, m = 2$, neighbours 5, 120	
W	$2x^3 - 2x^2 + 2y$
A	0
B	$-4x^3 + 4x^2y + 4xy + 4x - 4y^2 - 4y$
$i = 163, m = 2$, neighbours 76, 120	
W	$2x^3 - 2x^2y + 2x^2 - 3xy - x + 2y^2 - y$
A	0
B	$-6x^2 + 3xy - 3x + 3y$
$i = 164, m = 2$, neighbours 26, 34	
W	$2x^4 + x^3 - 2x^2y - x^2 - 2xy - 3x + 2y^2 + 2y - 3$
A	0
B	$3x^3 + 9x^2 + 3x - 6y - 3$
$i = 165, m = 2$, neighbours 3, 29	
W	$x^3 - x^2y + 4x^2 + 2xy + 3x + 2y^2 + 1$
A	0
B	$-3x^3 - 3x^2y + 3x + 3$
$i = 166, m = 2$, neighbours 26, 59	
W	$-x^2y + x^2 - 2xy - x + 2y^2 - 2$
A	0
B	$-3x^2y - 3x^2 - 3x + 6y$
$i = 167, m = 2$, neighbours 59, 80	
W	$x^2y + 2x^2 + 2xy + x + 2y^2 + 3y + 1$
A	0
B	$-3x^2y + 3x + 3y + 3$
$i = 168, m = 2$, neighbours 43, 80	
W	$-x^3 + x^2y + 3x^2 + 3xy + 2x + 2y^2 + 3y + 1$
A	0
B	$-3x^3 - 3x^2y - 3x^2 - 3xy + 3y + 3$
$i = 169, m = 2$, neighbours 32, 59	
W	$x^2y - x^2 + 2y^2 + 4y + 3$
A	0

B	$3x^2y + 3x^2 + 3$
$i = 170, m = 2$, neighbours 32, 76	
W	$-2x^2 - 2xy - 4x + 2y^2 + 2y - 2$
A	0
B	$-4x^3 + 4x^2y - 4x^2 - 4xy - 4x + 4y^2 + 4y + 4$
$i = 171, m = 2$, neighbours 5, 34	
W	$x^3 + x^2y - x^2 - 2xy - 3x + y^2 + 5y + 3$
A	0
B	$-3x^3 + 3x^2y + 3x^2 - 3x - 3y^2 - 3y + 3$
$i = 172, m = 2$, neighbours 53, 111	
W	$x^3 + 2x^2 - 2xy + 3$
A	0
B	$3x^3 + 6x + 6y + 9$
$i = 173, m = 4$, neighbours 53, 123	
W	$2x^4 + 2x^3 + 2x^2 - 6x + 4y^2 + 6y$
A	$6x^4 - 6x^3 - 6x^2 + 12xy + 6x + 6y$
B	$12x^3 + 12x^2 - 24xy - 6x + 6y^3 - 6y + 18$
$i = 174, m = 2$, neighbours 5, 42	
W	$2x^2 - 4x + y^2 + 2$
A	0
B	$-3y^2$
$i = 175, m = 2$, neighbours 18, 111	
W	$6x^2 - 6y$
A	0
B	$6xy + 18x$
$i = 176, m = 2$, neighbours 5, 18, 111, 118, 191, 198	
W	2
A	$10584x^4 - 21168x^3y + 35280x^2y^2 + 10584x^2 - 28224xy^3 - 42336xy - 5292x$ $+ 21168y^4 - 5292y^3 + 109368y^2 + 5292y - 68796$
B	$-7056x^3y^5 + 10584x^3y^4 - 77616x^3y^3 - 7056x^3y^2 + 95256x^3y$ $- 5292x^3 + 3528x^2y^6 - 17640x^2y^5 + 67032x^2y^4 - 8820x^2y^3$ $- 38808x^2y^2 + 15876x^2y + 18522x^2 + 14112xy^6 - 21168xy^5 + 24696xy^4$ $+ 68796xy^3 - 33516xy^2 + 63504xy + 42336x - 10584y^7 + 13230y^6 - 65268y^5$ $- 21168y^4 - 12348y^3 - 156114y^2 + 26460y + 103192$
$i = 177, m = 2$, neighbours 6, 111	
W	$6x^2 - 3xy - 3y - 6$
A	0
B	$3xy + 24x - 9y$
$i = 178, m = 2$, neighbours 18, 68	
W	$-6xy - 9x + 3y$
A	0
B	$9x + 6y^2 + 9y$
$i = 179, m = 2$, neighbours 18, 70	
W	$6x^2 + 3xy + 3x + 3y^2 + 3y$
A	0
B	$3xy + 9x - 3y^2 - 3y$
$i = 180, m = 2$, neighbours 38, 70	
W	$-2xy - 2y - 2$
A	0
B	$4xy - 4x + 4y^2 + 4y$
$i = 181, m = 2$, neighbours 43, 123	
W	$-x^3 + x^2y + 3x^2 - 2x + 2y^2 + y$
A	0
B	$-3x^3 - 3x^2y + 3x^2 + 3y$
$i = 182, m = 2$, neighbours 3, 62	
W	$-2xy + 5x + 2y^2 + 1$
A	0
B	$6x^2 + 3x - 3$
$i = 183, m = 2$, neighbours 5, 6	

W	$6x^2 - 6xy - 6x + 3y^2 + 6y$
A	0
B	$-6xy + 6x + 3y^2 - 6y - 6$
$i = 184, m = 2, \text{neighbours } 5, 62$	
W	$2x^2 - 4x + 4y + 2$
A	0
B	$-8x^2 + 8xy + 8x - 4y^2 - 8y$
$i = 185, m = 15, \text{neighbours } 0$	
W	15
A	0
B	$-3y - 6$
$i = 186, m = 2, \text{neighbours } 1, 187$	
W	$-3x + 6$
A	0
B	$-15x - 12$
$i = 187, m = 42, \text{neighbours } 46, 186, 188, 189, 196, 200, 212, 220$	
W	42
A	$-7056000x^5 + 17640000x^4y - 31752000x^4 + 75411000x^3y^2 - 63415800x^3y$ $-5450760x^3 + 61740000x^2y^3 - 15986250x^2y^2 - 85311450x^2y + 40519080x^2$ $+11025000xy^4 + 70471800xy^3 - 72495990xy^2 - 6049638xy + 38218824x + 43328250y^4$ $-12877200y^3 - 104472900y^2 + 63757134y + 61117308$
B	$-2751000x^3y^5 + 3481800x^3y^4 - 11746224x^3y^3 - 7003794x^3y^2$ $+16453710x^3y - 974736x^3 - 2940000x^2y^6 + 971250x^2y^5 - 2591610x^2y^4$ $-14335902x^2y^3 + 13964538x^2y^2 + 10960908x^2y - 8914752x^2 - 525000xy^7$ $-3355800xy^6 + 2389590xy^5 - 7431732xy^4 - 6929538xy^3 + 16572444xy^2$ $-4345236xy - 8692740x - 2063250y^7 + 613200y^6 + 2777250y^5 - 8605884y^4$ $+4119192y^3 + 11881926y^2 - 12099150y - 8731128$
$i = 188, m = 2, \text{neighbours } 44, 187$	
W	$-6x + 9$
A	0
B	$-12x - 3$
$i = 189, m = 2, \text{neighbours } 44, 187$	
W	$-3x - 6$
A	0
B	$-15x + 12$
$i = 190, m = 2, \text{neighbours } 84, 191$	
W	$3x - 6$
A	0
B	$-27x + 12$
$i = 191, m = 7, \text{neighbours } 176, 190, 192, 199, 223, 225$	
W	7
A	10584
B	$-4536x^3y + 9240x^3 - 5544x^2y + 6636x^2 + 1512xy^3 - 3024xy^2 + 2226xy - 3150x$ $-2184y^3 + 1344y^2 - 1008y - 6055$
$i = 192, m = 1, \text{neighbours } 191$	
W	1
A	0
B	-2
$i = 193, m = 2, \text{neighbours } 2, 194$	
W	$-30x - 30y - 30$
A	0
B	$-90x - 30y + 30$
$i = 194, m = 30, \text{neighbours } 193, 195$	
W	30
A	0
B	$-900x - 300y + 297$
$i = 195, m = 10, \text{neighbours } 2, 194$	
W	10
A	0

B	$9000x^2 + 9000xy + 3000y^2 + 3000y + 2990$
$i = 196, m = 2, \text{neighbours } 10, 187$	
W	$-6x + 9y$
A	0
B	$-12x - 3y$
$i = 197, m = 2, \text{neighbours } 3, 198$	
W	$-18x - 6y - 12$
A	0
B	$6x + 30y - 24$
$i = 198, m = 42, \text{neighbours } 176, 197$	
W	42
A	0
B	$84x + 420y - 357$
$i = 199, m = 2, \text{neighbours } 76, 191$	
W	$-4x + 4y + 1$
A	0
B	$6x - 6y + 9$
$i = 200, m = 2, \text{neighbours } 12, 187$	
W	$-9x + 3y + 6$
A	0
B	$3x - 15y + 12$
$i = 201, m = 2, \text{neighbours } 15, 202$	
W	$3x - 3y + 3$
A	0
B	$3x + 3y + 9$
$i = 202, m = 12, \text{neighbours } 37, 201, 205, 207, 209, 214, 215, 219, 221, 222$	
W	12
A	$1728x^8y + 2592x^8 + 4608x^7y^2 + 8352x^7y + 2160x^7 + 432x^6y^3$ $+ 6840x^6y^2 - 8568x^6y - 26784x^6 - 4176x^5y^4 - 1080x^5y^3$ $- 21744x^5y^2 - 50328x^5y - 9072x^5 - 144x^4y^5 + 936x^4y^4 - 288x^4y^3$ $- 82368x^4y^2 - 73656x^4y + 68040x^4 + 1296x^3y^6 - 5544x^3y^5$ $- 23760x^3y^4 - 121392x^3y^3 - 134100x^3y^2 + 34452x^3y + 7128x^3$ $- 288x^2y^7 - 3672x^2y^6 - 9540x^2y^5 - 42588x^2y^4 - 182232x^2y^3$ $- 9432x^2y^2 + 260712x^2y - 22680x^2 + 5256xy^7 + 26028xy^6 + 61560xy^5$ $- 4284xy^4 + 51300xy^3 + 285876xy^2 + 121824xy - 4860x - 504y^8 - 5652y^7$ $+ 19152y^6 + 89424y^5 + 58536y^4 + 294660y^3 + 434358y^2 - 128034y - 93960$
B	$-432x^3y^9 - 840x^3y^8 + 9312x^3y^7 + 93792x^3y^6 + 196848x^3y^5$ $+ 270504x^3y^4 + 355644x^3y^3 + 113232x^3y^2 - 110772x^3y + 432x^3$ $+ 96x^2y^{10} + 1992x^2y^9 + 7500x^2y^8 + 55428x^2y^7 + 190824x^2y^6$ $+ 247032x^2y^5 + 332640x^2y^4 + 301116x^2y^3 - 258300x^2y^2$ $- 278208x^2y + 24300x^2 - 1752xy^{10} - 15588xy^9 - 59352xy^8 - 31548xy^7$ $+ 78288xy^6 + 66768xy^5 + 7416xy^4 - 297624xy^3 - 486822xy^2 - 77526xy + 36180x$ $+ 168y^{11} + 3420y^{10} - 840y^9 - 39324y^8 - 76884y^7 - 168492y^6 - 175962y^5$ $- 77790y^4 - 439230y^3 - 391680y^2 + 159354y + 93912$
$i = 203, m = 2, \text{neighbours } 87, 204$	
W	$6x + 6$
A	0
B	$6x + 12y + 18$
$i = 204, m = 6, \text{neighbours } 105, 203, 206$	
W	6
A	144
B	$72x^3y + 60x^3 + 48x^2y^2 - 48x^2 - 120xy^2 - 180xy - 48x - 60y^3 - 96y^2 - 48y - 39$
$i = 205, m = 2, \text{neighbours } 15, 202$	
W	$3y + 3$
A	0
B	$6x - 3y + 9$
$i = 206, m = 2, \text{neighbours } 4, 204$	
W	$12x + 6$
A	0
B	$12x + 12y + 6$

$i = 207, m = 4$, neighbours 62, 97, 202
 $W \quad 9x^2 - 18x - 9y^2 + 9$
 $A \quad 0$
 $B \quad 405x^3 - 297x^2y - 27x^2 + 243xy^2 - 54xy + 135x - 63y^3 + 81y^2 - 81y + 63$

$i = 208, m = 4$, neighbours 38, 97
 $W \quad -6xy - 6x + 2y^2 - 2$
 $A \quad 0$
 $B \quad -54x^2y + 54x^2 - 12xy^2 - 24xy - 12x - 22y^3 + 50y^2 - 50y + 22$

$i = 209, m = 2$, neighbours 21, 202
 $W \quad 6y$
 $A \quad 0$
 $B \quad -12x^2 + 12y$

$i = 210, m = 8$, neighbours 29, 120
 $W \quad -8xy - 8x - 8y$
 $A \quad 0$
 $B \quad 64x^6 + 128x^5 - 128x^4y + 256x^4 - 64x^3y + 320x^3 + 128x^2y^2 + 320x^2$
 $\quad + 64xy + 192x - 64y^3 - 64y^2$

$i = 211, m = 2$, neighbours 80
 $W \quad -2x^2 - x - y - 1$
 $A \quad 0$
 $B \quad -3x - 3y - 3$

$i = 212, m = 2$, neighbours 32, 187
 $W \quad -3x^2 + 6y + 15$
 $A \quad 0$
 $B \quad 15x^2 + 12y + 9$

$i = 213, m = 2$, neighbours 6, 34
 $W \quad 6x^3 + 6x^2 + 6x + 6$
 $A \quad 0$
 $B \quad 36x^5 - 12x^4y + 48x^4 - 36x^3y + 72x^3 + 12x^2y^2 - 48x^2y + 36xy^2 + 36xy$
 $\quad + 36x - 12y^3 - 24y^2 - 36y$

$i = 214, m = 2$, neighbours 53, 202
 $W \quad -3x^2 - 3x - 3$
 $A \quad 0$
 $B \quad 3x^2 - 3x - 6y - 9$

$i = 215, m = 2$, neighbours 42, 202
 $W \quad 3x - 3$
 $A \quad 0$
 $B \quad 3x + 6y - 3$

$i = 216, m = 28$, neighbours 111
 $W \quad 14x - 42$
 $A \quad 0$
 $B \quad 42x + 42$

$i = 217, m = 2$, neighbours 18, 218
 $W \quad -30x - 6y$
 $A \quad 0$
 $B \quad -18x + 30y$

$i = 218, m = 14$, neighbours 112, 217
 $W \quad 14$
 $A \quad 0$
 $B \quad 84x - 140y - 7$

$i = 219, m = 2$, neighbours 68, 202
 $W \quad 3$
 $A \quad 0$
 $B \quad -6y - 9$

$i = 220, m = 2$, neighbours 70, 187
 $W \quad 6x - 3y - 3$
 $A \quad 0$
 $B \quad -12x - 15y - 15$

$i = 221, m = 2$, neighbours 123, 202

W	$6y$
A	0
B	$-12x^2 + 12x + 12y$
$i = 222, m = 2, \text{neighbours } 5, 202$	
W	$3y$
A	0
B	$6x - 3y - 6$
$i = 223, m = 2, \text{neighbours } 76, 191$	
W	$-4x + 4y + 1$
A	0
B	$-6x + 6y - 9$
$i = 224, m = 2, \text{neighbours } 6$	
W	$3x + 3$
A	0
B	$9x - 6y - 3$
$i = 225, m = 2, \text{neighbours } 72, 191$	
W	$-4x + 5y - 9$
A	0
B	$-6x - 3y - 3$

Table 2: The data of the identities (3).

$\{i, j\} = \{0, 1\}, k_{ij} = 7$	
α	$-3x - 2$
β	$-3xy^2 + 3xy - 6x - 2y^2 + 2y - 4$
γ	$6x^3 - 2x^2 - 4x + 9$
$\{i, j\} = \{0, 3\}, k_{ij} = 2^2 \cdot 7 \cdot 13$	
α	$-x + 19$
β	$2x^4 + 2x^3y - 42x^3 - 38x^2y + 70x^2 - xy^2 - 5xy + 125x + 19y^2 + 96y - 192$
γ	$2x^3 - 38x^2 - 6x + 115$
$\{i, j\} = \{0, 4\}, k_{ij} = 7^2$	
α	$-80x + 24$
β	$80x^4 + 40x^3y - 64x^3 - 32x^2y + 12x^2 - 80xy^2 + 86xy - 110x + 24y^2 + y + 48$
γ	$40x^3 - 32x^2 + 6x + 25$
$\{i, j\} = \{0, 6\}, k_{ij} = 2 \cdot 3 \cdot 13$	
α	$-162x + 108$
β	$-324x^4 + 108x^3y + 540x^3 - 180x^2y - 432x^2 - 162xy^2 + 306xy - 234x + 108y^2 - 138y + 216$
γ	$36x^3 - 60x^2 + 48x - 10$
$\{i, j\} = \{0, 23\}, k_{ij} = 2^2 \cdot 3 \cdot 5$	
α	$-15x^3 + 21x^2 - 21x + 39$
β	$-30x^5 + 42x^4 - 15x^3y^2 + 45x^3y - 12x^3 + 21x^2y^2 - 63x^2y + 21x^2 - 21xy^2 + 3xy - 42x + 39y^2 - 18y + 78$
γ	$30x^3 - 42x^2 - 18x + 21$
$\{i, j\} = \{0, 87\}, k_{ij} = 2^2 \cdot 3$	
α	$21x + 27$
β	$-42x^4 - 42x^3y - 54x^3 + 30x^2y + 42x^2 + 21xy^2 - 39xy - 9x + 27y^2 - 42y + 24$
γ	$-42x^3 + 30x^2 - 18x - 15$
$\{i, j\} = \{0, 185\}, k_{ij} = 3 \cdot 5^2$	
α	0
β	0
γ	1
$\{i, j\} = \{1, 7\}, k_{ij} = 2 \cdot 7 \cdot 13$	
α	$-9x + 1$
β	$9xy^2 + 30xy - 14x - 28y^2 + 9y + 218$
γ	$18x^3 + 34x^2 - 9xy^2 - 12xy + 100x + y^2 - 29y + 39$
$\{i, j\} = \{1, 9\}, k_{ij} = 7$	
α	$-3x - 2$

β	$6x^3 - 2x^2 - 4x + 2y^3 + 2y + 13$
γ	$-2xy^2 + 2xy - 4x + y^2 - y + 2$
$\{i, j\}$	$\{1, 11\}, k_{ij} = 2 \cdot 3 \cdot 7^2$
α	$-30x - 48$
β	$60x^3 + 36x^2 - 96x + 18y^3 - 96y^2 - 54y + 438$
γ	$18xy^2 - 60xy - 174x - 30y^2 - 96y - 102$
$\{i, j\}$	$\{1, 14\}, k_{ij} = 3 \cdot 7 \cdot 13$
α	$-15x + 18$
β	$30x^3 - 66x^2 + 36x - 18y^3 - 66y^2 + 54y + 273$
γ	$18xy^2 + 30xy - 114x + 33y^2 - 36y - 27$
$\{i, j\}$	$\{1, 17\}, k_{ij} = 2^4 \cdot 3 \cdot 7$
α	$-40x - 8$
β	$80x^3 - 64x^2 - 16x - 16y^3 - 40y^2 - 64y + 360$
γ	$-8xy^2 - 32xy - 80x - 24y^2 - 40y - 72$
$\{i, j\}$	$\{1, 20\}, k_{ij} = 2^2$
α	$-x$
β	$2x^3 - x^2y^2 + x^2y - 3x^2 + xy - x + y^2 - y + 4$
γ	$-xy^2 + xy - x + y - 1$
$\{i, j\}$	$\{1, 22\}, k_{ij} = 3$
α	$-x$
β	$2x^3 + x^2y^2 - x^2 + 3$
γ	$-xy^2 - x$
$\{i, j\}$	$\{1, 25\}, k_{ij} = 2$
α	$-x$
β	$2x^3 + x^2y^2 + x^2y - x^2 + xy + x - y^2 - y + 2$
γ	$-xy^2 - xy - x - y - 1$
$\{i, j\}$	$\{1, 28\}, k_{ij} = 7$
α	$-2x + 1$
β	$4x^3 - 2x^2y^2 - x^2y - 5x^2 + xy^2 - 3xy + 5x - 2y^2 - y + 7$
γ	$-2xy^2 - xy + x + y^2 - 3y + 3$
$\{i, j\}$	$\{1, 31\}, k_{ij} = 7$
α	$-2x + 1$
β	$4x^3 + 2x^2y^2 + x^2y - 7x^2 - 3xy^2 + 2xy - y^2 - 4y + 10$
γ	$-2xy^2 - xy + x + y^2 - 3y + 3$
$\{i, j\}$	$\{1, 33\}, k_{ij} = 7$
α	$-3x - 2$
β	$6x^3 + 3x^2y^2 - 3x^2y + 4x^2 + 2xy^2 - 2xy + 9$
γ	$-3xy^2 + 3xy - 6x - 2y^2 + 2y - 4$
$\{i, j\}$	$\{1, 35\}, k_{ij} = 2 \cdot 3^2$
α	$6x + 12$
β	$-12x^3 - 12x^2 + 24x - 6y^3 + 24y^2 - 6y - 30$
γ	$-6xy^2 + 18xy + 12x + 6y^2 + 18y - 12$
$\{i, j\}$	$\{1, 40\}, k_{ij} = 13$
α	$-4x - 1$
β	$8x^3 - 6x^2 - 2x + 4y^2 + y + 13$
γ	$-4xy^2 - xy - x - y^2 + 3y + 3$
$\{i, j\}$	$\{1, 48\}, k_{ij} = 2 \cdot 3 \cdot 13$
α	$-14x - 4$
β	$28x^3 - 20x^2 - 8x - 28y^2 - 12y + 98$
γ	$14xy^2 + 6xy - 28x + 4y^2 + 24y - 8$
$\{i, j\}$	$\{1, 53\}, k_{ij} = 2^9 \cdot 3$
α	$1152xy + 2944x + 1280y + 1664$
β	$-x^4 + 1152x^3y^2 - 4224x^3y - 2433x^3 + 1280x^2y^2 - 1407x^2y + 4223x^2$
γ	$-1152xy^3 + 1920xy^2 - 897xy + 3325x + 1024y^3 + 767y^2 - 3971y + 1533$
γ	$x^2 - 1152xy^2 + 1920xy - 3455x - 1280y^2 + 1152y - 1663$
$\{i, j\}$	$\{1, 56\}, k_{ij} = 2 \cdot 7 \cdot 13$
α	$-9x + 1$
β	$18x^3 - 20x^2 + 9xy^2 + 12xy - 64x - 28y^2 - 7y + 202$
γ	$9xy^2 + 12xy - 66x - y^2 + 29y - 23$

$$\begin{aligned}
&\{i, j\} = \{1, 58\}, k_{ij} = 2^2 \\
&\alpha \quad -x \\
&\beta \quad 2x^3 - 2x^2 + y^2 - y + 4 \\
&\gamma \quad xy^2 - xy + x - y + 1 \\
&\{i, j\} = \{1, 60\}, k_{ij} = 2 \cdot 3^2 \\
&\alpha \quad 6x + 12 \\
&\beta \quad -12x^3 - 12x^2 + 24x - 6y^3 + 24y^2 - 6y - 30 \\
&\gamma \quad 6xy^2 - 18xy - 12x - 6y^2 - 18y + 12 \\
&\{i, j\} = \{1, 61\}, k_{ij} = 13 \\
&\alpha \quad -4x - 1 \\
&\beta \quad 8x^3 - 6x^2 - 4xy^2 - xy - 3x + 3y^2 + 4y + 16 \\
&\gamma \quad 4xy^2 + xy + x + y^2 - 3y - 3 \\
&\{i, j\} = \{1, 186\}, k_{ij} = 3 \cdot 7 \\
&\alpha \quad 0 \\
&\beta \quad 25 \\
&\gamma \quad 5x + 1 \\
&\{i, j\} = \{2, 7\}, k_{ij} = 2^5 \cdot 3^2 \cdot 7 \cdot 13 \\
&\alpha \quad -447x^3 - 13401x + 149y^3 + 1192y^2 - 1949y - 178 \\
&\beta \quad 298x^5 - 298x^4y - 298x^3y + 11780x^3 - 9984x^2y - 5623x^2 + 447xy^3 \\
&\quad + 3576xy^2 + 4120xy - 4110x - 149y^4 - 1937y^3 + 8059y^2 - 5578y + 7969 \\
&\gamma \quad 596x^3y^2 + 596x^3y - 8836x^3 + 8934xy^2 - 14766xy - 6945x - 894y^4 + 5067y^3 \\
&\quad - 3062y^2 + 15139y - 18773 \\
&\{i, j\} = \{2, 8\}, k_{ij} = 2^4 \\
&\alpha \quad -13y - 30 \\
&\beta \quad 13y^2 - 22y + 49 \\
&\gamma \quad 26x^3y + 60x^3 + 26x^2y + 60x^2 - 39xy^2 + 92xy - 87x - 39y^3 + 27y^2 - 42y - 57 \\
&\{i, j\} = \{2, 41\}, k_{ij} = 2^3 \cdot 3^2 \\
&\alpha \quad -6y - 12 \\
&\beta \quad 6y^2 + 24y + 30 \\
&\gamma \quad 12x^3y + 24x^3 - 12x^2y - 24x^2 - 18xy^2 - 60xy - 66x - 18y^3 - 54y^2 - 24y + 78 \\
&\{i, j\} = \{2, 45\}, k_{ij} = 2^5 \cdot 3^2 \\
&\alpha \quad 72x - 144 \\
&\beta \quad -36x^3 + 36x^2y + 96x^2 - 36xy^2 - 96xy + 108x + 36y^3 + 96y^2 - 108y + 264 \\
&\gamma \quad -36x^4 + 36x^3y + 36x^3 - 36x^2y - 324x^2 - 36xy^3 - 96xy^2 + 228xy - 636x - 36y^4 \\
&\quad - 96y^3 + 216y^2 - 468y + 456 \\
&\{i, j\} = \{2, 48\}, k_{ij} = 2 \cdot 3^2 \cdot 7 \cdot 11^2 \\
&\alpha \quad 726x \\
&\beta \quad -18x^3 + 35x^2y + 299x^2 - 69xy^2 - 600xy + 2001x + 136y^3 + 1200y^2 - 4002y + 10180 \\
&\gamma \quad -699x^4 + 324x^3y - 99x^3 - 102x^2y^2 + 348x^2y - 2943x^2 - 67xy^3 \\
&\quad - 1190xy^2 + 4132xy - 14311x - 136y^4 - 1200y^3 + 3866y^2 - 11244y + 5066 \\
&\{i, j\} = \{2, 56\}, k_{ij} = 2^5 \cdot 3^2 \cdot 7 \cdot 13 \\
&\alpha \quad -447x^3 - 13401x + 149y^3 + 1192y^2 - 1949y - 178 \\
&\beta \quad 298x^5 - 298x^4y - 298x^3y + 11780x^3 - 9984x^2y - 5623x^2 + 447xy^3 \\
&\quad + 3576xy^2 + 4120xy - 4110x - 149y^4 - 1937y^3 + 8059y^2 - 5578y + 7969 \\
&\gamma \quad -596x^3y^2 - 596x^3y + 8836x^3 - 8934xy^2 + 14766xy + 6945x + 894y^4 - 5067y^3 \\
&\quad + 3062y^2 - 15139y + 18773 \\
&\{i, j\} = \{2, 57\}, k_{ij} = 2^4 \\
&\alpha \quad -13y - 30 \\
&\beta \quad 13y^2 - 22y + 49 \\
&\gamma \quad -26x^3y - 60x^3 - 26x^2y - 60x^2 + 39xy^2 - 92xy + 87x + 39y^3 - 27y^2 + 42y + 57 \\
&\{i, j\} = \{2, 82\}, k_{ij} = 3 \cdot 7^3 \\
&\alpha \quad 300x + 168y + 654 \\
&\beta \quad -200x^3 + 88x^2y - 374x^2 - 24xy^2 + 352xy + 333x - 180y^2 - 434y + 67 \\
&\gamma \quad -93x^3 + 4x^2y^2 + 133x^2y - 353x^2 - 8xy^3 - 117xy^2 + 228xy + 113x + 12y^4 \\
&\quad - 40y^3 + 379y^2 - 791y - 1000 \\
&\{i, j\} = \{2, 89\}, k_{ij} = 2^5 \cdot 3^2 \cdot 7 \cdot 13^2 \\
&\alpha \quad -52x^3 + 12x^2y - 164454x^2 - 30652xy - 148564x - 133819y - 101024 \\
&\beta \quad 35x^5 - 43x^4y + 109637x^4 + 32x^3y^2 - 89565x^3y + 98517x^3 - 18x^2y^3 \\
&\quad + 53737x^2y^2 - 46582x^2y + 58925x^2 + 8xy^4 - 25280xy^3 + 45085xy^2 - 51610xy
\end{aligned}$$

$$\begin{aligned}
& +249921x - 3y^5 + 9476y^4 + 4040y^3 + 9684y^2 + 43383y + 538509 \\
\gamma \quad & x^5 - x^4y + x^4 + 2x^3y^2 - 1054x^3y - 1543x^3 - 2x^2y^3 + 3161x^2y^2 \\
& +1409x^2y + 87399x^2 + 2xy^4 - 6321xy^3 + 415xy^2 - 57328xy + 507329x - 3y^5 \\
& +9479y^4 - 5448y^3 - 90244y^2 + 131123y + 105267 \\
\{i, j\} = \{2, 90\}, k_{ij} = 2^4 \cdot 3 \cdot 7 \cdot 13 \\
\alpha \quad & 729x^3 - 39x^2 - 2457x - 335y^2 - 671y + 673 \\
\beta \quad & -1458x^5 + 4985x^4 + 12350x^3 - 4966x^2y^2 + 4285x^2y + 19257x^2 - 5971xy^2 \\
& -5971xy + 12046x + 335y^3 + 336y^2 - 1009y + 2349 \\
\gamma \quad & 2916x^4 + 1458x^3y - 17793x^3 - 78x^2y - 18717x^2 + 9203xy^2 - 19876xy - 10528x \\
& +4966y^3 + 3459y^2 - 3660y - 2674 \\
\{i, j\} = \{2, 91\}, k_{ij} = 2^{10} \cdot 3^3 \cdot 13 \\
\alpha \quad & 25920x^2 + 10518xy - 232980x - 27648y + 82944 \\
\beta \quad & -17280x^4 + 10266x^3y + 155316x^3 - 4544x^2y^2 - 117026x^2y - 21188x^2 \\
& +1248xy^3 + 43772xy^2 - 1136xy + 78576x + 27648y^2 - 110592y + 110592 \\
\gamma \quad & 3x^4 + 57x^3y - 21261x^3 - 189x^2y^2 + 27468x^2y - 104241x^2 + 400xy^3 \\
& -18633xy^2 + 68877xy - 38842x - 624y^4 - 26521y^3 + 106118y^2 - 112213y + 134355 \\
\{i, j\} = \{2, 92\}, k_{ij} = 2^5 \cdot 3^2 \cdot 7 \cdot 13^2 \\
\alpha \quad & -52x^3 + 12x^2y - 164454x^2 - 30652xy - 148564x - 133819y - 101024 \\
\beta \quad & 35x^5 - 43x^4y + 109637x^4 + 32x^3y^2 - 89565x^3y + 98517x^3 - 18x^2y^3 \\
& +53737x^2y^2 - 46582x^2y + 58925x^2 + 8xy^4 - 25280xy^3 + 45085xy^2 - 51610xy \\
& +249921x - 3y^5 + 9476y^4 + 4040y^3 + 9684y^2 + 43383y + 538509 \\
\gamma \quad & -x^5 + x^4y - x^4 - 2x^3y^2 + 1054x^3y + 1543x^3 + 2x^2y^3 - 3161x^2y^2 \\
& -1409x^2y - 87399x^2 - 2xy^4 + 6321xy^3 - 415xy^2 + 57328xy - 507329x + 3y^5 \\
& -9479y^4 + 5448y^3 + 90244y^2 - 131123y - 105267 \\
\{i, j\} = \{2, 93\}, k_{ij} = 2^5 \cdot 3^2 \cdot 7^2 \\
\alpha \quad & -84x^2 + 42xy - 1722x + 420y - 798 \\
\beta \quad & -28x^3y + 142x^3 + 56x^2y^2 - 696x^2y - 133x^2 - 56xy^3 + 1109xy^2 - 1113xy \\
& -418x - 825y^3 + 2489y^2 - 4173y + 7843 \\
\gamma \quad & 84x^4 - 84x^3y + 1551x^3 - 1182x^2y + 1815x^2 + 56xy^3 - 270xy^2 - 636xy + 2281x \\
& +825y^3 - 2084y^2 + 4132y - 8663 \\
\{i, j\} = \{2, 94\}, k_{ij} = 2^4 \cdot 3^4 \cdot 5 \cdot 7 \cdot 13 \\
\alpha \quad & -16648x^2 - 3148xy - 154664x - 25874y - 34152 \\
\beta \quad & 11100x^4 - 9038x^3y + 98889x^3 + 5411x^2y^2 - 84629x^2y + 10051x^2 \\
& -2541xy^3 + 53222xy^2 - 48124xy + 27426x + 949y^4 - 26796y^3 + 79598y^2 + 9714y \\
& +148720 \\
\gamma \quad & -4x^4 + 114x^3y + 12657x^3 - 329x^2y^2 - 14675x^2y + 39700x^2 + 643xy^3 \\
& +6059xy^2 - 29779xy - 99505x - 949y^4 + 25847y^3 - 29775y^2 + 23330y - 271708 \\
\{i, j\} = \{2, 95\}, k_{ij} = 2^7 \cdot 3^2 \cdot 7 \cdot 13^2 \\
\alpha \quad & -2760x^3 + 1194x^2y - 10270257x^2 - 1911027xy + 14549247x + 802545y - 6831510 \\
\beta \quad & 1808x^5 - 2588x^4y + 6846906x^4 + 1974x^3y^2 - 5529682x^3y - 10015812x^3 \\
& -1106x^2y^3 + 3160473x^2y^2 + 7264923x^2y - 2849206x^2 + 432xy^4 \\
& -1185142xy^3 - 1921928xy^2 + 6310245xy - 9754202x - 432y^4 + 1186248y^3 \\
& -133231y^2 - 9916770y + 19944130 \\
\gamma \quad & -48x^5 + 72x^4y + 78x^4 - 89x^3y^2 + 65623x^3y - 473552x^3 + 97x^2y^3 \\
& -197250x^2y^2 + 604848x^2y - 7919983x^2 - 121xy^4 + 394972xy^3 - 395162xy^2 \\
& +5199695xy - 28975245x + 216y^5 - 593016y^4 - 631057y^3 + 8163160y^2 - 9429917y \\
& +1913216 \\
\{i, j\} = \{2, 193\}, k_{ij} = 2^4 \cdot 3^3 \cdot 5 \\
\alpha \quad & -3120x - 2390y - 2200 \\
\beta \quad & 2070x^3 - 510x^2y + 1650x^2 - 130xy^2 - 2480xy + 1800x + 190y^3 + 2180y^2 - 2920y \\
& +8920 \\
\gamma \quad & -x^4 - 3x^3y + 18x^3 + 6x^2y^2 - 21x^2y + 255x^2 - 13xy^3 + 17xy^2 - 166xy \\
& +928x + 19y^4 + 17y^3 - 258y^2 + 321y - 16 \\
\{i, j\} = \{2, 195\}, k_{ij} = 2 \cdot 5 \\
\alpha \quad & 0 \\
\beta \quad & 0 \\
\gamma \quad & 1 \\
\{i, j\} = \{3, 46\}, k_{ij} = 3^3 \cdot 7 \\
\alpha \quad & 24x - 12y + 57 \\
\beta \quad & -48x^3 + 48x^2y - 31x^2 - 26xy + 8x - 24y^3 + 47y^2 - 31y - 28
\end{aligned}$$

γ	$24x^3y - 35x^3 - 24x^2y^2 + 20x^2y + 36x^2 + 14xy^2 - 46xy + 63x + 24y^4 - 35y^3 + 10y^2 - 28y - 46$
$\{i, j\} = \{3, 55\}, k_{ij} = 2^3 \cdot 3^4 \cdot 7$	
α	$-238x^2 - 714xy + 182x - 280y + 742$
β	$476x^4 + 476x^3y - 440x^3 - 476x^2y^2 + 401x^2y - 1326x^2 + 714xy^2 + 451xy - 278x - 476y^3 + 681y^2 + 533y + 1488$
γ	$-476x^3y + 400x^3 + 476x^2y^2 - 123x^2y - 206x^2 - 277xy^2 - 311xy - 1676x - 476y^4 + 401y^3 + 519y^2 + 1875y + 822$
$\{i, j\} = \{3, 100\}, k_{ij} = 2^2 \cdot 3^5 \cdot 5 \cdot 7^2$	
α	$-598x^2 - 1890xy - 5953x - 54826y + 87271$
β	$1195x^4 + 1958x^3y + 20210x^3 - 1905x^2y^2 + 47861x^2y - 112460x^2 - 22xy^3 - 21230xy^2 - 32013xy - 22784x + 643y^4 - 8865y^3 + 109078y^2 + 53534y - 23673$
γ	$-x^4 - 628x^3y + 9500x^3 + 619x^2y^2 - 30690x^2y + 73987x^2 + 22xy^3 + 21169xy^2 - 10187xy - 189619x - 643y^4 + 9508y^3 - 63760y^2 - 22862y + 22955$
$\{i, j\} = \{3, 101\}, k_{ij} = 2^3 \cdot 3^3 \cdot 7$	
α	$-340x^2 - 226xy - 267x - 287y + 386$
β	$680x^4 - 228x^3y - 146x^3 - 226x^2y^2 + 825x^2y - 961x^2 + 226xy^3 - 464xy^2 + 682xy + 1045x + 113y^3 + 55y^2 - 155y + 354$
γ	$-113x^3y + 345x^3 + 114x^2y^2 + 5x^2y - 402x^2 - 350xy^2 + 692xy - 849x - 113y^4 + 345y^3 - 289y^2 - 277y + 984$
$\{i, j\} = \{3, 102\}, k_{ij} = 2^2 \cdot 3^3$	
α	$21x + 39$
β	$-42x^3 + 6x^2 + 30x - 87$
γ	$42x^4 + 42x^3y - 6x^3 - 6x^2y - 30x^2 - 21xy^2 - 51xy + 45x - 39y^2 + 48y - 78$
$\{i, j\} = \{3, 115\}, k_{ij} = 2^4 \cdot 3^3 \cdot 7$	
α	$-228x^2 - 152xy + 284x - 432y + 1296$
β	$456x^4 - 152x^3y - 1024x^3 - 152x^2y^2 + 1566x^2y - 1246x^2 + 152xy^3 - 626xy^2 + 156xy + 1386x + 432y^2 - 864y - 864$
γ	$76x^3y - 389x^3 - 76x^2y^2 - 118x^2y + 556x^2 + 508xy^2 - 993xy + 562x + 76y^4 - 389y^3 + 436y^2 + 446y - 1335$
$\{i, j\} = \{3, 116\}, k_{ij} = 2^6 \cdot 3^3$	
α	$-84x^3 - 252x^2y + 264x^2 + 396xy - 312x - 72y + 348$
β	$168x^5 + 168x^4y - 290x^4 - 168x^3y^2 - 267x^3y - 434x^3 + 252x^2y^2 + 1665x^2y + 1006x^2 + 168xy^3 - 1274xy^2 - 939xy + 984x - 168y^4 + 237y^3 + 1020y^2 + 552y + 684$
γ	$168x^3y - 406x^3 - 168x^2y^2 - 403x^2y + 1586x^2 + 809xy^2 - 1043xy - 2074x + 168y^4 - 405y^3 - 543y^2 - 117y - 1080$
$\{i, j\} = \{3, 117\}, k_{ij} = 2^6 \cdot 3^4 \cdot 7 \cdot 13$	
α	$1045x^3y^2 - 2090x^3y + 38335x^3 - 20142x - 1045y^5 + 836y^4 + 15641y^3 - 46060y^2 + 320579y - 194179$
β	$-2090x^7 + 2090x^6 - 74580x^5 + 2090x^4y^3 - 1672x^4y^2 - 38806x^4y + 128685x^4 + 2090x^3y^3 - 418x^3y^2 - 112444x^3y - 13821x^3 - 6270x^2y^3 + 99306x^2y^2 - 104484x^2y - 228174x^2 - 1045xy^5 - 2299xy^4 + 18149xy^3 - 2352xy^2 + 161211xy + 440567x + 1045y^6 + 209y^5 - 24837y^4 + 119902y^3 - 255810y^2 - 433213y + 219067$
γ	$2090x^6 - 2090x^3y^3 - 2508x^3y^2 + 115476x^3y - 52015x^3 + 6270xy^3 - 63696xy^2 - 125352xy + 411318x + 9405y^4 - 96589y^3 + 80631y^2 + 57639y - 417607$
$\{i, j\} = \{3, 136\}, k_{ij} = 2^4 \cdot 3^3 \cdot 7 \cdot 13$	
α	$-547x^3 - 1749x + 3829y^3 - 6564y^2 - 2727y + 9718$
β	$1094x^5 - 1094x^4y - 1094x^4 - 6564x^3y^2 + 19692x^3y - 4908x^3 - 9330x^2y - 12887x^2 + 3829xy^3 + 2766xy^2 + 984xy - 14629x - 3829y^4 + 8390y^3 - 33505y^2 + 46835y + 10158$
γ	$-6564x^3y^2 + 16410x^3y - 8406x^3 + 3498xy^2 - 13992xy - 34065x + 5655y^3 - 33312y^2 + 36747y - 19107$
$\{i, j\} = \{3, 158\}, k_{ij} = 2^4 \cdot 3^4$	
α	$-162x^2y + 1044x^2 + 1404xy + 612x + 162y^3 + 846y^2 - 702y + 1044$
β	$-207x^3 + 207x^2y - 342x^2 + 162xy^3 + 90xy^2 - 675xy - 234x - 162y^4 - 1008y^3 - 306y^2 - 1512y - 4185$
γ	$-324x^3y + 2088x^3 + 324x^2y^2 + 720x^2y + 1017x^2 + 648xy^2 - 333xy + 1539x - 162y^3 + 1143y^2 - 1611y + 2349$

$$\begin{aligned}
& \{i, j\} = \{3, 165\}, k_{ij} = 2^4 \cdot 3^3 \cdot 5 \cdot 13^2 \\
& \alpha \quad -10x^2y^2 + 81201x^2y - 361676x^2 + 32xy^2 - 322167xy - 167855x + 10y^4 \\
& \quad -81139y^3 - 41658y^2 - 146202y + 120324 \\
& \beta \quad 10x^5 - 10x^4y - 15x^4 + 10x^3y^2 + 15x^3y + 24049x^3 - 10x^2y^3 \\
& \quad -16x^2y^2 - 24049x^2y + 646496x^2 + 10xy^4 - 81141xy^3 + 37110xy^2 - 105786xy \\
& \quad +191342x - 10y^5 + 81129y^4 + 122787y^3 + 268989y^2 + 229760y + 538870 \\
& \gamma \quad 10x^4 - 20x^3y^2 + 162392x^3y - 723357x^3 + 20x^2y^3 - 162318x^2y^2 \\
& \quad +79003x^2y - 311656x^2 + 30xy^3 - 162335xy^2 + 181664xy + 187831x - 20y^4 \\
& \quad +81193y^3 - 306923y^2 + 627392y - 534802 \\
& \{i, j\} = \{3, 182\}, k_{ij} = 2 \cdot 3^3 \cdot 13 \\
& \alpha \quad 28x^2 + 84xy + 44x + 6y + 211 \\
& \beta \quad -56x^4 - 56x^3y - 66x^3 + 56x^2y^2 + 50x^2y - 37x^2 - 56xy^2 - 83xy - 16x \\
& \quad -34y^2 - 228y - 67 \\
& \gamma \quad -28x^3y + 17x^3 + 28x^2y^2 + 16x^2y - 140x^2 - 34xy^2 + 8xy + 124x - 28y^4 \\
& \quad +17y^3 + 133y^2 + 68y - 136 \\
& \{i, j\} = \{3, 197\}, k_{ij} = 2^3 \cdot 3^2 \cdot 7 \\
& \alpha \quad 122x + 64y + 142 \\
& \beta \quad -242x^3 + 106x^2y - 22x^2 + 64xy^2 - 120xy + 136x - 80y^3 + 60y^2 - 174y - 250 \\
& \gamma \quad -x^4 + 9x^3y - 14x^3 - 9x^2y^2 + 3x^2y + 8x^2 + 11xy^2 - 13xy + 39x + 8y^4 \\
& \quad -14y^3 + 6y^2 - 41 \\
& \{i, j\} = \{4, 36\}, k_{ij} = 3 \cdot 7 \\
& \alpha \quad 8 \\
& \beta \quad -4x^2 + 2xy + 4x - 3y - 3 \\
& \gamma \quad 2xy^2 - 12x + 5y^2 - 6y + 2 \\
& \{i, j\} = \{4, 49\}, k_{ij} = 2^2 \cdot 3^2 \cdot 7^3 \\
& \alpha \quad -4544x^2 - 2140xy - 2528x - 1350y + 4576 \\
& \beta \quad 2272x^4 - 66x^3y - 1008x^3 - 527x^2y^2 + 631x^2y - 1788x^2 + 264xy^3 \\
& \quad +902xy^2 - 770xy + 4261x - 264y^3 + 2323y^2 - 2477y + 1358 \\
& \gamma \quad 32x^3y - 120x^3 - 64x^2y^2 + 1592x^2y - 2110x^2 + xy^3 - 2698xy^2 + 5808xy \\
& \quad -1209x + 264y^4 - 973y^3 - 3185y^2 + 3331y - 2738 \\
& \{i, j\} = \{4, 93\}, k_{ij} = 2^2 \cdot 3 \cdot 7^3 \\
& \alpha \quad -32x^2 - 228xy + 404x - 566y + 1778 \\
& \beta \quad 16x^4 + 108x^3y - 226x^3 - 60x^2y^2 + 370x^2y - 953x^2 - 155xy^2 + 394xy + 266x \\
& \quad +26y^4 - 168y^3 + 615y^2 - 1042y - 79 \\
& \gamma \quad -4x^3y + 16x^3 + 8x^2y^2 - 190x^2y + 580x^2 + 305xy^2 - 1193xy + 1801x - 26y^4 \\
& \quad +142y^3 + 67y^2 - 1041y + 1139 \\
& \{i, j\} = \{4, 132\}, k_{ij} = 3^2 \cdot 7^3 \cdot 19 \\
& \alpha \quad 16x^3 + 40x^2y + 12612x^2 - 7542xy + 21720x - 14573y + 30053 \\
& \beta \quad -8x^5 - 16x^4y - 6298x^4 + 10x^3y^2 + 6938x^3y - 4560x^3 - x^2y^3 \\
& \quad -1875x^2y^2 + 4041x^2y - 8666x^2 - 2xy^4 - 828xy^3 + 349xy^2 - 5646xy + 12818x \\
& \quad +y^5 + 825y^4 - 3711y^3 + 11793y^2 - 30681y + 3982 \\
& \gamma \quad -102x^3y + 460x^3 + 204x^2y^2 - 4316x^2y + 12124x^2 + xy^4 - 2xy^3 \\
& \quad +6789xy^2 - 24842xy + 52983x - y^5 - 825y^4 + 3709y^3 + 1132y^2 - 4877y + 35488 \\
& \{i, j\} = \{4, 147\}, k_{ij} = 2^2 \cdot 3^2 \cdot 7^3 \\
& \alpha \quad 2896x^3 + 8752x^2 + 4240x - 462y^2 - 1090y + 4998 \\
& \beta \quad -1448x^5 - 864x^4 - 724x^3y + 1584x^3 + 77x^2y^2 - 4756x^2y + 4015x^2 \\
& \quad -1494xy^2 - 1060xy - 1806x + 385y^3 + 2456y^2 - 5432y + 509 \\
& \gamma \quad 1448x^3y - 4128x^3 + 4376x^2y - 2892x^2 - 724xy^2 + 8708xy - 11638x - 77y^3 \\
& \quad +1828y^2 + 40y - 3155 \\
& \{i, j\} = \{4, 148\}, k_{ij} = 3^4 \cdot 7 \\
& \alpha \quad -60x^2 - 26xy - 60x - 61y + 201 \\
& \beta \quad 30x^4 - 2x^3y - 6x^2y^2 + 25x^2y - 108x^2 + 3xy^3 - 3xy^2 + 22xy + 123x + 6y^3 \\
& \quad +27y^2 - 126y - 36 \\
& \gamma \quad 2x^3y - 2x^2y^2 + 8x^2y - 30x^2 - 7xy^2 + 35xy - 279x + 3y^4 - 14y^3 + 54y^2 - 98y \\
& \quad -72 \\
& \{i, j\} = \{4, 149\}, k_{ij} = 2^4 \cdot 3^4 \cdot 5^5 \cdot 7^4 \cdot 19 \\
& \alpha \quad -319428x^4 + 6072x^3y + 720624564x^3 - 22464927x^2y - 368651726117x^2 \\
& \quad -122758687622xy - 393989807992x - 85086845103y - 34358274432 \\
& \beta \quad 159714x^6 - 82893x^5y - 360471996x^5 + 1808x^4y^2 + 191510503x^4y \\
& \quad +184686294202x^4 + 19295x^3y^3 - 6091387x^3y^2 - 31063667342x^3y
\end{aligned}$$

$$\begin{aligned}
& +12400316383x^3 - 10005x^2y^4 - 43845183x^2y^3 - 30203343969x^2y^2 \\
& +19652636475x^2y - 43106278365x^2 + 1286xy^5 + 23279144xy^4 + 21671478392xy^3 \\
& +42018757155xy^2 - 18140507035xy + 100706291736x + 3y^6 - 3054455y^5 \\
& -3090840080y^4 + 21663320051y^3 + 49766183942y^2 - 42974129343y + 115928432176 \\
\gamma \quad & -232x^5y + 924x^5 + 373x^4y^2 + 378419x^4y - 1509195x^4 - x^3y^3 \\
& -665776x^3y^2 - 382466816x^3y + 1534363675x^3 - 886x^2y^4 + 3520x^2y^3 \\
& +772824126x^2y^2 - 13737631413x^2y + 30155188802x^2 + 1307xy^5 + 1901928xy^4 \\
& -7567562xy^3 + 21323310851xy^2 - 61277547912xy + 154050497864x + 3y^6 - 3054446y^5 \\
& -3100003418y^4 + 12363309788y^3 + 1778431550y^2 + 22380582894y + 85951670560 \\
\{i, j\} = \{4, 206\}, k_{ij} = 2^3 \cdot 3^5 \cdot 7^2 \\
\alpha \quad & 17880x + 4056y + 45228 \\
\beta \quad & -8940x^3 + 2408x^2y - 13522x^2 + 1064xy^2 + 4310xy + 19346x - 1068y^3 + 1394y^2 \\
& -18940y - 19744 \\
\gamma \quad & 34x^3y - 152x^3 - 67x^2y^2 + 2413x^2y - 3513x^2 + xy^3 - 4214xy^2 + 10042xy \\
& +1065x + 267y^4 - 1229y^3 - 6039y^2 + 11328y - 10342 \\
\{i, j\} = \{5, 8\}, k_{ij} = 2^2 \cdot 3^2 \\
\alpha \quad & -y + 6 \\
\beta \quad & y^2 - 6y + 1 \\
\gamma \quad & 2x^3y - 12x^3 + 2x^2y - 12x^2 - xy^2 + 8xy - 13x + y^3 - 5y^2 - 2y - 17 \\
\{i, j\} = \{5, 35\}, k_{ij} = 2^3 \cdot 3^3 \\
\alpha \quad & -26x^2 + 17xy - 39x + 51y + 101 \\
\beta \quad & 52x^4 + 18x^3y + 182x^3 - 17x^2y^2 + 84x^2y + 73x^2 - 17xy^3 - 37xy^2 + 46xy \\
& -38x - 17y^3 - 88y^2 - 72y + 165 \\
\gamma \quad & 17x^3y + 37x^3 + 18x^2y^2 + 114x^2y + 65x^2 + 77xy^2 + 25xy - 132x - 17y^4 \\
& -37y^3 - 39y^2 + 95y + 252 \\
\{i, j\} = \{5, 48\}, k_{ij} = 3^3 \cdot 31^2 \\
\alpha \quad & 56x^2 + 10165x + 7668 \\
\beta \quad & -34x^3 + 66x^2y + 652x^2 - 132xy^2 - 1306xy + 3038x - y^4 + 264y^3 + 2613y^2 - 6080y \\
& +9851 \\
\gamma \quad & -56x^5 + 28x^4y - 10120x^4 - 14x^3y^2 + 4996x^3y - 2968x^3 + 7x^2y^3 \\
& -2375x^2y^2 + 48x^2y + 606x^2 - 3xy^4 + 925xy^3 - 1287xy^2 + 2539xy - 7683x \\
& +y^5 - 264y^4 - 2612y^3 - 1853y^2 - 4531y - 6908 \\
\{i, j\} = \{5, 60\}, k_{ij} = 2^3 \cdot 3^3 \\
\alpha \quad & -26x^2 + 17xy - 39x + 51y + 101 \\
\beta \quad & 52x^4 + 18x^3y + 182x^3 - 17x^2y^2 + 84x^2y + 73x^2 - 17xy^3 - 37xy^2 + 46xy \\
& -38x - 17y^3 - 88y^2 - 72y + 165 \\
\gamma \quad & -17x^3y - 37x^3 - 18x^2y^2 - 114x^2y - 65x^2 - 77xy^2 - 25xy + 132x + 17y^4 \\
& +37y^3 + 39y^2 - 95y - 252 \\
\{i, j\} = \{5, 162\}, k_{ij} = 2^3 \cdot 3^3 \\
\alpha \quad & 2x^3y - 2x^3 + 52x^2y - 48x^2 - 10xy - 14x + 74y + 100 \\
\beta \quad & 244x^3 + 60x^2 - 2xy^3 + 10xy^2 - 114xy + 82x - 52y^3 - 60y^2 - 238y - 84 \\
\gamma \quad & -4x^5 - 100x^4 - 4x^3y + 116x^3 - 96x^2y - 114x^2 - 2xy^3 - 2xy^2 - 46xy + 38x \\
& -52y^3 - 38y^2 - 250y - 308 \\
\{i, j\} = \{5, 171\}, k_{ij} = 2^4 \cdot 3^2 \\
\alpha \quad & 31x^3 - 71x^2 + 52x + 24y^2 - 4y + 12 \\
\beta \quad & -62x^5 - 27x^4 - 124x^3y - 162x^3 + 152x^2y^2 + 497x^2y + 166x^2 - 21xy^2 \\
& -208xy - 225x - 176y^3 - 385y^2 - 101y + 287 \\
\gamma \quad & 62x^3y - 45x^3 - 142x^2y - 111x^2 + 155xy^2 + 557xy + 452x - 152y^3 - 413y^2 - 365y \\
& -179 \\
\{i, j\} = \{5, 174\}, k_{ij} = 2^3 \cdot 3^2 \\
\alpha \quad & 9xy - 18x - 11y + 30 \\
\beta \quad & -18x^3y + 36x^3 + 22x^2y + 12x^2 + 38xy - 12x + 21y - 18 \\
\gamma \quad & 18x^4 - 18x^3y - 4x^3 + 22x^2y - 4x^2 + 9xy^2 + 20xy + 14x - 11y^2 + 51y - 8 \\
\{i, j\} = \{5, 176\}, k_{ij} = 2 \\
\alpha \quad & 0 \\
\beta \quad & 0 \\
\gamma \quad & 1 \\
\{i, j\} = \{5, 183\}, k_{ij} = 2^4 \cdot 3^4 \cdot 13 \\
\alpha \quad & 4743xy - 8730x + 3825y + 11538 \\
\beta \quad & -36x^3y + 8010x^3 - 75x^2y^2 - 422x^2y + 7868x^2 - 148xy^3 - 847xy^2 + 1638xy
\end{aligned}$$

$$\begin{aligned}
& -3634x - 298y^4 - 1693y^3 + 3274y^2 - 6163y - 6274 \\
\gamma \quad & 4725x^4 + 2343x^3y - 2737x^3 + 1117x^2y^2 + 1255x^2y + 2943x^2 + 446xy^3 \\
& -567xy^2 + 3883xy + 2877x + 298y^4 + 2587y^3 + 662y^2 - 458y + 5746 \\
\{i, j\} = \{5, 184\}, k_{ij} = 2^4 \cdot 3^3 \cdot 13 \\
\alpha \quad & -753x^2 + 502xy + 754x - 1310y + 935 \\
\beta \quad & 1506x^4 + 1397x^3 - 1004x^2y^2 + 1543x^2y - 127x^2 - 502xy^3 - 680xy^2 + 792xy \\
& + 552x + 502y^4 - 395y^3 - 114y^2 - 42y + 2546 \\
\gamma \quad & -502x^3y - 107x^3 - 502x^2y^2 - 1182x^2y + 455x^2 - 1075xy^2 - 362xy + 1343x \\
& + 502y^4 + 107y^3 - 815y^2 + 491y + 265 \\
\{i, j\} = \{5, 222\}, k_{ij} = 2^3 \cdot 3^5 \cdot 7 \\
\alpha \quad & -x^2 + 2xy - 458x - 1851y + 2727 \\
\beta \quad & 2x^4 - 2x^3y + 920x^3 - 2x^2y^2 + 2770x^2y - 2720x^2 - xy^3 + 923xy^2 + 180xy \\
& - 173x + y^4 - 921y^3 + 1372y^2 + 737y + 8139 \\
\gamma \quad & -x^3y^2 + 923x^3y - 448x^3 + 923x^2y^2 + 1560x^2y - 3541x^2 - xy^3 \\
& + 2006xy^2 - 511xy - 6193x + y^5 - 922y^4 + 445y^3 + 3025y^2 + 2347y + 1356 \\
\{i, j\} = \{6, 37\}, k_{ij} = 2^3 \cdot 3^5 \cdot 13 \\
\alpha \quad & -126x + 8271 \\
\beta \quad & 3x^3 - 3x^2y - 264x^2 + 3xy^2 + 264xy + 140x - 3y^3 - 264y^2 - 139y + 153 \\
\gamma \quad & -225x^4 + 171x^3y + 14166x^3 - 108x^2y^2 - 9405x^2y + 1269x^2 + 45xy^3 \\
& + 3852xy^2 - 4581xy + 459x - 9y^4 - 801y^3 + 7053y^2 - 750y + 8313 \\
\{i, j\} = \{6, 54\}, k_{ij} = 2^2 \cdot 3^2 \\
\alpha \quad & -9x \\
\beta \quad & x^3 + 2x^2y - x^2 + 2xy^2 - xy - 2x + y^3 - y^2 - y + 12 \\
\gamma \quad & 9x^4 + 9x^3 - 3x^2y^2 + 6x^2y + 15x^2 + 3xy^2 + 3xy - 96x + 3y^4 - 3y^2 + 30y + 33 \\
\{i, j\} = \{6, 130\}, k_{ij} = 2^2 \cdot 3^5 \cdot 7 \\
\alpha \quad & -1620x^3 + 324x + 180y^3 - 360y^2 - 198y + 54 \\
\beta \quad & 360x^5 + 360x^4y + 240x^3y^2 - 120x^3y - 192x^3 - 462x^2y + 282x^2 + 60xy^3 \\
& - 404xy^2 + 646xy + 490x + 60y^4 - 226y^3 + 1342y^2 - 488y + 248 \\
\gamma \quad & -360x^3y^2 + 360x^3y - 513x^3 + 108xy^2 + 2970xy - 1701x - 180y^4 + 429y^3 \\
& - 1935y^2 - 630y - 2949 \\
\{i, j\} = \{6, 149\}, k_{ij} = 2^5 \cdot 3^5 \cdot 5 \cdot 7^3 \cdot 19 \\
\alpha \quad & 268125x^3y + 214509x^3 - 16635888x^2y - 13384404x^2 + 936120147xy + 774911595x \\
& + 1059415011y + 959062050 \\
\beta \quad & -2x^5 - 3x^4y + 13960x^4 - x^3y^2 + 2999x^3y - 5610478x^3 + 7279x^2y^2 \\
& - 4567677x^2y - 34070891x^2 - 2xy^5 + 13xy^4 + 20767xy^3 - 22871397xy^2 \\
& + 530987xy + 36915193x - 4y^6 + 86y^5 + 25651y^4 - 114355218y^3 + 2607161y^2 \\
& - 171639627y - 209580492 \\
\gamma \quad & -107250x^6 - 21450x^5y + 6714417x^5 - 4290x^4y^2 + 1354893x^4y \\
& - 378189429x^4 - 858x^3y^3 + 273381x^3y^2 - 76412910x^3y - 213840903x^3 \\
& - 168x^2y^4 + 1506x^2y^3 - 15461775x^2y^2 + 36314610x^2y + 31254345x^2 \\
& - 30xy^5 + 261xy^4 + 220920xy^3 + 183354291xy^2 - 126004860xy + 297792507x - 12y^6 \\
& + 222y^5 + 77619y^4 - 342832761y^3 + 38737509y^2 - 499291506y - 997531857 \\
\{i, j\} = \{6, 177\}, k_{ij} = 2^5 \cdot 3^5 \cdot 7 \cdot 13 \cdot 19 \\
\alpha \quad & -80290954211867721x^4y + 103601231241119640x^4 - 516650661713970x^3y \\
& - 4024925654800185x^3 + 5606044372633824x^2y - 7233605642108160x^2 \\
& - 810949986236691x - 12215032320300y^4 + 2004904316186583y^3 \\
& + 193354807004753211y^2 + 1509555749608280067y + 2012386926646835760 \\
\beta \quad & -165761969985791424x^6 + 897853275281056530x^5 - 24062141980350x^4y \\
& - 1331090958553582128x^4 - 16041427986900x^3y^2 + 8020713993450x^3y \\
& - 48725666097185472x^3 - 82895113223049312x^2y^3 + 4054972934729952x^2y^2 \\
& - 109597089762605940x^2y - 169960593900077463x^2 - 637732611100xy^4 \\
& + 448988721607809215xy^3 - 21964279249410428xy^2 + 585775728896191406xy \\
& + 1306858451817202211x + 4071677440100y^5 - 633814821374761y^4 \\
& - 670865973953802362y^3 + 29457508419903323y^2 - 909280644199004335y \\
& - 2012386926642354192 \\
\gamma \quad & 160581908423735442x^7 - 1009078703277488550x^6 + 497285909957374272x^5y \\
& + 1561446468527978718x^5 - 1201774282396987824x^4y - 294626020279643325x^4 \\
& + 80315384276508321x^3y^3 - 4150281504056616x^3y^2 + 78383237143909671x^3y \\
& + 148920699924901662x^3 - 504539351638744275x^2y^3 \\
& + 23509218336851520x^2y^2 - 594901618786592019x^2y - 1449530085682094346x^2 \\
& + 248642954978687136xy^4 + 774755509609250127xy^3 + 285144126270965010xy^2
\end{aligned}$$

$$\begin{aligned}
& +1761908205570323139xy + 2330484486049247661x + 38558294794200y^5 \\
& -607048186440259386y^4 - 138200944079438550y^3 - 813436960866187134y^2 \\
& -2195670005547275304y - 741316179017466540 \\
\{i, j\} = \{6, 183\}, k_{ij} = 2^3 \cdot 3^6 \cdot 13 \\
\alpha & 10071xy - 50247x + 39447y - 4131 \\
\beta & -76x^3y + 9004x^3 - 157x^2y^2 - 560x^2y + 5040x^2 - 315xy^3 - 1120xy^2 \\
& + 4196xy - 8806x - 633y^4 - 2238y^3 + 8382y^2 - 32313y + 28269 \\
\gamma & 9729x^4 + 4500x^3y - 8820x^3 + 1425x^2y^2 + 3387x^2y + 17301x^2 - 954xy^3 \\
& - 6792xy^2 + 22368xy - 92346x + 1899y^4 + 12411y^3 - 27360y^2 + 52740y - 1701 \\
\{i, j\} = \{6, 213\}, k_{ij} = 2^2 \cdot 3^6 \\
\alpha & 567x + 81 \\
\beta & -126x^3 + 234x^2 - 342x + 387 \\
\gamma & 1134x^4 - 378x^3y - 1728x^3 + 702x^2y + 2376x^2 + 567xy^2 - 2160xy + 378x + 81y^2 \\
& + 999y - 756 \\
\{i, j\} = \{6, 224\}, k_{ij} = 2^2 \cdot 3^6 \\
\alpha & 69012x^4y + 441936x^4 - 5290920y^2 - 20422935y - 19986588 \\
\beta & -15336x^6y - 249210x^6 - 15336x^5y^2 - 241542x^5y + 3834x^5 - 10224x^4y^3 \\
& + 94640x^4y^2 + 223766x^4y - 1063930x^4 + 1322730x^2y^2 + 4890918x^2y \\
& - 686730x^2 + 1322730xy^3 + 4890918xy^2 + 3774762xy + 2230746x + 881820y^4 \\
& + 2819702y^3 - 2895329y^2 + 6768640y + 16121195 \\
\gamma & 453006x^7 + 279000x^6y + 139500x^6 - 558000x^5y^2 - 209250x^5y + 3487500x^5 \\
& - 15336x^4y^4 + 168798x^4y^3 + 598842x^4y^2 - 721836x^4y + 1011354x^4 \\
& + 16343694x^3 - 2488680x^2y - 1244340x^2 + 12824235xy^2 - 19476990xy - 48091635x \\
& + 1322730y^5 + 2245458y^4 - 10124907y^3 + 2463219y^2 + 10612302y - 11599095 \\
\{i, j\} = \{8, 12\}, k_{ij} = 2^5 \\
\alpha & -2y + 6 \\
\beta & 4x^3y - 12x^3 + 4x^2y - 12x^2 - 2xy^2 + 6xy - 8x + 2y^3 - 2y^2 + 2y - 18 \\
\gamma & 2y^2 - 2y - 4 \\
\{i, j\} = \{8, 43\}, k_{ij} = 2^3 \\
\alpha & -2y \\
\beta & 4x^3y + 4x^2y - 2xy^2 + 8xy - 8x - 2y^3 + 4y^2 - 2y \\
\gamma & 2y^2 - 4y + 8 \\
\{i, j\} = \{8, 62\}, k_{ij} = 2^2 \\
\alpha & -y \\
\beta & 2x^3y + 2x^2y - 3xy^2 - 4xy - 3x + 3y^3 + 3y^2 - 3 \\
\gamma & y^2 + 2y + 1 \\
\{i, j\} = \{9, 10\}, k_{ij} = 3^2 \cdot 7 \\
\alpha & -39x^3 + 7x^2y + 7x^2 - 27x + 7y^3 - 7y + 21 \\
\beta & 64x^5 - 78x^4y - 14x^3y^2 + 92x^3y - 24x^3 + 39x^2y^2 - 93x^2y + 146x^2 \\
& - 7xy^2 - 21xy - x - 7y^4 + 7y^3 + 13y^2 - 19y + 60 \\
\gamma & 78x^5 - 78x^4 + 54x^3 + 21x^2 - 36x - 7y^4 - 42y + 7 \\
\{i, j\} = \{10, 36\}, k_{ij} = 3^2 \\
\alpha & -x + 3 \\
\beta & 2x^3 + 2x^2y - 6x^2 - 6xy + 1 \\
\gamma & 2x^2y^2 - 7xy^2 - 2x + 3y^2 + y + 3 \\
\{i, j\} = \{10, 46\}, k_{ij} = 5 \cdot 7 \\
\alpha & 9 \\
\beta & -15x^2 - 18xy - 9y^2 + 3y - 4 \\
\gamma & -3x^3 + 6x^2y - 3x^2 + 6xy + x + 9y^3 - 3y^2 + y - 8 \\
\{i, j\} = \{10, 74\}, k_{ij} = 3^4 \cdot 19 \\
\alpha & -72x^4 + 192x - 192y^2 + 192y + 513 \\
\beta & 144x^6 + 144x^5y + 216x^4y + 216x^4 - 384x^3 - 640x^2y - 874x^2 - 64xy^2 - 938xy \\
& + 192y^3 - 448y^2 + 21y - 171 \\
\gamma & -216x^5 - 144x^4y^2 + 360x^4y + 72x^4 - 192x^3y + 64x^3 + 384xy^2 - 21x \\
& + 256y^3 - 342y^2 + 555y - 1089 \\
\{i, j\} = \{10, 96\}, k_{ij} = 2 \cdot 3^2 \\
\alpha & 3x^2 - 3y + 6 \\
\beta & -6x^4 - 6x^3y - x^2y^2 + 4x^2y - 12x^2 + 2xy^3 + 2xy^2 - 10xy - 3x + y^4 - y^3 \\
& - y^2 - 5y + 9 \\
\gamma & -x^3y - 2x^2y^2 + x^2y + xy^2 - 4xy + 9x + y^4 - 4y^2 - 3y + 3
\end{aligned}$$

$$\begin{aligned}
& \{i, j\} = \{10, 98\}, k_{ij} = 3^3 \cdot 7^2 \\
& \alpha \quad -248x^3 - 124x^2y + 176x^2 + 169xy - 236x - 147y + 441 \\
& \beta \quad 496x^5 + 496x^4y - 374x^4 - 542x^3y + 43x^3 - 248x^2y^3 - 146x^2y^2 \\
& \quad + 155x^2y - 484x^2 + 248xy^3 + 271xy^2 + 108xy - 128x + 248y^4 + 22y^3 + 213y^2 \\
& \quad - 197y - 120 \\
& \gamma \quad 248x^3y + 22x^3 + 248x^2y^2 - 148x^2y + 429x^2 - 170xy^2 + 363xy - 172x - 248y^4 \\
& \quad - 22y^3 - 66y^2 - 244y + 267 \\
& \{i, j\} = \{10, 99\}, k_{ij} = 3^2 \cdot 19 \\
& \alpha \quad -19x - 2y^2 - 11y + 57 \\
& \beta \quad 38x^3 + 38x^2y - 114x^2 + 2xy^2 - 103xy + 2y^3 + 11y^2 + 2y + 19 \\
& \gamma \quad -2x^3y - 11x^3 + 19xy^2 + x - 57y^2 - 2y - 24 \\
& \{i, j\} = \{10, 100\}, k_{ij} = 2^2 \cdot 3^2 \cdot 5^2 \cdot 7 \\
& \alpha \quad 400x^2 + 200xy - 700x - 700y + 2100 \\
& \beta \quad -800x^4 - 1000x^3y + 1460x^3 - 200x^2y^2 + 2080x^2y - 3940x^2 + 400xy^3 \\
& \quad + 700xy^2 - 3360xy + 1000x + 200y^4 + 60y^3 - 800y^2 - 600y + 700 \\
& \gamma \quad 200x^3y + 60x^3 + 200x^2y^2 - 920x^2y + 200x^2 - 980xy^2 + 1700xy + 1200x \\
& \quad - 200y^4 - 60y^3 + 1500y^2 - 1500y \\
& \{i, j\} = \{10, 101\}, k_{ij} = 2 \cdot 3^3 \cdot 7 \\
& \alpha \quad 24x^2 - 16xy - 28x - 56y + 126 \\
& \beta \quad -48x^4 - 16x^3y + 56x^3 + 16x^2y^2 + 116x^2y - 222x^2 + 16xy^3 + 78xy^2 - 81xy \\
& \quad - 114x + 8y^3 + 87y^2 - 228y + 60 \\
& \gamma \quad 8x^3y + 30x^3 + 8x^2y^2 - 9x^2y - 90x^2 - 39xy^2 + 12xy + 104x - 8y^4 - 31y^3 \\
& \quad + 102y^2 - 4y + 42 \\
& \{i, j\} = \{10, 103\}, k_{ij} = 3^4 \cdot 5 \cdot 7 \\
& \alpha \quad 30x^2 - 20xy - 875x + 805y + 945 \\
& \beta \quad -60x^4 - 20x^3y + 1750x^3 + 20x^2y^2 + 440x^2y - 2035x^2 + 20xy^3 - 260xy^2 \\
& \quad - 1795xy - 270x + 20y^3 - 1095y^2 - 3435y + 3105 \\
& \gamma \quad 10x^3y - 145x^3 + 10x^2y^2 - 530x^2y - 240x^2 - 385xy^2 + 1005xy + 2320x - 10y^4 \\
& \quad + 145y^3 + 1245y^2 - 1955y - 1680 \\
& \{i, j\} = \{10, 104\}, k_{ij} = 2 \cdot 3^2 \cdot 5 \cdot 7 \\
& \alpha \quad 120xy + 90x + 90y + 210 \\
& \beta \quad -60x^3 - 2x^2y^2 - 5x^2y - 392x^2 - 3xy^3 - 9xy^2 - 58xy + 184x - 8y^4 - 18y^3 \\
& \quad - 118y^2 - 366y - 360 \\
& \gamma \quad -120x^4 - 59x^3y - 28x^3 - 29x^2y^2 + 62x^2y - 184x^2 - 11xy^3 + 58xy^2 + 74xy \\
& \quad + 270x - 8y^4 - 26y^3 - 54y^2 - 210y - 480 \\
& \{i, j\} = \{10, 105\}, k_{ij} = 3^3 \cdot 7 \\
& \alpha \quad 20x^3 - 45x^2 + 63 \\
& \beta \quad -20xy^2 - 10xy - 63x + 5y^2 - 88y + 24 \\
& \gamma \quad -40x^5 + 50x^4 - 40x^3y + 50x^3 + 10x^2y - 96x^2 - 20xy^3 - 20xy^2 + 50xy - 48x \\
& \quad + 5y^3 - 25y^2 + 24y + 63 \\
& \{i, j\} = \{10, 106\}, k_{ij} = 2 \cdot 3^2 \cdot 5 \cdot 7 \\
& \alpha \quad 71x^3 + 35x^2y - 13x^2 - 4xy - 61x - 70y + 210 \\
& \beta \quad -142x^5 - 141x^4y - 41x^4 + x^3y - 20x^3 + 71x^2y^3 + 39x^2y^2 + 52x^2y \\
& \quad - 367x^2 + 110xy^2 - 142xy - 85x - 71y^4 + 67y^3 + 6y^2 + 3y + 43 \\
& \gamma \quad 71x^3y - 67x^3 + 70x^2y^2 - 33x^2y - 142x^2 + 35xy^2 - 206xy + 57x - 71y^4 \\
& \quad + 67y^3 - 64y^2 + 213y - 27 \\
& \{i, j\} = \{10, 108\}, k_{ij} = 2^2 \cdot 3^2 \cdot 13 \cdot 19 \\
& \alpha \quad 49x^3 - 148x^2y - 225x^2 + 978xy - 988x - 1895y + 2964 \\
& \beta \quad -98x^5 + 99x^4y + 494x^4 + 99x^3y^2 - 338x^3y + 1292x^3 + 149x^2y^2 \\
& \quad + 2390x^2y - 4868x^2 + 198xy^3 + 115xy^2 - 2920xy - 2643x + 99y^4 + 56y^3 - 200y^2 \\
& \quad - 2527y + 1663 \\
& \gamma \quad -99x^3y + 44x^3 - 98x^2y^2 + 1124x^2y - 684x^2 + 1080xy^2 - 2736xy + 1109x \\
& \quad + 99y^4 - 43y^3 - 2052y^2 + 2489y - 2721 \\
& \{i, j\} = \{10, 109\}, k_{ij} = 2^2 \cdot 3^3 \\
& \alpha \quad 35x^3 + 17x^2y - 76x^2 - xy + 39x + 39y + 36 \\
& \beta \quad -70x^5 - 69x^4y + 153x^4 + x^3y^2 + 104x^3y - 33x^3 + 34x^2y^3 \\
& \quad - 16x^2y^2 - 129x^2y - 227x^2 - 35xy^3 - 156xy^2 - 59xy + 3x - 34y^4 - 35y^3 \\
& \quad - 8y^2 + 59y + 177 \\
& \gamma \quad 35x^3y + x^3 + 35x^2y^2 - 85x^2y + 44x^2 - xy^3 - 87xy^2 + 76xy - 165x - 34y^4 \\
& \quad - y^3 + 32y^2 + 63y + 153
\end{aligned}$$

$$\begin{aligned}
& \{i, j\} = \{10, 110\}, k_{ij} = 3^4 \cdot 5 \cdot 7 \\
& \alpha \quad -30x^2 + 20xy + 245x - 595y + 945 \\
& \beta \quad 60x^4 + 20x^3y - 490x^3 - 20x^2y^2 + 316x^2y - 1202x^2 - 20xy^3 + 344xy^2 \\
& \quad -257xy - 780x - 20y^3 + 939y^2 - 1110y + 375 \\
& \gamma \quad -20x^3y + 344x^3 - 20x^2y^2 + 442x^2y - 405x^2 + 98xy^2 - 240xy + 265x + 20y^4 \\
& \quad -344y^3 + 165y^2 + 220y + 840 \\
& \{i, j\} = \{10, 113\}, k_{ij} = 3^5 \cdot 5 \cdot 19 \\
& \alpha \quad -3630x^2 - 1815xy - 5835x + 3975y + 7695 \\
& \beta \quad 7260x^4 + 10890x^3y + 11670x^3 + 3630x^2y^2 + 90x^2y - 14595x^2 - 3630xy^3 \\
& \quad -8745xy^2 - 7665xy + 1920x - 3630y^4 - 6465y^3 - 675y^2 + 7365y + 29115 \\
& \gamma \quad 3630x^3y - 795x^3 + 3630x^2y^2 - 8520x^2y + 1710x^2 - 7725xy^2 + 3420xy - 12390x \\
& \quad -3630y^4 + 795y^3 + 1710y^2 + 11640y + 9810 \\
& \{i, j\} = \{10, 114\}, k_{ij} = 2 \cdot 3^2 \cdot 7 \cdot 13 \\
& \alpha \quad 162x^2 + 82xy - 187x - 182y + 546 \\
& \beta \quad -325x^4 - 406x^3y + 333x^3 - 81x^2y^2 + 431x^2y - 782x^2 + 162xy^3 + 101xy^2 \\
& \quad -402xy - 881x + 81y^4 - 41y^3 + 28y^2 - 821y + 167 \\
& \gamma \quad x^4 - 82x^3y + 42x^3 - 81x^2y^2 + 226x^2y - 268x^2 + 184xy^2 - 422xy + 451x \\
& \quad + 81y^4 - 41y^3 - 154y^2 - 275y - 15 \\
& \{i, j\} = \{10, 196\}, k_{ij} = 2^3 \cdot 3^2 \cdot 5 \cdot 7 \\
& \alpha \quad 105y^2 - 210y + 840 \\
& \beta \quad -160x^4 + 420x^2y - 1552x^2 + 420xy^2 - 1680xy - 168y^2 - 700y - 120 \\
& \gamma \quad -40x^5 + 50x^4y + 32x^3 - 40x^2y - 175xy - 30x + 105y^4 - 378y^3 + 245y^2 - 15y + 210 \\
& \{i, j\} = \{11, 12\}, k_{ij} = 2^6 \cdot 7^2 \cdot 11 \\
& \alpha \quad -x^3 - 2x^2y - 6088x^2 + 4061xy + 3422x - 6517y + 9437 \\
& \beta \quad x^4y - x^4 + 2x^3y^2 + 4058x^3y - 9124x^3 + 3x^2y^3 + 4057x^2y^2 \\
& \quad -2917x^2y - 7201x^2 + 2xy^4 - xy^3 + 6204xy^2 - 7709xy + 11908x - y^5 - 4054y^4 \\
& \quad + 9119y^3 - 505y^2 + 5555y - 10604 \\
& \gamma \quad 2x^5 + 6x^4y + 12178x^4 + 5x^3y^2 + 4061x^3y + 5334x^3 + x^2y^3 \\
& \quad -4051x^2y^2 + 17360x^2y - 7469x^2 - xy^4 - 4052xy^3 + 5058xy^2 + 676xy + 7690x \\
& \quad -y^4 - 4055y^3 + 11582y^2 + 5695y + 6185 \\
& \{i, j\} = \{12, 41\}, k_{ij} = 2 \cdot 7 \\
& \alpha \quad y + 2 \\
& \beta \quad -y^2 - 2y + 2 \\
& \gamma \quad -2x^3y - 4x^3 + 2x^2y + 4x^2 + xy^2 - 6x - y^3 - 4y^2 - y + 6 \\
& \{i, j\} = \{12, 63\}, k_{ij} = 2^7 \cdot 3 \cdot 7^2 \\
& \alpha \quad -2x^2y - 4536x^2 + 3026xy + 7224x - 4200y + 840 \\
& \beta \quad 4x^4y + 9072x^4 + 4x^3y^2 + 3023x^3y - 5376x^3 + x^2y^3 - 3019x^2y^2 \\
& \quad + 5874x^2y - 16128x^2 - xy^4 - 3018xy^3 + 6203xy^2 - 16290xy - 6216x + y^4 \\
& \quad + 3017y^3 - 2012y^2 + 6213y + 16296 \\
& \gamma \quad -x^4 + x^3y^2 + 3022x^3y - 9242x^3 + 2x^2y^3 + 3020x^2y^2 - 2025x^2y \\
& \quad -6881x^2 + 2xy^4 + 7210xy^2 - 18973xy + 42324x - y^5 - 3016y^4 + 9228y^3 \\
& \quad -12082y^2 - 3871y + 21843 \\
& \{i, j\} = \{12, 85\}, k_{ij} = 2^9 \cdot 3^4 \cdot 5 \cdot 7 \cdot 19 \\
& \alpha \quad 71925x^3 - 4362552x + 1366575y^3 - 866250y^2 - 4359402y + 11597064 \\
& \beta \quad -143850x^5 - 143850x^4y - 143850x^4 - 431550x^3y + 8725104x^3 + 239400x^2y^2 \\
& \quad + 7264926x^2y - 8194211x^2 - 1127175xy^3 - 1232328xy^2 + 14579882xy - 20236534x \\
& \quad -1366575y^4 - 659925y^3 + 2391979y^2 - 22525678y + 2308411 \\
& \gamma \quad 863100x^3y^2 + 221550x^3y - 3101444x^3 + 2908368xy^2 - 6786192xy - 4994091x \\
& \quad + 79800y^4 + 693649y^3 + 5720346y^2 - 7273785y + 9520723 \\
& \{i, j\} = \{12, 115\}, k_{ij} = 2^8 \cdot 3 \cdot 7 \\
& \alpha \quad 108x^2 - 72xy - 900x - 768y + 2304 \\
& \beta \quad -216x^4 - 72x^3y + 1584x^3 + 72x^2y^2 + 2080x^2y - 1984x^2 + 72xy^3 \\
& \quad + 894xy^2 + 42xy - 2292x + 768y^2 - 1536y - 1536 \\
& \gamma \quad -36x^3y - 412x^3 - 36x^2y^2 + 163x^2y - 692x^2 + 575xy^2 - 1505xy + 902x + 36y^4 \\
& \quad + 411y^3 - 813y^2 - 1185y + 576 \\
& \{i, j\} = \{12, 131\}, k_{ij} = 2^7 \cdot 3 \cdot 5 \cdot 7 \cdot 13 \\
& \alpha \quad 12x^2 - 12018xy - 7842x - 8370y + 50058 \\
& \beta \quad -12x^3y^2 + 23964x^3y + 63738x^3 + 12x^2y^3 + 27x^2y^2 + 8350x^2y \\
& \quad + 20999x^2 + 4xy^3 + 13xy^2 - 4994xy + 20139x - 2y^3 + 28y^2 - 37104y + 31654 \\
& \gamma \quad 12x^5 - 24x^4y + 24021x^4 + 24x^3y^2 - 24013x^3y + 16678x^3 - 12x^2y^3
\end{aligned}$$

$$\begin{aligned}
& +13x^2y^2 - 8374x^2y + 23106x^2 - 4xy^3 + 12015xy^2 - 11159xy + 801x + 2y^3 \\
& +8336y^2 - 29594y + 7108 \\
\{i, j\} = \{12, 132\}, k_{ij} = 2^4 \cdot 7 \cdot 19 \\
\alpha & -170x^2 + 34xy - 10x - 100y + 512 \\
\beta & 340x^4 + 272x^3y + 360x^3 + 812x^2y - 556x^2 - 136xy^3 + 479xy^2 - 274xy - 58x \\
& -68y^4 + 87y^3 + 171y^2 - 323y + 638 \\
\gamma & -68x^3y + 224x^3 - 68x^2y^2 - 47x^2y + 232x^2 - 271xy^2 + 607xy - 306x + 68y^4 \\
& -223y^3 + 375y^2 - 703y + 46 \\
\{i, j\} = \{12, 133\}, k_{ij} = 2^8 \cdot 3 \cdot 7^2 \cdot 13 \\
\alpha & x^4 + 499x^3 - 1546x^2y - 582x^2 + 27583xy - 77090x - 7011y + 119298 \\
\beta & -2x^6 - 2x^5y - 999x^5 - x^4y^2 + 1062x^4y + 9977x^4 + 1040x^3y^2 \\
& -12908x^3y + 99674x^3 - 3x^2y^3 + 565x^2y^2 + 51655x^2y - 68269x^2 - xy^4 \\
& +2106xy^3 - 22757xy^2 + 47639xy - 36177x + 1043y^4 - 8796y^3 + 14108y^2 - 87798y \\
& +9224 \\
\gamma & x^4 - x^3y^2 - 1027x^3y + 9811x^3 - x^2y^3 - 1019x^2y^2 + 31188x^2y \\
& -55669x^2 - xy^4 + 21xy^3 + 21353xy^2 - 37697xy + 35275x + 1043y^4 - 9839y^3 \\
& +17979y^2 - 1243y + 122098 \\
\{i, j\} = \{12, 134\}, k_{ij} = 2^6 \cdot 3 \cdot 7 \\
\alpha & -75x^3 + 99x^2 + 414x - 459y^2 + 1026y - 837 \\
\beta & 150x^5 + 40x^4 + 150x^3y + 929x^3 + 605x^2y^2 - 2580x^2y + 2320x^2 - 371xy^2 \\
& +285xy + 2044x - 146y^3 + 1282y^2 + 1294y - 1745 \\
\gamma & -150x^3y + 88x^3 + 198x^2y + 1779x^2 - 225xy^2 - 66xy - 2097x - 605y^3 + 2454y^2 \\
& -2640y + 2800 \\
\{i, j\} = \{12, 135\}, k_{ij} = 2^8 \cdot 7 \cdot 19 \\
\alpha & -2x^2y - 3100x^2 + 2069xy + 1357x - 6205y + 10137 \\
\beta & 4x^4y + 6200x^4 + 4x^3y^2 + 2066x^3y + 3486x^3 + x^2y^3 - 2062x^2y^2 \\
& +12917x^2y - 7868x^2 - xy^4 - 2061xy^3 + 5033xy^2 - 1170xy + 3095x - y^4 - 2066y^3 \\
& +7111y^2 + 2455y - 819 \\
\gamma & -x^3y^2 - 2067x^3y + 5040x^3 - 2x^2y^3 - 2067x^2y^2 + 265x^2y + 5463x^2 \\
& -2xy^4 - 4773xy^2 + 7084xy - 4754x + y^5 + 2064y^4 - 5035y^3 + 1620y^2 + 748y + 4456 \\
\{i, j\} = \{12, 136\}, k_{ij} = 2^5 \cdot 3 \cdot 7 \cdot 13 \\
\alpha & 438x^2 - 24x - 546y + 1224 \\
\beta & -438x^3y + 987x^3 - 414x^2y + 903x^2 + 438xy^3 - 987xy^2 - 534xy + 3912x + 414y^3 \\
& -903y^2 - 1212y + 5064 \\
\gamma & 876x^4 - 438x^3y + 939x^3 - 438x^2y^2 - 303x^2y + 2364x^2 + 438xy^3 \\
& -963xy^2 - 570xy + 4434x + 414y^3 - 1035y^2 - 741y + 3345 \\
\{i, j\} = \{12, 137\}, k_{ij} = 2^6 \cdot 3 \cdot 7 \cdot 13 \\
\alpha & -1203x^2 + 802xy + 2955x - 3798y + 3428 \\
\beta & 2406x^4 - 1566x^3 - 1604x^2y^2 + 4273x^2y - 4109x^2 - 802xy^3 + 2117xy^2 \\
& -183xy + 670x + 802y^4 - 1935y^3 + 4757y^2 + 4858y + 3151 \\
\gamma & -802x^3y + 1938x^3 - 802x^2y^2 + 1317x^2y + 2843x^2 - 620xy^2 + 5407xy - 1738x \\
& +802y^4 - 1935y^3 + 2563y^2 + 2222y + 4037 \\
\{i, j\} = \{12, 200\}, k_{ij} = 2^6 \cdot 3^2 \cdot 7 \\
\alpha & -686x + 343y + 847 \\
\beta & 1372x^3 + 588x^2y - 164x^2 - 392xy^2 + 641xy - 835x - 490y^3 + 344y^2 - 1036y - 97 \\
\gamma & 98x^3y - 158x^3 + 98x^2y^2 - 27x^2y - 69x^2 + 130xy^2 - 147xy + 388x - 98y^4 \\
& +157y^3 - 79y^2 + 262y + 397 \\
\{i, j\} = \{13, 23\}, k_{ij} = 2^2 \\
\alpha & -x^3 - x^2 - x - 1 \\
\beta & -2x^5 - 2x^4 - x^3y^2 + x^3y - x^2y^2 + x^2y - x^2 - xy^2 + xy + 2x - y^2 + 2y + 4 \\
\gamma & 2x^3 + 2x^2 + 2x + 3 \\
\{i, j\} = \{13, 43\}, k_{ij} = 2^2 \cdot 3 \cdot 7 \cdot 19 \\
\alpha & -3x + 69 \\
\beta & 6x^4 + 6x^3y - 126x^3 - 138x^2y - 294x^2 - 3xy^2 - 21xy + 375x + 69y^2 + 486y + 972 \\
\gamma & 6x^3 - 138x^2 - 18x + 417 \\
\{i, j\} = \{13, 76\}, k_{ij} = 3 \cdot 7 \\
\alpha & x + 4 \\
\beta & 4x^4 - 2x^3y + 20x^3 - 10x^2y + 16x^2 + xy^2 - 7xy + 4y^2 + 5y + 8 \\
\gamma & -2x^3 - 10x^2 - 8x + 1 \\
\{i, j\} = \{13, 80\}, k_{ij} = 3 \cdot 19
\end{aligned}$$

α	$7x^3 + 36x^2 - 9x + 121$
β	$-14x^5 - 86x^4 + 7x^3y^2 - 7x^3y - 26x^3 + 36x^2y^2 - 22x^2y - 3x^2$ $-9xy^2 + 109xy + 62x + 121y^2 - 16y - 169$
γ	$-14x^3 - 58x^2 + 118x - 137$
$\{i, j\} = \{14, 15\}, k_{ij} = 2^3 \cdot 7 \cdot 13 \cdot 31^2$	
α	$-19x^3 + 17x^2y - 21813x^2 + 14423xy - 8027x - 11690y + 53146$
β	$-x^4y - 2x^3y^2 - 14489x^3y - 997x^3 + 10x^2y^3 - 14488x^2y^2$ $+125297x^2y + 48397x^2 + 23xy^4 - 91xy^3 + 126258xy^2 - 30022xy - 200526x + 12y^5$ $+14304y^4 + 984y^3 - 78272y^2 + 73525y - 58397$
γ	$38x^5 + 4x^4y + 43512x^4 - 33x^3y^2 + 14656x^3y - 114596x^3 - 35x^2y^3$ $-14264x^2y^2 - 105219x^2y + 108973x^2 - 12xy^4 - 14328xy^3 - 44049xy^2$ $+110651xy - 56061x - 24y^3 - 16918y^2 - 55042y + 218988$
$\{i, j\} = \{15, 47\}, k_{ij} = 2^2 \cdot 3^2 \cdot 7 \cdot 31^2$	
α	$-6x^3 + 5x^2y - 6302x^2 + 4175xy - 4706x + 6125y + 21355$
β	$12x^5 + 2x^4y + 12568x^4 - 10x^3y^2 + 4213x^3y - 28326x^3 - 11x^2y^3$ $-4138x^2y^2 - 37116x^2y + 20043x^2 - 4xy^4 - 4141xy^3 - 16022xy^2 + 25953xy$ $-44201x + 4y^4 + 4140y^3 - 2600y^2 - 49098y + 73747$
γ	$-x^4 - x^3y^2 - 4185x^3y - 7736x^3 + 3x^2y^3 - 4178x^2y^2 + 23848x^2y$ $+41553x^2 + 7xy^4 - 11xy^3 + 31553xy^2 + 33905xy - 9607x + 4y^5 + 4144y^4$ $+7681y^3 - 7642y^2 + 64430y - 43134$
$\{i, j\} = \{15, 48\}, k_{ij} = 2 \cdot 3^3 \cdot 7^2 \cdot 13$	
α	$-56x^2 - 6742x + 18166$
β	$22x^3 - 43x^2y - 837x^2 + 86xy^2 + 1673xy + 7859x + y^4 - 170y^3 - 3344y^2 - 15718y$ -8487
γ	$56x^5 - 28x^4y + 6703x^4 + 14x^3y^2 - 3305x^3y - 21132x^3 - 7x^2y^3$ $+1570x^2y^2 + 11028x^2y + 7887x^2 + 3xy^4 - 604xy^3 - 3501xy^2 - 86xy - 6778x$ $-y^5 + 171y^4 + 3170y^3 - 5108y^2 + 23627y + 5361$
$\{i, j\} = \{15, 64\}, k_{ij} = 2^5 \cdot 3^4 \cdot 13 \cdot 31^2$	
α	$-2465x^3 + 228942x - 32045y^3 + 550275y^2 - 2785071y - 1884651$
β	$4930x^5 + 4930x^4y - 14790x^4 - 29580x^3y - 428304x^3 + 90480x^2y^2$ $-1221850x^2y + 5101049x^2 + 122525xy^3 - 946666xy^2 + 6446164xy - 1135564x$ $+32045y^4 - 550275y^3 + 2540021y^2 + 3681848y - 12125397$
γ	$59160x^3y^2 - 1205820x^3y + 6437638x^3 + 457884xy^2 + 1373652xy - 20490354x$ $-90480y^4 + 300256y^3 - 2819790y^2 + 402840y - 37206000$
$\{i, j\} = \{15, 77\}, k_{ij} = 2 \cdot 3 \cdot 13 \cdot 31^2$	
α	$203x^2 - 127xy - 3782x + 8815y + 3111$
β	$-406x^4 - 151x^3y + 8784x^3 + 119x^2y^2 - 3235x^2y - 20495x^2 + 129xy^3$ $-4872xy^2 - 23208xy - 16255x + 129y^3 - 13804y^2 - 23773y + 10145$
γ	$-x^4 + 136x^3y - 5429x^3 + 141x^2y^2 - 7922x^2y + 33736x^2 + 10xy^3$ $-2517xy^2 + 37883xy + 1035x - 129y^4 + 5376y^3 + 4147y^2 + 10011y + 11730$
$\{i, j\} = \{15, 89\}, k_{ij} = 2 \cdot 3^3 \cdot 7 \cdot 13^3 \cdot 31^2$	
α	$325488x^3 + 23027688x + 1627440y^3 + 25062576y^2 + 24997092y + 75965682$
β	$-650976x^5 - 650976x^4y + 1952928x^4 + 1301952x^3y^2 + 35152704x^3y$ $-13805423x^3 + 13019520x^2y^2 - 68655286x^2y - 44387749x^2 + 11392080xy^3$ $-23494414xy^2 + 16262371xy + 126034793x + 4882320y^4 - 5441171y^3 + 68820629y^2$ $+170232821y + 69846765$
γ	$1301952x^3y^2 + 31246848x^3y + 36155809x^3 - 23027688xy^2 - 80596908xy$ $-11412450x + 6509760y^4 + 6601885y^3 + 103398111y^2 + 59625549y + 360640485$
$\{i, j\} = \{15, 92\}, k_{ij} = 2 \cdot 3^3 \cdot 7 \cdot 13^3 \cdot 31^2$	
α	$325488x^3 + 23027688x + 1627440y^3 + 25062576y^2 + 24997092y + 75965682$
β	$-650976x^5 - 650976x^4y + 1952928x^4 + 1301952x^3y^2 + 35152704x^3y$ $-13805423x^3 + 13019520x^2y^2 - 68655286x^2y - 44387749x^2 + 11392080xy^3$ $-23494414xy^2 + 16262371xy + 126034793x + 4882320y^4 - 5441171y^3 + 68820629y^2$ $+170232821y + 69846765$
γ	$-1301952x^3y^2 - 31246848x^3y - 36155809x^3 + 23027688xy^2 + 80596908xy$ $+11412450x - 6509760y^4 - 6601885y^3 - 103398111y^2 - 59625549y - 360640485$
$\{i, j\} = \{15, 104\}, k_{ij} = 2 \cdot 3^3 \cdot 7 \cdot 31^2$	
α	$105x^2y + 53x^2 - 66481xy + 8135x - 83884y + 77274$
β	$-x^4 - x^3y^2 + 530x^3y - 82561x^3 - x^2y^3 + 1042x^2y^2 + 3395x^2y$ $+102640x^2 - 2xy^4 + 2087xy^3 + 6814xy^2 + 3039xy - 104430x - 3y^5 + 4178y^4$

$$\begin{aligned}
& +13630y^3 + 6059y^2 + 31401y + 43812 \\
\gamma \quad & -105x^5 - 52x^4y + 66294x^4 - 26x^3y^2 + 32932x^3y - 9502x^3 - 12x^2y^3 \\
& +15699x^2y^2 + 8943x^2y + 44140x^2 - 5xy^4 + 6242xy^3 - 14760xy^2 - 39688xy \\
& -46736x - 3y^5 + 4175y^4 + 17796y^3 - 47495y^2 + 102070y + 80175 \\
\{i, j\} = \{15, 105\}, k_{ij} = 3 \cdot 7^2 \cdot 13 \\
\alpha \quad & -30x^3 + 260x^2 - 714x + 771 \\
\beta \quad & 30xy^2 + 344xy + 161x - 200y^2 - 687y - 134 \\
\gamma \quad & 60x^5 - 460x^4 + 60x^3y + 968x^3 - 400x^2y - 544x^2 + 30xy^3 + 30xy^2 + 254xy \\
& -875x - 200y^3 + 84y^2 - 734y - 1290 \\
\{i, j\} = \{15, 140\}, k_{ij} = 2^2 \cdot 3^3 \cdot 5^2 \cdot 7^2 \cdot 13 \cdot 31^2 \\
\alpha \quad & 170625x^3y + 34130x^3 - 9741676x^2y - 2080655x^2 + 103732096xy + 466673024x \\
& +463221500y + 1210684212 \\
\beta \quad & -10x^5 - 43x^4y + 2583x^4 - 11x^3y^2 + 46119x^3y - 876658378x^3 - 29x^2y^3 \\
& +15701x^2y^2 - 819420x^2y + 396307417x^2 - xy^5 - 67xy^4 + 74299xy^3 \\
& -4113628xy^2 - 115967481xy + 143477666x - 5y^6 + 32y^5 + 349931y^4 - 21162745y^3 \\
& -579886531y^2 - 86361217y - 992115630 \\
\gamma \quad & -68250x^6 - 13650x^5y + 3954007x^5 - 2730x^4y^2 + 802263x^4y - 44833368x^4 \\
& -546x^3y^3 + 162758x^3y^2 - 9637100x^3y - 322242690x^3 - 109x^2y^4 \\
& -1105x^2y^3 - 2085485x^2y^2 + 119722872x^2y + 332825412x^2 - 21xy^5 \\
& -143xy^4 + 1473859xy^3 + 29262804xy^2 - 18387649xy - 89194442x - 5y^6 + 12y^5 \\
& +349979y^4 - 19762769y^3 - 195716251y^2 + 337258063y - 332372718 \\
\{i, j\} = \{15, 141\}, k_{ij} = 2^2 \cdot 3 \cdot 7 \cdot 31^2 \\
\alpha \quad & 4x^3 - 5x^2y - 2090x^2 + 2805xy + 1268x + 617y + 8519 \\
\beta \quad & -143x^3y - 12928x^3 + x^2y^3 - 281x^2y^2 - 7830x^2y + 21572x^2 - 415xy^3 \\
& -2738xy^2 + 9691xy - 15129x + y^5 - 550y^4 + 2350y^3 - 2187y^2 + 10575y + 12639 \\
\gamma \quad & 8x^5 + 6x^4y - 4188x^4 + 4x^3y^2 - 2923x^3y - 6220x^3 + 3x^2y^3 \\
& -1806x^2y^2 + 562x^2y + 14426x^2 + xy^4 - 966xy^3 + 3386xy^2 + 4555xy + 6739x \\
& +y^5 - 551y^4 + 2902y^3 - 5020y^2 + 25360y - 8625 \\
\{i, j\} = \{15, 142\}, k_{ij} = 2 \cdot 3^2 \cdot 7^2 \cdot 11 \cdot 13 \cdot 31^2 \\
\alpha \quad & 1836368x^3 - 14090478x + 7345472y^3 - 6886380y^2 - 23884818y - 19233072 \\
\beta \quad & -3672736x^5 - 3672736x^4y + 11018208x^4 + 22036416x^3y + 6144540x^3 \\
& +36360074x^2y + 58487646x^2 - 7345472xy^3 - 6970918xy^2 - 62232785xy - 89386491x \\
& -7345472y^4 + 6886380y^3 - 15264809y^2 - 162618642y - 100425183 \\
\gamma \quad & -5509104x^2y^2 + 17904588x^3y - 36673057x^3 - 14090478xy^2 - 28180956xy \\
& +103606098x + 17946857y^3 + 30733989y^2 + 32853867y + 160060617 \\
\{i, j\} = \{15, 145\}, k_{ij} = 2^2 \cdot 3^4 \cdot 5^2 \cdot 7 \cdot 13^2 \cdot 31^2 \\
\alpha \quad & -15164944x^3 + 1106535348x - 106154608y^3 + 378302568y^2 + 1999353120y + 3903698388 \\
\beta \quad & 30329888x^5 + 30329888x^4y - 90989664x^4 - 181979328x^3y - 2031091368x^3 \\
& +73023552x^2y^2 - 1793834806x^2y + 211616501x^2 + 179178160xy^3 \\
& +614979578xy^2 + 7286025076xy + 3866314196x + 106154608y^4 - 341790792y^3 \\
& -1628812903y^2 + 1399751516y - 4952154189 \\
\gamma \quad & -90989664x^3y^2 + 505804008x^3y + 1637605237x^3 - 1106535348xy^2 \\
& -2766338370xy + 8388173250x + 36511776y^4 + 619361209y^3 + 4957277481y^2 \\
& -1747054593y + 12353957703 \\
\{i, j\} = \{15, 146\}, k_{ij} = 2^3 \cdot 3^4 \cdot 7 \cdot 13 \cdot 31^2 \\
\alpha \quad & -x^4 + 2x^3y - 1418x^3 + 613x^2y - 1375168x^2 + 267524xy - 938382x - 1681551y \\
& +3226425 \\
\beta \quad & 2x^6 - 2x^5y + 2830x^5 - 4x^4y^2 + 1610x^4y + 2741840x^4 - 2x^3y^3 \\
& -1142x^3y^2 + 2202239x^3y - 6357026x^3 + x^2y^4 - 2615x^2y^3 \\
& +13006x^2y^2 - 7355587x^2y + 6202604x^2 + 2xy^5 - 1810xy^4 - 1083699xy^3 \\
& -3192106xy^2 + 8908750xy - 6162872x + y^6 - 472y^5 - 543206y^4 + 1037036y^3 \\
& +4140731y^2 - 9031136y + 15662991 \\
\gamma \quad & 66x^4y + 73x^4 - x^3y^3 + 67x^3y^2 + 549093x^3y + 602825x^3 - 402x^2y^3 \\
& +548579x^2y^2 - 2759234x^2y - 2668520x^2 - 869xy^4 + 1507xy^3 - 3360105xy^2 \\
& -3630969xy + 3572767x + y^6 - 469y^5 - 544610y^4 - 598210y^3 - 965068y^2 - 5588807y \\
& +9637212 \\
\{i, j\} = \{15, 201\}, k_{ij} = 2^2 \cdot 3^4 \cdot 31^2 \\
\alpha \quad & 8x^2 - 13xy + 9703x - 12050y + 17394 \\
\beta \quad & -16x^4 + 9x^3y - 19358x^3 + 23x^2y^2 - 101x^2y + 21934x^2 + 16xy^3 \\
& +9560xy^2 + 34734xy - 10886x + 4y^4 + 4783y^3 + 27740y^2 - 14834y - 18924 \\
\gamma \quad & x^4y + x^3y^2 + 4813x^3y + 1400x^3 - 3x^2y^3 + 4812x^2y^2 - 28363x^2y
\end{aligned}$$

$$\begin{aligned}
& -1050x^2 - 8xy^4 + 19xy^3 - 29762xy^2 - 13807xy - 20797x - 4y^5 - 4771y^4 \\
& -1389y^3 - 12742y^2 - 14020y + 105318 \\
\{i, j\} &= \{15, 205\}, k_{ij} = 2 \cdot 3^6 \cdot 31^2 \\
\alpha & 123x^2 + 80682x + 206070 \\
\beta & x^3y - 631x^3 + x^2y^2 - 1258x^2y + 3745x^2 - xy^3 - 2511xy^2 + 7395xy - 130860x \\
& -5y^4 - 5012y^3 + 14600y^2 - 261474y + 338820 \\
\gamma & -123x^5 - 62x^4y - 80182x^4 - 31x^3y^2 - 39686x^3y - 86723x^3 - 15x^2y^3 \\
& -18856x^2y^2 + 16228x^2y + 190782x^2 - 6xy^4 - 7586xy^3 + 34688xy^2 + 127089xy \\
& -305436x - 5y^5 - 5027y^4 - 496y^3 - 71928y^2 + 166836y - 77844 \\
\{i, j\} &= \{16, 70\}, k_{ij} = 1 \\
\alpha & -x + 1 \\
\beta & -2x^4 - 2x^3y + 2x^2y + 2x^2 + xy^2 - xy + x - y^2 - 3 \\
\gamma & 2x^3 - 2x^2 + 1 \\
\{i, j\} &= \{17, 18\}, k_{ij} = 2^4 \cdot 3^4 \cdot 5 \cdot 7 \\
\alpha & 371100x^4y - 667980x^4 - 140280x^3y + 210420x^3 + 58050x^2y - 104490x^2 \\
& + 19350y^3 + 111330y^2 - 6720y + 15120 \\
\beta & 371100x^7 - 325830x^6 + 37110x^5y + 276630x^5 + 74220x^4y + 158565x^4 \\
& + 185550x^3y^3 + 19350x^3y^2 + 191115x^3y + 419304x^3 - 162915x^2y^3 \\
& - 9675x^2y^2 - 133650x^2y - 321027x^2 + 18555xy^4 + 111225xy^3 + 30138xy^2 \\
& + 213675xy + 324660x + 37110y^4 + 4890y^3 + 57036y^2 + 107100y - 5040 \\
\gamma & 24740x^6 + 61850x^5 - 68990x^4 + 21365x^3 + 12370x^2y^3 + 1290x^2y^2 \\
& + 14692x^2y + 56152x^2 + 30925xy^3 + 3225xy^2 + 36730xy + 48479x - 6450y^4 \\
& - 24740y^3 - 2580y^2 - 42488y - 73390 \\
\{i, j\} &= \{18, 19\}, k_{ij} = 2 \cdot 3^5 \\
\alpha & -54x + 162 \\
\beta & 12x^3 - 36x^2 - 4xy^2 + 12y^2 + 6 \\
\gamma & -6xy^3 - 27xy^2 - 27x + 18y^3 + 81y^2 + 9y + 81 \\
\{i, j\} &= \{18, 37\}, k_{ij} = 2^2 \cdot 3^4 \\
\alpha & 108 \\
\beta & -6x^2 + 6xy + 4x - 6y^2 - 5y \\
\gamma & 162x^3 - 108x^2y + 36x^2 + 36xy^2 - 81xy - 18y^3 + 93y^2 + 108 \\
\{i, j\} &= \{18, 46\}, k_{ij} = 2 \cdot 3^4 \cdot 7^2 \\
\alpha & 864x + 1728 \\
\beta & -36x^3 + 36x^2y - 168x^2 - 36xy^2 + 168xy - 270x + 36y^3 - 168y^2 + 270y - 372 \\
\gamma & -1404x^4 + 1080x^3y - 3348x^3 - 648x^2y^2 + 2916x^2y - 918x^2 + 216xy^3 \\
& - 1548xy^2 + 1404xy + 1566x - 108y^4 + 396y^3 - 2142y^2 - 1026y - 2754 \\
\{i, j\} &= \{18, 50\}, k_{ij} = 2^5 \cdot 3^4 \cdot 19 \\
\alpha & 53460x^3 + 38700x + 1584y^3 + 4596y^2 + 9252y + 16416 \\
\beta & -11880x^5 + 3960x^3y^2 - 8600x^3 - 396x^2y^2 - 7548x^2y - 3945x^2 \\
& - 1320xy^4 - 6116xy^3 + 2290xy^2 - 4059xy - 17820x + 132y^4 + 2516y^3 + 1315y^2 \\
& - 5882y + 4523 \\
\gamma & 1584x^3y^2 + 5190x^3y - 16929x^3 - 26469x - 1980y^5 - 12144y^4 - 14781y^3 \\
& - 8118y^2 - 36609y - 44172 \\
\{i, j\} &= \{18, 67\}, k_{ij} = 2^5 \cdot 3^4 \cdot 19 \\
\alpha & 53460x^3 + 38700x + 1584y^3 + 4596y^2 + 9252y + 16416 \\
\beta & -11880x^5 + 3960x^3y^2 - 8600x^3 - 396x^2y^2 - 7548x^2y - 3945x^2 \\
& - 1320xy^4 - 6116xy^3 + 2290xy^2 - 4059xy - 17820x + 132y^4 + 2516y^3 + 1315y^2 \\
& - 5882y + 4523 \\
\gamma & -1584x^3y^2 - 5190x^3y + 16929x^3 + 26469x + 1980y^5 + 12144y^4 + 14781y^3 \\
& + 8118y^2 + 36609y + 44172 \\
\{i, j\} &= \{18, 105\}, k_{ij} = 2^2 \cdot 3^6 \cdot 5 \cdot 7 \\
\alpha & -6x^6 + 15x^5 - 18x^4 + 4848x^3 - 924x^2 + 34020 \\
\beta & -6x^2 + xy^3 - 1610xy^2 + 321xy - 11340x + 2y^4 - 1294y^2 - 11661y - 2604 \\
\gamma & 12x^8 - 18x^7 + 12x^6y + 18x^6 - 6x^5y - 9672x^5 + 6x^4y^3 + 12x^4y^2 \\
& + 18x^4y - 7821x^4 - 9x^3y^3 + 6x^3y^2 - 9669x^3y - 7848x^3 + 6x^2y^4 \\
& + 21x^2y^3 + 24x^2y^2 - 17472x^2y - 80628x^2 - 3xy^4 - 4830xy^3 + 4845xy^2 \\
& - 33057xy + 7812x + 6y^5 - 3882y^3 - 963y^2 - 7812y + 34020 \\
\{i, j\} &= \{18, 144\}, k_{ij} = 2^6 \cdot 3^4 \cdot 5^2 \cdot 13 \\
\alpha & -26040x^2y - 15669x^2 - 1462005xy - 826011x - 826011y + 561600 \\
\beta & 10x^4 + x^3y^2 + 3x^3y - 9708x^3 - x^2y^2 - 3357x^2y - 143982x^2
\end{aligned}$$

$$\begin{aligned}
& +16906xy^2 - 7738xy + 10709x + y^5 + 46y^4 - 84562y^3 + 38820y^2 - 204881y - 148647 \\
\gamma \quad & 10416x^5 - 2085x^4y + 579798x^4 + 417x^3y^2 - 114963x^3y + 57546x^3 \\
& - 84x^2y^3 + 22791x^2y^2 + 19188x^2y - 26904x^2 + 15xy^4 + 609xy^3 \\
& + 139482xy^2 - 68100xy + 736947x - 3y^5 - 129y^4 + 254073y^3 - 52668y^2 + 211047y \\
& + 638811 \\
\{i, j\} = \{18, 146\}, k_{ij} = 2^4 \cdot 3^6 \cdot 7^2 \cdot 13 \\
\alpha \quad & 120x^2y + 396x^2 - 45732xy - 1300140x - 350964y + 2476656 \\
\beta \quad & -8x^4 + x^3y^2 + 1774x^3y + 264225x^3 - 4x^2y^3 - 1920x^2y^2 + 64566x^2y \\
& - 720234x^2 + 6xy^4 + 1825xy^3 - 63748xy^2 + 395594xy - 316141x - 9y^5 - 1733y^4 \\
& + 63169y^3 - 395284y^2 + 107387y + 207420 \\
\gamma \quad & 240x^5 - 231x^4y - 74085x^4 + 195x^3y^2 + 55503x^3y - 509598x^3 \\
& - 138x^2y^3 - 32757x^2y^2 + 993321x^2y - 948291x^2 + 45xy^4 + 10971xy^3 \\
& - 361890xy^2 - 704214xy + 189276x - 27y^5 - 5280y^4 + 173667y^3 - 1015815y^2 \\
& - 248628y - 474588 \\
\{i, j\} = \{18, 150\}, k_{ij} = 2^2 \cdot 3^5 \cdot 7 \\
\alpha \quad & 53613487866816x^4y^2 + 565920149705280x^4y + 625490691779520x^4 \\
& - 698538986964486x^3 - 2328455914493859x^2 - 17780414976xy^2 - 187682158080xy \\
& - 398471003572014x + 1980416464704y^6 - 55159423122240y^5 - 47744895780288y^4 \\
& + 38743570367766y^3 - 835426837461321y^2 - 132823667857842y + 2268 \\
\beta \quad & -35742325244544x^8 + 11914108414848x^7y - 35742325244544x^7 \\
& - 137674141682688x^6y - 174740256751104x^6 + 3971369471616x^5y^2 \\
& + 27799586301312x^5y + 202642618373022x^5 - 1321155688320x^4y^4 \\
& + 26816720304960x^4y^3 + 66271773286656x^4y^2 + 7769721036480x^4y \\
& + 407072494055781x^4 - 878045184x^3y^5 + 5919590946240x^3y^4 \\
& - 19543350894912x^3y^3 - 33619803986028x^3y^2 + 3216657656562x^3y \\
& + 89355467560128x^3 + 292681728x^2y^6 + 10146299904x^2y^5 \\
& - 68360845149696x^2y^4 - 75619795429016x^2y^3 - 4321897899480x^2y^2 \\
& - 358326564572166x^2y - 174265026184587x^2 - 292681728xy^6 + 1970270164800xy^5 \\
& + 12542161251072xy^4 + 32780768750720xy^3 - 75618035331678xy^2 + 98056686809898xy \\
& + 473130299033865x - 660724185024y^7 + 22328478112320y^6 + 31803685533888y^5 \\
& + 14931000206974y^4 + 273581427602639y^3 + 247081079739126y^2 + 357774369346515y \\
& + 621325698645414 \\
\gamma \quad & -107226975733632x^7 + 11914108414848x^4y^3 + 47656433659392x^4y^2 \\
& + 119141084148480x^4y + 142235197142742x^4 + 7902406656x^3y^4 \\
& - 53252611296192x^3y^3 + 16132324165632x^3y^2 - 48420188658126x^3y \\
& - 473321657971305x^3 - 77741612428977x^2y + 542002425054111x^2 \\
& - 14330255867748xy^3 - 81034178415534xy^2 + 28079671058055xy - 198831482843697x \\
& - 878045184y^7 + 5913444629952y^6 + 21866566328064y^5 + 6225915466248y^4 \\
& - 117065817633330y^3 + 310497136009884y^2 + 21814922836953y - 398471003571258 \\
\{i, j\} = \{18, 151\}, k_{ij} = 2^3 \cdot 3^4 \cdot 7^2 \cdot 11 \\
\alpha \quad & 14616x^4y + 10962x^4 - 17280x^3 + 25587x + 4872y^5 - 1218y^4 + 6792y^3 \\
& + 22668y^2 - 60273y + 116424 \\
\beta \quad & -7308x^6 - 3248x^5y^2 + 1624x^5y + 3840x^5 + 2436x^4y^2 - 1280x^3y^2 \\
& - 5686x^3 - 3654x^2y^3 - 960x^2y^2 + 6786x^2y - 41706x^2 - 1624xy^5 \\
& + 812xy^4 - 4208xy^3 - 4334xy^2 + 6620xy + 5760x + 1218y^5 + 320y^4 - 2262y^3 \\
& + 13902y^2 - 16370y + 21712 \\
\gamma \quad & -14616x^7 + 4872x^4y^3 + 4872x^4y^2 + 7308x^4y + 3654x^4 - 7308x^3y^3 \\
& - 1920x^3y^2 - 28536x^3y - 42300x^3 + 73665x + 2436y^6 + 2436y^5 + 7530y^4 \\
& + 18405y^3 + 1488y^2 + 55923y + 64395 \\
\{i, j\} = \{18, 175\}, k_{ij} = 2^3 \cdot 3^4 \cdot 7 \\
\alpha \quad & -5022x^3 - 15120x - 390y^2 - 504y + 1512 \\
\beta \quad & 1116x^5 - 372x^3y^2 + 3360x^3 - 390x^2y - 504x^2 + 124xy^4 + 558xy^3 \\
& - 1120xy^2 + 558xy + 1674x + 130y^3 + 168y^2 - 374y + 558 \\
\gamma \quad & 390x^3y + 1089x^3 + 1683x - 186y^5 - 279y^4 + 2517y^3 - 828y^2 + 168y + 7308 \\
\{i, j\} = \{18, 176\}, k_{ij} = 2 \\
\alpha \quad & 0 \\
\beta \quad & 0 \\
\gamma \quad & 1 \\
\{i, j\} = \{18, 178\}, k_{ij} = 2^3 \cdot 3^7 \cdot 7 \\
\alpha \quad & 20304x^4y^2 - 30456x^4y + 258876x^4 + 166644x^2y^2 - 249966x^2y \\
& + 2124711x^2 - 139590y^3 + 674649y^2 + 40824
\end{aligned}$$

$$\begin{aligned}
\beta & -4512x^6y^2 + 6768x^6y - 54144x^6 + 2256x^5y^2 + 3384x^5y + 1504x^4y^4 \\
& -2256x^4y^3 - 18984x^4y^2 + 55548x^4y - 444384x^4 + 18516x^3y^2 \\
& +27774x^3y + 12344x^2y^4 - 18516x^2y^3 + 10614x^2y^2 + 10152x^2y \\
& -96921x^2 - 1038xy^2 - 1557xy + 45838y^4 - 257979y^3 + 134145y^2 + 83322y - 697572 \\
\gamma & -10152x^7 - 83322x^5 - 2256x^4y^4 - 3384x^4y^3 - 6768x^4y^2 + 10152x^4y \\
& -86292x^4 + 4671x^3 - 18516x^2y^4 - 27774x^2y^3 - 55548x^2y^2 \\
& +83322x^2y - 708237x^2 - 31995x + 1038y^4 + 48087y^3 - 203553y^2 + 31995y - 13608 \\
\{i, j\} & = \{18, 179\}, k_{ij} = 2^4 \cdot 3^4 \cdot 5 \\
\alpha & -2400xy - 900x - 660y + 2160 \\
\beta & 101x^3y - 1097x^3 + 100x^2y^2 - 13x^2y + 735x^2 + 100xy^3 - 12xy^2 + 274xy \\
& -582x + 99y^4 - 12y^3 + 273y^2 + 114y + 950 \\
\gamma & 3891x^4 + 2991x^3y - 3645x^3 + 1788x^2y^2 - 2619x^2y + 1746x^2 + 597xy^3 \\
& -2475xy^2 + 1890xy - 2550x + 297y^4 - 333y^3 + 492y^2 + 2010y + 180 \\
\{i, j\} & = \{18, 217\}, k_{ij} = 2^3 \cdot 3^5 \cdot 7 \\
\alpha & 3024x^4 + 20412x^2 - 1512y^3 + 6804y^2 - 4032y + 4536 \\
\beta & -600x^6 - 120x^5y + 200x^4y^2 - 4050x^4 - 810x^3y - 162x^2y^2 - 2664x^2 \\
& +1080xy + 504y^4 - 2268y^3 + 888y^2 - 6048y - 6420 \\
\gamma & 108x^7 + 729x^5 - 60x^4y^3 + 504x^4 - 972x^3 - 405x^2y^3 + 3402x^2 + 576x \\
& +288y^3 + 1134y^2 + 288y + 756 \\
\{i, j\} & = \{20, 21\}, k_{ij} = 2^4 \cdot 3^5 \cdot 7^2 \cdot 13 \\
\alpha & -102x^8 - 4x^7 - 102x^6y + 204x^6 - 106x^5y - 207690x^5 + 207898x^4y \\
& -583369x^4 + 9417x^3y - 73030x^3 - 519784x^2y + 829826x^2 - 597096xy - 275184x \\
& +462914y + 825552 \\
\beta & -25x^5y + 57x^5 - x^4y + 2x^4 + 103963x^3y^2 - 234975x^3y + 2790680x^3 \\
& +4607x^2y^2 - 7228x^2y - 1789536x^2 + 102xy^5 + 207698xy^4 - 469691xy^3 \\
& +531045xy^2 + 2160855xy - 1603558x + 4y^5 + 9205y^4 - 14133y^3 + 665503y^2 \\
& -2758147y + 2214294 \\
\gamma & 204x^8 + 8x^7 - 408x^6 + 204x^5y + 415482x^5 + 102x^4y^3 - 204x^4y^2 \\
& -415311x^4y + 1167105x^4 + 4x^3y^3 - 212x^3y^2 - 434236x^3y + 145927x^3 \\
& +519784x^2y^2 - 362305x^2y + 1338108x^2 + 102xy^4 + 207698xy^3 + 127405xy^2 \\
& +806229xy + 2757951x + 4y^4 + 9205y^3 - 477047y^2 - 160049y - 3221061 \\
\{i, j\} & = \{21, 138\}, k_{ij} = 2^3 \cdot 3^6 \cdot 7 \\
\alpha & -282x^5 + 196x^4y + 462x^4 + 714x^3y - 7088x^3 - 1906x^2y + 6120x^2 - 1420xy \\
& -4536x - 2232y + 13608 \\
\beta & -6x^3y + 25x^3 - 40x^2y^2 + 11366x^2y - 43706x^2 + 86xy^3 + 426xy^2 - 8476xy \\
& +26368x - 196y^4 - 754y^3 + 14815y^2 - 12155y - 17766 \\
\gamma & -564x^6 + 392x^5y + 2052x^5 + 638x^4y - 18255x^4 - 604x^3y^2 + 6842x^3y \\
& +4762x^3 + 196x^2y^3 + 3288x^2y^2 - 27113x^2y + 7388x^2 - 86xy^3 + 908xy^2 \\
& +13920xy - 11028x + 196y^4 + 950y^3 - 11633y^2 - 13086y + 6912 \\
\{i, j\} & = \{21, 152\}, k_{ij} = 2^3 \cdot 3^5 \cdot 7 \cdot 13^2 \\
\alpha & 4x^3y^3 - 9508x^3y^2 - 96070x^3y - 28392x^3 + 10x^2y^4 + 41x^2y^3 \\
& +110882x^2y^2 - 236517x^2y + 85176x^2 + 7xy^4 - 4727xy^3 - 8053xy^2 - 316889xy \\
& -255528x + 5y^5 + 36y^4 + 57903y^3 + 99527y^2 - 255528y + 766584 \\
\beta & -x^5y - 8x^5 + 7x^4y + 12x^4 + x^3y^3 - 2373x^3y^2 - 32902x^3y + 88203x^3 \\
& +x^2y^4 + 2x^2y^3 + 28943x^2y^2 + 49683x^2y + 826523x^2 - 2xy^5 \\
& +4731xy^4 + 65782xy^3 + 47048xy^2 + 431494xy + 33650x - 5y^6 - 36y^5 - 57908y^4 \\
& -99578y^3 + 197517y^2 - 1039820y - 43053 \\
\gamma & x^5y + 8x^5 - 7x^4y - 12x^4 - 9x^3y^3 + 21391x^3y^2 + 225058x^3y \\
& -31419x^3 - 21x^2y^4 - 84x^2y^3 - 250720x^2y^2 + 423335x^2y - 996875x^2 \\
& +2xy^5 - 4755xy^4 - 32563xy^3 + 227019xy^2 + 82650xy + 477406x + 5y^6 + 4y^5 \\
& +57741y^4 - 274511y^3 + 211635y^2 - 207871y - 1461723 \\
\{i, j\} & = \{21, 153\}, k_{ij} = 3^5 \cdot 13 \\
\alpha & -138x^3y^2 + 706x^3y + 601x^3 - 71x^2y^2 + 354x^2y - 1276x^2 + 138xy^3 \\
& +1055xy^2 - 406xy - 351x + 71y^3 + 546y^2 + 1042y + 1053 \\
\beta & -35x^3y^2 - 230x^3y - 3796x^3 - 16x^2y^2 - 118x^2y + 3670x^2 - 138xy^4 \\
& -917xy^3 + 322xy^2 - 1338xy - 1365x - 71y^4 - 475y^3 - 1080y^2 - 546y - 5768 \\
\gamma & -311x^3y^2 + 1182x^3y - 2594x^3 - 158x^2y^2 + 590x^2y + 1118x^2 - 138xy^4 \\
& -54xy^3 + 149xy^2 - 1940xy + 527x - 71y^4 - 33y^3 - 1266y^2 + 3088y - 4179 \\
\{i, j\} & = \{21, 154\}, k_{ij} = 2 \cdot 3^5 \cdot 13 \\
\alpha & -93x^2y + 381x^2 + xy^2 + 330xy - 702x - 408y + 2106 \\
\beta & 92x^2y^2 - 842x^2y + 1093x^2 - xy^3 - 148xy^2 - 1383xy + 2259x - 2y^3 - 438y^2
\end{aligned}$$

$$\begin{aligned}
& +1324y - 5272 \\
\gamma & -92x^3y^2 + 1028x^3y - 1855x^3 + x^2y^3 + 238x^2y^2 - 305x^2y + 1000x^2 \\
& -91xy^3 + 1042xy^2 - 1296xy + 60x + y^4 + 423y^3 - 1715y^2 + 2840y - 441 \\
\{i, j\} & = \{21, 155\}, k_{ij} = 2^2 \cdot 3^4 \\
\alpha & -76x^7 + 84x^6 - 76x^5y + 266x^5 + 84x^4y - 294x^4 + 81x^3y - 45x^3 - 90x^2y \\
& +144x^2 - 17xy - 36x + 96y + 108 \\
\beta & 152x^7 - 168x^6 - 532x^5 + 664x^4 + 76x^3y^3 - 152x^3y^2 + 446x^3y \\
& +234x^3 - 84x^2y^3 + 168x^2y^2 - 492x^2y - 806x^2 - 76xy^3 + 88xy^2 - 420xy \\
& -298x + 84y^3 - 174y^2 + 378y + 868 \\
\gamma & 38x^3y^2 - 36x^3y - 134x^3 - 42x^2y^2 + 40x^2y - 157x^2 + 76xy^4 - 71xy^3 \\
& +456xy^2 + 315xy - 9x - 84y^4 + 78y^3 - 486y^2 - 964y - 396 \\
\{i, j\} & = \{21, 209\}, k_{ij} = 3^5 \\
\alpha & 24xy - 27x - 45y + 81 \\
\beta & -8xy^2 - 55xy + 64x + 15y^2 + 93y - 185 \\
\gamma & -8x^3y^2 - 7x^3y + 10x^3 + 15x^2y^2 + 3x^2y - 23x^2 + 8xy^3 - 41xy^2 + 44xy \\
& -15y^3 + 87y^2 - 115y - 27 \\
\{i, j\} & = \{22, 23\}, k_{ij} = 2^4 \cdot 3 \cdot 5^4 \\
\alpha & 1720x^5 + 1720x^4y - 5114x^4 - 360x^3y + 3806x^3 + 6020x^2y + 7826x^2 + 5214xy \\
& +4660x - 11900y + 5740 \\
\beta & 860x^3y^2 - 1952x^3y - 9508x^3 - 180x^2y^2 + 1180x^2y - 166x^2 - 1720xy^4 \\
& +3894xy^3 - 10488xy^2 + 6886xy + 4814x + 360y^4 - 2360y^3 + 4820y^2 - 400y - 27640 \\
\gamma & -3440x^5 - 3440x^4y + 10228x^4 - 2720x^3y - 4172x^3 - 4300x^2y^2 + 3580x^2y \\
& -18092x^2 - 1720xy^3 + 40xy^2 - 11108xy + 7920x + 360y^3 + 9180y^2 - 10460y + 12780 \\
\{i, j\} & = \{23, 24\}, k_{ij} = 2 \cdot 5^2 \cdot 11 \\
\alpha & 132xy + 77x - 176y - 11 \\
\beta & -132xy^2 + 187xy - 374x + 176y^2 - 341y + 682 \\
\gamma & 132x^3y^2 - 451x^3y + 220x^3 - 176x^2y^2 + 693x^2y - 660x^2 - 132xy^3 \\
& +187xy^2 - 110xy + 154x + 176y^3 - 341y^2 + 198y - 99 \\
\{i, j\} & = \{23, 90\}, k_{ij} = 2^3 \cdot 3 \cdot 5^2 \cdot 7 \cdot 13 \cdot 19 \\
\alpha & -48151x^2y^2 + 67593x^2y + 451066x^2 - 48069xy^2 + 384420xy + 328749x - 6y^4 \\
& -48067y^3 + 231220y^2 + 151426y + 518687 \\
\beta & 6x^5 - 6x^4y + 34x^4 + 6x^3y^2 - 34x^3y - 77262x^3 + 6x^2y^3 + 48x^2y^2 \\
& +77422x^2y - 959961x^2 - 6xy^4 - 48137xy^3 - 133770xy^2 + 302966xy - 1227896x \\
& +6y^5 + 48061y^4 - 279281y^3 + 127879y^2 - 454304y - 518661 \\
\gamma & -6x^6 + 12x^5y - 28x^5 - 18x^4y^2 + 62x^4y + 77296x^4 + 12x^3y^3 \\
& +96192x^3y^2 - 289922x^3y - 19445x^3 - 6x^2y^4 - 48043x^2y^3 \\
& +423566x^2y^2 - 917093x^2y + 512507x^2 - 6xy^4 - 96276xy^3 + 78092xy^2 \\
& +117916xy - 32403x + 6y^5 + 96200y^4 - 394901y^3 + 436130y^2 - 284993y - 138519 \\
\{i, j\} & = \{23, 98\}, k_{ij} = 2^4 \cdot 5^2 \cdot 7^2 \\
\alpha & 437x^5 - 437x^4y + 437x^4 - 439x^3y - 4040x^3 + 2390x^2y + 2673x^2 - 110xy - 6108x \\
& +505y + 5794 \\
\beta & x^3y + 1647x^3 + x^2y^2 + 1647x^2y + 2039x^2 + 439xy^3 + 5020xy^2 - 6791xy \\
& +6244x + 437y^4 - 3265y^3 + 518y^2 - 9527y + 2218 \\
\gamma & -874x^6 - 874x^5 + 3x^4y + 7307x^4 + 876x^3y^2 + 880x^3y - 6948x^3 \\
& -3265x^2y^2 + 1706x^2y - 4730x^2 - 439xy^3 - 4912xy^2 + 4612xy - 6168x - 437y^4 \\
& +2828y^3 + 1805y^2 + 6480y + 6286 \\
\{i, j\} & = \{23, 116\}, k_{ij} = 2^7 \cdot 3^2 \cdot 5^2 \\
\alpha & 1149x^7 - 310x^6 - 2202x^5 - 1149x^4y + 4809x^4 + 7682x^3y - 6591x^3 \\
& -5152x^2y + 6602x^2 + 2644xy - 11158x - 2825y + 11301 \\
\beta & -2298x^7 + 620x^6 - 2298x^5y + 6702x^5 + 2918x^4y - 11388x^4 - 1149x^3y^3 \\
& -5215x^3y + 5640x^3 + 310x^2y^3 + 2872x^2y^2 - 6561x^2y + 4626x^2 \\
& -1149xy^4 + 5649xy^3 - 9202xy^2 + 2442xy + 14758x + 310y^4 - 665y^3 - 1764y^2 \\
& +2277y - 5103 \\
\gamma & x^5 - 574x^3y^2 + 2249x^3y - 3931x^3 + 156x^2y^2 - 177x^2y + 944x^2 \\
& +1149xy^4 - 4500xy^3 + 2368xy^2 + 11914xy - 15586x - 310y^4 + 355y^3 + 4944y^2 \\
& -11149y + 7415 \\
\{i, j\} & = \{23, 124\}, k_{ij} = 2^4 \cdot 3 \cdot 5^2 \cdot 7 \\
\alpha & -1138x^2y^2 + 281x^2y + 1727x^2 - 1139xy^2 + 2104xy + 2403x - 1137y^3 - 1527y^2 \\
& -40y + 3178 \\
\beta & 3991x^3 + 3991x^2y - 7187x^2 + 1139xy^3 - 968xy^2 + 2088xy - 3320x + 1137y^4
\end{aligned}$$

$$\begin{aligned}
& +390y^3 + 1925y^2 - 1583y + 3511 \\
\gamma \quad & -3991x^4 + 2276x^3y^2 - 8544x^3y + 3733x^3 + 1137x^2y^3 - 5298x^2y^2 \\
& + 1428x^2y - 1486x^2 - 3414xy^3 + 4272xy^2 - 2457xy - 2422x - 2275y^4 + 1028y^3 \\
& + 768y^2 - 4986y - 4645 \\
\{i, j\} = \{23, 156\}, k_{ij} = 2^6 \cdot 3^2 \cdot 5^3 \cdot 7 \\
\alpha \quad & x^9 + 10x^8 - x^7y + 14x^7 - 10x^6y + 49x^6 - 5478x^5y + 92963x^5 - 40904x^4y \\
& + 97190x^4 - 60048x^3y + 45615x^3 - 178533x^2y + 54698x^2 + 158588xy + 115424x \\
& - 171898y + 74488 \\
\beta \quad & -2x^9 - 20x^8 - 26x^7 - 79x^6 - x^5y^3 + 2x^5y^2 + 10932x^5y - 185921x^5 \\
& - 10x^4y^3 + 20x^4y^2 + 81745x^4y - 194330x^4 - 11x^3y^3 + 13700x^3y^2 \\
& - 119122x^3y + 106929x^3 - 19x^2y^3 + 102260x^2y^2 + 65877x^2y - 130390x^2 \\
& - xy^4 - 5476xy^3 - 109286xy^2 - 136789xy - 85968x - 9y^4 - 35417y^3 + 128447y^2 \\
& - 42691y + 51068 \\
\gamma \quad & x^5y - 10x^5 + 4x^4y + 5x^4 + 2743x^3y^2 - 42395x^3y + 12308x^3 \\
& + 17703x^2y^2 + 21738x^2y - 134596x^2 - xy^5 - 5467xy^4 + 84709xy^3 \\
& - 177376xy^2 - 164366xy + 564260x - 9y^5 - 35417y^4 - 43460y^3 + 168287y^2 - 203361y \\
& - 229468 \\
\{i, j\} = \{23, 157\}, k_{ij} = 2^2 \cdot 3^2 \cdot 5^2 \cdot 7 \\
\alpha \quad & 314x^6 - 314x^5y + 1067x^5 - 113x^4y + 730x^4 - 1100x^3y + 762x^3 + 559x^2y \\
& + 1093x^2 + 804xy + 655x + 240y + 2135 \\
\beta \quad & -628x^6 + 628x^5y - 2134x^5 - 402x^4y - 832x^4 + 786x^3y^2 - 802x^3y \\
& + 216x^3 - 314x^2y^3 + 124x^2y^2 - 730x^2y - 3829x^2 - 427xy^3 + 206xy^2 \\
& - 653xy - 4100x - 113y^3 - 26y^2 - 2245y - 105 \\
\gamma \quad & 158x^3y^2 - 240x^3y - 80x^3 + 56x^2y^2 - 50x^2y - 1174x^2 - 314xy^4 \\
& + 482xy^3 + 30xy^2 - 1607xy - 1951x - 113y^4 + 101y^3 - 249y^2 - 2136y + 505 \\
\{i, j\} = \{23, 158\}, k_{ij} = 2^8 \cdot 3^4 \cdot 5^2 \\
\alpha \quad & -26x^2y^3 + 23636x^2y^2 + 16638x^2y + 242714x^2 + 60xy^3 - 70840xy^2 \\
& + 50854xy + 69914x - 20y^4 + 23471y^3 + 210658y^2 + 24623y + 242714 \\
\beta \quad & -3x^5 + 3x^4y + 23x^4 - 3x^3y^2 - 64x^3y + 168571x^3 + 9x^2y^3 \\
& + 24x^2y^2 - 168488x^2y - 348444x^2 - 14xy^4 + 23618xy^3 - 160348xy^2 \\
& + 336981xy - 652319x + 20y^5 - 23491y^4 - 187121y^3 + 115342y^2 - 537776y - 105712 \\
\gamma \quad & -3x^6 + 6x^5y + 23x^5 - 9x^4y^2 - 87x^4y + 168568x^4 - 34x^3y^3 \\
& + 47386x^3y^2 - 303777x^3y + 137004x^3 + 20x^2y^4 - 23654x^2y^3 \\
& + 1720x^2y^2 + 301621x^2y - 343900x^2 + 80xy^4 - 94525xy^3 + 99789xy^2 \\
& + 172269xy + 199863x - 46y^5 + 47236y^4 + 85770y^3 - 128451y^2 + 326905y - 104030 \\
\{i, j\} = \{23, 159\}, k_{ij} = 2^5 \cdot 5^2 \cdot 13 \\
\alpha \quad & -1213x^2y^2 - 4166x^2y - 864x^2 + 1205xy^2 + 9057xy + 6812x - 1210y^3 - 411y^2 \\
& - 1211y + 3199 \\
\beta \quad & 8020x^3 - 2x^2y^2 + 8022x^2y - 9948x^2 + 1218xy^3 - 3262xy^2 - 3602xy - 4089x \\
& + 1210y^4 - 799y^3 + 2010y^2 - 369y + 803 \\
\gamma \quad & -8020x^4 + 2428x^3y^2 - 7710x^3y + 3656x^3 + 1210x^2y^3 - 4430x^2y^2 \\
& - 2526x^2y + 2141x^2 - 2431xy^3 + 2734xy^2 - 3800xy - 2424x - 2423y^4 - 2167y^3 \\
& + 3173y^2 - 7654y - 9773 \\
\{i, j\} = \{23, 160\}, k_{ij} = 2^5 \cdot 5 \cdot 7 \\
\alpha \quad & 74x^5 - 74x^4y + 222x^4 - 74x^3y - 410x^3 + 102x^2y + 750x^2 + 250xy - 272x + 60y \\
& + 748 \\
\beta \quad & 391x^3 + 389x^2y - 2136x^2 + 74xy^3 + 160xy^2 - 580xy - 227x + 74y^4 - 324y^3 \\
& - 551y^2 + 783y - 1497 \\
\gamma \quad & -148x^6 - 148x^5 + 873x^4 + 148x^3y^2 + 132x^3y - 888x^3 - 176x^2y^2 \\
& - 829x^2y + 987x^2 - 74xy^3 - 484xy^2 + 32xy - 78x - 74y^4 + 176y^3 + 991y^2 + 159y \\
& + 373 \\
\{i, j\} = \{25, 26\}, k_{ij} = 2^{10} \cdot 3 \cdot 5^2 \cdot 13 \\
\alpha \quad & -5910454x^5 - 11786285x^4 + 2955227x^3y^2 - 2955227x^3y + 10673580x^3 \\
& + 2955227x^2y^2 + 1386036x^2y + 16500661x^2 + 4103840xy + 1442441x - 2955227y^3 \\
& + 4541968y^2 - 17917752y + 1740791 \\
\beta \quad & -11820908x^5 - 35393478x^4 + 5928004x^3y^2 + 20819998x^3y + 6562084x^3 \\
& + 5742754x^2y^2 + 35114962x^2y - 1103618x^2 - 4375886xy^3 + 23524007xy^2 \\
& - 18255845xy + 2508199x - 2955227y^5 + 8865681y^4 - 42978754y^3 + 30586100y^2 \\
& - 37361130y - 36725245 \\
\gamma \quad & 11820908x^5 + 23572570x^4 - 5910454x^3y^2 + 17731362x^3y + 2294656x^3 \\
& - 5928004x^2y^2 + 5890954x^2y - 12375038x^2 + 167700xy^2 - 14683344xy - 1844141x
\end{aligned}$$

$$\begin{aligned}
& +4323713y^3 - 17782062y^2 + 11063247y - 4223973 \\
\{i, j\} &= \{26, 41\}, k_{ij} = 2^5 \\
\alpha & -2y + 8 \\
\beta & 2y^2 - 8y + 2 \\
\gamma & -2x^3y^2 + 12x^3y - 18x^3 - 4x^2y + 16x^2 + 2xy^3 - 8xy^2 + 6xy - 16x + 2y^3 \\
& -10y^2 + 8y + 6 \\
\{i, j\} &= \{26, 45\}, k_{ij} = 2^5 \cdot 7 \\
\alpha & 34x^3 + 46x^2 + 102x + 146 \\
\beta & -17x^3 + 17x^2y + 60x^2 - 17xy^2 - 60xy - 26x + 17y^3 + 60y^2 + 26y + 9 \\
\gamma & -51x^6 + 34x^5y - 84x^5 - 17x^4y^2 + 76x^4y - 154x^4 - 85x^3y^2 + 138x^3y \\
& -138x^3 + 17x^2y^3 - 122x^2y^2 - 62x^2 + 68xy^3 + 2xy^2 + 76xy - 212x - 17y^4 \\
& -26y^3 - 103y^2 + 60y - 223 \\
\{i, j\} &= \{26, 46\}, k_{ij} = 2^2 \cdot 5 \cdot 37 \\
\alpha & -19x^3 - 7x^2 + 63x + 165 \\
\beta & 9x^3 - 10x^2y - 8x^2 + 9xy^2 + 8xy - 26x - 10y^3 - 7y^2 + 25y - 93 \\
\gamma & 29x^6 - 19x^5y + 42x^5 + 10x^4y^2 - 50x^4y - 50x^4 + 48x^3y^2 - 23x^3y \\
& -251x^3 - 10x^2y^3 + 57x^2y^2 + 199x^2y - 95x^2 - 39xy^3 - 112xy^2 + 150xy \\
& -103x + 10y^4 + 7y^3 - 180y^2 - 55y - 338 \\
\{i, j\} &= \{26, 51\}, k_{ij} = 2^7 \cdot 3 \cdot 7^2 \\
\alpha & -x^{10} + x^9 + x^8 - 3x^7 + 3x^6 + 2395x^5 - 2566x^4 - 3609x^3 + 3066x^2 - 548x \\
& + 2437 \\
\beta & -2392x^2y^2 + 881x^2y + 294x^2 + xy^4 + 169xy^2 + 2647xy + 4117x - y^3 + 1386y^2 \\
& -5083y + 10329 \\
\gamma & 2x^{11} - 2x^{10} - 2x^9 + 2x^8y + 5x^8 + x^7y^3 - x^7y - 2x^7 - x^6y^3 \\
& -3x^6y - 4792x^6 - x^5y^3 + 2x^5y^2 + 2x^5y + 5127x^5 + x^4y^4 \\
& + 2x^4y^3 - x^4y^2 + 2x^4y + 7226x^4 - x^3y^4 - 2x^3y^3 + 2389x^3y^2 \\
& -5673x^3y - 4038x^3 - 2x^2y^4 - 2392x^2y^3 + 2223x^2y^2 - 798x^2y \\
& -13061x^2 + xy^5 + 172xy^3 - 671xy^2 + 12520xy - 15239x - y^4 + 1386y^3 - 2647y^2 \\
& + 14151y + 1176 \\
\{i, j\} &= \{26, 106\}, k_{ij} = 2^{11} \cdot 3 \cdot 5 \cdot 7 \\
\alpha & -2x^7 + 2x^6y - 2x^6 + 2x^5y + 4284x^5 - 4280x^4y + 18x^4 - 4306x^3y \\
& + 21498x^3 + 13304x^2y - 11634x^2 - 4322xy - 15248x - 31092y + 75910 \\
\beta & -x^5 - x^4 + x^3y^2 + 2x^3y - 30069x^3 + 8x^2y^2 - 30058x^2y - 10047x^2 \\
& -2xy^4 + 4299xy^3 - 17565xy^2 - 12354xy - 5668x - 2y^5 + 4274y^4 + 3808y^3 \\
& + 36650y^2 + 4676y - 12690 \\
\gamma & -4x^8 - 4x^7 - 4x^6y + 8563x^6 - x^5y + 28x^5 - 2x^4y^3 - 22x^4y + 21481x^4 \\
& + 2x^3y^3 - 8575x^3y^2 + 18069x^3y - 59079x^3 + 6x^2y^3 + 461x^2y^2 \\
& -58594x^2y - 11862x^2 + 25xy^3 - 25722xy^2 - 69674xy + 65865x - 2y^5 + 4276y^4 \\
& -468y^3 + 6024y^2 + 47744y - 11808 \\
\{i, j\} &= \{26, 107\}, k_{ij} = 2^{15} \\
\alpha & 192xy - 1408x - 1856y + 6784 \\
\beta & -192xy^2 + 4736xy - 960x + 1856y^2 - 3456y + 1088 \\
\gamma & -192x^3y^2 + 5120x^3y - 3776x^3 + 1664x^2y^2 - 2432x^2y + 13696x^2 \\
& + 192xy^3 - 3072xy^2 + 2624xy - 2688x - 1472y^3 - 4544y^2 + 3328y + 11328 \\
\{i, j\} &= \{26, 121\}, k_{ij} = 2^{11} \cdot 13 \\
\alpha & 274x^5 - 274x^4y + 276x^4 - 276x^3y + 1668x^3 + 1780x^2y - 3524x^2 - 776xy - 670x \\
& -4222y + 11176 \\
\beta & -x^3y - 2460x^3 - x^2y^2 - 2457x^2y + 6532x^2 + 276xy^3 - 2391xy^2 + 787xy \\
& -5456x + 274y^4 - 683y^3 + 3284y^2 - 5261y - 5760 \\
\gamma & -548x^6 - 4x^5 + x^4y - 1420x^4 + 550x^3y^2 - 2527x^3y + 3570x^3 \\
& -681x^2y^2 + 5440x^2y - 5828x^2 - 2xy^3 + 2486xy^2 + 5129xy - 3194x - 274y^4 \\
& + 957y^3 - 293y^2 - 991y + 1144 \\
\{i, j\} &= \{26, 164\}, k_{ij} = 2^{13} \cdot 7 \cdot 13 \\
\alpha & 169x^2y^3 - 1008x^2y^2 + 48048x^2y - 275629x^2 - 42xy^3 + 198xy^2 - 840xy \\
& -123208x - 169y^4 + 966y^3 - 857y^2 + 6426y + 338789 \\
\beta & 32x^3y^2 - 24x^3y - 520x^3 - 43264x^2y^2 + 32842x^2y + 709358x^2 + 42xy^4 \\
& -30xy^3 - 93105xy^2 + 102719xy + 114465x + 169y^5 - 37y^4 - 2872y^3 + 85492y^2 \\
& -355186y - 228289 \\
\gamma & -32x^4y^2 + 24x^4y + 520x^4 - 338x^3y^3 + 45344x^3y^2 - 128986x^3y \\
& -159140x^3 - 42x^2y^4 + 1160x^2y^3 - 51x^2y^2 + 252533x^2y - 100507x^2
\end{aligned}$$

$$\begin{aligned}
& -169xy^5 + 459xy^4 - 45436xy^3 - 3086xy^2 + 320408xy - 102053x + 380y^5 - 2036y^4 \\
& + 48624y^3 - 75672y^2 + 145958y + 42606 \\
\{i, j\} &= \{26, 166\}, k_{ij} = 2^{11} \cdot 5^2 \cdot 7 \cdot 37 \\
\alpha & 1020x^9 + 408x^8y + 279841x^8 + 554371x^7 - 3059x^6y + 266869x^6 - 558655x^5y \\
& - 1258427x^5 - 820392x^4y + 426177x^4 - 277190x^3y + 3398535x^3 + 1986983x^2y \\
& - 2105835x^2 + 1608639xy - 2657077x - 3218356y + 4486570 \\
\beta & -2040x^9 - 816x^8y - 557642x^8 - 1224x^7y - 551100x^7 - 816x^6y^2 \\
& - 554380x^6y + 18382x^6 - 1020x^5y^3 + 2654x^5y + 2775253x^5 - 408x^4y^4 \\
& - 278821x^4y^3 - 410x^4y^2 - 282355x^4y - 3912759x^4 - 612x^3y^4 \\
& - 275958x^3y^3 - 555038x^3y^2 + 3102149x^3y - 4281336x^3 - 408x^2y^5 \\
& - 276374x^2y^4 + 563297x^2y^3 - 290930x^2y^2 - 4567403x^2y + 6157426x^2 \\
& + 816xy^4 + 277174xy^3 + 1459440xy^2 - 6658316xy - 51232x + 816y^5 + 553564y^4 \\
& - 567730y^3 + 2029972y^2 + 12439194y - 198910 \\
\gamma & -34x^5y + 239x^5 + 34x^4y^2 - 68x^4y - 4x^4 - x^3y^3 - 139284x^3y^2 \\
& + 983787x^3y - 3694566x^3 + x^2y^4 - 3x^2y^3 + 699975x^2y^2 - 371455x^2y \\
& + 8900535x^2 + 204xy^5 + 276790xy^4 - 1960253xy^3 + 7579439xy^2 + 3004611xy \\
& + 866415x - 408y^6 - 276374y^5 + 560239y^4 + 33545y^3 - 7164031y^2 + 6913244y \\
& - 3734803 \\
\{i, j\} &= \{27, 32\}, k_{ij} = 2^5 \cdot 3^2 \\
\alpha & -65xy - 142x - 169y - 398 \\
\beta & -65x^3y^2 + 248x^3y + 7x^3 - 169x^2y^2 + 616x^2y + 191x^2 - 65xy^3 \\
& + 53xy^2 - 29xy + 7x - 169y^3 + 109y^2 + 76y + 333 \\
\gamma & 65xy^2 - 118xy + 277x + 169y^2 - 278y + 605 \\
\{i, j\} &= \{28, 29\}, k_{ij} = 7^3 \\
\alpha & 199x^5 + 519x^4 + 832x^3 + 929x^2 + 511x + 345 \\
\beta & 398x^6 + 640x^5 + 228x^4 + 199x^3y^2 - 701x^3y + 588x^3 + 199x^2y^3 \\
& + 121x^2y^2 + 133x^2y + 1634x^2 + 121xy^3 + 495xy^2 + 466xy + 1237x + 192y^3 \\
& + 118y^2 - 574y + 271 \\
\gamma & 199x^2y^2 - 303x^2y + 437x^2 + 121xy^2 + 176xy + 743x + 192y^2 - 227y - 421 \\
\{i, j\} &= \{29, 65\}, k_{ij} = 5^2 \cdot 7^3 \\
\alpha & -x^9 - x^8 + 706x^7 - 2x^6y + 704x^6 - 715x^5y + 4257x^5 + 713x^4y + 5727x^4 \\
& + 806x^3y + 11028x^3 + 1809x^2y + 19777x^2 + 3710xy + 9143x - 5717y + 4191 \\
\beta & 2x^9 + 2x^7y - 1416x^7 + 5x^6 + x^5y^3 + 9x^5y - 5685x^5 - 2x^4y^2 \\
& - 17x^4y - 3650x^4 + x^3y^4 - 710x^3y^3 + 1425x^3y^2 - 6593x^3y - 8452x^3 \\
& + 2x^2y^3 - 1807x^2y^2 + 3460x^2y - 15927x^2 + xy^4 - 1424xy^3 - 4633xy^2 \\
& + 2340xy - 8460x - y^5 + 711y^4 - 2946y^3 + 6332y^2 + 4905y - 6418 \\
\gamma & -357x^3y^2 - 1578x^3y + 2879x^3 - x^2y^4 - 2x^2y^3 - 3052x^2y^2 \\
& + 3529x^2y - 20371x^2 + 714xy^4 + 3156xy^3 - 9022xy^2 - 1504xy - 10324x + y^6 \\
& - 711y^5 + 2947y^4 - 1327y^3 - 1115y^2 + 12053y + 2420 \\
\{i, j\} &= \{29, 78\}, k_{ij} = 2 \cdot 7^2 \\
\alpha & -54x^5 - 210x^4 - 354x^3 - 310x^2 - 182x + 4 \\
\beta & -54x^2y^2 + 22x^2y - 372x^2 - 48xy^2 - 38xy - 204x - 48y^2 - 132y - 30 \\
\gamma & -108x^6 - 204x^5 + 24x^4 - 54x^3y^2 + 130x^3y - 266x^3 - 54x^2y^3 \\
& + 6x^2y^2 - 126x^2y - 444x^2 - 48xy^3 - 76xy^2 - 126xy - 316x - 48y^3 - 80y^2 - 50y \\
& - 58 \\
\{i, j\} &= \{29, 150\}, k_{ij} = 2^3 \cdot 3^2 \cdot 7^3 \\
\alpha & -28x^{10} - 30x^9 + 14x^8y + 9654x^8 - 52x^7y + 29390x^7 - 19631x^6y + 105248x^6 \\
& + 9641x^5y + 66183x^5 + 7568x^4y + 116304x^4 + 153517x^3y + 44392x^3 \\
& + 140328x^2y - 157011x^2 + 67931xy - 187642x + 62351y - 106466 \\
\beta & 56x^{10} + 4x^9 + 28x^8y - 19424x^8 + 24x^7y - 39448x^7 + 28x^6y^3 \\
& - 28x^6y^2 + 19726x^6y - 132150x^6 + 2x^5y^3 + 20x^5y^2 - 59124x^5y \\
& + 107960x^5 + 14x^4y^4 - 9768x^4y^3 + 39120x^4y^2 - 78459x^4y - 36285x^4 \\
& + 12x^3y^4 - 19658x^3y^3 - 78482x^3y^2 + 120165x^3y - 158095x^3 \\
& - 14x^2y^5 + 9821x^2y^4 - 46683x^2y^3 + 127697x^2y^2 - 153010x^2y \\
& + 107607x^2 + 10xy^5 - 29405xy^4 - 23975xy^3 + 201745xy^2 - 345635xy + 502484x \\
& - 28y^6 + 19568y^5 - 83655y^4 + 141301y^3 + 90965y^2 - 250921y + 344094 \\
\gamma & x^5y + 5x^5 - x^4y^2 + 5x^4y - 7x^4 + 4918x^3y^2 + 20144x^3y + 9541x^3 \\
& + 7x^2y^4 + 30x^2y^3 + 38656x^2y^2 - 18985x^2y + 80086x^2 - 2xy^5 \\
& - 9808xy^4 - 40233xy^3 + 142139xy^2 - 186130xy + 206936x - 14y^6 + 9777y^5 \\
& - 36946y^4 + 57080y^3 + 76941y^2 - 101031y + 173760 \\
\{i, j\} &= \{29, 165\}, k_{ij} = 2^5 \cdot 7^2 \cdot 13^2
\end{aligned}$$

$$\begin{aligned}
\alpha & -33x^2y^3 + 87765x^2y^2 - 569929x^2y + 1024651x^2 - 156xy^2 + 449769xy \\
& + 1286720x - 11y^4 + 87638y^3 + 481737y^2 + 249789y + 686924 \\
\beta & -11x^5 + 11x^4y + 48x^4 - 11x^3y^2 - 48x^3y - 387982x^3 + 11x^2y^3 \\
& + 48x^2y^2 + 387982x^2y - 346311x^2 - 11xy^4 + 87838xy^3 - 143745xy^2 \\
& + 115128xy - 534500x + 11y^5 - 87649y^4 - 394044y^3 - 31137y^2 - 743130y - 1257378 \\
\gamma & -11x^6 + 22x^5y + 37x^5 - 33x^4y^2 - 63x^4y - 387967x^4 - 22x^3y^3 \\
& + 175619x^3y^2 - 363972x^3y + 1315109x^3 + 11x^2y^4 - 87759x^2y^3 \\
& + 219956x^2y^2 + 475443x^2y + 528820x^2 + 66xy^4 - 350934xy^3 + 100838xy^2 \\
& - 287023xy - 959426x - 44y^5 + 175436y^4 - 263264y^3 + 336645y^2 + 564217y - 538402 \\
\{i, j\} & = \{29, 210\}, k_{ij} = 2^4 \cdot 7^2 \\
\alpha & -352x^3 - 176x^2 - 400x + 288 \\
\beta & 352xy^2 + 1104xy + 1072x - 528y^2 - 1264y - 80 \\
\gamma & 704x^5 + 352x^4 - 352x^3y^2 - 400x^3y - 272x^3 + 176x^2y^2 + 512x^2y \\
& - 1216x^2 + 352xy^3 + 528xy^2 + 208xy - 832x - 528y^3 - 1504y^2 - 1872y - 1056 \\
\{i, j\} & = \{30, 53\}, k_{ij} = 2^9 \cdot 3 \\
\alpha & 1152xy + 2944x + 1280y + 1664 \\
\beta & x^4 - 1152x^3y^2 + 4224x^3y + 2433x^3 - 1280x^2y^2 + 1407x^2y - 4223x^2 \\
& + 1152xy^3 - 1920xy^2 + 897xy - 3325x - 1024y^3 - 767y^2 + 3971y - 1533 \\
\gamma & x^2 - 1152xy^2 + 1920xy - 3455x - 1280y^2 + 1152y - 1663 \\
\{i, j\} & = \{31, 32\}, k_{ij} = 2^7 \cdot 3^4 \cdot 7 \\
\alpha & -7x^8 - 6x^7 - 7x^6y + 22x^6 + x^5y + 14467x^5 + 14463x^4y - 51998x^4 \\
& + 12705x^3y - 23164x^3 - 52380x^2y - 20390x^2 + 34578xy + 100247x + 9832y + 1069 \\
\beta & 2x^5y - 5x^5 + x^4 - 7242x^3y^2 + 21497x^3y + 50246x^3 + 867x^2y^2 \\
& - 6100x^2y + 82212x^2 - 7xy^5 - 14463xy^4 + 42997xy^3 + 5466xy^2 - 234080xy \\
& - 196963x + y^5 + 1730y^4 - 12205y^3 - 10146y^2 + 162563y + 133725 \\
\gamma & 14x^8 + 12x^7 - 44x^6 - 14x^5y - 28927x^5 + 7x^4y^3 - 14x^4y^2 - 28919x^4y \\
& + 103986x^4 + 6x^3y^3 + 2x^3y^2 + 3494x^3y + 46287x^3 - 8x^2y^3 \\
& + 36212x^2y^2 - 20591x^2y - 23872x^2 - 7xy^4 - 14455xy^3 + 24609xy^2 - 51454xy \\
& - 187337x + y^4 + 1729y^3 - 23767y^2 + 30489y + 112273 \\
\{i, j\} & = \{32, 107\}, k_{ij} = 2^5 \cdot 3^3 \cdot 5 \\
\alpha & -975xy - 2130x - 2535y - 5970 \\
\beta & 975xy^2 - 1770xy + 4155x + 2535y^2 - 4170y + 9075 \\
\gamma & 975x^3y^2 - 3720x^3y - 105x^3 + 2535x^2y^2 - 9240x^2y - 2865x^2 + 975xy^3 \\
& - 795xy^2 + 435xy - 105x + 2535y^3 - 1635y^2 - 1140y - 4995 \\
\{i, j\} & = \{32, 117\}, k_{ij} = 2^8 \cdot 3^3 \cdot 7 \cdot 13 \\
\alpha & 34x^7 + 34x^6y + 18x^6 - 16x^5y + 13426x^5 + 13374x^4y + 40775x^4 + 13896x^3y \\
& - 127793x^3 - 84602x^2y + 25790x^2 + 18496xy + 152877x + 302155y + 264299 \\
\beta & -2x^5 + x^4y + 16x^4 - 5x^3y^2 - 23x^3y + 33307x^3 + 8x^2y^3 + 53x^2y^2 \\
& - 33312x^2y - 164856x^2 - 18xy^4 + 13457xy^3 + 29538xy^2 + 141629xy - 335470x \\
& + 34y^5 - 13366y^4 - 126046y^3 + 20889y^2 - 114659y + 56257 \\
\gamma & -68x^8 - 36x^7 + 68x^6y - 26918x^6 + 169x^5y - 81772x^5 - 34x^4y^3 \\
& - 128x^4y^2 + 54005x^4y + 195169x^4 + 34x^3y^3 - 27175x^3y^2 + 7532x^3y \\
& - 35812x^3 + 94x^2y^3 + 44615x^2y^2 - 148519x^2y - 19706x^2 + 18xy^4 \\
& - 13437xy^3 - 75051xy^2 - 284391xy - 224909x - 34y^5 + 13264y^4 + 165974y^3 \\
& + 121720y^2 + 497779y + 166338 \\
\{i, j\} & = \{32, 153\}, k_{ij} = 2^6 \cdot 3^2 \cdot 5^2 \cdot 7 \cdot 13 \\
\alpha & 92x^3y^3 - 50995x^3y^2 + 293348x^3y + 243635x^3 + 11x^2y^3 \\
& - 6324x^2y^2 + 100584x^2y - 598681x^2 - 92xy^4 + 49543xy^3 + 498982xy^2 \\
& + 1058788xy + 1012641x - 11y^4 + 6150y^3 - 1992y^2 - 399824y - 190071 \\
\beta & 6x^5y + 36x^5 + x^4y - 3x^4 + 24x^3y^3 - 12467x^3y^2 - 74750x^3y \\
& + 114162x^3 + 3x^2y^3 - 1551x^2y^2 + 6627x^2y + 617779x^2 + 92xy^5 \\
& - 49911xy^4 - 298946xy^3 - 77870xy^2 - 472013xy - 949284x + 11y^5 - 6194y^4 \\
& + 26815y^3 + 265812y^2 - 520335y + 626871 \\
\gamma & 6x^5y + 36x^5 + x^4y - 3x^4 + 208x^3y^3 - 114463x^3y^2 + 511904x^3y \\
& + 601396x^3 + 25x^2y^3 - 14200x^2y^2 + 207797x^2y - 579580x^2 + 92xy^5 \\
& - 50487xy^4 + 16397xy^3 - 63941xy^2 + 45191xy + 474710x + 11y^5 - 6263y^4 \\
& + 66032y^3 - 183933y^2 + 542498y + 1069935 \\
\{i, j\} & = \{32, 169\}, k_{ij} = 2^8 \cdot 3^3 \cdot 7^2 \cdot 37 \\
\alpha & -164x^9 - x^8y - 742x^8 + 336478x^7 + 5x^6y + 1533735x^6 + 335658x^5y \\
& - 842924x^5 + 1530007x^4y - 3841358x^4 + 1541075x^3y + 1503937x^3 + 4170016x^2y \\
& + 4432700x^2 - 1999094xy - 1764193x - 2313098y + 2038803
\end{aligned}$$

β	$328x^9 + 2x^8y + 1484x^8 - 328x^7y - 672956x^7 - 2x^6y^2 - 1494x^6y$ $-3067306x^6 + 164x^5y^3 + 821x^5y + 1686098x^5 + x^4y^4 + 742x^4y^3$ $+5x^4y^2 + 3579x^4y + 7344012x^4 - 164x^3y^4 - 336150x^3y^3$ $+673160x^3y^2 - 3085026x^3y - 3532363x^3 - x^2y^5 - 745x^2y^4$ $-1532240x^2y^3 + 3068397x^2y^2 - 8347374x^2y - 3422272x^2 - 164xy^3$ $+1663108xy^2 - 1683032xy + 1610308x - y^4 - 744y^3 + 781601y^2 - 208928y - 2101969$
γ	$-41x^5y + 188x^5 - 187x^4y + 509x^4 + 168615x^3y^2 - 770916x^3y + 3229726x^3$ $+767885x^2y^2 - 2086883x^2y - 13988045x^2 + 164xy^5 + 336478xy^4$ $-1541607xy^3 + 448558xy^2 + 10292745xy - 1577148x + y^6 + 747y^5 + 1533732y^4$ $-4172446y^3 + 2697048y^2 + 11833182y + 12720954$
$\{i, j\} = \{32, 170\}, k_{ij} = 2^8 \cdot 3^2 \cdot 13^2$	
α	$-x^2y^3 + 2882x^2y^2 - 49819x^2y - 95691x^2 + 2902xy^2 - 65027xy - 157886x$ $+2884y^3 + 70265y^2 + 29117y - 296357$
β	$x^4y + 7x^4 + x^3y^2 + 7x^3y + 39414x^3 + 8x^2y^2 + 39413x^2y + 147184x^2$ $-2903xy^3 + 73732xy^2 - 177621xy + 344914x - 2884y^4 - 61615y^3 + 161492y^2$ $-305043y + 424323$
γ	$x^5y + 7x^5 + 2x^4y^2 + 14x^4y + 39414x^4 + 5787x^3y^2 - 20803x^3y$ $-44191x^3 + 2885x^2y^3 + 58742x^2y^2 - 312436x^2y + 68570x^2 + xy^4$ $+2904xy^3 + 34800xy^2 - 51806xy - 133733x + y^5 + 2y^4 + 111426y^3 - 116791y^2$ $-74417y + 2739$
$\{i, j\} = \{32, 212\}, k_{ij} = 2^5 \cdot 3^4 \cdot 5^5 \cdot 7 \cdot 11 \cdot 13$	
α	$-253312500x^5 - 126656250x^3y^2 + 1705462500x^3 + 126656250x^2y^5$ $-303975000x^2y^4 - 1659866250x^2y^3 + 2416869000x^2y^2 + 5113154025x^2y$ $-17800115760x^2 - 63328125xy^4 + 126656250xy^3 + 916059375xy^2 - 1203234375xy$ $-5327568750x + 63328125y^7 - 151987500y^6 - 1146573750y^5 + 1968372000y^4$ $+6674578575y^3 - 14815574130y^2 - 52310714568y - 30006185706$
β	$506625000x^5 + 253312500x^3y^2 - 506625000x^3y - 3410925000x^3$ $-253312500x^2y^5 + 607950000x^2y^4 + 3319732500x^2y^3$ $-4833738000x^2y^2 - 10226308050x^2y + 35600231520x^2 + 126656250xy^4$ $-253312500xy^3 - 1832118750xy^2 + 4550831250xy + 10655137500x - 75390625y^8$ $+506625000y^7 - 595790625y^6 - 2168305625y^5 + 6999615375y^4 - 104657150y^3$ $+10737445135y^2 + 59990651011y + 44019671412$
γ	$50662500x^4 - 151987500x^3 + 15078125x^2y^8 - 126656250x^2y^7$ $+230615625x^2y^6 + 770700625x^2y^5 - 2851218375x^2y^4$ $-1682152400x^2y^3 + 5849333485x^2y^2 + 1765579321x^2y + 2827050000x^2$ $-25331250xy^5 + 60795000xy^4 + 331973250xy^3 - 483373800xy^2 - 1022630805xy$ $+4674493152x + 3015625y^9 - 34378125y^8 + 122116875y^7 + 15770750y^6$ $-956670300y^5 + 1191915545y^4 + 1183238387y^3 - 17660176504y^2 - 26616498345y$ -11310785706
$\{i, j\} = \{33, 34\}, k_{ij} = 2^3 \cdot 7 \cdot 13$	
α	$314x^5 - 52x^4 + 194x^3 - 202x^2 - 148x - 290$
β	$-628x^6 + 732x^5 - 492x^4 + 314x^3y^2 - 1040x^3y - 1328x^3 - 314x^2y^3$ $-366x^2y^2 + 1818x^2y - 406x^2 + 52xy^3 + 30xy^2 + 660xy + 2528x + 434y^3$ $-750y^2 - 2676y - 3830$
γ	$-314x^2y^2 + 412x^2y + 550x^2 + 52xy^2 - 360xy + 2364x + 434y^2 - 1328y - 744$
$\{i, j\} = \{34, 122\}, k_{ij} = 2^4 \cdot 3 \cdot 5 \cdot 13^2$	
α	$-521x^5 + 521x^4y - 1036x^4 + 499x^3y - 15344x^3 + 701x^2y + 27076x^2 + 8157xy$ $-1067x - 29634y - 39316$
β	$4x^3y + 9343x^3 + 6x^2y^2 + 9333x^2y - 36516x^2 - 499xy^3 + 7648xy^2 - 45480xy$ $+9005x - 521y^4 + 329y^3 - 2432y^2 - 21897y + 61348$
γ	$1042x^6 + 1030x^5 + 28x^4y + 19273x^4 - 1020x^3y^2 + 7468x^3y - 38460x^3$ $-186x^2y^2 - 8477x^2y + 13885x^2 + 499xy^3 - 15328xy^2 + 39216xy + 41038x$ $+521y^4 + 713y^3 + 31408y^2 + 35401y - 25536$
$\{i, j\} = \{34, 147\}, k_{ij} = 2^2 \cdot 3^2 \cdot 7 \cdot 13$	
α	$5x^4 + 460x^3 - 774x^2y + 815x^2 - 2268xy - 4202x - 2834y - 3940$
β	$x^3y - 1013x^3 - 2x^2y^2 + 2027x^2y - 157x^2 + 1813xy^2 - 1647xy + 6002x$ $+771y^3 + 673y^2 - 1712y + 5434$
γ	$-5x^5 + 2x^4y + 49x^4 - 265x^3y - 756x^3 - 776x^2y^2 + 3091x^2y + 1530x^2$ $-455xy^2 + 1587xy + 952x + 771y^3 - 619y^2 + 70y - 1206$
$\{i, j\} = \{34, 164\}, k_{ij} = 2^7 \cdot 3^2 \cdot 7 \cdot 13^2$	
α	$-177x^2y^3 + 1638x^2y^2 - 56981x^2y - 280616x^2 + 398xy^3 - 189xy^2$

$$\begin{aligned}
& -288866xy - 626208x + 177y^4 - 621y^3 + 5934y^2 - 436787y - 1325876 \\
\beta & -33x^3y^2 + 72x^3y - 439x^3 + 45295x^2y^2 - 98640x^2y + 599762x^2 - 44xy^4 \\
& + 97xy^3 + 393102xy^2 - 715561xy + 1059820x - 177y^5 + 2y^4 - 1645y^3 + 717532y^2 \\
& - 638858y + 1780607 \\
\gamma & 33x^4y^2 - 72x^4y + 439x^4 + 354x^3y^3 - 48670x^3y^2 + 212818x^3y \\
& - 39847x^3 + 44x^2y^4 - 1988x^2y^3 - 246675x^2y^2 + 654472x^2y \\
& + 311698x^2 + 177xy^5 - 488xy^4 + 53858xy^3 - 39456xy^2 - 63234xy - 23663x - 575y^5 \\
& + 2258y^4 + 136734y^3 + 25132y^2 + 786256y + 1377 \\
\{i, j\} &= \{34, 171\}, k_{ij} = 2^7 \cdot 3 \cdot 13 \\
\alpha & 4x^7 - 4x^6y - 4x^6 + 4x^5y - 3812x^5 + 3819x^4y - 14x^4 + 3849x^3y - 4132x^3 \\
& - 4187x^2y - 16756x^2 + 13613xy + 19028x - 12110y - 25906 \\
\beta & -x^4 - x^3y^2 - 7x^3y - 551x^3 - x^2y^3 - 3x^2y^2 - 560x^2y + 27047x^2 \\
& - 3826xy^3 - 20870xy^2 + 4140xy - 22441x + 4y^5 - 3822y^4 + 11861y^3 - 11302y^2 \\
& - 27469y + 44582 \\
\gamma & 8x^8 + 8x^6y - 7632x^6 + 6x^5y - 7649x^5 + 4x^4y^3 + 5x^4y^2 + 40x^4y \\
& - 8824x^4 - 4x^3y^3 + 7662x^3y^2 - 10095x^3y - 19096x^3 + 11x^2y^3 \\
& - 15654x^2y^2 + 10360x^2y - 13756x^2 - 3830xy^3 - 11091xy^2 - 47884xy - 14891x \\
& + 4y^5 - 3814y^4 + 4225y^3 - 7334y^2 - 40139y - 51036 \\
\{i, j\} &= \{34, 213\}, k_{ij} = 2^3 \cdot 13 \\
\alpha & 6x^3 + 14x^2 + 2x - 6 \\
\beta & -12x^3 - 28x^2 + 20x + 62 \\
\gamma & 12x^5 + 28x^4 + 6x^3y^2 - 24x^3y - 2x^3 + 14x^2y^2 - 56x^2y - 20x^2 \\
& + 2xy^2 + 16xy + 30x - 6y^2 + 74y + 32 \\
\{i, j\} &= \{35, 42\}, k_{ij} = 2^3 \cdot 3^2 \cdot 13 \cdot 19 \\
\alpha & 2x^3 + x^2y - 1563x^2 - 1037xy + 577x - 3264y + 3948 \\
\beta & -x^4y + x^3y^2 + 1039x^3y - 1692x^3 - 1039x^2y^2 + 4166x^2y - 2347x^2 \\
& - xy^4 + xy^3 - 2474xy^2 - 902xy - 2571x + y^5 + 1036y^4 - 1691y^3 + 3248y^2 - 1951y \\
& - 4808 \\
\gamma & -4x^5 + 2x^4y + 3118x^4 + x^3y^2 - 1043x^3y + 5086x^3 - 2x^2y^3 \\
& - 1034x^2y^2 - 260x^2y + 848x^2 + xy^4 + 1034xy^3 - 1693xy^2 - 2408xy - 1178x \\
& + y^4 + 1038y^3 + 3646y^2 + 217y + 1132 \\
\{i, j\} &= \{35, 73\}, k_{ij} = 2^2 \cdot 13 \cdot 19 \\
\alpha & -12x - 80 \\
\beta & 68xy^2 + 180xy + 312x + 124y^2 + 212y + 104 \\
\gamma & 24x^3 + 88x^2 - 336x + 68y^3 + 112y^2 + 132y + 444 \\
\{i, j\} &= \{36, 118\}, k_{ij} = 2^4 \cdot 3 \\
\alpha & 21x + 33 \\
\beta & 42x^4 - 42x^3y + 24x^3 + 18x^2y - 24x^2 + 21xy^2 + 6xy + 66x + 33y^2 - 51y + 84 \\
\gamma & -42x^3 + 18x^2 + 6x - 51 \\
\{i, j\} &= \{37, 38\}, k_{ij} = 2 \\
\alpha & y \\
\beta & 2x^3y - 2x^2y + 2x^2 - 2x + 3y \\
\gamma & 2x^3 - 2x^2 - y^2 - y + 2 \\
\{i, j\} &= \{37, 53\}, k_{ij} = 2^2 \cdot 3 \cdot 5^2 \cdot 13 \\
\alpha & -25x^3 + 75x^2 + 575x + 700 \\
\beta & -38x^6 + 25x^5y + 70x^5 - 12x^4y^2 - 4x^4y + 782x^4 - 62x^3y^2 - 313x^3y \\
& + 1631x^3 + 12x^2y^3 + 142x^2y^2 - 1085x^2y + 72x^2 + 13xy^3 + 705xy^2 \\
& - 640xy + 260x - 12y^4 - 128y^3 + 675y^2 + 1185y + 3361 \\
\gamma & 12x^3 - 13x^2y - 92x^2 + 13xy^2 + 91xy - 288x - 12y^3 - 92y^2 + 287y + 600 \\
\{i, j\} &= \{37, 62\}, k_{ij} = 2^3 \cdot 3^4 \cdot 13 \\
\alpha & -28x + 14y + 2757 \\
\beta & x^4 + 438x^3 + x^2y^2 + 583x^2y - 1511x^2 + 435xy^2 - 1008xy - 1261x + y^4 \\
& - 443y^3 + 1511y^2 - 1080y + 2163 \\
\gamma & 19x^3 + 10x^2y - 1692x^2 + 4xy^2 - 1358xy - 510x + y^3 - 458y^2 - 789y + 153 \\
\{i, j\} &= \{37, 76\}, k_{ij} = 3 \cdot 13^2 \\
\alpha & -16x + 176 \\
\beta & -32x^4 + 32x^3y + 352x^3 - 24x^2y^2 - 344x^2y + 2x^2 + 8xy^3 + 248xy^2 - 86xy \\
& - 35x - 88y^3 + 176y^2 - 21y + 109 \\
\gamma & 8xy^2 + 8xy + 2x - 88y^2 - 88y - 21 \\
\{i, j\} &= \{37, 84\}, k_{ij} = 2 \cdot 3 \cdot 13
\end{aligned}$$

α	27
β	$54x^3 - 54x^2y + 27xy - 9x + 27y^2 - 18y + 36$
γ	$-6y^2 + 3y - 1$
$\{i, j\}$	$\{37, 87\}, k_{ij} = 2^2 \cdot 3^2 \cdot 13$
α	$-2x + y + 21$
β	$2x^3y + 18x^3 - 2x^2y^2 - 34x^2y - 94x^2 + 17xy^2 + 35xy + 31x + 2y^4 + 18y^3$ $+ 59y^2 + 166y + 426$
γ	$4x^3 - 4x^2y - 36x^2 + 20xy + 2x + 2y^3 + 11y^2 - y + 135$
$\{i, j\}$	$\{37, 202\}, k_{ij} = 3$
α	0
β	0
γ	1
$\{i, j\}$	$\{38, 58\}, k_{ij} = 2$
α	1
β	$-2x^2 - 2x - y - 1$
γ	$2x^3 + 2x^2y + 2xy - 2x - 1$
$\{i, j\}$	$\{38, 155\}, k_{ij} = 2$
α	$-xy + y + 1$
β	$2x^5 - 4x^4 + 3x^2 + xy^2 + xy - x - y^2 - 2y - 1$
γ	$2x^3y - 2x^3 - 4x^2y + 2x^2 + 2xy - y - 3$
$\{i, j\}$	$\{38, 180\}, k_{ij} = 2^4$
α	$y - 1$
β	$-2x^3y + 4x^2y + 2x^2 - 6xy - 7x + 5y + 11$
γ	$2x^3y - 2x^3 - 4x^2y + 2x^2 + 6xy - y^2 - 4y - 8$
$\{i, j\}$	$\{38, 208\}, k_{ij} = 2$
α	0
β	2
γ	y
$\{i, j\}$	$\{39, 68\}, k_{ij} = 1$
α	0
β	-1
γ	0
$\{i, j\}$	$\{40, 72\}, k_{ij} = 13$
α	$-3x - y$
β	$x^4 + 3x^3y - 10x^3 + 3x^2y^2 - 17x^2y + 15x^2 - 8xy^2 + 2x - 2y^4 + 10y^3$ $- 14y^2 + 7y - 17$
γ	$5x^3 + 5x^2y - 6x^2 - 2xy - 2x - 2y^3 + 7y^2 + 3y + 10$
$\{i, j\}$	$\{41, 42\}, k_{ij} = 2^2 \cdot 3$
α	$-y - 2$
β	$2x^3y + 4x^3 - 2x^2y - 4x^2 - xy^2 - 2xy - x - y^3 - y^2 + 6y + 13$
γ	$y^2 + 4y + 5$
$\{i, j\}$	$\{41, 43\}, k_{ij} = 2 \cdot 7$
α	$y + 2$
β	$-2x^3y - 4x^3 + 2x^2y + 4x^2 + xy^2 - 6x + y^3 - 5y + 6$
γ	$-y^2 - 2y + 2$
$\{i, j\}$	$\{41, 62\}, k_{ij} = 2^2 \cdot 3^2$
α	$-13y - 32$
β	$26x^3y + 64x^3 - 26x^2y - 64x^2 - 39xy^2 + 86xy - 59x + 39y^3 - 21y^2 + 50y + 91$
γ	$13y^2 - 20y + 41$
$\{i, j\}$	$\{42, 99\}, k_{ij} = 2^2 \cdot 3^2 \cdot 19$
α	$4x^3 - 183x - 16y^3 + 48y^2 - 72y + 293$
β	$-8x^5 + 8x^4y - 16x^4 + 24x^3y + 342x^3 - 402x^2y + 110x^2 - 16xy^3 + 36xy^2$ $- 486xy - 134x + 16y^4 - 32y^3 + 46y^2 + 190y - 195$
γ	$-12x^3y^2 + 48x^3y - 114x^3 - 183xy^2 + 258x - 6y^3 - 405y^2 + 267y - 462$
$\{i, j\}$	$\{42, 126\}, k_{ij} = 2^3 \cdot 3^3 \cdot 19^3$
α	$216x^3 + 189x^2y - 224093x^2 - 279574xy + 56775x + 177100y + 414026$
β	$-432x^5 + 54x^4y + 447322x^4 + 352x^3y^2 + 111507x^3y + 781535x^3$ $- 339x^2y^3 - 334839x^2y^2 - 502025x^2y + 213375x^2 + 27xy^4 + 111370xy^3$ $+ 31138xy^2 - 78415xy + 34487x + 122y^5 + 111497y^4 + 148343y^3 - 266027y^2 - 700225y$

	-218555
γ	$-13x^4y + 9x^4 + 40x^3y^2 + 111598x^3y - 74096x^3 - 111529x^2y^2$ $+471619x^2y + 91731x^2 - 95xy^4 + 63xy^3 - 397515xy^2 - 151666xy - 597927x$ $+122y^5 + 111253y^4 - 74163y^3 + 59643y^2 - 5635y + 458021$
$\{i, j\} = \{42, 127\}, k_{ij} = 2^2 \cdot 3^4 \cdot 7^2 \cdot 19$	
α	$-15x^2 - 30450xy - 48498x + 75837y + 98787$
β	$-30x^3y^2 + 60795x^3y + 157980x^3 - 15x^2y^3 - 36x^2y^2 - 5186x^2y$ $+27594x^2 + xy^3 + 26xy^2 - 66558xy - 65997x + 16y^3 + 35y^2 - 39085y - 184698$
γ	$30x^5 + 45x^4y - 60954x^4 + 45x^3y^2 - 60925x^3y + 29838x^3 + 15x^2y^3$ $-10x^2y^2 + 5176x^2y - 6942x^2 - xy^3 + 30411xy^2 + 54271xy + 66711x - 16y^3$ $-75824y^2 + 91949y - 189981$
$\{i, j\} = \{42, 128\}, k_{ij} = 2^3 \cdot 3^5 \cdot 7$	
α	$x^4 + x^3y - 290x^3 - 859x^2y + 428x^2 + 2316xy - 3392x - 1836y + 5521$
β	$-2x^6 + 576x^5 + x^4y^2 + 569x^4y - 581x^4 - x^3y^3 - 572x^3y^2$ $-1441x^3y + 619x^3 + 2x^2y^3 + 1431x^2y^2 + 5291x^2y + 7847x^2 - xy^4$ $-572xy^3 - 4883xy^2 - 8130xy - 2306x + y^5 + 571y^4 + 1444y^3 + 2874y^2 - 2968y - 2955$
γ	$-x^3y^2 - 571x^3y - 879x^3 + 569x^2y^2 - 246x^2y - 6184x^2 + xy^3$ $+1124xy^2 + 6351xy + 5294x - y^5 - 570y^4 - 874y^3 - 165y^2 - 1123y - 1051$
$\{i, j\} = \{42, 130\}, k_{ij} = 2^5 \cdot 3^4 \cdot 7 \cdot 37$	
α	$98x^3 - 14x^2y - 148080x^2 - 29270xy + 209332x - 70662y + 50538$
β	$-196x^5 + 224x^4y + 295768x^4 - 34x^3y^2 - 236968x^3y + 173066x^3$ $-164x^2y^3 + 448x^2y^2 - 291120x^2y - 28942x^2 + 178xy^4 + 117882xy^3$ $+220088xy^2 - 160338xy - 286048x - 64y^5 - 58990y^4 - 36862y^3 + 162084y^2 + 331212y$ $+374568$
γ	$-3x^4y + x^4 + 10x^3y^2 + 29557x^3y - 10933x^3 - 29532x^2y^2 + 128561x^2y$ $+90509x^2 - 25xy^4 + 10xy^3 - 117604xy^2 - 85472xy - 250898x + 32y^5 + 29463y^4$ $-11000y^3 - 5184y^2 - 102402y - 72573$
$\{i, j\} = \{42, 145\}, k_{ij} = 2^2 \cdot 3^2 \cdot 7 \cdot 13^2$	
α	$-1300x^2 - 872xy - 5532x + 7234y + 13930$
β	$2600x^4 - 858x^3y + 16263x^3 - 872x^2y^2 - 21847x^2y + 5850x^2 + 866xy^3$ $+9285xy^2 - 6323xy + 2755x + 433y^3 - 3459y^2 - 32322y - 27884$
γ	$-x^4 + 435x^3y + 3783x^3 - 432x^2y^2 + 3150x^2y + 9801x^2 - 3xy^3 - 6927xy^2$ $+909xy - 27175x + 433y^4 + 3775y^3 - 10725y^2 - 2512y + 28682$
$\{i, j\} = \{42, 148\}, k_{ij} = 2^3 \cdot 3^3 \cdot 7$	
α	$-130x^2 - 87xy - 118x - 198y + 500$
β	$260x^4 - 87x^3y + 756x^3 - 86x^2y^2 - 360x^2y + 252x^2 + 86xy^3 - 83xy^2$ $-310xy - 122x + 172y^3 + 202y^2 - 20y + 12$
γ	$-x^4 + 87x^3y - 84x^3 - 87x^2y^2 + 398x^2y - 13x^2 - 313xy^2 - 212xy - 325x$ $+86y^4 - 84y^3 + 225y^2 + 18y - 63$
$\{i, j\} = \{42, 174\}, k_{ij} = 2^3 \cdot 3^2$	
α	$3xy - 18x - 5y + 30$
β	$-6x^3y + 36x^3 - 38x^2y + 12x^2 - 46xy - 12x - 33y - 18$
γ	$-6x^4 - 6x^3y + 4x^3 - 38x^2y + 4x^2 + 3xy^2 - 64xy + 4x - 5y^2 - 3y + 10$
$\{i, j\} = \{42, 215\}, k_{ij} = 2^2 \cdot 3^5 \cdot 11$	
α	$5049y + 1782$
β	$5069x^3 - 2534x^2y + 2817x^2 + 1267xy^2 - 1409xy - 906x - 634y^3 + 704y^2 + 451y$ -6881
γ	$-5069x^4 - 2495x^3y - 1312x^3 + 1188x^2y^2 - 1278x^2y + 159x^2 - 475xy^3$ $+1632xy^2 + 1392xy + 2411x + 317y^4 - 3035y^3 - 1965y^2 - 193y - 12227$
$\{i, j\} = \{43, 57\}, k_{ij} = 2^4 \cdot 3$	
α	$-12y$
β	$12y^2 - 24y + 48$
γ	$-24x^3y - 24x^2y + 12xy^2 - 48xy + 48x + 12y^3 - 24y^2 + 12y$
$\{i, j\} = \{43, 61\}, k_{ij} = 2^5 \cdot 13$	
α	$-54x^3 + 88x^2 - 40x + 110$
β	$-54xy^2 - 102xy + 96x + 34y^2 + 18y + 86$
γ	$108x^5 - 176x^4 - 108x^3y + 80x^3 + 176x^2y - 166x^2 + 54xy^3 + 162xy^2 + 76xy$ $-22x - 34y^3 - 162y^2 - 138y - 214$
$\{i, j\} = \{43, 95\}, k_{ij} = 2^6 \cdot 3 \cdot 7 \cdot 13$	
α	$20x^3 + 12x^2y - 23500x^2 + 11746xy + 70261x - 16126y - 46781$

β	$-40x^5 + 14x^4y + 46956x^4 + 20x^3y^2 - 46935x^3y - 60110x^3 - 20x^2y^3$
	$-7x^2y^2 + 52025x^2y - 44617x^2 - 2xy^4 + 23437xy^3 + 9868xy^2 + 34850xy$
	$-44726x + 2y^4 - 23417y^3 - 5513y^2 - 48690y + 120009$
γ	$-x^4y - 2x^4 - 2x^3y^2 + 11731x^3y + 16707x^3 + 9x^2y^3 - 11729x^2y^2$
	$+15940x^2y - 7194x^2 - 10xy^4 + 9xy^3 - 32637xy^2 - 46666xy - 37073x - y^5$
	$+11708y^4 + 16673y^3 + 53864y^2 - 51772y + 37806$
$\{i, j\} = \{43, 114\}, k_{ij} = 2^4 \cdot 3 \cdot 5 \cdot 7^2 \cdot 13$	
α	$-2138x^2 + 3570xy - 15397x - 10414y + 48911$
β	$-x^4 - 240x^3y + 14880x^3 + 235x^2y^2 + 9039x^2y - 17834x^2 - 478xy^3$
	$+5846xy^2 - 3913xy - 38568x + 713y^4 + 3193y^3 - 13912y^2 - 23816y + 6147$
γ	$4277x^4 - 2622x^3y + 20190x^3 + 1661x^2y^2 + 14238x^2y - 44917x^2 - 478xy^3$
	$+6799xy^2 - 14533xy - 36925x + 713y^4 + 3906y^3 - 20420y^2 + 14376y - 41995$
$\{i, j\} = \{43, 168\}, k_{ij} = 2^5 \cdot 7 \cdot 13 \cdot 19$	
α	$-14678x^3 + 5560x^2 + 21388x + 8200y^2 + 12366y + 5728$
β	$28495x^5 + 49648x^4 + 861x^3y^2 - 27634x^3y - 44615x^3 - 60768x^2y^2$
	$-59015x^2y - 110085x^2 - 54290xy^2 - 65804xy - 45939x - 8200y^3 + 57463y^2 + 72071y$
	$+60174$
γ	$-861x^4 + 29356x^3y + 33134x^3 - 11120x^2y - 56126x^2 + 861xy^3 - 41451xy^2$
	$+31635xy + 36221x - 61629y^3 - 73633y^2 - 153041y - 22030$
$\{i, j\} = \{43, 181\}, k_{ij} = 2^2 \cdot 3 \cdot 13$	
α	$-3x^2y - 38x^2 - 3xy^2 - 38xy - 16x + y^2 - 2y + 52$
β	$2x^3y + 27x^3 - 3x^2y^2 - 19x^2y + 145x^2 + 3xy^3 + 36xy^2 - 32xy - 4x - y^3$
	$+6y^2 - 27y$
γ	$-6x^3y - 76x^3 - 4x^2y - 81x^2 + 5xy^2 + 79xy + 141x - 3y^3 - 31y^2 + 20y - 19$
$\{i, j\} = \{44, 47\}, k_{ij} = 2^2 \cdot 3^2 \cdot 7$	
α	$12x + 3$
β	$-24x^3 - 30x^2 - 6x - 15y^3 - 9y^2 + 111y + 150$
γ	$15xy^2 - 21xy - 69x - 12y^2 - 21y + 93$
$\{i, j\} = \{44, 48\}, k_{ij} = 2 \cdot 3 \cdot 7$	
α	$2x$
β	$-4x^3 - 4x^2 + 4y^2 - 12y + 26$
γ	$-2xy^2 + 6xy - 12x - 4y + 16$
$\{i, j\} = \{44, 49\}, k_{ij} = 2 \cdot 3 \cdot 7$	
α	$-3x$
β	$6x^3 + 6x^2 + 3y^3 - 24y + 21$
γ	$3xy^2 + 6xy - 12x - 3y^2 + 3y + 21$
$\{i, j\} = \{44, 50\}, k_{ij} = 2^6 \cdot 3^2$	
α	$72x$
β	$-144x^3 - 144x^2 - 48y^3 - 72y^2 - 120y + 576$
γ	$-24xy^2 - 72xy - 168x + 48y^2 + 72y + 120$
$\{i, j\} = \{44, 51\}, k_{ij} = 7$	
α	$x + 2$
β	$-2x^3 - x^2y^2 - 7x^2 - 3xy^2 - 7x - 2y^2 - 1$
γ	$xy^2 + x + 2y^2 + 2$
$\{i, j\} = \{44, 52\}, k_{ij} = 2 \cdot 3$	
α	$x + 1$
β	$x^2y^2 - 2x^2y - 2x^2 + 2xy^2 - xy + x + 2y^2 + y + 4$
γ	$2x^2 + xy^2 - 2xy + y^2 + y + 1$
$\{i, j\} = \{44, 54\}, k_{ij} = 2$	
α	1
β	$-2x^2 - 2x - y - 1$
γ	$xy + x + y^2$
$\{i, j\} = \{44, 55\}, k_{ij} = 2 \cdot 3^2$	
α	$2x - 4$
β	$2xy^2 + 12xy + 6x + 4y + 34$
γ	$4x^3 - 12x^2 + 2xy^2 + 8xy + 22x - 4y^2 - 4y - 4$
$\{i, j\} = \{44, 58\}, k_{ij} = 2$	
α	1
β	$-2x^2 - 2x - y - 1$

$$\begin{aligned}
&\gamma \quad -xy - x - y^2 \\
&\{i, j\} = \{44, 63\}, k_{ij} = 7 \\
&\alpha \quad x + 2 \\
&\beta \quad -2x^3 - 6x^2 - 4x - 3y^3 - 3y + 1 \\
&\gamma \quad -3xy^2 - 3x + y^2 + 1 \\
&\{i, j\} = \{44, 64\}, k_{ij} = 3^2 \\
&\alpha \quad x - 2 \\
&\beta \quad -2x^3 + 2x^2 + 4x + 2y^3 + 2y^2 - 2y + 1 \\
&\gamma \quad -2xy^2 - 2xy + 2x + y^2 - 2y - 7 \\
&\{i, j\} = \{44, 65\}, k_{ij} = 7^3 \\
&\alpha \quad -18x - 1 \\
&\beta \quad 36x^3 + 18x^2y^2 - 35x^2y + 105x^2 + 37xy^2 - 91xy + 235x + 20y^2 - 77y + 247 \\
&\gamma \quad 18xy^2 - 35xy + 67x + y^2 - 21y + 99 \\
&\{i, j\} = \{44, 66\}, k_{ij} = 2 \cdot 3^2 \cdot 7 \\
&\alpha \quad -2x - 1 \\
&\beta \quad 4x^3 + 2x^2y^2 - 5x^2y + 17x^2 + 5xy^2 - 16xy + 54x + 4y^2 - 17y + 69 \\
&\gamma \quad 2xy^2 - 5xy + 11x + y^2 - 6y + 30 \\
&\{i, j\} = \{44, 67\}, k_{ij} = 2^6 \cdot 3^2 \\
&\alpha \quad 72x \\
&\beta \quad -144x^3 - 144x^2 - 48y^3 - 72y^2 - 120y + 576 \\
&\gamma \quad 24xy^2 + 72xy + 168x - 48y^2 - 72y - 120 \\
&\{i, j\} = \{44, 69\}, k_{ij} = 2 \cdot 3 \\
&\alpha \quad x + 1 \\
&\beta \quad -2x^3 - 4x^2 - 2x - 2y^3 + y^2 - y + 1 \\
&\gamma \quad 2xy^2 - xy + x - y^2 + 2y - 2 \\
&\{i, j\} = \{44, 71\}, k_{ij} = 2 \cdot 3 \cdot 7 \\
&\alpha \quad -2x - 8 \\
&\beta \quad 4x^3 + 20x^2 + 16x + 10y^3 + 16y^2 + 2y - 2 \\
&\gamma \quad 10xy^2 + 16xy + 2x - 2y^2 - 20y - 34 \\
&\{i, j\} = \{44, 188\}, k_{ij} = 3 \cdot 7 \\
&\alpha \quad 0 \\
&\beta \quad 16 \\
&\gamma \quad 4x - 5 \\
&\{i, j\} = \{44, 189\}, k_{ij} = 3 \cdot 7 \\
&\alpha \quad 0 \\
&\beta \quad 25 \\
&\gamma \quad 5x - 1 \\
&\{i, j\} = \{45, 53\}, k_{ij} = 7 \cdot 43 \\
&\alpha \quad 11x^2 - 15xy^2 + 87x - 9y^2 + 47 \\
&\beta \quad -33x^4 + 45x^3y^2 + 11x^3y - 240x^3 - 15x^2y^3 - 29x^2y^2 + 132x^2y \\
&\quad + 94x^2 + 15xy^4 - 54xy^3 - 99xy^2 + 268xy - 111x + 9y^4 + 48y^3 + 85y^2 - 207y - 800 \\
&\gamma \quad -22x^2 + 30xy^2 - 119x - 57y^2 + 320 \\
&\{i, j\} = \{45, 68\}, k_{ij} = 3 \cdot 7^2 \\
&\alpha \quad -24y - 33 \\
&\beta \quad 48x^3y + 66x^3 + 30x^2y + 78x^2 - 18xy + 12x - 24y - 33 \\
&\gamma \quad -48x^3 - 30x^2 + 18x + 24y^2 - 39y + 93 \\
&\{i, j\} = \{45, 123\}, k_{ij} = 2^2 \cdot 7 \cdot 11 \cdot 43 \\
&\alpha \quad -725x^4y - 1885x^4 + 3555xy + 9243x + 3393y^2 + 7549y + 13113 \\
&\beta \quad 3625x^7 - 725x^6y - 3335x^6 - 725x^5y^2 + 2465x^5y + 3190x^5 + 725x^4y^3 \\
&\quad - 290x^4y^2 - 4785x^4y - 19080x^4 + 3555x^3y - 3383x^3 + 1293x^2y^2 \\
&\quad - 2247x^2y + 5034x^2 - 162xy^3 - 77xy^2 - 2575xy - 14747x - 3393y^4 - 3025y^3 \\
&\quad - 8533y^2 - 9665y - 26095 \\
&\gamma \quad 1450x^4y + 145x^4 - 7110xy - 711x - 1131y^2 - 4948y + 470 \\
&\{i, j\} = \{46, 59\}, k_{ij} = 7 \cdot 37 \\
&\alpha \quad 18y + 5 \\
&\beta \quad -36x^3y - 10x^3 - 62x^2y - 46x^2 - 78xy - 108x - 66y - 191 \\
&\gamma \quad 36x^3 + 62x^2 + 78x - 18y^2 + 13y + 53 \\
&\{i, j\} = \{46, 80\}, k_{ij} = 7 \\
&\alpha \quad x - 1
\end{aligned}$$

β	$-x^4 - xy^2 - 2xy - 3x - 4y - 8$
γ	$-x + y + 2$
$\{i, j\}$	$\{46, 187\}, k_{ij} = 3 \cdot 7$
α	0
β	0
γ	1
$\{i, j\}$	$\{52, 53\}, k_{ij} = 2^3 \cdot 3^2 \cdot 5 \cdot 7^2$
α	$106720x^5 - 9102x^4 - 53360x^3y^2 - 53360x^3y + 188892x^3 + 124292x^2y$ $+ 84633x^2 - 53360xy^2 - 67852xy + 54518x - 53360y^3 - 135807y^2 - 29362y + 30858$
β	$-213440x^5 - 515396x^4 + 118235x^3y^2 - 318655x^3y - 1044109x^3$ $+ 5390x^2y^2 - 210241x^2y - 834274x^2 + 115190xy^3 + 135545xy^2 - 953294xy$ $- 1338753x - 53360y^5 + 53360y^4 + 277464y^3 - 669034y^2 - 1789173y - 1073430$
γ	$-213440x^5 + 18204x^4 + 106720x^3y^2 - 106720x^3y - 697944x^3$ $+ 11515x^2y^2 - 228875x^2y - 213475x^2 + 100595xy^2 - 134930xy - 486077x$ $+ 135807y^3 + 6125y^2 - 552337y - 382788$
$\{i, j\}$	$\{53, 66\}, k_{ij} = 2^3 \cdot 3^2 \cdot 5 \cdot 7$
α	$-237x^7 - 237x^6 + 237x^5y - 1185x^5 - 238x^4y - 1964x^4 - 623x^3y - 3424x^3$ $- 598x^2y - 5895x^2 - 1209xy - 4660x + 727y - 983$
β	$474x^7 + 1896x^5 + 2x^4y + 847x^4 + 237x^3y^3 - 474x^3y^2 + 1955x^3y$ $+ 3631x^3 + 596x^2y^2 - 1434x^2y + 3775x^2 + 475xy^3 + 624xy^2 + 218xy + 3022x$ $- 237y^4 + 1097y^3 - 1821y^2 - 3144y - 975$
γ	$120x^3y^2 + 376x^3y + 255x^3 + x^2y^3 + 806x^2y^2 + 143x^2y + 3563x^2$ $- 238xy^4 - 750xy^3 + 1175xy^2 + 1918xy + 4411x + 237y^5 - 860y^4 + 234y^3$ $+ 5831y^2 + 3481y + 4197$
$\{i, j\}$	$\{53, 88\}, k_{ij} = 2^6 \cdot 3^5 \cdot 5 \cdot 7^3 \cdot 13$
α	$-208x^{10} + 69x^9y + 75271x^9 + 230x^8y + 75019x^8 + 270x^7y - 51371688x^7$ $+ 176317x^6y - 68434950x^6 + 52006021x^5y - 222649043x^5 - 68692370x^4y$ $- 234955387x^4 - 178722157x^3y - 525742812x^3 - 316172138x^2y - 694788395x^2$ $- 78973493xy - 195729938x + 117607508y + 24483666$
β	$416x^{10} - 138x^9y - 150958x^9 + 94x^8y + 504x^8 - 138x^7y^2 - 151870x^7y$ $+ 102892790x^7 + 208x^6y^3 - 46x^6y^2 - 50091x^6y + 34353585x^6$ $- 69x^5y^4 - 75479x^5y^3 - 477x^5y^2 - 540503x^5y + 308202226x^5$ $+ 47x^4y^4 + 250x^4y^3 + 101703x^4y^2 + 69820268x^4y - 130158636x^4$ $- 69x^3y^5 - 75797x^3y^4 + 51598155x^3y^3 - 103034813x^3y^2$ $+ 305157091x^3y + 684257215x^3 - 23x^2y^5 - 25218x^2y^4 + 17214803x^2y^3$ $+ 164781928x^2y^2 - 185668054x^2y + 531763234x^2 - 54xy^5 - 42189xy^4$ $+ 51585565xy^3 - 15047150xy^2 + 169865500xy + 427446204x + 23y^6 + 25322y^5$ $- 17143233y^4 + 48053536y^3 - 90012750y^2 - 321771631y - 302488761$
γ	$-2x^6y - 5x^6 + x^5y^2 + 2102x^5y + 6554x^5 + 3x^4y^3 - 2093x^4y^2$ $+ 5875x^4y - 2937x^4 - 3x^3y^4 + 10x^3y^3 + 8623430x^3y^2 + 26900341x^3y$ $+ 98665956x^3 + 4x^2y^5 + 12646x^2y^4 + 39451x^2y^3 + 51003632x^2y^2$ $+ 86773542x^2y + 228641906x^2 + 8xy^6 - 8386xy^5 - 17223247xy^4 - 53692876xy^3$ $+ 47242181xy^2 - 11865470xy + 193805398x - 23y^7 - 25322y^6 + 17143164y^5$ $- 48129295y^4 + 24062218y^3 + 350978953y^2 + 367297537y + 393582375$
$\{i, j\}$	$\{53, 105\}, k_{ij} = 2 \cdot 3^2 \cdot 5 \cdot 7^3$
α	$210x^3 + 2450x^2 + 3535x + 7980$
β	$-3x^2 - 210xy^2 + 1365xy - 8327x - 2030y^2 - 4792y + 2310$
γ	$-420x^5 - 5317x^4 + 210x^3y^2 - 1785x^3y - 4057x^3 + 2450x^2y^2 - 3678x^2y$ $- 14207x^2 - 210xy^3 + 4270xy^2 - 12842xy - 11261x - 2030y^3 - 2902y^2 - 18156y + 534$
$\{i, j\}$	$\{53, 109\}, k_{ij} = 2^6 \cdot 3^2 \cdot 5$
α	$-277x^5 + 277x^4y - 1110x^4 + 837x^3y + 369x^3 + 597x^2y - 4745x^2 - 710xy - 2091x$ $+ 1651y - 99$
β	$-x^3y - 2267x^3 - x^2y^2 - 2266x^2y + 4084x^2 - 283xy^3 - 1512xy^2 + 2415xy$ $+ 1303x - 277y^4 + 1080y^3 + 2489y^2 - 2439y + 1059$
γ	$-554x^6 - 1666x^5 + 7x^4y + 691x^4 + 560x^3y^2 + 165x^3y - 6975x^3$ $- 241x^2y^2 - 4486x^2y + 3281x^2 - 560xy^3 - 1985xy^2 + 2127xy + 180x - 277y^4$ $+ 526y^3 + 6023y^2 + 2146y + 714$
$\{i, j\}$	$\{53, 134\}, k_{ij} = 2^3 \cdot 3^2 \cdot 7$
α	$682x^5 - 682x^4y + 2728x^4 - 2046x^3y + 5024x^3 - 1602x^2y + 7980x^2 - 814xy$ $+ 7128x + 1364y + 5456$
β	$-175x^3 - 175x^2y - 4155x^2 + 682xy^3 + 1452xy^2 - 714xy - 6568x + 682y^4$

$$\begin{aligned}
& -3172y^3 + 3005y^2 + 2036y - 8378 \\
\gamma & 1364x^6 + 4092x^5 + 5781x^4 - 1364x^3y^2 + 2402x^3y + 9259x^3 + 1126x^2y^2 \\
& + 4515x^2y + 7057x^2 + 1364xy^3 + 1558xy^2 + 9033xy + 8999x + 682y^4 - 1126y^3 \\
& - 1737y^2 + 4739y + 4635 \\
\{i, j\} &= \{53, 172\}, k_{ij} = 2^6 \cdot 3^4 \cdot 7^2 \\
\alpha & 73x^2y^3 + 55x^2y^2 - 74758x^2y + 26596x^2 - 341xy^2 - 176217xy - 223918x \\
& - 73y^4 - 128y^3 - 662y^2 - 310306y - 299907 \\
\beta & 74440x^2y^2 - 118213x^2y + 142403x^2 + 73xy^4 - 115xy^3 + 25416xy^2 - 32829xy \\
& + 862465x + 73y^5 - 91y^4 + 107y^3 + 109144y^2 + 444565y + 1171650 \\
\gamma & -146x^3y^3 - 74550x^3y^2 + 267729x^3y - 195595x^3 - 73x^2y^4 \\
& + 115x^2y^3 - 99174x^2y^2 + 503476x^2y - 557032x^2 - 73xy^5 + 164xy^4 \\
& + 74996xy^3 - 176789xy^2 - 65966xy - 1185514x + 341y^4 + 100219y^3 - 386011y^2 \\
& - 681226y - 787071 \\
\{i, j\} &= \{53, 173\}, k_{ij} = 2^5 \cdot 3^3 \cdot 5 \cdot 43 \\
\alpha & -x^3y^3 + 2524x^3y^2 - 31219x^3y - 84564x^3 - 1235x^2y^3 + 14043x^2y^2 \\
& - 12680x^2y - 20634x^2 + xy^4 - 1289xy^3 + 18995xy^2 + 28278xy + 131355x + 1233y^4 \\
& - 9045y^3 - 91285y^2 - 182970y - 48918 \\
\beta & -x^4y - 9x^4 + 606x^3y^2 - 5830x^3y - 36757x^3 + x^2y^4 - 6x^2y^3 \\
& + 47780x^2y^2 + 14592x^2y + 232644x^2 - xy^5 + 52xy^4 + 50xy^3 - 150512xy^2 \\
& - 83128xy - 270902x - 1233y^5 + 11508y^4 + 73321y^3 - 77476y^2 + 43423y + 110838 \\
\gamma & -x^4y - 9x^4 - 2x^3y^3 + 5654x^3y^2 - 68270x^3y - 205903x^3 + x^2y^4 \\
& - 2478x^2y^3 + 82127x^2y^2 - 84861x^2y - 51302x^2 - xy^5 + 56xy^4 \\
& - 5620xy^3 + 29476xy^2 - 134177xy - 69528x - 1236y^5 + 14094y^4 - 11305y^3 \\
& - 136380y^2 - 448146y - 135045 \\
\{i, j\} &= \{53, 214\}, k_{ij} = 2^5 \cdot 3^3 \cdot 5 \cdot 13 \\
\alpha & -255xy^2 + 44568xy + 160746x - 84y^3 - 630y^2 + 96072y + 157809 \\
\beta & -7x^3y - 53x^3 - 3x^2y^2 - 12x^2y - 79x^2 + 29xy^3 - 14607xy^2 - 108673xy \\
& - 245561x + 28y^4 + 156y^3 + 28811y^2 - 39543y - 17457 \\
\gamma & 7x^5y + 53x^5 + 3x^4y^2 + 26x^4y + 185x^4 - 29x^3y^3 + 15130x^3y^2 \\
& + 19656x^3y - 75455x^3 - 28x^2y^4 - 43x^2y^3 + 2245x^2y^2 - 23817x^2y \\
& - 127262x^2 - 85xy^4 + 15348xy^3 - 35835xy^2 - 212809xy - 90213x - 28y^5 - 153y^4 \\
& + 62905y^3 + 7785y^2 + 10587y + 121632 \\
\{i, j\} &= \{54, 87\}, k_{ij} = 2^2 \cdot 3 \cdot 5 \\
\alpha & 20x + 25y + 75 \\
\beta & -10x^3y - 41x^3 + 10x^2y^2 + 48x^2y + 81x^2 - 7xy^2 - 27xy - 16x - 10y^4 - 41y^3 \\
& - 54y^2 - 111y + 12 \\
\gamma & -40x^3 + 11x^2 + 20xy^2 + 20xy + 6x - 10y^3 - 36y^2 + 9y - 55 \\
\{i, j\} &= \{54, 118\}, k_{ij} = 2^2 \\
\alpha & -x \\
\beta & x^4 + 2x^3y - 2x^3 - x^2y - x^2 + xy^3 - 2xy^2 - 3x - y^3 + 4y - 5 \\
\gamma & x^3 + xy^2 - y^2 - y + 4 \\
\{i, j\} &= \{54, 120\}, k_{ij} = 2 \cdot 3^4 \\
\alpha & -63x^3 + 36x^2 - 36x + 54 \\
\beta & 94x^6 + 63x^5y - 143x^5 + 31x^4y^2 - 149x^4y - 12x^4 - 158x^3y^2 + 14x^3y \\
& - 61x^3 - 62x^2y^3 + 20x^2y^2 + 14x^2y + 176x^2 + 32xy^3 + 3xy^2 + 67xy - 36x \\
& + 31y^4 + 39y^3 + 68y^2 - 36y + 54 \\
\gamma & 32x^3 + 31x^2y + 39x^2 + 32xy^2 + 39xy + 13x + 31y^3 + 39y^2 + 14y - 36 \\
\{i, j\} &= \{58, 59\}, k_{ij} = 2^2 \\
\alpha & -y \\
\beta & -2x^3y + 2x^2y + 2x^2 - 2x - 3y \\
\gamma & 2x^3 - 2x^2 + y^2 - y + 4 \\
\{i, j\} &= \{59, 166\}, k_{ij} = 2^3 \cdot 7 \cdot 37 \\
\alpha & 300xy - 40x + 240y - 32 \\
\beta & 680x^5 + 464x^4 - 824x^3 - 2148x^2 - 300xy^2 + 340xy - 1412x - 240y^2 + 272y + 2168 \\
\gamma & 680x^3y + 600x^3 + 464x^2y - 200x^2 + 536xy - 624x - 1220y - 1564 \\
\{i, j\} &= \{59, 167\}, k_{ij} = 2 \cdot 3 \cdot 7 \\
\alpha & -2xy + x - 4y + 9 \\
\beta & -2x^5 - 12x^4 + 28x^3 + 13x^2 + 2xy^2 - 3xy - 23x + 4y^2 - 13y + 12 \\
\gamma & -2x^3y - 6x^3 - 12x^2y - 22x^2 + 26xy + 8x + y + 3 \\
\{i, j\} &= \{59, 169\}, k_{ij} = 3 \cdot 7
\end{aligned}$$

α	$-2xy - 3x + 3y + 8$
β	$2x^5 - 10x^4 - 6x^3 + 17x^2 + 2xy^2 + xy - x - 3y^2 - 5y + 5$
γ	$-2x^3y + 4x^3 + 10x^2y - 6x^2 + 6xy + 2x - 17y - 8$
$\{i, j\}$	$\{62, 91\}, k_{ij} = 2^6 \cdot 3^6 \cdot 13$
α	$-40554x^3 + 239598x + 6008y^4 - 4506y^3 - 94674y^2 - 147338y + 49800$
β	$27036x^5 + 27036x^4y + 18024x^3y^2 - 9012x^3y - 168744x^3 - 33492x^2y$ $+ 92550x^2 - 6008xy^4 + 10514xy^3 + 19752xy^2 + 165320xy + 96802x - 6008y^5$ $+ 10514y^4 + 84160y^3 + 39146y^2 - 64914y + 457128$
γ	$-6008x^3y^3 + 30040x^3y^2 + 7558x^3y - 84174x^3 - 79866xy^2 + 319464xy$ $- 425883x - 6008y^5 + 25534y^4 + 60039y^3 - 276698y^2 + 362425y - 216349$
$\{i, j\}$	$\{62, 137\}, k_{ij} = 2^3 \cdot 3^3 \cdot 7^2 \cdot 13$
α	$-12120x^3 + 22458x - 1010y^2 - 23234y - 55022$
β	$8080x^5 + 8080x^4y + 8080x^3y^2 - 2020x^3y - 15477x^3 - 5711x^2y + 45098x^2$ $- 8741xy^2 + 6551xy + 80851x + 4211y^3 + 103923y^2 - 160669y + 157416$
γ	$8080x^3y^2 + 2020x^3y + 6565x^3 - 14972xy^2 - 136673xy + 197281x + 3201y^3$ $+ 88101y^2 - 17265y + 145242$
$\{i, j\}$	$\{62, 141\}, k_{ij} = 2^2 \cdot 3^4 \cdot 7 \cdot 11$
α	$-2832x^3 + 19323x + 1652y^2 - 4393y - 3061$
β	$1888x^5 + 1888x^4y + 1888x^3y^2 - 2360x^3y - 5920x^3 - 20534x^2y + 10576x^2$ $- 15578xy^2 + 13523xy + 38087x + 608y^3 + 30461y^2 + 844y + 2933$
γ	$1888x^3y^2 - 5192x^3y + 22774x^3 - 12882xy^2 - 14661xy + 54297x + 2260y^3$ $+ 24416y^2 + 1568y - 15599$
$\{i, j\}$	$\{62, 159\}, k_{ij} = 2^4 \cdot 3^4 \cdot 5 \cdot 7 \cdot 13$
α	$7x^2y^2 + 53764x^2y + 14038x^2 - 14xy^3 - 107810xy^2 + 48508xy + 248229x + 7y^4$ $+ 54046y^3 - 8415y^2 - 53782y + 73619$
β	$7x^5 + 7x^4y - 7x^4 + 7x^3y^2 - 8x^3y - 268333x^3 - 7x^2y^3 + 92x^2y^2$ $- 268401x^2y + 400568x^2 - 7xy^4 - 54300xy^3 + 138515xy^2 + 123502xy - 363934x$ $- 7y^5 - 54039y^4 + 62454y^3 - 8700y^2 - 281096y + 368823$
γ	$-21x^4 - 14x^3y^2 - 107549x^3y - 28076x^3 + 14x^2y^3 + 108050x^2y^2$ $- 232624x^2y + 280500x^2 + 21xy^3 - 467xy^2 + 398818xy - 1012283x + 54653y^3$ $- 537546y^2 + 617058y - 227186$
$\{i, j\}$	$\{62, 182\}, k_{ij} = 2^6 \cdot 3^6 \cdot 13$
α	$194x^3 - 82x^2y - 307491x^2 + 173623xy + 343773x - 45319y + 112322$
β	$-130x^5 - 76x^4y + 204995x^4 - 34x^3y^2 + 90208x^3y - 211477x^3 - 10x^2y^3$ $+ 24645x^2y^2 - 224369x^2y - 149474x^2 - 5xy^3 - 161316xy^2 - 107312xy - 303726x$ $+ 5y^3 + 33029y^2 - 129619y + 119914$
γ	$x^5 + x^4y - 2x^4 + 2x^3y^2 - 1378x^3y - 26491x^3 + 2x^2y^3 - 4116x^2y^2$ $- 37113x^2y + 22539x^2 + 2xy^4 - 8210xy^3 - 31869xy^2 + 15203xy + 690267x + 5y^5$ $- 12285y^4 + 15737y^3 - 21995y^2 - 33340y - 149648$
$\{i, j\}$	$\{62, 184\}, k_{ij} = 2^5 \cdot 3^4 \cdot 13$
α	$-1176x^3 - 8544x - 294y^2 + 5280y - 3600$
β	$784x^5 + 784x^4y + 784x^3y^2 - 588x^3y + 4471x^3 + 2617x^2y + 2792x^2$ $+ 1735xy^2 - 6016xy + 7795x - 4549y^3 + 13334y^2 - 12425y + 12562$
γ	$784x^3y^2 - 980x^3y - 2891x^3 + 5696xy^2 - 24697xy + 21437x - 4843y^3 + 14065y^2$ $- 12377y + 31934$
$\{i, j\}$	$\{62, 207\}, k_{ij} = 2 \cdot 3^4$
α	$114x - 86y - 85$
β	$-75x^3 - 18x^2y + 60x^2 + 4xy^2 + 90xy - 52x + 7y^3 + 77y^2 - 88y + 343$
γ	$-x^4 - 3x^3 + 3x^2y^2 - 6x^2y + 78x^2 + 4xy^3 - 4xy^2 + 50xy - 427x + 7y^4$ $+ 5y^3 - 84y^2 + 162y + 74$
$\{i, j\}$	$\{68, 81\}, k_{ij} = 3 \cdot 7^3$
α	$-7y + 21$
β	$7y^2 - 42y + 112$
γ	$14x^3y - 42x^3 - 28x^2y + 84x^2 + 56xy - 168x - 105y + 315$
$\{i, j\}$	$\{68, 105\}, k_{ij} = 3^3$
α	$3xy + 4x - 4y - 7$
β	$-6x^3 + 2x^2 - 3xy^2 + 5xy + 3x + 4y^2 - 5y + 16$
γ	$-6x^3y - 8x^3 + 2x^2y + 6x^2 + 12xy + 24x + 9y + 14$
$\{i, j\}$	$\{68, 113\}, k_{ij} = 3^3 \cdot 19$
α	$-24y - 45$

$$\begin{aligned}
\beta & 48x^3y + 42x^3 - 90x^2y - 36x^2 + 126xy + 6x - 159y + 216 \\
\gamma & 48x^3y + 90x^3 - 90x^2y - 126x^2 + 126xy + 108x - 24y^2 - 156y + 27 \\
\{i, j\} &= \{68, 127\}, k_{ij} = 2^3 \cdot 3 \cdot 7^2 \cdot 19 \\
\alpha & -236y - 300 \\
\beta & -472x^3y - 127x^3 + x^2y^2 - 1070x^2y + 89x^2 + 2xy^2 - 2052xy + 1336x + 238y^2 \\
& -3416y + 6750 \\
\gamma & 472x^3y + 600x^3 - x^2y^2 + 1069x^2y + 981x^2 - 2xy^2 + 2050xy + 714x - 2y^2 \\
& + 3006y - 2994 \\
\{i, j\} &= \{68, 139\}, k_{ij} = 3^2 \cdot 7^2 \\
\alpha & -9y + 6 \\
\beta & -18x^3y + 48x^3 + 78x^2y - 12x^2 + 54xy - 153x - 285y + 141 \\
\gamma & 18x^3y - 12x^3 - 78x^2y - 144x^2 - 54xy + 36x + 9y^2 + 270y + 453 \\
\{i, j\} &= \{68, 178\}, k_{ij} = 2^3 \cdot 3 \cdot 7 \\
\alpha & -36y - 48 \\
\beta & -48x^3y + 8x^3 - 16x^2y + 40x^2 + 32xy + 32x + 36y^2 - 36y + 104 \\
\gamma & 24x^3y + 32x^3 + 8x^2y - 8x^2 - 16xy - 40x - 12y - 16 \\
\{i, j\} &= \{68, 219\}, k_{ij} = 3 \cdot 7 \\
\alpha & 0 \\
\beta & 28 \\
\gamma & 14y + 21 \\
\{i, j\} &= \{69, 70\}, k_{ij} = 2 \cdot 3 \\
\alpha & -17x^2 - 11xy - 4x - 11y - 5 \\
\beta & -11x^3y - 24x^3 + 12x^2y^2 + 11x^2y + 24x^2 + 12xy^2 - 9xy + 60x - 11y^4 - 23y^3 \\
& -16y^2 + 22y + 39 \\
\gamma & 34x^4 - 12x^3y - 26x^3 - 11x^2y^2 + 16x^2y + 2x^2 + 11xy^3 + 23xy^2 - 13xy + 17x \\
& -11y^3 - y^2 - 21y + 60 \\
\{i, j\} &= \{70, 125\}, k_{ij} = 2 \cdot 3^2 \\
\alpha & -6x - 132y^3 - 198y^2 - 84y \\
\beta & 88x^4y + 12x^3 + 57x^2y - 12x^2 - 132xy^3 - 228xy^2 - 96xy + 132y^4 + 153y^3 \\
& + 208y^2 + 103y + 18 \\
\gamma & -88x^5 - 176x^4y - 176x^4 - 132x^3y - 165x^3 - 138x^2y - 138x^2 + 12xy^2 - 262xy \\
& -178x + 219y^3 + 211y^2 - 133y - 113 \\
\{i, j\} &= \{70, 179\}, k_{ij} = 2^2 \cdot 3 \cdot 5 \\
\alpha & 250x^2 + 50xy + 170x - 150y - 90 \\
\beta & -500x^4 + 400x^3y + 160x^3 + 230x^2y - 110x^2 - 200xy^3 - 480xy^2 - 113xy + 351x \\
& + 100y^4 + 110y^3 + 183y^2 - 487y + 330 \\
\gamma & 100x^3y + 210x^3 - 100x^2y^2 - 100x^2y - 217x^2 - 110xy^2 + 73xy - 497x + 100y^4 \\
& + 210y^3 + 143y^2 - 244y - 367 \\
\{i, j\} &= \{70, 180\}, k_{ij} = 2^4 \cdot 7 \\
\alpha & -35xy + 42x - 70y + 42 \\
\beta & 70x^3y - 154x^3 + 154x^2y - 140x^2 + 21x - 56y + 119 \\
\gamma & -70x^4 - 70x^3y - 140x^3 - 154x^2y - 133x^2 + 35xy^2 - 42xy + 175x + 70y^2 + 14y + 133 \\
\{i, j\} &= \{70, 220\}, k_{ij} = 2^3 \cdot 3 \cdot 7 \\
\alpha & 324x - 57y + 111 \\
\beta & -644x^3 + 527x^2y - 73x^2 + 58xy^2 + 62xy + 452x - 290y^3 - 273y^2 - 300y + 1035 \\
\gamma & x^4 - 59x^3y - 125x^3 + 59x^2y^2 + 37x^2y + 90x^2 + 87xy^2 - 10xy + 322x - 58y^4 \\
& -124y^3 - 81y^2 + 102y + 240 \\
\{i, j\} &= \{71, 72\}, k_{ij} = 2 \cdot 3^5 \cdot 7 \\
\alpha & -x^3 - 776x^2 + 515xy - 1640x + 612y - 395 \\
\beta & x^3y^2 + 513x^3y - 1650x^3 + 513x^2y^2 - 2669x^2y - 1061x^2 + xy^3 \\
& -1021xy^2 - 2624xy - 3499x - y^5 - 509y^4 + 1650y^3 - 1563y^2 - 1675y - 4389 \\
\gamma & 2x^5 + 2x^4y + 1546x^4 + 516x^3y - 1364x^3 - x^2y^3 - 511x^2y^2 - 1154x^2y \\
& + 245x^2 - xy^4 - 511xy^3 + 111xy^2 - 317xy + 836x + y^4 + 510y^3 - 1753y^2 - 1531y \\
& -1397 \\
\{i, j\} &= \{72, 94\}, k_{ij} = 2^2 \cdot 3^4 \cdot 7^2 \cdot 13 \\
\alpha & 14480x^3 - 35451x + 36200y^3 - 200910y^2 + 286359y - 108222 \\
\beta & -28960x^5 - 28960x^4y + 86880x^4 + 21720x^3y^2 - 27150x^3y + 36842x^3 \\
& + 146824x^2y - 308902x^2 - 36200xy^3 + 95083xy^2 - 379269xy - 37264x - 36200y^4 \\
& + 75971y^3 - 255695y^2 - 6067y + 748032 \\
\gamma & -21720x^3y^2 + 114030x^3y - 139700x^3 - 35451xy^2 + 35451xy + 530604x + 16339y^3
\end{aligned}$$

$$\begin{aligned}
& +269649y^2 - 755721y + 856521 \\
\{i, j\} &= \{72, 102\}, k_{ij} = 2^2 \cdot 3^3 \\
\alpha & 21x + 39 \\
\beta & -42x^3 + 6x^2 + 30x - 87 \\
\gamma & -42x^4 + 42x^3y - 78x^3 - 6x^2y + 42x^2 - 21xy^2 - 51xy - 69x - 39y^2 + 48y - 252 \\
\{i, j\} &= \{72, 160\}, k_{ij} = 2^3 \cdot 3^2 \cdot 7 \\
\alpha & 413x^3 + 342x^2 - 807x + 354y^2 - 764y - 242 \\
\beta & -826x^5 - 911x^4 + 1652x^3y + 956x^3 - 68x^2y^2 - 3797x^2y + 928x^2 \\
& -1525xy^2 - 178xy + 2016x - 286y^3 + 2670y^2 - 1114y + 1471 \\
\gamma & -826x^3y + 2705x^3 - 684x^2y + 459x^2 + 1239xy^2 + 1245xy - 4615x - 68y^3 \\
& -2832y^2 + 7106y - 3183 \\
\{i, j\} &= \{72, 225\}, k_{ij} = 2^3 \cdot 3 \cdot 7 \\
\alpha & 70x + 98 \\
\beta & 6x^2 - xy^2 - 11xy - 4x + 2y^3 + 23y^2 + 9y - 2 \\
\gamma & 70x^4 - 35x^3y + 66x^3 + 17x^2y^2 - 24x^2y - 26x^2 - 7xy^3 + 22xy^2 + 6xy + 50x \\
& + 2y^4 + 15y^3 + 29y^2 + 11y + 120 \\
\{i, j\} &= \{73, 74\}, k_{ij} = 3^2 \cdot 19 \\
\alpha & 8x + 9 \\
\beta & -16x^3 + 30x^2 - 42x - y^3 - 7y^2 + 14y + 48 \\
\gamma & xy^2 + 6xy - 20x - 6y^2 + 21y - 51 \\
\{i, j\} &= \{73, 75\}, k_{ij} = 2^2 \cdot 3^4 \cdot 13^2 \\
\alpha & 4860x + 7128 \\
\beta & -9720x^3 + 14904x^2 + 8xy^2 - 8xy - 15558x - 4y^3 - 9708y^2 + 10363y + 5502 \\
\gamma & -4x^2y^2 + 4x^2y + 3x^2 + 4848xy^2 - 5172xy - 4203x + 7116y^2 - 14892y + 8433 \\
\{i, j\} &= \{73, 77\}, k_{ij} = 2 \cdot 3 \cdot 13 \\
\alpha & 5x + 3 \\
\beta & -10x^3 + 24x^2 - 42x + 2y^3 + 7y^2 - 7y + 23 \\
\gamma & -2xy^2 - 3xy + 13x - 9y^2 + 6y \\
\{i, j\} &= \{73, 78\}, k_{ij} = 2 \cdot 3^3 \cdot 7 \\
\alpha & 108x + 270 \\
\beta & 108x^2y^2 - 378x^2y + 270x^2 + 162xy^2 + 54y^2 - 540y - 540 \\
\gamma & 216x^2 + 108xy^2 - 378xy + 378x + 270y^2 - 378y - 594 \\
\{i, j\} &= \{73, 79\}, k_{ij} = 2^2 \cdot 3^3 \cdot 7^3 \\
\alpha & 972x + 918 \\
\beta & -x^3y^2 + x^3y - 1947x^3 + 970x^2y^2 - 1888x^2y + 7608x^2 - 55xy^2 + 1135xy \\
& -11505x - y^3 + 54y^2 - 1136y + 10851 \\
\gamma & -x^2y^2 + x^2y - 3x^2 + 969xy^2 - 1887xy + 3609x + 915y^2 - 753y - 1737 \\
\{i, j\} &= \{73, 81\}, k_{ij} = 2 \cdot 3 \\
\alpha & 0 \\
\beta & 6 \\
\gamma & -6x - 6 \\
\{i, j\} &= \{75, 76\}, k_{ij} = 2 \cdot 13^2 \\
\alpha & 30x + 44 \\
\beta & -30x^4 - 14x^3y - 14x^3 - 8x^2y^2 + 7x^2y + 46x^2 - 2xy^3 + 12xy^2 + 17xy - 138x \\
& -6y^4 - 3y^3 - 101y^2 + 149y - 6 \\
\gamma & -2x^2y - 2xy^2 + 2xy - 64x - 6y^3 + 3y^2 - 130y + 218 \\
\{i, j\} &= \{76, 81\}, k_{ij} = 2 \cdot 3^4 \cdot 7^2 \cdot 11 \\
\alpha & -594y + 1782 \\
\beta & 2xy - 5x + 594y^2 - 3560y + 9494 \\
\gamma & 1188x^3y - 3564x^3 - 2378x^2y + 7133x^2 - 590xy^2 + 8304xy - 23755x + 1186y^3 \\
& -6527y^2 + 6527y + 36239 \\
\{i, j\} &= \{76, 124\}, k_{ij} = 2^2 \cdot 3^2 \cdot 7 \\
\alpha & 72x^2 - 24xy^2 - 84xy - 24x + 24y^3 + 12y^2 + 60 \\
\beta & -12x^3 - 12x^2y - 48x^2 - 24xy^3 + 12xy^2 - 144x - 24y^4 - 36y^3 - 36y^2 - 168y \\
& -144 \\
\gamma & -144x^3 + 48x^2y^2 + 24x^2y + 60x^2 + 48xy + 84x + 24y^3 - 24y^2 - 36y + 216 \\
\{i, j\} &= \{76, 125\}, k_{ij} = 3^2 \\
\alpha & -3x + 3y \\
\beta & 6x^3 + x^2y^2 - 6x^2 + xy^3 - y^2 + 4y + 9
\end{aligned}$$

$$\begin{aligned}
&\gamma \quad -x^3y + 6x^3 - 2x^2y + xy^3 - 2xy^2 + 4xy - 10x + 2y^2 + 3y + 4 \\
&\{i, j\} = \{76, 151\}, k_{ij} = 3^3 \cdot 5 \cdot 7^2 \\
&\alpha \quad 240x^2 - 240xy + 240x - 480y + 2040 \\
&\beta \quad -480x^4 - 480x^3y - 432x^2y^2 - 708x^2y - 1206x^2 - 336xy^3 - 840xy^2 + 176xy \\
&\quad -414x - 192y^4 - 732y^3 + 1018y^2 - 1052y + 495 \\
&\gamma \quad -24x^3y + 798x^3 - 48x^2y^2 + 426x^2y - 460x^2 - 72xy^3 + 54xy^2 - 40xy - 55x \\
&\quad -96y^4 - 318y^3 + 380y^2 - 191y + 947 \\
&\{i, j\} = \{76, 163\}, k_{ij} = 3^2 \cdot 7 \cdot 13^2 \\
&\alpha \quad -541x^2 + 1082xy - 1496x - 823y + 2922 \\
&\beta \quad 1082x^4 + 216x^3y - 2328x^3 - 324x^2y^2 - 2695x^2y + 1803x^2 - 433xy^3 \\
&\quad -2342xy^2 + 2939xy + 1732x - 433y^3 + 320y^2 - 619y + 1881 \\
&\gamma \quad 108x^3y - 2119x^3 + 217x^2y^2 - 1389x^2y + 4260x^2 + 324xy^3 - 659xy^2 \\
&\quad +1356xy - 1797x + 433y^4 + 70y^3 - 1550y^2 - 1997y + 2047 \\
&\{i, j\} = \{76, 170\}, k_{ij} = 2 \cdot 3 \cdot 13^2 \\
&\alpha \quad 262x^2 + 92xy^2 - 532xy + 236x - 92y^3 + 270y^2 - 328y - 268 \\
&\beta \quad -173x^3 - 173x^2y - 452x^2 + 92xy^3 + 176xy^2 + 234xy + 381x + 92y^4 - 178y^3 \\
&\quad +150y^2 + 418y - 912 \\
&\gamma \quad 524x^3 + 184x^2y^2 - 540x^2y + 299x^2 - 184xy - 637x + 92y^3 + 349y^2 - 39y + 1365 \\
&\{i, j\} = \{76, 199\}, k_{ij} = 7 \\
&\alpha \quad 0 \\
&\beta \quad 4 \\
&\gamma \quad -2x + 2y + 1 \\
&\{i, j\} = \{76, 223\}, k_{ij} = 7 \\
&\alpha \quad 0 \\
&\beta \quad 4 \\
&\gamma \quad 2x - 2y - 1 \\
&\{i, j\} = \{79, 80\}, k_{ij} = 2^3 \cdot 3^4 \cdot 5 \cdot 7^4 \cdot 31 \\
&\alpha \quad -8665590102x^5 - 20397618138x^4 - 4332795051x^3y^2 - 32964500134x^3 \\
&\quad -8963257542x^2y^2 - 3582892608x^2y - 23869526342x^2 - 4928129931xy^2 \\
&\quad -5308432330x - 3811877031y^3 + 10573585683y^2 + 21622770878y + 7478464604 \\
&\beta \quad 17331180204x^5 + 55237886446x^4 + 818585460x^3y^3 + 15438182082x^3y^2 \\
&\quad +24812982840x^3y + 107688883200x^3 + 1116252900x^2y^3 + 27161867784x^2y^2 \\
&\quad +58163479632x^2y + 90302201646x^2 + 297667440xy^4 + 10968049416xy^3 \\
&\quad +20368705701xy^2 + 42968820660xy + 44306984906x - 4332795051y^5 + 10302761022y^4 \\
&\quad +601068179y^3 - 28196718537y^2 + 12656128730y + 16822953552 \\
&\gamma \quad 17331180204x^5 + 40795236276x^4 + 8665590102x^3y^2 - 17331180204x^3y \\
&\quad +80371650438x^3 + 818585460x^2y^3 + 24699107064x^2y^2 - 26147648424x^2y \\
&\quad +95275995684x^2 + 1934838360xy^3 + 25864204542xy^2 - 2028258948xy + 80319338484x \\
&\quad -520918020y^4 - 3561783672y^3 + 737420718y^2 - 32125127316y - 5371283820 \\
&\{i, j\} = \{80, 108\}, k_{ij} = 2 \cdot 3^2 \cdot 13 \cdot 19 \\
&\alpha \quad -x^7y + 104x^7 + 5x^6y - 622x^6 - 65x^5y + 24661x^5 + 16002x^4y + 55822x^4 \\
&\quad -94316x^3y + 212739x^3 + 80274x^2y + 152327x^2 + 18701xy - 8010x - 40049y - 35603 \\
&\beta \quad x^5 + 2x^4y + 47x^4 + 5x^3y^2 + 193x^3y + 70x^3 + 13x^2y^3 - 2515x^2y^2 \\
&\quad +209489x^2y - 345727x^2 + 34xy^4 - 4544xy^3 - 402958xy^2 - 89831xy - 162877x - y^6 \\
&\quad +91y^5 - 14313y^4 - 386986y^3 - 196367y^2 - 182099y + 111255 \\
&\gamma \quad -2x^8y + 208x^8 + 10x^7y - 1244x^7 + 2x^6y^2 - 338x^6y + 49323x^6 \\
&\quad +2x^5y^3 - 216x^5y^2 + 33041x^5y + 111796x^5 - x^4y^4 + 92x^4y^3 \\
&\quad +1576x^4y^2 - 236409x^4y + 425287x^4 + x^3y^4 - 57x^3y^3 - 84529x^3y^2 \\
&\quad +208492x^3y - 18112x^3 - x^2y^5 + 91x^2y^4 - 14266x^2y^3 \\
&\quad -313758x^2y^2 - 338999x^2y - 394584x^2 - xy^5 + 125xy^4 - 18823xy^3 \\
&\quad -775787xy^2 - 675818xy - 433890x - y^6 + 90y^5 - 14222y^4 - 401299y^3 - 623402y^2 \\
&\quad -374020y - 115339 \\
&\{i, j\} = \{80, 133\}, k_{ij} = 2^3 \cdot 7^2 \cdot 13 \\
&\alpha \quad 101x^7 - 662x^6 - 40x^5y + 34287x^5 + 23801x^4y - 30507x^4 - 134371x^3y \\
&\quad +382427x^3 + 37138x^2y - 266530x^2 - 189213xy - 545523x - 371598y - 366502 \\
&\beta \quad x^5 + 3x^4y + 45x^4 + 6x^3y^2 + 315x^3y - 354x^3 + 16x^2y^3 - 3824x^2y^2 \\
&\quad +365054x^2y - 585543x^2 + 40xy^4 - 7108xy^3 - 551828xy^2 + 304726xy + 433216x \\
&\quad +101y^5 - 22670y^4 - 372556y^3 + 558708y^2 + 14627y + 745689 \\
&\gamma \quad 202x^8 - 1122x^7 - 282x^6y + 67655x^6 - 202x^5y^2 + 48445x^5y + 5868x^5 \\
&\quad +101x^4y^3 + 1208x^4y^2 - 287814x^4y + 836589x^4 - 61x^3y^3
\end{aligned}$$

$$\begin{aligned}
& -118909x^3y^2 + 193477x^3y - 176452x^3 + 101x^2y^4 - 22412x^2y^3 \\
& -391315x^2y^2 - 532785x^2y - 505631x^2 + 141xy^4 - 29597xy^3 - 1150302xy^2 \\
& -823408xy - 280436x + 101y^5 - 22468y^4 - 417694y^3 - 603140y^2 - 24709y + 358913 \\
\{i, j\} = \{80, 143\}, k_{ij} = 2 \cdot 7 \cdot 13 \cdot 19 \\
\alpha & 48x^7 + 48x^6y + 132x^6 + 36x^5y + 26749x^5 + 26574x^4y + 107006x^4 \\
& + 26930x^3y - 62470x^3 - 48357x^2y - 110601x^2 + 46994xy + 43536x - 13832y - 10374 \\
\beta & -3x^5 + 3x^4y + 22x^4 - 10x^3y^2 - 84x^3y + 154296x^3 + 10x^2y^3 \\
& + 180x^2y^2 - 154237x^2y + 56426x^2 - 36xy^4 + 26577xy^3 + 99517xy^2 + 29771xy \\
& + 100516x + 48y^5 - 26372y^4 - 181519y^3 - 381301y^2 - 563111y - 421941 \\
\gamma & 96x^8 + 72x^7 - 96x^6y + 53255x^6 - 176x^5y + 107401x^5 + 48x^4y^3 \\
& + 190x^4y^2 - 106480x^4y - 238481x^4 - 48x^3y^3 + 53434x^3y^2 - 12899x^3y \\
& + 333009x^3 - 74x^2y^3 + 4904x^2y^2 + 249x^2y + 204962x^2 + 12xy^4 \\
& + 157xy^3 - 8588xy^2 - 233051xy - 166611x + 48y^5 - 26372y^4 - 181519y^3 \\
& - 395133y^2 - 545821y - 456521 \\
\{i, j\} = \{80, 156\}, k_{ij} = 2 \cdot 3 \cdot 7 \\
\alpha & x^{10} + 4x^9 - 564x^8 + 3x^7y - 1131x^7 - 562x^6y - 552x^6 - 1127x^5y - 4237x^5 \\
& - 2960x^4y + 1667x^4 - 4418x^3y + 2304x^3 - 1977x^2y + 4034x^2 - 2462xy + 5465x \\
& + 7801y + 7843 \\
\beta & -2x^{10} - 6x^9 + 2x^8y + 1138x^8 + 2x^7y + 1139x^7 - x^6y^3 - 5x^6y - 2304x^6 \\
& - 3x^5y^3 - 1156x^5y + 6782x^5 + x^4y^4 + 567x^4y^3 - 1141x^4y^2 \\
& + 5391x^4y - 6103x^4 + x^3y^4 + 570x^3y^3 + 1449x^3y^2 + 1015x^3y \\
& - 2699x^3 + x^2y^3 + 1972x^2y^2 - 2496x^2y + 1061x^2 - xy^4 + 4xy^3 \\
& + 906xy^2 + 8752xy + 3245x + y^5 + 571y^4 - 2398y^3 + 7508y^2 - 10816y - 10217 \\
\gamma & -2x^4 + 288x^3y^2 - 1982x^3y + 1958x^3 + 1054x^2y^2 + 4500x^2y - 2365x^2 \\
& + 574xy^4 - 3950xy^3 + 17292xy^2 - 1449xy - 6221x + y^6 + 572y^5 - 1827y^4 \\
& + 12911y^3 - 3266y^2 - 13274y + 13270 \\
\{i, j\} = \{80, 161\}, k_{ij} = 3 \cdot 7^2 \\
\alpha & -x^3y^3 + 1930x^3y^2 - 5098x^3y - 8969x^3 - 2x^2y^4 + 3x^2y^3 \\
& + 2492x^2y^2 - 6736x^2y - 7815x^2 - 3xy^4 + 962xy^3 + 805xy^2 - 728xy - 715x \\
& - y^5 - 2y^4 - 682y^3 + 2010y^2 + 5475y + 2931 \\
\beta & 480x^3y^2 - 24x^3y + 11236x^3 - 341x^2y^2 + 1684x^2y + 6277x^2 - 959xy^4 \\
& + 40xy^3 + 919xy^2 - 2847xy - 974x + y^6 + 682y^4 - 3369y^3 + 1269y^2 - 2064y - 8646 \\
\gamma & -2x^3y^3 + 4340x^3y^2 - 10220x^3y - 6702x^3 - 4x^2y^4 + 6x^2y^3 \\
& + 5123x^2y^2 - 11812x^2y + 1883x^2 - 967xy^4 + 6782xy^3 - 2756xy^2 - 8353xy \\
& + 19643x + y^6 - 6y^5 - 281y^4 + 5392y^3 - 2517y^2 + 6159y + 17480 \\
\{i, j\} = \{80, 167\}, k_{ij} = 2^2 \cdot 3 \cdot 7 \cdot 11 \\
\alpha & 47x^9 - 19x^8y - 12771x^8 - 25520x^7 + 141x^6y - 66614x^6 - 25550x^5y \\
& - 138071x^5 - 92637x^4y - 137323x^4 - 57556x^3y - 87807x^3 - 223574x^2y \\
& - 89387x^2 - 56768xy + 122175x + 176171y + 177095 \\
\beta & -94x^9 + 38x^8y + 25636x^8 + 56x^7y + 25592x^7 - 38x^6y^2 - 25862x^6y \\
& + 56505x^6 - 47x^5y^3 + 70x^5y + 129905x^5 + 19x^4y^4 + 12818x^4y^3 \\
& + 133x^4y^2 + 90645x^4y + 18744x^4 + 28x^3y^4 + 12816x^3y^3 \\
& + 25895x^3y^2 - 192736x^3y - 98368x^3 - 19x^2y^5 - 12893x^2y^4 \\
& + 54063x^2y^3 - 121835x^2y^2 + 184527x^2y + 261263x^2 - 9xy^4 + 58xy^3 \\
& + 21492xy^2 + 196990xy + 75241x + 19y^5 + 12893y^4 - 54054y^3 + 169171y^2 - 245188y \\
& - 229652 \\
\gamma & 2x^5y - 11x^5 + 2x^4y^2 - 5x^4y + 36x^4 - 6498x^3y^2 + 44766x^3y \\
& - 44573x^3 - 23875x^2y^2 - 101133x^2y + 52393x^2 - 10xy^5 - 12942xy^4 \\
& + 89272xy^3 - 390425xy^2 + 33421xy + 141152x - 19y^6 - 12912y^5 + 41161y^4 \\
& - 291288y^3 + 75093y^2 + 299593y - 300709 \\
\{i, j\} = \{80, 168\}, k_{ij} = 2 \cdot 7^2 \cdot 13 \cdot 19 \\
\alpha & -1801898x^5 - 79198x^4 + 945252x^3y^2 + 1846201x^3y - 272261x^3 \\
& + 2109607x^2y^2 + 370748x^2y + 787643x^2 - 726149xy^2 - 1702121xy + 2667002x \\
& - 945252y^3 - 1654429y^2 + 3097254y + 3830637 \\
\beta & 3603796x^5 + 158396x^4 - 1890504x^3y^2 - 1890504x^3y - 2970668x^3 \\
& - 4219214x^2y^2 + 1508584x^2y - 4106312x^2 + 1452298xy^2 + 1452298xy - 930470x \\
& + 945252y^4 - 2345682y^3 + 3743844y^2 - 1077836y - 2317324 \\
\gamma & 3603796x^5 + 1801898x^4y + 247002x^4 - 3692402x^3y^2 + 2072868x^3y \\
& - 26210x^3 - 6469294x^2y^2 - 162414x^2y - 3578412x^2 + 945252xy^4 \\
& - 2153910xy^3 - 5140190xy^2 + 3657213xy - 6266525x + 2109607y^4 - 4259341y^3 \\
& + 854719y^2 + 1740466y - 9150381
\end{aligned}$$

$$\begin{aligned}
&\{i, j\} = \{80, 211\}, k_{ij} = 2 \\
&\alpha \quad 92x^3 + 24x^2 + 6x + 2 \\
&\beta \quad -46x^3 + 46x^2y + 107x^2 - 46xy^2 - 106xy + 76x + 46y^3 + 107y^2 - 76y + 213 \\
&\gamma \quad 138x^6 - 92x^5y - 29x^5 + 46x^4y^2 + 15x^4y + 132x^4 + 46x^3y^2 - 222x^3y \\
&\quad + 375x^3 + 46x^2y^3 + 131x^2y^2 - 59x^2y + 458x^2 - 39xy^2 - 114xy + 305x \\
&\quad + 46y^4 + 153y^3 + 33y^2 + 135y + 217 \\
&\{i, j\} = \{82, 83\}, k_{ij} = 7^2 \\
&\alpha \quad 6x + 8y + 27 \\
&\beta \quad -2x^3y - 5x^3 + 4x^2y^2 + 4x^2y + 16x^2 - 2xy^3 - 2xy^2 - 8xy - 53x - 2y^4 + y^2 \\
&\quad + 30y - 59 \\
&\gamma \quad -12x^3 - 8x^2 + 4xy^2 + 2xy + 33x - 6y^2 - 27y + 9 \\
&\{i, j\} = \{83, 102\}, k_{ij} = 2^2 \cdot 5^2 \cdot 7 \\
&\alpha \quad 36x + 92 \\
&\beta \quad -72x^3 + 104x^2 - 128x + 60 \\
&\gamma \quad 72x^4 + 72x^3y - 32x^3 - 104x^2y + 24x^2 - 36xy^2 + 92xy - 4x - 92y^2 - 152y - 244 \\
&\{i, j\} = \{83, 103\}, k_{ij} = 2 \cdot 3^2 \cdot 5^2 \cdot 7^3 \\
&\alpha \quad 14x^3 + 16x^2y + 26583x^2 + 17655xy + 116668x - 5572y + 172344 \\
&\beta \quad -28x^5 - 4x^4y - 53082x^4 + 34x^3y^2 + 17783x^3y - 74006x^3 - 32x^2y^3 \\
&\quad + 17562x^2y^2 + 32134x^2y + 40811x^2 + 10xy^4 - 17627xy^3 + 272xy^2 - 96686xy \\
&\quad + 12346x + 10y^4 - 17590y^3 - 2824y^2 - 183012y - 120894 \\
&\gamma \quad -x^4y + x^3y^2 + 8819x^3y + 4249x^3 - 6x^2y^3 - 8822x^2y^2 - 18824x^2y \\
&\quad - 60614x^2 + 11xy^4 + 16xy^3 + 14619xy^2 + 28724xy + 234994x - 5y^5 + 8795y^4 \\
&\quad + 4183y^3 + 31719y^2 + 67469y + 196704 \\
&\{i, j\} = \{83, 138\}, k_{ij} = 2^5 \cdot 3^2 \cdot 5^2 \cdot 7^3 \\
&\alpha \quad -5x^4 + 35x^3y + 20244x^3 + 60497x^2y - 144137x^2 - 14081xy - 517959x - 257689y \\
&\quad - 238959 \\
&\beta \quad 10x^6 - 84x^5y - 40505x^5 + 79x^4y^2 - 39967x^4y + 303226x^4 - 25x^3y^3 \\
&\quad + 40032x^3y^2 - 266008x^3y + 527070x^3 + 34x^2y^3 + 59857x^2y^2 \\
&\quad + 127739x^2y - 845004x^2 - 79xy^4 + 39848xy^3 - 309379xy^2 + 285159xy - 464547x \\
&\quad + 25y^5 - 39944y^4 + 145753y^3 + 369019y^2 - 298802y + 1062159 \\
&\gamma \quad -4x^4y + 13x^4 + 5x^3y^2 + 40278x^3y - 106559x^3 - 30x^2y^3 \\
&\quad - 40213x^2y^2 - 191653x^2y + 492428x^2 + 54xy^4 + 75xy^3 + 297851xy^2 \\
&\quad - 395171xy + 203339x - 25y^5 + 39919y^4 - 105909y^3 - 97407y^2 + 3095y - 1908432 \\
&\{i, j\} = \{83, 139\}, k_{ij} = 3^2 \cdot 5^2 \cdot 7^4 \\
&\alpha \quad 3x^2 + 12537xy + 9810x + 6966y + 53151 \\
&\beta \quad 6x^3y^2 - 25056x^3y - 44685x^3 + 6x^2y^3 - 3x^2y^2 + 55783x^2y \\
&\quad + 63956x^2 - 14xy^3 - 64119xy - 126340x + 16y^3 + 30y^2 + 3428y + 126924 \\
&\gamma \quad -6x^5 - 12x^4y + 25065x^4 - 12x^3y^2 + 25073x^3y - 36203x^3 - 6x^2y^3 \\
&\quad + 23x^2y^2 - 55815x^2y + 68524x^2 + 14xy^3 - 12567xy^2 + 66856xy + 50409x - 16y^3 \\
&\quad - 6980y^2 - 49647y - 84333 \\
&\{i, j\} = \{84, 85\}, k_{ij} = 2^2 \cdot 3^2 \cdot 5^3 \cdot 7 \\
&\alpha \quad -74466x^4 + 37233x^3 - 38475xy + 29070x + 29070y - 12282 \\
&\beta \quad 16548x^6 + 8274x^5 + 124110x^4y - 79982x^4 - 6276x^3 - 5130x^2y^4 \\
&\quad - 456x^2y^3 - 1080x^2y^2 + 4884x^2y + 23799x^2 - 1710xy^4 + 1797xy^3 \\
&\quad + 990xy^2 - 5673xy + 15160x + 1949y^3 + 1350y^2 - 31301y + 22847 \\
&\gamma \quad 372330x^5 - 149760x^4 - 15390x^3y^3 - 36450x^3y^2 - 27540x^3y + 15120x^3 \\
&\quad - 84825x - 14535y^3 - 4050y^2 - 18585y + 195 \\
&\{i, j\} = \{84, 86\}, k_{ij} = 2^3 \cdot 3^2 \cdot 5^3 \cdot 7^2 \cdot 13 \\
&\alpha \quad 4424436x^4 - 4326318x^3 + 4017600xy + 6785280x - 1696320y - 1142178 \\
&\beta \quad -983208x^6 - 21804x^5 + 1843515x^4y + 2763952x^4 + 323496x^3 + 535680x^2y^4 \\
&\quad - 190464x^2y^3 + 146880x^2y^2 - 545664x^2y - 2483214x^2 + 357120xy^4 \\
&\quad + 427938xy^3 - 85680xy^2 + 221313xy - 1148940x + 178560y^4 + 491426y^3 - 134640y^2 \\
&\quad + 3053026y + 3053178 \\
&\gamma \quad 5530545x^5 - 1008630x^4 + 1607040x^3y^3 - 1239300x^3y^2 + 2093040x^3y \\
&\quad - 2576475x^3 + 9288675x + 1160640y^3 - 332100y^2 + 2100240y - 2927940 \\
&\{i, j\} = \{84, 88\}, k_{ij} = 2^4 \cdot 3 \cdot 7^3 \cdot 13 \\
&\alpha \quad 81x^6 + 27x^5 - 18x^4 - 144x^3 + 75x^2 - 36426x + 20715 \\
&\beta \quad -3x^3y^2 + 5x^3y - 9x^3 + 12146x^2y^2 - 30513x^2y + 60116x^2 - 3xy^4 \\
&\quad + 6xy^3 + 13334xy^2 - 38850xy + 89362x - 2y^4 + 2y^3 + 8730y^2 - 28145y + 94990 \\
&\gamma \quad 54x^7 - 18x^6 - 18x^5 + 18x^4y + 3x^4 + 27x^3y^3 + 9x^3y + 126x^3
\end{aligned}$$

$$\begin{aligned}
& -9x^2y^3 - 9x^2y^2 + 6x^2y - 24393x^2 - 9xy^3 + 36459xy^2 - 91548xy + 210342x \\
& + 6y^3 - 20739y^2 + 36027y - 44361 \\
\{i, j\} &= \{84, 190\}, k_{ij} = 2 \cdot 7 \\
\alpha & 0 \\
\beta & 27x^2 + 15x + 6 \\
\gamma & 27x^3 - 1 \\
\{i, j\} &= \{86, 87\}, k_{ij} = 2^3 \cdot 3^2 \cdot 5^5 \cdot 7^4 \cdot 13 \\
\alpha & 7156187703x^4y + 7204869252x^4 + 12290393493x^3y + 16387191324x^3 \\
& - 1798560288x^2y - 1810795392x^2 - 1798560288y^3 - 7488101796y^2 - 8990756670y \\
& + 4270598724 \\
\beta & 4770791802x^7 + 11341669164x^6 + 1557809568x^5y + 4701868108x^5 + 43272488x^4y \\
& + 6697914321x^4 + 2385395901x^3y^3 - 1199040192x^3y^2 + 2845866933x^3y \\
& + 13923854925x^3 + 5670834582x^2y^3 - 791203392x^2y^2 + 177601950x^2y \\
& + 11560177640x^2 + 778904784xy^4 + 2558930822xy^3 - 3353024940xy^2 + 7078112850xy \\
& + 8748836340x + 621156340y^4 + 1719386382y^3 - 3099730892y^2 + 7050058242y - 62717274 \\
\gamma & 4673428704x^6 - 1427992104x^5 + 12692753416x^4 - 14207495904x^3 \\
& + 2336714352x^2y^3 - 1174569984x^2y^2 - 395958672x^2y + 13696650636x^2 \\
& - 713996052xy^3 + 358896384xy^2 + 120987372xy + 422467784x + 1798560288y^4 \\
& + 1471264592y^3 - 739544064y^2 + 9517753244y - 1929623724 \\
\{i, j\} &= \{87, 105\}, k_{ij} = 2 \cdot 3^2 \cdot 5 \\
\alpha & 3x^3 + 6x + y^2 + 6y + 20 \\
\beta & -2x^3y + 2x^3 - 24x - y^3 - 3y^2 - 24y + 10 \\
\gamma & -6x^5 - 6x^4 - 4x^3y - 20x^3 - 18x^2y - 69x^2 - xy^3 - 41xy - 60x - 15y^2 - 62y - 4 \\
\{i, j\} &= \{87, 126\}, k_{ij} = 2^4 \cdot 19^3 \\
\alpha & x^3 - 1603x^2y - 813x^2 - 323315xy + 77752x + 190858y + 646008 \\
\beta & -2x^5 + 2x^4y + 29x^4 - 5x^3y^2 + 2547x^3y - 475607x^3 + 12x^2y^3 \\
& - 4966x^2y^2 - 59174x^2y + 465183x^2 - 24xy^4 + 9916xy^3 + 118239xy^2 - 21584xy \\
& - 349672x + 51y^5 - 19802y^4 - 236484y^3 + 43356y^2 - 413080y - 243923 \\
\gamma & -1603x^5 + 799x^4y - 321625x^4 - 395x^3y^2 + 159411x^3y + 8841x^3 \\
& + 189x^2y^3 - 75843x^2y^2 - 258309x^2y - 372913x^2 - 75xy^4 + 29619xy^3 \\
& + 70688xy^2 + 566170xy - 194990x + 51y^5 - 19700y^4 - 275986y^3 - 278358y^2 - 344084y \\
& - 1096511 \\
\{i, j\} &= \{87, 131\}, k_{ij} = 2^5 \cdot 3 \cdot 5 \cdot 7 \cdot 13 \\
\alpha & 27x^2 - 27345xy - 70797x - 71025y - 92664 \\
\beta & -27x^3y^2 + 54636x^3y + 32427x^3 + 27x^2y^3 + 42x^2y^2 - 54501x^2y \\
& - 31574x^2 - 27xy^3 - 41xy^2 + 53155xy + 139968x + 26y^3 + 170y^2 - 208884y + 192779 \\
\gamma & 27x^5 - 54597x^4 - 54705x^3y - 32512x^3 - 27x^2y^3 - 42x^2y^2 + 54474x^2y \\
& + 87533x^2 + 27xy^3 + 27387xy^2 - 37201xy + 179856x - 26y^3 + 70829y^2 + 159276y \\
& + 256665 \\
\{i, j\} &= \{87, 142\}, k_{ij} = 2^4 \cdot 3^2 \cdot 7^2 \cdot 13 \\
\alpha & -28820x^3 + 246984x + 115280y^3 + 216150y^2 + 149385y + 312543 \\
\beta & 57640x^5 - 57640x^4y - 172920x^4 + 172920x^3y - 148128x^3 - 530422x^2y \\
& - 272282x^2 + 115280xy^3 + 548860xy^2 + 988942xy + 388950x - 115280y^4 + 129690y^3 \\
& + 709930y^2 + 2159118y + 1521858 \\
\gamma & -86460x^3y^2 - 43230x^3y + 319580x^3 - 246984xy^2 - 493968xy - 1059423x \\
& - 339275y^3 - 1511562y^2 - 2130717y - 1803825 \\
\{i, j\} &= \{87, 143\}, k_{ij} = 2 \cdot 7 \cdot 13 \cdot 19 \\
\alpha & -1566x^3 - 2064x^2 - 1027x - 307y^2 - 4072y - 2565 \\
\beta & 3132x^5 + 9537x^4 + 6264x^3y + 11143x^3 - 6361x^2y^2 - 4027x^2y - 1803x^2 \\
& + 1259xy^2 + 5367xy - 581x - 6054y^3 - 13429y^2 - 12296y - 2663 \\
\gamma & 3132x^3y + 14805x^3 + 4128x^2y + 17486x^2 + 1566xy^2 + 2696xy + 3764x - 6361y^3 \\
& - 22941y^2 - 21374y - 19142 \\
\{i, j\} &= \{87, 144\}, k_{ij} = 2^5 \cdot 3^2 \cdot 5^5 \cdot 13 \\
\alpha & -39375x^3y - 23625x^3 - 3158634x^2y - 1886492x^2 - 184605397xy - 80763375x \\
& - 31446162y + 76989468 \\
\beta & -2x^4y + 5302x^4 - 2x^3y^2 + 9074x^3y - 58195800x^3 + 3x^2y^3 \\
& - 1467x^2y^2 - 1743933x^2y + 52589781x^2 + xy^5 - 19xy^4 + 2241xy^3 \\
& + 8683722xy^2 - 3522248xy - 45947143x - y^6 + 85y^5 + 14249y^4 - 43429009y^3 \\
& + 17714662y^2 - 95264356y - 73089468 \\
\gamma & 15750x^6 - 3150x^5y + 1255894x^5 + 630x^4y^2 - 249666x^4y + 73235548x^4
\end{aligned}$$

	$-126x^3y^3 + 49631x^3y^2 - 14529668x^3y - 10657866x^3 + 25x^2y^4$
	$-1987x^2y^3 + 2880102x^2y^2 + 19375827x^2y + 21574189x^2 - 5xy^5 + 371xy^4$
	$+53681xy^3 + 2039241xy^2 - 34051584xy + 46528135x + y^6 - 85y^5 - 14246y^4$
	$+43428745y^3 + 13689545y^2 + 54349294y + 174136830$
$\{i, j\} = \{87, 203\}, k_{ij} = 2^6 \cdot 3^2 \cdot 5$	
α	$4x^2 - 4xy + 7202x + 1432y + 10074$
β	$-8x^4 + 16x^3y - 14380x^3 - 8x^2y^2 + 8626x^2y + 14694x^2 - 10xy^3$
	$+2860xy^2 - 8738xy - 15614x + 12y^4 - 5718y^3 - 9496y^2 + 19588y + 55044$
γ	$1433x^3y + 4161x^3 + x^2y^3 - 1431x^2y^2 - 6992x^2y - 5149x^2 + xy^4$
	$+2830xy^2 + 7414xy - 14302x - 3y^5 + 1425y^4 + 4158y^3 - 2242y^2 - 18325y - 32079$
$\{i, j\} = \{96, 97\}, k_{ij} = 2^3 \cdot 3^3 \cdot 5^2$	
α	$312xy + 516x - 516y + 1596$
β	$507x^4 - 390x^3y - 15x^3 + 234x^2y^2 - 501x^2y - 786x^2 - 78xy^3 + 453xy^2$
	$+270xy + 639x + 39y^4 - 129y^3 + 156y^2 + 363y + 1311$
γ	$-13x^3y - 2x^3 + 13x^2y^2 - 30x^2y - 358x^2 - 13xy^3 + 30xy^2 - 194xy - 174x$
	$+13y^4 - 30y^3 + 194y^2 - 45y + 204$
$\{i, j\} = \{97, 105\}, k_{ij} = 2^5 \cdot 3^4 \cdot 13$	
α	$-3x^6 + 6x^5 + 3222x^3 - 5175x^2 - 636x + 12966$
β	$-x^2 + xy^3 - 1074xy^2 - 214xy - 647x + y^4 + 651y^2 - 4751y - 1734$
γ	$6x^8 - 6x^7 + 6x^6y - 6x^6 - 6447x^5 + 3x^4y^3 + 6x^4y^2 - 3x^4y + 3906x^4$
	$-3x^3y^3 + 6x^3y^2 - 6456x^3y + 5160x^3 + 3x^2y^4 + 3x^2y^3$
	$+3x^2y^2 - 2550x^2y - 23985x^2 - 3222xy^3 + 3213xy^2 + 1314xy - 22653x + 3y^5$
	$-6y^4 + 1956y^3 - 5193y^2 + 25257y + 9117$
$\{i, j\} = \{97, 140\}, k_{ij} = 2^7 \cdot 3^3 \cdot 5^3 \cdot 7^2 \cdot 13$	
α	$-27300x^3y - 5460x^3 + 3299736x^2y + 685347x^2 - 146958243xy - 123946887x$
	$-130662030y + 185698830$
β	$x^4y - 985x^4 + x^3y^3 - 1262x^3y + 20444767x^3 - 2x^2y^3 - 994x^2y^2$
	$+320945x^2y - 21427163x^2 + xy^5 - 11xy^4 - 3324xy^3 + 1606799xy^2 + 7487640xy$
	$-8928310x - 63y^5 - 8347y^4 + 8072388y^3 + 37437587y^2 + 1600631y + 31218729$
γ	$10920x^6 + 2184x^5y - 1329069x^5 + 435x^4y^2 - 267648x^4y + 59903664x^4$
	$+87x^3y^3 - 53892x^3y^2 + 12208272x^3y + 1166214x^3 + 15x^2y^4$
	$-5370x^2y^3 + 2493726x^2y^2 - 25219608x^2y + 21850164x^2 + 3xy^5 - 972xy^4$
	$-141084xy^3 - 21875034xy^2 - 24388614xy - 29245734x - 189y^5 - 25419y^4$
	$+24165948y^3 + 29931411y^2 + 37320165y + 125189559$
$\{i, j\} = \{97, 207\}, k_{ij} = 2^2 \cdot 3^3 \cdot 5^2$	
α	$-1404x + 540$
β	$-x^2y + 10x^2 + 32xy + 4x + 3y^3 + 93y^2 + 9y - 3$
γ	$936x^4 + 315x^3y - 702x^3 + 105x^2y^2 - 435x^2y + 222x^2 + 27xy^3 - 471xy^2$
	$+213xy + 393x + 9y^4 + 270y^3 + 288y^2 + 504y + 1089$
$\{i, j\} = \{97, 208\}, k_{ij} = 2 \cdot 3^3 \cdot 5^2$	
α	$-54x + 270$
β	$12x^3 - 60x^2 - 4xy^2 + 8xy - 4x + 20y^2 - 40y + 26$
γ	$-6xy^3 - 9xy^2 - 45xy - 48x + 30y^3 + 45y^2 + 234y + 231$
$\{i, j\} = \{105, 123\}, k_{ij} = 2^4 \cdot 3^3$	
α	$24x^3 - 96x^2 + 144$
β	$-48x^5 + 144x^4 + 8x^3y^2 - 56x^3y + 56x^3 - 32x^2y^2 + 136x^2y - 128x^2$
	$-8xy^3 - 40xy^2 + 8xy - 96x + 24y^3 + 112y^2 + 48y + 144$
γ	$-8xy^2 + 8xy + 88x + 24y^2 - 32y + 48$
$\{i, j\} = \{105, 204\}, k_{ij} = 2^2 \cdot 3$	
α	0
β	0
γ	1
$\{i, j\} = \{110, 111\}, k_{ij} = 2^4 \cdot 3^2 \cdot 7$	
α	$-8x + 40$
β	$12xy^2 + 16xy + 24x - 4y^2 - 24y - 64$
γ	$16x^3 - 80x^2 - 48x - 12y^3 + 8y^2 + 8y + 296$
$\{i, j\} = \{111, 135\}, k_{ij} = 2^4 \cdot 3 \cdot 7 \cdot 19$	
α	$-132x + 156$
β	$264x^3 - 312x^2 - 792x - 72y^3 + 12y^2 + 228y + 2220$
γ	$72xy^2 + 204xy + 384x + 60y^2 - 96y - 744$

$$\begin{aligned}
&\{i, j\} = \{111, 154\}, k_{ij} = 2^4 \cdot 3^3 \cdot 13 \\
&\alpha \quad -360x - 648 \\
&\beta \quad 720x^3 - 180x^2y^2 - 756x^2y + 468x^2 - 504xy^2 - 432xy + 576x + 216y^2 + 2592y \\
&\quad + 2520 \\
&\gamma \quad 180xy^2 + 756xy + 828x + 324y^2 - 324y - 3564 \\
&\{i, j\} = \{111, 172\}, k_{ij} = 2^4 \cdot 3^2 \cdot 7^2 \\
&\alpha \quad -72x + 312 \\
&\beta \quad 144x^3 - 36x^2y^2 + 132x^2y - 948x^2 + 156xy^2 - 180xy - 204x + 36y^2 - 132y + 2268 \\
&\gamma \quad 36xy^2 - 132xy + 324x - 156y^2 + 180y - 228 \\
&\{i, j\} = \{111, 175\}, k_{ij} = 2^3 \cdot 3^2 \cdot 7 \\
&\alpha \quad -6x - 54 \\
&\beta \quad 12x^3 + 108x^2 - 36x + 18y^3 + 18y + 222 \\
&\gamma \quad -9xy^2 + 27xy - 90x + 3y^2 - 9y + 30 \\
&\{i, j\} = \{111, 176\}, k_{ij} = 2 \\
&\alpha \quad 0 \\
&\beta \quad 0 \\
&\gamma \quad 1 \\
&\{i, j\} = \{111, 177\}, k_{ij} = 2^6 \cdot 3^3 \cdot 7 \cdot 13 \cdot 19 \\
&\alpha \quad -1296x + 35856 \\
&\beta \quad 2592x^3 - 71712x^2 - 7776x - 2664y^3 - 34560y^2 - 13032y + 960048 \\
&\gamma \quad 2664xy^2 + 13248xy - 92952x + 9288y^2 - 34560y - 1080 \\
&\{i, j\} = \{111, 216\}, k_{ij} = 2^3 \cdot 3 \cdot 7 \\
&\alpha \quad 0 \\
&\beta \quad 42 \\
&\gamma \quad -3x + 3 \\
&\{i, j\} = \{112, 218\}, k_{ij} = 2 \\
&\alpha \quad 0 \\
&\beta \quad -1 \\
&\gamma \quad 0 \\
&\{i, j\} = \{118, 119\}, k_{ij} = 2^5 \cdot 3 \\
&\alpha \quad -6x^3y + 6x^3 + 48x^2y - 48x^2 + 30xy - 6x - 30y + 72 \\
&\beta \quad -36x^3 - 156x^2 + 6xy^3 - 18xy^2 + 258xy + 18x - 48y^3 - 120y^2 - 102y - 120 \\
&\gamma \quad -12x^5 + 108x^4 - 12x^3y - 60x^3 + 120x^2y + 138x^2 - 6xy^3 - 6xy^2 - 246xy + 222x \\
&\quad + 48y^3 + 102y^2 - 42y + 168 \\
&\{i, j\} = \{118, 121\}, k_{ij} = 2^7 \cdot 13 \\
&\alpha \quad 114x^3 - 200x^2 - 14x + 70y^2 - 104y + 716 \\
&\beta \quad -228x^5 + 479x^4 + 456x^3y + 260x^3 - 333x^2y^2 - 1133x^2y + 1298x^2 \\
&\quad - 412xy^2 - 468xy - 606x + 263y^3 + 967y^2 - 1265y - 1711 \\
&\gamma \quad -228x^3y + 377x^3 + 400x^2y - 1339x^2 + 342xy^2 - 422xy - 2127x - 333y^3 - 600y^2 \\
&\quad + 850y - 1227 \\
&\{i, j\} = \{118, 122\}, k_{ij} = 2^5 \cdot 13 \\
&\alpha \quad -7x^3 + 400x^2 - 255x + 100y^2 - 26y - 558 \\
&\beta \quad 14x^5 - 780x^4 - 28x^3y + 583x^3 - 227x^2y^2 + 2389x^2y - 2651x^2 + 148xy^2 \\
&\quad - 1020xy - 1910x + 127y^3 - 1588y^2 - 127y - 772 \\
&\gamma \quad 14x^3y - 48x^3 - 800x^2y + 1521x^2 - 21xy^2 - 171xy + 1915x - 227y^3 + 1968y^2 \\
&\quad - 2871y + 2862 \\
&\{i, j\} = \{118, 176\}, k_{ij} = 2 \\
&\alpha \quad 0 \\
&\beta \quad 0 \\
&\gamma \quad 1 \\
&\{i, j\} = \{119, 120\}, k_{ij} = 2^5 \cdot 3^2 \cdot 5 \\
&\alpha \quad x^4y - x^4 + 400x^3 - 2x^2y - 319x^2 + 482xy - 721x - 160y + 480 \\
&\beta \quad 2x^6 - 2x^5 + 2x^4y + 2x^4 - 876x^3y^2 - 1283x^3y - 396x^3 - x^2y^4 \\
&\quad - 883x^2y^2 + 2959x^2y - 2075x^2 + 875xy^3 - 79xy^2 + 1677xy + 1518x + y^5 \\
&\quad + 884y^3 - 3679y^2 - 405y + 1683 \\
&\gamma \quad -800x^3 + 2x^2y + 1438x^2 - 876xy^3 - 403xy^2 - 1680xy + 1199x - y^5 - 884y^3 \\
&\quad + 3839y^2 - 75y - 1523 \\
&\{i, j\} = \{120, 152\}, k_{ij} = 2^2 \cdot 3^2 \cdot 7 \cdot 13^2 \\
&\alpha \quad -x^3y^3 - 754x^3y^2 - 610x^3y - 1018x^3 + x^2y^4 - x^2y^3
\end{aligned}$$

$$\begin{aligned}
& +1331x^2y^2 + 843x^2y - 631x^2 + 754xy^3 + 1553xy^2 - 3717xy - 4732x - y^5 \\
& -571y^3 - 631y^2 - 4732y + 14196 \\
\beta \quad & 378x^3y^2 + 396x^3y + 12964x^3 - 288x^2y^2 - 316x^2y - 3819x^2 - 754xy^4 \\
& -794xy^3 + 3080xy^2 - 466xy - 2940x + y^6 + 572y^4 + 634y^3 + 5303y^2 - 11852y + 6625 \\
\gamma \quad & 2x^3y^3 + 1130x^3y^2 + 824x^3y - 10928x^3 - 2x^2y^4 + 2x^2y^3 \\
& -2752x^2y^2 - 1766x^2y - 7883x^2 + 756xy^4 + 1172xy^3 - 4660xy^2 + 22820xy \\
& + 3259x - y^6 + 186y^4 - 517y^3 - 7257y^2 + 9223y - 27240 \\
\{i, j\} = \{120, 161\}, k_{ij} = 2^4 \cdot 3^2 \cdot 5^2 \cdot 7^2 \\
\alpha \quad & -5x^3y^3 - 7480x^3y^2 + 32578x^3y - 21303x^3 + 7x^2y^4 - 30x^2y^3 \\
& + 4936x^2y^2 - 21346x^2y - 116691x^2 - 2xy^4 + 7481xy^3 - 7356xy^2 - 43204xy \\
& - 129168x - 7y^5 + 14y^4 + 2626y^3 + 26651y^2 - 19600y + 58800 \\
\beta \quad & x^5y - 2x^5 + 3x^4 - 2x^3y^3 + 3747x^3y^2 - 7517x^3y + 248389x^3 \\
& - 2x^2y^4 + 3x^2y^3 + 1299x^2y^2 + 13299x^2y + 56915x^2 + 2xy^5 \\
& - 7492xy^4 + 14904xy^3 + 28420xy^2 + 39758xy - 41861x + 7y^6 - 14y^5 - 2619y^4 \\
& - 26644y^3 + 16932y^2 - 93329y - 60353 \\
\gamma \quad & x^5y - 2x^5 + x^4y + x^4 - 12x^3y^3 - 11214x^3y^2 + 57643x^3y + 205782x^3 \\
& + 12x^2y^4 - 59x^2y^3 + 14919x^2y^2 - 36913x^2y + 71924x^2 + 2xy^5 \\
& - 7506xy^4 + 11148xy^3 + 80216xy^2 - 274222xy + 210890x + 7y^6 - 10y^5 - 10148y^4 \\
& - 9201y^3 + 97830y^2 - 110677y + 328704 \\
\{i, j\} = \{120, 162\}, k_{ij} = 2^2 \cdot 3^3 \\
\alpha \quad & -12x^3 + 12x^2 - 18xy + 18x - 12y + 36 \\
\beta \quad & 24x^3 - 48x^2 + 30xy^3 - 6xy^2 + 18xy + 30x + 48y^3 - 102y^2 + 24y - 78 \\
\gamma \quad & -30x^3y^2 - 24x^3y + 42x^3 - 48x^2y^2 + 114x^2y + 6x^2 + 30xy^3 - 24xy^2 \\
& + 72xy - 66x + 48y^3 - 114y^2 + 60y - 90 \\
\{i, j\} = \{120, 163\}, k_{ij} = 2^3 \cdot 3^4 \cdot 5^3 \cdot 7 \cdot 13^2 \\
\alpha \quad & 674000x^4y + 3053452x^4 + 3370000x^3y + 32286452x^3 + 20719860x^2y \\
& - 31929242x^2 - 674000xy^3 - 23860382xy^2 + 27489777xy - 35346847x - 9348930y^2 \\
& + 14052847y + 31941000 \\
\beta \quad & -1348000x^4y - 6106904x^4 - 10784000x^3y - 64021616x^3 - 41439720x^2y \\
& - 56392612x^2 + 1348000xy^3 + 47720764xy^2 - 54979554xy + 42681146x - 674000y^4 \\
& - 13163452y^3 + 51134764y^2 - 93571050y + 9853644 \\
\gamma \quad & -2696000x^5y - 2777808x^5 - 93800904x^4 + 10110000x^3y^2 - 6989148x^3y \\
& - 25786210x^3 + 75644528x^2y^2 - 44615152x^2y + 63628598x^2 + 31176956xy^2 \\
& - 15973216xy + 26918153x + 674000y^5 + 22512382y^4 - 65187611y^3 + 70978980y^2 \\
& + 4140299y - 74099541 \\
\{i, j\} = \{120, 210\}, k_{ij} = 3^4 \\
\alpha \quad & -9x + 27 \\
\beta \quad & 9xy^2 + 18xy + 9x - 27y^2 - 54y - 18 \\
\gamma \quad & -9x^3y^2 - 18x^3y + 9x^3 + 18x^2y^2 + 36x^2y - 45x^2 + 9xy^3 + 36xy^2 + 63xy \\
& + 9x - 27y^3 - 27y^2 - 18y + 27 \\
\{i, j\} = \{123, 128\}, k_{ij} = 2^4 \cdot 3^8 \cdot 5 \cdot 7 \\
\alpha \quad & -8x^8 + 4x^7y + 16921x^7 + 33895x^6y - 81289x^6 - 67802x^5y + 161649x^5 \\
& - 50793x^4y - 210523x^4 + 7083x^3y - 28592x^3 - 122973x^2y + 40952x^2 + 506766xy \\
& - 511470x - 408240y + 1224720 \\
\beta \quad & 16x^8 - 8x^7y - 33810x^7 - 67822x^6y + 94942x^6 + 8x^5y^2 + 33794x^5y \\
& - 99596x^5 + 8x^4y^3 + 67814x^4y^2 + 108246x^4y + 110009x^4 - 4x^3y^4 \\
& - 16905x^3y^3 + 67814x^3y^2 + 312246x^3y + 407353x^3 - 33911x^2y^4 \\
& + 47487x^2y^3 - 126989x^2y^2 + 626505x^2y + 856956x^2 + 4xy^5 + 16901xy^4 \\
& - 83523xy^3 - 324020xy^2 - 94400xy + 123570x + 33903y^5 + 3274y^4 + 135363y^3 \\
& - 1039410y^2 - 1107378y + 717930 \\
\gamma \quad & -2x^5y - 6x^5 + 4x^4y^2 + 17x^4 + 8494x^3y^2 + 26514x^3y + 151985x^3 \\
& + 12x^2y^4 + 39x^2y^3 + 51352x^2y^2 + 236190x^2y - 1627552x^2 - 4xy^5 \\
& + 17002xy^4 + 52894xy^3 - 50657xy^2 - 568903xy - 290064x - 33903y^5 - 3274y^4 \\
& - 135363y^3 + 1447650y^2 - 117342y - 309690 \\
\{i, j\} = \{123, 129\}, k_{ij} = 2^2 \cdot 3^3 \\
\alpha \quad & 9xy - 18x - 12y + 36 \\
\beta \quad & -9xy^2 + 18xy - 9x + 12y^2 - 36y + 12 \\
\gamma \quad & -9x^3y^2 + 36x^3y - 45x^3 + 21x^2y^2 - 60x^2y + 57x^2 - 9xy^3 + 6xy^2 + 21xy \\
& + 24x + 12y^3 - 36y^2 + 15y + 18 \\
\{i, j\} = \{123, 157\}, k_{ij} = 2^5 \cdot 3^3 \cdot 7 \\
\alpha \quad & 170x^3y^2 + 334x^3y - 579x^3 - 170x^2y^3 - 420x^2y^2 + 689x^2y + 243x^2
\end{aligned}$$

	$+340xy^3 + 160xy^2 - 1366xy - 672x - 170y^4 - 80y^3 + 837y^2 - 647y + 2016$
β	$-84x^3y^2 - 124x^3y + 378x^3 - 96x^2y^2 - 392x^2y + 4724x^2 - 234xy^2$
	$+708xy + 1305x + 170y^5 + 250y^4 - 757y^3 + 308y^2 - 1343y - 57$
γ	$256x^3y^2 + 544x^3y - 780x^3 - 340x^2y^3 - 512x^2y^2 + 1778x^2y$
	$+3674x^2 + 256xy^3 - 442xy^2 + 792xy - 5435x + 170y^5 - 90y^4 - 929y^3 + 1270y^2$
	$+1807y + 75$
$\{i, j\} = \{123, 173\}, k_{ij} = 2^8 \cdot 3^3 \cdot 43$	
α	$-x^3y^3 + 2057x^3y^2 + 1064x^3y + 18173x^3 - 2066x^2y^3 - 8239x^2y^2$
	$+18297x^2y - 61114x^2 - xy^4 + 6197xy^3 + 6210xy^2 - 5323xy - 6595x - 2070y^4$
	$-2063y^3 + 20523y^2 - 6595y + 99072$
β	$-x^4y + 2x^4 - 1036x^3y^2 - 2078x^3y + 9206x^3 - x^2y^4 - 2x^2y^3$
	$+8330x^2y^2 - 32782x^2y + 156314x^2 + xy^5 + 10xy^4 - 41xy^3 - 42635xy^2$
	$+13338xy + 55167x + 2070y^5 + 4131y^4 - 18473y^3 - 1394y^2 - 64884y - 44019$
γ	$-x^4y + 2x^4 - 2x^3y^3 + 3078x^3y^2 + 51x^3y + 45550x^3 - x^2y^4$
	$-4136x^2y^3 - 2999x^2y^2 + 8019x^2y + 61228x^2 + xy^5 + 9xy^4 + 7183xy^3$
	$-49907xy^2 + 83449xy - 154669x + 2068y^5 - 20y^4 - 16262y^3 + 48522y^2 + 10955y$
	$+79287$
$\{i, j\} = \{123, 181\}, k_{ij} = 2^7 \cdot 3^3 \cdot 13$	
α	$3117x^2y^2 - 2738x^2y - 2496x^2 - xy^3 - 6241xy^2 - 6001xy - 4992x + y^4$
	$+3120y^3 - 4703y^2 + 7491y + 14976$
β	$-4171x^3 + x^2y^3 + 2x^2y^2 + 4171x^2y + 2980x^2 + 3119xy^3 + 1343xy^2$
	$-13146xy - 528x - y^5 - 3119y^4 + 4706y^3 - 3222y^2 - 9400y + 49929$
γ	$-4171x^4 + x^3y^3 + 6236x^3y^2 + 2866x^3y + 2159x^3 - x^2y^4$
	$-3120x^2y^3 - 7773x^2y^2 - 36954x^2y - 18484x^2 + 3xy^4 + 9364xy^3$
	$+18040xy^2 + 20555xy + 65433x - y^5 - 4y^4 - 7392y^3 - 10233y^2 - 43311y - 37449$
$\{i, j\} = \{123, 221\}, k_{ij} = 2^2 \cdot 3^4$	
α	$45xy - 54x + 108$
β	$-15xy^2 - 102xy + 309x - 276y + 90$
γ	$-15x^3y^2 - 12x^3y + 201x^3 + 15x^2y^2 - 84x^2y - 111x^2 + 15xy^3 - 78xy^2$
	$+273xy + 18x - 84y^2 + 45y + 54$
$\{i, j\} = \{176, 191\}, k_{ij} = 2$	
α	0
β	1
γ	0
$\{i, j\} = \{176, 198\}, k_{ij} = 2$	
α	0
β	1
γ	0
$\{i, j\} = \{186, 187\}, k_{ij} = 3 \cdot 7$	
α	0
β	0
γ	1
$\{i, j\} = \{187, 188\}, k_{ij} = 3 \cdot 7$	
α	0
β	1
γ	0
$\{i, j\} = \{187, 189\}, k_{ij} = 3 \cdot 7$	
α	0
β	1
γ	0
$\{i, j\} = \{187, 196\}, k_{ij} = 3 \cdot 7$	
α	0
β	1
γ	0
$\{i, j\} = \{187, 200\}, k_{ij} = 3 \cdot 7$	
α	0
β	1
γ	0
$\{i, j\} = \{187, 212\}, k_{ij} = 3 \cdot 7$	
α	0

β	1
γ	0
$\{i, j\}$	$= \{187, 220\}, k_{ij} = 3 \cdot 7$
α	0
β	1
γ	0
$\{i, j\}$	$= \{190, 191\}, k_{ij} = 7$
α	0
β	0
γ	1
$\{i, j\}$	$= \{191, 192\}, k_{ij} = 7$
α	0
β	1
γ	0
$\{i, j\}$	$= \{191, 199\}, k_{ij} = 7$
α	0
β	1
γ	0
$\{i, j\}$	$= \{191, 223\}, k_{ij} = 7$
α	0
β	1
γ	0
$\{i, j\}$	$= \{191, 225\}, k_{ij} = 7$
α	0
β	1
γ	0
$\{i, j\}$	$= \{193, 194\}, k_{ij} = 2^2 \cdot 3 \cdot 5^2$
α	0
β	0
γ	1
$\{i, j\}$	$= \{194, 195\}, k_{ij} = 2^2 \cdot 3 \cdot 5^2$
α	0
β	1
γ	0
$\{i, j\}$	$= \{197, 198\}, k_{ij} = 2^2 \cdot 3 \cdot 7$
α	0
β	0
γ	1
$\{i, j\}$	$= \{201, 202\}, k_{ij} = 3$
α	0
β	0
γ	1
$\{i, j\}$	$= \{202, 205\}, k_{ij} = 3$
α	0
β	1
γ	0
$\{i, j\}$	$= \{202, 207\}, k_{ij} = 3$
α	0
β	1
γ	0
$\{i, j\}$	$= \{202, 209\}, k_{ij} = 3$
α	0
β	1
γ	0
$\{i, j\}$	$= \{202, 214\}, k_{ij} = 3$
α	0
β	1
γ	0
$\{i, j\}$	$= \{202, 215\}, k_{ij} = 3$

α	0
β	1
γ	0
$\{i, j\} = \{202, 219\}, k_{ij} = 3$	
α	0
β	1
γ	0
$\{i, j\} = \{202, 221\}, k_{ij} = 3$	
α	0
β	1
γ	0
$\{i, j\} = \{202, 222\}, k_{ij} = 3$	
α	0
β	1
γ	0
$\{i, j\} = \{203, 204\}, k_{ij} = 2^2 \cdot 3$	
α	0
β	0
γ	1
$\{i, j\} = \{204, 206\}, k_{ij} = 2^2 \cdot 3$	
α	0
β	1
γ	0
$\{i, j\} = \{217, 218\}, k_{ij} = 2^2 \cdot 7$	
α	0
β	0
γ	1

Table 3: The polynomials J_i , the constants c_i and, for the linear vertices, $\eta = r_i(\xi)$ of the first certificate. Here $\langle A, B \rangle = A + B\omega$.

i	J_i	c_i	chart
0	$\langle -y, 0 \rangle$	$\langle 2, 0 \rangle$	$y = \langle 0, 0 \rangle$
1	$\langle -y, 1 \rangle$	$\langle 1, -1 \rangle$	$y = \langle 0, 1 \rangle$
2	$\langle -y - 1, 0 \rangle$	$\langle 1, -1 \rangle$	$y = \langle -1, 0 \rangle$
3	$\langle x + 1, 1 \rangle$	$\langle 0, -3 \rangle$	$x = \langle -1, -1 \rangle$
4	$\langle x + 1, 0 \rangle$	$\langle 0, -1 \rangle$	$x = \langle -1, 0 \rangle$
5	$\langle x, 1 \rangle$	$\langle 11, -2 \rangle$	$x = \langle 0, -1 \rangle$
6	$\langle x, -1 \rangle$	$\langle 1, 0 \rangle$	$x = \langle 0, 1 \rangle$
7	$\langle x - 1, -1 \rangle$	$\langle 1, 0 \rangle$	$x = \langle 1, 1 \rangle$
8	$\langle 2x + 1, 0 \rangle$	$\langle 1, 0 \rangle$	–
9	$\langle 3x + 2, 1 \rangle$	$\langle 0, -3 \rangle$	$3x = \langle -2, -1 \rangle$
10	$\langle 3x - 1, 1 \rangle$	$\langle -6, -53 \rangle$	$3x = \langle 1, -1 \rangle$
11	$\langle 3x - 2, -1 \rangle$	$\langle 25, 20 \rangle$	$3x = \langle 2, 1 \rangle$
12	$\langle -x - y, 0 \rangle$	$\langle -2, -1 \rangle$	$y = \langle -x, 0 \rangle$
13	$\langle -2x - y, -x + 1 \rangle$	$\langle -1, -1 \rangle$	$y = \langle -2x, -x + 1 \rangle$
14	$\langle x - y, -y \rangle$	$\langle -24, -15 \rangle$	$y = \langle 0, -x \rangle$
15	$\langle x, y \rangle$	$\langle -2, 1 \rangle$	$y = \langle x, x \rangle$
16	$\langle x + 1, -y + 1 \rangle$	$\langle -6, -4 \rangle$	$y = \langle -x, -x - 1 \rangle$
17	$\langle x + 1, -y \rangle$	$\langle 0, -3 \rangle$	$y = \langle -x - 1, -x - 1 \rangle$
18	$\langle x + y + 1, 0 \rangle$	$\langle 1, -1 \rangle$	$y = \langle -x - 1, 0 \rangle$
19	$\langle x + y, y + 1 \rangle$	$\langle 4, 20 \rangle$	$y = \langle -1, x - 1 \rangle$
20	$\langle x - y, -1 \rangle$	$\langle -7, 14 \rangle$	$y = \langle x, -1 \rangle$
21	$\langle x + y, y - 1 \rangle$	$\langle -2, 4 \rangle$	$y = \langle 1, x + 1 \rangle$
22	$\langle x - y - 1, -y \rangle$	$\langle 4, 0 \rangle$	$y = \langle 0, -x + 1 \rangle$
23	$\langle x - y - 1, 0 \rangle$	$\langle -12, 0 \rangle$	$y = \langle x - 1, 0 \rangle$
24	$\langle x - 1, -y \rangle$	$\langle -2, 11 \rangle$	$y = \langle -x + 1, -x + 1 \rangle$

i	J_i	c_i	chart
25	$\langle x + y - 1, 0 \rangle$	$\langle 2, -2 \rangle$	$y = \langle -x + 1, 0 \rangle$
26	$\langle x + y - 1, y \rangle$	$\langle -2, 1 \rangle$	$y = \langle 0, x - 1 \rangle$
27	$\langle x - y - 1, -y - 1 \rangle$	$\langle 13, 11 \rangle$	$y = \langle -1, -x \rangle$
28	$\langle x - 1, y - 1 \rangle$	$\langle 5, 8 \rangle$	$y = \langle x, x - 1 \rangle$
29	$\langle x + 2, y + 1 \rangle$	$\langle 0, -1 \rangle$	$y = \langle x + 1, x + 2 \rangle$
30	$\langle x - y + 1, -y - 1 \rangle$	$\langle -1, 2 \rangle$	$y = \langle -1, -x - 2 \rangle$
31	$\langle 2x - y, 0 \rangle$	$\langle 28, -7 \rangle$	$y = \langle 2x, 0 \rangle$
32	$\langle 2x + y + 1, 0 \rangle$	$\langle -6, -4 \rangle$	$y = \langle -2x - 1, 0 \rangle$
33	$\langle 2x + y + 1, y \rangle$	$\langle -4, -2 \rangle$	$y = \langle 0, 2x + 1 \rangle$
34	$\langle 3x - y, -2y \rangle$	$\langle 1, 3 \rangle$	$\langle 1, -1 \rangle y = \langle -3x, -3x \rangle$
35	$\langle 3x + y, 2y \rangle$	$\langle -3, -8 \rangle$	$\langle 1, -1 \rangle y = \langle 3x, 3x \rangle$
36	$\langle 3x - 2y - 1, -y - 2 \rangle$	$\langle 3, -6 \rangle$	$\langle 1, -1 \rangle y = \langle -2, -3x - 1 \rangle$
37	$\langle -x^2 - y, 0 \rangle$	$\langle -21, -14 \rangle$	$y = \langle -x^2, 0 \rangle$
38	$\langle x^2 - y, x^2 + 1 \rangle$	$\langle -1, -2 \rangle$	$y = \langle x^2, x^2 + 1 \rangle$
39	$\langle x^2 + x - y, x \rangle$	$\langle 0, -3 \rangle$	$y = \langle x^2 + x, x \rangle$
40	$\langle -y, x^2 + x \rangle$	$\langle 1, -1 \rangle$	$y = \langle 0, x^2 + x \rangle$
41	$\langle x^2 - x - y, 0 \rangle$	$\langle -3, 0 \rangle$	$y = \langle x^2 - x, 0 \rangle$
42	$\langle -x - y, -x^2 \rangle$	$\langle -5, -1 \rangle$	$y = \langle -x, -x^2 \rangle$
43	$\langle x^2 - y - 1, 0 \rangle$	$\langle -1, -1 \rangle$	$y = \langle x^2 - 1, 0 \rangle$
44	$\langle x^2 + x - y + 1, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x^2 + x + 1, 0 \rangle$
45	$\langle x^2 + x - y, x^2 - 1 \rangle$	$\langle 1, -1 \rangle$	$y = \langle x^2 + x, x^2 - 1 \rangle$
46	$\langle -x - y - 1, x^2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x - 1, x^2 \rangle$
47	$\langle -x^2 - x - y - 1, -x^2 \rangle$	$\langle -3, -3 \rangle$	$y = \langle -x^2 - x - 1, -x^2 \rangle$
48	$\langle -x^2 - y - 1, -x^2 + 1 \rangle$	$\langle -5, -1 \rangle$	$y = \langle -x^2 - 1, -x^2 + 1 \rangle$
49	$\langle -y - 2, x^2 - 1 \rangle$	$\langle -3, -8 \rangle$	$y = \langle -2, x^2 - 1 \rangle$
50	$\langle x^2 - y - 1, x^2 + x + 1 \rangle$	$\langle 2, -2 \rangle$	$y = \langle x^2 - 1, x^2 + x + 1 \rangle$
51	$\langle x - 2, 0 \rangle$	$\langle 0, -21 \rangle$	$x = \langle 2, 0 \rangle$
52	$\langle x + 1, y + 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x, x + 1 \rangle$
53	$\langle x - 1, y \rangle$	$\langle 1, -1 \rangle$	$y = \langle x - 1, x - 1 \rangle$
54	$\langle x + 1, -y - 1 \rangle$	$\langle -6, -16 \rangle$	$y = \langle -x - 2, -x - 1 \rangle$
55	$\langle 3x - y + 1, y + 2 \rangle$	$\langle -6, -6 \rangle$	$\langle 1, -1 \rangle y = \langle 3x + 1, 2 \rangle$
56	$\langle 3x + y - 1, 2y - 2 \rangle$	$\langle 2, 3 \rangle$	$\langle 1, -1 \rangle y = \langle 3x + 1, 3x - 1 \rangle$
57	$\langle x^2 - y, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x^2, 0 \rangle$
58	$\langle -y - 1, -x^2 - 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -1, -x^2 - 1 \rangle$
59	$\langle -x - y - 1, x - 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x - 1, x - 1 \rangle$
60	$\langle x - y + 1, 1 \rangle$	$\langle 7, 0 \rangle$	$y = \langle x + 1, 1 \rangle$
61	$\langle x^2 + x - y, 0 \rangle$	$\langle -1, -2 \rangle$	$y = \langle x^2 + x, 0 \rangle$
62	$\langle x, y + 1 \rangle$	$\langle -8, -5 \rangle$	$y = \langle x - 1, x \rangle$
63	$\langle x - y, 0 \rangle$	$\langle 1, -1 \rangle$	$y = \langle x, 0 \rangle$
64	$\langle -y + 1, 0 \rangle$	$\langle -1, -1 \rangle$	$y = \langle 1, 0 \rangle$
65	$\langle 2x + 1, -y \rangle$	$\langle -1, 2 \rangle$	$y = \langle -2x - 1, -2x - 1 \rangle$
66	$\langle -x - y + 1, -x - 1 \rangle$	$\langle 4, 0 \rangle$	$y = \langle -x + 1, -x - 1 \rangle$
67	$\langle x + 1, 2 \rangle$	$\langle 1, -1 \rangle$	$x = \langle -1, -2 \rangle$
68	$\langle -y + 2, 2 \rangle$	$\langle 0, -1 \rangle$	$y = \langle 2, 2 \rangle$
69	$\langle -y - 2, -1 \rangle$	$\langle -3, -3 \rangle$	$y = \langle -2, -1 \rangle$
70	$\langle 2x + 2, 1 \rangle$	$\langle -750, 25 \rangle$	—
71	$\langle -x^2 - y - 1, 0 \rangle$	$\langle 0, -3 \rangle$	$y = \langle -x^2 - 1, 0 \rangle$

Table 4: The polynomials J_i , the constants c_i and, for the linear vertices, $\eta = r_i(\xi)$ of the second certificate. Here $\langle A, B \rangle = A + B\omega$.

i	J_i	c_i	chart
0	$\langle -y, 0 \rangle$	$\langle -1, -1 \rangle$	$y = \langle 0, 0 \rangle$
1	$\langle -y + 1, 1 \rangle$	$\langle -20, -4 \rangle$	$y = \langle 1, 1 \rangle$
2	$\langle -y - 1, 0 \rangle$	$\langle 0, -1 \rangle$	$y = \langle -1, 0 \rangle$
3	$\langle x + 1, 1 \rangle$	$\langle -1, 2 \rangle$	$x = \langle -1, -1 \rangle$

i	J_i	c_i	chart
4	$\langle x+1, 0 \rangle$	$\langle 1, 0 \rangle$	$x = \langle -1, 0 \rangle$
5	$\langle x, 1 \rangle$	$\langle -6, 3 \rangle$	$x = \langle 0, -1 \rangle$
6	$\langle x, -1 \rangle$	$\langle 1, 0 \rangle$	$x = \langle 0, 1 \rangle$
7	$\langle x-1, 0 \rangle$	$\langle 0, -3 \rangle$	$x = \langle 1, 0 \rangle$
8	$\langle x-1, -1 \rangle$	$\langle 1, 0 \rangle$	$x = \langle 1, 1 \rangle$
9	$\langle x+2, 1 \rangle$	$\langle -4, -3 \rangle$	$x = \langle -2, -1 \rangle$
10	$\langle 2x+1, 0 \rangle$	$\langle 2, 0 \rangle$	—
11	$\langle 3x-1, 1 \rangle$	$\langle -3431, -2089 \rangle$	$3x = \langle 1, -1 \rangle$
12	$\langle 2x-y, x+1 \rangle$	$\langle 0, -3 \rangle$	$y = \langle 2x, x+1 \rangle$
13	$\langle -2x-y, -x+1 \rangle$	$\langle 1, -1 \rangle$	$y = \langle -2x, -x+1 \rangle$
14	$\langle x-y, -y \rangle$	$\langle -11, -13 \rangle$	$y = \langle 0, -x \rangle$
15	$\langle x-y, 0 \rangle$	$\langle 4, 0 \rangle$	$y = \langle x, 0 \rangle$
16	$\langle x, y \rangle$	$\langle 1, 0 \rangle$	$y = \langle x, x \rangle$
17	$\langle x-y+1, 1 \rangle$	$\langle 5, 8 \rangle$	$y = \langle x+1, 1 \rangle$
18	$\langle x+y+1, 0 \rangle$	$\langle -2, -2 \rangle$	$y = \langle -x-1, 0 \rangle$
19	$\langle x-y, -1 \rangle$	$\langle 1, 5 \rangle$	$y = \langle x, -1 \rangle$
20	$\langle x+y, y-1 \rangle$	$\langle 4, 0 \rangle$	$y = \langle 1, x+1 \rangle$
21	$\langle x-y-1, 0 \rangle$	$\langle 0, -4 \rangle$	$y = \langle x-1, 0 \rangle$
22	$\langle x-1, -y \rangle$	$\langle 12, 18 \rangle$	$y = \langle -x+1, -x+1 \rangle$
23	$\langle x+y-1, 0 \rangle$	$\langle 8, -2 \rangle$	$y = \langle -x+1, 0 \rangle$
24	$\langle x+y-1, y \rangle$	$\langle 8, 3 \rangle$	$y = \langle 0, x-1 \rangle$
25	$\langle x-y-1, -y-1 \rangle$	$\langle 0, -4 \rangle$	$y = \langle -1, -x \rangle$
26	$\langle x-1, y-1 \rangle$	$\langle 0, -1 \rangle$	$y = \langle x, x-1 \rangle$
27	$\langle x+2, -y+1 \rangle$	$\langle 4, 0 \rangle$	$y = \langle -x-1, -x-2 \rangle$
28	$\langle x+2, y+1 \rangle$	$\langle -3, -2 \rangle$	$y = \langle x+1, x+2 \rangle$
29	$\langle x+y+2, y+1 \rangle$	$\langle -2, -1 \rangle$	$y = \langle -1, x+1 \rangle$
30	$\langle x-y+1, -y-1 \rangle$	$\langle -4, 8 \rangle$	$y = \langle -1, -x-2 \rangle$
31	$\langle x+y+1, y-1 \rangle$	$\langle 2, 0 \rangle$	$y = \langle 1, x+2 \rangle$
32	$\langle 2x-y, 0 \rangle$	$\langle -4, 8 \rangle$	$y = \langle 2x, 0 \rangle$
33	$\langle 2x+y+1, 0 \rangle$	$\langle -14, 7 \rangle$	$y = \langle -2x-1, 0 \rangle$
34	$\langle 2x+y+1, y \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, 2x+1 \rangle$
35	$\langle 3x-y, -2y \rangle$	$\langle 13, 11 \rangle$	$\langle 1, -1 \rangle y = \langle -3x, -3x \rangle$
36	$\langle 3x+y, 2y \rangle$	$\langle -96, -60 \rangle$	$\langle 1, -1 \rangle y = \langle 3x, 3x \rangle$
37	$\langle -x^2-y, 0 \rangle$	$\langle 63, 21 \rangle$	$y = \langle -x^2, 0 \rangle$
38	$\langle x^2-y, x^2+1 \rangle$	$\langle 1, -1 \rangle$	$y = \langle x^2, x^2+1 \rangle$
39	$\langle x^2-x-y, 0 \rangle$	$\langle 2, -2 \rangle$	$y = \langle x^2-x, 0 \rangle$
40	$\langle -x-y, -x^2 \rangle$	$\langle -3, -3 \rangle$	$y = \langle -x, -x^2 \rangle$
41	$\langle x^2+x-y+1, 0 \rangle$	$\langle -1, -1 \rangle$	$y = \langle x^2+x+1, 0 \rangle$
42	$\langle -y-1, x^2+x \rangle$	$\langle 0, -3 \rangle$	$y = \langle -1, x^2+x \rangle$
43	$\langle -x-y-1, x^2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x-1, x^2 \rangle$
44	$\langle -x^2-x-y-1, -x^2 \rangle$	$\langle 0, -3 \rangle$	$y = \langle -x^2-x-1, -x^2 \rangle$
45	$\langle -x^2-y-1, -x^2+1 \rangle$	$\langle -3, 0 \rangle$	$y = \langle -x^2-1, -x^2+1 \rangle$
46	$\langle -y-2, x^2-1 \rangle$	$\langle -11, -13 \rangle$	$y = \langle -2, x^2-1 \rangle$
47	$\langle x^2-y-1, x^2+x+1 \rangle$	$\langle 2, 0 \rangle$	$y = \langle x^2-1, x^2+x+1 \rangle$
48	$\langle -y, -1 \rangle$	$\langle 0, -4 \rangle$	$y = \langle 0, -1 \rangle$
49	$\langle x+y, 0 \rangle$	$\langle -1, -1 \rangle$	$y = \langle -x, 0 \rangle$
50	$\langle x+1, -y-1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x-2, -x-1 \rangle$
51	$\langle -y-1, -x^2-1 \rangle$	$\langle 0, -3 \rangle$	$y = \langle -1, -x^2-1 \rangle$
52	$\langle -2x^2-y, 0 \rangle$	$\langle 2, -2 \rangle$	$y = \langle -2x^2, 0 \rangle$
53	$\langle x-y-1, x \rangle$	$\langle 10, 8 \rangle$	$y = \langle x-1, x \rangle$
54	$\langle -x-y-1, x-1 \rangle$	$\langle 1, -1 \rangle$	$y = \langle -x-1, x-1 \rangle$
55	$\langle x-y+2, -y+1 \rangle$	$\langle 0, -4 \rangle$	$y = \langle 1, -x-1 \rangle$
56	$\langle x^2-y+1, x^2+1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x^2+1, x^2+1 \rangle$
57	$\langle x^2+x-y, 0 \rangle$	$\langle 2, 0 \rangle$	$y = \langle x^2+x, 0 \rangle$
58	$\langle x-y-1, -x-1 \rangle$	$\langle 0, -4 \rangle$	$y = \langle x-1, -x-1 \rangle$
59	$\langle x^2-y, x+1 \rangle$	$\langle 0, -1 \rangle$	$y = \langle x^2, x+1 \rangle$
60	$\langle x+2, 0 \rangle$	$\langle 7, 0 \rangle$	$x = \langle -2, 0 \rangle$

i	J_i	c_i	chart
61	$\langle -y, x^2 - x \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, x^2 - x \rangle$
62	$\langle x + 1, 2 \rangle$	$\langle 28, -28 \rangle$	$x = \langle -1, -2 \rangle$
63	$\langle -y - 2, -1 \rangle$	$\langle 26, 22 \rangle$	$y = \langle -2, -1 \rangle$
64	$\langle 2x + 2, 1 \rangle$	$\langle 4, 10 \rangle$	–
65	$\langle -y + 2, 1 \rangle$	$\langle 2, 0 \rangle$	$y = \langle 2, 1 \rangle$
66	$\langle x + y - 1, y + 1 \rangle$	$\langle -2, 4 \rangle$	$y = \langle -1, x - 2 \rangle$
67	$\langle -x - y, -x - 1 \rangle$	$\langle 0, -1 \rangle$	$y = \langle -x, -x - 1 \rangle$
68	$\langle x^2 - y - 1, 0 \rangle$	$\langle 0, -1 \rangle$	$y = \langle x^2 - 1, 0 \rangle$

Table 5: The data of the identities (147) and (148); the entry $j:e$ means $\rho_{ij} = e \neq 0$.

The first certificate.

$i = 0$, $D_i = 4$, ρ_{ij} : 20:1, 22:2, 29:2, 30:2, 43:2, 48:1, 49:1, 60:1, 71:1

w_i $\langle 4t^3 + 22t^2 + 10t,$
 $4t^3 + 22t^2 + 10t \rangle$

$i = 1$, $D_i = 124$, ρ_{ij} : 11:2, 24:2, 25:2, 30:2, 35:2, 37:1, 45:1, 48:1, 56:2

w_i $\langle -1386t^3 + 4488t^2 - 9732t + 6246,$
 $612t^3 + 516t^2 - 3336t + 5112 \rangle$

$i = 2$, $D_i = 8$, ρ_{ij} : 3:2, 6:1, 9:2, 11:1, 12:2, 17:2, 28:2, 33:1, 36:2, 41:2, 43:1, 48:1, 49:2, 50:2, 65:1, 70:2

w_i $\langle -204t^3 - 162t^2 - 9t + 51,$
 $156t^3 - 60t^2 - 219t - 129 \rangle$

$i = 3$, $D_i = 8$, ρ_{ij} : 2:1, 13:1, 33:1, 35:2, 37:2, 41:2, 43:2, 45:1, 53:1, 55:2, 58:2

w_i $\langle 96t^2 - 120t + 72,$
 $168t^2 + 168t + 432 \rangle$

$i = 4$, $D_i = 4$, ρ_{ij} : 13:1, 18:2, 38:2, 41:1, 46:2, 54:1

w_i $\langle 8t^2 - 8t - 16,$
 $-4t^2 - 12t - 16 \rangle$

$i = 5$, $D_i = 8$, ρ_{ij} : 17:1, 25:2, 26:1, 30:1, 35:1, 36:1, 37:2, 38:2, 40:2, 46:2, 48:2, 53:1, 58:1, 68:1

w_i $\langle 168t^2 - 1160t - 280,$
 $-504t^2 - 904t + 280 \rangle$

$i = 6$, $D_i = 2$, ρ_{ij} : 2:2, 17:2, 22:1, 24:2, 40:2, 47:1, 49:1

w_i $\langle 2t + 8,$
 $-2t^2 - 2t + 4 \rangle$

$i = 7$, $D_i = 16$, ρ_{ij} : 16:1, 17:2, 19:2, 21:2, 26:2, 29:2, 33:1, 37:2, 39:1, 46:2, 47:2, 49:2, 53:2, 63:1, 69:2

w_i $\langle 6224t^2 + 5584t + 448,$
 $2000t^2 + 8320t + 9296 \rangle$

$i = 8$, $D_i = 32$, ρ_{ij} : 12:2, 13:2, 20:1, 22:1, 24:1, 25:1, 26:1, 28:2, 31:2, 33:2, 53:1, 60:1, 63:2, 71:2

T $\langle -30y^2 + 151y + 231,$
 $-30y^2 + 151y + 231 \rangle$

U $\langle 786432x^{19} - 2097152x^{18}y - 1835008x^{18} - 1572864x^{17}y^2 + 12582912x^{17}y - 1376256x^{17}$
 $+ 5701632x^{16}y^3 - 7536640x^{16}y^2 - 20971520x^{16}y + 7077888x^{16} + 1458176x^{15}y^4$
 $- 33357824x^{15}y^3 + 55689216x^{15}y^2 + 5111808x^{15}y - 12582912x^{15} - 5947392x^{14}y^5$
 $+ 24117248x^{14}y^4 + 35176448x^{14}y^3 - 90472448x^{14}y^2 + 12845056x^{14}y + 23298048x^{14}$
 $- 2310144x^{13}y^6 + 35127296x^{13}y^5 - 138510336x^{13}y^4 + 79396864x^{13}y^3$
 $+ 50757632x^{13}y^2 - 24936448x^{13}y - 24739840x^{13} + 5390336x^{12}y^7 - 23265280x^{12}y^6$
 $- 15728640x^{12}y^5 + 203792384x^{12}y^4 - 159776768x^{12}y^3 - 28286976x^{12}y^2$
 $+ 43843584x^{12}y + 14286848x^{12} + 2654208x^{11}y^8 - 21831680x^{11}y^7 + 124157952x^{11}y^6$
 $- 206544896x^{11}y^5 - 4939776x^{11}y^4 + 103104512x^{11}y^3 - 7610368x^{11}y^2$
 $- 9502720x^{11}y - 9420800x^{11} - 6782976x^{10}y^9 + 25092096x^{10}y^8 - 3833856x^{10}y^7$
 $- 194437120x^{10}y^6 + 372695040x^{10}y^5 - 136077312x^{10}y^4 - 103063552x^{10}y^3$
 $+ 76840960x^{10}y^2 - 38109184x^{10}y - 11763712x^{10} - 1736704x^9y^{10} + 8232960x^9y^9$
 $- 40927232x^9y^8 + 144293888x^9y^7 - 90136576x^9y^6 - 118095872x^9y^5$
 $+ 133672960x^9y^4 - 37109760x^9y^3 + 5754880x^9y^2 + 38273024x^9y + 29245440x^9$
 $+ 6094848x^8y^{11} - 35823616x^8y^{10} + 40927232x^8y^9 + 96444416x^8y^8$
 $- 383983616x^8y^7 + 389709824x^8y^6 - 10252288x^8y^5 - 277397504x^8y^4$
 $+ 242528256x^8y^3 - 40353792x^8y^2 - 69468160x^8y - 18685952x^8 + 1572864x^7y^{12}$
 $- 1359872x^7y^{11} - 12644352x^7y^{10} + 10731520x^7y^9 + 27865088x^7y^8$
 $+ 37019648x^7y^7 - 127016960x^7y^6 + 112508928x^7y^5 - 2662400x^7y^4$
 $- 105816064x^7y^3 - 8343552x^7y^2 + 39591936x^7y + 23158784x^7 - 3309568x^6y^{13}$
 $+ 30097408x^6y^{12} - 71528448x^6y^{11} + 4374528x^6y^{10} + 241594368x^6y^9$
 $- 473202688x^6y^8 + 280264704x^6y^7 + 249782272x^6y^6 - 535666688x^6y^5$
 $+ 313247744x^6y^4 + 124506112x^6y^3 - 92301312x^6y^2 + 46399488x^6y - 581632x^6$
 $- 1081344x^5y^{14} + 2408448x^5y^{13} + 20180992x^5y^{12} - 68681728x^5y^{11}$
 $+ 84729856x^5y^{10} - 33439744x^5y^9 - 2371584x^5y^8 - 25475072x^5y^7$
 $- 19124224x^5y^6 + 117995520x^5y^5 - 83822592x^5y^4 - 86147072x^5y^3$
 $- 14014464x^5y^2 - 38936576x^5y - 34328576x^5 + 1081344x^4y^{15} - 14712832x^4y^{14}$
 $+ 56004608x^4y^{13} - 60571648x^4y^{12} - 93298688x^4y^{11} + 367818752x^4y^{10}$

$$\begin{aligned}
& -453439488x^4y^9 + 52412416x^4y^8 + 565223424x^4y^7 - 710201344x^4y^6 \\
& + 154077184x^4y^5 + 316758016x^4y^4 - 287426560x^4y^3 + 107776000x^4y^2 \\
& + 93356032x^4y + 17498112x^4 - 278528x^3y^{16} - 2621440x^3y^{15} - 6725632x^3y^{14} \\
& + 52541440x^3y^{13} - 100524032x^3y^{12} + 90585088x^3y^{11} + 22460416x^3y^{10} \\
& - 130998272x^3y^9 + 156952576x^3y^8 - 147566592x^3y^7 + 70627328x^3y^6 \\
& + 110798848x^3y^5 - 114799616x^3y^4 + 94633984x^3y^3 + 38788096x^3y^2 \\
& - 94298112x^3y - 34578432x^3 - 147456x^2y^{17} + 4636672x^2y^{16} - 27795456x^2y^{15} \\
& + 56047616x^2y^{14} - 8216576x^2y^{13} - 195385344x^2y^{12} + 427919360x^2y^{11} \\
& - 329877504x^2y^{10} - 258750464x^2y^9 + 872785920x^2y^8 - 710391808x^2y^7 \\
& - 145014784x^2y^6 + 636879872x^2y^5 - 520671232x^2y^4 - 96900096x^2y^3 \\
& + 119506944x^2y^2 - 49840128x^2y + 17971200x^2 - 49152xy^{18} + 1032192xy^{17} - 745472xy^{16} \\
& - 21329920xy^{15} + 67756032xy^{14} - 101437440xy^{13} + 46094336xy^{12} + 137558016xy^{11} \\
& - 295718912xy^{10} + 297752576xy^9 - 72988672xy^8 - 202264576xy^7 + 365700096xy^6 \\
& - 211044352xy^5 + 29814784xy^4 + 297596928xy^3 - 13037568xy^2 + 28956672xy + 42743808x \\
& + 8192y^{18} + 618496y^{17} - 1396736y^{16} - 4001792y^{15} + 19931136y^{14} - 45551616y^{13} \\
& + 62887936y^{12} - 22740992y^{11} - 83862528y^{10} + 219298816y^9 - 232091392y^8 + 19939712y^7 \\
& + 266604928y^6 - 247556896y^5 + 102826384y^4 + 193098680y^3 - 35891272y^2 + 15468360y \\
& + 15404760, \\
& 786432x^{19} - 262144x^{18}y - 4456448x^{18} - 3473408x^{17}y^2 + 9306112x^{17}y + 8323072x^{17} \\
& + 917504x^{16}y^3 + 13697024x^{16}y^2 - 34603008x^{16}y - 8519680x^{16} + 6029312x^{15}y^4 \\
& - 29458432x^{15}y^3 + 5947392x^{15}y^2 + 48496640x^{15}y + 8847360x^{15} - 1114112x^{14}y^5 \\
& - 19333120x^{14}y^4 + 104316928x^{14}y^3 - 71073792x^{14}y^2 - 39518208x^{14}y - 1736704x^{14} \\
& - 5963776x^{13}y^6 + 36831232x^{13}y^5 - 49053696x^{13}y^4 - 97976320x^{13}y^3 \\
& + 80986112x^{13}y^2 + 27557888x^{13}y - 12288000x^{13} + 720896x^{12}y^7 + 16744448x^{12}y^6 \\
& - 129073152x^{12}y^5 + 244547584x^{12}y^4 - 24313856x^{12}y^3 - 48504832x^{12}y^2 \\
& + 16973824x^{12}y + 14450688x^{12} + 5767168x^{11}y^8 - 31555584x^{11}y^7 + 54968320x^{11}y^6 \\
& + 72491008x^{11}y^5 - 256409600x^{11}y^4 + 49053696x^{11}y^3 + 31301632x^{11}y^2 \\
& - 53149696x^{11}y - 20267008x^{11} - 851968x^{10}y^9 - 1835008x^{10}y^8 + 73531392x^{10}y^7 \\
& - 263839744x^{10}y^6 + 239665152x^{10}y^5 + 48242688x^{10}y^4 - 93134848x^{10}y^3 \\
& + 78225408x^{10}y^2 + 26705920x^{10}y + 20152320x^{10} - 6553600x^9y^{10} + 32735232x^9y^9 \\
& - 42352640x^9y^8 - 60375040x^9y^7 + 311345152x^9y^6 - 281501696x^9y^5 \\
& + 15925248x^9y^4 + 149381120x^9y^3 - 105828352x^9y^2 - 20611072x^9y + 4849664x^9 \\
& + 1048576x^8y^{11} - 14286848x^8y^{10} + 15237120x^8y^9 + 97173504x^8y^8 \\
& - 271106048x^8y^7 + 125755392x^8y^6 + 106119168x^8y^5 - 264798208x^8y^4 \\
& + 64950272x^8y^3 + 1597440x^8y^2 - 15810560x^8y - 9838592x^8 + 5242880x^7y^{12} \\
& - 32145408x^7y^{11} + 50036736x^7y^{10} + 51912704x^7y^9 - 272519168x^7y^8 \\
& + 411410432x^7y^7 - 162398208x^7y^6 - 166182912x^7y^5 + 411643904x^7y^4 \\
& - 89645056x^7y^3 - 22806528x^7y^2 + 82403328x^7y + 19996672x^7 - 458752x^6y^{13} \\
& + 13533184x^6y^{12} - 58671104x^6y^{11} + 52854784x^6y^{10} + 100446208x^6y^9 \\
& - 231956480x^6y^8 + 77627392x^6y^7 + 202833920x^6y^6 - 293924864x^6y^5 \\
& + 11624448x^6y^4 + 26619904x^6y^3 - 83314688x^6y^2 - 41402368x^6y - 35340288x^6 \\
& - 2293760x^5y^{14} + 21463040x^5y^{13} - 49987584x^5y^{12} - 4972544x^5y^{11} \\
& + 197779456x^5y^{10} - 386088960x^5y^9 + 318476288x^5y^8 + 73490432x^5y^7 \\
& - 555683840x^5y^6 + 472176640x^5y^5 - 21032960x^5y^4 - 216674304x^5y^3 \\
& + 167839744x^5y^2 + 34365440x^5y - 258048x^5 - 65536x^4y^{15} - 4751360x^4y^{14} \\
& + 41975808x^4y^{13} - 93470720x^4y^{12} + 35774464x^4y^{11} + 145629184x^4y^{10} \\
& - 244604928x^4y^9 + 99207168x^4y^8 + 259432448x^4y^7 - 213858304x^4y^6 \\
& + 68440064x^4y^5 + 172523520x^4y^4 - 41093120x^4y^3 + 24104960x^4y^2 - 12042240x^4y \\
& + 5087232x^4 + 524288x^3y^{16} - 9207808x^3y^{15} + 33931264x^3y^{14} - 26165248x^3y^{13} \\
& - 89018368x^3y^{12} + 289294336x^3y^{11} - 385736704x^3y^{10} + 97886208x^3y^9 \\
& + 460077056x^3y^8 - 821839872x^3y^7 + 317247488x^3y^6 + 203620352x^3y^5 \\
& - 619427840x^3y^4 + 126494720x^3y^3 + 28898304x^3y^2 - 113098752x^3y - 12718080x^3 \\
& + 65536x^2y^{17} + 589824x^2y^{16} - 15876096x^2y^{15} + 62889984x^2y^{14} \\
& - 80216064x^2y^{13} - 19552256x^2y^{12} + 212615168x^2y^{11} - 308037632x^2y^{10} \\
& + 70141952x^2y^9 + 293699584x^2y^8 - 280240128x^2y^7 + 106309632x^2y^6 \\
& + 230256640x^2y^5 - 5427200x^2y^4 + 126817280x^2y^3 + 61114368x^2y^2 + 62853120x^2y \\
& + 53139456x^2 - 65536xy^{18} + 2392064xy^{17} - 16121856xy^{16} + 28696576xy^{15} + 13824000xy^{14} \\
& - 155291648xy^{13} + 337055744xy^{12} - 266459136xy^{11} - 228978688xy^{10} + 841895936xy^9 \\
& - 814143488xy^8 + 53059584xy^7 + 814833664xy^6 - 817756160xy^5 + 41816064xy^4 \\
& + 262127616xy^3 - 280175616xy^2 - 45398016xy - 2156544x - 32768y^{18} + 1179648y^{17} - 8142848y^{16} \\
& + 17170432y^{15} - 4343808y^{14} - 62808064y^{13} + 166397952y^{12} - 151951360y^{11} - 89010176y^{10} \\
& + 417712128y^9 - 418197504y^8 - 17203200y^7 + 412454912y^6 - 428344320y^5 - 62740480y^4 \\
& + 109991936y^3 - 147140096y^2 - 49459200y - 15556608) \\
V \quad & (16384x^3y^{19} - 876544x^3y^{18} + 9228288x^3y^{17} - 31561728x^3y^{16} \\
& + 41086976x^3y^{15} + 34109440x^3y^{14} - 234307584x^3y^{13} + 351354880x^3y^{12} \\
& - 92768256x^3y^{11} - 547226624x^3y^{10} + 860447744x^3y^9 - 288635136x^3y^8 \\
& - 594746752x^3y^7 + 833924736x^3y^6 - 250815712x^3y^5 - 421169040x^3y^4 \\
& + 128054344x^3y^3 - 191730104x^3y^2 - 117102408x^3y - 24362712x^3 - 245760x^2y^{19} \\
& + 1798144x^2y^{18} + 92160x^2y^{17} - 15765504x^2y^{16} + 54581248x^2y^{15} \\
& - 103686144x^2y^{14} + 78893056x^2y^{13} + 100315136x^2y^{12} - 391422976x^2y^{11} \\
& + 547781120x^2y^{10} - 213815808x^2y^9 - 508989312x^2y^8 + 897709248x^2y^7 \\
& - 553489216x^2y^6 - 321814416x^2y^5 + 642623944x^2y^4 - 159764004x^2y^3)
\end{aligned}$$

$$\begin{aligned}
& +25078492x^2y^2 + 242207652x^2y + 55063404x^2 + 106496xy^{20} - 2375680xy^{19} \\
& +13686784xy^{18} - 30601216xy^{17} + 18126848xy^{16} + 96096256xy^{15} - 276908032xy^{14} \\
& +284622848xy^{13} + 134912000xy^{12} - 789054976xy^{11} + 852209408xy^{10} - 19530496xy^9 \\
& -982728256xy^8 + 1050429856xy^7 + 40771488xy^6 - 488242744xy^5 + 518124828xy^4 \\
& +7694610xy^3 - 160360814xy^2 - 69807570xy - 75860406x + 24576y^{21} - 618496y^{20} + 1421312y^{19} \\
& +3522560y^{18} - 20389888y^{17} + 53841920y^{16} - 70682624y^{15} + 8499200y^{14} + 157662208y^{13} \\
& -385352704y^{12} + 384176896y^{11} + 12250752y^{10} - 692573824y^9 + 923891360y^8 - 429315984y^7 \\
& -745416336y^6 + 676143276y^5 - 518474582y^4 - 561645157y^3 + 103204059y^2 - 37637307y \\
& -33887889, \\
& 163840x^3y^{18} + 2326528x^3y^{17} - 19464192x^3y^{16} + 45654016x^3y^{15} \\
& -45271040x^3y^{14} - 22081536x^3y^{13} + 134250496x^3y^{12} - 176214016x^3y^{11} \\
& +56741888x^3y^{10} - 36229120x^3y^9 - 126242816x^3y^8 + 64589824x^3y^7 \\
& -263047168x^3y^6 + 125821952x^3y^5 + 140548096x^3y^4 - 126752768x^3y^3 \\
& +102554624x^3y^2 + 121528320x^3y + 14284800x^3 - 393216x^2y^{19} + 5308416x^2y^{18} \\
& -19275776x^2y^{17} + 23977984x^2y^{16} + 31250432x^2y^{15} - 194514944x^2y^{14} \\
& +327348224x^2y^{13} - 113758208x^2y^{12} - 494129152x^2y^{11} + 1068113920x^2y^{10} \\
& -554184704x^2y^9 - 466405376x^2y^8 + 1491740672x^2y^7 - 564395008x^2y^6 \\
& -188336128x^2y^5 + 854424576x^2y^4 - 197864448x^2y^3 - 68632576x^2y^2 \\
& +83816448x^2y - 14635008x^2 - 32768xy^{20} - 98304xy^{19} + 6004736xy^{18} - 24346624xy^{17} \\
& +43266048xy^{16} - 26812416xy^{15} - 71745536xy^{14} + 198991872xy^{13} - 223518720xy^{12} \\
& +68130816xy^{11} + 148242432xy^{10} - 449499136xy^9 + 225241088xy^8 - 402350080xy^7 \\
& -463237120xy^6 + 130145280xy^5 - 534814720xy^4 - 279484416xy^3 + 51305984xy^2 - 140058624xy \\
& -71313408x + 32768y^{21} - 1245184y^{20} + 8208384y^{19} - 18251776y^{18} + 8849408y^{17} \\
& +70164480y^{16} - 213565440y^{15} + 227823616y^{14} + 111069184y^{13} - 764954624y^{12} \\
& +963160064y^{11} - 133576704y^{10} - 1247393792y^9 + 1700172800y^8 - 298170368y^7 \\
& -919012352y^6 + 1494913536y^5 + 127688704y^4 - 167279104y^3 + 490879488y^2 + 163934208y \\
& +46669824) \\
i = 9, D_i = 18, e_i = 0, \rho_{ij}: 2:1, 24:1, 28:1, 40:2 \\
w_i \quad \langle -162t, \\
\quad -54 \rangle \\
i = 10, D_i = 1944, e_i = 1, \rho_{ij}: 12:2, 16:1, 24:2, 25:2, 29:1, 30:2, 47:2, 54:2, 62:2, 66:1, 71:2 \\
w_i \quad \langle 162030456t^2 + 1057685688t + 1142138880, \\
\quad -254619288t^2 - 489713040t - 314980488 \rangle \\
i = 11, D_i = 54, e_i = 0, \rho_{ij}: 1:1, 2:2, 24:2, 29:2, 30:1, 50:1, 60:1, 71:2 \\
w_i \quad \langle -55890t^2 - 70470t + 10530, \\
\quad -53460t^2 - 9720t + 84240 \rangle \\
i = 12, D_i = 16, \rho_{ij}: 2:1, 8:1, 10:1, 13:1, 20:1, 23:2, 27:2, 29:1, 31:2, 36:1, 38:1, 44:1, 45:2, 48:1, 50:1, 54:1, 60:1, 68:2, 69:1 \\
w_i \quad \langle -4332t^3 - 2352t^2 + 2952t + 3492, \\
\quad -330t^3 + 3576t^2 + 4860t + 1134 \rangle \\
i = 13, D_i = 512, \rho_{ij}: 3:2, 4:2, 8:1, 12:2, 17:1, 21:2, 30:1, 34:1, 37:2, 40:2, 46:2, 48:2, 49:1, 51:1, 61:2, 65:1, 68:2 \\
w_i \quad \langle 763530t^3 + 2122898t^2 - 289012t - 984240, \\
\quad 1054548t^3 + 978868t^2 - 1068926t - 798090 \rangle \\
i = 14, D_i = 128, \rho_{ij}: 19:2, 20:1, 23:2, 25:1, 26:1, 41:2, 42:1, 43:2, 47:2, 57:2, 58:2, 61:2, 67:1, 69:1, 70:2 \\
w_i \quad \langle -35061t^3 + 77424t^2 - 33492t - 30897, \\
\quad -39645t^3 + 35286t^2 + 35211t - 54693 \rangle \\
i = 15, D_i = 16, \rho_{ij}: 30:1, 38:2, 39:2, 41:1, 42:1, 45:2, 54:1, 56:2, 59:1, 67:2, 70:1 \\
w_i \quad \langle -99t^3 - 102t^2 - 339t - 171, \\
\quad -195t^3 - 48t^2 - 228t + 9 \rangle \\
i = 16, D_i = 8, \rho_{ij}: 7:2, 10:2, 20:1, 29:1, 30:2, 36:1, 41:1, 46:2, 47:1, 48:2, 58:1, 65:1, 70:2 \\
w_i \quad \langle 621t^3 + 369t^2 - 888t - 78, \\
\quad -594t^3 - 27t^2 - 732t - 3 \rangle \\
i = 17, D_i = 4, \rho_{ij}: 2:1, 5:2, 6:1, 7:1, 13:2, 36:1, 39:2, 58:2 \\
w_i \quad \langle 12t^3 - 10t^2 - 17t - 3, \\
\quad 3t^3 + 7t^2 - 4t - 12 \rangle \\
i = 18, D_i = 4, \rho_{ij}: 4:1, 26:1, 27:2, 37:1, 41:2, 45:2, 63:2 \\
w_i \quad \langle -32t^3 - 24t^2 - 16t + 8, \\
\quad 8t^3 + 24t^2 - 8t - 8 \rangle \\
i = 19, D_i = 32, \rho_{ij}: 7:1, 14:1, 25:2, 30:1, 36:1, 37:1, 40:2, 41:1, 42:2, 45:1, 47:2, 56:1, 58:2 \\
w_i \quad \langle 430t^3 + 1671t^2 - 848t + 3147, \\
\quad -139t^3 + 4071t^2 - 1468t - 204 \rangle \\
i = 20, D_i = 64, \rho_{ij}: 0:2, 8:2, 12:2, 14:2, 16:2, 22:2, 28:2, 31:1, 37:1, 39:2, 49:2, 52:2, 53:1, 64:2, 66:1, 71:1 \\
w_i \quad \langle 14364t^3 + 1449t^2 - 1281t + 2289, \\
\quad 10227t^3 - 567t^2 + 8778t - 2499 \rangle \\
i = 21, D_i = 4, \rho_{ij}: 7:1, 13:1, 29:2, 30:2, 33:1, 41:1, 48:1, 49:2, 55:1, 57:2, 59:2 \\
w_i \quad \langle -86t^3 - 90t^2 + 48t + 64, \\
\quad -94t^3 + 12t^2 + 72t + 26 \rangle \\
i = 22, D_i = 4, \rho_{ij}: 0:1, 6:2, 8:2, 20:1, 44:2, 51:1, 58:2, 60:2, 66:1, 68:1, 71:2 \\
w_i \quad \langle 43t^3 + 135t^2 - 25t + 165, \\
\quad 119t^3 - 23t^2 + 34t + 44 \rangle
\end{aligned}$$

$$\begin{aligned}
i = 23, D_i = 4, \rho_{ij}: & 12:1, 14:1, 29:1, 37:2, 41:1, 46:1, 49:2, 61:2, 71:1 \\
w_i & \langle 63t^3 + 3t^2 + 15t + 9, \\
& 33t^3 + 21t^2 + 81t + 15 \rangle \\
i = 24, D_i = 128, \rho_{ij}: & 1:1, 6:1, 8:2, 9:2, 10:1, 11:1, 30:2, 32:1, 37:2, 42:2, 44:2, 45:1, 50:1, 60:2, 61:2, 67:1, 70:2 \\
w_i & \langle -228774t^3 + 86052t^2 - 31911t + 147741, \\
& -437829t^3 + 440025t^2 - 241014t - 174228 \rangle \\
i = 25, D_i = 64, \rho_{ij}: & 1:1, 5:1, 8:2, 10:1, 14:2, 19:1, 26:2, 34:1, 38:2, 41:2, 43:1, 45:1, 46:1, 47:1, 58:2, 59:1, 60:1, 66:1, 71:1 \\
w_i & \langle -5705t^3 + 15855t^2 - 16975t + 4095, \\
& 791t^3 - 30513t^2 + 57169t - 47649 \rangle \\
i = 26, D_i = 16, \rho_{ij}: & 5:2, 7:1, 8:2, 14:2, 18:2, 25:1, 29:2, 30:1, 31:1, 32:1, 33:2, 35:2, 37:1, 42:2, 44:2, 47:1, 54:1, 70:2 \\
w_i & \langle -10401t^3 - 2121t^2 + 10281t + 7227, \\
& 6891t^3 - 1491t^2 + 9528t + 5322 \rangle \\
i = 27, D_i = 2, \rho_{ij}: & 12:1, 18:1, 34:1, 36:2, 48:1, 53:1, 55:1, 71:1 \\
w_i & \langle 6t^3 - 36t^2 + 6t, \\
& -24t^3 - 24t^2 + 18t \rangle \\
i = 28, D_i = 2, \rho_{ij}: & 2:1, 8:1, 9:2, 20:1, 33:2, 35:1, 63:2, 67:1 \\
w_i & \langle -30t^3 - 12t^2 + 22t + 20, \\
& -6t^3 + 6t^2 - 4t + 4 \rangle \\
i = 29, D_i = 64, \rho_{ij}: & 0:1, 7:1, 10:2, 11:1, 12:2, 16:2, 21:1, 23:2, 26:1, 32:2, 33:1, 35:1, 41:2, 49:2, 54:1, 57:2, 59:2, 61:2, 63:1, \\
& 66:1 \\
w_i & \langle 17104269t^3 + 29127567t^2 + 16719111t - 7350507, \\
& 9352203t^3 + 26780208t^2 + 33463467t + 9597537 \rangle \\
i = 30, D_i = 256, \rho_{ij}: & 0:1, 1:1, 5:2, 10:1, 11:2, 13:2, 15:2, 16:1, 19:2, 21:1, 24:1, 26:2, 31:2, 37:2, 41:2, 42:1, 48:2, 51:2, 57:2, \\
& 61:1, 63:1, 65:1 \\
w_i & \langle 164236653t^3 + 317433465t^2 + 245467683t - 19494999, \\
& 122685948t^3 + 123055011t^2 - 64937682t - 152840214 \rangle \\
i = 31, D_i = 16, \rho_{ij}: & 8:1, 12:1, 20:2, 26:2, 30:1, 37:2, 41:1, 45:1, 48:1, 57:2, 60:1, 70:1 \\
w_i & \langle 124488t^3 - 8064t^2 + 35028t + 46116, \\
& 210336t^3 - 14952t^2 + 58716t + 77868 \rangle \\
i = 32, D_i = 4, \rho_{ij}: & 24:2, 26:2, 29:1, 51:2, 56:1, 62:1, 64:1, 71:1 \\
w_i & \langle -240t^3 - 216t^2 - 120t + 48, \\
& -120t^3 - 108t^2 - 60t + 24 \rangle \\
i = 33, D_i = 8, \rho_{ij}: & 2:2, 3:2, 7:2, 8:1, 21:2, 26:1, 28:1, 29:2, 36:2, 42:1, 47:1, 58:1 \\
w_i & \langle -112t^3 + 1236t^2 + 468t + 604, \\
& 40t^3 - 60t^2 + 132t + 212 \rangle \\
i = 34, D_i = 18, e_i = 1, \rho_{ij}: & 13:2, 25:2, 27:2, 38:1, 66:1, 67:2 \\
w_i & \langle -270t^3 + 432t + 162, \\
& -54t^3 + 540t + 486 \rangle \\
i = 35, D_i = 144, e_i = 0, \rho_{ij}: & 1:1, 3:1, 5:2, 26:1, 28:2, 29:2, 38:1, 39:2, 41:1, 54:2, 59:2, 63:1 \\
w_i & \langle 955557t^3 + 110160t^2 + 93636t + 606771, \\
& -459594t^3 + 26892t^2 + 335502t + 803844 \rangle \\
i = 36, D_i = 24, e_i = 1, \rho_{ij}: & 2:1, 5:2, 12:2, 16:2, 17:2, 19:2, 27:1, 33:1, 48:1, 55:2, 58:2, 62:1, 66:1, 67:2, 69:2 \\
w_i & \langle 2960955t^3 - 23180499t^2 + 7029261t + 9234729, \\
& -8592480t^3 - 10367352t^2 + 12968667t + 1093743 \rangle \\
i = 37, D_i = 16, \rho_{ij}: & 1:2, 3:1, 5:1, 7:1, 13:1, 18:2, 19:2, 20:2, 23:1, 24:1, 26:2, 30:1, 31:1, 40:2, 46:1, 47:2, 49:2, 52:2, 60:1 \\
w_i & \langle -1344t^5 - 224t^4 + 224t^3 + 336t^2 + 560t + 1344, \\
& 448t^5 + 560t^4 - 224t^3 + 112t^2 + 1904t + 3024 \rangle \\
i = 38, D_i = 4, \rho_{ij}: & 4:1, 5:1, 12:2, 15:1, 25:1, 34:2, 35:2, 46:1, 57:1, 58:1, 60:2, 64:1, 65:2, 68:1 \\
w_i & \langle 84t^5 + 132t^4 + 120t^3 + 360t^2 + 120t + 240, \\
& 84t^5 - 36t^4 + 132t^3 + 288t^2 + 324t + 312 \rangle \\
i = 39, D_i = 32, \rho_{ij}: & 7:2, 15:1, 17:1, 20:1, 35:1, 43:1, 61:1, 63:1, 66:1 \\
w_i & \langle 32t^5 + 32t^4 - 32t^3 + 64t^2, \\
& -32t^5 - 32t^4 - 64t^3 + 32t^2 - 96t \rangle \\
i = 40, D_i = 16, \rho_{ij}: & 5:1, 6:1, 9:1, 13:1, 19:1, 37:1, 46:1, 49:1, 55:1 \\
w_i & \langle -32t^4 - 32t^3 + 16t^2 + 32t - 48, \\
& -16t^4 - 16t^3 - 16t^2 - 32t - 48 \rangle \\
i = 41, D_i = 33554432, \rho_{ij}: & 2:1, 3:1, 4:2, 14:1, 15:2, 16:2, 18:1, 19:2, 21:2, 23:2, 25:1, 29:1, 30:1, 31:2, 35:2, 42:1, 44:2, 50:2, \\
& 52:2, 54:1, 56:2, 66:2, 67:2, 69:2 \\
w_i & \langle 15181232996352t^5 + 4005191221248t^4 + 5271132831744t^3 + 8879710666752t^2 + 12772964302848t \\
& + 3609383141376, \\
& 22672595484672t^5 - 38578604212224t^4 + 11559367606272t^3 - 3445905948672t^2 \\
& - 1207959552000t - 25112271126528 \rangle \\
i = 42, D_i = 32768, \rho_{ij}: & 14:2, 15:2, 19:1, 24:1, 26:1, 30:2, 33:2, 41:2, 48:2, 56:2, 58:2, 59:1, 61:2 \\
w_i & \langle 48070656t^5 + 2326528t^4 - 24248320t^3 - 8060928t^2 + 6389760t + 43352064, \\
& 7962624t^5 - 74678272t^4 - 7602176t^3 - 31752192t^2 - 19365888t - 18579456 \rangle \\
i = 43, D_i = 4, \rho_{ij}: & 0:1, 2:2, 3:1, 14:1, 25:2, 39:2, 45:2, 47:2, 52:1, 62:1 \\
w_i & \langle -16t^5 - 32t^4 - 24t^3 + 80t^2 + 16t + 24, \\
& -8t^5 - 16t^4 + 20t^3 + 112t^2 + 32t + 28 \rangle
\end{aligned}$$

$$\begin{aligned}
i = 44, D_i = 16, \rho_{ij}: 12:2, 22:1, 24:1, 26:1, 41:1, 51:2, 53:1, 63:2 \\
w_i \quad \langle 32t^5 - 16t^4 - 48t^3 - 48t^2 - 80t - 32, \\
\quad 32t^5 - 16t^4 - 48t^3 - 48t^2 - 80t - 32 \rangle \\
i = 45, D_i = 32768, \rho_{ij}: 1:2, 3:2, 12:1, 15:1, 18:1, 19:2, 24:2, 25:2, 31:2, 43:1, 46:2, 48:1, 55:2, 61:1, 65:1 \\
w_i \quad \langle 26181632t^5 + 41156608t^4 + 7372800t^3 - 19988480t^2 - 4784128t + 10354688, \\
\quad 7438336t^5 - 3506176t^4 - 41189376t^3 - 47644672t^2 - 25886720t - 4554752 \rangle \\
i = 46, D_i = 8192, \rho_{ij}: 4:1, 5:1, 7:1, 13:1, 16:1, 23:2, 25:2, 37:2, 38:2, 40:2, 45:1, 65:2, 71:1 \\
w_i \quad \langle 2514944t^5 + 3563520t^4 - 794624t^3 - 1843200t^2 - 2015232t + 770048, \\
\quad 2940928t^5 + 3375104t^4 + 5152768t^3 + 4210688t^2 + 1359872t - 327680 \rangle \\
i = 47, D_i = 1024, \rho_{ij}: 6:2, 7:1, 10:1, 14:1, 16:2, 19:1, 25:2, 26:2, 33:2, 37:1, 43:1, 58:1, 64:2 \\
w_i \quad \langle -3072t^5 + 166912t^4 - 1320960t^3 - 1376256t^2 - 795648t + 272384, \\
\quad -657408t^5 - 722944t^4 - 1320960t^3 - 1118208t^2 - 411648t + 117760 \rangle \\
i = 48, D_i = 4, \rho_{ij}: 0:2, 1:2, 2:2, 5:1, 12:2, 13:1, 16:1, 21:2, 27:2, 30:1, 31:2, 36:2, 42:1, 45:2, 60:1, 67:2 \\
w_i \quad \langle -9668t^5 + 5132t^4 - 1392t^3 + 26476t^2 + 13004t - 24048, \\
\quad -10468t^5 + 18532t^4 + 18840t^3 - 11152t^2 + 4432t - 19560 \rangle \\
i = 49, D_i = 4, \rho_{ij}: 0:2, 2:1, 6:2, 7:1, 13:2, 20:1, 21:1, 23:1, 29:1, 37:1, 40:2, 50:1, 62:1, 71:2 \\
w_i \quad \langle -108t^5 + 768t^4 - 624t^3 - 1916t^2 + 772t + 180, \\
\quad 272t^5 + 1012t^4 - 712t^3 - 256t^2 + 164t - 1200 \rangle \\
i = 50, D_i = 128, \rho_{ij}: 2:1, 11:2, 12:2, 24:2, 41:1, 49:2, 52:1, 60:1, 65:1, 69:1 \\
w_i \quad \langle -1024t^5 - 1792t^4 + 2688t^3 + 14464t^2 + 12928t + 6144, \\
\quad -1664t^5 - 7424t^4 - 9216t^3 - 256t^2 + 7040t + 5760 \rangle \\
i = 51, D_i = 4, \rho_{ij}: 13:2, 22:2, 30:1, 32:1, 44:1, 53:2, 56:1, 59:2, 61:2 \\
w_i \quad \langle -156t^2 - 1520t - 1840, \\
\quad 156t^2 + 1520t + 1840 \rangle \\
i = 52, D_i = 4, \rho_{ij}: 20:1, 37:1, 41:1, 43:2, 50:2, 64:2 \\
w_i \quad \langle t^3 - 12t^2 - 6t + 9, \\
\quad 3t^3 - 5t^2 - 6t + 4 \rangle \\
i = 53, D_i = 8, \rho_{ij}: 3:2, 5:2, 7:1, 8:2, 20:2, 27:2, 44:2, 51:1, 55:2, 60:2, 71:2 \\
w_i \quad \langle 368t^3 - 282t^2 - 101t + 567, \\
\quad 115t^3 + 357t^2 - 706t + 528 \rangle \\
i = 54, D_i = 4, \rho_{ij}: 4:2, 10:1, 12:2, 15:2, 26:2, 29:2, 35:1, 41:2, 56:1, 67:2 \\
w_i \quad \langle 210t^3 + 1794t^2 + 3560t + 2028, \\
\quad -84t^3 + 1662t^2 + 2680t + 1110 \rangle \\
i = 55, D_i = 6, e_i = 2, \rho_{ij}: 3:1, 21:2, 27:2, 36:1, 40:2, 45:1, 53:1, 60:1, 68:2, 71:1 \\
w_i \quad \langle 2268t^3 + 8748t^2 - 3402t - 4050, \\
\quad -2754t^3 + 4374t^2 + 1458t - 2754 \rangle \\
i = 56, D_i = 72, e_i = 2, \rho_{ij}: 1:1, 15:1, 19:2, 32:2, 41:1, 42:1, 51:2, 54:2, 59:2, 61:1 \\
w_i \quad \langle -22356t^3 - 106434t^2 + 5346t + 73386, \\
\quad -52245t^3 - 47385t^2 + 133893t + 89667 \rangle \\
i = 57, D_i = 2, \rho_{ij}: 14:1, 21:1, 29:1, 30:1, 31:1, 38:2 \\
w_i \quad \langle 0, \\
\quad -2t^3 + 2t^2 + 4t \rangle \\
i = 58, D_i = 8, \rho_{ij}: 3:1, 5:2, 14:1, 16:2, 17:1, 19:1, 22:1, 25:1, 33:2, 36:1, 38:2, 42:1, 47:2, 64:2, 66:2 \\
w_i \quad \langle -48t^5 + 16t^4 - 248t^3 - 144t^2 - 392t - 336, \\
\quad -120t^5 - 136t^4 - 256t^3 - 240t^2 - 184t - 408 \rangle \\
i = 59, D_i = 16, \rho_{ij}: 15:2, 21:1, 25:2, 29:1, 35:1, 42:2, 51:1, 56:1, 61:2, 71:2 \\
w_i \quad \langle 552t^3 + 1512t^2 - 862t - 898, \\
\quad -66t^3 + 2070t^2 + 112t - 956 \rangle \\
i = 60, D_i = 128, \rho_{ij}: 0:2, 8:2, 11:2, 12:2, 22:1, 24:1, 25:2, 31:2, 37:2, 38:1, 48:2, 50:2, 53:1, 55:2, 68:2, 71:1 \\
w_i \quad \langle 39318t^3 + 66762t^2 + 13548t + 29106, \\
\quad -13632t^3 + 40770t^2 + 3894t + 8946 \rangle \\
i = 61, D_i = 8192, \rho_{ij}: 13:1, 14:1, 23:1, 24:1, 29:1, 30:2, 39:2, 42:1, 45:2, 51:1, 56:2, 59:1, 66:2, 67:2 \\
w_i \quad \langle -229376t^5 - 2179072t^4 - 417792t^3 - 221184t^2 - 434176t - 1499136, \\
\quad -729088t^5 - 622592t^4 + 245760t^2 - 647168t - 344064 \rangle \\
i = 62, D_i = 2, \rho_{ij}: 10:1, 32:2, 36:2, 43:2, 49:2 \\
w_i \quad \langle 30t^3 - 4t^2, \\
\quad -18t^3 + 22t^2 \rangle \\
i = 63, D_i = 4, \rho_{ij}: 7:2, 8:1, 18:1, 28:1, 29:2, 30:2, 35:2, 39:2, 44:1, 64:1 \\
w_i \quad \langle -8t^3 + 16t^2 - 16t - 24, \\
\quad -16t^3 + 32t^2 - 32t - 48 \rangle \\
i = 64, D_i = 4, \rho_{ij}: 20:1, 32:2, 38:2, 47:1, 52:1, 58:1, 63:2, 67:1 \\
w_i \quad \langle 4t^3 + 4t^2 - 4t - 4, \\
\quad 8t^2 - 8 \rangle \\
i = 65, D_i = 16, \rho_{ij}: 2:2, 13:2, 16:2, 30:2, 38:1, 45:2, 46:1, 50:2 \\
w_i \quad \langle -88t^3 - 512t^2 - 304t - 224, \\
\quad -272t^3 - 768t^2 - 344t - 224 \rangle \\
i = 66, D_i = 16, \rho_{ij}: 10:2, 20:2, 22:2, 25:2, 29:2, 34:2, 36:2, 39:2, 41:1, 58:1, 61:1 \\
w_i \quad \langle -75948t^3 - 9684t^2 + 216018t - 115182,
\end{aligned}$$

$$4878t^3 - 165870t^2 + 169740t + 21708)$$

$i = 67, D_i = 4, \rho_{ij}: 14:2, 15:1, 24:2, 28:2, 34:1, 36:1, 41:1, 48:1, 54:1, 61:1, 64:2$
 $w_i \quad \langle -296t^2 - 872t + 48, -160t^2 - 712t - 864 \rangle$

$i = 68, D_i = 2, \rho_{ij}: 5:2, 12:1, 13:1, 22:2, 38:2, 55:1, 60:1$
 $w_i \quad \langle 2t^3 - 4t + 4, 4t^3 - 6t^2 - 5t + 5 \rangle$

$i = 69, D_i = 2, \rho_{ij}: 7:1, 12:2, 14:2, 36:1, 41:1, 50:2$
 $w_i \quad \langle -4t^3 + 6t^2 - 2t + 15, 4t^3 - 6t^2 - t + 9 \rangle$

$i = 70, D_i = 8, \rho_{ij}: 2:1, 14:1, 15:2, 16:1, 24:1, 26:1, 31:2$
 $T \quad \langle -80y^2 + 40y + 175, -100y^2 - 20y + 140 \rangle$
 $U \quad \langle 768000x^4y + 768000x^4 - 768000x^3y^2 - 1510400x^3y - 742400x^3 + 1753600x^2y^3 + 2496000x^2y^2 - 76800x^2y - 819200x^2 - 1561600xy^4 - 3308800xy^3 + 633600xy^2 + 2790400xy + 409600x + 396800y^6 + 435200y^5 - 2803200y^4 - 6304000y^3 - 5290400y^2 - 2271200y - 702600, -25600x^4y - 25600x^4 + 25600x^3y^2 + 844800x^3y + 819200x^3 + 736000x^2y^3 - 83200x^2y^2 - 2380800x^2y - 1561600x^2 - 742400xy^4 - 1280000xy^3 + 1766400xy^2 + 3084800xy + 780800x + 384000y^6 + 1574400y^5 - 502400y^4 - 4520000y^3 - 1440000y^2 + 348800y - 1377000 \rangle$
 $V \quad \langle -396800x^3y^6 + 1516800x^3y^5 + 5190400x^3y^4 + 4972800x^3y^3 + 1168800x^3y^2 - 864800x^3y - 78200x^3 - 1356800x^2y^6 - 2665600x^2y^5 - 3009600x^2y^4 + 279200x^2y^3 + 2210400x^2y^2 + 2424800x^2y + 2104300x^2 + 1184000xy^7 - 259200xy^6 + 953600xy^5 - 837600xy^4 - 7542000xy^3 - 5595400xy^2 - 2764600xy - 1536200x - 793600y^8 + 841600y^7 + 1961600y^6 + 804000y^5 + 9774400y^4 + 16158000y^3 + 5239700y^2 + 43100y + 1950975, -384000x^3y^6 - 646400x^3y^5 + 3004800x^3y^4 + 4961600x^3y^3 - 1203200x^3y^2 - 2230400x^3y + 1005800x^3 - 352000x^2y^6 - 2195200x^2y^5 - 6900800x^2y^4 - 5456800x^2y^3 + 1274800x^2y^2 + 1061200x^2y - 911800x^2 + 953600xy^7 + 902400xy^6 + 4238400xy^5 + 7550400xy^4 + 2044800xy^3 - 1310600xy^2 + 1066600xy + 1355750x - 768000y^8 - 723200y^7 - 264000y^6 - 4868000y^5 - 2734000y^4 + 5655000y^3 - 19000y^2 - 4343000y - 392475 \rangle$

$i = 71, D_i = 8, \rho_{ij}: 0:2, 8:1, 10:1, 11:1, 20:2, 22:1, 23:2, 25:2, 27:2, 32:2, 46:2, 49:1, 53:1, 55:2, 59:1, 60:2$
 $w_i \quad \langle 600t^5 - 2808t^4 - 5952t^3 - 7320t^2 + 3384t + 2112, 2112t^5 + 600t^4 - 696t^3 - 5952t^2 + 1128t + 1272 \rangle$

The second certificate.

$i = 0, D_i = 8, \rho_{ij}: 4:1, 10:2, 30:1, 34:2, 38:2, 40:2, 53:2, 56:2, 59:1, 61:2, 68:1$
 $w_i \quad \langle -4t^3 - 10t^2 + 56t + 30, -68t^3 - 32t^2 + 40t - 12 \rangle$

$i = 1, D_i = 2, \rho_{ij}: 5:2, 17:1, 20:1, 27:1, 35:2, 38:1, 43:2, 59:1$
 $w_i \quad \langle -22t^3 - 38t^2 - 20t, 4t^3 - 16t^2 - 46t - 42 \rangle$

$i = 2, D_i = 4, \rho_{ij}: 6:2, 8:2, 12:2, 13:2, 18:1, 39:1, 43:1, 45:2, 46:1, 49:1, 62:1, 64:1$
 $w_i \quad \langle -4t^3 - 6t^2 - 11t + 1, -14t^3 - 4t^2 - 3t + 1 \rangle$

$i = 3, D_i = 8, \rho_{ij}: 13:1, 19:1, 24:2, 26:1, 28:2, 29:2, 30:2, 37:1, 39:1, 41:2, 45:1, 47:1, 52:2, 55:1, 57:1, 61:1, 63:2$
 $w_i \quad \langle -72t^2 + 56t + 696, -304t^2 - 560t - 48 \rangle$

$i = 4, D_i = 4, \rho_{ij}: 0:2, 15:1, 17:1, 26:1, 38:2, 43:1, 44:1, 49:2, 56:2$
 $w_i \quad \langle 8t^2 + 16t + 48, 16t^2 + 16t + 16 \rangle$

$i = 5, D_i = 4, \rho_{ij}: 1:1, 12:1, 14:2, 20:1, 31:1, 36:1, 42:1, 47:2, 50:2, 52:2, 58:2, 68:2$
 $w_i \quad \langle 12t^2 + 260t + 616, 204t^2 + 304t + 392 \rangle$

$i = 6, D_i = 4, \rho_{ij}: 2:1, 17:2, 19:1, 22:2, 37:1, 46:1, 48:1, 52:2, 66:2, 67:2, 68:1$
 $w_i \quad \langle 8t^2 - 36t + 24, -20t^2 - 4t + 24 \rangle$

$i = 7, D_i = 2, \rho_{ij}: 15:1, 20:2, 35:1, 54:2, 58:2$
 $w_i \quad \langle 2t^2 + 2t + 12, -2t^2 - 2t + 12 \rangle$

$i = 8, D_i = 4, \rho_{ij}: 2:1, 14:2, 15:2, 17:1, 25:2, 27:1, 29:1, 30:2, 34:1, 37:1, 45:1, 47:2, 52:2, 53:1, 56:1, 63:2$
 $w_i \quad \langle -84t^2 - 288t - 156, -8t^2 - 92t - 188 \rangle$

$i = 9, D_i = 4, \rho_{ij}: 15:1, 24:2, 29:1, 31:2, 44:2, 48:2, 50:1, 54:2, 59:1$
 $w_i \quad \langle 44t^2 + 240t - 1160, 20t^2 + 544t - 1156 \rangle$

$i = 10, D_i = 8, \rho_{ij}: 0:1, 12:1, 13:1, 18:2, 22:2, 23:2, 24:2, 31:1, 32:1, 49:2, 50:1, 54:2$
 $T \quad \langle -2y^2 - 5y + 42, 0 \rangle$
 $U \quad \langle 9216x^{13}y + 90624x^{12}y^2 - 12288x^{12}y + 425472x^{11}y^3 - 101376x^{11}y^2 - 24576x^{11}y$

$$\begin{aligned}
& +1263616x^{10}y^4 - 376320x^{10}y^3 - 227328x^{10}y^2 + 22016x^{10}y + 2640384x^9y^5 \\
& - 812544x^9y^4 - 973824x^9y^3 + 119040x^9y^2 + 1536x^9y + 4067840x^8y^6 \\
& - 1071872x^8y^5 - 2492416x^8y^4 + 173056x^8y^3 + 62720x^8y^2 + 90112x^8y \\
& + 4681728x^7y^7 - 754432x^7y^6 - 4115968x^7y^5 - 448256x^7y^4 + 260608x^7y^3 \\
& + 712192x^7y^2 - 47872x^7y + 3941376x^6y^8 - 3072x^6y^7 - 4348160x^6y^6 \\
& - 2500864x^6y^5 + 171776x^6y^4 + 2612480x^6y^3 - 311168x^6y^2 - 128000x^6y \\
& + 2208768x^5y^9 + 405248x^5y^8 - 2465024x^5y^7 - 5361664x^5y^6 - 1302784x^5y^5 \\
& + 5925888x^5y^4 - 424064x^5y^3 - 610048x^5y^2 + 94976x^5y + 489472x^4y^{10} \\
& - 96512x^4y^9 + 234880x^4y^8 - 6658304x^4y^7 - 4380672x^4y^6 + 8961792x^4y^5 \\
& + 1197440x^4y^4 - 1193344x^4y^3 + 132992x^4y^2 - 135808x^4y - 433152x^3y^{11} \\
& - 1091584x^3y^{10} + 1652096x^3y^9 - 4913920x^3y^8 - 6847744x^3y^7 + 8835840x^3y^6 \\
& + 5217792x^3y^5 - 468096x^3y^4 - 1012352x^3y^3 - 1022912x^3y^2 + 103040x^3y \\
& - 529408x^2y^{12} - 1608192x^2y^{11} + 1228288x^2y^{10} - 1752960x^2y^9 - 5793536x^2y^8 \\
& + 5013888x^2y^7 + 8596352x^2y^6 + 3733056x^2y^5 - 3692800x^2y^4 - 3703168x^2y^3 \\
& + 825536x^2y^2 + 179328x^2y - 267264xy^{13} - 1226240xy^{12} + 259200xy^{11} + 144512xy^{10} \\
& - 1295616xy^9 + 1130944xy^8 + 7131648xy^7 + 10854592xy^6 - 5060224xy^5 - 9330176xy^4 \\
& + 1596160xy^3 + 818496xy^2 - 104256xy + 1024y^{15} - 3584y^{14} - 127488y^{13} - 485120y^{12} \\
& + 471872y^{11} - 387968y^{10} + 60672y^9 + 1323840y^8 + 1870528y^7 + 4499648y^6 - 3377056y^5 \\
& - 3687568y^4 + 1619080y^3 + 51312y^2 - 49056y - 28224, \\
& 6144x^{12}y^2 - 6144x^{12}y + 58368x^{11}y^3 - 55296x^{11}y^2 + 33792x^{11}y + 266240x^{10}y^4 \\
& - 216576x^{10}y^3 + 247296x^{10}y^2 - 38912x^{10}y + 773120x^9y^5 - 465920x^9y^4 \\
& + 749568x^9y^3 - 303104x^9y^2 - 62464x^9y + 1591296x^8y^6 - 513536x^8y^5 \\
& + 1085952x^8y^4 - 1009152x^8y^3 - 345088x^8y^2 + 139776x^8y + 2439168x^7y^7 \\
& + 47616x^7y^6 + 283648x^7y^5 - 1916928x^7y^4 - 796416x^7y^3 + 838400x^7y^2 \\
& - 38400x^7y + 2838528x^6y^8 + 1184768x^6y^7 - 1789184x^6y^6 - 2511104x^6y^5 \\
& - 1013504x^6y^4 + 2100224x^6y^3 - 308992x^6y^2 - 53760x^6y + 2494464x^5y^9 \\
& + 2132480x^5y^8 - 3568128x^5y^7 - 2983424x^5y^6 - 902400x^5y^5 + 2937600x^5y^4 \\
& - 771328x^5y^3 + 1024x^5y^2 + 97536x^5y + 1587200x^4y^{10} + 2012160x^4y^9 \\
& - 3418368x^4y^8 - 3774720x^4y^7 - 1128192x^4y^6 + 2771456x^4y^5 - 490240x^4y^4 \\
& + 798336x^4y^3 + 313216x^4y^2 - 163840x^4y + 628736x^3y^{11} + 802816x^3y^{10} \\
& - 1780992x^3y^9 - 4086272x^3y^8 - 2372352x^3y^7 + 2243840x^3y^6 + 1465600x^3y^5 \\
& + 2380288x^3y^4 + 84736x^3y^3 - 985600x^3y^2 + 54784x^3y + 30720x^2y^{12} \\
& - 576512x^2y^{11} - 518656x^2y^{10} - 2794880x^2y^9 - 4036224x^2y^8 + 1645824x^2y^7 \\
& + 4683008x^2y^6 + 3992320x^2y^5 - 840704x^2y^4 - 2391552x^2y^3 + 445312x^2y^2 \\
& + 50304x^2y - 153600xy^{13} - 1213440xy^{12} - 425216xy^{11} - 598784xy^{10} - 4224000xy^9 \\
& + 669952xy^8 + 7471360xy^7 + 5705088xy^6 - 1592576xy^5 - 3569408xy^4 + 753216xy^3 \\
& - 183360xy^2 - 72576xy - 76800y^{13} - 614400y^{12} + 10112y^{11} - 168576y^{10} - 1683712y^9 \\
& + 1069184y^8 + 3152640y^7 + 1961664y^6 - 1767616y^5 - 1159104y^4 + 877696y^3 - 333888y^2 \\
& - 28800y) \\
V \quad & \langle -1024x^3y^{15} - 47104x^3y^{14} - 291328x^3y^{13} + 498560x^3y^{12} - 252928x^3y^{11} \\
& + 3557504x^3y^{10} + 820992x^3y^9 - 26880x^3y^8 + 13225600x^3y^7 - 6434048x^3y^6 \\
& - 13306464x^3y^5 + 2818704x^3y^4 + 558392x^3y^3 + 287088x^3y^2 + 211488x^3y \\
& + 28224x^3 + 20480x^2y^{15} + 164352x^2y^{14} + 582912x^2y^{13} - 455808x^2y^{12} \\
& + 4682368x^2y^{11} + 1577088x^2y^{10} - 3898368x^2y^9 + 11206656x^2y^8 \\
& - 3156928x^2y^7 - 16434560x^2y^6 - 2295504x^2y^5 + 1576440x^2y^4 + 3605956x^2y^3 \\
& + 940872x^2y^2 - 349968x^2y - 14112x^2 + 16384xy^{16} + 241664xy^{15} + 678912xy^{14} \\
& - 487808xy^{13} + 3358720xy^{12} + 2176000xy^{11} - 4338816xy^{10} + 3863936xy^9 - 1858112xy^8 \\
& - 11954848xy^7 - 10119488xy^6 - 470680xy^5 + 11191300xy^4 + 2539390xy^3 - 2207652xy^2 \\
& - 41880xy + 7056x + 4608y^{17} + 126464y^{16} + 485632y^{15} - 333632y^{14} + 1255552y^{13} + 920768y^{12} \\
& - 2353536y^{11} - 767296y^{10} - 6005504y^9 - 2464992y^8 - 6423664y^7 - 11739936y^6 \\
& + 13768508y^5 + 9492326y^4 - 4882723y^3 - 97446y^2 + 201852y + 10584, \\
& - 116736x^3y^{14} - 922624x^3y^{13} - 30208x^3y^{12} + 713984x^3y^{11} - 1848064x^3y^{10} \\
& + 1706368x^3y^9 + 6643072x^3y^8 + 4660480x^3y^7 - 3302144x^3y^6 - 2939136x^3y^5 \\
& + 395776x^3y^4 - 2288512x^3y^3 + 416256x^3y^2 + 243072x^3y - 46080x^2y^{15} \\
& - 397312x^2y^{14} + 116224x^2y^{13} + 739712x^2y^{12} + 556160x^2y^{11} + 1966720x^2y^{10} \\
& + 4100096x^2y^9 + 5395072x^2y^8 - 1959936x^2y^7 - 4872320x^2y^6 - 3422336x^2y^5 \\
& - 3439232x^2y^4 + 2690368x^2y^3 + 1015104x^2y^2 - 225792x^2y - 10240xy^{16} \\
& - 39936xy^{15} + 461824xy^{14} + 323072xy^{13} + 1862144xy^{12} + 3043968xy^{11} + 643968xy^{10} \\
& + 3964672xy^9 + 210944xy^8 - 7181440xy^7 - 8239232xy^6 - 3281280xy^5 + 5940096xy^4 \\
& + 1644416xy^3 - 1104384xy^2 + 73728xy - 1024y^{17} + 73728y^{16} + 614400y^{15} + 66688y^{14} \\
& + 1013376y^{13} + 3516800y^{12} - 928896y^{11} - 957056y^{10} + 2020288y^9 - 4592576y^8 \\
& - 10260480y^7 - 4995072y^6 + 6794816y^5 + 2625344y^4 - 2299200y^3 + 1030464y^2 + 86400y) \\
i = 11, D_i = 5832, e_i = 2, \rho_{ij}: 16:1, 22:1, 24:2, 26:2, 28:2, 30:1, 35:1, 40:1, 42:1, 44:2, 46:1, 50:2, 53:1, 54:2, 55:1, 59:1, 65:2 \\
w_i \quad & \langle -69703801560t^2 - 266811725520t - 147740598000, \\
& - 320267604240t^2 - 354818355120t + 70894229400) \\
i = 12, D_i = 8, \rho_{ij}: 2:1, 5:2, 10:2, 13:1, 20:2, 21:2, 26:1, 28:2, 38:2, 46:2, 49:1, 55:1, 56:1 \\
w_i \quad & \langle 2002t^3 - 658t^2 - 2206t + 1450, \\
& - 634t^3 + 4714t^2 - 2120t - 1048) \\
i = 13, D_i = 8, \rho_{ij}: 2:1, 3:2, 10:2, 12:2, 23:1, 30:2, 37:2, 38:1, 44:2, 52:1, 55:2, 57:1, 64:2, 67:2 \\
w_i \quad & \langle 15006t^3 + 28758t^2 - 12910t - 17134, \\
& 12402t^3 - 3174t^2 - 17156t - 4784)
\end{aligned}$$

$i = 14, D_i = 4, \rho_{ij}: 5:1, 8:1, 17:1, 19:1, 24:2, 26:2, 31:2, 32:2, 43:1, 44:1, 45:1, 51:1, 64:1, 67:1, 68:1$
$w_i \quad \langle 158t^3 + 86t^2 + 21t + 30, 13t^3 - 32t^2 - 126t - 39 \rangle$
$i = 15, D_i = 2, \rho_{ij}: 4:2, 7:2, 8:1, 9:2, 22:1, 28:1, 41:1, 42:1, 48:1, 52:2, 64:2$
$w_i \quad \langle 20t^3 + 16t^2 - 32t - 36, 4t^3 + 8t^2 - 12 \rangle$
$i = 16, D_i = 128, \rho_{ij}: 11:2, 21:1, 22:2, 24:1, 29:2, 30:1, 31:1, 36:2, 38:2, 39:2, 40:1, 47:2, 50:1, 61:1, 62:2, 63:2, 64:1, 66:2, 68:1$
$w_i \quad \langle 4086t^3 + 36450t^2 - 31203t - 40230, 18081t^3 + 54486t^2 - 11565t + 10881 \rangle$
$i = 17, D_i = 64, \rho_{ij}: 1:2, 4:2, 6:1, 8:2, 14:2, 23:1, 25:2, 27:1, 37:1, 38:2, 47:2, 49:1, 51:2, 60:1, 64:1$
$w_i \quad \langle 1625t^3 + 3133t^2 + 3252t + 3465, 2782t^3 + 9203t^2 + 7719t + 4179 \rangle$
$i = 18, D_i = 1, \rho_{ij}: 2:2, 10:1, 47:1, 61:2, 62:1, 63:2$
$w_i \quad \langle -5t^2 - 2t + 1, -4t^2 - 2t \rangle$
$i = 19, D_i = 32, \rho_{ij}: 3:2, 6:2, 14:2, 30:1, 31:1, 35:2, 37:2, 42:2, 46:1, 49:1, 53:2, 60:1, 65:1, 67:2$
$w_i \quad \langle -1245t^3 - 4749t^2 - 6072t - 2793, 1125t^3 - 414t^2 - 1737t - 1386 \rangle$
$i = 20, D_i = 8, \rho_{ij}: 1:2, 5:2, 7:1, 12:1, 26:2, 28:2, 30:1, 34:1, 36:1, 39:2, 41:1, 43:1, 44:1, 59:1$
$w_i \quad \langle 254t^3 - 121t^2 - 164t + 91, 151t^3 - 209t^2 + 8t - 70 \rangle$
$i = 21, D_i = 2, \rho_{ij}: 12:1, 16:2, 30:2, 43:1, 46:2, 49:1, 52:2, 60:2, 68:1$
$w_i \quad \langle 12t^3 + 12t^2, 12t^3 + 36t^2 + 24t \rangle$
$i = 22, D_i = 64, \rho_{ij}: 6:1, 10:1, 11:2, 15:2, 16:1, 23:2, 28:2, 30:2, 37:1, 40:2, 41:2, 47:1, 54:2, 57:2, 59:2, 63:2, 64:1$
$w_i \quad \langle -1450404t^3 + 1590210t^2 - 540159t - 1298127, 193677t^3 - 453045t^2 + 709920t - 429762 \rangle$
$i = 23, D_i = 16, \rho_{ij}: 10:1, 13:2, 17:2, 22:1, 32:1, 36:2, 38:1, 39:1, 43:2, 51:1, 52:1, 61:2, 62:1, 65:2, 66:1, 68:2$
$w_i \quad \langle -5397t^3 + 12131t^2 - 13923t + 6643, -1197t^3 + 5803t^2 - 8715t + 3899 \rangle$
$i = 24, D_i = 512, \rho_{ij}: 3:1, 9:1, 10:1, 11:1, 14:1, 16:2, 27:1, 28:1, 34:1, 37:2, 39:1, 40:2, 41:2, 44:2, 56:2, 59:2, 61:2, 63:1, 64:2$
$w_i \quad \langle -949136t^3 + 1073362t^2 - 86497t - 1146505, -1667551t^3 + 2290913t^2 - 1086149t - 1194839 \rangle$
$i = 25, D_i = 2, \rho_{ij}: 8:1, 17:1, 27:2, 31:1, 37:1, 45:1, 49:1, 62:2$
$w_i \quad \langle 4t^3 + 2t^2 + 12, -2t^3 + 6t + 2 \rangle$
$i = 26, D_i = 512, \rho_{ij}: 3:2, 4:2, 11:1, 12:2, 14:1, 20:1, 27:2, 29:1, 30:1, 34:2, 36:2, 40:2, 43:1, 47:2, 57:1, 59:2, 61:2, 62:1, 64:1, 65:2$
$w_i \quad \langle -79531t^3 + 105647t^2 - 32086t + 95594, -139118t^3 - 498341t^2 - 766412t - 236417 \rangle$
$i = 27, D_i = 16, \rho_{ij}: 1:2, 8:2, 17:2, 24:2, 25:1, 26:1, 39:2, 47:2, 49:2, 53:1, 55:1, 56:1, 60:1, 63:1$
$w_i \quad \langle -4385t^3 - 14782t^2 - 6551t - 1549, 2832t^3 - 14445t^2 - 20294t - 10202 \rangle$
$i = 28, D_i = 16, \rho_{ij}: 3:1, 11:1, 12:1, 15:2, 20:1, 22:1, 24:2, 31:2, 32:1, 33:1, 40:1, 47:1, 54:2, 55:1, 56:1, 61:2, 67:1$
$w_i \quad \langle 38127t^3 + 78303t^2 + 67689t - 1413, 6399t^3 + 39126t^2 + 74883t + 35577 \rangle$
$i = 29, D_i = 8, \rho_{ij}: 3:1, 8:2, 9:2, 16:1, 26:2, 36:2, 43:2, 59:2, 64:1$
$w_i \quad \langle 167t^3 + 281t^2 + 184t + 108, 19t^3 - 128t^2 - 316t - 183 \rangle$
$i = 30, D_i = 64, \rho_{ij}: 0:2, 3:1, 8:1, 11:2, 13:1, 16:2, 19:2, 20:2, 21:1, 22:1, 26:2, 32:1, 33:1, 40:1, 45:2, 49:1, 56:1, 59:1, 61:1, 63:1, 65:2, 66:1, 67:1$
$w_i \quad \langle -63781095t^3 - 97367094t^2 - 38875947t + 39268575, -17164455t^3 + 17538057t^2 + 85210863t + 63985293 \rangle$
$i = 31, D_i = 4, \rho_{ij}: 5:2, 9:1, 10:2, 14:1, 16:2, 19:2, 25:2, 28:1, 34:2, 39:2, 53:1, 60:1, 62:2$
$w_i \quad \langle 129t^3 - 94t^2 - 137t - 91, -67t^3 - 87t^2 + 45t + 7 \rangle$
$i = 32, D_i = 8, \rho_{ij}: 10:2, 14:1, 23:2, 28:2, 30:2, 33:2, 35:1, 39:2, 41:1, 60:1$
$w_i \quad \langle 64344t^3 - 5208t^2 + 18144t + 24192, 0 \rangle$
$i = 33, D_i = 8, \rho_{ij}: 28:2, 30:2, 32:1, 40:1, 45:1, 53:2, 54:2$
$w_i \quad \langle 2016t^3 + 2576t^2 + 1372t - 196, 1624t^3 + 2128t^2 + 1148t - 196 \rangle$
$i = 34, D_i = 8, \rho_{ij}: 0:1, 8:2, 20:2, 24:2, 26:1, 31:1, 51:1, 58:2$
$w_i \quad \langle -56t^3 - 168t^2 - 96t - 76, -24t^3 - 88t^2 - 116t - 60 \rangle$
$i = 35, D_i = 96, e_i = 1, \rho_{ij}: 1:1, 7:2, 11:2, 19:1, 32:2, 38:2, 42:1, 47:2, 50:2, 59:2, 65:1$
$w_i \quad \langle -426303t^3 - 259848t^2 - 30456t + 455139, 584253t^3 - 15228t^2 - 140130t + 317601 \rangle$

$$\begin{aligned}
i = 36, D_i = 288, e_i = 0, \rho_{ij}: & 5:2, 16:1, 20:2, 23:1, 26:1, 29:1, 37:2, 39:1, 40:1, 47:1, 50:2, 51:1, 54:2, 55:2, 62:1 \\
w_i & \langle -83241108t^3 - 4318596t^2 + 16367022t + 20276406, \\
& -70206831t^3 - 7753644t^2 - 2670570t - 30596859 \rangle \\
i = 37, D_i = 8, \rho_{ij}: & 3:2, 6:2, 8:2, 13:1, 17:2, 19:1, 22:2, 24:1, 25:2, 36:1, 41:2, 43:1, 46:1, 52:2, 61:2, 64:2, 65:1, 66:2 \\
w_i & \langle 6664t^5 + 21504t^4 + 4984t^3 + 8344t^2 + 18816t + 29736, \\
& -3136t^5 - 2688t^4 + 1064t^3 + 392t^2 - 5880t - 4536 \rangle \\
i = 38, D_i = 4, \rho_{ij}: & 0:1, 1:2, 4:1, 12:1, 13:2, 16:1, 17:1, 23:2, 35:1, 47:1, 51:1, 61:1, 65:1 \\
w_i & \langle -12t^5 + 8t^4 + 72t^3 + 56t^2 + 8t - 60, \\
& -12t^5 + 4t^4 + 12t^3 + 76t^2 + 88t + 24 \rangle \\
i = 39, D_i = 2048, \rho_{ij}: & 2:2, 3:2, 16:1, 20:1, 23:2, 24:2, 27:1, 31:1, 32:1, 36:2, 40:1, 41:1, 47:1, 50:1 \\
w_i & \langle -643072t^5 + 860160t^4 - 307200t^3 + 28672t^2 - 61440t + 565248, \\
& -401408t^5 + 1155072t^4 - 221184t^3 + 229376t^2 + 245760t + 761856 \rangle \\
i = 40, D_i = 65536, \rho_{ij}: & 0:1, 11:2, 16:2, 22:1, 24:1, 26:1, 28:2, 30:2, 33:2, 36:2, 39:2, 49:2, 51:2, 54:2, 63:1, 65:1 \\
w_i & \langle -1753350144t^5 + 132317184t^4 + 868024320t^3 + 390463488t^2 - 221184000t - 1533542400, \\
& -157286400t^5 + 2913533952t^4 + 204668928t^3 + 1207173120t^2 + 750256128t + 879427584 \rangle \\
i = 41, D_i = 512, \rho_{ij}: & 3:1, 15:2, 20:2, 22:1, 24:1, 32:2, 37:1, 39:2, 67:1, 68:2 \\
w_i & \langle -12288t^5 - 29696t^4 - 26112t^3 - 22016t^2 - 21504t - 17408, \\
& -12288t^5 - 29696t^4 - 26112t^3 - 22016t^2 - 21504t - 17408 \rangle \\
i = 42, D_i = 32, \rho_{ij}: & 5:2, 11:2, 15:2, 19:1, 35:2, 47:1 \\
w_i & \langle -96t^5 - 384t^4 - 288t^3 + 288t + 96, \\
& -384t^4 - 576t^3 - 288t^2 - 192t - 96 \rangle \\
i = 43, D_i = 131072, \rho_{ij}: & 1:1, 2:2, 4:2, 14:2, 20:2, 21:2, 23:1, 26:2, 29:1, 37:2, 49:2, 56:2, 63:2, 65:2, 68:1 \\
w_i & \langle 1899102208t^5 + 2422865920t^4 + 1414266880t^3 + 725352448t^2 - 276430848t + 167378944, \\
& 1214382080t^5 + 1161428992t^4 + 3852599296t^3 + 3562536960t^2 + 1635516416t - 484835328 \rangle \\
i = 44, D_i = 512, \rho_{ij}: & 4:2, 9:1, 11:1, 13:1, 14:2, 20:2, 24:1, 49:2, 50:2, 51:1, 54:1, 67:1, 68:1 \\
w_i & \langle 200704t^5 + 213504t^4 + 483840t^3 + 429568t^2 + 175104t - 49152, \\
& 237056t^5 + 316416t^4 + 81408t^3 - 8704t^2 - 95232t + 38400 \rangle \\
i = 45, D_i = 4, \rho_{ij}: & 2:1, 3:2, 8:2, 14:2, 25:2, 30:1, 33:2, 49:1, 52:1, 54:2, 57:2, 58:2, 63:1 \\
w_i & \langle 768t^5 + 796t^4 - 228t^3 - 2532t^2 - 644t + 624, \\
& 660t^5 - 568t^4 - 1548t^3 - 1452t^2 + 164t + 1800 \rangle \\
i = 46, D_i = 4, \rho_{ij}: & 2:2, 6:2, 11:2, 12:1, 19:2, 21:1, 37:2, 54:2, 57:2, 66:1, 67:1 \\
w_i & \langle -388t^5 + 1040t^4 + 1748t^3 + 636t^2 - 536t - 1956, \\
& -848t^5 - 344t^4 + 928t^3 + 2592t^2 + 344t - 1296 \rangle \\
i = 47, D_i = 262144, \rho_{ij}: & 3:2, 5:1, 8:1, 16:1, 17:1, 18:2, 22:2, 26:1, 27:1, 28:2, 35:1, 36:2, 38:2, 39:2, 42:2, 50:2, 61:1, 62:1 \\
w_i & \langle -516947968t^5 - 998244352t^4 - 634388480t^3 + 853540864t^2 + 1365245952t + 799014912, \\
& -241172480t^5 - 1197473792t^4 - 1866465280t^3 - 1422917632t^2 - 116391936t + 220200960 \rangle \\
i = 48, D_i = 1, \rho_{ij}: & 6:2, 9:1, 15:2, 51:2 \\
w_i & \langle 1, \\
& -t - 1 \rangle \\
i = 49, D_i = 64, \rho_{ij}: & 2:2, 4:1, 10:1, 12:2, 17:2, 19:2, 21:2, 25:2, 27:1, 30:2, 40:1, 43:1, 44:1, 45:2, 58:2, 61:2, 62:1, 65:2, 68:1 \\
w_i & \langle 12186t^3 + 23304t^2 + 16932t + 3042, \\
& 35004t^3 + 9648t^2 - 20712t - 28692 \rangle \\
i = 50, D_i = 4, \rho_{ij}: & 5:1, 9:2, 10:2, 11:1, 16:2, 35:1, 36:1, 39:2, 44:1, 47:1 \\
w_i & \langle -108t^3 + 30t^2 + 208t + 150, \\
& -246t^3 - 420t^2 - 400t - 174 \rangle \\
i = 51, D_i = 8, \rho_{ij}: & 14:2, 17:1, 23:2, 34:2, 36:2, 38:2, 40:1, 44:2, 48:1, 66:2, 67:1, 68:2 \\
w_i & \langle 672t^5 + 1320t^4 + 24t^3 + 1104t^2 + 96t + 336, \\
& -192t^5 + 288t^4 - 960t^3 - 1080t^2 - 1872t - 2040 \rangle \\
i = 52, D_i = 4, \rho_{ij}: & 3:1, 5:1, 6:1, 8:1, 13:2, 15:1, 21:1, 23:2, 37:1, 45:2, 54:2 \\
w_i & \langle -24t^5 + 52t^4 - 40t^3 - 20t^2 + 18t + 6, \\
& -12t^5 + 26t^4 - 20t^3 - 10t^2 + 9t + 3 \rangle \\
i = 53, D_i = 2, \rho_{ij}: & 0:1, 8:2, 11:2, 19:1, 27:2, 31:2, 33:1, 58:1, 68:1 \\
w_i & \langle 22t^3 - 26t^2 + 58t + 18, \\
& 26t^3 - 4t^2 + 38t + 6 \rangle \\
i = 54, D_i = 16, \rho_{ij}: & 7:1, 9:1, 10:1, 11:1, 22:1, 28:1, 33:1, 36:1, 40:1, 44:2, 45:1, 46:1, 52:1, 55:2, 63:1, 67:2 \\
w_i & \langle -5264t^3 - 13472t^2 + 3398t + 7322, \\
& 482t^3 - 11830t^2 - 3560t + 3892 \rangle \\
i = 55, D_i = 2, \rho_{ij}: & 3:2, 11:2, 12:2, 13:1, 27:2, 28:2, 36:1, 54:1, 57:2, 63:1, 64:1 \\
w_i & \langle -2055t^3 - 1083t^2 + 3462t + 4668, \\
& -3357t^3 - 6246t^2 - 4362t + 921 \rangle \\
i = 56, D_i = 4, \rho_{ij}: & 0:1, 4:1, 8:2, 12:2, 24:1, 27:2, 28:2, 30:2, 43:1, 59:2 \\
w_i & \langle -16t^4 + 28t^3 + 32t^2 + 8t - 52, \\
& -20t^4 + 28t^3 - 52t^2 + 12t - 8 \rangle \\
i = 57, D_i = 16, \rho_{ij}: & 3:2, 13:2, 22:1, 26:2, 45:1, 46:1, 55:1 \\
w_i & \langle 32t^5 + 64t^4 + 32t^3 + 96t^2 + 96t + 64, \\
& -32t^5 - 32t^4 - 32t^3 + 96t^2 + 96t + 32 \rangle \\
i = 58, D_i = 1, \rho_{ij}: & 5:1, 7:1, 34:1, 45:1, 49:1, 53:2, 62:2, 63:2 \\
w_i & \langle 2t^3 + 62t^2 - 14t - 26,
\end{aligned}$$

$$12t^3 + 36t^2 - 34t - 10)$$

$i = 59, D_i = 512, \rho_{ij}: 0:2, 1:2, 9:2, 11:2, 20:2, 22:1, 24:1, 26:1, 29:1, 30:2, 35:1, 56:1$
 $w_i \quad \langle 27136t^5 + 6144t^4 - 68096t^3 + 19968t^2 + 68608t + 56832,$
 $\quad 45056t^5 - 21504t^4 - 53248t^3 - 12288t^2 + 11264t - 18432 \rangle$

$i = 60, D_i = 2, \rho_{ij}: 17:2, 19:2, 21:1, 27:2, 31:2, 32:2$
 $w_i \quad \langle -14t^2 + 84t + 420,$
 $\quad 0 \rangle$

$i = 61, D_i = 8192, \rho_{ij}: 0:1, 3:2, 16:2, 18:1, 23:1, 24:1, 26:1, 28:1, 30:2, 37:1, 38:2, 47:2, 49:1, 63:1, 65:1$
 $w_i \quad \langle 704512t^5 - 2277376t^4 + 589824t^3 + 253952t^2 - 1359872t - 1499136,$
 $\quad 1056768t^5 - 1359872t^4 + 286720t^3 + 475136t^2 + 204800t - 811008 \rangle$

$i = 62, D_i = 2, \rho_{ij}: 2:2, 16:1, 18:2, 23:2, 25:1, 26:2, 31:1, 36:2, 47:2, 49:2, 58:1$
 $w_i \quad \langle 728t^2 + 336t - 3024,$
 $\quad -56t^2 - 168t - 6048 \rangle$

$i = 63, D_i = 8, \rho_{ij}: 3:1, 8:1, 16:1, 18:1, 22:1, 24:2, 27:2, 30:2, 40:2, 43:1, 45:2, 54:2, 55:2, 58:1, 61:2, 64:1$
 $w_i \quad \langle -244t^3 + 3772t^2 + 1720t - 5760,$
 $\quad 934t^3 + 410t^2 - 706t - 7536 \rangle$

$i = 64, D_i = 128, \rho_{ij}: 2:2, 13:1, 14:2, 15:1, 16:2, 17:2, 22:2, 24:1, 26:2, 29:2, 37:1, 55:2, 63:2, 66:1, 67:1$
 $T \quad \langle 1980y^2 - 2535y - 4025,$
 $\quad 2860y^2 + 3875y + 245 \rangle$

$U \quad \langle 16777216x^{17}y^4 + 96468992x^{17}y^3 + 186646528x^{17}y^2 + 150994944x^{17}y + 44040192x^{17}$
 $\quad -46137344x^{16}y^5 - 218103808x^{16}y^4 - 356515840x^{16}y^3 - 306184192x^{16}y^2$
 $\quad -184549376x^{16}y - 62914560x^{16} + 94371840x^{15}y^6 + 425721856x^{15}y^5$
 $\quad +402653184x^{15}y^4 - 840957952x^{15}y^3 - 2313158656x^{15}y^2 - 2055208960x^{15}y$
 $\quad -654311424x^{15} - 150994944x^{14}y^7 - 660602880x^{14}y^6 - 482344960x^{14}y^5$
 $\quad +2063597568x^{14}y^4 + 5559549952x^{14}y^3 + 6254755840x^{14}y^2 + 3590324224x^{14}y$
 $\quad +858783744x^{14} + 226492416x^{13}y^8 + 950009856x^{13}y^7 - 178257920x^{13}y^6$
 $\quad -6934233088x^{13}y^5 - 13784580096x^{13}y^4 - 10387193856x^{13}y^3 - 1129316352x^{13}y^2$
 $\quad +2528116736x^{13}y + 1022361600x^{13} - 320864256x^{12}y^9 - 1256194048x^{12}y^8$
 $\quad +815792128x^{12}y^7 + 10658775040x^{12}y^6 + 20943208448x^{12}y^5 + 18442354688x^{12}y^4$
 $\quad +1693450240x^{12}y^3 - 13894680576x^{12}y^2 - 12736004096x^{12}y - 3554672640x^{12}$
 $\quad +383778816x^{11}y^{10} + 1414529024x^{11}y^9 - 1714421760x^{11}y^8 - 15454961664x^{11}y^7$
 $\quad -30557601792x^{11}y^6 - 24486346752x^{11}y^5 + 6397362176x^{11}y^4 + 32332840960x^{11}y^3$
 $\quad +31185174528x^{11}y^2 + 15623782400x^{11}y + 3735552000x^{11} - 427819008x^{10}y^{11}$
 $\quad -1627389952x^{10}y^{10} + 1678770176x^{10}y^9 + 17559453696x^{10}y^8 + 39623589888x^{10}y^7$
 $\quad +41184919552x^{10}y^6 + 1585446912x^{10}y^5 - 53999566848x^{10}y^4 - 68973232128x^{10}y^3$
 $\quad -36365664256x^{10}y^2 - 6193938432x^{10}y + 541065216x^{10} + 434110464x^9y^{12}$
 $\quad +1736441856x^9y^{11} - 923795456x^9y^{10} - 16682844160x^9y^9 - 42760404992x^9y^8$
 $\quad -50635735040x^9y^7 - 9135194112x^9y^6 + 66293596160x^9y^5 + 102823886848x^9y^4$
 $\quad +61119922176x^9y^3 + 2476212224x^9y^2 - 14007926784x^9y - 5091360768x^9$
 $\quad -434110464x^8y^{13} - 1800404992x^8y^{12} + 292552704x^8y^{11} + 13248757760x^8y^{10}$
 $\quad +39788216320x^8y^9 + 66090696704x^8y^8 + 29607591936x^8y^7 - 90532478976x^8y^6$
 $\quad -156116189184x^8y^5 - 83912294400x^8y^4 + 14194049024x^8y^3 + 48091103232x^8y^2$
 $\quad +32023511040x^8y + 8170242048x^8 + 396361728x^7y^{14} + 1754267648x^7y^{13}$
 $\quad +1431830528x^7y^{12} - 3763863552x^7y^{11} - 26921140224x^7y^{10} - 63717244928x^7y^9$
 $\quad -20894973952x^7y^8 + 101532565504x^7y^7 + 153298665472x^7y^6 + 99469230080x^7y^5$
 $\quad -10122952704x^7y^4 - 96499400704x^7y^3 - 88460230656x^7y^2 - 37784125440x^7y$
 $\quad -7736131584x^7 - 320864256x^6y^{15} - 1681915904x^6y^{14} - 2906652672x^6y^{13}$
 $\quad -2226651136x^6y^{12} + 13410238464x^6y^{11} + 32044482560x^6y^{10} - 13446938624x^6y^9$
 $\quad -91510538240x^6y^8 - 143227617280x^6y^7 - 131739418624x^6y^6 + 7847804928x^6y^5$
 $\quad +146862768128x^6y^4 + 159365988352x^6y^3 + 94927847424x^6y^2 + 26329743360x^6y$
 $\quad +375128064x^6 + 257949696x^5y^{16} + 1615855616x^5y^{15} + 4397727744x^5y^{14}$
 $\quad +7506231296x^5y^{13} + 3827826688x^5y^{12} + 11751129088x^5y^{11} + 41164996608x^5y^{10}$
 $\quad +60435988480x^5y^9 + 102377193472x^5y^8 + 110153695232x^5y^7 + 1723596800x^5y^6$
 $\quad -135319519232x^5y^5 - 212799062016x^5y^4 - 157527572480x^5y^3 - 37503762432x^5y^2$
 $\quad +9533915136x^5y + 4703256576x^5 - 195035136x^4y^{17} - 1360003072x^4y^{16}$
 $\quad -4506255360x^4y^{15} - 9432465408x^4y^{14} - 19044761600x^4y^{13} - 47942205440x^4y^{12}$
 $\quad -55351967744x^4y^{11} - 41144025088x^4y^{10} - 66016247808x^4y^9 - 65819115520x^4y^8$
 $\quad +9604169728x^4y^7 + 162379333632x^4y^6 + 263113146368x^4y^5 + 167090847744x^4y^4$
 $\quad +20804009984x^4y^3 - 45415661568x^4y^2 - 39003881472x^4y - 8953528320x^4$
 $\quad +150994944x^3y^{18} + 1053818880x^3y^{17} + 3711434752x^3y^{16} + 10799546368x^3y^{15}$
 $\quad +24019992576x^3y^{14} + 37952946176x^3y^{13} + 36564893696x^3y^{12} + 38745145344x^3y^{11}$
 $\quad +23736877056x^3y^{10} - 9098231808x^3y^9 - 53860106240x^3y^8 - 164130193408x^3y^7$
 $\quad -210115100672x^3y^6 - 136507686912x^3y^5 - 7947419648x^3y^4 + 116322598912x^3y^3$
 $\quad +116953055232x^3y^2 + 49946296320x^3y + 11869618176x^3 - 94371840x^2y^{19}$
 $\quad -756023296x^2y^{18} - 2971664384x^2y^{17} - 9625927680x^2y^{16} - 17586454528x^2y^{15}$
 $\quad -20809515008x^2y^{14} - 28443148288x^2y^{13} - 8565293056x^2y^{12} + 50653298688x^2y^{11}$
 $\quad +73066086400x^2y^{10} + 117670936576x^2y^9 + 180723908608x^2y^8 + 172029378560x^2y^7$
 $\quad +110590296064x^2y^6 - 57451216896x^2y^5 - 217695911936x^2y^4 - 226622308352x^2y^3$
 $\quad -143203172352x^2y^2 - 46374322176x^2y - 2914320384x^2 + 56623104xy^{20} + 514850816xy^{19}$
 $\quad +1826095104xy^{18} + 4719640576xy^{17} + 6681264128xy^{16} + 3553624064xy^{15} - 1618870272xy^{14}$
 $\quad -36394762240xy^{13} - 70318161920xy^{12} - 99797696512xy^{11} - 158550458368xy^{10}$

$$\begin{aligned}
& -139702304768xy^9 - 94869651456xy^8 - 49168908288xy^7 + 69795577856xy^6 + 217864404992xy^5 \\
& + 329279340544xy^4 + 252689383424xy^3 + 77877018624xy^2 + 3302424576xy - 2578120704x \\
& - 20971520y^{21} - 5242880y^{20} + 501743616y^{19} + 2106064896y^{18} + 6703022080y^{17} \\
& + 11826626560y^{16} + 7173242880y^{15} - 11482038272y^{14} - 64306790400y^{13} - 147882815488y^{12} \\
& - 247765291008y^{11} - 294541817856y^{10} - 223054084608y^9 - 154393403904y^8 - 142591716864y^7 \\
& - 92942141824y^6 + 41948354816y^5 + 91115426016y^4 - 54970230344y^3 - 75679076744y^2 \\
& + 38603793640y + 34444686984, \\
& 2097152x^{17}y^4 + 41943040x^{17}y^3 + 127926272x^{17}y^2 + 138412032x^{17}y + 50331648x^{17} \\
& + 4194304x^{16}y^5 + 12582912x^{16}y^4 + 184549376x^{16}y^3 + 549453824x^{16}y^2 \\
& + 574619648x^{16}y + 201326592x^{16} - 23068672x^{15}y^6 - 111149056x^{15}y^5 \\
& - 627048448x^{15}y^4 - 1883242496x^{15}y^3 - 2774532096x^{15}y^2 - 1990197248x^{15}y \\
& - 559939584x^{15} + 60817408x^{14}y^7 + 440401920x^{14}y^6 + 1941962752x^{14}y^5 \\
& + 4421844992x^{14}y^4 + 4634705920x^{14}y^3 + 1242562560x^{14}y^2 - 1224736768x^{14}y \\
& - 692060160x^{14} - 111149056x^{13}y^8 - 852492288x^{13}y^7 - 3812622336x^{13}y^6 \\
& - 9495904256x^{13}y^5 - 10354688000x^{13}y^4 + 927989760x^{13}y^3 + 11889803264x^{13}y^2 \\
& + 9258926080x^{13}y + 2227175424x^{13} + 174063616x^{12}y^9 + 1382023168x^{12}y^8 \\
& + 6322913280x^{12}y^7 + 14364442624x^{12}y^6 + 10868490240x^{12}y^5 - 10574888960x^{12}y^4 \\
& - 25501368320x^{12}y^3 - 20043530240x^{12}y^2 - 8809086976x^{12}y - 2073034752x^{12} \\
& - 255852544x^{11}y^{10} - 2012217344x^{11}y^9 - 9157214208x^{11}y^8 - 20676870144x^{11}y^7 \\
& - 16047407104x^{11}y^6 + 18216910848x^{11}y^5 + 49922179072x^{11}y^4 + 42091937792x^{11}y^3 \\
& + 13228310528x^{11}y^2 - 467664896x^{11}y - 537919488x^{11} + 325058560x^{10}y^{11} \\
& + 2446327808x^{10}y^{10} + 10940841984x^{10}y^9 + 24274534400x^{10}y^8 + 18630049792x^{10}y^7 \\
& - 21227372544x^{10}y^6 - 59291729920x^{10}y^5 - 54928605184x^{10}y^4 - 22293774336x^{10}y^3 \\
& + 6040846336x^{10}y^2 + 14548992000x^{10}y + 6253707264x^{10} - 369098752x^9y^{12} \\
& - 2811232256x^9y^{11} - 12251037696x^9y^{10} - 27860664320x^9y^9 - 26736066560x^9y^8 \\
& + 13767802880x^9y^7 + 66536341504x^9y^6 + 77390675968x^9y^5 + 36248748032x^9y^4 \\
& - 22776119296x^9y^3 - 51371835392x^9y^2 - 32688832512x^9y - 7035420672x^9 \\
& + 369098752x^8y^{13} + 2681208832x^8y^{12} + 12177113088x^8y^{11} + 29565648896x^8y^{10} \\
& + 33615249408x^8y^9 + 8722055168x^8y^8 - 36822319104x^8y^7 - 91099234304x^8y^6 \\
& - 83238060032x^8y^5 + 26291208192x^8y^4 + 109025689600x^8y^3 + 82461327360x^8y^2 \\
& + 26685210624x^8y + 3189768192x^8 - 343932928x^7y^{14} - 2378170368x^7y^{13} \\
& - 11138498560x^7y^{12} - 26468679680x^7y^{11} - 34340339712x^7y^{10} - 32877051904x^7y^9 \\
& + 19226165248x^7y^8 + 126548705280x^7y^7 + 110979710976x^7y^6 - 50905743360x^7y^5 \\
& - 156475064320x^7y^4 - 135372996608x^7y^3 - 55359307776x^7y^2 - 6039797760x^7y \\
& - 42467328x^7 + 293601280x^6y^{15} + 1831862272x^6y^{14} + 8721530880x^6y^{13} \\
& + 22290628608x^6y^{12} + 37440454656x^6y^{11} + 42680975360x^6y^{10} - 29533143040x^6y^9 \\
& - 139250106368x^6y^8 - 109166723072x^6y^7 + 32805748736x^6y^6 + 158121328640x^6y^5 \\
& + 177996038144x^6y^4 + 81622728704x^6y^3 + 2689597440x^6y^2 - 14998044672x^6y \\
& - 8543010816x^6 - 211812352x^5y^{16} - 1210056704x^5y^{15} - 5571608576x^5y^{14} \\
& - 14416084992x^5y^{13} - 27243053056x^5y^{12} - 18698731520x^5y^{11} + 59057111040x^5y^{10} \\
& + 125436690432x^5y^9 + 86632824832x^5y^8 - 31531204608x^5y^7 - 163609051136x^5y^6 \\
& - 179082100736x^5y^5 - 84346535936x^5y^4 + 15542779904x^5y^3 + 71146536960x^5y^2 \\
& + 50394562560x^5y + 10581442560x^5 + 130023424x^4y^{17} + 694157312x^4y^{16} \\
& + 3073900544x^4y^{15} + 8151105536x^4y^{14} + 13098287104x^4y^{13} - 17907318784x^4y^{12} \\
& - 87031021568x^4y^{11} - 11057024048x^4y^{10} - 81553522688x^4y^9 + 11660689408x^4y^8 \\
& + 1246232781888x^4y^7 + 172625756160x^4y^6 + 134160580608x^4y^5 - 41339846656x^4y^4 \\
& - 172494159872x^4y^3 - 131148546048x^4y^2 - 46945271808x^4y - 5103157248x^4 \\
& - 60817408x^3y^{18} - 184549376x^3y^{17} - 1205862400x^3y^{16} - 1363935232x^3y^{15} \\
& + 8168144896x^3y^{14} + 39259996160x^3y^{13} + 73571237888x^3y^{12} + 96013647872x^3y^{11} \\
& + 8222130456x^3y^{10} - 15371599872x^3y^9 - 114916589568x^3y^8 - 186125123584x^3y^7 \\
& - 124210249728x^3y^6 + 119736500224x^3y^5 + 259725066240x^3y^4 + 220556951552x^3y^3 \\
& + 103890419712x^3y^2 + 19052494848x^3y + 4391829504x^3 + 23068672x^2y^{19} \\
& - 27262976x^2y^{18} + 47710208x^2y^{17} - 3369861120x^2y^{16} - 17456168960x^2y^{15} \\
& - 38979239936x^2y^{14} - 71780270080x^2y^{13} - 96507789312x^2y^{12} - 42405462016x^2y^{11} \\
& + 37837864960x^2y^{10} + 111058092032x^2y^9 + 192813989888x^2y^8 + 107623088128x^2y^7 \\
& - 112936878080x^2y^6 - 276531511296x^2y^5 - 290118893568x^2y^4 - 154269122560x^2y^3 \\
& - 38960037888x^2y^2 + 1061683200x^2y + 7523794944x^2 + 2097152xy^{20} + 141557760xy^{19} \\
& + 894435328xy^{18} + 4103864320xy^{17} + 14377156608xy^{16} + 29165617152xy^{15} + 49708662784xy^{14} \\
& + 41423470592xy^{13} - 9808642048xy^{12} - 44073746432xy^{11} - 124173942784xy^{10} \\
& - 163105734656xy^9 - 43645140992xy^8 + 132335468544xy^7 + 264714977280xy^6 \\
& + 236265668608xy^5 + 138550116352xy^4 + 33185923072xy^3 - 65210351616xy^2 - 57353306112xy \\
& - 13320585216x - 12582912y^{21} - 82837504y^{20} + 93847552y^{19} + 1245708288y^{18} + 6073876480y^{17} \\
& + 21218852864y^{16} + 46055587840y^{15} + 80701128704y^{14} + 99215458304y^{13} + 64453406720y^{12} \\
& - 35377221632y^{11} - 194133082112y^{10} - 288058756096y^9 - 273028360192y^8 - 194987303168y^7 \\
& - 27664374592y^6 + 219170870592y^5 + 386819759456y^4 + 181225809704y^3 - 136273010904y^2 \\
& - 130363879816y - 22158199080) \\
V \quad & (-16777216x^3y^{21} - 270532608x^3y^{20} - 1451229184x^3y^{19} - 3345743872x^3y^{18} \\
& - 6540754944x^3y^{17} - 5923405824x^3y^{16} + 18849267712x^3y^{15} + 78853832704x^3y^{14} \\
& + 159232278528x^3y^{13} + 276458418176x^3y^{12} + 389782851584x^3y^{11} \\
& + 384576485376x^3y^{10} + 260735580672x^3y^9 + 130326225408x^3y^8 + 24111934976x^3y^7 \\
& - 210302957184x^3y^6 - 445962262784x^3y^5 - 340550029536x^3y^4 - 11459221944x^3y^3 \\
& + 103469027720x^3y^2 - 594355432x^3y - 26949203592x^3 + 20971520x^2y^{22}
\end{aligned}$$

$ \begin{aligned} &+97517568x^2y^{21} + 416284672x^2y^{20} + 306184192x^2y^{19} - 2083782656x^2y^{18} \\ &-12028477440x^2y^{17} - 47473164288x^2y^{16} - 111756787712x^2y^{15} \\ &-184002461696x^2y^{14} - 261059092480x^2y^{13} - 350782040064x^2y^{12} \\ &-358291834880x^2y^{11} - 190138718208x^2y^{10} - 1921078272x^2y^9 + 121171456512x^2y^8 \\ &+270284628608x^2y^7 + 408543388704x^2y^6 + 430825901664x^2y^5 + 202697802288x^2y^4 \\ &-94914632156x^2y^3 - 80093479196x^2y^2 + 39830822316x^2y + 24410446620x^2 \\ &-10485760xy^{23} - 17825792xy^{22} + 99090432xy^{21} + 639369216xy^{20} + 5246156800xy^{19} \\ &+18380423168xy^{18} + 43421466624xy^{17} + 90707165184xy^{16} + 144634945536xy^{15} \\ &+179021848576xy^{14} + 213587304448xy^{13} + 165511444480xy^{12} - 1793617920xy^{11} \\ &-190334547456xy^{10} - 370531626368xy^9 - 425590838400xy^8 - 387454363200xy^7 \\ &-339567492432xy^6 - 168283587280xy^5 + 86051162464xy^4 + 222174865940xy^3 \\ &+104250356868xy^2 - 34419331284xy - 24091680708x - 176160768y^{22} - 928514048y^{21} \\ &-4247781376y^{20} - 12614762496y^{19} - 24925863936y^{18} - 46156251136y^{17} - 62093295616y^{16} \\ &-48171892736y^{15} + 14385356800y^{14} + 156462458880y^{13} + 347757568512y^{12} \\ &+588734559744y^{11} + 772314578688y^{10} + 692996210752y^9 + 452832864832y^8 + 218771453568y^7 \\ &+1935365536y^6 - 311716320016y^5 - 579391985700y^4 - 392686553237y^3 - 32148727085y^2 \\ &+45134237833y + 4992094221, \\ &-2097152x^3y^{21} - 38797312x^3y^{20} - 1067974656x^3y^{19} - 4631298048x^3y^{18} \\ &-14770503680x^3y^{17} - 39465648128x^3y^{16} - 69133434880x^3y^{15} - 72240431104x^3y^{14} \\ &-43260297216x^3y^{13} + 32058052608x^3y^{12} + 190683648000x^3y^{11} \\ &+332216254464x^3y^{10} + 326489459712x^3y^9 + 178794670080x^3y^8 + 22007521536x^3y^7 \\ &-169166120128x^3y^6 - 407606465856x^3y^5 - 356522794336x^3y^4 + 33890605272x^3y^3 \\ &+304353386712x^3y^2 + 193807118728x^3y + 27617020200x^3 + 12582912x^2y^{22} \\ &+94371840x^2y^{21} + 662175744x^2y^{20} + 2732851200x^2y^{19} + 6958481408x^2y^{18} \\ &+15333195776x^2y^{17} + 15213592576x^2y^{16} - 16733323264x^2y^{15} - 77257916416x^2y^{14} \\ &-155530821632x^2y^{13} - 279406960640x^2y^{12} - 419063132160x^2y^{11} \\ &-397101145088x^2y^{10} - 211730072320x^2y^9 - 39232619776x^2y^8 + 142159083648x^2y^7 \\ &+266354118048x^2y^6 + 211168499488x^2y^5 - 14777247968x^2y^4 - 318774566800x^2y^3 \\ &-349793065264x^2y^2 - 127911880272x^2y - 10683550032x^2 - 6291456xy^{23} - 74448896xy^{22} \\ &-195559424xy^{21} - 1083703296xy^{20} - 1910243328xy^{19} + 3513319424xy^{18} + 20400242688xy^{17} \\ &+62678499328xy^{16} + 141662732288xy^{15} + 228935565312xy^{14} + 338993926144xy^{13} \\ &+430612448256xy^{12} + 363199314944xy^{11} + 198239296000xy^{10} - 21022446720xy^9 \\ &-224127530624xy^8 - 232352959872xy^7 - 87699642592xy^6 + 139980205376xy^5 \\ &+348263609176xy^4 + 437373443254xy^3 + 305457663270xy^2 + 67362759762xy - 11488848102x \\ &+19922944y^{23} + 27787264y^{22} - 74973184y^{21} - 1444937728y^{20} - 10093592576y^{19} \\ &-30399660032y^{18} - 72500576256y^{17} - 145953505280y^{16} - 234520260608y^{15} \\ &-317835563008y^{14} - 322709493760y^{13} - 217610507776y^{12} - 13692427776y^{11} \\ &+294414183936y^{10} + 471608551168y^9 + 342521633152y^8 + 7374809696y^7 - 304502376264y^6 \\ &-486332260312y^5 - 606308748556y^4 - 430900475641y^3 - 10225324137y^2 + 141400477445y \\ &+48850241769) \end{aligned} $					
$i = 65$,	$D_i = 8$,	$\rho_{ij}:$	11:1, 19:2, 23:1, 26:1, 30:1, 35:2, 37:2, 38:2, 40:2, 43:1, 49:1, 61:2		
w_i	$ \begin{aligned} &\langle 420t^3 - 390t^2 + 606t + 1650, \\ &-144t^3 - 612t^2 + 1056t + 312 \rangle \end{aligned} $				
$i = 66$,	$D_i = 2$,	$\rho_{ij}:$	6:1, 16:1, 23:2, 30:2, 37:1, 46:2, 51:1, 64:2, 67:1		
w_i	$ \begin{aligned} &\langle 65t^3 - 46t^2 + 23t + 73, \\ &-39t^3 + 127t^2 - 151t + 71 \rangle \end{aligned} $				
$i = 67$,	$D_i = 32$,	$\rho_{ij}:$	6:1, 13:1, 14:2, 19:1, 28:2, 30:2, 41:2, 44:2, 46:2, 51:2, 54:1, 64:2, 66:2		
w_i	$ \begin{aligned} &\langle -2478t^3 + 4134t^2 + 288t + 1416, \\ &-3600t^3 + 1434t^2 + 2424t - 1674 \rangle \end{aligned} $				
$i = 68$,	$D_i = 8$,	$\rho_{ij}:$	0:2, 5:1, 6:2, 14:2, 16:2, 21:2, 23:1, 41:1, 43:2, 44:2, 49:2, 51:1, 53:2		
w_i	$ \begin{aligned} &\langle 232t^5 - 192t^4 - 72t^3 - 184t^2 - 48t - 24, \\ &40t^5 - 32t^4 - 256t^3 - 1160t^2 - 304t - 256 \rangle \end{aligned} $				

Table 6: The edges (i, j) with a linear endpoint, the linear endpoint k used, and the norm of $R_{ij} = \text{Res}_t(f_k, g_{kl})$.

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
<i>The first certificate.</i>					
(0, 20)	0	3^2	(20, 52)	20	$3 \cdot 7^3$
(0, 22)	0	$2^2 \cdot 3^2$	(20, 53)	20	3
(0, 29)	0	$3 \cdot 7 \cdot 13$	(20, 64)	20	$3 \cdot 7$
(0, 30)	0	$3 \cdot 7 \cdot 13$	(20, 66)	20	$7 \cdot 13^2$
(0, 43)	0	$2^6 \cdot 3^2$	(20, 71)	20	$3 \cdot 37$
(0, 48)	0	$2^2 \cdot 3 \cdot 7$	(21, 29)	21	3^3
(0, 49)	0	$2^2 \cdot 3 \cdot 7$	(21, 30)	21	$2^6 \cdot 3^2 \cdot 7$
(0, 60)	0	3^2	(21, 33)	21	$3 \cdot 7$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(0, 71)	0	$2^2 \cdot 13^2$	(21, 41)	21	$2^8 \cdot 3$
(1, 11)	1	$2^2 \cdot 3^4 \cdot 7^3$	(21, 48)	21	$3^4 \cdot 19$
(1, 24)	1	$2^2 \cdot 3$	(21, 49)	21	$3^4 \cdot 7^2 \cdot 13$
(1, 25)	1	$7 \cdot 37$	(21, 55)	21	$2^4 \cdot 3^6 \cdot 13$
(1, 30)	1	$2^2 \cdot 3^2 \cdot 7 \cdot 37$	(21, 57)	21	$2^2 \cdot 3^4$
(1, 35)	1	$2^2 \cdot 3^4 \cdot 7^3$	(21, 59)	21	$3^2 \cdot 19$
(1, 37)	1	37	(22, 44)	22	$3^4 \cdot 43$
(1, 45)	1	$2^2 \cdot 7 \cdot 37$	(22, 51)	22	$31 \cdot 43$
(1, 48)	1	$3^2 \cdot 7$	(22, 58)	22	$3 \cdot 37$
(1, 56)	1	3^{10}	(22, 60)	22	3
(2, 3)	2	7	(22, 66)	22	$2^4 \cdot 13 \cdot 37$
(2, 6)	2	7	(22, 68)	22	$2^4 \cdot 13$
(2, 9)	2	$3^4 \cdot 13$	(22, 71)	22	$2^4 \cdot 31$
(2, 11)	2	$3^4 \cdot 5^2 \cdot 7$	(23, 29)	23	$3^2 \cdot 13 \cdot 19$
(2, 12)	2	2^4	(23, 37)	23	19^2
(2, 17)	2	7	(23, 41)	23	$2^4 \cdot 3^4$
(2, 28)	2	$2^2 \cdot 7 \cdot 13$	(23, 46)	23	$3^2 \cdot 13 \cdot 19$
(2, 33)	2	$2^2 \cdot 7^2$	(23, 49)	23	$2^2 \cdot 3^4$
(2, 36)	2	$3^4 \cdot 13$	(23, 61)	23	$2^4 \cdot 13^2$
(2, 41)	2	1	(23, 71)	23	$2^2 \cdot 3^2$
(2, 43)	2	1	(24, 30)	24	$2^2 \cdot 3 \cdot 7 \cdot 13^2$
(2, 48)	2	13	(24, 32)	24	$3^5 \cdot 13$
(2, 49)	2	13	(24, 37)	24	7^4
(2, 50)	2	1	(24, 42)	24	$7 \cdot 37$
(2, 65)	2	$2^2 \cdot 7^2$	(24, 44)	24	$3^7 \cdot 13$
(2, 70)	2	$2^2 \cdot 5^2$	(24, 45)	24	$2^8 \cdot 13$
(3, 13)	3	$2^2 \cdot 7 \cdot 13$	(24, 50)	24	$2^2 \cdot 3^3 \cdot 7$
(3, 33)	3	$3 \cdot 7$	(24, 60)	24	$3^4 \cdot 7$
(3, 35)	3	$2^2 \cdot 3^4 \cdot 7$	(24, 61)	24	$2^4 \cdot 37^2$
(3, 37)	3	13	(24, 67)	24	$2^4 \cdot 7^2$
(3, 41)	3	$3 \cdot 13$	(24, 70)	24	$2^2 \cdot 5^2 \cdot 19$
(3, 43)	3	$2^2 \cdot 3$	(25, 26)	25	$2^2 \cdot 3^2$
(3, 45)	3	$7 \cdot 13$	(25, 34)	25	$2^2 \cdot 3^5 \cdot 7^3$
(3, 53)	3	3	(25, 38)	25	$3 \cdot 5^2 \cdot 7^2$
(3, 55)	3	$2^4 \cdot 3^3$	(25, 41)	25	$2^4 \cdot 3^2 \cdot 7^2$
(3, 58)	3	3^2	(25, 43)	25	$2^2 \cdot 3^6 \cdot 7^2$
(4, 13)	4	$2^2 \cdot 3$	(25, 45)	25	$2^4 \cdot 5^2 \cdot 37$
(4, 18)	4	2^4	(25, 46)	25	7^3
(4, 38)	4	$2^2 \cdot 7$	(25, 47)	25	7^3
(4, 41)	4	$2^2 \cdot 7^2$	(25, 58)	25	$3 \cdot 5^2 \cdot 7^2$
(4, 46)	4	$3 \cdot 7$	(25, 59)	25	$2^2 \cdot 7^2$
(4, 54)	4	2^2	(25, 60)	25	$2^2 \cdot 3 \cdot 7^3$
(5, 17)	5	$2^2 \cdot 7$	(25, 66)	25	$2^2 \cdot 7^2$
(5, 25)	5	7	(25, 71)	25	$2^2 \cdot 5^4$
(5, 26)	5	$3 \cdot 7$	(26, 29)	26	$2^2 \cdot 3 \cdot 7 \cdot 13^2$
(5, 30)	5	$2^4 \cdot 3 \cdot 7$	(26, 30)	26	$2^2 \cdot 3 \cdot 7 \cdot 13$
(5, 35)	5	$2^6 \cdot 3^3$	(26, 31)	26	$2^2 \cdot 3 \cdot 7 \cdot 13 \cdot 31$
(5, 36)	5	$2^4 \cdot 3^4 \cdot 7$	(26, 32)	26	$3^5 \cdot 13$
(5, 37)	5	7	(26, 33)	26	$3^2 \cdot 7$
(5, 38)	5	7	(26, 35)	26	$2^2 \cdot 3^5 \cdot 7$
(5, 40)	5	2^6	(26, 37)	26	7^4
(5, 46)	5	$2^2 \cdot 3$	(26, 42)	26	$7^2 \cdot 13$
(5, 48)	5	$3^2 \cdot 7^2$	(26, 44)	26	$3^7 \cdot 13$
(5, 53)	5	2^2	(26, 47)	26	$2^2 \cdot 3 \cdot 13 \cdot 19$
(5, 58)	5	$2^2 \cdot 7$	(26, 54)	26	19^3
(5, 68)	5	3^3	(26, 70)	26	$2^2 \cdot 7^2 \cdot 31$
(6, 17)	6	2^2	(27, 34)	27	$2^2 \cdot 3^4 \cdot 7$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(6, 22)	6	3	(27, 36)	27	$2^4 \cdot 3^4$
(6, 24)	6	$2^2 \cdot 3 \cdot 7$	(27, 48)	27	$7^2 \cdot 13$
(6, 40)	6	2^2	(27, 53)	27	$2^2 \cdot 13^2$
(6, 47)	6	7	(27, 55)	27	$2^4 \cdot 3^4 \cdot 7 \cdot 13$
(6, 49)	6	7	(27, 71)	27	2^4
(7, 16)	7	3^3	(28, 33)	28	$7^2 \cdot 19$
(7, 17)	7	13	(28, 35)	28	$2^2 \cdot 3^4 \cdot 7$
(7, 19)	7	7^2	(28, 63)	28	2^6
(7, 21)	7	3^2	(28, 67)	28	$2^2 \cdot 7 \cdot 19$
(7, 26)	7	$2^2 \cdot 3$	(29, 32)	29	$3^2 \cdot 7^2 \cdot 13$
(7, 29)	7	$2^4 \cdot 3 \cdot 7$	(29, 33)	29	$7 \cdot 19$
(7, 33)	7	7^3	(29, 35)	29	$2^2 \cdot 3^5 \cdot 7 \cdot 13$
(7, 37)	7	7	(29, 41)	29	$3^2 \cdot 7^2 \cdot 13$
(7, 39)	7	$7 \cdot 13$	(29, 49)	29	$3^3 \cdot 7^4$
(7, 46)	7	$3^2 \cdot 7^2$	(29, 54)	29	$2^4 \cdot 3 \cdot 7^2 \cdot 13$
(7, 47)	7	$2^2 \cdot 3$	(29, 57)	29	$3^3 \cdot 7^2$
(7, 49)	7	$3^2 \cdot 7^2$	(29, 59)	29	$2^2 \cdot 3 \cdot 7 \cdot 13^2$
(7, 53)	7	1	(29, 61)	29	$3 \cdot 13^2 \cdot 37$
(7, 63)	7	2^2	(29, 63)	29	$2^2 \cdot 3 \cdot 13$
(7, 69)	7	7^2	(29, 66)	29	$2^2 \cdot 3 \cdot 7^2 \cdot 37$
(8, 12)	12	$2^4 \cdot 13^2$	(30, 31)	30	$3^4 \cdot 7 \cdot 13$
(8, 13)	13	$2^4 \cdot 7^2$	(30, 37)	30	$3 \cdot 19 \cdot 37$
(8, 20)	20	$2^4 \cdot 3 \cdot 37$	(30, 41)	30	$3^2 \cdot 7^2 \cdot 13$
(8, 22)	22	$2^6 \cdot 3^2 \cdot 13$	(30, 42)	30	$3^2 \cdot 13 \cdot 31$
(8, 24)	24	$2^4 \cdot 3^3$	(30, 48)	30	$3^3 \cdot 7^4$
(8, 25)	25	$2^6 \cdot 3^2 \cdot 5^2$	(30, 51)	30	$3 \cdot 7 \cdot 37$
(8, 26)	26	$2^4 \cdot 3^3$	(30, 57)	30	$3^3 \cdot 7^2$
(8, 28)	28	$2^4 \cdot 7^2$	(30, 61)	30	$3 \cdot 13^2 \cdot 37$
(8, 31)	31	$2^2 \cdot 5^2$	(30, 63)	30	$2^2 \cdot 3 \cdot 13$
(8, 33)	33	$2^2 \cdot 3^2 \cdot 7^2$	(30, 65)	30	$7 \cdot 19$
(8, 53)	53	$2^6 \cdot 3^2 \cdot 13$	(31, 37)	31	$3^2 \cdot 5^2 \cdot 7^2$
(8, 60)	60	$2^4 \cdot 3 \cdot 37$	(31, 41)	31	$2^2 \cdot 3^4 \cdot 5^2 \cdot 13^2$
(8, 63)	63	2^{10}	(31, 45)	31	$2^8 \cdot 7 \cdot 13$
(8, 71)	71	37^2	(31, 48)	31	$2^4 \cdot 3^4 \cdot 7^2$
(9, 24)	9	$2^2 \cdot 3^{10} \cdot 7$	(31, 57)	31	$3^4 \cdot 5^2 \cdot 7^2$
(9, 28)	9	$3^{10} \cdot 13$	(31, 60)	31	$3 \cdot 7$
(9, 40)	9	$2^2 \cdot 3^{14} \cdot 7 \cdot 13^2$	(31, 70)	31	$2^2 \cdot 3^2 \cdot 7 \cdot 31$
(10, 12)	10	$2^2 \cdot 3^{10} \cdot 13^2$	(32, 51)	32	$3^2 \cdot 31^2$
(10, 16)	10	$3^{10} \cdot 7^2 \cdot 19$	(32, 56)	32	$3^6 \cdot 7^2 \cdot 13$
(10, 24)	10	$3^{10} \cdot 7^2 \cdot 19$	(32, 62)	32	7^4
(10, 25)	10	$3^{10} \cdot 5^2 \cdot 7^2$	(32, 64)	32	$2^{10} \cdot 3^2$
(10, 29)	10	$2^2 \cdot 3^{10} \cdot 7^2 \cdot 13$	(32, 71)	32	$3^2 \cdot 31^2$
(10, 30)	10	$3^{10} \cdot 7^2 \cdot 13^2$	(33, 36)	33	$2^6 \cdot 3^4 \cdot 7^2 \cdot 19$
(10, 47)	10	$2^2 \cdot 3^{14} \cdot 7^3 \cdot 13$	(33, 42)	33	7^3
(10, 54)	10	$3^{10} \cdot 5^2 \cdot 7^2 \cdot 13$	(33, 47)	33	$7 \cdot 19$
(10, 62)	10	$2^2 \cdot 3^{10} \cdot 13^2$	(33, 58)	33	$2^4 \cdot 3 \cdot 7$
(10, 66)	10	$3^{10} \cdot 7 \cdot 13^2$	(34, 38)	34	$2^4 \cdot 3^{11} \cdot 7$
(10, 71)	10	$2^2 \cdot 3^{14} \cdot 5^2 \cdot 13$	(34, 66)	34	$2^2 \cdot 3^8 \cdot 7 \cdot 13$
(11, 24)	11	$2^2 \cdot 3^{10} \cdot 5^2 \cdot 13$	(34, 67)	34	$2^2 \cdot 3^4 \cdot 7$
(11, 29)	11	$3^{10} \cdot 7^2 \cdot 13^2$	(35, 38)	35	$2^6 \cdot 3^{11} \cdot 7$
(11, 30)	11	$2^2 \cdot 3^{10} \cdot 7^2 \cdot 13$	(35, 39)	35	$2^2 \cdot 3^{13}$
(11, 50)	11	$2^6 \cdot 3^{14} \cdot 7^3 \cdot 13$	(35, 41)	35	$3^{13} \cdot 5^2 \cdot 7$
(11, 60)	11	$3^{10} \cdot 7 \cdot 13^2$	(35, 54)	35	$2^4 \cdot 3^8 \cdot 5^2 \cdot 13$
(11, 71)	11	$2^2 \cdot 3^{14} \cdot 5^2 \cdot 13$	(35, 59)	35	$2^2 \cdot 3^7 \cdot 7 \cdot 13$
(12, 13)	12	$2^2 \cdot 3$	(35, 63)	35	$2^8 \cdot 3^9$
(12, 20)	12	$2^2 \cdot 3 \cdot 13^2$	(36, 48)	36	$2^2 \cdot 3^{12} \cdot 7 \cdot 13$
(12, 23)	12	$2^8 \cdot 3^2$	(36, 55)	36	$2^2 \cdot 3^{17}$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(12, 27)	12	$2^2 \cdot 13^2$	(36, 58)	36	$2^6 \cdot 3^{10} \cdot 13$
(12, 29)	12	$2^2 \cdot 3^4$	(36, 62)	36	$2^4 \cdot 3^7 \cdot 7 \cdot 13$
(12, 31)	12	3^{10}	(36, 66)	36	$2^{10} \cdot 3^7$
(12, 36)	12	$2^6 \cdot 3^7 \cdot 13$	(36, 67)	36	$2^2 \cdot 3^3 \cdot 7 \cdot 13 \cdot 19$
(12, 38)	12	$2^6 \cdot 31$	(36, 69)	36	$2^6 \cdot 3^7 \cdot 7^2$
(12, 44)	12	$2^4 \cdot 3^4$	(37, 40)	37	$3^2 \cdot 13$
(12, 45)	12	2^4	(37, 46)	37	$7^2 \cdot 19$
(12, 48)	12	$3^4 \cdot 13$	(37, 47)	37	$7^2 \cdot 19$
(12, 50)	12	$2^4 \cdot 3^3 \cdot 13$	(37, 49)	37	$3 \cdot 7 \cdot 19$
(12, 54)	12	$2^2 \cdot 3^2 \cdot 13$	(37, 52)	37	$2^2 \cdot 7$
(12, 60)	12	$2^2 \cdot 3 \cdot 13^2$	(37, 60)	37	7^3
(12, 68)	12	$3 \cdot 13 \cdot 31$	(38, 46)	38	7
(12, 69)	12	$2^4 \cdot 3^2$	(38, 57)	38	$2^2 \cdot 5^2$
(13, 17)	13	$2^2 \cdot 7^2$	(38, 58)	38	$2^2 \cdot 3^6 \cdot 5^2$
(13, 21)	13	$2^2 \cdot 3^4 \cdot 7$	(38, 60)	38	$3 \cdot 7^2$
(13, 30)	13	$2^2 \cdot 3 \cdot 7 \cdot 13^2$	(38, 64)	38	2^2
(13, 34)	13	$2^2 \cdot 3^4 \cdot 7 \cdot 13$	(38, 65)	38	$2^4 \cdot 3 \cdot 7$
(13, 37)	13	$7 \cdot 13 \cdot 19$	(38, 68)	38	$7 \cdot 31$
(13, 40)	13	$2^2 \cdot 13$	(39, 43)	39	$2^2 \cdot 3$
(13, 46)	13	$3 \cdot 7 \cdot 13$	(39, 61)	39	3^2
(13, 48)	13	$2^4 \cdot 3^2 \cdot 7^2$	(39, 63)	39	$2^2 \cdot 3^2$
(13, 49)	13	$2^4 \cdot 3^2 \cdot 19$	(39, 66)	39	$2^4 \cdot 3^2 \cdot 13$
(13, 51)	13	$3 \cdot 7^3$	(40, 46)	40	2^2
(13, 61)	13	$2^8 \cdot 7 \cdot 13^2$	(40, 49)	40	$2^2 \cdot 3 \cdot 13^2$
(13, 65)	13	$2^2 \cdot 7$	(40, 55)	40	$2^2 \cdot 3^6 \cdot 13^2$
(13, 68)	13	$2^2 \cdot 3^5$	(41, 42)	41	3^4
(14, 19)	14	$2^2 \cdot 7^2 \cdot 13$	(41, 44)	41	3^{10}
(14, 20)	14	$3 \cdot 7$	(41, 50)	41	$2^4 \cdot 3^5 \cdot 7^2$
(14, 23)	14	3^2	(41, 52)	41	$2^4 \cdot 3 \cdot 19$
(14, 25)	14	$5^2 \cdot 7^2$	(41, 54)	41	$2^4 \cdot 3^4 \cdot 5^2$
(14, 26)	14	$2^6 \cdot 3 \cdot 19$	(41, 56)	41	$2^4 \cdot 3^6 \cdot 13$
(14, 41)	14	$3^5 \cdot 7$	(41, 66)	41	$2^2 \cdot 3^3 \cdot 7^3$
(14, 42)	14	$3^4 \cdot 13$	(41, 67)	41	$2^2 \cdot 3^3 \cdot 19$
(14, 43)	14	13^2	(41, 69)	41	3^7
(14, 47)	14	$3^3 \cdot 13 \cdot 19$	(42, 48)	42	$3^2 \cdot 7^2 \cdot 19$
(14, 57)	14	3^4	(42, 56)	42	$3^6 \cdot 7^2$
(14, 58)	14	7^2	(42, 58)	42	7
(14, 61)	14	$3^2 \cdot 7$	(42, 59)	42	$7 \cdot 13 \cdot 19$
(14, 67)	14	$3^2 \cdot 7^2$	(42, 61)	42	$3^2 \cdot 13 \cdot 37$
(14, 69)	14	$3^2 \cdot 7^2$	(43, 45)	43	$2^8 \cdot 13$
(14, 70)	14	$2^2 \cdot 3^2 \cdot 5^2$	(43, 47)	43	$2^4 \cdot 7 \cdot 13$
(15, 30)	15	$3^5 \cdot 7$	(43, 52)	43	$3 \cdot 7^2$
(15, 38)	15	7^2	(43, 62)	43	$2^2 \cdot 7$
(15, 39)	15	3^4	(44, 51)	44	$3^4 \cdot 43^2$
(15, 41)	15	$3^5 \cdot 7$	(44, 53)	44	$3^4 \cdot 43$
(15, 42)	15	$3^4 \cdot 7^2$	(44, 63)	44	$2^2 \cdot 13^2$
(15, 45)	15	$5^2 \cdot 7$	(45, 46)	45	$2^4 \cdot 7^3$
(15, 54)	15	$2^2 \cdot 3^2 \cdot 5^2$	(45, 48)	45	$2^6 \cdot 7^2$
(15, 56)	15	3^6	(45, 55)	45	$2^6 \cdot 3^6 \cdot 7^3$
(15, 59)	15	$2^2 \cdot 3^4 \cdot 7$	(45, 61)	45	2^{12}
(15, 67)	15	$3^4 \cdot 7$	(45, 65)	45	7^3
(15, 70)	15	$2^2 \cdot 3^3 \cdot 7$	(46, 65)	46	$7 \cdot 19$
(16, 20)	16	$3^4 \cdot 7$	(46, 71)	46	$2^2 \cdot 3$
(16, 29)	16	$2^6 \cdot 3^2 \cdot 7$	(47, 58)	47	7
(16, 30)	16	3^3	(47, 64)	47	$3 \cdot 19$
(16, 36)	16	$2^4 \cdot 3^6 \cdot 13$	(48, 60)	48	$3^2 \cdot 13^2$
(16, 41)	16	$2^8 \cdot 3$	(48, 67)	48	$2^2 \cdot 3 \cdot 13^2$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(16, 46)	16	3^5	(49, 50)	49	3^5
(16, 47)	16	$3^2 \cdot 7^3$	(49, 62)	49	$7^2 \cdot 13$
(16, 48)	16	$3^4 \cdot 7^2 \cdot 13$	(49, 71)	49	$2^6 \cdot 3^2$
(16, 58)	16	$2^4 \cdot 3^3$	(50, 52)	50	$2^4 \cdot 3 \cdot 7$
(16, 65)	16	$3 \cdot 7$	(50, 60)	50	$3^2 \cdot 7 \cdot 13^2$
(16, 70)	16	$2^2 \cdot 19^2$	(50, 65)	50	$2^2 \cdot 3^3 \cdot 7$
(17, 36)	17	$2^8 \cdot 3^4 \cdot 13$	(50, 69)	50	$2^2 \cdot 3^3 \cdot 7^2$
(17, 39)	17	$2^8 \cdot 13$	(51, 53)	51	$31 \cdot 43$
(17, 58)	17	$2^2 \cdot 7 \cdot 13$	(51, 56)	51	$3^4 \cdot 7 \cdot 37$
(18, 26)	18	$2^4 \cdot 7^2$	(51, 59)	51	$3 \cdot 7^3$
(18, 27)	18	3^4	(51, 61)	51	$7^2 \cdot 37^2$
(18, 37)	18	5^2	(52, 64)	52	$2^2 \cdot 7 \cdot 19$
(18, 41)	18	$2^6 \cdot 5^2$	(53, 55)	53	$2^4 \cdot 3^4 \cdot 13$
(18, 45)	18	$2^8 \cdot 7^2$	(53, 60)	53	$3 \cdot 13^2$
(18, 63)	18	2^{10}	(53, 71)	53	$2^4 \cdot 31$
(19, 25)	19	$7 \cdot 37$	(54, 56)	54	$3^4 \cdot 7^5$
(19, 30)	19	$2^4 \cdot 7 \cdot 31$	(54, 67)	54	$2^2 \cdot 3 \cdot 7$
(19, 36)	19	$3^4 \cdot 7^4 \cdot 13$	(55, 60)	55	$3^8 \cdot 7$
(19, 37)	19	$2^4 \cdot 13$	(55, 68)	55	$2^2 \cdot 3^{11}$
(19, 40)	19	$7^2 \cdot 13$	(55, 71)	55	$2^6 \cdot 3^{12}$
(19, 41)	19	$2^4 \cdot 7 \cdot 19$	(56, 59)	56	$2^2 \cdot 3^{12}$
(19, 42)	19	$13 \cdot 31$	(56, 61)	56	$2^4 \cdot 3^{10} \cdot 37$
(19, 45)	19	$7^2 \cdot 13 \cdot 37$	(58, 64)	58	2^2
(19, 47)	19	$7 \cdot 13 \cdot 19$	(58, 66)	58	$2^2 \cdot 7 \cdot 37$
(19, 56)	19	$3^4 \cdot 7$	(59, 61)	59	$2^8 \cdot 7 \cdot 13^2$
(19, 58)	19	$2^2 \cdot 7 \cdot 13$	(59, 71)	59	$2^6 \cdot 3$
(20, 22)	20	$3 \cdot 13^2$	(60, 68)	60	$3 \cdot 7$
(20, 28)	20	7	(60, 71)	60	$3 \cdot 37$
(20, 31)	20	$3 \cdot 7$	(61, 66)	61	$2^4 \cdot 3 \cdot 13 \cdot 37$
(20, 37)	20	7^3	(61, 67)	61	$2^4 \cdot 3 \cdot 7^2$
(20, 39)	20	$3 \cdot 7 \cdot 13^2$	(63, 64)	63	2^6
(20, 49)	20	$3^2 \cdot 13^2$	(64, 67)	64	$2^2 \cdot 7 \cdot 19$
<i>The second certificate.</i>					
(0, 4)	0	2^4	(19, 67)	19	$3^4 \cdot 7$
(0, 10)	0	$2^2 \cdot 3^2 \cdot 7^2$	(20, 26)	20	$2^2 \cdot 7 \cdot 13$
(0, 30)	0	$3 \cdot 7 \cdot 13$	(20, 28)	20	3^3
(0, 34)	0	$2^2 \cdot 3^2 \cdot 7^2$	(20, 30)	20	$2^6 \cdot 3^2 \cdot 7$
(0, 38)	0	$3^2 \cdot 7$	(20, 34)	20	$3 \cdot 7$
(0, 40)	0	$3^2 \cdot 7$	(20, 36)	20	$2^4 \cdot 3^5$
(0, 53)	0	7	(20, 39)	20	$2^8 \cdot 3$
(0, 56)	0	$2^2 \cdot 13^2$	(20, 41)	20	$2^2 \cdot 3^4$
(0, 59)	0	$3^3 \cdot 13$	(20, 43)	20	$3^2 \cdot 7^3$
(0, 61)	0	$2^2 \cdot 3^4$	(20, 44)	20	3^5
(0, 68)	0	$2^6 \cdot 3^2$	(20, 59)	20	$2^6 \cdot 3^3$
(1, 5)	1	7	(21, 30)	21	$3^2 \cdot 13 \cdot 19$
(1, 17)	1	7	(21, 43)	21	$3^2 \cdot 13 \cdot 19$
(1, 20)	1	7	(21, 46)	21	$2^2 \cdot 3^4$
(1, 27)	1	$2^2 \cdot 7^4$	(21, 49)	21	$2^8 \cdot 3^2$
(1, 35)	1	$2^2 \cdot 3^4 \cdot 7 \cdot 13$	(21, 52)	21	$2^{10} \cdot 3^4$
(1, 38)	1	2^2	(21, 60)	21	3^2
(1, 43)	1	$7^2 \cdot 13$	(21, 68)	21	$2^2 \cdot 3^2$
(1, 59)	1	13^2	(22, 23)	22	$2^2 \cdot 3^2$
(2, 6)	2	7	(22, 28)	22	$2^2 \cdot 3 \cdot 7 \cdot 13$
(2, 8)	2	1	(22, 30)	22	$2^2 \cdot 3 \cdot 7 \cdot 13^2$
(2, 12)	2	13	(22, 37)	22	7^4
(2, 13)	2	$5^2 \cdot 7$	(22, 40)	22	$7 \cdot 37$
(2, 18)	2	1	(22, 41)	22	$3^7 \cdot 13$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(2, 39)	2	1	(22, 47)	22	$2^2 \cdot 3^3 \cdot 7$
(2, 43)	2	7	(22, 54)	22	$2^{10} \cdot 3$
(2, 45)	2	13	(22, 57)	22	$2^4 \cdot 37^2$
(2, 46)	2	13	(22, 59)	22	$2^2 \cdot 3^2 \cdot 13$
(2, 49)	2	2^4	(22, 63)	22	$2^2 \cdot 3 \cdot 7^2 \cdot 19$
(2, 62)	2	$2^2 \cdot 7 \cdot 13$	(22, 64)	22	$2^2 \cdot 5^2 \cdot 19$
(2, 64)	2	$2^2 \cdot 5^2$	(23, 32)	23	$2^2 \cdot 5^4 \cdot 7^2$
(3, 13)	3	$2^2 \cdot 7 \cdot 13$	(23, 36)	23	$2^2 \cdot 3^5 \cdot 7^3$
(3, 19)	3	3	(23, 38)	23	$3 \cdot 5^2 \cdot 7^2$
(3, 24)	3	$2^2 \cdot 3 \cdot 7$	(23, 39)	23	$2^4 \cdot 3^2 \cdot 7^2$
(3, 26)	3	7^2	(23, 43)	23	7^3
(3, 28)	3	2^2	(23, 51)	23	$3 \cdot 5^2 \cdot 7^2$
(3, 29)	3	7	(23, 52)	23	$2^{10} \cdot 7^2$
(3, 30)	3	$7 \cdot 13$	(23, 61)	23	$2^2 \cdot 3^2 \cdot 7$
(3, 37)	3	13	(23, 62)	23	$2^2 \cdot 7^2$
(3, 39)	3	$3 \cdot 13$	(23, 65)	23	$3 \cdot 7$
(3, 41)	3	3^2	(23, 66)	23	$5^2 \cdot 7^2$
(3, 45)	3	7	(23, 68)	23	$2^2 \cdot 3^6 \cdot 7^2$
(3, 47)	3	$2^2 \cdot 3$	(24, 27)	24	$2^2 \cdot 7$
(3, 52)	3	7^2	(24, 28)	24	$2^2 \cdot 3 \cdot 7 \cdot 13^2$
(3, 55)	3	13	(24, 34)	24	$3^2 \cdot 7$
(3, 57)	3	7	(24, 37)	24	7^4
(3, 61)	3	$3 \cdot 5^2$	(24, 39)	24	$2^2 \cdot 3^3 \cdot 13$
(3, 63)	3	$3 \cdot 5^2$	(24, 40)	24	$7^2 \cdot 13$
(4, 15)	4	2^2	(24, 41)	24	$3^7 \cdot 13$
(4, 17)	4	$3 \cdot 7$	(24, 44)	24	$2^2 \cdot 3 \cdot 13 \cdot 19$
(4, 26)	4	$2^2 \cdot 7$	(24, 56)	24	$2^4 \cdot 3 \cdot 7^3$
(4, 38)	4	$2^2 \cdot 7$	(24, 59)	24	$3^2 \cdot 5^2 \cdot 19$
(4, 43)	4	$3 \cdot 7$	(24, 61)	24	$2^4 \cdot 3^4$
(4, 44)	4	$3 \cdot 7$	(24, 63)	24	$3 \cdot 13 \cdot 31$
(4, 49)	4	$2^2 \cdot 3^2$	(24, 64)	24	$2^2 \cdot 7^2 \cdot 31$
(4, 56)	4	$2^2 \cdot 3$	(25, 27)	25	$7 \cdot 13$
(5, 12)	5	$3 \cdot 7$	(25, 31)	25	2^{12}
(5, 14)	5	3^2	(25, 37)	25	$7 \cdot 31$
(5, 20)	5	3^3	(25, 45)	25	$7^2 \cdot 13$
(5, 31)	5	$2^2 \cdot 7^2$	(25, 49)	25	$2^2 \cdot 13^2$
(5, 36)	5	$2^6 \cdot 3^3$	(25, 62)	25	$13 \cdot 31$
(5, 42)	5	7	(26, 27)	26	$2^4 \cdot 7^2$
(5, 47)	5	$2^2 \cdot 3$	(26, 29)	26	$2^2 \cdot 5^2 \cdot 7 \cdot 13$
(5, 50)	5	7^3	(26, 30)	26	$13 \cdot 37$
(5, 52)	5	3^3	(26, 34)	26	$7^2 \cdot 19$
(5, 58)	5	$2^4 \cdot 3 \cdot 7$	(26, 36)	26	$2^2 \cdot 3^4 \cdot 7$
(5, 68)	5	$2^2 \cdot 7$	(26, 40)	26	7^2
(6, 17)	6	3	(26, 43)	26	$7^4 \cdot 13$
(6, 19)	6	3^2	(26, 47)	26	$2^4 \cdot 7$
(6, 22)	6	$2^2 \cdot 3 \cdot 7$	(26, 57)	26	$2^4 \cdot 7 \cdot 13$
(6, 37)	6	13	(26, 59)	26	$2^2 \cdot 13 \cdot 43$
(6, 46)	6	7	(26, 61)	26	$7 \cdot 37$
(6, 48)	6	2^2	(26, 62)	26	$2^2 \cdot 7 \cdot 19$
(6, 52)	6	7^2	(26, 64)	26	$2^2 \cdot 5^2 \cdot 31$
(6, 66)	6	$2^2 \cdot 7 \cdot 13$	(26, 65)	26	$7^2 \cdot 31$
(6, 67)	6	$3 \cdot 7$	(27, 39)	27	$3^3 \cdot 5^2$
(6, 68)	6	$2^2 \cdot 3$	(27, 47)	27	$2^2 \cdot 3^4 \cdot 7^2 \cdot 13$
(7, 15)	7	2^6	(27, 49)	27	$2^2 \cdot 3^2 \cdot 13$
(7, 20)	7	$2^4 \cdot 3$	(27, 53)	27	2^{12}
(7, 35)	7	$2^4 \cdot 3^4$	(27, 55)	27	$3 \cdot 7 \cdot 13$
(7, 54)	7	2^4	(27, 56)	27	$7^3 \cdot 13^2$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(7, 58)	7	2^2	(27, 60)	27	$5^2 \cdot 7^2$
(8, 14)	8	$3 \cdot 7$	(27, 63)	27	$3^2 \cdot 7^2$
(8, 15)	8	2^2	(28, 31)	28	$3 \cdot 7$
(8, 17)	8	$3 \cdot 7$	(28, 32)	28	$3^4 \cdot 7 \cdot 13$
(8, 25)	8	7	(28, 33)	28	$3^2 \cdot 7^2 \cdot 13$
(8, 27)	8	$2^2 \cdot 7^2$	(28, 40)	28	$3^2 \cdot 7^3 \cdot 13$
(8, 29)	8	7^3	(28, 47)	28	$2^2 \cdot 3^2 \cdot 7^2 \cdot 13$
(8, 30)	8	3^3	(28, 54)	28	$2^2 \cdot 3 \cdot 7 \cdot 13^2$
(8, 34)	8	7^3	(28, 55)	28	$3^5 \cdot 7 \cdot 13$
(8, 37)	8	7	(28, 56)	28	$3 \cdot 13 \cdot 31$
(8, 45)	8	3^2	(28, 61)	28	$3^2 \cdot 7^3 \cdot 31$
(8, 47)	8	$2^2 \cdot 3^4 \cdot 7$	(28, 67)	28	$2^6 \cdot 3^2 \cdot 7$
(8, 52)	8	3^3	(29, 36)	29	$2^2 \cdot 3^5 \cdot 7$
(8, 53)	8	$2^2 \cdot 7$	(29, 43)	29	$7 \cdot 13 \cdot 19$
(8, 56)	8	3^2	(29, 59)	29	$2^2 \cdot 13^3$
(8, 63)	8	7^2	(29, 64)	29	$2^2 \cdot 3^2 \cdot 5^2$
(9, 15)	9	$2^2 \cdot 3 \cdot 13$	(30, 32)	30	$3^4 \cdot 7 \cdot 13$
(9, 24)	9	$13 \cdot 19$	(30, 33)	30	$3^2 \cdot 7^2 \cdot 13$
(9, 29)	9	$3 \cdot 7 \cdot 13$	(30, 40)	30	$3^2 \cdot 13 \cdot 31$
(9, 31)	9	$2^2 \cdot 3^2 \cdot 13$	(30, 45)	30	$3^3 \cdot 7^4$
(9, 44)	9	$13 \cdot 19$	(30, 49)	30	$2^2 \cdot 3^4$
(9, 48)	9	2^8	(30, 56)	30	$3^2 \cdot 7^2 \cdot 13 \cdot 31$
(9, 50)	9	$3^3 \cdot 19$	(30, 59)	30	$2^2 \cdot 3 \cdot 13^2 \cdot 31$
(9, 54)	9	$2^2 \cdot 7$	(30, 61)	30	$2^2 \cdot 3^2 \cdot 31 \cdot 37$
(9, 59)	9	$2^4 \cdot 13 \cdot 19$	(30, 63)	30	$2^2 \cdot 3 \cdot 13$
(10, 12)	12	$2^4 \cdot 7 \cdot 13$	(30, 65)	30	$2^4 \cdot 3 \cdot 19 \cdot 31$
(10, 13)	13	$2^4 \cdot 7^2$	(30, 66)	30	$2^8 \cdot 3 \cdot 7$
(10, 18)	18	2^{10}	(30, 67)	30	3^3
(10, 22)	22	$2^4 \cdot 3^3$	(31, 34)	31	$2^2 \cdot 7^2$
(10, 23)	23	$2^6 \cdot 3^2 \cdot 5^2$	(31, 39)	31	$3^3 \cdot 5^2$
(10, 24)	24	$2^4 \cdot 3^3$	(31, 53)	31	$7 \cdot 13$
(10, 31)	31	$2^4 \cdot 7^2$	(31, 60)	31	$5^2 \cdot 7^2$
(10, 32)	32	$2^2 \cdot 5^2$	(31, 62)	31	$3 \cdot 7$
(10, 49)	49	$2^4 \cdot 13^2$	(32, 33)	32	$2^2 \cdot 3^2 \cdot 7^2 \cdot 13^2$
(10, 50)	50	$2^4 \cdot 7 \cdot 13$	(32, 35)	32	$3^6 \cdot 7^4$
(10, 54)	54	$2^4 \cdot 7^2$	(32, 39)	32	$2^2 \cdot 3^4 \cdot 5^2 \cdot 13^2$
(11, 16)	11	$3^{10} \cdot 5^2 \cdot 7^2$	(32, 41)	32	$3^2 \cdot 7^2$
(11, 22)	11	$3^{10} \cdot 7^2 \cdot 19$	(32, 60)	32	$5^2 \cdot 7^2$
(11, 24)	11	$2^2 \cdot 3^{10} \cdot 5^2 \cdot 13$	(33, 40)	33	7^4
(11, 26)	11	$3^{10} \cdot 7^2 \cdot 43$	(33, 45)	33	$3^4 \cdot 7^4$
(11, 28)	11	$2^2 \cdot 3^{10} \cdot 7^2 \cdot 13$	(33, 53)	33	7^4
(11, 30)	11	$3^{10} \cdot 7^2 \cdot 13^2$	(33, 54)	33	$3^2 \cdot 7^2$
(11, 35)	11	$2^2 \cdot 3^{13} \cdot 7^3$	(34, 51)	34	$2^4 \cdot 3 \cdot 7$
(11, 40)	11	$3^{14} \cdot 7^3 \cdot 13$	(34, 58)	34	$2^6 \cdot 7^2 \cdot 19$
(11, 42)	11	$2^8 \cdot 3^{14} \cdot 7^2$	(35, 38)	35	$2^4 \cdot 3^{11} \cdot 7$
(11, 44)	11	$2^2 \cdot 3^{14} \cdot 7^3 \cdot 13$	(35, 42)	35	$2^8 \cdot 3^{10} \cdot 7^2 \cdot 13$
(11, 46)	11	$3^{14} \cdot 7^2 \cdot 13 \cdot 19$	(35, 47)	35	$2^6 \cdot 3^{12} \cdot 13$
(11, 50)	11	$3^{10} \cdot 5^2 \cdot 7^2 \cdot 13$	(35, 50)	35	$2^2 \cdot 3^8 \cdot 7$
(11, 53)	11	$2^2 \cdot 3^{10} \cdot 13^2$	(35, 59)	35	$2^6 \cdot 3^{12} \cdot 13$
(11, 54)	11	$3^{10} \cdot 5^2 \cdot 7$	(35, 65)	35	$2^2 \cdot 3^{12} \cdot 7$
(11, 55)	11	$3^{10} \cdot 5^2 \cdot 13 \cdot 19$	(36, 37)	36	$2^2 \cdot 3^{13} \cdot 7 \cdot 31$
(11, 59)	11	$3^{14} \cdot 5^2 \cdot 13 \cdot 43$	(36, 39)	36	$3^{13} \cdot 5^2 \cdot 7$
(11, 65)	11	$3^4 \cdot 7^3 \cdot 13$	(36, 40)	36	$3^{12} \cdot 13 \cdot 37$
(12, 13)	12	$2^8 \cdot 3^4 \cdot 13$	(36, 47)	36	$2^4 \cdot 3^{12} \cdot 7 \cdot 13$
(12, 20)	12	$2^4 \cdot 3^2 \cdot 13$	(36, 50)	36	$2^4 \cdot 3^8 \cdot 5^2 \cdot 13$
(12, 21)	12	$2^2 \cdot 3^2$	(36, 51)	36	$2^4 \cdot 3^{11} \cdot 7$
(12, 26)	12	$7 \cdot 13$	(36, 54)	36	$2^2 \cdot 3^7 \cdot 7 \cdot 13$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(12, 28)	12	$2^4 \cdot 3 \cdot 7$	(36, 55)	36	$2^8 \cdot 3^8 \cdot 37$
(12, 38)	12	$2^6 \cdot 13$	(36, 62)	36	$2^2 \cdot 3^4 \cdot 7 \cdot 31$
(12, 46)	12	$2^2 \cdot 3^2 \cdot 7 \cdot 13$	(37, 41)	37	$2^8 \cdot 7^4$
(12, 49)	12	$2^6 \cdot 3^3 \cdot 13$	(37, 43)	37	$7^2 \cdot 19$
(12, 55)	12	$2^2 \cdot 3^4 \cdot 7^2$	(37, 46)	37	$3 \cdot 7 \cdot 19$
(12, 56)	12	$2^2 \cdot 3 \cdot 7$	(37, 52)	37	3^4
(13, 23)	13	$2^2 \cdot 7^2$	(37, 61)	37	$3^2 \cdot 7$
(13, 30)	13	$2^2 \cdot 3 \cdot 7 \cdot 13^2$	(37, 64)	37	$3 \cdot 7 \cdot 19 \cdot 31$
(13, 37)	13	$7 \cdot 13 \cdot 19$	(37, 65)	37	$3 \cdot 19^2 \cdot 31$
(13, 38)	13	$2^2 \cdot 7 \cdot 13$	(37, 66)	37	$3 \cdot 13^2 \cdot 19$
(13, 44)	13	$2^2 \cdot 3^3 \cdot 7^2$	(38, 47)	38	$2^2 \cdot 3$
(13, 52)	13	$2^6 \cdot 3^4 \cdot 7$	(38, 51)	38	$2^2 \cdot 3^6 \cdot 5^2$
(13, 55)	13	$2^4 \cdot 3 \cdot 7 \cdot 13$	(38, 61)	38	$3 \cdot 7^2$
(13, 57)	13	$2^8 \cdot 7 \cdot 13^2$	(38, 65)	38	3
(13, 64)	13	$2^2 \cdot 5^2 \cdot 7 \cdot 19$	(39, 40)	39	3^4
(13, 67)	13	$3^2 \cdot 19$	(39, 41)	39	3^{10}
(14, 17)	14	$3 \cdot 7$	(39, 47)	39	$2^4 \cdot 3^5 \cdot 7^2$
(14, 19)	14	$3 \cdot 7$	(39, 50)	39	$2^4 \cdot 3^4 \cdot 5^2$
(14, 24)	14	$2^6 \cdot 3 \cdot 19$	(40, 49)	40	3^4
(14, 26)	14	$5^2 \cdot 7^2$	(40, 51)	40	7
(14, 31)	14	$2^2 \cdot 3^3 \cdot 7$	(40, 54)	40	$7 \cdot 13 \cdot 19$
(14, 32)	14	3^6	(40, 63)	40	$3^2 \cdot 13 \cdot 19$
(14, 43)	14	$3 \cdot 7^2$	(40, 65)	40	$3^2 \cdot 7^2 \cdot 31$
(14, 44)	14	$3^3 \cdot 13 \cdot 19$	(41, 67)	41	$2^2 \cdot 3^4$
(14, 45)	14	3^5	(41, 68)	41	$3^4 \cdot 7^2$
(14, 51)	14	7^2	(42, 47)	42	$2^4 \cdot 13$
(14, 64)	14	$2^2 \cdot 3^2 \cdot 5^2$	(43, 49)	43	$2^4 \cdot 3$
(14, 67)	14	3^2	(43, 56)	43	3^5
(14, 68)	14	13^2	(43, 63)	43	$2^4 \cdot 7^2 \cdot 19$
(15, 22)	15	$2^2 \cdot 5^2 \cdot 13$	(43, 65)	43	$2^2 \cdot 7 \cdot 13 \cdot 19$
(15, 28)	15	$2^2 \cdot 3 \cdot 13$	(43, 68)	43	$2^4 \cdot 7 \cdot 13$
(15, 41)	15	$2^2 \cdot 13^2$	(44, 49)	44	$2^4 \cdot 3$
(15, 42)	15	2^6	(44, 50)	44	$7 \cdot 13 \cdot 19$
(15, 48)	15	2^2	(44, 51)	44	7
(15, 52)	15	$2^{10} \cdot 3^2$	(44, 54)	44	$3 \cdot 7 \cdot 13$
(15, 64)	15	$2^2 \cdot 3 \cdot 5^2$	(44, 67)	44	$3^2 \cdot 7^3$
(16, 21)	16	3^2	(44, 68)	44	$2^4 \cdot 7 \cdot 13$
(16, 22)	16	$2^6 \cdot 3 \cdot 19$	(45, 49)	45	$3^4 \cdot 13$
(16, 24)	16	$3 \cdot 7$	(45, 52)	45	$2^6 \cdot 3^8$
(16, 29)	16	$3^3 \cdot 19$	(45, 54)	45	$2^4 \cdot 3^2 \cdot 19$
(16, 30)	16	$3^5 \cdot 7$	(45, 57)	45	$2^{12} \cdot 3$
(16, 31)	16	$3^2 \cdot 7^2$	(45, 58)	45	$2^2 \cdot 3^2 \cdot 7 \cdot 13$
(16, 36)	16	3^6	(45, 63)	45	$3 \cdot 7 \cdot 19$
(16, 38)	16	7^2	(46, 54)	46	$2^4 \cdot 3^2 \cdot 7^2$
(16, 39)	16	$3^5 \cdot 7$	(46, 57)	46	$2^{12} \cdot 3$
(16, 40)	16	$3^4 \cdot 7^2$	(46, 66)	46	$3^2 \cdot 7^2 \cdot 19$
(16, 47)	16	3^5	(46, 67)	46	$3^4 \cdot 19$
(16, 50)	16	$2^2 \cdot 3^2 \cdot 5^2$	(47, 50)	47	$2^4 \cdot 3 \cdot 13$
(16, 61)	16	$3^4 \cdot 13$	(47, 61)	47	$2^2 \cdot 3^4 \cdot 43$
(16, 62)	16	$3^4 \cdot 7$	(47, 62)	47	$3^2 \cdot 7^2$
(16, 63)	16	$3^3 \cdot 19$	(48, 51)	48	2^2
(16, 64)	16	$2^2 \cdot 3^3 \cdot 7$	(49, 58)	49	$2^6 \cdot 3^3 \cdot 13$
(16, 66)	16	$3^4 \cdot 7$	(49, 61)	49	$2^4 \cdot 3^4$
(16, 68)	16	13^2	(49, 62)	49	$2^2 \cdot 3^2 \cdot 13$
(17, 23)	17	$2^2 \cdot 3 \cdot 7^3$	(49, 65)	49	$2^2 \cdot 3^2 \cdot 13$
(17, 25)	17	13^2	(49, 68)	49	2^{12}
(17, 27)	17	$7^2 \cdot 13^2$	(51, 66)	51	$2^4 \cdot 5^2 \cdot 31$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(17, 37)	17	7^3	(51, 67)	51	$2^4 \cdot 3^3$
(17, 38)	17	$3 \cdot 7^2$	(51, 68)	51	$2^4 \cdot 3 \cdot 7$
(17, 47)	17	$3^2 \cdot 7 \cdot 13^2$	(52, 54)	52	$2^{10} \cdot 3^4 \cdot 7$
(17, 49)	17	$2^2 \cdot 3 \cdot 13^2$	(53, 58)	53	$2^4 \cdot 7 \cdot 13$
(17, 51)	17	$3^4 \cdot 7 \cdot 31$	(53, 68)	53	$2^2 \cdot 7$
(17, 60)	17	$3 \cdot 7^3$	(54, 55)	54	$3 \cdot 5^2 \cdot 7 \cdot 37$
(17, 64)	17	$2^2 \cdot 7 \cdot 31$	(54, 63)	54	$2^2 \cdot 19 \cdot 37$
(18, 47)	18	$7 \cdot 43$	(54, 67)	54	$2^2 \cdot 3^4 \cdot 7$
(18, 61)	18	$2^4 \cdot 7 \cdot 43$	(55, 57)	55	$2^8 \cdot 3 \cdot 37$
(18, 62)	18	$2^4 \cdot 7^2$	(55, 63)	55	$3 \cdot 7 \cdot 37^2$
(18, 63)	18	7^2	(55, 64)	55	$2^2 \cdot 3 \cdot 7 \cdot 19$
(19, 30)	19	$3 \cdot 7$	(56, 59)	56	$3 \cdot 13 \cdot 31$
(19, 31)	19	$7^2 \cdot 13^2$	(58, 62)	58	$2^2 \cdot 7 \cdot 13 \cdot 19$
(19, 35)	19	$2^2 \cdot 3^5 \cdot 7^3$	(58, 63)	58	$2^6 \cdot 7^2$
(19, 37)	19	7^3	(61, 63)	61	$2^4 \cdot 3^3 \cdot 5^2$
(19, 42)	19	7^2	(61, 65)	61	$2^2 \cdot 3^3 \cdot 7 \cdot 13$
(19, 46)	19	$3^2 \cdot 13^2$	(63, 64)	63	$2^2 \cdot 3^2 \cdot 7 \cdot 31$
(19, 49)	19	$2^2 \cdot 3 \cdot 13^2$	(64, 66)	66	$2^2 \cdot 3^2 \cdot 19 \cdot 31$
(19, 53)	19	13^2	(64, 67)	67	$2^2 \cdot 19^2$
(19, 60)	19	$3 \cdot 7^3$	(66, 67)	66	$7^2 \cdot 19$
(19, 65)	19	$3 \cdot 13^2$			

Table 7: The terminal residue classes $x \equiv a, y \equiv b \pmod{p^k}$ of Step 6, grouped by p, k and the joint value $((\mathcal{B}_p^{(g)})_g; B_p)$.

p	k	value	classes (a, b)
2	5	$(1, 0; -1)$	(29, 3), (29, 19)
2	6	$(1, 0; -1)$	(29, 11), (29, 27), (29, 43), (29, 59), (61, 7), (61, 39)
2	7	$(1, 0; -1)$	(3, 88), (7, 116), (11, 16), (13, 54), (15, 44), (19, 72), (21, 78), (23, 100), (25, 13), (25, 45), (25, 77), (25, 109), (27, 0), (29, 15), (29, 47), (29, 79), (29, 111), (31, 28), (35, 56), (39, 84), (43, 112), (47, 12), (51, 40), (53, 46), (55, 68), (57, 5), (57, 37), (57, 69), (57, 101), (59, 96), (61, 23), (61, 55), (61, 87), (61, 119), (63, 124), (67, 24), (71, 52), (75, 80), (79, 108), (83, 8), (85, 14), (87, 36), (89, 29), (89, 61), (89, 73), (89, 93), (89, 125), (91, 64), (93, 31), (93, 63), (93, 95), (93, 127), (95, 92), (99, 120), (103, 20), (107, 48), (111, 76), (115, 104), (117, 110), (119, 4), (121, 21), (121, 53), (121, 85), (121, 117), (123, 32), (127, 60)
2	8	$(1, 0; -1)$	(5, 158), (29, 230), (37, 254), (57, 65), (57, 193), (69, 94), (77, 118), (89, 9), (89, 137), (101, 190), (109, 214), (121, 113), (121, 241), (125, 6), (133, 30), (157, 102), (165, 126), (197, 222), (205, 246), (217, 105), (229, 62), (253, 134)
2	9	$(1, 0; -1)$	(25, 89), (25, 217), (25, 250), (25, 345), (25, 473), (45, 22), (61, 326), (73, 138), (93, 422), (145, 98), (153, 57), (153, 185), (153, 313), (153, 441), (173, 406), (185, 33), (185, 161), (185, 289), (185, 417), (189, 198), (193, 498), (201, 10), (217, 41), (217, 169), (217, 297), (217, 425), (221, 294), (237, 86), (249, 81), (249, 209), (249, 337), (249, 465), (281, 25), (281, 153), (281, 281), (281, 409), (281, 506), (301, 278), (313, 1), (313, 129), (313, 257), (313, 385), (317, 70), (329, 394), (349, 166), (377, 49), (377, 177), (377, 305), (377, 433), (401, 354), (409, 121), (409, 249), (409, 377), (409, 505), (429, 150), (441, 97), (441, 225), (441, 353), (441, 481), (445, 454), (449, 242), (457, 266), (473, 233), (473, 489), (477, 38), (493, 342), (505, 17), (505, 145), (505, 273), (505, 401)

p	k	value	classes (a, b)
2	10	$(1, 0; -1)$	(9, 970), (17, 226), (57, 858), (65, 626), (137, 330), (153, 634), (169, 938), (185, 218), (209, 802), (257, 178), (265, 714), (273, 994), (313, 602), (321, 370), (393, 74), (409, 378), (425, 682), (441, 986), (473, 570), (521, 458), (529, 738), (569, 346), (577, 114), (649, 842), (665, 122), (681, 426), (697, 730), (721, 290), (769, 690), (777, 202), (785, 482), (825, 90), (833, 882), (905, 586), (921, 890), (937, 170), (953, 474), (985, 58)
2	11	$(1, 0; -1)$	(1, 1458), (33, 18), (41, 554), (49, 834), (81, 1442), (89, 1466), (97, 1234), (105, 1770), (113, 2), (121, 1050), (129, 1842), (161, 402), (177, 1218), (217, 1850), (225, 1618), (233, 106), (241, 386), (249, 1434), (289, 786), (297, 1322), (305, 1602), (337, 162), (345, 186), (353, 2002), (361, 490), (369, 770), (377, 1818), (385, 562), (417, 1170), (433, 1986), (465, 546), (481, 338), (489, 874), (497, 1154), (505, 154), (513, 946), (545, 1554), (553, 42), (561, 322), (593, 930), (601, 954), (609, 722), (617, 1258), (625, 1538), (633, 538), (641, 1330), (673, 1938), (689, 706), (729, 1338), (737, 1106), (745, 1642), (753, 1922), (761, 922), (801, 274), (809, 810), (817, 1090), (849, 1698), (857, 1722), (865, 1490), (873, 2026), (881, 258), (889, 1306), (897, 50), (929, 658), (945, 1474), (977, 34), (993, 1874), (1001, 362), (1009, 642), (1017, 1690), (1025, 434), (1057, 1042), (1065, 1578), (1073, 1858), (1105, 418), (1113, 442), (1121, 210), (1129, 746), (1137, 1026), (1145, 26), (1153, 818), (1185, 1426), (1201, 194), (1241, 826), (1249, 594), (1257, 1130), (1265, 1410), (1273, 410), (1313, 1810), (1321, 298), (1329, 578), (1361, 1186), (1369, 1210), (1377, 978), (1385, 1514), (1393, 1794), (1401, 794), (1409, 1586), (1441, 146), (1457, 962), (1489, 1570), (1505, 1362), (1513, 1898), (1521, 130), (1529, 1178), (1537, 1970), (1569, 530), (1577, 1066), (1585, 1346), (1617, 1954), (1625, 1978), (1633, 1746), (1641, 234), (1649, 514), (1657, 1562), (1665, 306), (1697, 914), (1713, 1730), (1753, 314), (1761, 82), (1769, 618), (1777, 898), (1785, 1946), (1825, 1298), (1833, 1834), (1841, 66), (1873, 674), (1881, 698), (1889, 466), (1897, 1002), (1905, 1282), (1913, 282), (1921, 1074), (1953, 1682), (1969, 450), (2001, 1058), (2017, 850), (2025, 1386), (2033, 1666), (2041, 666)
3	4	$(0, 0; 1)$	(0, 24), (9, 15), (18, 6), (27, 78), (36, 69), (45, 60), (54, 51), (63, 42), (72, 33)
3	4	$(0, 1; 1)$	(3, 48), (7, 21), (12, 39), (16, 21), (21, 30), (25, 21), (30, 21), (34, 21), (39, 12), (43, 21), (48, 3), (52, 21), (57, 75), (61, 21), (66, 66), (70, 21), (75, 57), (79, 21)
3	4	$(1, 0; 1)$	(1, 48), (5, 10), (8, 22), (10, 48), (11, 7), (14, 46), (17, 58), (19, 48), (23, 1), (26, 13), (28, 48), (32, 37), (35, 49), (37, 48), (38, 34), (41, 73), (44, 4), (46, 48), (50, 28), (53, 40), (55, 48), (59, 64), (62, 76), (64, 48), (65, 61), (68, 19), (71, 31), (73, 48), (77, 55), (80, 67)
3	4	$(2, 2; 1)$	(4, 21), (13, 21), (22, 21), (31, 21), (40, 21), (49, 21), (58, 21), (67, 21), (76, 21)
3	5	$(0, 2; 1)$	(6, 72), (15, 63), (24, 54), (33, 45), (51, 27), (60, 18), (78, 0), (87, 234), (105, 216), (114, 207), (132, 189), (141, 180), (159, 162), (168, 153), (186, 135), (195, 126), (213, 108), (222, 99), (231, 90), (240, 81)
3	5	$(1, 0; 1)$	(2, 52), (20, 205), (29, 160), (47, 70), (56, 25), (74, 178), (83, 133), (101, 43), (110, 241), (128, 151), (137, 106), (155, 16), (164, 214), (182, 124), (191, 79), (209, 232), (218, 187), (236, 97)

p	k	value	classes (a, b)
3	6	$(0, 2; 1)$	$(42, 36), (69, 9), (96, 711), (123, 684), (150, 657), (177, 630),$ $(204, 603), (285, 522), (312, 495), (339, 468), (366, 441),$ $(393, 414), (420, 387), (447, 360), (528, 279), (555, 252),$ $(582, 225), (609, 198), (636, 171), (663, 144), (690, 117)$
5	2	$(0, 0; 1)$	$(0, 6), (10, 16), (13, 16)$
5	3	$(0, 0; 1)$	$(2, 34), (3, 96), (5, 111), (7, 59), (8, 56), (12, 109), (15, 71),$ $(17, 59), (18, 101), (20, 26), (22, 34), (23, 61), (27, 34), (28, 21),$ $(30, 11), (32, 59), (33, 106), (37, 109), (40, 96), (42, 59), (43, 26),$ $(45, 51), (47, 34), (48, 111), (52, 34), (53, 71), (55, 36), (57, 59),$ $(58, 31), (62, 109), (65, 121), (67, 59), (68, 76), (70, 76),$ $(72, 34), (73, 36), (77, 34), (78, 121), (80, 61), (82, 59), (83, 81),$ $(87, 109), (90, 21), (92, 59), (93, 1), (95, 101), (97, 34), (98, 86),$ $(102, 34), (103, 46), (105, 86), (107, 59), (108, 6), (112, 109),$ $(115, 46), (117, 59), (118, 51), (120, 1), (122, 34), (123, 11)$
7	2	$(1, 0; -1)$	$(38, 28)$
7	2	$(1, 0; 1)$	$(0, 40), (7, 47), (14, 5), (16, 20), (21, 12), (35, 26), (42, 33),$ $(44, 6)$
7	2	$(2, 0; 1)$	$(13, 16), (20, 16), (48, 16)$
7	3	$(1, 0; -1)$	$(10, 224), (23, 4), (23, 53), (23, 102), (23, 151), (23, 200),$ $(23, 249), (23, 298), (24, 126), (45, 175), (66, 322), (72, 11),$ $(72, 60), (72, 109), (72, 144), (72, 158), (72, 193), (72, 207),$ $(72, 256), (72, 305), (72, 340), (73, 126), (94, 175), (108, 224),$ $(115, 322), (121, 25), (121, 32), (121, 74), (121, 123), (121, 130),$ $(121, 172), (121, 221), (121, 270), (121, 319), (122, 126),$ $(129, 322), (143, 175), (164, 322), (171, 126), (192, 175),$ $(206, 224), (213, 322), (219, 18), (219, 39), (219, 67), (219, 88),$ $(219, 116), (219, 165), (219, 186), (219, 214), (219, 235),$ $(219, 263), (219, 284), (219, 312), (219, 333), (220, 126),$ $(241, 175), (255, 224), (262, 322), (269, 126), (276, 322),$ $(290, 175), (304, 224), (311, 322), (318, 126), (339, 175)$
7	3	$(1, 0; 1)$	$(9, 146), (23, 139), (28, 68), (30, 307), (37, 132), (51, 125),$ $(58, 293), (77, 117), (79, 111), (86, 279), (100, 272), (107, 97),$ $(121, 90), (126, 166), (128, 258), (135, 83), (149, 76), (156, 244),$ $(170, 237), (175, 215), (177, 62), (184, 230), (198, 223),$ $(205, 48), (219, 41), (224, 264), (226, 209), (233, 34), (247, 27),$ $(254, 195), (268, 188), (273, 313), (275, 13), (282, 181),$ $(296, 174), (303, 342), (317, 335), (324, 160), (331, 328)$
7	3	$(2, 0; 1)$	$(5, 127), (6, 212), (19, 274), (26, 127), (27, 310), (33, 176),$ $(34, 65), (40, 78), (41, 310), (47, 176), (55, 212), (68, 274),$ $(75, 127), (76, 310), (82, 176), (83, 65), (89, 78), (90, 310),$ $(96, 176), (104, 212), (117, 274), (124, 127), (125, 310),$ $(131, 176), (132, 65), (138, 78), (139, 310), (145, 176),$ $(153, 212), (166, 274), (173, 127), (174, 310), (180, 176),$ $(181, 65), (187, 78), (188, 310), (194, 176), (202, 212),$ $(215, 274), (222, 127), (229, 176), (230, 65), (236, 78),$ $(237, 310), (243, 176), (251, 212), (264, 274), (271, 127),$ $(272, 310), (278, 176), (279, 65), (285, 78), (286, 310),$ $(292, 176), (299, 127), (300, 212), (313, 274), (320, 127),$ $(321, 310), (327, 176), (328, 65), (334, 78), (335, 310), (341, 176)$

p	k	value	classes (a, b)
7	4	$(1, 0; -1)$	$(3, 1890), (17, 665), (31, 1351), (52, 861), (59, 567), (72, 291),$ $(72, 634), (72, 977), (72, 1320), (72, 1663), (72, 2006),$ $(72, 2349), (80, 1694), (101, 2233), (121, 326), (121, 669),$ $(121, 1012), (121, 1355), (121, 1698), (121, 2041), (121, 2384),$ $(150, 1204), (157, 1596), (178, 2380), (199, 175), (219, 137),$ $(219, 480), (219, 823), (219, 1166), (219, 1509), (219, 1852),$ $(219, 2195), (227, 322), (248, 1547), (297, 518), (325, 1008),$ $(346, 1890), (360, 665), (374, 1351), (402, 567), (415, 95),$ $(415, 438), (415, 781), (415, 1124), (415, 1467), (415, 1810),$ $(415, 2153), (423, 1694), (444, 2233), (493, 1204), (500, 1596),$ $(521, 2380), (542, 175), (570, 322), (591, 1547), (640, 518),$ $(668, 1008), (689, 1890), (703, 665), (717, 1351), (745, 567),$ $(758, 242), (758, 585), (758, 928), (758, 1271), (758, 1614),$ $(758, 1957), (758, 2300), (766, 1694), (787, 2233), (807, 81),$ $(807, 424), (807, 767), (807, 1110), (807, 1453), (807, 1796),$ $(807, 2139), (836, 1204), (843, 1596), (864, 2380), (885, 175),$ $(913, 322), (934, 1547), (983, 518), (1011, 1008), (1032, 1890),$ $(1046, 665), (1060, 1351), (1088, 567), (1101, 46), (1101, 389),$ $(1101, 732), (1101, 1075), (1101, 1418), (1101, 1761),$ $(1101, 2104), (1109, 1694), (1130, 2233), (1179, 1204),$ $(1186, 1596), (1207, 2380), (1228, 175), (1256, 322),$ $(1277, 1547), (1326, 518), (1354, 1008), (1375, 1890),$ $(1389, 665), (1403, 1351), (1424, 861), (1431, 567), (1452, 1694),$ $(1473, 2233), (1493, 179), (1493, 522), (1493, 865), (1493, 1208),$ $(1493, 1551), (1493, 1894), (1493, 2237), (1522, 1204),$ $(1529, 1596), (1550, 2380), (1571, 175), (1599, 322),$ $(1620, 1547), (1669, 518), (1697, 1008), (1718, 1890),$ $(1732, 665), (1746, 1351), (1767, 861), (1774, 567), (1795, 1694),$ $(1816, 2233), (1836, 228), (1836, 571), (1836, 914), (1836, 1257),$ $(1836, 1600), (1836, 1943), (1836, 2286), (1865, 1204),$ $(1872, 1596), (1893, 2380), (1914, 175), (1942, 322),$ $(1963, 1547), (2012, 518), (2040, 1008), (2061, 1890),$ $(2075, 665), (2089, 1351), (2138, 1694), (2159, 2233),$ $(2179, 277), (2179, 620), (2179, 963), (2179, 1306), (2179, 1649),$ $(2179, 1992), (2179, 2335), (2208, 1204), (2215, 1596),$ $(2236, 2380), (2257, 175), (2285, 322), (2306, 1547), (2355, 518),$ $(2383, 1008)$
7	4	$(1, 0; 1)$	$(2, 2036), (72, 286), (322, 19), (345, 664), (415, 1315),$ $(665, 362), (688, 1693), (758, 2344), (1008, 705), (1031, 321),$ $(1101, 972), (1374, 1350), (1444, 2001), (1694, 1391),$ $(1717, 2379), (1787, 629), (2037, 1734), (2060, 1007),$ $(2130, 1658), (2380, 2077)$

p	k	value	classes (a, b)
7	4	$(2, 0; 1)$	$(12, 274), (54, 1842), (61, 1989), (103, 2185), (110, 1303),$ $(152, 127), (159, 617), (201, 470), (208, 2332), (223, 2368),$ $(250, 813), (257, 1646), (306, 960), (355, 274), (397, 1842),$ $(404, 1989), (446, 2185), (453, 1303), (495, 127), (502, 617),$ $(544, 470), (551, 2332), (566, 2368), (593, 813), (600, 1646),$ $(649, 960), (698, 274), (740, 1842), (747, 1989), (789, 2185),$ $(838, 127), (845, 617), (887, 470), (894, 2332), (909, 2368),$ $(936, 813), (943, 1646), (992, 960), (1041, 274), (1083, 1842),$ $(1090, 1989), (1132, 2185), (1181, 127), (1188, 617), (1230, 470),$ $(1237, 2332), (1252, 2368), (1279, 813), (1286, 1646),$ $(1335, 960), (1384, 274), (1426, 1842), (1433, 1989),$ $(1475, 2185), (1524, 127), (1531, 617), (1573, 470), (1580, 2332),$ $(1595, 2368), (1622, 813), (1629, 1646), (1678, 960), (1727, 274),$ $(1769, 1842), (1776, 1989), (1818, 2185), (1825, 1303),$ $(1867, 127), (1874, 617), (1916, 470), (1923, 2332), (1938, 2368),$ $(1965, 813), (1972, 1646), (2021, 960), (2070, 274), (2112, 1842),$ $(2119, 1989), (2161, 2185), (2210, 127), (2217, 617), (2259, 470),$ $(2266, 2332), (2281, 2368), (2308, 813), (2315, 1646),$ $(2364, 960)$
7	5	$(1, 0; -1)$	$(395, 10465), (738, 3262), (1081, 12866), (2110, 8064),$ $(2117, 10171), (2796, 10465), (3139, 3262), (3482, 12866),$ $(4511, 8064), (4518, 10171), (5197, 10465), (5540, 3262),$ $(5883, 12866), (6912, 8064), (6919, 10171), (7598, 10465),$ $(7941, 3262), (8284, 12866), (9313, 8064), (9320, 10171),$ $(9999, 10465), (10342, 3262), (10685, 12866), (11714, 8064),$ $(11721, 10171), (12400, 10465), (12743, 3262), (13086, 12866),$ $(14115, 8064), (14122, 10171), (14801, 10465), (15144, 3262),$ $(15487, 12866), (16516, 8064), (16523, 10171)$
7	5	$(1, 0; 1)$	$(1351, 1048), (3752, 3449), (6153, 5850), (8554, 8251),$ $(10955, 10652), (13356, 13053), (15757, 15454)$
7	5	$(2, 0; 1)$	$(796, 8506), (1139, 3704), (1482, 15709), (2168, 6105),$ $(3197, 8506), (3540, 3704), (3883, 15709), (4569, 6105),$ $(5598, 8506), (5941, 3704), (6284, 15709), (6970, 6105),$ $(7999, 8506), (8342, 3704), (8685, 15709), (9371, 6105),$ $(10400, 8506), (10743, 3704), (11086, 15709), (11772, 6105),$ $(12801, 8506), (13144, 3704), (13487, 15709), (14173, 6105),$ $(15202, 8506), (15545, 3704), (15888, 15709), (16574, 6105)$
11	1	$(0, 0; 1)$	$(0, 3), (3, 3), (4, 5), (4, 7), (4, 10), (8, 8), (9, 0)$
13	1	$(0, 1; 1)$	$(0, 2), (12, 10)$

p	k	value	classes (a, b)
13	2	(0, 1; 1)	(0, 87), (2, 118), (3, 97), (4, 162), (5, 108), (6, 59), (8, 78), (9, 36), (13, 35), (15, 118), (16, 6), (17, 45), (18, 108), (19, 59), (21, 52), (21, 73), (22, 62), (26, 152), (28, 118), (29, 84), (30, 97), (31, 108), (32, 59), (34, 26), (34, 86), (35, 88), (39, 100), (41, 118), (42, 162), (43, 149), (44, 108), (45, 59), (46, 129), (47, 0), (47, 99), (48, 114), (52, 48), (54, 118), (55, 71), (56, 32), (57, 108), (58, 59), (60, 112), (60, 143), (61, 140), (65, 165), (67, 118), (69, 84), (70, 108), (71, 59), (72, 116), (73, 117), (73, 125), (74, 166), (78, 113), (80, 118), (81, 58), (82, 136), (83, 108), (84, 59), (86, 91), (86, 138), (87, 23), (91, 61), (93, 118), (94, 136), (95, 19), (96, 108), (97, 59), (98, 103), (99, 151), (100, 49), (104, 9), (106, 118), (107, 45), (108, 71), (109, 108), (110, 59), (112, 39), (112, 164), (117, 126), (119, 118), (120, 123), (121, 123), (122, 108), (123, 59), (125, 8), (125, 13), (126, 101), (130, 74), (132, 118), (133, 32), (134, 6), (136, 59), (138, 21), (138, 156), (139, 127), (143, 22), (145, 118), (146, 110), (147, 58), (148, 108), (149, 59), (151, 34), (151, 130), (152, 153), (156, 139), (158, 118), (159, 19), (160, 110), (161, 108), (162, 59), (163, 155), (164, 47), (165, 10)

p	k	value	classes (a, b)
13	3	(0, 1; 1)	(7, 1416), (8, 369), (8, 1243), (20, 142), (21, 2072), (33, 51), (34, 2085), (47, 408), (59, 1221), (60, 1435), (68, 1501), (73, 772), (85, 532), (86, 616), (99, 967), (99, 1755), (111, 181), (112, 1825), (113, 751), (124, 1780), (125, 993), (135, 784), (137, 168), (138, 668), (150, 1936), (151, 850), (164, 611), (164, 1539), (176, 233), (177, 538), (177, 1412), (189, 1156), (190, 44), (202, 1065), (203, 57), (216, 577), (228, 38), (229, 1604), (237, 318), (242, 941), (254, 1546), (255, 785), (268, 1136), (268, 1417), (280, 1195), (281, 1994), (282, 1089), (293, 597), (294, 1162), (304, 784), (306, 1182), (307, 837), (319, 753), (320, 1019), (333, 273), (333, 1708), (345, 1247), (346, 707), (346, 1581), (358, 2170), (359, 213), (371, 2079), (372, 226), (385, 746), (397, 1052), (398, 1773), (406, 1332), (411, 1110), (423, 363), (424, 954), (437, 1079), (437, 1305), (449, 12), (450, 2163), (451, 1427), (462, 1611), (463, 1331), (473, 784), (475, 2196), (476, 1006), (488, 1767), (489, 1188), (502, 1877), (502, 2132), (514, 64), (515, 876), (515, 1750), (527, 987), (528, 382), (540, 896), (541, 395), (554, 915), (566, 2066), (567, 1942), (575, 149), (580, 1279), (592, 1377), (593, 1123), (606, 741), (606, 1474), (618, 1026), (619, 135), (620, 1765), (631, 428), (632, 1500), (642, 784), (644, 1013), (645, 1175), (657, 584), (658, 1357), (671, 1794), (671, 2046), (683, 1078), (684, 1045), (684, 1919), (696, 2001), (697, 551), (709, 1910), (710, 564), (723, 1084), (735, 883), (736, 2111), (744, 1163), (749, 1448), (761, 194), (762, 1292), (775, 403), (775, 1643), (787, 2040), (788, 304), (789, 2103), (800, 1442), (801, 1669), (811, 784), (813, 2027), (814, 1344), (826, 1598), (827, 1526), (840, 18), (840, 1456), (852, 2092), (853, 1214), (853, 2088), (865, 818), (866, 720), (878, 727), (879, 733), (892, 1253), (904, 1897), (905, 83), (913, 2177), (918, 1617), (930, 1208), (931, 1461), (944, 65), (944, 1812), (956, 857), (957, 473), (958, 244), (969, 259), (970, 1838), (980, 784), (982, 844), (983, 1513), (995, 415), (996, 1695), (1009, 187), (1009, 1118), (1021, 909), (1022, 60), (1022, 1383), (1034, 1832), (1035, 889), (1047, 1741), (1048, 902), (1061, 1422), (1073, 714), (1074, 252), (1082, 994), (1087, 1786), (1099, 25), (1100, 1630), (1113, 1924), (1113, 1981), (1125, 1871), (1126, 642), (1127, 582), (1138, 1273), (1139, 2007), (1149, 784), (1151, 1858), (1152, 1682), (1164, 1429), (1165, 1864), (1178, 356), (1178, 780), (1190, 1923), (1191, 229), (1191, 1552), (1203, 649), (1204, 1058), (1216, 558), (1217, 1071), (1230, 1591), (1242, 1728), (1243, 421), (1251, 2008), (1256, 1955), (1268, 1039), (1269, 1799), (1282, 1586), (1282, 2150), (1294, 688), (1295, 811), (1296, 920), (1307, 90), (1308, 2176), (1318, 784), (1320, 675), (1321, 1851), (1333, 246), (1334, 2033), (1347, 442), (1347, 525), (1359, 740), (1360, 398), (1360, 1721), (1372, 1663), (1373, 1227), (1385, 1572), (1386, 1240), (1399, 1760), (1411, 545), (1412, 590), (1420, 825), (1425, 2124), (1437, 2053), (1438, 1968), (1451, 122), (1451, 1248), (1463, 1702), (1464, 980), (1465, 1258), (1476, 1104), (1477, 148), (1487, 784), (1489, 1689), (1490, 2020), (1502, 1260), (1503, 5), (1516, 104), (1516, 694), (1528, 1754), (1529, 567), (1529, 1890), (1541, 480), (1542, 1396), (1554, 389), (1555, 1409), (1568, 1929), (1580, 1559), (1581, 759), (1589, 1839), (1594, 96), (1606, 870), (1607, 2137), (1620, 291), (1620, 910), (1632, 519), (1633, 1149), (1634, 1596), (1645, 2118), (1646, 317), (1656, 784), (1658, 506), (1659, 2189), (1671, 77), (1672, 174), (1685, 863), (1685, 1963), (1697, 571), (1698, 736), (1698, 2059), (1710, 1494), (1711, 1565), (1723, 1403), (1724, 1578), (1737, 2098), (1749, 376), (1750, 928), (1758, 656), (1763, 265), (1775, 1884), (1776, 109), (1789, 460), (1789, 572), (1801, 1533), (1802, 1318), (1803, 1934), (1814, 935), (1815, 486), (1825, 784), (1827, 1520), (1828, 161), (1840, 1001), (1841, 242), (1854, 1022)

p	k	value	classes (a, b)
19	1	(0, 0; 1)	(0, 5), (0, 9), (1, 6), (6, 1), (11, 1), (12, 17), (16, 17)
19	2	(0, 0; 1)	(3, 162), (4, 82), (10, 82), (14, 13), (15, 134), (22, 333), (23, 82), (33, 260), (34, 305), (41, 143), (42, 82), (48, 25), (52, 146), (53, 115), (60, 314), (61, 82), (67, 177), (71, 32), (72, 286), (79, 124), (80, 82), (86, 329), (90, 279), (91, 96), (98, 295), (99, 82), (105, 120), (109, 165), (110, 267), (117, 105), (118, 82), (124, 272), (128, 51), (129, 77), (136, 276), (137, 82), (143, 63), (147, 298), (148, 248), (155, 86), (156, 82), (162, 215), (166, 184), (167, 58), (174, 257), (175, 82), (181, 6), (186, 229), (193, 67), (194, 82), (200, 158), (204, 317), (205, 39), (212, 238), (213, 82), (219, 310), (223, 203), (224, 210), (231, 48), (232, 82), (238, 101), (242, 89), (243, 20), (250, 219), (251, 82), (257, 253), (261, 336), (262, 191), (269, 14), (269, 29), (269, 33), (269, 52), (269, 71), (269, 90), (269, 109), (269, 128), (269, 147), (269, 166), (269, 185), (269, 204), (269, 223), (269, 242), (269, 261), (269, 280), (269, 299), (269, 318), (269, 337), (269, 356), (270, 82), (276, 44), (280, 222), (281, 1), (288, 200), (289, 82), (295, 196), (299, 108), (300, 172), (307, 10), (308, 82), (314, 348), (318, 355), (319, 343), (327, 82), (333, 139), (337, 241), (338, 153), (345, 352), (346, 82), (352, 291), (356, 127), (357, 324)
19	3	(0, 0; 1)	(29, 2039), (185, 6207), (326, 5957), (390, 4927), (546, 4041), (687, 2347), (751, 956), (907, 1875), (1048, 5596), (1112, 3844), (1268, 6568), (1409, 1986), (1629, 4402), (1770, 5235), (1834, 2761), (1990, 2236), (2131, 1625), (2195, 5649), (2351, 70), (2492, 4874), (2556, 1678), (2712, 4763), (2853, 1264), (2917, 4566), (3073, 2597), (3214, 4513), (3278, 595), (3434, 431), (3575, 903), (3639, 3483), (3795, 5124), (3936, 4152), (4000, 6371), (4156, 2958), (4297, 542), (4361, 2400), (4517, 792), (4658, 3791), (4722, 5288), (4878, 5485), (5019, 181), (5083, 1317), (5239, 3319), (5380, 3430), (5444, 4205), (5600, 1153), (5741, 6679), (5805, 234), (5961, 5846), (6102, 3069), (6166, 3122), (6322, 3680), (6463, 6318), (6527, 6010), (6683, 1514), (6824, 2708)
19	4	(0, 0; 1)	(1473, 27309), (8332, 82181), (15191, 6732), (22050, 61604), (28909, 116476), (35768, 41027), (42627, 95899), (49486, 20450), (56345, 75322), (63204, 130194), (70063, 54745), (76922, 109617), (83781, 34168), (90640, 89040), (97499, 13591), (104358, 68463), (111217, 123335), (118076, 47886), (124935, 102758)
31	1	(0, 0; 1)	(1, 9), (1, 26), (1, 27), (2, 9), (3, 10), (5, 21), (7, 8), (9, 2), (11, 14), (12, 20), (13, 25), (14, 0), (15, 11), (15, 13), (18, 29), (20, 29), (21, 1), (24, 23), (28, 8), (30, 19)

p	k	value	classes (a, b)
31	2	(0, 0; 1)	(2, 585), (2, 894), (10, 393), (11, 809), (12, 241), (12, 638), (15, 658), (33, 740), (33, 956), (41, 83), (42, 654), (43, 80), (43, 923), (46, 658), (60, 247), (64, 57), (64, 895), (72, 734), (73, 499), (74, 483), (74, 644), (77, 658), (91, 495), (95, 89), (95, 119), (103, 424), (104, 344), (105, 365), (105, 886), (108, 658), (122, 743), (126, 181), (126, 244), (134, 114), (135, 189), (136, 86), (136, 328), (139, 658), (153, 30), (157, 243), (157, 399), (165, 765), (166, 34), (167, 731), (167, 768), (170, 658), (188, 305), (188, 554), (196, 455), (197, 840), (198, 173), (198, 489), (201, 658), (219, 367), (219, 709), (227, 145), (228, 685), (229, 210), (229, 576), (232, 658), (250, 429), (250, 864), (258, 796), (259, 530), (260, 18), (260, 892), (263, 658), (281, 58), (281, 491), (289, 486), (290, 375), (291, 421), (291, 613), (294, 658), (312, 213), (312, 553), (320, 176), (321, 220), (322, 334), (322, 824), (325, 658), (343, 368), (343, 615), (351, 827), (352, 65), (353, 55), (353, 266), (356, 658), (374, 523), (374, 677), (382, 517), (383, 871), (384, 669), (384, 737), (387, 658), (401, 92), (405, 678), (405, 739), (413, 207), (414, 716), (415, 111), (415, 458), (418, 658), (432, 340), (436, 801), (436, 833), (444, 858), (445, 561), (446, 179), (446, 514), (449, 658), (463, 588), (467, 27), (467, 863), (475, 548), (476, 406), (477, 861), (477, 917), (480, 658), (498, 182), (506, 238), (507, 251), (508, 359), (508, 582), (511, 658), (525, 123), (529, 26), (529, 337), (537, 889), (538, 96), (539, 303), (539, 762), (542, 658), (560, 88), (560, 492), (568, 579), (569, 902), (570, 24), (570, 204), (573, 658), (587, 619), (591, 150), (591, 647), (599, 269), (600, 747), (601, 607), (601, 706), (604, 658), (622, 212), (622, 802), (630, 920), (631, 592), (632, 49), (632, 427), (635, 658), (653, 274), (653, 957), (661, 610), (662, 437), (663, 148), (663, 452), (666, 658), (684, 151), (684, 336), (692, 300), (693, 282), (694, 830), (694, 855), (697, 658), (715, 306), (715, 398), (723, 951), (724, 127), (725, 297), (725, 551), (728, 658), (746, 460), (746, 461), (754, 641), (755, 933), (756, 272), (756, 700), (759, 658), (777, 522), (777, 616), (785, 331), (786, 778), (787, 142), (787, 954), (790, 658), (808, 584), (808, 771), (816, 21), (817, 623), (818, 545), (818, 675), (821, 658), (839, 646), (839, 926), (847, 672), (848, 468), (849, 396), (849, 948), (852, 658), (870, 120), (870, 708), (878, 362), (879, 313), (880, 117), (880, 390), (883, 658), (897, 216), (901, 275), (901, 770), (909, 52), (910, 158), (911, 793), (911, 799), (914, 658), (932, 430), (932, 832), (940, 703), (941, 3), (942, 235), (942, 520), (945, 658), (959, 712)

p	k	value	classes (a, b)
31	3	(0, 0; 1)	(29, 3843), (184, 18537), (215, 26473), (246, 14228), (277, 11593), (308, 18568), (339, 5362), (370, 1766), (494, 23900), (498, 27833), (556, 3254), (618, 21048), (649, 14569), (680, 17700), (711, 650), (742, 23001), (773, 25171), (804, 7160), (835, 28550), (866, 29759), (928, 1425), (990, 11531), (1145, 26225), (1176, 4370), (1207, 21916), (1238, 19281), (1269, 26256), (1300, 13050), (1331, 9454), (1455, 1797), (1459, 29755), (1517, 10942), (1579, 28736), (1610, 22257), (1641, 25388), (1672, 8338), (1703, 898), (1734, 3068), (1765, 14848), (1796, 6447), (1827, 7656), (1889, 9113), (1951, 19219), (2106, 4122), (2137, 12058), (2168, 29604), (2199, 26969), (2230, 4153), (2261, 20738), (2292, 17142), (2416, 9485), (2420, 1886), (2478, 18630), (2540, 6633), (2571, 154), (2602, 3285), (2633, 16026), (2664, 8586), (2695, 10756), (2726, 22536), (2757, 14135), (2788, 15344), (2850, 16801), (2912, 26907), (3067, 11810), (3098, 19746), (3129, 7501), (3160, 4866), (3191, 11841), (3222, 28426), (3253, 24830), (3377, 17173), (3381, 3808), (3439, 26318), (3501, 14321), (3532, 7842), (3563, 10973), (3594, 23714), (3625, 16274), (3656, 18444), (3687, 433), (3718, 21823), (3749, 23032), (3811, 24489), (3873, 4804), (4028, 19498), (4059, 27434), (4090, 15189), (4121, 12554), (4152, 19529), (4183, 6323), (4214, 2727), (4338, 24861), (4342, 5730), (4400, 4215), (4462, 22009), (4493, 15530), (4524, 18661), (4555, 1611), (4586, 23962), (4617, 26132), (4648, 8121), (4679, 29511), (4710, 929), (4772, 2386), (4834, 12492), (4989, 27186), (5020, 5331), (5051, 22877), (5082, 20242), (5113, 27217), (5144, 14011), (5175, 10415), (5299, 2758), (5303, 7652), (5361, 11903), (5423, 29697), (5454, 23218), (5485, 26349), (5516, 9299), (5547, 1859), (5578, 4029), (5609, 15809), (5640, 7408), (5671, 8617), (5733, 10074), (5795, 20180), (5950, 5083), (5981, 13019), (6012, 774), (6043, 27930), (6074, 5114), (6105, 21699), (6136, 18103), (6260, 10446), (6264, 9574), (6322, 19591), (6384, 7594), (6415, 1115), (6446, 4246), (6477, 16987), (6508, 9547), (6539, 11717), (6570, 23497), (6601, 15096), (6632, 16305), (6694, 17762), (6756, 27868), (6911, 12771), (6942, 20707), (6973, 8462), (7004, 5827), (7035, 12802), (7066, 29387), (7097, 25791), (7221, 18134), (7225, 11496), (7283, 27279), (7345, 15282), (7376, 8803), (7407, 11934), (7438, 24675), (7469, 17235), (7500, 19405), (7531, 1394), (7562, 22784), (7593, 23993), (7655, 25450), (7717, 5765), (7872, 20459), (7903, 28395), (7934, 16150), (7965, 13515), (7996, 20490), (8027, 7284), (8058, 3688), (8182, 25822), (8186, 13418), (8244, 5176), (8306, 22970), (8337, 16491), (8368, 19622), (8399, 2572), (8430, 24923), (8461, 27093), (8492, 9082), (8523, 681), (8554, 1890), (8616, 3347), (8678, 13453), (8833, 28147), (8864, 6292), (8895, 23838), (8926, 21203), (8957, 28178), (8988, 14972), (9019, 11376), (9143, 3719), (9147, 15340), (9205, 12864), (9267, 867), (9298, 24179), (9329, 27310), (9360, 10260), (9391, 2820), (9422, 4990), (9453, 16770), (9484, 8369), (9515, 9578), (9577, 11035), (9639, 21141), (9794, 6044), (9825, 13980), (9856, 1735), (9887, 28891), (9918, 6075), (9949, 22660), (9980, 19064), (10104, 11407), (10108, 17262), (10166, 20552), (10228, 8555), (10259, 2076), (10290, 5207), (10321, 17948), (10352, 10508), (10383, 12678), (10414, 24458), (10445, 16057), (10476, 17266), (10538, 18723), (10600, 28829), (10755, 13732), (10786, 21668), (10817, 9423), (10848, 6788), (10879, 13763), (10949, 557), (10941, 26752), (11065, 19095), (11069, 19184), (11127, 28240), (11189, 16243), (11220, 9764), (11251, 12895), (11282, 25636), (11313, 18196), (11344, 20366), (11375, 2355), (11406, 23745), (11437, 24954), (11499, 26411), (11561, 6726), (11716, 21420), (11747, 20256), (11778, 17111)

p	k	value	classes (a, b)
37	1	$(0, 0; 1)$	$(1, 4), (2, 0), (2, 31), (3, 7), (5, 0), (5, 6), (5, 31), (6, 27), (7, 9),$ $(7, 29), (7, 36), (9, 12), (10, 18), (12, 13), (12, 35), (13, 13),$ $(13, 26), (13, 35), (14, 0), (14, 6), (14, 31), (15, 12), (16, 0),$ $(16, 6), (16, 31), (17, 3), (18, 1), (18, 28), (19, 12), (20, 26),$ $(20, 35), (21, 21), (22, 4), (25, 22), (26, 3), (28, 5), (29, 13),$ $(29, 26), (29, 35), (31, 12), (32, 2), (32, 11), (32, 24), (33, 34),$ $(34, 15), (35, 18), (36, 19)$

p	k	value	classes (a, b)
37	2	(0, 0; 1)	(2, 820), (3, 948), (4, 1246), (8, 84), (12, 63), (18, 563), (20, 1160), (23, 205), (39, 1338), (40, 430), (41, 358), (45, 158), (49, 211), (55, 563), (57, 716), (60, 871), (76, 487), (77, 1281), (78, 839), (82, 232), (86, 359), (92, 563), (94, 272), (97, 168), (113, 1005), (114, 763), (115, 1320), (119, 306), (123, 507), (129, 563), (131, 1197), (134, 834), (150, 154), (151, 245), (152, 432), (156, 380), (160, 655), (166, 563), (168, 753), (171, 131), (187, 672), (188, 1096), (189, 913), (193, 454), (197, 803), (203, 563), (205, 309), (208, 797), (224, 1190), (225, 578), (226, 25), (230, 528), (234, 951), (240, 563), (242, 1234), (245, 94), (261, 339), (262, 60), (263, 506), (267, 602), (271, 1099), (277, 563), (279, 790), (282, 760), (298, 857), (299, 911), (300, 987), (304, 676), (308, 1247), (314, 563), (316, 346), (319, 57), (335, 6), (336, 393), (337, 99), (341, 750), (345, 26), (351, 563), (353, 1271), (356, 723), (372, 524), (373, 1244), (378, 824), (382, 174), (388, 563), (390, 827), (393, 20), (409, 1042), (410, 726), (411, 1061), (415, 898), (419, 322), (425, 563), (427, 383), (430, 686), (446, 191), (447, 208), (448, 173), (452, 972), (456, 470), (462, 563), (464, 1308), (467, 1352), (483, 709), (484, 1059), (485, 654), (489, 1046), (493, 618), (499, 563), (501, 864), (504, 649), (520, 1227), (521, 541), (522, 1135), (526, 1120), (530, 766), (536, 563), (538, 420), (541, 1315), (557, 376), (558, 23), (559, 247), (563, 1194), (567, 914), (573, 563), (575, 1345), (578, 612), (594, 894), (595, 874), (596, 728), (600, 1268), (604, 1062), (610, 563), (612, 901), (615, 1278), (631, 43), (632, 356), (633, 1209), (637, 1342), (641, 1210), (647, 563), (649, 457), (652, 575), (668, 561), (669, 1207), (670, 321), (674, 47), (678, 1358), (684, 563), (686, 13), (689, 1241), (705, 1079), (706, 689), (707, 802), (711, 121), (715, 137), (721, 563), (723, 938), (726, 538), (742, 228), (743, 171), (744, 1283), (748, 195), (752, 285), (758, 563), (760, 494), (763, 1204), (779, 746), (780, 1022), (781, 395), (785, 269), (789, 433), (795, 563), (797, 50), (800, 501), (816, 1264), (817, 504), (818, 876), (822, 343), (826, 581), (832, 563), (834, 975), (837, 1167), (853, 413), (854, 1355), (855, 1357), (859, 417), (863, 729), (869, 563), (871, 531), (874, 464), (890, 931), (891, 837), (892, 469), (896, 491), (900, 877), (906, 563), (908, 87), (911, 1130), (927, 80), (928, 319), (929, 950), (933, 565), (937, 1025), (943, 563), (945, 1012), (948, 427), (964, 598), (965, 1170), (966, 62), (970, 639), (974, 1173), (980, 563), (982, 568), (985, 1093), (1001, 1116), (1002, 652), (1003, 543), (1007, 713), (1011, 1321), (1017, 563), (1019, 124), (1022, 390), (1038, 265), (1039, 134), (1040, 1024), (1044, 787), (1048, 100), (1054, 563), (1056, 1049), (1059, 1056), (1075, 783), (1076, 985), (1077, 136), (1081, 861), (1085, 248), (1091, 563), (1096, 353), (1112, 1301), (1113, 467), (1114, 617), (1118, 935), (1122, 396), (1128, 563), (1130, 161), (1133, 1019), (1149, 450), (1150, 1318), (1151, 1098), (1155, 1009), (1159, 544), (1165, 563), (1167, 1086), (1170, 316), (1186, 968), (1187, 800), (1188, 210), (1192, 1083), (1196, 692), (1202, 563), (1204, 642), (1207, 982), (1223, 117), (1224, 282), (1225, 691), (1229, 1157), (1233, 840), (1239, 563), (1241, 198), (1244, 279), (1260, 635), (1261, 1133), (1262, 1172), (1266, 1231), (1270, 988), (1276, 563), (1278, 1123), (1281, 945), (1297, 1153), (1298, 615), (1299, 284), (1303, 1305), (1307, 1136), (1313, 563), (1315, 679), (1318, 242), (1334, 302), (1335, 97), (1336, 765), (1340, 10), (1344, 1284), (1350, 563), (1352, 235), (1355, 908)

p	k	value	classes (a, b)
37	3	$(0, 0; 1)$	(374, 40281), (1093, 23878), (1743, 7425), (2462, 7450), (3112, 25222), (3831, 41675), (4481, 43019), (5200, 25247), (5850, 10163), (6569, 8819), (7219, 27960), (7938, 43044), (8588, 45757), (9307, 26616), (9957, 12901), (10676, 10188), (11326, 30698), (12045, 44413), (12695, 48495), (13414, 27985), (14064, 15639), (14783, 11557), (15433, 33436), (16152, 45782), (16802, 580), (17521, 29354), (18171, 18377), (18890, 12926), (19540, 36174), (20259, 47151), (20909, 3318), (21628, 30723), (22278, 21115), (22997, 14295), (23647, 38912), (24366, 48520), (25016, 6056), (25735, 32092), (26385, 23853), (27104, 15664), (27754, 41650), (28473, 49889), (29123, 8794), (29842, 33461), (30492, 26591), (31211, 17033), (31861, 44388), (32580, 605), (33230, 11532), (33949, 34830), (34599, 29329), (35318, 18402), (35968, 47126), (36687, 1974), (37337, 14270), (38056, 36199), (38706, 32067), (39425, 19771), (40075, 49864), (40794, 3343), (41444, 17008), (42163, 37568), (42813, 34805), (43532, 21140), (44182, 1949), (44901, 4712), (45551, 19746), (46270, 38937), (46920, 37543), (47639, 22509), (48289, 4687), (49008, 6081), (49658, 22484), (50377, 40306)
43	1	$(0, 0; 1)$	(1, 21), (1, 29), (1, 36), (2, 14), (2, 22), (3, 8), (6, 20), (6, 33), (7, 8), (8, 2), (10, 4), (11, 3), (13, 11), (16, 27), (17, 15), (17, 37), (18, 35), (19, 1), (19, 18), (19, 24), (20, 30), (22, 32), (23, 31), (26, 16), (28, 16), (29, 12), (32, 11), (33, 1), (33, 18), (33, 24), (34, 38), (35, 30), (36, 39), (37, 32), (38, 38), (41, 2), (42, 35)
43	2	$(0, 0; 1)$	(2, 1641), (17, 249), (40, 131), (45, 910), (60, 1711), (83, 131), (88, 179), (103, 1324), (126, 131), (131, 1297), (146, 937), (169, 131), (174, 566), (189, 550), (212, 131), (217, 1684), (232, 163), (255, 131), (260, 953), (275, 1625), (298, 131), (303, 222), (318, 1238), (341, 131), (346, 1340), (361, 851), (384, 131), (389, 609), (404, 464), (427, 131), (432, 1727), (447, 77), (470, 131), (475, 996), (490, 1539), (513, 131), (518, 265), (533, 1152), (556, 131), (561, 1383), (576, 765), (599, 131), (604, 652), (619, 378), (642, 131), (647, 1770), (662, 1840), (685, 131), (690, 1039), (705, 1453), (728, 131), (733, 308), (748, 1066), (771, 131), (776, 1426), (791, 679), (814, 131), (819, 695), (834, 292), (857, 131), (862, 1813), (877, 1754), (900, 131), (905, 1082), (920, 1367), (943, 131), (948, 351), (963, 980), (986, 131), (991, 1469), (1006, 593), (1029, 131), (1034, 738), (1049, 206), (1072, 131), (1077, 7), (1092, 1668), (1115, 131), (1120, 1125), (1135, 1281), (1158, 131), (1163, 394), (1178, 894), (1201, 131), (1206, 1512), (1221, 507), (1244, 131), (1249, 781), (1264, 120), (1287, 131), (1292, 50), (1307, 1582), (1330, 131), (1335, 1168), (1350, 1195), (1373, 131), (1378, 437), (1393, 808), (1416, 131), (1421, 1555), (1436, 421), (1459, 131), (1464, 824), (1479, 34), (1502, 131), (1507, 93), (1522, 1496), (1545, 131), (1550, 1211), (1565, 1109), (1588, 131), (1593, 480), (1608, 722), (1631, 131), (1636, 1598), (1651, 335), (1674, 131), (1679, 867), (1694, 1797), (1717, 131), (1722, 136), (1737, 1410), (1760, 131), (1765, 1254), (1780, 1023), (1803, 131), (1808, 523), (1823, 636), (1846, 131)